Model for "URu2Si2"

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General Condition

- Basis type: `jml`
- SAMB selection:
  - Type: `[Q, G]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_{1g}, A_{2g}, B_{1g}, B_{2g}, E_g, A_{1u}, A_{2u}, B_{1u}, B_{2u}, E_u]
  - Spin (s): `[0, 1]`
- Atomic selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_{1g}, A_{2g}, B_{1g}, B_{2g}, E_g, A_{1u}, A_{2u}, B_{1u}, B_{2u}, E_u]
  - Spin (s): `[0, 1]`
- Site-cluster selection:
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_{1g}, A_{2g}, B_{1g}, B_{2g}, E_g, A_{1u}, A_{2u}, B_{1u}, B_{2u}, E_u]
- Bond-cluster selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_{1g}, A_{2g}, B_{1g}, B_{2g}, E_g, A_{1u}, A_{2u}, B_{1u}, B_{2u}, E_u]
- Max. neighbor: `10`
- Search cell range: `(-2, 3), (-2, 3), (-2, 3)`
- Toroidal priority: `false`

Group and Unit Cell

- Group: SG No. 139 $D_{4\mathrm{h}}^{17}$ $I4/mmm$ [tetragonal]
- Associated point group: PG No. 139 $D_{4\mathrm{h}}$ $4/mmm$ [tetragonal]
- Unit cell:
  $a =$ `4.12600`, $b =$ `4.12600`, $c =$ `9.58000`, $\alpha =$ `90.0`, $\beta =$ `90.0`, $\gamma =$ `90.0`
- Lattice vectors (conventional cell):
  $\boldsymbol{a}_1 =$ `[ 4.12600, 0.00000, 0.00000]`
  $\boldsymbol{a}_2 =$ `[ 0.00000, 4.12600, 0.00000]`
  $\boldsymbol{a}_3 =$ `[ 0.00000, 0.00000, 9.58000]`

- Plus sets:
  $+[0,\,0,\,0]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$

Symmetry Operation

Table 1: Symmetry operation

#	SO	#	SO	#	SO	#	SO	#	SO
1	$\{1\|0\}$	2	$\{2_{001}\|0\}$	3	$\{4^{+}_{001}\|0\}$	4	$\{4^{-}_{001}\|0\}$	5	$\{2_{010}\|0\}$
6	$\{2_{100}\|0\}$	7	$\{2_{110}\|0\}$	8	$\{2_{1-10}\|0\}$	9	$\{-1\|0\}$	10	$\{\mathrm{m}_{001}\|0\}$
11	$\{-4^{+}_{001}\|0\}$	12	$\{-4^{-}_{001}\|0\}$	13	$\{\mathrm{m}_{010}\|0\}$	14	$\{\mathrm{m}_{100}\|0\}$	15	$\{\mathrm{m}_{110}\|0\}$
16	$\{\mathrm{m}_{1-10}\|0\}$								

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{Q}_0(A_{1g})$	A_{1g}	0	$Q,\,T$	-	-	1
2	$\mathbb{Q}_2(A_{1g})$	A_{1g}	2	$Q,\,T$	-	-	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
3	$\mathbb{Q}_4(A_{1g},1)$	A_{1g}	4	$Q,\,T$	1	-	$\frac{\sqrt{21}\left(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4\right)}{6}$
4	$\mathbb{Q}_4(A_{1g},2)$	A_{1g}	4	$Q,\,T$	2	-	$-\frac{\sqrt{15}\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{12}$
5	$\mathbb{G}_5(A_{1g})$	A_{1g}	5	$G,\,M$	-	-	$\frac{3\sqrt{35}xyz(x-y)(x+y)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
6	$\mathbb{Q}_6(A_{1g},1)$	A_{1g}	6	Q, T	1	-	$\frac{\sqrt{2}\left(2x^6-15x^4y^2-15x^4z^2-15x^2y^4+180x^2y^2z^2-15x^2z^4+2y^6-15y^4z^2-15y^2z^4+2z^6\right)}{8}$
7	$\mathbb{Q}_6(A_{1g},2)$	A_{1g}	6	Q, T	2	-	$-\frac{\sqrt{14}\left(x^6-15x^4z^2+15x^2z^4+y^6-15y^4z^2+15y^2z^4-2z^6\right)}{8}$
8	$\mathbb{G}_7(A_{1g})$	A_{1g}	7	G, M	-	-	$-\frac{\sqrt{231}xyz(x-y)(x+y)\left(3x^2+3y^2-10z^2\right)}{4}$
9	$\mathbb{G}_0(A_{1u})$	A_{1u}	0	G, M	-	-	1
10	$\mathbb{G}_2(A_{1u})$	A_{1u}	2	G, M	-	-	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
11	$\mathbb{G}_4(A_{1u},1)$	A_{1u}	4	G, M	1	-	$\frac{\sqrt{21}\left(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4\right)}{6}$
12	$\mathbb{G}_4(A_{1u},2)$	A_{1u}	4	G, M	2	-	$-\frac{\sqrt{15}\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{12}$
13	$\mathbb{Q}_5(A_{1u})$	A_{1u}	5	Q, T	-	-	$\frac{3\sqrt{35}xyz(x-y)(x+y)}{2}$
14	$\mathbb{G}_6(A_{1u},1)$	A_{1u}	6	G, M	1	-	$\frac{\sqrt{2}\left(2x^6-15x^4y^2-15x^4z^2-15x^2y^4+180x^2y^2z^2-15x^2z^4+2y^6-15y^4z^2-15y^2z^4+2z^6\right)}{8}$
15	$\mathbb{G}_6(A_{1u},2)$	A_{1u}	6	G, M	2	-	$-\frac{\sqrt{14}\left(x^6-15x^4z^2+15x^2z^4+y^6-15y^4z^2+15y^2z^4-2z^6\right)}{8}$
16	$\mathbb{Q}_7(A_{1u})$	A_{1u}	7	Q, T	-	-	$-\frac{\sqrt{231}xyz(x-y)(x+y)\left(3x^2+3y^2-10z^2\right)}{4}$
17	$\mathbb{G}_8(A_{1u},1)$	A_{1u}	8	G, M	1	-	$\frac{\sqrt{33}\left(x^8-14x^6y^2-14x^6z^2+35x^4y^4+35x^4z^4-14x^2y^6-14x^2z^6+y^8-14y^6z^2+35y^4z^4-14y^2z^6+z^8\right)}{8}$
18	$\mathbb{G}_1(A_{2g})$	A_{2g}	1	G, M	-	-	z
19	$\mathbb{G}_3(A_{2g})$	A_{2g}	3	G, M	-	-	$-\frac{z\left(3x^2+3y^2-2z^2\right)}{2}$
20	$\mathbb{Q}_4(A_{2g})$	A_{2g}	4	Q, T	-	-	$\frac{\sqrt{35}xy(x-y)(x+y)}{2}$
21	$\mathbb{G}_5(A_{2g},1)$	A_{2g}	5	G, M	1	-	$\frac{z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{8}$
22	$\mathbb{G}_5(A_{2g},2)$	A_{2g}	5	G, M	2	-	$\frac{3\sqrt{35}z\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{8}$
23	$\mathbb{Q}_6(A_{2g})$	A_{2g}	6	Q, T	-	-	$-\frac{3\sqrt{7}xy(x-y)(x+y)\left(x^2+y^2-10z^2\right)}{4}$
24	$\mathbb{G}_7(A_{2g},1)$	A_{2g}	7	G, M	1	-	$-\frac{z\left(35x^6+105x^4y^2-210x^4z^2+105x^2y^4-420x^2y^2z^2+168x^2z^4+35y^6-210y^4z^2+168y^2z^4-16z^6\right)}{16}$
25	$\mathbb{G}_7(A_{2g},2)$	A_{2g}	7	G, M	2	-	$-\frac{\sqrt{231}z\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)\left(3x^2+3y^2-10z^2\right)}{16}$
26	$\mathbb{Q}_1(A_{2u})$	A_{2u}	1	Q, T	-	-	z

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
27	$\mathbb{Q}_3(A_{2u})$	A_{2u}	3	Q, T	-	-	$-\frac{z\left(3x^2+3y^2-2z^2\right)}{2}$
28	$\mathbb{G}_4(A_{2u})$	A_{2u}	4	G, M	-	-	$\frac{\sqrt{35}xy(x-y)(x+y)}{2}$
29	$\mathbb{Q}_5(A_{2u},1)$	A_{2u}	5	Q, T	1	-	$\frac{z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{8}$
30	$\mathbb{Q}_5(A_{2u},2)$	A_{2u}	5	Q, T	2	-	$\frac{3\sqrt{35}z\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{8}$
31	$\mathbb{G}_6(A_{2u})$	A_{2u}	6	G, M	-	-	$-\frac{3\sqrt{7}xy(x-y)(x+y)\left(x^2+y^2-10z^2\right)}{4}$
32	$\mathbb{Q}_7(A_{2u},1)$	A_{2u}	7	Q, T	1	-	$-\frac{z\left(35x^6+105x^4y^2-210x^4z^2+105x^2y^4-420x^2y^2z^2+168x^2z^4+35y^6-210y^4z^2+168y^2z^4-16z^6\right)}{16}$
33	$\mathbb{Q}_7(A_{2u},2)$	A_{2u}	7	Q, T	2	-	$-\frac{\sqrt{231}z\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)\left(3x^2+3y^2-10z^2\right)}{16}$
34	$\mathbb{G}_8(A_{2u},1)$	A_{2u}	8	G, M	1	-	$\frac{3\sqrt{715}xy(x-y)(x+y)\left(x^2-2xy-y^2\right)\left(x^2+2xy-y^2\right)}{16}$
35	$\mathbb{Q}_2(B_{1g})$	B_{1g}	2	Q, T	-	-	$\frac{\sqrt{3}(x-y)(x+y)}{2}$
36	$\mathbb{G}_3(B_{1g})$	B_{1g}	3	G, M	-	-	$\sqrt{15}xyz$
37	$\mathbb{Q}_4(B_{1g})$	B_{1g}	4	Q, T	-	-	$\frac{\sqrt{5}(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{4}$
38	$\mathbb{G}_5(B_{1g})$	B_{1g}	5	G, M	-	-	$\frac{\sqrt{105}xyz\left(x^2+y^2-2z^2\right)}{2}$
39	$\mathbb{Q}_6(B_{1g},1)$	B_{1g}	6	Q, T	1	-	$-\frac{\sqrt{2310}(x-y)(x+y)(x-z)(x+z)(y-z)(y+z)}{8}$
40	$\mathbb{Q}_6(B_{1g},2)$	B_{1g}	6	Q, T	2	-	$\frac{\sqrt{42}(x-y)(x+y)\left(x^4-9x^2y^2-5x^2z^2+y^4-5y^2z^2+5z^4\right)}{8}$
41	$\mathbb{G}_7(B_{1g},1)$	B_{1g}	7	G, M	1	-	$\frac{\sqrt{91}xyz\left(3x^4-5x^2y^2-5x^2z^2+3y^4-5y^2z^2+3z^4\right)}{2}$
42	$\mathbb{G}_7(B_{1g},2)$	B_{1g}	7	G, M	2	-	$-\frac{\sqrt{77}xyz\left(3x^4-20x^2y^2+10x^2z^2+3y^4+10y^2z^2-6z^4\right)}{4}$
43	$\mathbb{G}_2(B_{1u})$	B_{1u}	2	G, M	-	-	$\frac{\sqrt{3}(x-y)(x+y)}{2}$
44	$\mathbb{Q}_3(B_{1u})$	B_{1u}	3	Q, T	-	-	$\sqrt{15}xyz$
45	$\mathbb{G}_4(B_{1u})$	B_{1u}	4	G, M	-	-	$\frac{\sqrt{5}(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{4}$
46	$\mathbb{Q}_5(B_{1u})$	B_{1u}	5	Q, T	-	-	$\frac{\sqrt{105}xyz\left(x^2+y^2-2z^2\right)}{2}$
47	$\mathbb{G}_6(B_{1u},1)$	B_{1u}	6	G, M	1	-	$-\frac{\sqrt{2310}(x-y)(x+y)(x-z)(x+z)(y-z)(y+z)}{8}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
48	$\mathbb{G}_6(B_{1u},2)$	B_{1u}	6	G, M	2	-	$\frac{\sqrt{42}(x-y)(x+y)\left(x^4-9x^2y^2-5x^2z^2+y^4-5y^2z^2+5z^4\right)}{8}$
49	$\mathbb{Q}_7(B_{1u},1)$	B_{1u}	7	Q, T	1	-	$\frac{\sqrt{91}xyz\left(3x^4-5x^2y^2-5x^2z^2+3y^4-5y^2z^2+3z^4\right)}{2}$
50	$\mathbb{Q}_7(B_{1u},2)$	B_{1u}	7	Q, T	2	-	$-\frac{\sqrt{77}xyz\left(3x^4-20x^2y^2+10x^2z^2+3y^4+10y^2z^2-6z^4\right)}{4}$
51	$\mathbb{Q}_2(B_{2g})$	B_{2g}	2	Q, T	-	-	$\sqrt{3}xy$
52	$\mathbb{G}_3(B_{2g})$	B_{2g}	3	G, M	-	-	$\frac{\sqrt{15}z(x-y)(x+y)}{2}$
53	$\mathbb{Q}_4(B_{2g})$	B_{2g}	4	Q, T	-	-	$-\frac{\sqrt{5}xy\left(x^2+y^2-6z^2\right)}{2}$
54	$\mathbb{G}_5(B_{2g})$	B_{2g}	5	G, M	-	-	$-\frac{\sqrt{105}z(x-y)(x+y)\left(x^2+y^2-2z^2\right)}{4}$
55	$\mathbb{Q}_6(B_{2g},1)$	B_{2g}	6	Q, T	1	-	$\frac{\sqrt{462}xy\left(x^2-3y^2\right)\left(3x^2-y^2\right)}{16}$
56	$\mathbb{Q}_6(B_{2g},2)$	B_{2g}	6	Q, T	2	-	$\frac{\sqrt{210}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{16}$
57	$\mathbb{G}_7(B_{2g},1)$	B_{2g}	7	G, M	1	-	$\frac{\sqrt{6006}z(x-y)(x+y)\left(x^2-4xy+y^2\right)\left(x^2+4xy+y^2\right)}{32}$
58	$\mathbb{G}_7(B_{2g},2)$	B_{2g}	7	G, M	2	-	$\frac{\sqrt{42}z(x-y)(x+y)\left(15x^4+30x^2y^2-80x^2z^2+15y^4-80y^2z^2+48z^4\right)}{32}$
59	$\mathbb{G}_2(B_{2u})$	B_{2u}	2	G, M	-	-	$\sqrt{3}xy$
60	$\mathbb{Q}_3(B_{2u})$	B_{2u}	3	Q, T	-	-	$\frac{\sqrt{15}z(x-y)(x+y)}{2}$
61	$\mathbb{G}_4(B_{2u})$	B_{2u}	4	G, M	-	-	$-\frac{\sqrt{5}xy\left(x^2+y^2-6z^2\right)}{2}$
62	$\mathbb{Q}_5(B_{2u})$	B_{2u}	5	Q, T	-	-	$-\frac{\sqrt{105}z(x-y)(x+y)\left(x^2+y^2-2z^2\right)}{4}$
63	$\mathbb{G}_6(B_{2u},1)$	B_{2u}	6	G, M	1	-	$\frac{\sqrt{462}xy\left(x^2-3y^2\right)\left(3x^2-y^2\right)}{16}$
64	$\mathbb{G}_6(B_{2u},2)$	B_{2u}	6	G, M	2	-	$\frac{\sqrt{210}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{16}$
65	$\mathbb{Q}_7(B_{2u},1)$	B_{2u}	7	Q, T	1	-	$\frac{\sqrt{6006}z(x-y)(x+y)\left(x^2-4xy+y^2\right)\left(x^2+4xy+y^2\right)}{32}$
66	$\mathbb{Q}_7(B_{2u},2)$	B_{2u}	7	Q, T	2	-	$\frac{\sqrt{42}z(x-y)(x+y)\left(15x^4+30x^2y^2-80x^2z^2+15y^4-80y^2z^2+48z^4\right)}{32}$
67	$\mathbb{G}_{1,1}(E_g)$	E_g	1	G, M	-	1	x
68	$\mathbb{G}_{1,2}(E_g)$					2	$-y$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
69	$\mathbb{Q}_{2,1}(E_g)$	E_g	2	Q, T	-	1	$\sqrt{3}yz$
70	$\mathbb{Q}_{2,2}(E_g)$					2	$\sqrt{3}xz$
71	$\mathbb{G}_{3,1}(E_g,1)$	E_g	3	G, M	1	1	$\frac{x\left(2x^2-3y^2-3z^2\right)}{2}$
72	$\mathbb{G}_{3,2}(E_g,1)$					2	$\frac{y\left(3x^2-2y^2+3z^2\right)}{2}$
73	$\mathbb{G}_{3,1}(E_g,2)$	E_g	3	G, M	2	1	$\frac{\sqrt{15}x(y-z)(y+z)}{2}$
74	$\mathbb{G}_{3,2}(E_g,2)$					2	$-\frac{\sqrt{15}y(x-z)(x+z)}{2}$
75	$\mathbb{Q}_{4,1}(E_g,1)$	E_g	4	Q, T	1	1	$\frac{\sqrt{35}yz(y-z)(y+z)}{2}$
76	$\mathbb{Q}_{4,2}(E_g,1)$					2	$\frac{\sqrt{35}xz(x-z)(x+z)}{2}$
77	$\mathbb{Q}_{4,1}(E_g,2)$	E_g	4	Q, T	2	1	$\frac{\sqrt{5}yz\left(6x^2-y^2-z^2\right)}{2}$
78	$\mathbb{Q}_{4,2}(E_g,2)$					2	$-\frac{\sqrt{5}xz\left(x^2-6y^2+z^2\right)}{2}$
79	$\mathbb{G}_{5,1}(E_g,1)$	E_g	5	G, M	1	1	$\frac{x\left(8x^4-40x^2y^2-40x^2z^2+15y^4+30y^2z^2+15z^4\right)}{8}$
80	$\mathbb{G}_{5,2}(E_g,1)$					2	$-\frac{y\left(15x^4-40x^2y^2+30x^2z^2+8y^4-40y^2z^2+15z^4\right)}{8}$
81	$\mathbb{G}_{5,1}(E_g,2)$	E_g	5	G, M	2	1	$\frac{3\sqrt{35}x\left(y^2-2yz-z^2\right)\left(y^2+2yz-z^2\right)}{8}$
82	$\mathbb{G}_{5,2}(E_g,2)$					2	$-\frac{3\sqrt{35}y\left(x^2-2xz-z^2\right)\left(x^2+2xz-z^2\right)}{8}$
83	$\mathbb{G}_{5,1}(E_g,3)$	E_g	5	G, M	3	1	$\frac{\sqrt{105}x(y-z)(y+z)\left(2x^2-y^2-z^2\right)}{4}$
84	$\mathbb{G}_{5,2}(E_g,3)$					2	$\frac{\sqrt{105}y(x-z)(x+z)\left(x^2-2y^2+z^2\right)}{4}$
85	$\mathbb{Q}_{6,1}(E_g,1)$	E_g	6	Q, T	1	1	$\frac{3\sqrt{7}yz(y-z)(y+z)\left(10x^2-y^2-z^2\right)}{4}$
86	$\mathbb{Q}_{6,2}(E_g,1)$					2	$-\frac{3\sqrt{7}xz(x-z)(x+z)\left(x^2-10y^2+z^2\right)}{4}$
87	$\mathbb{Q}_{6,1}(E_g,2)$	E_g	6	Q, T	2	1	$\frac{\sqrt{462}yz\left(y^2-3z^2\right)\left(3y^2-z^2\right)}{16}$
88	$\mathbb{Q}_{6,2}(E_g,2)$					2	$\frac{\sqrt{462}xz\left(x^2-3z^2\right)\left(3x^2-z^2\right)}{16}$
89	$\mathbb{Q}_{6,1}(E_g,3)$	E_g	6	Q, T	3	1	$\frac{\sqrt{210}yz\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{16}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
90	$\mathbb{Q}_{6,2}(E_g,3)$					2	$\frac{\sqrt{210}xz\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{16}$
91	$\mathbb{G}_{7,1}(E_g,1)$	E_g	7	G, M	1	1	$\frac{x\left(16x^6-168x^4y^2-168x^4z^2+210x^2y^4+420x^2y^2z^2+210x^2z^4-35y^6-105y^4z^2-105y^2z^4-35z^6\right)}{16}$
92	$\mathbb{G}_{7,2}(E_g,1)$					2	$\frac{y\left(35x^6-210x^4y^2+105x^4z^2+168x^2y^4-420x^2y^2z^2+105x^2z^4-16y^6+168y^4z^2-210y^2z^4+35z^6\right)}{16}$
93	$\mathbb{G}_{7,1}(E_g,2)$	E_g	7	G, M	2	1	$\frac{\sqrt{231}x\left(10x^2-3y^2-3z^2\right)\left(y^2-2yz-z^2\right)\left(y^2+2yz-z^2\right)}{16}$
94	$\mathbb{G}_{7,2}(E_g,2)$					2	$\frac{\sqrt{231}y\left(x^2-2xz-z^2\right)\left(x^2+2xz-z^2\right)\left(3x^2-10y^2+3z^2\right)}{16}$
95	$\mathbb{G}_{7,1}(E_g,3)$	E_g	7	G, M	3	1	$\frac{\sqrt{6006}x(y-z)(y+z)\left(y^2-4yz+z^2\right)\left(y^2+4yz+z^2\right)}{32}$
96	$\mathbb{G}_{7,2}(E_g,3)$					2	$-\frac{\sqrt{6006}y(x-z)(x+z)\left(x^2-4xz+z^2\right)\left(x^2+4xz+z^2\right)}{32}$
97	$\mathbb{G}_{7,1}(E_g,4)$	E_g	7	G, M	4	1	$\frac{\sqrt{42}x(y-z)(y+z)\left(48x^4-80x^2y^2-80x^2z^2+15y^4+30y^2z^2+15z^4\right)}{32}$
98	$\mathbb{G}_{7,2}(E_g,4)$					2	$-\frac{\sqrt{42}y(x-z)(x+z)\left(15x^4-80x^2y^2+30x^2z^2+48y^4-80y^2z^2+15z^4\right)}{32}$
99	$\mathbb{Q}_{1,1}(E_u)$	E_u	1	Q, T	-	1	x
100	$\mathbb{Q}_{1,2}(E_u)$					2	y
101	$\mathbb{G}_{2,1}(E_u)$	E_u	2	G, M	-	1	$\sqrt{3}yz$
102	$\mathbb{G}_{2,2}(E_u)$					2	$-\sqrt{3}xz$
103	$\mathbb{Q}_{3,1}(E_u,1)$	E_u	3	Q, T	1	1	$\frac{x\left(2x^2-3y^2-3z^2\right)}{2}$
104	$\mathbb{Q}_{3,2}(E_u,1)$					2	$-\frac{y\left(3x^2-2y^2+3z^2\right)}{2}$
105	$\mathbb{Q}_{3,1}(E_u,2)$	E_u	3	Q, T	2	1	$\frac{\sqrt{15}x(y-z)(y+z)}{2}$
106	$\mathbb{Q}_{3,2}(E_u,2)$					2	$\frac{\sqrt{15}y(x-z)(x+z)}{2}$
107	$\mathbb{G}_{4,1}(E_u,1)$	E_u	4	G, M	1	1	$\frac{\sqrt{35}yz(y-z)(y+z)}{2}$
108	$\mathbb{G}_{4,2}(E_u,1)$					2	$-\frac{\sqrt{35}xz(x-z)(x+z)}{2}$
109	$\mathbb{G}_{4,1}(E_u,2)$	E_u	4	G, M	2	1	$\frac{\sqrt{5}yz\left(6x^2-y^2-z^2\right)}{2}$
110	$\mathbb{G}_{4,2}(E_u,2)$					2	$\frac{\sqrt{5}xz\left(x^2-6y^2+z^2\right)}{2}$

continued …

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
111	$\mathbb{Q}_{5,1}(E_u,1)$	E_u	5	Q, T	1	1	$\frac{x\left(8x^4-40x^2y^2-40x^2z^2+15y^4+30y^2z^2+15z^4\right)}{8}$
112	$\mathbb{Q}_{5,2}(E_u,1)$					2	$\frac{y\left(15x^4-40x^2y^2+30x^2z^2+8y^4-40y^2z^2+15z^4\right)}{8}$
113	$\mathbb{Q}_{5,1}(E_u,2)$	E_u	5	Q, T	2	1	$\frac{3\sqrt{35}x\left(y^2-2yz-z^2\right)\left(y^2+2yz-z^2\right)}{8}$
114	$\mathbb{Q}_{5,2}(E_u,2)$					2	$\frac{3\sqrt{35}y\left(x^2-2xz-z^2\right)\left(x^2+2xz-z^2\right)}{8}$
115	$\mathbb{Q}_{5,1}(E_u,3)$	E_u	5	Q, T	3	1	$\frac{\sqrt{105}x(y-z)(y+z)\left(2x^2-y^2-z^2\right)}{4}$
116	$\mathbb{Q}_{5,2}(E_u,3)$					2	$-\frac{\sqrt{105}y(x-z)(x+z)\left(x^2-2y^2+z^2\right)}{4}$
117	$\mathbb{G}_{6,1}(E_u,1)$	E_u	6	G, M	1	1	$\frac{3\sqrt{7}yz(y-z)(y+z)\left(10x^2-y^2-z^2\right)}{4}$
118	$\mathbb{G}_{6,2}(E_u,1)$					2	$\frac{3\sqrt{7}xz(x-z)(x+z)\left(x^2-10y^2+z^2\right)}{4}$
119	$\mathbb{G}_{6,1}(E_u,2)$	E_u	6	G, M	2	1	$\frac{\sqrt{462}yz\left(y^2-3z^2\right)\left(3y^2-z^2\right)}{16}$
120	$\mathbb{G}_{6,2}(E_u,2)$					2	$-\frac{\sqrt{462}xz\left(x^2-3z^2\right)\left(3x^2-z^2\right)}{16}$
121	$\mathbb{G}_{6,1}(E_u,3)$	E_u	6	G, M	3	1	$\frac{\sqrt{210}yz\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{16}$
122	$\mathbb{G}_{6,2}(E_u,3)$					2	$-\frac{\sqrt{210}xz\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{16}$
123	$\mathbb{Q}_{7,1}(E_u,1)$	E_u	7	Q, T	1	1	$\frac{x\left(16x^6-168x^4y^2-168x^4z^2+210x^2y^4+420x^2y^2z^2+210x^2z^4-35y^6-105y^4z^2-105y^2z^4-35z^6\right)}{16}$
124	$\mathbb{Q}_{7,2}(E_u,1)$					2	$-\frac{y\left(35x^6-210x^4y^2+105x^4z^2+168x^2y^4-420x^2y^2z^2+105x^2z^4-16y^6+168y^4z^2-210y^2z^4+35z^6\right)}{16}$
125	$\mathbb{Q}_{7,1}(E_u,2)$	E_u	7	Q, T	2	1	$\frac{\sqrt{231}x\left(10x^2-3y^2-3z^2\right)\left(y^2-2yz-z^2\right)\left(y^2+2yz-z^2\right)}{16}$
126	$\mathbb{Q}_{7,2}(E_u,2)$					2	$-\frac{\sqrt{231}y\left(x^2-2xz-z^2\right)\left(x^2+2xz-z^2\right)\left(3x^2-10y^2+3z^2\right)}{16}$
127	$\mathbb{Q}_{7,1}(E_u,3)$	E_u	7	Q, T	3	1	$\frac{\sqrt{6006}x(y-z)(y+z)\left(y^2-4yz+z^2\right)\left(y^2+4yz+z^2\right)}{32}$
128	$\mathbb{Q}_{7,2}(E_u,3)$					2	$\frac{\sqrt{6006}y(x-z)(x+z)\left(x^2-4xz+z^2\right)\left(x^2+4xz+z^2\right)}{32}$
129	$\mathbb{Q}_{7,1}(E_u,4)$	E_u	7	Q, T	4	1	$\frac{\sqrt{42}x(y-z)(y+z)\left(48x^4-80x^2y^2-80x^2z^2+15y^4+30y^2z^2+15z^4\right)}{32}$
130	$\mathbb{Q}_{7,2}(E_u,4)$					2	$\frac{\sqrt{42}y(x-z)(x+z)\left(15x^4-80x^2y^2+30x^2z^2+48y^4-80y^2z^2+15z^4\right)}{32}$

Basis in full matrix

Table 3: dimension = 24

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
0	$\lvert\frac{3}{2},\frac{3}{2};d\rangle$ @U(1)	1	$\lvert\frac{3}{2},\frac{1}{2};d\rangle$ @U(1)	2	$\lvert\frac{3}{2},-\frac{1}{2};d\rangle$ @U(1)	3	$\lvert\frac{3}{2},-\frac{3}{2};d\rangle$ @U(1)	4	$\lvert\frac{5}{2},\frac{5}{2};d\rangle$ @U(1)
5	$\lvert\frac{5}{2},\frac{3}{2};d\rangle$ @U(1)	6	$\lvert\frac{5}{2},\frac{1}{2};d\rangle$ @U(1)	7	$\lvert\frac{5}{2},-\frac{1}{2};d\rangle$ @U(1)	8	$\lvert\frac{5}{2},-\frac{3}{2};d\rangle$ @U(1)	9	$\lvert\frac{5}{2},-\frac{5}{2};d\rangle$ @U(1)
10	$\lvert\frac{5}{2},\frac{5}{2};f\rangle$ @U(1)	11	$\lvert\frac{5}{2},\frac{3}{2};f\rangle$ @U(1)	12	$\lvert\frac{5}{2},\frac{1}{2};f\rangle$ @U(1)	13	$\lvert\frac{5}{2},-\frac{1}{2};f\rangle$ @U(1)	14	$\lvert\frac{5}{2},-\frac{3}{2};f\rangle$ @U(1)
15	$\lvert\frac{5}{2},-\frac{5}{2};f\rangle$ @U(1)	16	$\lvert\frac{7}{2},\frac{7}{2};f\rangle$ @U(1)	17	$\lvert\frac{7}{2},\frac{5}{2};f\rangle$ @U(1)	18	$\lvert\frac{7}{2},\frac{3}{2};f\rangle$ @U(1)	19	$\lvert\frac{7}{2},\frac{1}{2};f\rangle$ @U(1)
20	$\lvert\frac{7}{2},-\frac{1}{2};f\rangle$ @U(1)	21	$\lvert\frac{7}{2},-\frac{3}{2};f\rangle$ @U(1)	22	$\lvert\frac{7}{2},-\frac{5}{2};f\rangle$ @U(1)	23	$\lvert\frac{7}{2},-\frac{7}{2};f\rangle$ @U(1)		

SAMB: 1428 (all 1428)

- `U` : 'U' site-cluster
  * bra: $\langle\frac{3}{2},\frac{3}{2};d\lvert$, $\langle\frac{3}{2},\frac{1}{2};d\lvert$, $\langle\frac{3}{2},-\frac{1}{2};d\lvert$, $\langle\frac{3}{2},-\frac{3}{2};d\lvert$, $\langle\frac{5}{2},\frac{5}{2};d\lvert$, $\langle\frac{5}{2},\frac{3}{2};d\lvert$, $\langle\frac{5}{2},\frac{1}{2};d\lvert$, $\langle\frac{5}{2},-\frac{1}{2};d\lvert$, $\langle\frac{5}{2},-\frac{3}{2};d\lvert$, $\langle\frac{5}{2},-\frac{5}{2};d\lvert$
  * ket: $\lvert\frac{3}{2},\frac{3}{2};d\rangle$, $\lvert\frac{3}{2},\frac{1}{2};d\rangle$, $\lvert\frac{3}{2},-\frac{1}{2};d\rangle$, $\lvert\frac{3}{2},-\frac{3}{2};d\rangle$, $\lvert\frac{5}{2},\frac{5}{2};d\rangle$, $\lvert\frac{5}{2},\frac{3}{2};d\rangle$, $\lvert\frac{5}{2},\frac{1}{2};d\rangle$, $\lvert\frac{5}{2},-\frac{1}{2};d\rangle$, $\lvert\frac{5}{2},-\frac{3}{2};d\rangle$, $\lvert\frac{5}{2},-\frac{5}{2};d\rangle$
  * wyckoff: `2a`

  $\boxed{\text{z1}}$ $\mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

  $\boxed{\text{z2}}$ $\mathbb{Q}_2^{(c)}(A_{1g}) = \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

  $\boxed{\text{z3}}$ $\mathbb{Q}_4^{(c)}(A_{1g},1) = \mathbb{Q}_4^{(a)}(A_{1g},1)\mathbb{Q}_0^{(s)}(A_{1g})$

  $\boxed{\text{z4}}$ $\mathbb{Q}_4^{(c)}(A_{1g},2) = \mathbb{Q}_4^{(a)}(A_{1g},2)\mathbb{Q}_0^{(s)}(A_{1g})$

  $\boxed{\text{z5}}$ $\mathbb{Q}_2^{(1,-1;c)}(A_{1g}) = \mathbb{Q}_2^{(1,-1;a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

  $\boxed{\text{z6}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{1g},1) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z7}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{1g},2) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z8}}$ $\mathbb{Q}_0^{(1,1;c)}(A_{1g}) = \mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z9}}$ $\mathbb{Q}_2^{(1,1;c)}(A_{1g}) = \mathbb{Q}_2^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z104}}$ $\mathbb{Q}_4^{(c)}(A_{2g}) = \mathbb{Q}_4^{(a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z105}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{2g}) = \mathbb{Q}_4^{(1,-1;a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z106}}$ $\mathbb{G}_1^{(1,0;c)}(A_{2g}) = \mathbb{G}_1^{(1,0;a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z107}}$ $\mathbb{G}_3^{(1,0;c)}(A_{2g}) = \mathbb{G}_3^{(1,0;a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z183}}$ $\mathbb{Q}_2^{(c)}(B_{1g}) = \mathbb{Q}_2^{(a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z184}}$ $\mathbb{Q}_4^{(c)}(B_{1g}) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z185}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{1g}) = \mathbb{Q}_2^{(1,-1;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z186}}$ $\mathbb{Q}_4^{(1,-1;c)}(B_{1g}) = \mathbb{Q}_4^{(1,-1;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z187}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{1g}) = \mathbb{Q}_2^{(1,1;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z188}}$ $\mathbb{G}_3^{(1,0;c)}(B_{1g}) = \mathbb{G}_3^{(1,0;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z279}}$ $\mathbb{Q}_2^{(c)}(B_{2g}) = \mathbb{Q}_2^{(a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z280}}$ $\mathbb{Q}_4^{(c)}(B_{2g}) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z281}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{2g}) = \mathbb{Q}_2^{(1,-1;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z282}}$ $\mathbb{Q}_4^{(1,-1;c)}(B_{2g}) = \mathbb{Q}_4^{(1,-1;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z283}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{2g}) = \mathbb{Q}_2^{(1,1;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z284}}$ $\mathbb{G}_3^{(1,0;c)}(B_{2g}) = \mathbb{G}_3^{(1,0;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z363}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z364}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z365}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,1) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g,1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z366}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,1) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g,1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z367}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g,2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z368}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g,2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z369}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z370}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z371}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,1) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z372}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,1) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z373}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z374}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z375}}$ $\mathbb{Q}_{2,1}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,1;a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z376}}$ $\mathbb{Q}_{2,2}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,1;a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z377}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z378}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z379}}$ $\mathbb{G}_{3,1}^{(1,0;c)}(E_g,1) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z380}}$ $\mathbb{G}_{3,2}^{(1,0;c)}(E_g,1) = \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z381}}$ $\mathbb{G}_{3,1}^{(1,0;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z382}}$ $\mathbb{G}_{3,2}^{(1,0;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

- `U` : 'U' site-cluster
  * bra: $\langle\frac{3}{2},\frac{3}{2};d|$, $\langle\frac{3}{2},\frac{1}{2};d|$, $\langle\frac{3}{2},-\frac{1}{2};d|$, $\langle\frac{3}{2},-\frac{3}{2};d|$, $\langle\frac{5}{2},\frac{5}{2};d|$, $\langle\frac{5}{2},\frac{3}{2};d|$, $\langle\frac{5}{2},\frac{1}{2};d|$, $\langle\frac{5}{2},-\frac{1}{2};d|$, $\langle\frac{5}{2},-\frac{3}{2};d|$, $\langle\frac{5}{2},-\frac{5}{2};d|$
  * ket: $|\frac{5}{2},\frac{5}{2};f\rangle$, $|\frac{5}{2},\frac{3}{2};f\rangle$, $|\frac{5}{2},\frac{1}{2};f\rangle$, $|\frac{5}{2},-\frac{1}{2};f\rangle$, $|\frac{5}{2},-\frac{3}{2};f\rangle$, $|\frac{5}{2},-\frac{5}{2};f\rangle$, $|\frac{7}{2},\frac{7}{2};f\rangle$, $|\frac{7}{2},\frac{5}{2};f\rangle$, $|\frac{7}{2},\frac{3}{2};f\rangle$, $|\frac{7}{2},\frac{1}{2};f\rangle$, $|\frac{7}{2},-\frac{1}{2};f\rangle$, $|\frac{7}{2},-\frac{3}{2};f\rangle$, $|\frac{7}{2},-\frac{5}{2};f\rangle$, $|\frac{7}{2},-\frac{7}{2};f\rangle$
  * wyckoff: `2a`

$\boxed{\text{z689}}$ $\mathbb{Q}_5^{(c)}(A_{1u}) = \mathbb{Q}_5^{(a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z690}}$ $\mathbb{Q}_5^{(1,-1;c)}(A_{1u}) = \mathbb{Q}_5^{(1,-1;a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z691}}$ $\mathbb{Q}_5^{(1,0;c)}(A_{1u}) = \mathbb{Q}_5^{(1,0;a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z692}}$ $\mathbb{G}_2^{(c)}(A_{1u}) = \mathbb{G}_2^{(a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z693}}$ $\mathbb{G}_4^{(c)}(A_{1u},1) = \mathbb{G}_4^{(a)}(A_{1u},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z694}}$ $\mathbb{G}_4^{(c)}(A_{1u},2) = \mathbb{G}_4^{(a)}(A_{1u},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z695}}$ $\mathbb{G}_2^{(1,-1;c)}(A_{1u}) = \mathbb{G}_2^{(1,-1;a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z696}}$ $\mathbb{G}_4^{(1,-1;c)}(A_{1u},1) = \mathbb{G}_4^{(1,-1;a)}(A_{1u},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z697}}$ $\mathbb{G}_4^{(1,-1;c)}(A_{1u},2) = \mathbb{G}_4^{(1,-1;a)}(A_{1u},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z698}}$ $\mathbb{G}_6^{(1,-1;c)}(A_{1u},1) = \mathbb{G}_6^{(1,-1;a)}(A_{1u},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z699}}$ $\mathbb{G}_6^{(1,-1;c)}(A_{1u},2) = \mathbb{G}_6^{(1,-1;a)}(A_{1u},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z700}}$ $\mathbb{G}_2^{(1,0;c)}(A_{1u}) = \mathbb{G}_2^{(1,0;a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z701}}$ $\mathbb{G}_4^{(1,0;c)}(A_{1u},1) = \mathbb{G}_4^{(1,0;a)}(A_{1u},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z702}}$ $\mathbb{G}_4^{(1,0;c)}(A_{1u},2) = \mathbb{G}_4^{(1,0;a)}(A_{1u},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z703}}$ $\mathbb{G}_0^{(1,1;c)}(A_{1u}) = \mathbb{G}_0^{(1,1;a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z704}}$ $\mathbb{G}_2^{(1,1;c)}(A_{1u}) = \mathbb{G}_2^{(1,1;a)}(A_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z705}}$ $\mathbb{G}_4^{(1,1;c)}(A_{1u},1) = \mathbb{G}_4^{(1,1;a)}(A_{1u},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z706}}$ $\mathbb{G}_4^{(1,1;c)}(A_{1u},2) = \mathbb{G}_4^{(1,1;a)}(A_{1u},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z785}}$ $\mathbb{Q}_1^{(c)}(A_{2u}) = \mathbb{Q}_1^{(a)}(A_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z786}}$ $\mathbb{Q}_3^{(c)}(A_{2u}) = \mathbb{Q}_3^{(a)}(A_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z787}}$ $\mathbb{Q}_5^{(c)}(A_{2u},1) = \mathbb{Q}_5^{(a)}(A_{2u},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z788}}$ $\mathbb{Q}_5^{(c)}(A_{2u},2) = \mathbb{Q}_5^{(a)}(A_{2u},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z789}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_{2u}) = \mathbb{Q}_3^{(1,-1;a)}(A_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z790}}$ $\mathbb{Q}_5^{(1,-1;c)}(A_{2u},1) = \mathbb{Q}_5^{(1,-1;a)}(A_{2u},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z791}}$ $\mathbb{Q}_5^{(1,-1;c)}(A_{2u},2) = \mathbb{Q}_5^{(1,-1;a)}(A_{2u},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z792}}$ $\mathbb{Q}_1^{(1,0;c)}(A_{2u}) = \mathbb{Q}_1^{(1,0;a)}(A_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z793}}$ $\mathbb{Q}_3^{(1,0;c)}(A_{2u}) = \mathbb{Q}_3^{(1,0;a)}(A_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z794}}$ $\mathbb{Q}_5^{(1,0;c)}(A_{2u},1) = \mathbb{Q}_5^{(1,0;a)}(A_{2u},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z795}}$ $\mathbb{Q}_5^{(1,0;c)}(A_{2u},2) = \mathbb{Q}_5^{(1,0;a)}(A_{2u},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z796}}$ $\mathbb{Q}_1^{(1,1;c)}(A_{2u}) = \mathbb{Q}_1^{(1,1;a)}(A_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z797}}$ $\mathbb{Q}_3^{(1,1;c)}(A_{2u}) = \mathbb{Q}_3^{(1,1;a)}(A_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z798}}$ $\mathbb{G}_4^{(c)}(A_{2u}) = \mathbb{G}_4^{(a)}(A_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z799}}$ $\mathbb{G}_4^{(1,-1;c)}(A_{2u}) = \mathbb{G}_4^{(1,-1;a)}(A_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z800}}$ $\mathbb{G}_6^{(1,-1;c)}(A_{2u}) = \mathbb{G}_6^{(1,-1;a)}(A_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z801}}$ $\mathbb{G}_4^{(1,0;c)}(A_{2u}) = \mathbb{G}_4^{(1,0;a)}(A_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z802}}$ $\mathbb{G}_4^{(1,1;c)}(A_{2u}) = \mathbb{G}_4^{(1,1;a)}(A_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z881}}$ $\mathbb{Q}_3^{(c)}(B_{1u}) = \mathbb{Q}_3^{(a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z882}}$ $\mathbb{Q}_5^{(c)}(B_{1u}) = \mathbb{Q}_5^{(a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z883}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{1u}) = \mathbb{Q}_3^{(1,-1;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z884}}$ $\mathbb{Q}_5^{(1,-1;c)}(B_{1u}) = \mathbb{Q}_5^{(1,-1;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z885}}$ $\mathbb{Q}_3^{(1,0;c)}(B_{1u}) = \mathbb{Q}_3^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z886}}$ $\mathbb{Q}_5^{(1,0;c)}(B_{1u}) = \mathbb{Q}_5^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z887}}$ $\mathbb{Q}_3^{(1,1;c)}(B_{1u}) = \mathbb{Q}_3^{(1,1;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z888}}$ $\mathbb{G}_2^{(c)}(B_{1u}) = \mathbb{G}_2^{(a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z889}}$ $\mathbb{G}_4^{(c)}(B_{1u}) = \mathbb{G}_4^{(a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z890}}$ $\mathbb{G}_2^{(1,-1;c)}(B_{1u}) = \mathbb{G}_2^{(1,-1;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z891}}$ $\mathbb{G}_4^{(1,-1;c)}(B_{1u}) = \mathbb{G}_4^{(1,-1;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z892}}$ $\mathbb{G}_6^{(1,-1;c)}(B_{1u},1) = \mathbb{G}_6^{(1,-1;a)}(B_{1u},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z893}}$ $\mathbb{G}_6^{(1,-1;c)}(B_{1u},2) = \mathbb{G}_6^{(1,-1;a)}(B_{1u},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z894}}$ $\mathbb{G}_2^{(1,0;c)}(B_{1u}) = \mathbb{G}_2^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z895}}$ $\mathbb{G}_4^{(1,0;c)}(B_{1u}) = \mathbb{G}_4^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z896}}$ $\mathbb{G}_2^{(1,1;c)}(B_{1u}) = \mathbb{G}_2^{(1,1;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z897}}$ $\mathbb{G}_4^{(1,1;c)}(B_{1u}) = \mathbb{G}_4^{(1,1;a)}(B_{1u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z976}}$ $\mathbb{Q}_3^{(c)}(B_{2u}) = \mathbb{Q}_3^{(a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z977}}$ $\mathbb{Q}_5^{(c)}(B_{2u}) = \mathbb{Q}_5^{(a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z978}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{2u}) = \mathbb{Q}_3^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z979}}$ $\mathbb{Q}_5^{(1,-1;c)}(B_{2u}) = \mathbb{Q}_5^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z980}}$ $\mathbb{Q}_3^{(1,0;c)}(B_{2u}) = \mathbb{Q}_3^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z981}}$ $\mathbb{Q}_5^{(1,0;c)}(B_{2u}) = \mathbb{Q}_5^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z982}}$ $\mathbb{Q}_3^{(1,1;c)}(B_{2u}) = \mathbb{Q}_3^{(1,1;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z983}}$ $\mathbb{G}_2^{(c)}(B_{2u}) = \mathbb{G}_2^{(a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z984}}$ $\mathbb{G}_4^{(c)}(B_{2u}) = \mathbb{G}_4^{(a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z985}}$ $\mathbb{G}_2^{(1,-1;c)}(B_{2u}) = \mathbb{G}_2^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z986}}$ $\mathbb{G}_4^{(1,-1;c)}(B_{2u}) = \mathbb{G}_4^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z987}}$ $\mathbb{G}_6^{(1,-1;c)}(B_{2u},1) = \mathbb{G}_6^{(1,-1;a)}(B_{2u},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z988}}$ $\mathbb{G}_6^{(1,-1;c)}(B_{2u},2) = \mathbb{G}_6^{(1,-1;a)}(B_{2u},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z989}}$ $\mathbb{G}_2^{(1,0;c)}(B_{2u}) = \mathbb{G}_2^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z990}}$ $\mathbb{G}_4^{(1,0;c)}(B_{2u}) = \mathbb{G}_4^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z991}}$ $\mathbb{G}_2^{(1,1;c)}(B_{2u}) = \mathbb{G}_2^{(1,1;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z992}}$ $\mathbb{G}_4^{(1,1;c)}(B_{2u}) = \mathbb{G}_4^{(1,1;a)}(B_{2u})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z1071}}$ $\mathbb{Q}_{1,1}^{(c)}(E_u) = \dfrac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1072}}$ $\mathbb{Q}_{1,2}^{(c)}(E_u) = \dfrac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1073}}$ $\mathbb{Q}_{3,1}^{(c)}(E_u,1) = \dfrac{\sqrt{2}\mathbb{Q}_{3,1}^{(a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1074}}$ $\mathbb{Q}_{3,2}^{(c)}(E_u,1) = \dfrac{\sqrt{2}\mathbb{Q}_{3,2}^{(a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1075}}$ $\mathbb{Q}_{3,1}^{(c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{Q}_{3,1}^{(a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1076}}$ $\mathbb{Q}_{3,2}^{(c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{Q}_{3,2}^{(a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1077}}$ $\mathbb{Q}_{5,1}^{(c)}(E_u,1) = \dfrac{\sqrt{2}\mathbb{Q}_{5,1}^{(a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1078}}$ $\mathbb{Q}_{5,2}^{(c)}(E_u,1) = \dfrac{\sqrt{2}\mathbb{Q}_{5,2}^{(a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1079}}$ $\mathbb{Q}_{5,1}^{(c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{Q}_{5,1}^{(a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1080}}$ $\mathbb{Q}_{5,2}^{(c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{Q}_{5,2}^{(a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1081}}$ $\mathbb{Q}_{5,1}^{(c)}(E_u,3) = \dfrac{\sqrt{2}\mathbb{Q}_{5,1}^{(a)}(E_u,3)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1082}}$ $\mathbb{Q}_{5,2}^{(c)}(E_u,3) = \dfrac{\sqrt{2}\mathbb{Q}_{5,2}^{(a)}(E_u,3)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1083}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(E_u,1) = \dfrac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1084}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(E_u,1) = \dfrac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1085}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1086}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1087}}$ $\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,1) = \dfrac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1088}}$ $\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,1) = \dfrac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1089}}$ $\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1090}}$ $\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1091}}$ $\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,3) = \dfrac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1092}}$ $\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,3) = \dfrac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1093}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(E_u) = \dfrac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1094}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(E_u) = \dfrac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E_u)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1095}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1096}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1097}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1098}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1099}}$ $\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1100}}$ $\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1101}}$ $\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1102}}$ $\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1103}}$ $\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,3) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1104}}$ $\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,3) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1105}}$ $\mathbb{Q}_{1,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1106}}$ $\mathbb{Q}_{1,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1107}}$ $\mathbb{Q}_{3,1}^{(1,1;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1108}}$ $\mathbb{Q}_{3,2}^{(1,1;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1109}}$ $\mathbb{Q}_{3,1}^{(1,1;c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1110}}$ $\mathbb{Q}_{3,2}^{(1,1;c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1111}}$ $\mathbb{G}_{2,1}^{(c)}(E_u) = \dfrac{\sqrt{2}\mathbb{G}_{2,1}^{(a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1112}}$ $\mathbb{G}_{2,2}^{(c)}(E_u) = \dfrac{\sqrt{2}\mathbb{G}_{2,2}^{(a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1113}}$ $\mathbb{G}_{4,1}^{(c)}(E_u,1) = \dfrac{\sqrt{2}\mathbb{G}_{4,1}^{(a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1114}}$ $\mathbb{G}_{4,2}^{(c)}(E_u,1) = \dfrac{\sqrt{2}\mathbb{G}_{4,2}^{(a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1115}}$ $\mathbb{G}_{4,1}^{(c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{G}_{4,1}^{(a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1116}}$ $\mathbb{G}_{4,2}^{(c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{G}_{4,2}^{(a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1117}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = \dfrac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1118}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \dfrac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1119}}$ $\mathbb{G}_{4,1}^{(1,-1;c)}(E_u,1) = \dfrac{\sqrt{2}\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1120}}$ $\mathbb{G}_{4,2}^{(1,-1;c)}(E_u,1) = \dfrac{\sqrt{2}\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1121}}$ $\mathbb{G}_{4,1}^{(1,-1;c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1122}}$ $\mathbb{G}_{4,2}^{(1,-1;c)}(E_u,2) = \dfrac{\sqrt{2}\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1123}}$ $\mathbb{G}_{6,1}^{(1,-1;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1124}}$ $\mathbb{G}_{6,2}^{(1,-1;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1125}}$ $\mathbb{G}_{6,1}^{(1,-1;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1126}}$ $\mathbb{G}_{6,2}^{(1,-1;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1127}}$ $\mathbb{G}_{6,1}^{(1,-1;c)}(E_u,3) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1128}}$ $\mathbb{G}_{6,2}^{(1,-1;c)}(E_u,3) = \frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1129}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1130}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1131}}$ $\mathbb{G}_{4,1}^{(1,0;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,0;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1132}}$ $\mathbb{G}_{4,2}^{(1,0;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,0;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1133}}$ $\mathbb{G}_{4,1}^{(1,0;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,0;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1134}}$ $\mathbb{G}_{4,2}^{(1,0;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,0;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1135}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1136}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1137}}$ $\mathbb{G}_{4,1}^{(1,1;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,1;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1138}}$ $\mathbb{G}_{4,2}^{(1,1;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,1;a)}(E_u,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1139}}$ $\mathbb{G}_{4,1}^{(1,1;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,1;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z1140}}$ $\mathbb{G}_{4,2}^{(1,1;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,1;a)}(E_u,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

- `U` : 'U' site-cluster
  * bra: $\langle\frac{5}{2},\frac{5}{2};f|$, $\langle\frac{5}{2},\frac{3}{2};f|$, $\langle\frac{5}{2},\frac{1}{2};f|$, $\langle\frac{5}{2},-\frac{1}{2};f|$, $\langle\frac{5}{2},-\frac{3}{2};f|$, $\langle\frac{5}{2},-\frac{5}{2};f|$, $\langle\frac{7}{2},\frac{7}{2};f|$, $\langle\frac{7}{2},\frac{5}{2};f|$, $\langle\frac{7}{2},\frac{3}{2};f|$, $\langle\frac{7}{2},\frac{1}{2};f|$, $\langle\frac{7}{2},-\frac{1}{2};f|$, $\langle\frac{7}{2},-\frac{3}{2};f|$, $\langle\frac{7}{2},-\frac{5}{2};f|$, $\langle\frac{7}{2},-\frac{7}{2};f|$
  * ket: $|\frac{5}{2},\frac{5}{2};f\rangle$, $|\frac{5}{2},\frac{3}{2};f\rangle$, $|\frac{5}{2},\frac{1}{2};f\rangle$, $|\frac{5}{2},-\frac{1}{2};f\rangle$, $|\frac{5}{2},-\frac{3}{2};f\rangle$, $|\frac{5}{2},-\frac{5}{2};f\rangle$, $|\frac{7}{2},\frac{7}{2};f\rangle$, $|\frac{7}{2},\frac{5}{2};f\rangle$, $|\frac{7}{2},\frac{3}{2};f\rangle$, $|\frac{7}{2},\frac{1}{2};f\rangle$, $|\frac{7}{2},-\frac{1}{2};f\rangle$, $|\frac{7}{2},-\frac{3}{2};f\rangle$, $|\frac{7}{2},-\frac{5}{2};f\rangle$, $|\frac{7}{2},-\frac{7}{2};f\rangle$
  * wyckoff: `2a`

$\boxed{\text{z10}}$ $\mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z11}}$ $\mathbb{Q}_2^{(c)}(A_{1g}) = \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z12}}$ $\mathbb{Q}_4^{(c)}(A_{1g},1) = \mathbb{Q}_4^{(a)}(A_{1g},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z13}}$ $\mathbb{Q}_4^{(c)}(A_{1g},2) = \mathbb{Q}_4^{(a)}(A_{1g},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z14}}$ $\mathbb{Q}_6^{(c)}(A_{1g},1) = \mathbb{Q}_6^{(a)}(A_{1g},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z15}}$ $\mathbb{Q}_6^{(c)}(A_{1g},2) = \mathbb{Q}_6^{(a)}(A_{1g},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z16}}$ $\mathbb{Q}_2^{(1,-1;c)}(A_{1g}) = \mathbb{Q}_2^{(1,-1;a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z17}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{1g},1) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z18}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{1g},2) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z19}}$ $\mathbb{Q}_6^{(1,-1;c)}(A_{1g},1) = \mathbb{Q}_6^{(1,-1;a)}(A_{1g},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z20}}$ $\mathbb{Q}_6^{(1,-1;c)}(A_{1g},2) = \mathbb{Q}_6^{(1,-1;a)}(A_{1g},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z21}}$ $\mathbb{Q}_0^{(1,1;c)}(A_{1g}) = \mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z22}}$ $\mathbb{Q}_2^{(1,1;c)}(A_{1g}) = \mathbb{Q}_2^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z23}}$ $\mathbb{Q}_4^{(1,1;c)}(A_{1g},1) = \mathbb{Q}_4^{(1,1;a)}(A_{1g},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z24}}$ $\mathbb{Q}_4^{(1,1;c)}(A_{1g},2) = \mathbb{Q}_4^{(1,1;a)}(A_{1g},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z25}}$ $\mathbb{G}_5^{(1,0;c)}(A_{1g}) = \mathbb{G}_5^{(1,0;a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z108}}$ $\mathbb{Q}_4^{(c)}(A_{2g}) = \mathbb{Q}_4^{(a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z109}}$ $\mathbb{Q}_6^{(c)}(A_{2g}) = \mathbb{Q}_6^{(a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z110}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{2g}) = \mathbb{Q}_4^{(1,-1;a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z111}}$ $\mathbb{Q}_6^{(1,-1;c)}(A_{2g}) = \mathbb{Q}_6^{(1,-1;a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z112}}$ $\mathbb{Q}_4^{(1,1;c)}(A_{2g}) = \mathbb{Q}_4^{(1,1;a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z113}}$ $\mathbb{G}_1^{(1,0;c)}(A_{2g}) = \mathbb{G}_1^{(1,0;a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z114}}$ $\mathbb{G}_3^{(1,0;c)}(A_{2g}) = \mathbb{G}_3^{(1,0;a)}(A_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z115}}$ $\mathbb{G}_5^{(1,0;c)}(A_{2g},1) = \mathbb{G}_5^{(1,0;a)}(A_{2g},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z116}}$ $\mathbb{G}_5^{(1,0;c)}(A_{2g},2) = \mathbb{G}_5^{(1,0;a)}(A_{2g},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z189}}$ $\mathbb{Q}_2^{(c)}(B_{1g}) = \mathbb{Q}_2^{(a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z190}}$ $\mathbb{Q}_4^{(c)}(B_{1g}) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z191}}$ $\mathbb{Q}_6^{(c)}(B_{1g},1) = \mathbb{Q}_6^{(a)}(B_{1g},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z192}}$ $\mathbb{Q}_6^{(c)}(B_{1g},2) = \mathbb{Q}_6^{(a)}(B_{1g},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z193}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{1g}) = \mathbb{Q}_2^{(1,-1;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z194}}$ $\mathbb{Q}_4^{(1,-1;c)}(B_{1g}) = \mathbb{Q}_4^{(1,-1;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z195}}$ $\mathbb{Q}_6^{(1,-1;c)}(B_{1g},1) = \mathbb{Q}_6^{(1,-1;a)}(B_{1g},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z196}}$ $\mathbb{Q}_6^{(1,-1;c)}(B_{1g},2) = \mathbb{Q}_6^{(1,-1;a)}(B_{1g},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z197}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{1g}) = \mathbb{Q}_2^{(1,1;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z198}}$ $\mathbb{Q}_4^{(1,1;c)}(B_{1g}) = \mathbb{Q}_4^{(1,1;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z199}}$ $\mathbb{G}_3^{(1,0;c)}(B_{1g}) = \mathbb{G}_3^{(1,0;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z200}}$ $\mathbb{G}_5^{(1,0;c)}(B_{1g}) = \mathbb{G}_5^{(1,0;a)}(B_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z285}}$ $\mathbb{Q}_2^{(c)}(B_{2g}) = \mathbb{Q}_2^{(a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z286}}$ $\mathbb{Q}_4^{(c)}(B_{2g}) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z287}}$ $\mathbb{Q}_6^{(c)}(B_{2g},1) = \mathbb{Q}_6^{(a)}(B_{2g},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z288}}$ $\mathbb{Q}_6^{(c)}(B_{2g},2) = \mathbb{Q}_6^{(a)}(B_{2g},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z289}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{2g}) = \mathbb{Q}_2^{(1,-1;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z290}}$ $\mathbb{Q}_4^{(1,-1;c)}(B_{2g}) = \mathbb{Q}_4^{(1,-1;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z291}}$ $\mathbb{Q}_6^{(1,-1;c)}(B_{2g},1) = \mathbb{Q}_6^{(1,-1;a)}(B_{2g},1)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z292}}$ $\mathbb{Q}_6^{(1,-1;c)}(B_{2g},2) = \mathbb{Q}_6^{(1,-1;a)}(B_{2g},2)\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z293}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{2g}) = \mathbb{Q}_2^{(1,1;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z294}}$ $\mathbb{Q}_4^{(1,1;c)}(B_{2g}) = \mathbb{Q}_4^{(1,1;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z295}}$ $\mathbb{G}_3^{(1,0;c)}(B_{2g}) = \mathbb{G}_3^{(1,0;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z296}}$ $\mathbb{G}_5^{(1,0;c)}(B_{2g}) = \mathbb{G}_5^{(1,0;a)}(B_{2g})\mathbb{Q}_0^{(s)}(A_{1g})$

$\boxed{\text{z383}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g) = \dfrac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z384}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g) = \dfrac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z385}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g,1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z386}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g,1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z387}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g,2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z388}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g,2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z389}}$ $\mathbb{Q}_{6,1}^{(c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{Q}_{6,1}^{(a)}(E_g,1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z390}}$ $\mathbb{Q}_{6,2}^{(c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g,1)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z391}}$ $\mathbb{Q}_{6,1}^{(c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{Q}_{6,1}^{(a)}(E_g,2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z392}}$ $\mathbb{Q}_{6,2}^{(c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g,2)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z393}}$ $\mathbb{Q}_{6,1}^{(c)}(E_g,3) = \dfrac{\sqrt{2}\mathbb{Q}_{6,1}^{(a)}(E_g,3)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z394}}$ $\mathbb{Q}_{6,2}^{(c)}(E_g,3) = \dfrac{\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g,3)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z395}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = \dfrac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z396}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = \dfrac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_{1g})}{2}$

$\boxed{\text{z397}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z398}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z399}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z400}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z401}}$ $\mathbb{Q}_{6,1}^{(1,-1;c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z402}}$ $\mathbb{Q}_{6,2}^{(1,-1;c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z403}}$ $\mathbb{Q}_{6,1}^{(1,-1;c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z404}}$ $\mathbb{Q}_{6,2}^{(1,-1;c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z405}}$ $\mathbb{Q}_{6,1}^{(1,-1;c)}(E_g,3) = \dfrac{\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z406}}$ $\mathbb{Q}_{6,2}^{(1,-1;c)}(E_g,3) = \dfrac{\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z407}}$ $\mathbb{Q}_{2,1}^{(1,1;c)}(E_g) = \dfrac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,1;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z408}}$ $\mathbb{Q}_{2,2}^{(1,1;c)}(E_g) = \dfrac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,1;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z409}}$ $\mathbb{Q}_{4,1}^{(1,1;c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z410}}$ $\mathbb{Q}_{4,2}^{(1,1;c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z411}}$ $\mathbb{Q}_{4,1}^{(1,1;c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z412}}$ $\mathbb{Q}_{4,2}^{(1,1;c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z413}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(E_g) = \dfrac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z414}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(E_g) = \dfrac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z415}}$ $\mathbb{G}_{3,1}^{(1,0;c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z416}}$ $\mathbb{G}_{3,2}^{(1,0;c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z417}}$ $\mathbb{G}_{3,1}^{(1,0;c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z418}}$ $\mathbb{G}_{3,2}^{(1,0;c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z419}}$ $\mathbb{G}_{5,1}^{(1,0;c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z420}}$ $\mathbb{G}_{5,2}^{(1,0;c)}(E_g,1) = \dfrac{\sqrt{2}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,1)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z421}}$ $\mathbb{G}_{5,1}^{(1,0;c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z422}}$ $\mathbb{G}_{5,2}^{(1,0;c)}(E_g,2) = \dfrac{\sqrt{2}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,2)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z423}}$ $\mathbb{G}_{5,1}^{(1,0;c)}(E_g,3) = \dfrac{\sqrt{2}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

$\boxed{\text{z424}}$ $\mathbb{G}_{5,2}^{(1,0;c)}(E_g,3) = \dfrac{\sqrt{2}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,3)\mathbb{Q}_0^{(s)}(A_{1g})}{2}$

- `U;U_001_1` : 'U'-'U' bond-cluster
  * bra: $\langle \frac{3}{2}, \frac{3}{2}; d|$, $\langle \frac{3}{2}, \frac{1}{2}; d|$, $\langle \frac{3}{2}, -\frac{1}{2}; d|$, $\langle \frac{3}{2}, -\frac{3}{2}; d|$, $\langle \frac{5}{2}, \frac{5}{2}; d|$, $\langle \frac{5}{2}, \frac{3}{2}; d|$, $\langle \frac{5}{2}, \frac{1}{2}; d|$, $\langle \frac{5}{2}, -\frac{1}{2}; d|$, $\langle \frac{5}{2}, -\frac{3}{2}; d|$, $\langle \frac{5}{2}, -\frac{5}{2}; d|$
  * ket: $|\frac{3}{2}, \frac{3}{2}; d\rangle$, $|\frac{3}{2}, \frac{1}{2}; d\rangle$, $|\frac{3}{2}, -\frac{1}{2}; d\rangle$, $|\frac{3}{2}, -\frac{3}{2}; d\rangle$, $|\frac{5}{2}, \frac{5}{2}; d\rangle$, $|\frac{5}{2}, \frac{3}{2}; d\rangle$, $|\frac{5}{2}, \frac{1}{2}; d\rangle$, $|\frac{5}{2}, -\frac{1}{2}; d\rangle$, $|\frac{5}{2}, -\frac{3}{2}; d\rangle$, $|\frac{5}{2}, -\frac{5}{2}; d\rangle$
  * wyckoff: `4b@4c`

$\boxed{\text{z26}}$ $\mathbb{Q}_0^{(c)}(A_{1g}, a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z27}}$ $\mathbb{Q}_0^{(c)}(A_{1g}, b) = \mathbb{Q}_2^{(a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z28}}$ $\mathbb{Q}_2^{(c)}(A_{1g}, a) = \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z29}}$ $\mathbb{Q}_2^{(c)}(A_{1g}, b) = -\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z30}}$ $\mathbb{Q}_4^{(c)}(A_{1g}, 1) = \mathbb{Q}_4^{(a)}(A_{1g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z31}}$ $\mathbb{Q}_4^{(c)}(A_{1g}, 2) = \mathbb{Q}_4^{(a)}(A_{1g}, 2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z32}}$ $\mathbb{Q}_0^{(1,-1;c)}(A_{1g}) = \mathbb{Q}_2^{(1,-1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z33}}$ $\mathbb{Q}_2^{(1,-1;c)}(A_{1g}, a) = \mathbb{Q}_2^{(1,-1;a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z34}}$ $\mathbb{Q}_2^{(1,-1;c)}(A_{1g}, b) = -\mathbb{Q}_4^{(1,-1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z35}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 1) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z36}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{1g}, 2) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z37}}$ $\mathbb{Q}_2^{(1,0;c)}(A_{1g}) = -\mathbb{G}_3^{(1,0;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z38}}$ $\mathbb{Q}_0^{(1,1;c)}(A_{1g}, a) = \mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z39}}$ $\mathbb{Q}_0^{(1,1;c)}(A_{1g}, b) = \mathbb{Q}_2^{(1,1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z40}}$ $\mathbb{Q}_2^{(1,1;c)}(A_{1g}) = \mathbb{Q}_2^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z117}}$ $\mathbb{Q}_5^{(1,-1;c)}(A_{1u}) = \frac{\sqrt{14}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,1}^{(b)}(E_u)}{7} - \frac{\sqrt{42}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,1}^{(b)}(E_u)}{14} - \frac{\sqrt{14}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 2)\mathbb{T}_{1,2}^{(b)}(E_u)}{7} + \frac{\sqrt{42}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g, 3)\mathbb{T}_{1,2}^{(b)}(E_u)}{14}$

$\boxed{\text{z118}}$ $$\mathbb{Q}_5^{(1,0;c)}(A_{1u}) = -\frac{\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z119}}$ $$\mathbb{G}_0^{(c)}(A_{1u}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z120}}$ $$\mathbb{G}_2^{(c)}(A_{1u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z121}}$ $$\mathbb{G}_4^{(c)}(A_{1u},1) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z122}}$ $$\mathbb{G}_0^{(1,-1;c)}(A_{1u}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z123}}$ $$\mathbb{G}_2^{(1,-1;c)}(A_{1u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z124}}$ $$\begin{aligned}\mathbb{G}_4^{(1,-1;c)}(A_{1u},1a) = {} & \frac{7\sqrt{170}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{9\sqrt{238}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{952} + \frac{3\sqrt{714}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{476} \\ & - \frac{7\sqrt{170}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{9\sqrt{238}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{952} - \frac{3\sqrt{714}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{476}\end{aligned}$$

$\boxed{\text{z125}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_{1u},1b) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z126}}$ $$\begin{aligned}\mathbb{G}_4^{(1,-1;c)}(A_{1u},2) = {} & -\frac{3\sqrt{102}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{\sqrt{3570}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{\sqrt{1190}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{68} \\ & + \frac{3\sqrt{102}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{\sqrt{3570}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{\sqrt{1190}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{68}\end{aligned}$$

$\boxed{\text{z201}}$ $$\mathbb{G}_2^{(1,0;c)}(A_{1u}) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z202}}$ $$\mathbb{G}_4^{(1,0;c)}(A_{1u},1) = \frac{\sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z203}}$ $$\mathbb{G}_0^{(1,1;c)}(A_{1u}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z204}}$ $\mathbb{G}_2^{(1,1;c)}(A_{1u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$

$\boxed{\text{z205}}$ $\mathbb{G}_4^{(1,1;c)}(A_{1u},1) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$

$\boxed{\text{z206}}$ $\mathbb{Q}_4^{(c)}(A_{2g},a) = \mathbb{Q}_2^{(a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z207}}$ $\mathbb{Q}_4^{(c)}(A_{2g},b) = \mathbb{Q}_4^{(a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z208}}$ $\mathbb{Q}_4^{(c)}(A_{2g},c) = -\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z209}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{2g},a) = \mathbb{Q}_2^{(1,-1;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z210}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{2g},b) = \mathbb{Q}_4^{(1,-1;a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z211}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{2g},c) = -\mathbb{Q}_4^{(1,-1;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z212}}$ $\mathbb{Q}_4^{(1,0;c)}(A_{2g}) = \mathbb{G}_3^{(1,0;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z213}}$ $\mathbb{Q}_4^{(1,1;c)}(A_{2g}) = \mathbb{Q}_2^{(1,1;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z214}}$ $\mathbb{G}_1^{(1,0;c)}(A_{2g}) = \mathbb{G}_1^{(1,0;a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z215}}$ $\mathbb{G}_3^{(1,0;c)}(A_{2g}) = \mathbb{G}_3^{(1,0;a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z297}}$ $\mathbb{Q}_1^{(c)}(A_{2u}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$

$\boxed{\text{z298}}$ $\mathbb{Q}_3^{(c)}(A_{2u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$

$\boxed{\text{z299}}$ $\mathbb{Q}_1^{(1,-1;c)}(A_{2u}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$

$\boxed{\text{z300}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_{2u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$

$\boxed{\text{z301}}$ $$\mathbb{Q}_5^{(1,-1;c)}(A_{2u},1) = \frac{\sqrt{30}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{30}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z302}}$ $$\mathbb{Q}_5^{(1,-1;c)}(A_{2u},2) = \frac{3\sqrt{10}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{11\sqrt{14}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{112} - \frac{\sqrt{42}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{56} + \frac{3\sqrt{10}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{11\sqrt{14}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{112} - \frac{\sqrt{42}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{56}$$

$\boxed{\text{z303}}$ $$\mathbb{Q}_1^{(1,0;c)}(A_{2u}) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z304}}$ $$\mathbb{Q}_3^{(1,0;c)}(A_{2u}) = -\frac{\sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z305}}$ $$\mathbb{Q}_5^{(1,0;c)}(A_{2u},2) = \frac{\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z306}}$ $$\mathbb{Q}_1^{(1,1;c)}(A_{2u}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z425}}$ $$\mathbb{Q}_3^{(1,1;c)}(A_{2u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z426}}$ $$\mathbb{G}_4^{(c)}(A_{2u}) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z427}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_{2u},a) = -\frac{\sqrt{2}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{3\sqrt{70}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{56} + \frac{\sqrt{210}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{28} - \frac{\sqrt{2}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{3\sqrt{70}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{56} + \frac{\sqrt{210}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{28}$$

$\boxed{\text{z428}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_{2u},b) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z429}}$ $$\mathbb{G}_4^{(1,1;c)}(A_{2u}) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z430}}$ $\mathbb{Q}_2^{(c)}(B_{1g},a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z431}}$ $\mathbb{Q}_2^{(c)}(B_{1g},b) = \mathbb{Q}_2^{(a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z432}}$ $\mathbb{Q}_2^{(c)}(B_{1g},c) = -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z433}}$ $\mathbb{Q}_2^{(c)}(B_{1g},d) = \frac{\sqrt{21}\mathbb{Q}_4^{(a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{6} - \frac{\sqrt{15}\mathbb{Q}_4^{(a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{6}$

$\boxed{\text{z434}}$ $\mathbb{Q}_4^{(c)}(B_{1g},a) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z435}}$ $\mathbb{Q}_4^{(c)}(B_{1g},b) = \frac{\sqrt{15}\mathbb{Q}_4^{(a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{6} + \frac{\sqrt{21}\mathbb{Q}_4^{(a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{6}$

$\boxed{\text{z436}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{1g},a) = \mathbb{Q}_2^{(1,-1;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z437}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{1g},b) = -\mathbb{Q}_2^{(1,-1;a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z438}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{1g},c) = \frac{\sqrt{21}\mathbb{Q}_4^{(1,-1;a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{6} - \frac{\sqrt{15}\mathbb{Q}_4^{(1,-1;a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{6}$

$\boxed{\text{z439}}$ $\mathbb{Q}_4^{(1,-1;c)}(B_{1g},a) = \mathbb{Q}_4^{(1,-1;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z440}}$ $\mathbb{Q}_4^{(1,-1;c)}(B_{1g},b) = \frac{\sqrt{15}\mathbb{Q}_4^{(1,-1;a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{6} + \frac{\sqrt{21}\mathbb{Q}_4^{(1,-1;a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{6}$

$\boxed{\text{z441}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{1g},a) = \mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z442}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{1g},b) = \mathbb{Q}_2^{(1,1;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z443}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{1g},c) = -\mathbb{Q}_2^{(1,1;a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z444}}$ $\mathbb{G}_3^{(1,0;c)}(B_{1g}) = \mathbb{G}_3^{(1,0;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z445}}$ $\mathbb{Q}_3^{(c)}(B_{1u}) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z446}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{1u}) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z447}}$ $$\mathbb{Q}_5^{(1,-1;c)}(B_{1u}) = -\frac{\sqrt{78}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{13} - \frac{\sqrt{26}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{26} - \frac{\sqrt{78}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{13} - \frac{\sqrt{26}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{26}$$

$\boxed{\text{z448}}$ $$\mathbb{Q}_3^{(1,0;c)}(B_{1u},a) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z449}}$ $$\mathbb{Q}_3^{(1,0;c)}(B_{1u},b) = \frac{\sqrt{2}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z450}}$ $$\mathbb{Q}_5^{(1,0;c)}(B_{1u}) = \frac{\sqrt{2}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z451}}$ $$\mathbb{Q}_3^{(1,1;c)}(B_{1u}) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z452}}$ $$\mathbb{G}_2^{(c)}(B_{1u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z453}}$ $$\mathbb{G}_2^{(c)}(B_{1u},b) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z454}}$ $$\mathbb{G}_2^{(1,-1;c)}(B_{1u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z455}}$ $$\mathbb{G}_2^{(1,-1;c)}(B_{1u},b) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z456}}$ $$\mathbb{G}_4^{(1,-1;c)}(B_{1u}) = \frac{\sqrt{65}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{3\sqrt{91}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{3\sqrt{273}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{104} + \frac{\sqrt{65}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{3\sqrt{91}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} + \frac{3\sqrt{273}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{104}$$

$\boxed{\text{z457}}$ $$\mathbb{G}_6^{(1,-1;c)}(B_{1u},1) = \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{3\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$\boxed{\text{z458}}$ $$\mathbb{G}_2^{(1,1;c)}(B_{1u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z459}}$ $\mathbb{G}_2^{(1,1;c)}(B_{1u},b) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z460}}$ $\mathbb{Q}_2^{(c)}(B_{2g},a) = \mathbb{Q}_2^{(a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z461}}$ $\mathbb{Q}_2^{(c)}(B_{2g},b) = \mathbb{Q}_4^{(a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z462}}$ $\mathbb{Q}_4^{(c)}(B_{2g}) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z463}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{2g},a) = \mathbb{Q}_2^{(1,-1;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z464}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{2g},b) = \mathbb{Q}_4^{(1,-1;a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z707}}$ $\mathbb{Q}_4^{(1,-1;c)}(B_{2g}) = \mathbb{Q}_4^{(1,-1;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z708}}$ $\mathbb{Q}_2^{(1,0;c)}(B_{2g},a) = \mathbb{G}_1^{(1,0;a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z709}}$ $\mathbb{Q}_2^{(1,0;c)}(B_{2g},b) = -\mathbb{G}_3^{(1,0;a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z710}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{2g}) = \mathbb{Q}_2^{(1,1;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z711}}$ $\mathbb{G}_3^{(1,0;c)}(B_{2g}) = \mathbb{G}_3^{(1,0;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z712}}$ $\mathbb{Q}_3^{(c)}(B_{2u}) = \frac{\sqrt{30}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$

$\boxed{\text{z713}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{2u}) = \frac{\sqrt{30}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$

$\boxed{\text{z714}}$ $\mathbb{Q}_5^{(1,-1;c)}(B_{2u}) = -\frac{\sqrt{2730}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} + \frac{\sqrt{78}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} - \frac{5\sqrt{26}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{52}$
$+ \frac{\sqrt{2730}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{104} - \frac{\sqrt{78}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{104} + \frac{5\sqrt{26}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{52}$

$\boxed{\text{z715}}$ $\mathbb{Q}_3^{(1,0;c)}(B_{2u},a) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$

$\boxed{\text{z716}}$ $\mathbb{Q}_3^{(1,0;c)}(B_{2u},b) = -\frac{\sqrt{14}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{3\sqrt{2}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{14}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{3\sqrt{2}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$

$\boxed{\text{z717}}$ $$\mathbb{Q}_5^{(1,0;c)}(B_{2u}) = \frac{3\sqrt{2}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{14}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{3\sqrt{2}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{14}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z718}}$ $$\mathbb{Q}_3^{(1,1;c)}(B_{2u}) = \frac{\sqrt{30}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z719}}$ $$\mathbb{G}_2^{(c)}(B_{2u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z720}}$ $$\mathbb{G}_2^{(c)}(B_{2u},b) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{30}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z721}}$ $$\mathbb{G}_2^{(1,-1;c)}(B_{2u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z803}}$ $$\mathbb{G}_2^{(1,-1;c)}(B_{2u},b) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{30}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z804}}$ $$\mathbb{G}_4^{(1,-1;c)}(B_{2u}) = \frac{\sqrt{65}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} - \frac{15\sqrt{91}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} - \frac{\sqrt{273}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{104}$$
$$- \frac{\sqrt{65}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{15\sqrt{91}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{\sqrt{273}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{104}$$

$\boxed{\text{z805}}$ $$\mathbb{G}_6^{(1,-1;c)}(B_{2u},1) = \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$
$$- \frac{3\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{15}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z806}}$ $$\mathbb{G}_2^{(1,1;c)}(B_{2u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z807}}$ $$\mathbb{G}_2^{(1,1;c)}(B_{2u},b) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{30}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z808}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z809}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z810}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g,b) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z811}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z812}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g,c) = -\frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} + \frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z813}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g,c) = \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{3\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z814}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z815}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z816}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,1b) = -\frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z817}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,1b) = \frac{3\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z898}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z899}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z900}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z901}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z902}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,b) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z903}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z904}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,c) = -\frac{\sqrt{14}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} + \frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z905}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,c) = \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{3\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z906}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z907}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z908}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,1b) = -\frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z909}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,1b) = \frac{3\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z910}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z911}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z912}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,a) = -\frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z993}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z994}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,b) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{\sqrt{30}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z995}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,b) = -\frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} + \frac{\sqrt{30}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z996}}$ $\mathbb{Q}_{4,1}^{(1,0;c)}(E_g,1) = -\frac{\sqrt{30}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z997}}$ $\mathbb{Q}_{4,2}^{(1,0;c)}(E_g,1) = \frac{\sqrt{30}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} + \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z998}}$ $\mathbb{Q}_{2,1}^{(1,1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,1;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z999}}$ $\mathbb{Q}_{2,2}^{(1,1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,1;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1000}}$ $\mathbb{Q}_{2,1}^{(1,1;c)}(E_g,b) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,1;a)}(E_g)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1001}}$ $\mathbb{Q}_{2,2}^{(1,1;c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,1;a)}(E_g)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1002}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1003}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1004}}$ $\mathbb{G}_{3,1}^{(1,0;c)}(E_g,1) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1005}}$ $\mathbb{G}_{3,2}^{(1,0;c)}(E_g,1) = \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1006}}$ $\mathbb{G}_{3,1}^{(1,0;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1007}}$ $\mathbb{G}_{3,2}^{(1,0;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1141}}$ $\mathbb{Q}_{1,1}^{(c)}(E_u) = -\frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z1142}}$ $\mathbb{Q}_{1,2}^{(c)}(E_u) = \frac{\sqrt{2}\mathbb{M}_{1}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$

$\boxed{\text{z1143}}$ $\mathbb{Q}_{3,1}^{(c)}(E_u,1) = \frac{\sqrt{3}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$

$\boxed{\text{z1144}}$ $\mathbb{Q}_{3,2}^{(c)}(E_u,1) = -\frac{\sqrt{3}\mathbb{M}_{3}^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3}^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$

$\boxed{\text{z1145}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_u,2) = \frac{\sqrt{165}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{44} + \frac{2\sqrt{11}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{11} + \frac{3\sqrt{11}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{44}$$

$\boxed{\text{z1146}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_u,2) = -\frac{\sqrt{165}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{44} - \frac{2\sqrt{11}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{11} + \frac{3\sqrt{11}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{44}$$

$\boxed{\text{z1147}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(E_u) = -\frac{\sqrt{2}\mathbb{M}_1^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1148}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{2}\mathbb{M}_1^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1149}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E_u,1) = \frac{\sqrt{3}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z1150}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E_u,1) = -\frac{\sqrt{3}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z1151}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E_u,2) = \frac{\sqrt{165}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{44} + \frac{2\sqrt{11}\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{11} + \frac{3\sqrt{11}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{44}$$

$\boxed{\text{z1152}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E_u,2) = -\frac{\sqrt{165}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{44} - \frac{2\sqrt{11}\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{11} + \frac{3\sqrt{11}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{44}$$

$\boxed{\text{z1153}}$ $$\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,1) = -\frac{\sqrt{30}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{42}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$\boxed{\text{z1154}}$ $$\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,1) = \frac{\sqrt{30}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z1155}}$ $$\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,2) = -\frac{\sqrt{46}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{23} - \frac{3\sqrt{1610}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{368} + \frac{11\sqrt{46}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{368} + \frac{\sqrt{138}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{23} - \frac{\sqrt{138}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{184}$$

$\boxed{\text{z1156}}$ $$\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,2) = -\frac{\sqrt{46}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{23} + \frac{3\sqrt{1610}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{368} - \frac{11\sqrt{46}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{368} - \frac{\sqrt{138}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{23} - \frac{\sqrt{138}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{184}$$

$\boxed{\text{z1157}}$ $$\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,3) = \frac{\sqrt{28014}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{322} - \frac{7\sqrt{20010}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{5336} + \frac{3\sqrt{28014}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{1288} + \frac{5\sqrt{9338}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{9338} + \frac{39\sqrt{9338}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{18676}$$

$\boxed{\text{z1158}}$ $$\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,3) = \frac{\sqrt{28014}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{322} + \frac{7\sqrt{20010}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{5336} - \frac{3\sqrt{28014}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{1288} - \frac{5\sqrt{9338}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{9338} + \frac{39\sqrt{9338}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{18676}$$

$\boxed{\text{z1159}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(E_u) = -\frac{\sqrt{14}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{14} + \frac{\sqrt{42}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{14} + \frac{\sqrt{42}\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{14}$$

$\boxed{\text{z1160}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(E_u) = -\frac{\sqrt{14}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{14} - \frac{\sqrt{42}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{14} + \frac{\sqrt{42}\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{14}$$

$\boxed{\text{z1161}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,1a) = -\frac{\sqrt{42}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{28} + \frac{3\sqrt{14}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{28} - \frac{\sqrt{14}\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{7}$$

$\boxed{\text{z1162}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,1a) = -\frac{\sqrt{42}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{28} - \frac{3\sqrt{14}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{28} - \frac{\sqrt{14}\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{7}$$

$\boxed{\text{z1163}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,1b) = \frac{\sqrt{273}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{39} - \frac{\sqrt{195}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{78} + \frac{\sqrt{455}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{52} + \frac{\sqrt{65}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{26} - \frac{\sqrt{65}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{52}$$

$\boxed{\text{z1164}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,1b) = \frac{\sqrt{273}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{39} - \frac{\sqrt{195}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{78} - \frac{\sqrt{455}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{52} - \frac{\sqrt{65}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{26} - \frac{\sqrt{65}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{52}$$

$\boxed{\text{z1165}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,2a) = -\frac{\sqrt{6}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{2}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z1166}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,2a) = -\frac{\sqrt{6}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{2}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z1167}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,2b) = \frac{\sqrt{5655}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{1131} + \frac{73\sqrt{7917}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{15834} - \frac{21\sqrt{377}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{1508} + \frac{31\sqrt{2639}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{5278} - \frac{83\sqrt{2639}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{10556}$$

$\boxed{\text{z1168}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,2b)=\frac{\sqrt{5655}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{1131}+\frac{73\sqrt{7917}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{15834}+\frac{21\sqrt{377}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{1508}-\frac{31\sqrt{2639}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{5278}-\frac{83\sqrt{2639}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{10556}$$

$\boxed{\text{z1169}}$ $$\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,1)=\frac{5\sqrt{58}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{116}-\frac{41\sqrt{2030}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8120}-\frac{7\sqrt{870}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{435}+\frac{59\sqrt{6090}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{24360}+\frac{11\sqrt{6090}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{3045}$$

$\boxed{\text{z1170}}$ $$\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,1)=\frac{5\sqrt{58}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{116}-\frac{41\sqrt{2030}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8120}+\frac{7\sqrt{870}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{435}-\frac{59\sqrt{6090}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{24360}+\frac{11\sqrt{6090}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{3045}$$

$\boxed{\text{z1171}}$ $$\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,2)=\frac{5\sqrt{62}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{124}-\frac{3\sqrt{2170}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{1240}-\frac{2\sqrt{930}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{465}-\frac{23\sqrt{6510}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{3720}-\frac{2\sqrt{6510}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{465}$$

$\boxed{\text{z1172}}$ $$\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,2)=\frac{5\sqrt{62}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{124}-\frac{3\sqrt{2170}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{1240}+\frac{2\sqrt{930}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{465}+\frac{23\sqrt{6510}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{3720}-\frac{2\sqrt{6510}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{465}$$

$\boxed{\text{z1173}}$ $$\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,3)=\frac{\sqrt{930}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{93}+\frac{5\sqrt{1302}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{372}+\frac{\sqrt{62}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{62}-\frac{\sqrt{434}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{124}+\frac{\sqrt{434}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{62}$$

$\boxed{\text{z1174}}$ $$\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,3)=\frac{\sqrt{930}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{93}+\frac{5\sqrt{1302}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{372}-\frac{\sqrt{62}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{62}+\frac{\sqrt{434}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{124}+\frac{\sqrt{434}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{62}$$

$\boxed{\text{z1175}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(E_u)=-\frac{\sqrt{2}\mathbb{M}_1^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1176}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(E_u)=\frac{\sqrt{2}\mathbb{M}_1^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1177}}$ $$\mathbb{Q}_{3,1}^{(1,1;c)}(E_u,1)=\frac{\sqrt{3}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}-\frac{\sqrt{5}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

z1178 $$\mathbb{Q}_{3,2}^{(1,1;c)}(E_u,1) = -\frac{\sqrt{3}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

z1179 $$\mathbb{Q}_{3,1}^{(1,1;c)}(E_u,2) = \frac{\sqrt{165}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{44} + \frac{2\sqrt{11}\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{11} + \frac{3\sqrt{11}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{44}$$

z1180 $$\mathbb{Q}_{3,2}^{(1,1;c)}(E_u,2) = -\frac{\sqrt{165}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{44} - \frac{2\sqrt{11}\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{11} + \frac{3\sqrt{11}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{44}$$

z1181 $$\mathbb{G}_{2,1}^{(c)}(E_u) = -\frac{\sqrt{110}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} + \frac{\sqrt{66}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22}$$

z1182 $$\mathbb{G}_{2,2}^{(c)}(E_u) = \frac{\sqrt{110}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22}$$

z1183 $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = -\frac{\sqrt{110}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} + \frac{\sqrt{66}\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22}$$

z1184 $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{110}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22}$$

z1185 $$\begin{aligned}\mathbb{G}_{4,1}^{(1,-1;c)}(E_u,1) = &-\frac{3\sqrt{100282}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{6916} + \frac{49\sqrt{71630}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{28652} + \frac{3\sqrt{100282}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{6916}\\ &+ \frac{79\sqrt{300846}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{200564} + \frac{83\sqrt{300846}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{100282}\end{aligned}$$

z1186 $$\begin{aligned}\mathbb{G}_{4,2}^{(1,-1;c)}(E_u,1) = &-\frac{3\sqrt{100282}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{6916} - \frac{49\sqrt{71630}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{28652} - \frac{3\sqrt{100282}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{6916}\\ &- \frac{79\sqrt{300846}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{200564} + \frac{83\sqrt{300846}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{100282}\end{aligned}$$

z1187 $$\begin{aligned}\mathbb{G}_{4,1}^{(1,-1;c)}(E_u,2) = &\frac{7\sqrt{2470}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{988} + \frac{3\sqrt{3458}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{988}\\ &- \frac{7\sqrt{2470}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{988} + \frac{5\sqrt{7410}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{988} - \frac{\sqrt{7410}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{494}\end{aligned}$$

z1188 $$\begin{aligned}\mathbb{G}_{4,2}^{(1,-1;c)}(E_u,2) = &\frac{7\sqrt{2470}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{988} - \frac{3\sqrt{3458}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{988}\\ &+ \frac{7\sqrt{2470}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{988} - \frac{5\sqrt{7410}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{988} - \frac{\sqrt{7410}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{494}\end{aligned}$$

$\boxed{\text{z1189}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(E_u) = -\frac{\sqrt{110}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} + \frac{\sqrt{66}\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22}$

$\boxed{\text{z1190}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{110}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22}$

- `U;U_001_1` : 'U'-'U' bond-cluster
  * bra: $\langle \frac{3}{2}, \frac{3}{2}; d|$, $\langle \frac{3}{2}, \frac{1}{2}; d|$, $\langle \frac{3}{2}, -\frac{1}{2}; d|$, $\langle \frac{3}{2}, -\frac{3}{2}; d|$, $\langle \frac{5}{2}, \frac{5}{2}; d|$, $\langle \frac{5}{2}, \frac{3}{2}; d|$, $\langle \frac{5}{2}, \frac{1}{2}; d|$, $\langle \frac{5}{2}, -\frac{1}{2}; d|$, $\langle \frac{5}{2}, -\frac{3}{2}; d|$, $\langle \frac{5}{2}, -\frac{5}{2}; d|$
  * ket: $|\frac{5}{2}, \frac{5}{2}; f\rangle$, $|\frac{5}{2}, \frac{3}{2}; f\rangle$, $|\frac{5}{2}, \frac{1}{2}; f\rangle$, $|\frac{5}{2}, -\frac{1}{2}; f\rangle$, $|\frac{5}{2}, -\frac{3}{2}; f\rangle$, $|\frac{5}{2}, -\frac{5}{2}; f\rangle$, $|\frac{7}{2}, \frac{7}{2}; f\rangle$, $|\frac{7}{2}, \frac{5}{2}; f\rangle$, $|\frac{7}{2}, \frac{3}{2}; f\rangle$, $|\frac{7}{2}, \frac{1}{2}; f\rangle$, $|\frac{7}{2}, -\frac{1}{2}; f\rangle$, $|\frac{7}{2}, -\frac{3}{2}; f\rangle$, $|\frac{7}{2}, -\frac{5}{2}; f\rangle$, $|\frac{7}{2}, -\frac{7}{2}; f\rangle$
  * wyckoff: `4b@4c`

$\boxed{\text{z41}}$ $\mathbb{Q}_0^{(c)}(A_{1g}) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{1,2}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z42}}$ $\mathbb{Q}_2^{(c)}(A_{1g}, a) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z43}}$ $\mathbb{Q}_2^{(c)}(A_{1g}, b) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$

$\boxed{\text{z44}}$ $\mathbb{Q}_4^{(c)}(A_{1g}, 1a) = \frac{\sqrt{2}\mathbb{M}_{4,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{4,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z45}}$ $\mathbb{Q}_4^{(c)}(A_{1g}, 1b) = \frac{\sqrt{770}\mathbb{T}_{5,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{44} + \frac{3\sqrt{22}\mathbb{T}_{5,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{44} + \frac{\sqrt{770}\mathbb{T}_{5,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{44} + \frac{3\sqrt{22}\mathbb{T}_{5,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{44}$

$\boxed{\text{z46}}$ $\mathbb{Q}_4^{(c)}(A_{1g}, 1c) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$

$\boxed{\text{z47}}$ $\mathbb{Q}_4^{(c)}(A_{1g}, 2a) = \frac{\sqrt{2}\mathbb{M}_{4,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{4,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z48}}$ $\mathbb{Q}_4^{(c)}(A_{1g}, 2b) = -\frac{3\sqrt{7590}\mathbb{T}_{5,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2024} + \frac{5\sqrt{10626}\mathbb{T}_{5,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2024} + \frac{\sqrt{3542}\mathbb{T}_{5,1}^{(a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{92}$
$- \frac{3\sqrt{7590}\mathbb{T}_{5,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2024} + \frac{5\sqrt{10626}\mathbb{T}_{5,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2024} + \frac{\sqrt{3542}\mathbb{T}_{5,2}^{(a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{92}$

$\boxed{\text{z49}}$ $\mathbb{Q}_6^{(c)}(A_{1g}, 1) = \frac{3\sqrt{322}\mathbb{T}_{5,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{184} - \frac{7\sqrt{230}\mathbb{T}_{5,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{184} + \frac{\sqrt{690}\mathbb{T}_{5,1}^{(a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{92}$
$+ \frac{3\sqrt{322}\mathbb{T}_{5,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{184} - \frac{7\sqrt{230}\mathbb{T}_{5,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{184} + \frac{\sqrt{690}\mathbb{T}_{5,2}^{(a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{92}$

$\boxed{\text{z50}}$ $$\mathbb{Q}_2^{(1,-1;c)}(A_{1g},a) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z51}}$ $$\mathbb{Q}_2^{(1,-1;c)}(A_{1g},b) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z52}}$ $$\mathbb{Q}_4^{(1,-1;c)}(A_{1g},1a) = \frac{\sqrt{2}\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z53}}$ $$\mathbb{Q}_4^{(1,-1;c)}(A_{1g},1b) = \frac{\sqrt{770}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{44} + \frac{3\sqrt{22}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{44} + \frac{\sqrt{770}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{44} + \frac{3\sqrt{22}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{44}$$

$\boxed{\text{z54}}$ $$\mathbb{Q}_4^{(1,-1;c)}(A_{1g},1c) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z55}}$ $$\mathbb{Q}_4^{(1,-1;c)}(A_{1g},2a) = \frac{\sqrt{2}\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z56}}$ $$\begin{aligned}\mathbb{Q}_4^{(1,-1;c)}(A_{1g},2b) = &-\frac{3\sqrt{7590}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2024} + \frac{5\sqrt{10626}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2024} + \frac{\sqrt{3542}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{92}\\ &- \frac{3\sqrt{7590}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2024} + \frac{5\sqrt{10626}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2024} + \frac{\sqrt{3542}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{92}\end{aligned}$$

$\boxed{\text{z57}}$ $$\mathbb{Q}_6^{(1,-1;c)}(A_{1g},1a) = -\frac{\sqrt{2}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z58}}$ $$\begin{aligned}\mathbb{Q}_6^{(1,-1;c)}(A_{1g},1b) = &\frac{3\sqrt{322}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{184} - \frac{7\sqrt{230}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{184} + \frac{\sqrt{690}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{92}\\ &+ \frac{3\sqrt{322}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{184} - \frac{7\sqrt{230}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{184} + \frac{\sqrt{690}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{92}\end{aligned}$$

$\boxed{\text{z59}}$ $$\mathbb{Q}_6^{(1,-1;c)}(A_{1g},2) = -\frac{3\sqrt{143}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{52} - \frac{\sqrt{65}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{52} - \frac{3\sqrt{143}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{52} - \frac{\sqrt{65}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{52}$$

$\boxed{\text{z60}}$ $$\mathbb{Q}_0^{(1,0;c)}(A_{1g}) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,0;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,0;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z61}}$ $$\mathbb{Q}_2^{(1,0;c)}(A_{1g},a) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,0;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,0;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z62}}$ $$\mathbb{Q}_2^{(1,0;c)}(A_{1g},b) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z63}}$ $$\mathbb{Q}_4^{(1,0;c)}(A_{1g},1a) = \frac{\sqrt{2}\mathbb{M}_{4,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{4,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z64}}$ $$\mathbb{Q}_4^{(1,0;c)}(A_{1g},1b) = \frac{\sqrt{770}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{44} + \frac{3\sqrt{22}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{44} + \frac{\sqrt{770}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{44} + \frac{3\sqrt{22}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{44}$$

$\boxed{\text{z65}}$ $$\mathbb{Q}_4^{(1,0;c)}(A_{1g},1c) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z66}}$ $$\mathbb{Q}_4^{(1,0;c)}(A_{1g},2a) = \frac{\sqrt{2}\mathbb{M}_{4,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{4,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z67}}$ $$\begin{aligned}\mathbb{Q}_4^{(1,0;c)}(A_{1g},2b) = &-\frac{3\sqrt{7590}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2024} + \frac{5\sqrt{10626}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2024} + \frac{\sqrt{3542}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{92}\\ &- \frac{3\sqrt{7590}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2024} + \frac{5\sqrt{10626}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2024} + \frac{\sqrt{3542}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{92}\end{aligned}$$

$\boxed{\text{z68}}$ $$\begin{aligned}\mathbb{Q}_6^{(1,0;c)}(A_{1g},1) = &\frac{3\sqrt{322}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{184} - \frac{7\sqrt{230}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{184} + \frac{\sqrt{690}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{92}\\ &+ \frac{3\sqrt{322}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{184} - \frac{7\sqrt{230}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{184} + \frac{\sqrt{690}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{92}\end{aligned}$$

$\boxed{\text{z69}}$ $$\mathbb{Q}_0^{(1,1;c)}(A_{1g}) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z70}}$ $$\mathbb{Q}_2^{(1,1;c)}(A_{1g},a) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z71}}$ $$\mathbb{Q}_2^{(1,1;c)}(A_{1g},b) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z72}}$ $$\mathbb{Q}_4^{(1,1;c)}(A_{1g},1a) = \frac{\sqrt{2}\mathbb{M}_{4,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{4,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z73}}$ $$\mathbb{Q}_4^{(1,1;c)}(A_{1g},1b) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z74}}$ $\mathbb{Q}_4^{(1,1;c)}(A_{1g},2) = \frac{\sqrt{2}\mathbb{M}_{4,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{4,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z75}}$ $\mathbb{G}_5^{(1,-1;c)}(A_{1g}) = -\frac{\sqrt{65}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{52} + \frac{3\sqrt{143}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{52} - \frac{\sqrt{65}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{52} + \frac{3\sqrt{143}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{52}$

$\boxed{\text{z127}}$ $\mathbb{Q}_5^{(c)}(A_{1u},a) = \mathbb{Q}_5^{(a)}(A_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z128}}$ $\mathbb{Q}_5^{(c)}(A_{1u},b) = \mathbb{Q}_5^{(a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z129}}$ $\mathbb{Q}_5^{(c)}(A_{1u},c) = \mathbb{Q}_3^{(a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z130}}$ $\mathbb{Q}_5^{(c)}(A_{1u},d) = -\mathbb{G}_4^{(a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z131}}$ $\mathbb{Q}_5^{(1,-1;c)}(A_{1u},a) = \mathbb{Q}_5^{(1,-1;a)}(A_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z132}}$ $\mathbb{Q}_5^{(1,-1;c)}(A_{1u},b) = \mathbb{Q}_5^{(1,-1;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z133}}$ $\mathbb{Q}_5^{(1,-1;c)}(A_{1u},c) = \mathbb{Q}_3^{(1,-1;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z134}}$ $\mathbb{Q}_5^{(1,-1;c)}(A_{1u},d) = -\mathbb{G}_4^{(1,-1;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z135}}$ $\mathbb{Q}_5^{(1,-1;c)}(A_{1u},e) = \frac{\sqrt{11}\mathbb{G}_6^{(1,-1;a)}(B_{1u},1)\mathbb{Q}_2^{(b)}(B_{1g})}{4} + \frac{\sqrt{5}\mathbb{G}_6^{(1,-1;a)}(B_{1u},2)\mathbb{Q}_2^{(b)}(B_{1g})}{4}$

$\boxed{\text{z136}}$ $\mathbb{Q}_7^{(1,-1;c)}(A_{1u}) = -\frac{\sqrt{5}\mathbb{G}_6^{(1,-1;a)}(B_{1u},1)\mathbb{Q}_2^{(b)}(B_{1g})}{4} + \frac{\sqrt{11}\mathbb{G}_6^{(1,-1;a)}(B_{1u},2)\mathbb{Q}_2^{(b)}(B_{1g})}{4}$

$\boxed{\text{z137}}$ $\mathbb{Q}_5^{(1,0;c)}(A_{1u},a) = \mathbb{Q}_5^{(1,0;a)}(A_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z138}}$ $\mathbb{Q}_5^{(1,0;c)}(A_{1u},b) = \mathbb{Q}_5^{(1,0;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z139}}$ $\mathbb{Q}_5^{(1,0;c)}(A_{1u},c) = \mathbb{Q}_3^{(1,0;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z140}}$ $\mathbb{Q}_5^{(1,0;c)}(A_{1u},d) = -\mathbb{G}_4^{(1,0;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z141}}$ $\mathbb{Q}_5^{(1,1;c)}(A_{1u},a) = \mathbb{Q}_3^{(1,1;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z142}}$ $\mathbb{Q}_5^{(1,1;c)}(A_{1u},b) = -\mathbb{G}_4^{(1,1;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z143}}$ $\mathbb{G}_0^{(c)}(A_{1u}) = \mathbb{G}_2^{(a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z144}}$ $\mathbb{G}_2^{(c)}(A_{1u}) = \mathbb{G}_2^{(a)}(A_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z145}}$ $\mathbb{G}_4^{(c)}(A_{1u},1) = \mathbb{G}_4^{(a)}(A_{1u},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z146}}$ $\mathbb{G}_4^{(c)}(A_{1u},2) = \mathbb{G}_4^{(a)}(A_{1u},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z147}}$ $\mathbb{G}_0^{(1,-1;c)}(A_{1u}) = \mathbb{G}_2^{(1,-1;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z148}}$ $\mathbb{G}_2^{(1,-1;c)}(A_{1u}) = \mathbb{G}_2^{(1,-1;a)}(A_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z149}}$ $\mathbb{G}_4^{(1,-1;c)}(A_{1u},1) = \mathbb{G}_4^{(1,-1;a)}(A_{1u},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z150}}$ $\mathbb{G}_4^{(1,-1;c)}(A_{1u},2) = \mathbb{G}_4^{(1,-1;a)}(A_{1u},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z151}}$ $\mathbb{G}_6^{(1,-1;c)}(A_{1u},1) = \mathbb{G}_6^{(1,-1;a)}(A_{1u},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z152}}$ $\mathbb{G}_6^{(1,-1;c)}(A_{1u},2) = \mathbb{G}_6^{(1,-1;a)}(A_{1u},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z153}}$ $\mathbb{G}_0^{(1,0;c)}(A_{1u}) = \mathbb{G}_2^{(1,0;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z154}}$ $\mathbb{G}_2^{(1,0;c)}(A_{1u}) = \mathbb{G}_2^{(1,0;a)}(A_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z155}}$ $\mathbb{G}_4^{(1,0;c)}(A_{1u},1) = \mathbb{G}_4^{(1,0;a)}(A_{1u},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z156}}$ $\mathbb{G}_4^{(1,0;c)}(A_{1u},2) = \mathbb{G}_4^{(1,0;a)}(A_{1u},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z157}}$ $\mathbb{G}_0^{(1,1;c)}(A_{1u},a) = \mathbb{G}_0^{(1,1;a)}(A_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z158}}$ $\mathbb{G}_0^{(1,1;c)}(A_{1u},b) = \mathbb{G}_2^{(1,1;a)}(B_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z159}}$ $\mathbb{G}_2^{(1,1;c)}(A_{1u}) = \mathbb{G}_2^{(1,1;a)}(A_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z160}}$ $\mathbb{G}_4^{(1,1;c)}(A_{1u},1) = \mathbb{G}_4^{(1,1;a)}(A_{1u},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z161}}$ $\mathbb{G}_4^{(1,1;c)}(A_{1u},2) = \mathbb{G}_4^{(1,1;a)}(A_{1u},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z216}}$ $$\mathbb{Q}_4^{(c)}(A_{2g},a) = \frac{\mathbb{M}_{4,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{M}_{4,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z217}}$ $$\mathbb{Q}_4^{(c)}(A_{2g},b) = -\frac{\sqrt{1610}\mathbb{T}_{5,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{92} - \frac{3\sqrt{46}\mathbb{T}_{5,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{92} + \frac{\sqrt{138}\mathbb{T}_{5,1}^{(a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{23} + \frac{\sqrt{1610}\mathbb{T}_{5,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{92} + \frac{3\sqrt{46}\mathbb{T}_{5,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{92} - \frac{\sqrt{138}\mathbb{T}_{5,2}^{(a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{23}$$

$\boxed{\text{z218}}$ $$\mathbb{Q}_4^{(c)}(A_{2g},c) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{T}_{3,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z219}}$ $$\mathbb{Q}_6^{(c)}(A_{2g}) = -\frac{9\sqrt{322}\mathbb{T}_{5,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{368} + \frac{13\sqrt{230}\mathbb{T}_{5,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{368} - \frac{\sqrt{690}\mathbb{T}_{5,1}^{(a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{184} + \frac{9\sqrt{322}\mathbb{T}_{5,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{368} - \frac{13\sqrt{230}\mathbb{T}_{5,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{368} + \frac{\sqrt{690}\mathbb{T}_{5,2}^{(a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{184}$$

$\boxed{\text{z220}}$ $$\mathbb{Q}_4^{(1,-1;c)}(A_{2g},a) = \frac{\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z221}}$ $$\mathbb{Q}_4^{(1,-1;c)}(A_{2g},b) = -\frac{\sqrt{1610}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{92} - \frac{3\sqrt{46}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{92} + \frac{\sqrt{138}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{23} + \frac{\sqrt{1610}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{92} + \frac{3\sqrt{46}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{92} - \frac{\sqrt{138}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{23}$$

$\boxed{\text{z222}}$ $$\mathbb{Q}_4^{(1,-1;c)}(A_{2g},c) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z223}}$ $$\mathbb{Q}_6^{(1,-1;c)}(A_{2g},a) = \frac{\sqrt{26}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{52} - \frac{\sqrt{429}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} + \frac{5\sqrt{195}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} - \frac{\sqrt{26}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{52} + \frac{\sqrt{429}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{104} - \frac{5\sqrt{195}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{104}$$

$\boxed{\text{z224}}$ $$\mathbb{Q}_6^{(1,-1;c)}(A_{2g},b) = -\frac{9\sqrt{322}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{368} + \frac{13\sqrt{230}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{368} - \frac{\sqrt{690}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{184} + \frac{9\sqrt{322}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{368} - \frac{13\sqrt{230}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{368} + \frac{\sqrt{690}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{184}$$

$\boxed{\text{z225}}$ $$\mathbb{Q}_4^{(1,0;c)}(A_{2g},a) = \frac{\mathbb{M}_{4,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z226}}$ $$\mathbb{Q}_4^{(1,0;c)}(A_{2g},b) = -\frac{\sqrt{1610}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{92} - \frac{3\sqrt{46}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{92} + \frac{\sqrt{138}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{23}$$
$$+ \frac{\sqrt{1610}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{92} + \frac{3\sqrt{46}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{92} - \frac{\sqrt{138}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{23}$$

$\boxed{\text{z227}}$ $$\mathbb{Q}_4^{(1,0;c)}(A_{2g},c) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z228}}$ $$\mathbb{Q}_6^{(1,0;c)}(A_{2g}) = -\frac{9\sqrt{322}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{368} + \frac{13\sqrt{230}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{368} - \frac{\sqrt{690}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{184}$$
$$+ \frac{9\sqrt{322}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{368} - \frac{13\sqrt{230}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{368} + \frac{\sqrt{690}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{184}$$

$\boxed{\text{z229}}$ $$\mathbb{Q}_4^{(1,1;c)}(A_{2g},a) = \frac{\mathbb{M}_{4,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z230}}$ $$\mathbb{Q}_4^{(1,1;c)}(A_{2g},b) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z231}}$ $$\mathbb{G}_1^{(c)}(A_{2g},a) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z232}}$ $$\mathbb{G}_1^{(c)}(A_{2g},b) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{T}_{1,2}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z233}}$ $$\mathbb{G}_3^{(c)}(A_{2g},a) = -\frac{\sqrt{7}\mathbb{M}_{4,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\mathbb{M}_{4,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z234}}$ $$\mathbb{G}_3^{(c)}(A_{2g},b) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{T}_{3,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z235}}$ $$\mathbb{G}_5^{(c)}(A_{2g},1) = \frac{\sqrt{30}\mathbb{T}_{5,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{T}_{5,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{T}_{5,1}^{(a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{T}_{5,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{42}\mathbb{T}_{5,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{14}\mathbb{T}_{5,2}^{(a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z236}}$ $$\mathbb{G}_1^{(1,-1;c)}(A_{2g}) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z237}}$ $$\mathbb{G}_3^{(1,-1;c)}(A_{2g},a) = -\frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z238}}$ $$\mathbb{G}_3^{(1,-1;c)}(A_{2g},b) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z239}}$ $$\mathbb{G}_5^{(1,-1;c)}(A_{2g},1a) = \frac{\sqrt{6}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{3\sqrt{11}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{6}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{3\sqrt{11}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{5}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{16}$$

$\boxed{\text{z240}}$ $$\mathbb{G}_5^{(1,-1;c)}(A_{2g},1b) = \frac{\sqrt{30}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{42}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{14}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z241}}$ $$\mathbb{G}_5^{(1,-1;c)}(A_{2g},2) = -\frac{\sqrt{4290}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} + \frac{7\sqrt{65}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} + \frac{3\sqrt{143}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} + \frac{\sqrt{4290}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{104} - \frac{7\sqrt{65}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} - \frac{3\sqrt{143}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{208}$$

$\boxed{\text{z242}}$ $$\mathbb{G}_1^{(1,0;c)}(A_{2g},a) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,0;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,0;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z243}}$ $$\mathbb{G}_1^{(1,0;c)}(A_{2g},b) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,0;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,0;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z244}}$ $$\mathbb{G}_3^{(1,0;c)}(A_{2g},a) = -\frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\mathbb{M}_{4,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z245}}$ $$\mathbb{G}_3^{(1,0;c)}(A_{2g},b) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z246}}$ $$\mathbb{G}_5^{(1,0;c)}(A_{2g},1) = \frac{\sqrt{30}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{42}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{14}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z247}}$ $$\mathbb{G}_1^{(1,1;c)}(A_{2g},a) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z248}}$ $\mathbb{G}_1^{(1,1;c)}(A_{2g},b) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$

$\boxed{\text{z249}}$ $\mathbb{G}_3^{(1,1;c)}(A_{2g},a) = -\frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\mathbb{M}_{4,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$

$\boxed{\text{z250}}$ $\mathbb{G}_3^{(1,1;c)}(A_{2g},b) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$

$\boxed{\text{z307}}$ $\mathbb{Q}_1^{(c)}(A_{2u},a) = \mathbb{Q}_1^{(a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z308}}$ $\mathbb{Q}_1^{(c)}(A_{2u},b) = \mathbb{Q}_3^{(a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z309}}$ $\mathbb{Q}_1^{(c)}(A_{2u},c) = -\mathbb{G}_2^{(a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z310}}$ $\mathbb{Q}_3^{(c)}(A_{2u},a) = \mathbb{Q}_5^{(a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z311}}$ $\mathbb{Q}_3^{(c)}(A_{2u},b) = \mathbb{Q}_3^{(a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z312}}$ $\mathbb{Q}_3^{(c)}(A_{2u},c) = -\mathbb{G}_4^{(a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z313}}$ $\mathbb{Q}_5^{(c)}(A_{2u},1) = \mathbb{Q}_5^{(a)}(A_{2u},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z314}}$ $\mathbb{Q}_5^{(c)}(A_{2u},2) = \mathbb{Q}_5^{(a)}(A_{2u},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z315}}$ $\mathbb{Q}_1^{(1,-1;c)}(A_{2u},a) = \mathbb{Q}_3^{(1,-1;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z316}}$ $\mathbb{Q}_1^{(1,-1;c)}(A_{2u},b) = -\mathbb{G}_2^{(1,-1;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z317}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_{2u},a) = \mathbb{Q}_5^{(1,-1;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z318}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_{2u},b) = \mathbb{Q}_3^{(1,-1;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z319}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_{2u},c) = -\mathbb{G}_4^{(1,-1;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z320}}$ $\mathbb{Q}_5^{(1,-1;c)}(A_{2u},1a) = \mathbb{Q}_5^{(1,-1;a)}(A_{2u},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z321}}$ $\mathbb{Q}_5^{(1,-1;c)}(A_{2u},1b) = -\mathbb{G}_6^{(1,-1;a)}(B_{2u},2)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z322}}$ $\mathbb{Q}_5^{(1,-1;c)}(A_{2u},2a) = \mathbb{Q}_5^{(1,-1;a)}(A_{2u},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z323}}$ $\mathbb{Q}_5^{(1,-1;c)}(A_{2u},2b) = -\mathbb{G}_6^{(1,-1;a)}(B_{2u},1)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z324}}$ $\mathbb{Q}_1^{(1,0;c)}(A_{2u},a) = \mathbb{Q}_1^{(1,0;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z325}}$ $\mathbb{Q}_1^{(1,0;c)}(A_{2u},b) = \mathbb{Q}_3^{(1,0;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z326}}$ $\mathbb{Q}_1^{(1,0;c)}(A_{2u},c) = -\mathbb{G}_2^{(1,0;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z327}}$ $\mathbb{Q}_3^{(1,0;c)}(A_{2u},a) = \mathbb{Q}_5^{(1,0;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z328}}$ $\mathbb{Q}_3^{(1,0;c)}(A_{2u},b) = \mathbb{Q}_3^{(1,0;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z329}}$ $\mathbb{Q}_3^{(1,0;c)}(A_{2u},c) = -\mathbb{G}_4^{(1,0;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z330}}$ $\mathbb{Q}_5^{(1,0;c)}(A_{2u},1) = \mathbb{Q}_5^{(1,0;a)}(A_{2u},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z331}}$ $\mathbb{Q}_5^{(1,0;c)}(A_{2u},2) = \mathbb{Q}_5^{(1,0;a)}(A_{2u},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z332}}$ $\mathbb{Q}_1^{(1,1;c)}(A_{2u},a) = \mathbb{Q}_1^{(1,1;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z333}}$ $\mathbb{Q}_1^{(1,1;c)}(A_{2u},b) = \mathbb{Q}_3^{(1,1;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z334}}$ $\mathbb{Q}_1^{(1,1;c)}(A_{2u},c) = -\mathbb{G}_2^{(1,1;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z335}}$ $\mathbb{Q}_3^{(1,1;c)}(A_{2u},a) = \mathbb{Q}_3^{(1,1;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z336}}$ $\mathbb{Q}_3^{(1,1;c)}(A_{2u},b) = -\mathbb{G}_4^{(1,1;a)}(B_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z337}}$ $\mathbb{G}_4^{(c)}(A_{2u}) = \mathbb{G}_4^{(a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z338}}$ $\mathbb{G}_4^{(1,-1;c)}(A_{2u}) = \mathbb{G}_4^{(1,-1;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z339}}$ $\mathbb{G}_6^{(1,-1;c)}(A_{2u}) = \mathbb{G}_6^{(1,-1;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z340}}$ $\mathbb{G}_4^{(1,0;c)}(A_{2u}) = \mathbb{G}_4^{(1,0;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z341}}$ $\mathbb{G}_4^{(1,1;c)}(A_{2u}) = \mathbb{G}_4^{(1,1;a)}(A_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z465}}$ $\mathbb{Q}_2^{(c)}(B_{1g},a) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z466}}$ $\mathbb{Q}_2^{(c)}(B_{1g},b) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{T}_{1,2}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z467}}$ $\mathbb{Q}_2^{(c)}(B_{1g},c) = \frac{3\sqrt{3}\mathbb{T}_{3,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{5}\mathbb{T}_{3,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{3\sqrt{3}\mathbb{T}_{3,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$

$\boxed{\text{z468}}$ $\mathbb{Q}_4^{(c)}(B_{1g},a) = -\frac{\sqrt{7}\mathbb{M}_{4,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\mathbb{M}_{4,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$

$\boxed{\text{z469}}$ $\mathbb{Q}_4^{(c)}(B_{1g},b) = \frac{5\sqrt{170}\mathbb{T}_{5,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} - \frac{3\sqrt{238}\mathbb{T}_{5,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{\sqrt{714}\mathbb{T}_{5,1}^{(a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{68}$
$- \frac{5\sqrt{170}\mathbb{T}_{5,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} + \frac{3\sqrt{238}\mathbb{T}_{5,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{\sqrt{714}\mathbb{T}_{5,2}^{(a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{68}$

$\boxed{\text{z470}}$ $\mathbb{Q}_4^{(c)}(B_{1g},c) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{3\sqrt{3}\mathbb{T}_{3,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{3\sqrt{3}\mathbb{T}_{3,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$

$\boxed{\text{z471}}$ $\mathbb{Q}_6^{(c)}(B_{1g},1) = \frac{5\sqrt{167790}\mathbb{T}_{5,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{6392} - \frac{21\sqrt{4794}\mathbb{T}_{5,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{6392} - \frac{\sqrt{1598}\mathbb{T}_{5,1}^{(a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{68}$
$- \frac{5\sqrt{167790}\mathbb{T}_{5,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{6392} + \frac{21\sqrt{4794}\mathbb{T}_{5,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{6392} + \frac{\sqrt{1598}\mathbb{T}_{5,2}^{(a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{68}$

$\boxed{\text{z472}}$ $\mathbb{Q}_6^{(c)}(B_{1g},2) = \frac{3\sqrt{658}\mathbb{T}_{5,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{188} + \frac{5\sqrt{470}\mathbb{T}_{5,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{188} - \frac{3\sqrt{658}\mathbb{T}_{5,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{188} - \frac{5\sqrt{470}\mathbb{T}_{5,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{188}$

$\boxed{\text{z473}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{1g},a) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z474}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{1g},b) = \frac{3\sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{3\sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$

$\boxed{\text{z475}}$ $\mathbb{Q}_4^{(1,-1;c)}(B_{1g},a) = -\frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$

$\boxed{\text{z476}}$ $$\mathbb{Q}_4^{(1,-1;c)}(B_{1g},b) = \frac{5\sqrt{170}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} - \frac{3\sqrt{238}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{\sqrt{714}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{68} - \frac{5\sqrt{170}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} + \frac{3\sqrt{238}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{\sqrt{714}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{68}$$

$\boxed{\text{z477}}$ $$\mathbb{Q}_4^{(1,-1;c)}(B_{1g},c) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{3\sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{3\sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$\boxed{\text{z478}}$ $$\mathbb{Q}_6^{(1,-1;c)}(B_{1g},1a) = \frac{3\sqrt{35}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{28} - \frac{\sqrt{77}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{28} - \frac{3\sqrt{35}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{28} + \frac{\sqrt{77}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{28}$$

$\boxed{\text{z479}}$ $$\mathbb{Q}_6^{(1,-1;c)}(B_{1g},1b) = \frac{5\sqrt{167790}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{6392} - \frac{21\sqrt{4794}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{6392} - \frac{\sqrt{1598}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{68} - \frac{5\sqrt{167790}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{6392} + \frac{21\sqrt{4794}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{6392} + \frac{\sqrt{1598}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{68}$$

$\boxed{\text{z480}}$ $$\mathbb{Q}_6^{(1,-1;c)}(B_{1g},2a) = \frac{\sqrt{266}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{38} - \frac{\sqrt{4389}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{266} - \frac{3\sqrt{1995}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{266} - \frac{\sqrt{266}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{38} + \frac{\sqrt{4389}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{266} + \frac{3\sqrt{1995}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{266}$$

$\boxed{\text{z481}}$ $$\mathbb{Q}_6^{(1,-1;c)}(B_{1g},2b) = \frac{3\sqrt{658}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{188} + \frac{5\sqrt{470}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{188} - \frac{3\sqrt{658}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{188} - \frac{5\sqrt{470}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{188}$$

$\boxed{\text{z482}}$ $$\mathbb{Q}_2^{(1,0;c)}(B_{1g},a) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,0;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,0;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z483}}$ $$\mathbb{Q}_2^{(1,0;c)}(B_{1g},b) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,0;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,0;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z484}}$ $$\mathbb{Q}_2^{(1,0;c)}(B_{1g},c) = \frac{3\sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{3\sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$\boxed{\text{z485}}$ $$\mathbb{Q}_4^{(1,0;c)}(B_{1g},a) = -\frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\mathbb{M}_{4,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z486}}$ $$\mathbb{Q}_4^{(1,0;c)}(B_{1g},b) = \frac{5\sqrt{170}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} - \frac{3\sqrt{238}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{\sqrt{714}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{68} - \frac{5\sqrt{170}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} + \frac{3\sqrt{238}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{\sqrt{714}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{68}$$

$\boxed{\text{z487}}$ $$\mathbb{Q}_4^{(1,0;c)}(B_{1g},c) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{3\sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{3\sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$\boxed{\text{z488}}$ $$\mathbb{Q}_6^{(1,0;c)}(B_{1g},1) = \frac{5\sqrt{167790}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{6392} - \frac{21\sqrt{4794}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{6392} - \frac{\sqrt{1598}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{68} - \frac{5\sqrt{167790}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{6392} + \frac{21\sqrt{4794}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{6392} + \frac{\sqrt{1598}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{68}$$

$\boxed{\text{z489}}$ $$\mathbb{Q}_6^{(1,0;c)}(B_{1g},2) = \frac{3\sqrt{658}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{188} + \frac{5\sqrt{470}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{188} - \frac{3\sqrt{658}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{188} - \frac{5\sqrt{470}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{188}$$

$\boxed{\text{z490}}$ $$\mathbb{Q}_2^{(1,1;c)}(B_{1g},a) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z491}}$ $$\mathbb{Q}_2^{(1,1;c)}(B_{1g},b) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z492}}$ $$\mathbb{Q}_2^{(1,1;c)}(B_{1g},c) = \frac{3\sqrt{3}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{3\sqrt{3}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$\boxed{\text{z493}}$ $$\mathbb{Q}_4^{(1,1;c)}(B_{1g},a) = -\frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\mathbb{M}_{4,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z494}}$ $$\mathbb{Q}_4^{(1,1;c)}(B_{1g},b) = \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{3\sqrt{3}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{3\sqrt{3}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$\boxed{\text{z495}}$ $$\mathbb{G}_3^{(c)}(B_{1g}) = \frac{\mathbb{M}_{4,1}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,1}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,2}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{M}_{4,2}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z496}}$ $$\mathbb{G}_3^{(1,-1;c)}(B_{1g}) = \frac{\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z497}}$ $$\mathbb{G}_5^{(1,-1;c)}(B_{1g}) = \frac{\sqrt{114}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{19} + \frac{\sqrt{209}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{76} + \frac{3\sqrt{95}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{76} - \frac{\sqrt{114}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{19} - \frac{\sqrt{209}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{76} - \frac{3\sqrt{95}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{76}$$

$\boxed{\text{z498}}$ $$\mathbb{G}_3^{(1,0;c)}(B_{1g}) = \frac{\mathbb{M}_{4,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z499}}$ $\mathbb{G}_3^{(1,1;c)}(B_{1g}) = \frac{\mathbb{M}_{4,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\mathbb{M}_{4,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$

$\boxed{\text{z500}}$ $\mathbb{Q}_3^{(c)}(B_{1u},a) = \mathbb{Q}_5^{(a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z501}}$ $\mathbb{Q}_3^{(c)}(B_{1u},b) = \mathbb{Q}_3^{(a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z502}}$ $\mathbb{Q}_3^{(c)}(B_{1u},c) = \mathbb{G}_2^{(a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z503}}$ $\mathbb{Q}_3^{(c)}(B_{1u},d) = -\mathbb{G}_4^{(a)}(A_{1u},2)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z504}}$ $\mathbb{Q}_5^{(c)}(B_{1u},a) = \mathbb{Q}_5^{(a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z505}}$ $\mathbb{Q}_5^{(c)}(B_{1u},b) = -\mathbb{G}_4^{(a)}(A_{1u},1)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z506}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{1u},a) = \mathbb{Q}_5^{(1,-1;a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z507}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{1u},b) = \mathbb{Q}_3^{(1,-1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z508}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{1u},c) = \mathbb{G}_2^{(1,-1;a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z509}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{1u},d) = -\mathbb{G}_4^{(1,-1;a)}(A_{1u},2)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z510}}$ $\mathbb{Q}_5^{(1,-1;c)}(B_{1u},a) = \mathbb{Q}_5^{(1,-1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z511}}$ $\mathbb{Q}_5^{(1,-1;c)}(B_{1u},b) = -\mathbb{G}_4^{(1,-1;a)}(A_{1u},1)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z512}}$ $\mathbb{Q}_5^{(1,-1;c)}(B_{1u},c) = \frac{\sqrt{14}\mathbb{G}_6^{(1,-1;a)}(A_{1u},1)\mathbb{Q}_2^{(b)}(B_{1g})}{4} + \frac{\sqrt{2}\mathbb{G}_6^{(1,-1;a)}(A_{1u},2)\mathbb{Q}_2^{(b)}(B_{1g})}{4}$

$\boxed{\text{z513}}$ $\mathbb{Q}_7^{(1,-1;c)}(B_{1u},1) = -\frac{\sqrt{2}\mathbb{G}_6^{(1,-1;a)}(A_{1u},1)\mathbb{Q}_2^{(b)}(B_{1g})}{4} + \frac{\sqrt{14}\mathbb{G}_6^{(1,-1;a)}(A_{1u},2)\mathbb{Q}_2^{(b)}(B_{1g})}{4}$

$\boxed{\text{z514}}$ $\mathbb{Q}_3^{(1,0;c)}(B_{1u},a) = \mathbb{Q}_5^{(1,0;a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z515}}$ $\mathbb{Q}_3^{(1,0;c)}(B_{1u},b) = \mathbb{Q}_3^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z516}}$ $\mathbb{Q}_3^{(1,0;c)}(B_{1u},c) = \mathbb{G}_2^{(1,0;a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z517}}$ $\mathbb{Q}_3^{(1,0;c)}(B_{1u},d) = -\mathbb{G}_4^{(1,0;a)}(A_{1u},2)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z518}}$ $\mathbb{Q}_5^{(1,0;c)}(B_{1u},a) = \mathbb{Q}_5^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z519}}$ $\mathbb{Q}_5^{(1,0;c)}(B_{1u},b) = -\mathbb{G}_4^{(1,0;a)}(A_{1u},1)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z520}}$ $\mathbb{Q}_3^{(1,1;c)}(B_{1u},a) = \mathbb{Q}_3^{(1,1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z521}}$ $\mathbb{Q}_3^{(1,1;c)}(B_{1u},b) = \mathbb{G}_2^{(1,1;a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z522}}$ $\mathbb{Q}_3^{(1,1;c)}(B_{1u},c) = -\mathbb{G}_4^{(1,1;a)}(A_{1u},2)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z523}}$ $\mathbb{Q}_5^{(1,1;c)}(B_{1u}) = -\mathbb{G}_4^{(1,1;a)}(A_{1u},1)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z524}}$ $\mathbb{G}_2^{(c)}(B_{1u}) = \mathbb{G}_2^{(a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z525}}$ $\mathbb{G}_4^{(c)}(B_{1u}) = \mathbb{G}_4^{(a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z526}}$ $\mathbb{G}_2^{(1,-1;c)}(B_{1u}) = \mathbb{G}_2^{(1,-1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z527}}$ $\mathbb{G}_4^{(1,-1;c)}(B_{1u}) = \mathbb{G}_4^{(1,-1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z528}}$ $\mathbb{G}_6^{(1,-1;c)}(B_{1u},1) = \mathbb{G}_6^{(1,-1;a)}(B_{1u},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z529}}$ $\mathbb{G}_6^{(1,-1;c)}(B_{1u},2) = \mathbb{G}_6^{(1,-1;a)}(B_{1u},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z530}}$ $\mathbb{G}_2^{(1,0;c)}(B_{1u}) = \mathbb{G}_2^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z531}}$ $\mathbb{G}_4^{(1,0;c)}(B_{1u}) = \mathbb{G}_4^{(1,0;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z532}}$ $\mathbb{G}_2^{(1,1;c)}(B_{1u},a) = \mathbb{G}_0^{(1,1;a)}(A_{1u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z533}}$ $\mathbb{G}_2^{(1,1;c)}(B_{1u},b) = \mathbb{G}_2^{(1,1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z534}}$ $\mathbb{G}_4^{(1,1;c)}(B_{1u}) = \mathbb{G}_4^{(1,1;a)}(B_{1u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z535}}$ $$\mathbb{Q}_2^{(c)}(B_{2g},a) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z536}}$ $$\mathbb{Q}_2^{(c)}(B_{2g},b) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{1,2}^{(a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z537}}$ $$\mathbb{Q}_2^{(c)}(B_{2g},c) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z538}}$ $$\mathbb{Q}_4^{(c)}(B_{2g},a) = \frac{\sqrt{7}\mathbb{M}_{4,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{5\mathbb{M}_{4,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{5\mathbb{M}_{4,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z539}}$ $$\mathbb{Q}_4^{(c)}(B_{2g},b) = \frac{\sqrt{170}\mathbb{T}_{5,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{68} - \frac{3\sqrt{238}\mathbb{T}_{5,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{68} + \frac{\sqrt{170}\mathbb{T}_{5,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{68} - \frac{3\sqrt{238}\mathbb{T}_{5,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{68}$$

$\boxed{\text{z540}}$ $$\mathbb{Q}_4^{(c)}(B_{2g},c) = -\frac{\sqrt{5}\mathbb{T}_{3,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z541}}$ $$\mathbb{Q}_6^{(c)}(B_{2g},1) = \frac{3\sqrt{7}\mathbb{T}_{5,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{T}_{5,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{T}_{5,1}^{(a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{3\sqrt{7}\mathbb{T}_{5,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{T}_{5,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{T}_{5,2}^{(a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z542}}$ $$\begin{aligned}\mathbb{Q}_6^{(c)}(B_{2g},2) = {} & \frac{3\sqrt{1785}\mathbb{T}_{5,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{272} + \frac{5\sqrt{51}\mathbb{T}_{5,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{272} + \frac{\sqrt{17}\mathbb{T}_{5,1}^{(a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} \\ & + \frac{3\sqrt{1785}\mathbb{T}_{5,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{272} + \frac{5\sqrt{51}\mathbb{T}_{5,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{272} + \frac{\sqrt{17}\mathbb{T}_{5,2}^{(a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}\end{aligned}$$

$\boxed{\text{z543}}$ $$\mathbb{Q}_2^{(1,-1;c)}(B_{2g},a) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z544}}$ $$\mathbb{Q}_2^{(1,-1;c)}(B_{2g},b) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z545}}$ $$\mathbb{Q}_4^{(1,-1;c)}(B_{2g},a) = \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{5\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{5\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z546}}$ $$\mathbb{Q}_4^{(1,-1;c)}(B_{2g},b) = \frac{\sqrt{170}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{68} - \frac{3\sqrt{238}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{68} + \frac{\sqrt{170}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{68} - \frac{3\sqrt{238}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{68}$$

$\boxed{\text{z547}}$ $$\mathbb{Q}_4^{(1,-1;c)}(B_{2g},c) = -\frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z548}}$ $$\mathbb{Q}_6^{(1,-1;c)}(B_{2g},1a) = \frac{\sqrt{11}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{\sqrt{6}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{32} - \frac{\sqrt{330}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{32} + \frac{\sqrt{11}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{6}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{32} - \frac{\sqrt{330}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{32}$$

$\boxed{\text{z549}}$ $$\mathbb{Q}_6^{(1,-1;c)}(B_{2g},1b) = \frac{3\sqrt{7}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{3\sqrt{7}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z550}}$ $$\mathbb{Q}_6^{(1,-1;c)}(B_{2g},2a) = -\frac{5\sqrt{285}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{152} - \frac{3\sqrt{2090}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{608} - \frac{37\sqrt{38}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{608} - \frac{5\sqrt{285}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{152} - \frac{3\sqrt{2090}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{608} - \frac{37\sqrt{38}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{608}$$

$\boxed{\text{z551}}$ $$\mathbb{Q}_6^{(1,-1;c)}(B_{2g},2b) = \frac{3\sqrt{1785}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{272} + \frac{5\sqrt{51}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{272} + \frac{\sqrt{17}\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{3\sqrt{1785}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{272} + \frac{5\sqrt{51}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{272} + \frac{\sqrt{17}\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z552}}$ $$\mathbb{Q}_2^{(1,0;c)}(B_{2g},a) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,0;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,0;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z553}}$ $$\mathbb{Q}_2^{(1,0;c)}(B_{2g},b) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,0;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,0;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z554}}$ $$\mathbb{Q}_2^{(1,0;c)}(B_{2g},c) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z555}}$ $$\mathbb{Q}_4^{(1,0;c)}(B_{2g},a) = \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{5\mathbb{M}_{4,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{5\mathbb{M}_{4,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z556}}$ $$\mathbb{Q}_4^{(1,0;c)}(B_{2g},b) = \frac{\sqrt{170}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{68} - \frac{3\sqrt{238}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{68} + \frac{\sqrt{170}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{68} - \frac{3\sqrt{238}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{68}$$

$\boxed{\text{z557}}$ $$\mathbb{Q}_4^{(1,0;c)}(B_{2g},c) = -\frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z558}}$ $$\mathbb{Q}_6^{(1,0;c)}(B_{2g},1) = \frac{3\sqrt{7}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{3\sqrt{7}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z559}}$ $$\mathbb{Q}_6^{(1,0;c)}(B_{2g},2) = \frac{3\sqrt{1785}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{272} + \frac{5\sqrt{51}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{272} + \frac{\sqrt{17}\mathbb{T}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{3\sqrt{1785}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{272} + \frac{5\sqrt{51}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{272} + \frac{\sqrt{17}\mathbb{T}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z560}}$ $$\mathbb{Q}_2^{(1,1;c)}(B_{2g},a) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z561}}$ $$\mathbb{Q}_2^{(1,1;c)}(B_{2g},b) = \frac{\sqrt{2}\mathbb{T}_{1,1}^{(1,1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{1,2}^{(1,1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z562}}$ $$\mathbb{Q}_2^{(1,1;c)}(B_{2g},c) = -\frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z563}}$ $$\mathbb{Q}_4^{(1,1;c)}(B_{2g},a) = \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{5\mathbb{M}_{4,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{5\mathbb{M}_{4,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z564}}$ $$\mathbb{Q}_4^{(1,1;c)}(B_{2g},b) = -\frac{\sqrt{5}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{T}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z565}}$ $$\mathbb{G}_3^{(c)}(B_{2g}) = -\frac{5\mathbb{M}_{4,1}^{(a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{7}\mathbb{M}_{4,1}^{(a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{5\mathbb{M}_{4,2}^{(a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z566}}$ $$\mathbb{G}_3^{(1,-1;c)}(B_{2g}) = -\frac{5\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{5\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z567}}$ $$\mathbb{G}_5^{(1,-1;c)}(B_{2g}) = -\frac{\sqrt{114}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{76} + \frac{7\sqrt{209}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{152} - \frac{3\sqrt{95}\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{152} - \frac{\sqrt{114}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{76} + \frac{7\sqrt{209}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{152} - \frac{3\sqrt{95}\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{152}$$

$\boxed{\text{z568}}$ $$\mathbb{G}_3^{(1,0;c)}(B_{2g}) = -\frac{5\mathbb{M}_{4,1}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{5\mathbb{M}_{4,2}^{(1,0;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,0;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z569}}$ $\mathbb{G}_3^{(1,1;c)}(B_{2g}) = -\frac{5\mathbb{M}_{4,1}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{7}\mathbb{M}_{4,1}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{5\mathbb{M}_{4,2}^{(1,1;a)}(E_u,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{\sqrt{7}\mathbb{M}_{4,2}^{(1,1;a)}(E_u,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$

$\boxed{\text{z570}}$ $\mathbb{Q}_3^{(c)}(B_{2u},a) = \frac{\sqrt{85}\mathbb{Q}_5^{(a)}(A_{2u},1)\mathbb{Q}_2^{(b)}(B_{1g})}{34} + \frac{3\sqrt{119}\mathbb{Q}_5^{(a)}(A_{2u},2)\mathbb{Q}_2^{(b)}(B_{1g})}{34}$

$\boxed{\text{z571}}$ $\mathbb{Q}_3^{(c)}(B_{2u},b) = \mathbb{Q}_1^{(a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z572}}$ $\mathbb{Q}_3^{(c)}(B_{2u},c) = \mathbb{Q}_3^{(a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z573}}$ $\mathbb{Q}_3^{(c)}(B_{2u},d) = -\mathbb{Q}_3^{(a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z574}}$ $\mathbb{Q}_3^{(c)}(B_{2u},e) = -\mathbb{G}_4^{(a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z575}}$ $\mathbb{Q}_5^{(c)}(B_{2u},a) = \mathbb{Q}_5^{(a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z576}}$ $\mathbb{Q}_5^{(c)}(B_{2u},b) = -\frac{3\sqrt{119}\mathbb{Q}_5^{(a)}(A_{2u},1)\mathbb{Q}_2^{(b)}(B_{1g})}{34} + \frac{\sqrt{85}\mathbb{Q}_5^{(a)}(A_{2u},2)\mathbb{Q}_2^{(b)}(B_{1g})}{34}$

$\boxed{\text{z577}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{2u},a) = \frac{\sqrt{85}\mathbb{Q}_5^{(1,-1;a)}(A_{2u},1)\mathbb{Q}_2^{(b)}(B_{1g})}{34} + \frac{3\sqrt{119}\mathbb{Q}_5^{(1,-1;a)}(A_{2u},2)\mathbb{Q}_2^{(b)}(B_{1g})}{34}$

$\boxed{\text{z578}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{2u},b) = \mathbb{Q}_3^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z579}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{2u},c) = -\mathbb{Q}_3^{(1,-1;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z580}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{2u},d) = -\mathbb{G}_4^{(1,-1;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z581}}$ $\mathbb{Q}_5^{(1,-1;c)}(B_{2u},a) = \mathbb{Q}_5^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z582}}$ $\mathbb{Q}_5^{(1,-1;c)}(B_{2u},b) = -\frac{3\sqrt{119}\mathbb{Q}_5^{(1,-1;a)}(A_{2u},1)\mathbb{Q}_2^{(b)}(B_{1g})}{34} + \frac{\sqrt{85}\mathbb{Q}_5^{(1,-1;a)}(A_{2u},2)\mathbb{Q}_2^{(b)}(B_{1g})}{34}$

$\boxed{\text{z583}}$ $\mathbb{Q}_5^{(1,-1;c)}(B_{2u},c) = -\mathbb{G}_6^{(1,-1;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z584}}$ $\mathbb{Q}_3^{(1,0;c)}(B_{2u},a) = \frac{\sqrt{85}\mathbb{Q}_5^{(1,0;a)}(A_{2u},1)\mathbb{Q}_2^{(b)}(B_{1g})}{34} + \frac{3\sqrt{119}\mathbb{Q}_5^{(1,0;a)}(A_{2u},2)\mathbb{Q}_2^{(b)}(B_{1g})}{34}$

$\boxed{\text{z585}}$ $\mathbb{Q}_3^{(1,0;c)}(B_{2u},b) = \mathbb{Q}_1^{(1,0;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z586}}$ $\mathbb{Q}_3^{(1,0;c)}(B_{2u},c) = \mathbb{Q}_3^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z587}}$ $\mathbb{Q}_3^{(1,0;c)}(B_{2u},d) = -\mathbb{Q}_3^{(1,0;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z588}}$ $\mathbb{Q}_3^{(1,0;c)}(B_{2u},e) = -\mathbb{G}_4^{(1,0;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z589}}$ $\mathbb{Q}_5^{(1,0;c)}(B_{2u},a) = \mathbb{Q}_5^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z590}}$ $\mathbb{Q}_5^{(1,0;c)}(B_{2u},b) = -\frac{3\sqrt{119}\mathbb{Q}_5^{(1,0;a)}(A_{2u},1)\mathbb{Q}_2^{(b)}(B_{1g})}{34} + \frac{\sqrt{85}\mathbb{Q}_5^{(1,0;a)}(A_{2u},2)\mathbb{Q}_2^{(b)}(B_{1g})}{34}$

$\boxed{\text{z591}}$ $\mathbb{Q}_3^{(1,1;c)}(B_{2u},a) = \mathbb{Q}_1^{(1,1;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z592}}$ $\mathbb{Q}_3^{(1,1;c)}(B_{2u},b) = \mathbb{Q}_3^{(1,1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z593}}$ $\mathbb{Q}_3^{(1,1;c)}(B_{2u},c) = -\mathbb{Q}_3^{(1,1;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z594}}$ $\mathbb{Q}_3^{(1,1;c)}(B_{2u},d) = -\mathbb{G}_4^{(1,1;a)}(A_{2u})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z595}}$ $\mathbb{G}_2^{(c)}(B_{2u}) = \mathbb{G}_2^{(a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z596}}$ $\mathbb{G}_4^{(c)}(B_{2u}) = \mathbb{G}_4^{(a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z597}}$ $\mathbb{G}_2^{(1,-1;c)}(B_{2u}) = \mathbb{G}_2^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z598}}$ $\mathbb{G}_4^{(1,-1;c)}(B_{2u}) = \mathbb{G}_4^{(1,-1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z599}}$ $\mathbb{G}_6^{(1,-1;c)}(B_{2u},1) = \mathbb{G}_6^{(1,-1;a)}(B_{2u},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z600}}$ $\mathbb{G}_6^{(1,-1;c)}(B_{2u},2) = \mathbb{G}_6^{(1,-1;a)}(B_{2u},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z601}}$ $\mathbb{G}_2^{(1,0;c)}(B_{2u}) = \mathbb{G}_2^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z602}}$ $\mathbb{G}_4^{(1,0;c)}(B_{2u}) = \mathbb{G}_4^{(1,0;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z603}}$ $\mathbb{G}_2^{(1,1;c)}(B_{2u}) = \mathbb{G}_2^{(1,1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z604}}$ $\mathbb{G}_4^{(1,1;c)}(B_{2u}) = \mathbb{G}_4^{(1,1;a)}(B_{2u})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z722}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g,a) = \frac{\sqrt{30}\mathbb{M}_2^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10} + \frac{\sqrt{10}\mathbb{M}_2^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10} + \frac{\sqrt{10}\mathbb{M}_2^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10}$

$\boxed{\text{z723}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g,a) = -\frac{\sqrt{30}\mathbb{M}_2^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10} + \frac{\sqrt{10}\mathbb{M}_2^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10} - \frac{\sqrt{10}\mathbb{M}_2^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10}$

$\boxed{\text{z724}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g,b) = \frac{\sqrt{2}\mathbb{T}_1^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z725}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g,b) = \frac{\sqrt{2}\mathbb{T}_1^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$

$\boxed{\text{z726}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g,c) = -\frac{\sqrt{78}\mathbb{T}_3^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{26} + \frac{\sqrt{130}\mathbb{T}_3^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{26} - \frac{\sqrt{130}\mathbb{T}_3^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{26}$

$\boxed{\text{z727}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g,c) = -\frac{\sqrt{78}\mathbb{T}_3^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{26} + \frac{\sqrt{130}\mathbb{T}_3^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{26} + \frac{\sqrt{130}\mathbb{T}_3^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{26}$

$\boxed{\text{z728}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,1a) = -\frac{\sqrt{15}\mathbb{M}_4^{(a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{9} - \frac{\sqrt{21}\mathbb{M}_4^{(a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{18} - \frac{\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{12} + \frac{\sqrt{7}\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{6} - \frac{\sqrt{7}\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{12}$

$\boxed{\text{z729}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,1a) = \frac{\sqrt{15}\mathbb{M}_4^{(a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{9} + \frac{\sqrt{21}\mathbb{M}_4^{(a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{18} - \frac{\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{12} + \frac{\sqrt{7}\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{6} + \frac{\sqrt{7}\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{12}$

$\boxed{\text{z730}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,1b) = -\frac{3\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{35}\mathbb{T}_5^{(a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{3\mathbb{T}_5^{(a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{3\sqrt{3}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{3}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$

$\boxed{\text{z731}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,1b) = \frac{3\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{35}\mathbb{T}_5^{(a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{3\mathbb{T}_5^{(a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{3\sqrt{3}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{3}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$

$\boxed{\text{z732}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,1c) = -\frac{7\sqrt{1430}\mathbb{T}_3^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{572} - \frac{5\sqrt{858}\mathbb{T}_3^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{286} - \frac{3\sqrt{858}\mathbb{T}_3^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{572}$

$\boxed{\text{z733}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,1c) = -\frac{7\sqrt{1430}\mathbb{T}_3^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{572} - \frac{5\sqrt{858}\mathbb{T}_3^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{286} + \frac{3\sqrt{858}\mathbb{T}_3^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{572}$

$\boxed{\text{z734}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,2a) = \frac{\sqrt{21945}\mathbb{M}_4^{(a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{1881} - \frac{47\sqrt{627}\mathbb{M}_4^{(a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{3762} + \frac{\sqrt{1463}\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{132} - \frac{25\sqrt{209}\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1254} - \frac{83\sqrt{209}\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2508}$

$\boxed{\text{z735}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_g,2a) = -\frac{\sqrt{21945}\mathbb{M}_4^{(a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{1881} + \frac{47\sqrt{627}\mathbb{M}_4^{(a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{3762} + \frac{\sqrt{1463}\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{132} - \frac{25\sqrt{209}\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1254} + \frac{83\sqrt{209}\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2508}$$

$\boxed{\text{z736}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_g,2b) = \frac{\sqrt{8463}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{\sqrt{6045}\mathbb{T}_5^{(a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{496} - \frac{\sqrt{8463}\mathbb{T}_5^{(a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} + \frac{35\sqrt{2821}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{6448} + \frac{\sqrt{2821}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1612}$$

$\boxed{\text{z737}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_g,2b) = -\frac{\sqrt{8463}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{208} + \frac{\sqrt{6045}\mathbb{T}_5^{(a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{496} - \frac{\sqrt{8463}\mathbb{T}_5^{(a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{35\sqrt{2821}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{6448} - \frac{\sqrt{2821}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1612}$$

$\boxed{\text{z738}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_g,2c) = -\frac{\sqrt{330}\mathbb{T}_3^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{44} + \frac{\sqrt{22}\mathbb{T}_3^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} + \frac{5\sqrt{22}\mathbb{T}_3^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{44}$$

$\boxed{\text{z739}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_g,2c) = -\frac{\sqrt{330}\mathbb{T}_3^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{44} + \frac{\sqrt{22}\mathbb{T}_3^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} - \frac{5\sqrt{22}\mathbb{T}_3^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{44}$$

$\boxed{\text{z740}}$ $$\mathbb{Q}_{6,1}^{(c)}(E_g,1) = -\frac{\sqrt{439270}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1417} + \frac{3\sqrt{614978}\mathbb{T}_5^{(a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{27032} - \frac{5\sqrt{439270}\mathbb{T}_5^{(a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{11336} + \frac{10\sqrt{1317810}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{43927} - \frac{53\sqrt{1317810}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{175708}$$

$\boxed{\text{z741}}$ $$\mathbb{Q}_{6,2}^{(c)}(E_g,1) = \frac{\sqrt{439270}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1417} + \frac{3\sqrt{614978}\mathbb{T}_5^{(a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{27032} - \frac{5\sqrt{439270}\mathbb{T}_5^{(a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{11336} + \frac{10\sqrt{1317810}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{43927} + \frac{53\sqrt{1317810}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{175708}$$

$\boxed{\text{z742}}$ $$\mathbb{Q}_{6,1}^{(c)}(E_g,2) = -\frac{5\sqrt{109}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{872} + \frac{27\sqrt{3815}\mathbb{T}_5^{(a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{3488} - \frac{67\sqrt{109}\mathbb{T}_5^{(a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{3488} - \frac{23\sqrt{327}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{872} + \frac{\sqrt{327}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1744}$$

$\boxed{\text{z743}}$ $$\mathbb{Q}_{6,2}^{(c)}(E_g,2) = \frac{5\sqrt{109}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{872} + \frac{27\sqrt{3815}\mathbb{T}_5^{(a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{3488} - \frac{67\sqrt{109}\mathbb{T}_5^{(a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{3488} - \frac{23\sqrt{327}\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{872} - \frac{\sqrt{327}\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1744}$$

$\boxed{\text{z744}}$ $$\mathbb{Q}_{6,1}^{(c)}(E_g,3) = \frac{\sqrt{3}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{\sqrt{105}\mathbb{T}_5^{(a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{32} + \frac{7\sqrt{3}\mathbb{T}_5^{(a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{32} + \frac{\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{7\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{16}$$

$\boxed{\text{z745}}$ $$\mathbb{Q}_{6,2}^{(c)}(E_g,3) = -\frac{\sqrt{3}\mathbb{T}_5^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{105}\mathbb{T}_5^{(a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{32} + \frac{7\sqrt{3}\mathbb{T}_5^{(a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{32} + \frac{\mathbb{T}_5^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{7\mathbb{T}_5^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{16}$$

$\boxed{\text{z746}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,a) = \frac{\sqrt{30}\mathbb{M}_2^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10} + \frac{\sqrt{10}\mathbb{M}_2^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10} + \frac{\sqrt{10}\mathbb{M}_2^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10}$$

$\boxed{\text{z747}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,a) = -\frac{\sqrt{30}\mathbb{M}_2^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10} + \frac{\sqrt{10}\mathbb{M}_2^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10} - \frac{\sqrt{10}\mathbb{M}_2^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10}$$

$\boxed{\text{z748}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,b) = -\frac{\sqrt{78}\mathbb{T}_3^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{26} + \frac{\sqrt{130}\mathbb{T}_3^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{26} - \frac{\sqrt{130}\mathbb{T}_3^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{26}$$

$\boxed{\text{z749}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,b) = -\frac{\sqrt{78}\mathbb{T}_3^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{26} + \frac{\sqrt{130}\mathbb{T}_3^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{26} + \frac{\sqrt{130}\mathbb{T}_3^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{26}$$

$\boxed{\text{z750}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,1a) = -\frac{\sqrt{15}\mathbb{M}_4^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{9} - \frac{\sqrt{21}\mathbb{M}_4^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{18} - \frac{\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{12} + \frac{\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{6} - \frac{\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{12}$$

$\boxed{\text{z751}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,1a) = \frac{\sqrt{15}\mathbb{M}_4^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{9} + \frac{\sqrt{21}\mathbb{M}_4^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{18} - \frac{\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{12} + \frac{\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{6} + \frac{\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{12}$$

$\boxed{\text{z752}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,1b) = -\frac{3\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{35}\mathbb{T}_5^{(1,-1;a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{3\mathbb{T}_5^{(1,-1;a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{3\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z753}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,1b) = \frac{3\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{35}\mathbb{T}_5^{(1,-1;a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{3\mathbb{T}_5^{(1,-1;a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{3\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z754}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,1c) = -\frac{7\sqrt{1430}\mathbb{T}_3^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{572} - \frac{5\sqrt{858}\mathbb{T}_3^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{286} - \frac{3\sqrt{858}\mathbb{T}_3^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{572}$$

$\boxed{\text{z755}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,1c) = -\frac{7\sqrt{1430}\mathbb{T}_3^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{572} - \frac{5\sqrt{858}\mathbb{T}_3^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{286} + \frac{3\sqrt{858}\mathbb{T}_3^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{572}$$

$\boxed{\text{z756}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,2a) = \frac{\sqrt{21945}\mathbb{M}_4^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{1881} - \frac{47\sqrt{627}\mathbb{M}_4^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{3762} + \frac{\sqrt{1463}\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{132} - \frac{25\sqrt{209}\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1254} - \frac{83\sqrt{209}\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2508}$$

$\boxed{\text{z818}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,2a) = -\frac{\sqrt{21945}\mathbb{M}_4^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{1881} + \frac{47\sqrt{627}\mathbb{M}_4^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{3762} + \frac{\sqrt{1463}\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{132} - \frac{25\sqrt{209}\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1254} + \frac{83\sqrt{209}\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2508}$$

$\boxed{\text{z819}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,2b) = \frac{\sqrt{8463}\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{\sqrt{6045}\mathbb{T}_5^{(1,-1;a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{496} - \frac{\sqrt{8463}\mathbb{T}_5^{(1,-1;a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} + \frac{35\sqrt{2821}\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{6448} + \frac{\sqrt{2821}\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1612}$$

$\boxed{\text{z820}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,2b) = -\frac{\sqrt{8463}\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{208} + \frac{\sqrt{6045}\mathbb{T}_5^{(1,-1;a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{496} - \frac{\sqrt{8463}\mathbb{T}_5^{(1,-1;a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{35\sqrt{2821}\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{6448} - \frac{\sqrt{2821}\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1612}$$

$\boxed{\text{z821}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,2c) = -\frac{\sqrt{330}\mathbb{T}_3^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{44} + \frac{\sqrt{22}\mathbb{T}_3^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} + \frac{5\sqrt{22}\mathbb{T}_3^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{44}$$

$\boxed{\text{z822}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,2c) = -\frac{\sqrt{330}\mathbb{T}_3^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{44} + \frac{\sqrt{22}\mathbb{T}_3^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} - \frac{5\sqrt{22}\mathbb{T}_3^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{44}$$

$\boxed{\text{z823}}$ $$\mathbb{Q}_{6,1}^{(1,-1;c)}(E_g,1a) = \frac{\sqrt{203}\mathbb{M}_6^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{29} + \frac{\sqrt{29}\mathbb{M}_6^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{58} - \frac{\sqrt{58}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{116} - \frac{\sqrt{87}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{58} - \frac{\sqrt{957}\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{232} + \frac{5\sqrt{435}\mathbb{M}_6^{(1,-1;a)}(B_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{232}$$

$\boxed{\text{z824}}$ $$\mathbb{Q}_{6,2}^{(1,-1;c)}(E_g,1a) = -\frac{\sqrt{203}\mathbb{M}_6^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{29} - \frac{\sqrt{29}\mathbb{M}_6^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{58} - \frac{\sqrt{58}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{116} - \frac{\sqrt{87}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{58} + \frac{\sqrt{957}\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{232} - \frac{5\sqrt{435}\mathbb{M}_6^{(1,-1;a)}(B_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{232}$$

$\boxed{\text{z825}}$ $$\mathbb{Q}_{6,1}^{(1,-1;c)}(E_g,1b) = -\frac{\sqrt{439270}\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1417} + \frac{3\sqrt{614978}\mathbb{T}_5^{(1,-1;a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{27032} - \frac{5\sqrt{439270}\mathbb{T}_5^{(1,-1;a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{11336} + \frac{10\sqrt{1317810}\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{43927} - \frac{53\sqrt{1317810}\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{175708}$$

$\boxed{\text{z826}}$ $$\mathbb{Q}_{6,2}^{(1,-1;c)}(E_g,1b) = \frac{\sqrt{439270}\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1417} + \frac{3\sqrt{614978}\mathbb{T}_5^{(1,-1;a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{27032} - \frac{5\sqrt{439270}\mathbb{T}_5^{(1,-1;a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{11336} + \frac{10\sqrt{1317810}\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{43927} + \frac{53\sqrt{1317810}\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{175708}$$

$\boxed{\text{z827}}$ $$\mathbb{Q}_{6,1}^{(1,-1;c)}(E_g,2a) = -\frac{\sqrt{1659119}\mathbb{M}_6^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{21547} + \frac{85\sqrt{237017}\mathbb{M}_6^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{86188} + \frac{31\sqrt{474034}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{172376} - \frac{\sqrt{323205}\mathbb{M}_6^{(1,-1;a)}(B_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{1486} + \frac{31\sqrt{711051}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{86188} + \frac{51\sqrt{64641}\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{344752} + \frac{19\sqrt{3555255}\mathbb{M}_6^{(1,-1;a)}(B_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{344752}$$

$\boxed{\text{z828}}$ $$\mathbb{Q}_{6,2}^{(1,-1;c)}(E_g,2a) = \frac{\sqrt{1659119}\mathbb{M}_6^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{21547} - \frac{85\sqrt{237017}\mathbb{M}_6^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{86188} + \frac{31\sqrt{474034}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{172376} - \frac{\sqrt{323205}\mathbb{M}_6^{(1,-1;a)}(B_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{1486} + \frac{31\sqrt{711051}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{86188} - \frac{51\sqrt{64641}\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{344752} - \frac{19\sqrt{3555255}\mathbb{M}_6^{(1,-1;a)}(B_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{344752}$$

$\boxed{\text{z829}}$ $$\mathbb{Q}_{6,1}^{(1,-1;c)}(E_g,2b) = -\frac{5\sqrt{109}\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{872} + \frac{27\sqrt{3815}\mathbb{T}_5^{(1,-1;a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{3488} - \frac{67\sqrt{109}\mathbb{T}_5^{(1,-1;a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{3488} - \frac{23\sqrt{327}\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{872} + \frac{\sqrt{327}\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1744}$$

$\boxed{\text{z830}}$ $$\mathbb{Q}_{6,2}^{(1,-1;c)}(E_g,2b) = \frac{5\sqrt{109}\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{872} + \frac{27\sqrt{3815}\mathbb{T}_5^{(1,-1;a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{3488} - \frac{67\sqrt{109}\mathbb{T}_5^{(1,-1;a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{3488} - \frac{23\sqrt{327}\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{872} - \frac{\sqrt{327}\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1744}$$

$\boxed{\text{z831}}$ $$\mathbb{Q}_{6,1}^{(1,-1;c)}(E_g,3a) = -\frac{39\sqrt{598141005}\mathbb{M}_6^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{5696581} + \frac{343\sqrt{85448715}\mathbb{M}_6^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{22786324} - \frac{43\sqrt{170897430}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1111528} + \frac{349\sqrt{517871}\mathbb{M}_6^{(1,-1;a)}(B_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{1035742} + \frac{655\sqrt{28482905}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{22786324} + \frac{33\sqrt{2589355}\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{202096} + \frac{479\sqrt{5696581}\mathbb{M}_6^{(1,-1;a)}(B_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{5361488}$$

$\boxed{\text{z832}}$ $$\mathbb{Q}_{6,2}^{(1,-1;c)}(E_g,3a) = \frac{39\sqrt{598141005}\mathbb{M}_6^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{5696581} - \frac{343\sqrt{85448715}\mathbb{M}_6^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{22786324} - \frac{43\sqrt{170897430}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1111528} + \frac{349\sqrt{517871}\mathbb{M}_6^{(1,-1;a)}(B_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{1035742} + \frac{655\sqrt{28482905}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{22786324} - \frac{33\sqrt{2589355}\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{202096} - \frac{479\sqrt{5696581}\mathbb{M}_6^{(1,-1;a)}(B_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{5361488}$$

$\boxed{\text{z833}}$ $$\mathbb{Q}_{6,1}^{(1,-1;c)}(E_g,3b) = \frac{\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{\sqrt{105}\mathbb{T}_5^{(1,-1;a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{32} + \frac{7\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{32} + \frac{\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{7\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{16}$$

$\boxed{\text{z834}}$ $$\mathbb{Q}_{6,2}^{(1,-1;c)}(E_g,3b) = -\frac{\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{105}\mathbb{T}_5^{(1,-1;a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{32} + \frac{7\sqrt{3}\mathbb{T}_5^{(1,-1;a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{32} + \frac{\mathbb{T}_5^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{7\mathbb{T}_5^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{16}$$

$\boxed{\text{z835}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,a) = \frac{\sqrt{30}\mathbb{M}_2^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10} + \frac{\sqrt{10}\mathbb{M}_2^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10} + \frac{\sqrt{10}\mathbb{M}_2^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10}$$

$\boxed{\text{z836}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,a) = -\frac{\sqrt{30}\mathbb{M}_2^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10} + \frac{\sqrt{10}\mathbb{M}_2^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10} - \frac{\sqrt{10}\mathbb{M}_2^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10}$$

$\boxed{\text{z837}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,b) = \frac{\sqrt{2}\mathbb{T}_1^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z838}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,b)=\frac{\sqrt{2}\mathbb{T}_1^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z839}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,c)=-\frac{\sqrt{78}\mathbb{T}_3^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{26}+\frac{\sqrt{130}\mathbb{T}_3^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{26}-\frac{\sqrt{130}\mathbb{T}_3^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{26}$$

$\boxed{\text{z840}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,c)=-\frac{\sqrt{78}\mathbb{T}_3^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{26}+\frac{\sqrt{130}\mathbb{T}_3^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{26}+\frac{\sqrt{130}\mathbb{T}_3^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{26}$$

$\boxed{\text{z841}}$ $$\mathbb{Q}_{4,1}^{(1,0;c)}(E_g,1a)=-\frac{\sqrt{15}\mathbb{M}_4^{(1,0;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{9}-\frac{\sqrt{21}\mathbb{M}_4^{(1,0;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{18}-\frac{\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{12}+\frac{\sqrt{7}\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{6}-\frac{\sqrt{7}\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{12}$$

$\boxed{\text{z842}}$ $$\mathbb{Q}_{4,2}^{(1,0;c)}(E_g,1a)=\frac{\sqrt{15}\mathbb{M}_4^{(1,0;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{9}+\frac{\sqrt{21}\mathbb{M}_4^{(1,0;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{18}-\frac{\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{12}+\frac{\sqrt{7}\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{6}+\frac{\sqrt{7}\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{12}$$

$\boxed{\text{z843}}$ $$\mathbb{Q}_{4,1}^{(1,0;c)}(E_g,1b)=-\frac{3\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{16}+\frac{\sqrt{35}\mathbb{T}_5^{(1,0;a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16}+\frac{3\mathbb{T}_5^{(1,0;a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16}+\frac{3\sqrt{3}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{16}+\frac{\sqrt{3}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z844}}$ $$\mathbb{Q}_{4,2}^{(1,0;c)}(E_g,1b)=\frac{3\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{16}+\frac{\sqrt{35}\mathbb{T}_5^{(1,0;a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16}+\frac{3\mathbb{T}_5^{(1,0;a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16}+\frac{3\sqrt{3}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{16}-\frac{\sqrt{3}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z845}}$ $$\mathbb{Q}_{4,1}^{(1,0;c)}(E_g,1c)=-\frac{7\sqrt{1430}\mathbb{T}_3^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{572}-\frac{5\sqrt{858}\mathbb{T}_3^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{286}-\frac{3\sqrt{858}\mathbb{T}_3^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{572}$$

$\boxed{\text{z846}}$ $$\mathbb{Q}_{4,2}^{(1,0;c)}(E_g,1c)=-\frac{7\sqrt{1430}\mathbb{T}_3^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{572}-\frac{5\sqrt{858}\mathbb{T}_3^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{286}+\frac{3\sqrt{858}\mathbb{T}_3^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{572}$$

$\boxed{\text{z847}}$ $$\mathbb{Q}_{4,1}^{(1,0;c)}(E_g,2a)=\frac{\sqrt{21945}\mathbb{M}_4^{(1,0;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{1881}-\frac{47\sqrt{627}\mathbb{M}_4^{(1,0;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{3762}+\frac{\sqrt{1463}\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{132}$$
$$-\frac{25\sqrt{209}\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1254}-\frac{83\sqrt{209}\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2508}$$

$\boxed{\text{z848}}$ $$\mathbb{Q}_{4,2}^{(1,0;c)}(E_g,2a)=-\frac{\sqrt{21945}\mathbb{M}_4^{(1,0;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{1881}+\frac{47\sqrt{627}\mathbb{M}_4^{(1,0;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{3762}+\frac{\sqrt{1463}\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{132}$$
$$-\frac{25\sqrt{209}\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1254}+\frac{83\sqrt{209}\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2508}$$

$\boxed{\text{z849}}$ $$\mathbb{Q}_{4,1}^{(1,0;c)}(E_g,2b)=\frac{\sqrt{8463}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{208}+\frac{\sqrt{6045}\mathbb{T}_5^{(1,0;a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{496}-\frac{\sqrt{8463}\mathbb{T}_5^{(1,0;a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{208}$$
$$+\frac{35\sqrt{2821}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{6448}+\frac{\sqrt{2821}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1612}$$

$\boxed{\text{z850}}$ $$\mathbb{Q}_{4,2}^{(1,0;c)}(E_g,2b) = -\frac{\sqrt{8463}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{208} + \frac{\sqrt{6045}\mathbb{T}_5^{(1,0;a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{496} - \frac{\sqrt{8463}\mathbb{T}_5^{(1,0;a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{35\sqrt{2821}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{6448} - \frac{\sqrt{2821}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1612}$$

$\boxed{\text{z851}}$ $$\mathbb{Q}_{4,1}^{(1,0;c)}(E_g,2c) = -\frac{\sqrt{330}\mathbb{T}_3^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{44} + \frac{\sqrt{22}\mathbb{T}_3^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} + \frac{5\sqrt{22}\mathbb{T}_3^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{44}$$

$\boxed{\text{z852}}$ $$\mathbb{Q}_{4,2}^{(1,0;c)}(E_g,2c) = -\frac{\sqrt{330}\mathbb{T}_3^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{44} + \frac{\sqrt{22}\mathbb{T}_3^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} - \frac{5\sqrt{22}\mathbb{T}_3^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{44}$$

$\boxed{\text{z913}}$ $$\mathbb{Q}_{6,1}^{(1,0;c)}(E_g,1) = -\frac{\sqrt{439270}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1417} + \frac{3\sqrt{614978}\mathbb{T}_5^{(1,0;a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{27032} - \frac{5\sqrt{439270}\mathbb{T}_5^{(1,0;a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{11336} + \frac{10\sqrt{1317810}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{43927} - \frac{53\sqrt{1317810}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{175708}$$

$\boxed{\text{z914}}$ $$\mathbb{Q}_{6,2}^{(1,0;c)}(E_g,1) = \frac{\sqrt{439270}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1417} + \frac{3\sqrt{614978}\mathbb{T}_5^{(1,0;a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{27032} - \frac{5\sqrt{439270}\mathbb{T}_5^{(1,0;a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{11336} + \frac{10\sqrt{1317810}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{43927} + \frac{53\sqrt{1317810}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{175708}$$

$\boxed{\text{z915}}$ $$\mathbb{Q}_{6,1}^{(1,0;c)}(E_g,2) = -\frac{5\sqrt{109}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{872} + \frac{27\sqrt{3815}\mathbb{T}_5^{(1,0;a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{3488} - \frac{67\sqrt{109}\mathbb{T}_5^{(1,0;a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{3488} - \frac{23\sqrt{327}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{872} + \frac{\sqrt{327}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1744}$$

$\boxed{\text{z916}}$ $$\mathbb{Q}_{6,2}^{(1,0;c)}(E_g,2) = \frac{5\sqrt{109}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{872} + \frac{27\sqrt{3815}\mathbb{T}_5^{(1,0;a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{3488} - \frac{67\sqrt{109}\mathbb{T}_5^{(1,0;a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{3488} - \frac{23\sqrt{327}\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{872} - \frac{\sqrt{327}\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1744}$$

$\boxed{\text{z917}}$ $$\mathbb{Q}_{6,1}^{(1,0;c)}(E_g,3) = \frac{\sqrt{3}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{\sqrt{105}\mathbb{T}_5^{(1,0;a)}(A_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{32} + \frac{7\sqrt{3}\mathbb{T}_5^{(1,0;a)}(A_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{32} + \frac{\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{7\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{16}$$

$\boxed{\text{z918}}$ $$\mathbb{Q}_{6,2}^{(1,0;c)}(E_g,3) = -\frac{\sqrt{3}\mathbb{T}_5^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{105}\mathbb{T}_5^{(1,0;a)}(A_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{32} + \frac{7\sqrt{3}\mathbb{T}_5^{(1,0;a)}(A_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{32} + \frac{\mathbb{T}_5^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{7\mathbb{T}_5^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{16}$$

$\boxed{\text{z919}}$ $$\mathbb{Q}_{2,1}^{(1,1;c)}(E_g,a) = \frac{\sqrt{30}\mathbb{M}_2^{(1,1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10} + \frac{\sqrt{10}\mathbb{M}_2^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10} + \frac{\sqrt{10}\mathbb{M}_2^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10}$$

$\boxed{\text{z920}}$ $$\mathbb{Q}_{2,2}^{(1,1;c)}(E_g,a) = -\frac{\sqrt{30}\mathbb{M}_2^{(1,1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10} + \frac{\sqrt{10}\mathbb{M}_2^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10} - \frac{\sqrt{10}\mathbb{M}_2^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10}$$

$\boxed{\text{z921}}$ $$\mathbb{Q}_{2,1}^{(1,1;c)}(E_g,b) = \frac{\sqrt{2}\mathbb{T}_1^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z922}}$ $$\mathbb{Q}_{2,2}^{(1,1;c)}(E_g,b) = \frac{\sqrt{2}\mathbb{T}_1^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z923}}$ $$\mathbb{Q}_{2,1}^{(1,1;c)}(E_g,c) = -\frac{\sqrt{78}\mathbb{T}_3^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{26} + \frac{\sqrt{130}\mathbb{T}_3^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{26} - \frac{\sqrt{130}\mathbb{T}_3^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{26}$$

$\boxed{\text{z924}}$ $$\mathbb{Q}_{2,2}^{(1,1;c)}(E_g,c) = -\frac{\sqrt{78}\mathbb{T}_3^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{26} + \frac{\sqrt{130}\mathbb{T}_3^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{26} + \frac{\sqrt{130}\mathbb{T}_3^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{26}$$

$\boxed{\text{z925}}$ $$\mathbb{Q}_{4,1}^{(1,1;c)}(E_g,1a) = -\frac{\sqrt{15}\mathbb{M}_4^{(1,1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{9} - \frac{\sqrt{21}\mathbb{M}_4^{(1,1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{18} - \frac{\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{12} + \frac{\sqrt{7}\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{6} - \frac{\sqrt{7}\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{12}$$

$\boxed{\text{z926}}$ $$\mathbb{Q}_{4,2}^{(1,1;c)}(E_g,1a) = \frac{\sqrt{15}\mathbb{M}_4^{(1,1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{9} + \frac{\sqrt{21}\mathbb{M}_4^{(1,1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{18} - \frac{\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{12} + \frac{\sqrt{7}\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{6} + \frac{\sqrt{7}\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{12}$$

$\boxed{\text{z927}}$ $$\mathbb{Q}_{4,1}^{(1,1;c)}(E_g,1b) = -\frac{7\sqrt{1430}\mathbb{T}_3^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{572} - \frac{5\sqrt{858}\mathbb{T}_3^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{286} - \frac{3\sqrt{858}\mathbb{T}_3^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{572}$$

$\boxed{\text{z928}}$ $$\mathbb{Q}_{4,2}^{(1,1;c)}(E_g,1b) = -\frac{7\sqrt{1430}\mathbb{T}_3^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{572} - \frac{5\sqrt{858}\mathbb{T}_3^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{286} + \frac{3\sqrt{858}\mathbb{T}_3^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{572}$$

$\boxed{\text{z929}}$ $$\mathbb{Q}_{4,1}^{(1,1;c)}(E_g,2a) = \frac{\sqrt{21945}\mathbb{M}_4^{(1,1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{1881} - \frac{47\sqrt{627}\mathbb{M}_4^{(1,1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{3762} + \frac{\sqrt{1463}\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{132} - \frac{25\sqrt{209}\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{1254} - \frac{83\sqrt{209}\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2508}$$

$\boxed{\text{z930}}$ $$\mathbb{Q}_{4,2}^{(1,1;c)}(E_g,2a) = -\frac{\sqrt{21945}\mathbb{M}_4^{(1,1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{1881} + \frac{47\sqrt{627}\mathbb{M}_4^{(1,1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{3762} + \frac{\sqrt{1463}\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{132} - \frac{25\sqrt{209}\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{1254} + \frac{83\sqrt{209}\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2508}$$

$\boxed{\text{z931}}$ $$\mathbb{Q}_{4,1}^{(1,1;c)}(E_g,2b) = -\frac{\sqrt{330}\mathbb{T}_3^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{44} + \frac{\sqrt{22}\mathbb{T}_3^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} + \frac{5\sqrt{22}\mathbb{T}_3^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{44}$$

$\boxed{\text{z932}}$ $$\mathbb{Q}_{4,2}^{(1,1;c)}(E_g,2b) = -\frac{\sqrt{330}\mathbb{T}_3^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{44} + \frac{\sqrt{22}\mathbb{T}_3^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} - \frac{5\sqrt{22}\mathbb{T}_3^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{44}$$

$\boxed{\text{z933}}$ $$\mathbb{G}_{1,1}^{(c)}(E_g) = -\frac{\sqrt{5}\mathbb{M}_2^{(a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{5} + \frac{\sqrt{15}\mathbb{M}_2^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10} + \frac{\sqrt{15}\mathbb{M}_2^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10}$$

$\boxed{\text{z934}}$ $$\mathbb{G}_{1,2}^{(c)}(E_g) = \frac{\sqrt{5}\mathbb{M}_2^{(a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{5} + \frac{\sqrt{15}\mathbb{M}_2^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10} - \frac{\sqrt{15}\mathbb{M}_2^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10}$$

$\boxed{\text{z935}}$ $$\mathbb{G}_{3,1}^{(c)}(E_g,1a) = \frac{\mathbb{M}_2^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\mathbb{M}_2^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z936}}$ $$\mathbb{G}_{3,2}^{(c)}(E_g,1a) = \frac{\mathbb{M}_2^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\mathbb{M}_2^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z937}}$ $$\mathbb{G}_{3,1}^{(c)}(E_g,1b) = \frac{97\sqrt{8778}\mathbb{M}_4^{(a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{21318} - \frac{\sqrt{6270}\mathbb{M}_4^{(a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{1254} + \frac{\sqrt{14630}\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{374} + \frac{71\sqrt{2090}\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{7106} + \frac{5\sqrt{2090}\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{7106}$$

$\boxed{\text{z938}}$ $$\mathbb{G}_{3,2}^{(c)}(E_g,1b) = -\frac{97\sqrt{8778}\mathbb{M}_4^{(a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{21318} + \frac{\sqrt{6270}\mathbb{M}_4^{(a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{1254} + \frac{\sqrt{14630}\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{374} + \frac{71\sqrt{2090}\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{7106} - \frac{5\sqrt{2090}\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{7106}$$

$\boxed{\text{z939}}$ $$\mathbb{G}_{3,1}^{(c)}(E_g,2) = \frac{\sqrt{105}\mathbb{M}_4^{(a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{68} + \frac{\sqrt{3}\mathbb{M}_4^{(a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{3\sqrt{7}\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{34} + \frac{2\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{17} - \frac{8\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{17}$$

$\boxed{\text{z940}}$ $$\mathbb{G}_{3,2}^{(c)}(E_g,2) = -\frac{\sqrt{105}\mathbb{M}_4^{(a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{68} - \frac{\sqrt{3}\mathbb{M}_4^{(a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{3\sqrt{7}\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{34} + \frac{2\mathbb{M}_4^{(a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{17} + \frac{8\mathbb{M}_4^{(a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{17}$$

$\boxed{\text{z941}}$ $$\mathbb{G}_{5,1}^{(c)}(E_g,1) = \frac{\sqrt{15}\mathbb{M}_4^{(a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{12} - \frac{\sqrt{21}\mathbb{M}_4^{(a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{12} - \frac{\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z942}}$ $$\mathbb{G}_{5,2}^{(c)}(E_g,1) = -\frac{\sqrt{15}\mathbb{M}_4^{(a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{12} + \frac{\sqrt{21}\mathbb{M}_4^{(a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{12} - \frac{\mathbb{M}_4^{(a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z943}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(E_g) = -\frac{\sqrt{5}\mathbb{M}_2^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{5} + \frac{\sqrt{15}\mathbb{M}_2^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10} + \frac{\sqrt{15}\mathbb{M}_2^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10}$$

$\boxed{\text{z944}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(E_g) = \frac{\sqrt{5}\mathbb{M}_2^{(1,-1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{5} + \frac{\sqrt{15}\mathbb{M}_2^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10} - \frac{\sqrt{15}\mathbb{M}_2^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10}$$

$\boxed{\text{z945}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(E_g,1a) = \frac{\mathbb{M}_2^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\mathbb{M}_2^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z946}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(E_g,1a)=\frac{\mathbb{M}_2^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}+\frac{\mathbb{M}_2^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z947}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(E_g,1b)=\frac{97\sqrt{8778}\mathbb{M}_4^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{21318}-\frac{\sqrt{6270}\mathbb{M}_4^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{1254}+\frac{\sqrt{14630}\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{374}$$
$$+\frac{71\sqrt{2090}\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{7106}+\frac{5\sqrt{2090}\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{7106}$$

$\boxed{\text{z1008}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(E_g,1b)=-\frac{97\sqrt{8778}\mathbb{M}_4^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{21318}+\frac{\sqrt{6270}\mathbb{M}_4^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{1254}+\frac{\sqrt{14630}\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{374}$$
$$+\frac{71\sqrt{2090}\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{7106}-\frac{5\sqrt{2090}\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{7106}$$

$\boxed{\text{z1009}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(E_g,2)=\frac{\sqrt{105}\mathbb{M}_4^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{68}+\frac{\sqrt{3}\mathbb{M}_4^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}-\frac{3\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{34}+\frac{2\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{17}-\frac{8\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{17}$$

$\boxed{\text{z1010}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(E_g,2)=-\frac{\sqrt{105}\mathbb{M}_4^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{68}-\frac{\sqrt{3}\mathbb{M}_4^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}-\frac{3\sqrt{7}\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{34}+\frac{2\mathbb{M}_4^{(1,-1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{17}+\frac{8\mathbb{M}_4^{(1,-1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{17}$$

$\boxed{\text{z1011}}$ $$\mathbb{G}_{5,1}^{(1,-1;c)}(E_g,1a)=\frac{\sqrt{15}\mathbb{M}_4^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{12}-\frac{\sqrt{21}\mathbb{M}_4^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{12}-\frac{\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1012}}$ $$\mathbb{G}_{5,2}^{(1,-1;c)}(E_g,1a)=-\frac{\sqrt{15}\mathbb{M}_4^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{12}+\frac{\sqrt{21}\mathbb{M}_4^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{12}-\frac{\mathbb{M}_4^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1013}}$ $$\mathbb{G}_{5,1}^{(1,-1;c)}(E_g,1b)=\frac{10397\sqrt{688120917}\mathbb{M}_6^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{917494556}-\frac{4691\sqrt{4816846419}\mathbb{M}_6^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{917494556}+\frac{7\sqrt{9633692838}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{22377916}$$
$$-\frac{349\sqrt{729825215}\mathbb{M}_6^{(1,-1;a)}(B_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{83408596}+\frac{9863\sqrt{1605615473}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{917494556}$$
$$+\frac{99\sqrt{145965043}\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4068712}-\frac{7\sqrt{8028077365}\mathbb{M}_6^{(1,-1;a)}(B_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{107940536}$$

$\boxed{\text{z1014}}$ $$\mathbb{G}_{5,2}^{(1,-1;c)}(E_g,1b)=-\frac{10397\sqrt{688120917}\mathbb{M}_6^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{917494556}+\frac{4691\sqrt{4816846419}\mathbb{M}_6^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{917494556}+\frac{7\sqrt{9633692838}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{22377916}$$
$$-\frac{349\sqrt{729825215}\mathbb{M}_6^{(1,-1;a)}(B_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{83408596}+\frac{9863\sqrt{1605615473}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{917494556}$$
$$-\frac{99\sqrt{145965043}\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4068712}+\frac{7\sqrt{8028077365}\mathbb{M}_6^{(1,-1;a)}(B_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{107940536}$$

$\boxed{\text{z1015}}$ $$\mathbb{G}_{5,1}^{(1,-1;c)}(E_g,2) = -\frac{1179\sqrt{9445843995}\mathbb{M}_6^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{359841676} - \frac{1709\sqrt{1349406285}\mathbb{M}_6^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{359841676} + \frac{2477\sqrt{2698812570}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{359841676} + \frac{1201\sqrt{989564609}\mathbb{M}_6^{(1,-1;a)}(B_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{359841676} + \frac{481\sqrt{449802095}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{359841676} + \frac{841\sqrt{4947823045}\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{719683352} + \frac{35727\sqrt{89960419}\mathbb{M}_6^{(1,-1;a)}(B_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{719683352}$$

$\boxed{\text{z1016}}$ $$\mathbb{G}_{5,2}^{(1,-1;c)}(E_g,2) = \frac{1179\sqrt{9445843995}\mathbb{M}_6^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{359841676} + \frac{1709\sqrt{1349406285}\mathbb{M}_6^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{359841676} + \frac{2477\sqrt{2698812570}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{359841676} + \frac{1201\sqrt{989564609}\mathbb{M}_6^{(1,-1;a)}(B_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{359841676} + \frac{481\sqrt{449802095}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{359841676} - \frac{841\sqrt{4947823045}\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{719683352} - \frac{35727\sqrt{89960419}\mathbb{M}_6^{(1,-1;a)}(B_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{719683352}$$

$\boxed{\text{z1017}}$ $$\mathbb{G}_{5,1}^{(1,-1;c)}(E_g,3) = -\frac{4\sqrt{21049}\mathbb{M}_6^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{3007} - \frac{16\sqrt{3007}\mathbb{M}_6^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{3007} - \frac{12\sqrt{6014}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{3007} - \frac{15\sqrt{496155}\mathbb{M}_6^{(1,-1;a)}(B_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{24056} - \frac{5\sqrt{9021}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{24056} - \frac{5\sqrt{99231}\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{6014} + \frac{2\sqrt{45105}\mathbb{M}_6^{(1,-1;a)}(B_{2u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{3007}$$

$\boxed{\text{z1018}}$ $$\mathbb{G}_{5,2}^{(1,-1;c)}(E_g,3) = \frac{4\sqrt{21049}\mathbb{M}_6^{(1,-1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{3007} + \frac{16\sqrt{3007}\mathbb{M}_6^{(1,-1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{3007} - \frac{12\sqrt{6014}\mathbb{M}_6^{(1,-1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{3007} - \frac{15\sqrt{496155}\mathbb{M}_6^{(1,-1;a)}(B_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{24056} - \frac{5\sqrt{9021}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{24056} + \frac{5\sqrt{99231}\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{6014} - \frac{2\sqrt{45105}\mathbb{M}_6^{(1,-1;a)}(B_{2u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{3007}$$

$\boxed{\text{z1019}}$ $$\mathbb{G}_{7,1}^{(1,-1;c)}(E_g,1) = \frac{\sqrt{5}\mathbb{M}_6^{(1,-1;a)}(B_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{\sqrt{11}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1020}}$ $$\mathbb{G}_{7,2}^{(1,-1;c)}(E_g,1) = \frac{\sqrt{5}\mathbb{M}_6^{(1,-1;a)}(B_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{11}\mathbb{M}_6^{(1,-1;a)}(B_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\mathbb{M}_6^{(1,-1;a)}(B_{2u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1021}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(E_g) = -\frac{\sqrt{5}\mathbb{M}_2^{(1,0;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{5} + \frac{\sqrt{15}\mathbb{M}_2^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10} + \frac{\sqrt{15}\mathbb{M}_2^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10}$$

$\boxed{\text{z1022}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(E_g) = \frac{\sqrt{5}\mathbb{M}_2^{(1,0;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{5} + \frac{\sqrt{15}\mathbb{M}_2^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10} - \frac{\sqrt{15}\mathbb{M}_2^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10}$$

$\boxed{\text{z1023}}$ $$\mathbb{G}_{3,1}^{(1,0;c)}(E_g,1a) = \frac{\mathbb{M}_2^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\mathbb{M}_2^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1024}}$ $$\mathbb{G}_{3,2}^{(1,0;c)}(E_g,1a)=\frac{\mathbb{M}_2^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}+\frac{\mathbb{M}_2^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1025}}$ $$\mathbb{G}_{3,1}^{(1,0;c)}(E_g,1b)=\frac{97\sqrt{8778}\mathbb{M}_4^{(1,0;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{21318}-\frac{\sqrt{6270}\mathbb{M}_4^{(1,0;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{1254}+\frac{\sqrt{14630}\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{374}$$
$$+\frac{71\sqrt{2090}\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{7106}+\frac{5\sqrt{2090}\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{7106}$$

$\boxed{\text{z1026}}$ $$\mathbb{G}_{3,2}^{(1,0;c)}(E_g,1b)=-\frac{97\sqrt{8778}\mathbb{M}_4^{(1,0;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{21318}+\frac{\sqrt{6270}\mathbb{M}_4^{(1,0;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{1254}+\frac{\sqrt{14630}\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{374}$$
$$+\frac{71\sqrt{2090}\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{7106}-\frac{5\sqrt{2090}\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{7106}$$

$\boxed{\text{z1027}}$ $$\mathbb{G}_{3,1}^{(1,0;c)}(E_g,2)=\frac{\sqrt{105}\mathbb{M}_4^{(1,0;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{68}+\frac{\sqrt{3}\mathbb{M}_4^{(1,0;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}-\frac{3\sqrt{7}\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{34}+\frac{2\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{17}-\frac{8\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{17}$$

$\boxed{\text{z1028}}$ $$\mathbb{G}_{3,2}^{(1,0;c)}(E_g,2)=-\frac{\sqrt{105}\mathbb{M}_4^{(1,0;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{68}-\frac{\sqrt{3}\mathbb{M}_4^{(1,0;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}-\frac{3\sqrt{7}\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{34}+\frac{2\mathbb{M}_4^{(1,0;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{17}+\frac{8\mathbb{M}_4^{(1,0;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{17}$$

$\boxed{\text{z1029}}$ $$\mathbb{G}_{5,1}^{(1,0;c)}(E_g,1)=\frac{\sqrt{15}\mathbb{M}_4^{(1,0;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{12}-\frac{\sqrt{21}\mathbb{M}_4^{(1,0;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{12}-\frac{\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1030}}$ $$\mathbb{G}_{5,2}^{(1,0;c)}(E_g,1)=-\frac{\sqrt{15}\mathbb{M}_4^{(1,0;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{12}+\frac{\sqrt{21}\mathbb{M}_4^{(1,0;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{12}-\frac{\mathbb{M}_4^{(1,0;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1031}}$ $$\mathbb{G}_{1,1}^{(1,1;c)}(E_g,a)=\frac{\sqrt{2}\mathbb{M}_0^{(1,1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1032}}$ $$\mathbb{G}_{1,2}^{(1,1;c)}(E_g,a)=-\frac{\sqrt{2}\mathbb{M}_0^{(1,1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1033}}$ $$\mathbb{G}_{1,1}^{(1,1;c)}(E_g,b)=-\frac{\sqrt{5}\mathbb{M}_2^{(1,1;a)}(A_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{5}+\frac{\sqrt{15}\mathbb{M}_2^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10}+\frac{\sqrt{15}\mathbb{M}_2^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10}$$

$\boxed{\text{z1034}}$ $$\mathbb{G}_{1,2}^{(1,1;c)}(E_g,b)=\frac{\sqrt{5}\mathbb{M}_2^{(1,1;a)}(A_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{5}+\frac{\sqrt{15}\mathbb{M}_2^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{10}-\frac{\sqrt{15}\mathbb{M}_2^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{10}$$

$\boxed{\text{z1035}}$ $$\mathbb{G}_{3,1}^{(1,1;c)}(E_g,1a)=\frac{\mathbb{M}_2^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}-\frac{\mathbb{M}_2^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1036}}$ $$\mathbb{G}_{3,2}^{(1,1;c)}(E_g,1a)=\frac{\mathbb{M}_2^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}+\frac{\mathbb{M}_2^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1037}}$ $$\mathbb{G}_{3,1}^{(1,1;c)}(E_g,1b)=\frac{97\sqrt{8778}\mathbb{M}_4^{(1,1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{21318}-\frac{\sqrt{6270}\mathbb{M}_4^{(1,1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{1254}+\frac{\sqrt{14630}\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{374}+\frac{71\sqrt{2090}\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{7106}+\frac{5\sqrt{2090}\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{7106}$$

$\boxed{\text{z1038}}$ $$\mathbb{G}_{3,2}^{(1,1;c)}(E_g,1b)=-\frac{97\sqrt{8778}\mathbb{M}_4^{(1,1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{21318}+\frac{\sqrt{6270}\mathbb{M}_4^{(1,1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{1254}+\frac{\sqrt{14630}\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{374}+\frac{71\sqrt{2090}\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{7106}-\frac{5\sqrt{2090}\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{7106}$$

$\boxed{\text{z1039}}$ $$\mathbb{G}_{3,1}^{(1,1;c)}(E_g,2)=\frac{\sqrt{105}\mathbb{M}_4^{(1,1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{68}+\frac{\sqrt{3}\mathbb{M}_4^{(1,1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}-\frac{3\sqrt{7}\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{34}+\frac{2\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,1}^{(b)}(E_u)}{17}-\frac{8\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{17}$$

$\boxed{\text{z1040}}$ $$\mathbb{G}_{3,2}^{(1,1;c)}(E_g,2)=-\frac{\sqrt{105}\mathbb{M}_4^{(1,1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{68}-\frac{\sqrt{3}\mathbb{M}_4^{(1,1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}-\frac{3\sqrt{7}\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{34}+\frac{2\mathbb{M}_4^{(1,1;a)}(B_{1u})\mathbb{T}_{1,2}^{(b)}(E_u)}{17}+\frac{8\mathbb{M}_4^{(1,1;a)}(B_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{17}$$

$\boxed{\text{z1041}}$ $$\mathbb{G}_{5,1}^{(1,1;c)}(E_g,1)=\frac{\sqrt{15}\mathbb{M}_4^{(1,1;a)}(A_{1u},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{12}-\frac{\sqrt{21}\mathbb{M}_4^{(1,1;a)}(A_{1u},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{12}-\frac{\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1042}}$ $$\mathbb{G}_{5,2}^{(1,1;c)}(E_g,1)=-\frac{\sqrt{15}\mathbb{M}_4^{(1,1;a)}(A_{1u},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{12}+\frac{\sqrt{21}\mathbb{M}_4^{(1,1;a)}(A_{1u},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{12}-\frac{\mathbb{M}_4^{(1,1;a)}(A_{2u})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1191}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_u,a)=\frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z1192}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_u,a)=\frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z1193}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_u,b)=\frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$\boxed{\text{z1194}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_u,b)=-\frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$\boxed{\text{z1195}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_u,c)=\frac{3\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8}-\frac{\sqrt{5}\mathbb{Q}_{3,1}^{(a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$\boxed{\text{z1196}}$ $\mathbb{Q}_{1,2}^{(c)}(E_u,c) = -\frac{3\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} + \frac{\sqrt{5}\mathbb{Q}_{3,2}^{(a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1197}}$ $\mathbb{Q}_{1,1}^{(c)}(E_u,d) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(a)}(E_u)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1198}}$ $\mathbb{Q}_{1,2}^{(c)}(E_u,d) = -\frac{\sqrt{2}\mathbb{G}_{2,2}^{(a)}(E_u)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1199}}$ $\mathbb{Q}_{3,1}^{(c)}(E_u,1a) = \frac{\sqrt{2010}\mathbb{Q}_{5,1}^{(a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{67} - \frac{\sqrt{938}\mathbb{Q}_{5,1}^{(a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{134}$

$\boxed{\text{z1200}}$ $\mathbb{Q}_{3,2}^{(c)}(E_u,1a) = -\frac{\sqrt{2010}\mathbb{Q}_{5,2}^{(a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{67} + \frac{\sqrt{938}\mathbb{Q}_{5,2}^{(a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{134}$

$\boxed{\text{z1201}}$ $\mathbb{Q}_{3,1}^{(c)}(E_u,1b) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(a)}(E_u,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1202}}$ $\mathbb{Q}_{3,2}^{(c)}(E_u,1b) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(a)}(E_u,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1203}}$ $\mathbb{Q}_{3,1}^{(c)}(E_u,1c) = \frac{\sqrt{5}\mathbb{Q}_{3,1}^{(a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} + \frac{3\sqrt{3}\mathbb{Q}_{3,1}^{(a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1204}}$ $\mathbb{Q}_{3,2}^{(c)}(E_u,1c) = -\frac{\sqrt{5}\mathbb{Q}_{3,2}^{(a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{3\sqrt{3}\mathbb{Q}_{3,2}^{(a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1205}}$ $\mathbb{Q}_{3,1}^{(c)}(E_u,1d) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1206}}$ $\mathbb{Q}_{3,2}^{(c)}(E_u,1d) = -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1207}}$ $\mathbb{Q}_{3,1}^{(c)}(E_u,2a) = \frac{17\sqrt{2345}\mathbb{Q}_{5,1}^{(a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{4288} - \frac{3\sqrt{67}\mathbb{Q}_{5,1}^{(a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} + \frac{85\sqrt{201}\mathbb{Q}_{5,1}^{(a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2144}$

$\boxed{\text{z1208}}$ $\mathbb{Q}_{3,2}^{(c)}(E_u,2a) = -\frac{17\sqrt{2345}\mathbb{Q}_{5,2}^{(a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{4288} + \frac{3\sqrt{67}\mathbb{Q}_{5,2}^{(a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} - \frac{85\sqrt{201}\mathbb{Q}_{5,2}^{(a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2144}$

$\boxed{\text{z1209}}$ $\mathbb{Q}_{3,1}^{(c)}(E_u,2b) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(a)}(E_u,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1210}}$ $\mathbb{Q}_{3,2}^{(c)}(E_u,2b) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(a)}(E_u,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1211}}$ $\mathbb{Q}_{3,1}^{(c)}(E_u,2c) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1212}}$ $\mathbb{Q}_{3,2}^{(c)}(E_u,2c) = -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1213}}$ $\mathbb{Q}_{5,1}^{(c)}(E_u,1a) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(a)}(E_u,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1214}}$ $\mathbb{Q}_{5,2}^{(c)}(E_u,1a) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(a)}(E_u,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1215}}$ $\mathbb{Q}_{5,1}^{(c)}(E_u,1b) = \frac{3\sqrt{7}\mathbb{Q}_{5,1}^{(a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} + \frac{17\sqrt{5}\mathbb{Q}_{5,1}^{(a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} + \frac{3\sqrt{15}\mathbb{Q}_{5,1}^{(a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{32}$

$\boxed{\text{z1216}}$ $\mathbb{Q}_{5,2}^{(c)}(E_u,1b) = -\frac{3\sqrt{7}\mathbb{Q}_{5,2}^{(a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} - \frac{17\sqrt{5}\mathbb{Q}_{5,2}^{(a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} - \frac{3\sqrt{15}\mathbb{Q}_{5,2}^{(a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{32}$

$\boxed{\text{z1217}}$ $\mathbb{Q}_{5,1}^{(c)}(E_u,2) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(a)}(E_u,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1218}}$ $\mathbb{Q}_{5,2}^{(c)}(E_u,2) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(a)}(E_u,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1219}}$ $\mathbb{Q}_{5,1}^{(c)}(E_u,3) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(a)}(E_u,3)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1220}}$ $\mathbb{Q}_{5,2}^{(c)}(E_u,3) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(a)}(E_u,3)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1221}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(E_u,a) = \frac{3\sqrt{3}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{\sqrt{5}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1222}}$ $\mathbb{Q}_{1,2}^{(1,-1;c)}(E_u,a) = -\frac{3\sqrt{3}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} + \frac{\sqrt{5}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1223}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(E_u,b) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1224}}$ $\mathbb{Q}_{1,2}^{(1,-1;c)}(E_u,b) = -\frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1225}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(E_u,1a) = \frac{\sqrt{2010}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{67} - \frac{\sqrt{938}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{Q}_2^{(b)}(B_{1g})}{134}$

$\boxed{\text{z1226}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(E_u,1a) = -\frac{\sqrt{2010}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{67} + \frac{\sqrt{938}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{Q}_2^{(b)}(B_{1g})}{134}$

$\boxed{\text{z1227}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(E_u,1b) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1228}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(E_u,1b) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1229}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(E_u,1c) = \frac{\sqrt{5}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} + \frac{3\sqrt{3}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1230}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(E_u,1c) = -\frac{\sqrt{5}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} - \frac{3\sqrt{3}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1231}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(E_u,1d) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1232}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(E_u,1d) = -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1233}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(E_u,2a) = \frac{17\sqrt{2345}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{4288} - \frac{3\sqrt{67}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{64} + \frac{85\sqrt{201}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{Q}_2^{(b)}(B_{1g})}{2144}$

$\boxed{\text{z1234}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(E_u,2a) = -\frac{17\sqrt{2345}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{4288} + \frac{3\sqrt{67}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{64} - \frac{85\sqrt{201}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{Q}_2^{(b)}(B_{1g})}{2144}$

$\boxed{\text{z1235}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(E_u,2b) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1236}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(E_u,2b) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1237}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(E_u,2c) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1238}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(E_u,2c) = -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1239}}$ $\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,1a) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1240}}$ $\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,1a) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1241}}$ $\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,1b) = \frac{3\sqrt{7}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} + \frac{17\sqrt{5}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} + \frac{3\sqrt{15}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{32}$

$\boxed{\text{z1242}}$ $\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,1b) = -\frac{3\sqrt{7}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} - \frac{17\sqrt{5}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} - \frac{3\sqrt{15}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{32}$

$\boxed{\text{z1243}}$ $\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,1c) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1244}}$ $\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,1c) = -\frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1245}}$ $\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,2a) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1246}}$ $\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,2a) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1247}}$ $\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,2b) = \frac{4\sqrt{321}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{107} + \frac{\sqrt{2354}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{214}$

$\boxed{\text{z1248}}$ $\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,2b) = -\frac{4\sqrt{321}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{107} - \frac{\sqrt{2354}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{214}$

$\boxed{\text{z1249}}$ $\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,3a) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,3)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1250}}$ $\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,3a) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,3)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1251}}$ $\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,3b) = \frac{\sqrt{2354}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{214} - \frac{4\sqrt{321}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{107}$

$\boxed{\text{z1252}}$ $\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,3b) = -\frac{\sqrt{2354}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{214} + \frac{4\sqrt{321}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{107}$

$\boxed{\text{z1253}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(E_u,a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1254}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(E_u,a) = \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1255}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(E_u,b) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1256}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(E_u,b) = -\frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1257}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(E_u,c) = \frac{3\sqrt{3}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} - \frac{\sqrt{5}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1258}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(E_u,c) = -\frac{3\sqrt{3}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} + \frac{\sqrt{5}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1259}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(E_u,d) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,0;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1260}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(E_u,d) = -\frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,0;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1261}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,1a) = \frac{\sqrt{2010}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{67} - \frac{\sqrt{938}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{Q}_2^{(b)}(B_{1g})}{134}$

$\boxed{\text{z1262}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,1a) = -\frac{\sqrt{2010}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{67} + \frac{\sqrt{938}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{Q}_2^{(b)}(B_{1g})}{134}$

$\boxed{\text{z1263}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,1b) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1264}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,1b) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1265}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,1c) = \frac{\sqrt{5}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} + \frac{3\sqrt{3}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1266}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,1c) = -\frac{\sqrt{5}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{3\sqrt{3}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1267}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,1d) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,0;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1268}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,1d) = -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,0;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1269}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,2a) = \frac{17\sqrt{2345}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{4288} - \frac{3\sqrt{67}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} + \frac{85\sqrt{201}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2144}$

$\boxed{\text{z1270}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,2a) = -\frac{17\sqrt{2345}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{4288} + \frac{3\sqrt{67}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} - \frac{85\sqrt{201}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2144}$

$\boxed{\text{z1271}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,2b) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1272}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,2b) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1273}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,2c) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,0;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1274}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,2c) = -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,0;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1275}}$ $\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,1a) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1276}}$ $\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,1a) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1277}}$ $\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,1b) = \frac{3\sqrt{7}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} + \frac{17\sqrt{5}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} + \frac{3\sqrt{15}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{32}$

$\boxed{\text{z1278}}$ $\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,1b) = -\frac{3\sqrt{7}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} - \frac{17\sqrt{5}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{64} - \frac{3\sqrt{15}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{32}$

$\boxed{\text{z1279}}$ $\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1280}}$ $\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1281}}$ $\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,3) = \frac{\sqrt{2}\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1282}}$ $\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,3) = \frac{\sqrt{2}\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1283}}$ $\mathbb{Q}_{1,1}^{(1,1;c)}(E_u,a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1284}}$ $\mathbb{Q}_{1,2}^{(1,1;c)}(E_u,a) = \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1285}}$ $\mathbb{Q}_{1,1}^{(1,1;c)}(E_u,b) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,1;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1286}}$ $\mathbb{Q}_{1,2}^{(1,1;c)}(E_u,b) = -\frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,1;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1287}}$ $\mathbb{Q}_{1,1}^{(1,1;c)}(E_u,c) = \frac{3\sqrt{3}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} - \frac{\sqrt{5}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1288}}$ $\mathbb{Q}_{1,2}^{(1,1;c)}(E_u,c) = -\frac{3\sqrt{3}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} + \frac{\sqrt{5}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1289}}$ $\mathbb{Q}_{1,1}^{(1,1;c)}(E_u,d) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,1;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1290}}$ $\mathbb{Q}_{1,2}^{(1,1;c)}(E_u,d) = -\frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,1;a)}(E_u)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1291}}$ $\mathbb{Q}_{3,1}^{(1,1;c)}(E_u,1a) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1292}}$ $\mathbb{Q}_{3,2}^{(1,1;c)}(E_u,1a) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1293}}$ $\mathbb{Q}_{3,1}^{(1,1;c)}(E_u,1b) = \frac{\sqrt{5}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} + \frac{3\sqrt{3}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1294}}$ $\mathbb{Q}_{3,2}^{(1,1;c)}(E_u,1b) = -\frac{\sqrt{5}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{3\sqrt{3}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1295}}$ $\mathbb{Q}_{3,1}^{(1,1;c)}(E_u,1c) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,1;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1296}}$ $\mathbb{Q}_{3,2}^{(1,1;c)}(E_u,1c) = -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,1;a)}(E_u,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1297}}$ $\mathbb{Q}_{3,1}^{(1,1;c)}(E_u,2a) = \frac{\sqrt{2}\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1298}}$ $\mathbb{Q}_{3,2}^{(1,1;c)}(E_u,2a) = \frac{\sqrt{2}\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1299}}$ $\mathbb{Q}_{3,1}^{(1,1;c)}(E_u,2b) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,1;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1300}}$ $\mathbb{Q}_{3,2}^{(1,1;c)}(E_u,2b) = -\frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,1;a)}(E_u,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1301}}$ $\mathbb{G}_{2,1}^{(c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(a)}(E_u)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1302}}$ $\mathbb{G}_{2,2}^{(c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(a)}(E_u)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1303}}$ $\mathbb{G}_{4,1}^{(c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(a)}(E_u,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1304}}$ $\mathbb{G}_{4,2}^{(c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(a)}(E_u,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1305}}$ $\mathbb{G}_{4,1}^{(c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(a)}(E_u,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1306}}$ $\mathbb{G}_{4,2}^{(c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(a)}(E_u,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1307}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1308}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1309}}$ $\mathbb{G}_{4,1}^{(1,-1;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1310}}$ $\mathbb{G}_{4,2}^{(1,-1;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1311}}$ $\mathbb{G}_{4,1}^{(1,-1;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1312}}$ $\mathbb{G}_{4,2}^{(1,-1;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1313}}$ $\mathbb{G}_{6,1}^{(1,-1;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1314}}$ $\mathbb{G}_{6,2}^{(1,-1;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1315}}$ $\mathbb{G}_{6,1}^{(1,-1;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1316}}$ $\mathbb{G}_{6,2}^{(1,-1;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1317}}$ $\mathbb{G}_{6,1}^{(1,-1;c)}(E_u,3) = \frac{\sqrt{2}\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1318}}$ $\mathbb{G}_{6,2}^{(1,-1;c)}(E_u,3) = \frac{\sqrt{2}\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1319}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1320}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,0;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1321}}$ $\mathbb{G}_{4,1}^{(1,0;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,0;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1322}}$ $\mathbb{G}_{4,2}^{(1,0;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,0;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1323}}$ $\mathbb{G}_{4,1}^{(1,0;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,0;a)}(E_u,2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1324}}$ $\mathbb{G}_{4,2}^{(1,0;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,0;a)}(E_u,2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1325}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1326}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1327}}$ $\mathbb{G}_{4,1}^{(1,1;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,1;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1328}}$ $\mathbb{G}_{4,2}^{(1,1;c)}(E_u,1) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,1;a)}(E_u,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1329}}$ $\mathbb{G}_{4,1}^{(1,1;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{4,1}^{(1,1;a)}(E_u,2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1330}}$ $\mathbb{G}_{4,2}^{(1,1;c)}(E_u,2) = \frac{\sqrt{2}\mathbb{G}_{4,2}^{(1,1;a)}(E_u,2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

- `U;U_001_1` : 'U'-'U' bond-cluster
  * bra: $\langle\frac{5}{2},\frac{5}{2};f|$, $\langle\frac{5}{2},\frac{3}{2};f|$, $\langle\frac{5}{2},\frac{1}{2};f|$, $\langle\frac{5}{2},-\frac{1}{2};f|$, $\langle\frac{5}{2},-\frac{3}{2};f|$, $\langle\frac{5}{2},-\frac{5}{2};f|$, $\langle\frac{7}{2},\frac{7}{2};f|$, $\langle\frac{7}{2},\frac{5}{2};f|$, $\langle\frac{7}{2},\frac{3}{2};f|$, $\langle\frac{7}{2},\frac{1}{2};f|$, $\langle\frac{7}{2},-\frac{1}{2};f|$, $\langle\frac{7}{2},-\frac{3}{2};f|$, $\langle\frac{7}{2},-\frac{5}{2};f|$, $\langle\frac{7}{2},-\frac{7}{2};f|$
  * ket: $|\frac{5}{2},\frac{5}{2};f\rangle$, $|\frac{5}{2},\frac{3}{2};f\rangle$, $|\frac{5}{2},\frac{1}{2};f\rangle$, $|\frac{5}{2},-\frac{1}{2};f\rangle$, $|\frac{5}{2},-\frac{3}{2};f\rangle$, $|\frac{5}{2},-\frac{5}{2};f\rangle$, $|\frac{7}{2},\frac{7}{2};f\rangle$, $|\frac{7}{2},\frac{5}{2};f\rangle$, $|\frac{7}{2},\frac{3}{2};f\rangle$, $|\frac{7}{2},\frac{1}{2};f\rangle$, $|\frac{7}{2},-\frac{1}{2};f\rangle$, $|\frac{7}{2},-\frac{3}{2};f\rangle$, $|\frac{7}{2},-\frac{5}{2};f\rangle$, $|\frac{7}{2},-\frac{7}{2};f\rangle$
  * wyckoff: `4b@4c`

  $\boxed{\text{z76}}$ $\mathbb{Q}_0^{(c)}(A_{1g},a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

  $\boxed{\text{z77}}$ $\mathbb{Q}_0^{(c)}(A_{1g},b) = \mathbb{Q}_2^{(a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

  $\boxed{\text{z78}}$ $\mathbb{Q}_2^{(c)}(A_{1g},a) = \mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

  $\boxed{\text{z79}}$ $\mathbb{Q}_2^{(c)}(A_{1g},b) = -\mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z80}}$ $\mathbb{Q}_4^{(c)}(A_{1g},1a) = \mathbb{Q}_4^{(a)}(A_{1g},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z81}}$ $\mathbb{Q}_4^{(c)}(A_{1g},1b) = \mathbb{Q}_6^{(a)}(B_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z82}}$ $\mathbb{Q}_4^{(c)}(A_{1g},2a) = \mathbb{Q}_4^{(a)}(A_{1g},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z83}}$ $\mathbb{Q}_4^{(c)}(A_{1g},2b) = -\mathbb{Q}_6^{(a)}(B_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z84}}$ $\mathbb{Q}_6^{(c)}(A_{1g},1) = \mathbb{Q}_6^{(a)}(A_{1g},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z85}}$ $\mathbb{Q}_6^{(c)}(A_{1g},2) = \mathbb{Q}_6^{(a)}(A_{1g},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z86}}$ $\mathbb{Q}_0^{(1,-1;c)}(A_{1g}) = \mathbb{Q}_2^{(1,-1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z87}}$ $\mathbb{Q}_2^{(1,-1;c)}(A_{1g},a) = \mathbb{Q}_2^{(1,-1;a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z88}}$ $\mathbb{Q}_2^{(1,-1;c)}(A_{1g},b) = -\mathbb{Q}_4^{(1,-1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z89}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{1g},1a) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z90}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{1g},1b) = \mathbb{Q}_6^{(1,-1;a)}(B_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z91}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{1g},2a) = \mathbb{Q}_4^{(1,-1;a)}(A_{1g},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z92}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{1g},2b) = -\mathbb{Q}_6^{(1,-1;a)}(B_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z93}}$ $\mathbb{Q}_6^{(1,-1;c)}(A_{1g},1) = \mathbb{Q}_6^{(1,-1;a)}(A_{1g},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z94}}$ $\mathbb{Q}_6^{(1,-1;c)}(A_{1g},2) = \mathbb{Q}_6^{(1,-1;a)}(A_{1g},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z95}}$ $\mathbb{Q}_2^{(1,0;c)}(A_{1g}) = -\mathbb{G}_3^{(1,0;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z96}}$ $\mathbb{Q}_4^{(1,0;c)}(A_{1g},1) = \mathbb{G}_5^{(1,0;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z97}}$ $\mathbb{Q}_0^{(1,1;c)}(A_{1g},a) = \mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z98}}$ $\mathbb{Q}_0^{(1,1;c)}(A_{1g},b) = \mathbb{Q}_2^{(1,1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z99}}$ $\mathbb{Q}_2^{(1,1;c)}(A_{1g},a) = \mathbb{Q}_2^{(1,1;a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z100}}$ $\mathbb{Q}_2^{(1,1;c)}(A_{1g},b) = -\mathbb{Q}_4^{(1,1;a)}(B_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z101}}$ $\mathbb{Q}_4^{(1,1;c)}(A_{1g},1) = \mathbb{Q}_4^{(1,1;a)}(A_{1g},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z102}}$ $\mathbb{Q}_4^{(1,1;c)}(A_{1g},2) = \mathbb{Q}_4^{(1,1;a)}(A_{1g},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z103}}$ $\mathbb{G}_5^{(1,0;c)}(A_{1g}) = \mathbb{G}_5^{(1,0;a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z162}}$ $$\mathbb{Q}_5^{(c)}(A_{1u}) = \frac{\sqrt{14}\mathbb{M}_{5,1}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{7} - \frac{\sqrt{42}\mathbb{M}_{5,1}^{(a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{14} - \frac{\sqrt{14}\mathbb{M}_{5,2}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{7} + \frac{\sqrt{42}\mathbb{M}_{5,2}^{(a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{14}$$

$\boxed{\text{z163}}$ $$\mathbb{Q}_5^{(1,-1;c)}(A_{1u}) = \frac{\sqrt{14}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{7} - \frac{\sqrt{42}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{14} - \frac{\sqrt{14}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{7} + \frac{\sqrt{42}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{14}$$

$\boxed{\text{z164}}$ $$\begin{aligned}\mathbb{Q}_7^{(1,-1;c)}(A_{1u}) = {} & \frac{\sqrt{10}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{10} - \frac{3\sqrt{65}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{40} + \frac{\sqrt{55}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{40} \\ & - \frac{\sqrt{10}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{10} + \frac{3\sqrt{65}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{40} - \frac{\sqrt{55}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{40}\end{aligned}$$

$\boxed{\text{z165}}$ $$\mathbb{Q}_5^{(1,0;c)}(A_{1u},a) = -\frac{\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z166}}$ $$\mathbb{Q}_5^{(1,0;c)}(A_{1u},b) = -\frac{\sqrt{290}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{58} + \frac{2\sqrt{87}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{29} + \frac{\sqrt{290}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{58} - \frac{2\sqrt{87}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{29}$$

$\boxed{\text{z167}}$ $$\begin{aligned}\mathbb{Q}_7^{(1,0;c)}(A_{1u}) = {} & -\frac{3\sqrt{1914}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{232} + \frac{\sqrt{29}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{3\sqrt{1595}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{464} \\ & + \frac{3\sqrt{1914}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{232} - \frac{\sqrt{29}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{3\sqrt{1595}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{464}\end{aligned}$$

$\boxed{\text{z168}}$ $$\mathbb{Q}_5^{(1,1;c)}(A_{1u}) = \frac{\sqrt{14}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{7} - \frac{\sqrt{42}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{14} - \frac{\sqrt{14}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{7} + \frac{\sqrt{42}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{14}$$

$\boxed{\text{z169}}$ $$\mathbb{G}_0^{(c)}(A_{1u}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z170}}$ $$\mathbb{G}_2^{(c)}(A_{1u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z171}}$ $$\mathbb{G}_4^{(c)}(A_{1u},1a) = \frac{7\sqrt{170}\mathbb{M}_{5,1}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{9\sqrt{238}\mathbb{M}_{5,1}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{952} + \frac{3\sqrt{714}\mathbb{M}_{5,1}^{(a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{476} - \frac{7\sqrt{170}\mathbb{M}_{5,2}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{9\sqrt{238}\mathbb{M}_{5,2}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{952} - \frac{3\sqrt{714}\mathbb{M}_{5,2}^{(a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{476}$$

$\boxed{\text{z172}}$ $$\mathbb{G}_4^{(c)}(A_{1u},1b) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z173}}$ $$\mathbb{G}_4^{(c)}(A_{1u},2) = -\frac{3\sqrt{102}\mathbb{M}_{5,1}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{\sqrt{3570}\mathbb{M}_{5,1}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{\sqrt{1190}\mathbb{M}_{5,1}^{(a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{68} + \frac{3\sqrt{102}\mathbb{M}_{5,2}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{\sqrt{3570}\mathbb{M}_{5,2}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{\sqrt{1190}\mathbb{M}_{5,2}^{(a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{68}$$

$\boxed{\text{z174}}$ $$\mathbb{G}_0^{(1,-1;c)}(A_{1u}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z175}}$ $$\mathbb{G}_2^{(1,-1;c)}(A_{1u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z176}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_{1u},1a) = \frac{7\sqrt{170}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{9\sqrt{238}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{952} + \frac{3\sqrt{714}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{476} - \frac{7\sqrt{170}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{9\sqrt{238}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{952} - \frac{3\sqrt{714}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{476}$$

$\boxed{\text{z177}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_{1u},1b) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z178}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_{1u},2) = -\frac{3\sqrt{102}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{\sqrt{3570}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{\sqrt{1190}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{68} + \frac{3\sqrt{102}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{\sqrt{3570}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{\sqrt{1190}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{68}$$

$\boxed{\text{z179}}$ $$\mathbb{G}_6^{(1,-1;c)}(A_{1u},1) = \frac{\sqrt{11690}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{334} - \frac{2\sqrt{55110}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{835} - \frac{3\sqrt{358215}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{6680} + \frac{11\sqrt{2505}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{6680} - \frac{\sqrt{11690}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{334} + \frac{2\sqrt{55110}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{835} + \frac{3\sqrt{358215}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{6680} - \frac{11\sqrt{2505}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{6680}$$

$\boxed{\text{z180}}$ $$\mathbb{G}_6^{(1,-1;c)}(A_{1u},2) = -\frac{55\sqrt{7014}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{10688} - \frac{35\sqrt{3674}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{10688} - \frac{17\sqrt{23881}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{10688} - \frac{383\sqrt{167}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{10688}$$
$$+ \frac{55\sqrt{7014}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{10688} + \frac{35\sqrt{3674}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{10688} + \frac{17\sqrt{23881}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{10688} + \frac{383\sqrt{167}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{10688}$$

$\boxed{\text{z181}}$ $$\mathbb{G}_8^{(1,-1;c)}(A_{1u},1) = \frac{\sqrt{858}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} + \frac{\sqrt{182}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} - \frac{\sqrt{7}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} - \frac{\sqrt{1001}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{64}$$
$$- \frac{\sqrt{858}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} - \frac{\sqrt{182}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} + \frac{\sqrt{7}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} + \frac{\sqrt{1001}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{64}$$

$\boxed{\text{z182}}$ $$\mathbb{G}_2^{(1,0;c)}(A_{1u}) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z251}}$ $$\mathbb{G}_4^{(1,0;c)}(A_{1u},1) = \frac{\sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z252}}$ $$\mathbb{G}_6^{(1,0;c)}(A_{1u},1) = -\frac{\sqrt{6}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{3\sqrt{11}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{5}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{16}$$
$$+ \frac{\sqrt{6}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{3\sqrt{11}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{16}$$

$\boxed{\text{z253}}$ $$\mathbb{G}_0^{(1,1;c)}(A_{1u}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z254}}$ $$\mathbb{G}_2^{(1,1;c)}(A_{1u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z255}}$ $$\mathbb{G}_4^{(1,1;c)}(A_{1u},1a) = \frac{7\sqrt{170}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{9\sqrt{238}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{952} + \frac{3\sqrt{714}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{476}$$
$$- \frac{7\sqrt{170}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{9\sqrt{238}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{952} - \frac{3\sqrt{714}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{476}$$

$\boxed{\text{z256}}$ $$\mathbb{G}_4^{(1,1;c)}(A_{1u},1b) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z257}}$ $$\mathbb{G}_4^{(1,1;c)}(A_{1u},2) = -\frac{3\sqrt{102}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{\sqrt{3570}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{136} + \frac{\sqrt{1190}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{68}$$
$$+ \frac{3\sqrt{102}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{\sqrt{3570}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{136} - \frac{\sqrt{1190}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{68}$$

$\boxed{\text{z258}}$ $\mathbb{Q}_4^{(c)}(A_{2g},a) = \mathbb{Q}_2^{(a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z259}}$ $\mathbb{Q}_4^{(c)}(A_{2g},b) = \mathbb{Q}_4^{(a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z260}}$ $\mathbb{Q}_4^{(c)}(A_{2g},c) = -\mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z261}}$ $\mathbb{Q}_4^{(c)}(A_{2g},d) = \frac{3\sqrt{1705}\mathbb{Q}_6^{(a)}(B_{2g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{124} + \frac{\sqrt{31}\mathbb{Q}_6^{(a)}(B_{2g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{124}$

$\boxed{\text{z262}}$ $\mathbb{Q}_6^{(c)}(A_{2g},a) = \mathbb{Q}_6^{(a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z263}}$ $\mathbb{Q}_6^{(c)}(A_{2g},b) = \frac{\sqrt{31}\mathbb{Q}_6^{(a)}(B_{2g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{124} - \frac{3\sqrt{1705}\mathbb{Q}_6^{(a)}(B_{2g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{124}$

$\boxed{\text{z264}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{2g},a) = \mathbb{Q}_2^{(1,-1;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z265}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{2g},b) = \mathbb{Q}_4^{(1,-1;a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z266}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{2g},c) = -\mathbb{Q}_4^{(1,-1;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z267}}$ $\mathbb{Q}_4^{(1,-1;c)}(A_{2g},d) = \frac{3\sqrt{1705}\mathbb{Q}_6^{(1,-1;a)}(B_{2g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{124} + \frac{\sqrt{31}\mathbb{Q}_6^{(1,-1;a)}(B_{2g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{124}$

$\boxed{\text{z268}}$ $\mathbb{Q}_6^{(1,-1;c)}(A_{2g},a) = \mathbb{Q}_6^{(1,-1;a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z269}}$ $\mathbb{Q}_6^{(1,-1;c)}(A_{2g},b) = \frac{\sqrt{31}\mathbb{Q}_6^{(1,-1;a)}(B_{2g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{124} - \frac{3\sqrt{1705}\mathbb{Q}_6^{(1,-1;a)}(B_{2g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{124}$

$\boxed{\text{z270}}$ $\mathbb{Q}_4^{(1,0;c)}(A_{2g},a) = -\mathbb{G}_5^{(1,0;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z271}}$ $\mathbb{Q}_4^{(1,0;c)}(A_{2g},b) = \mathbb{G}_3^{(1,0;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z272}}$ $\mathbb{Q}_4^{(1,1;c)}(A_{2g},a) = \mathbb{Q}_2^{(1,1;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z273}}$ $\mathbb{Q}_4^{(1,1;c)}(A_{2g},b) = \mathbb{Q}_4^{(1,1;a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z274}}$ $\mathbb{Q}_4^{(1,1;c)}(A_{2g},c) = -\mathbb{Q}_4^{(1,1;a)}(B_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z275}}$ $\mathbb{G}_1^{(1,0;c)}(A_{2g}) = \mathbb{G}_1^{(1,0;a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z276}}$ $\mathbb{G}_3^{(1,0;c)}(A_{2g}) = \mathbb{G}_3^{(1,0;a)}(A_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z277}}$ $\mathbb{G}_5^{(1,0;c)}(A_{2g},1) = \mathbb{G}_5^{(1,0;a)}(A_{2g},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z278}}$ $\mathbb{G}_5^{(1,0;c)}(A_{2g},2) = \mathbb{G}_5^{(1,0;a)}(A_{2g},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z342}}$ $$\mathbb{Q}_1^{(c)}(A_{2u}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z343}}$ $$\mathbb{Q}_3^{(c)}(A_{2u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z344}}$ $$\mathbb{Q}_5^{(c)}(A_{2u},1) = \frac{\sqrt{30}\mathbb{M}_{5,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{M}_{5,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_{5,1}^{(a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{30}\mathbb{M}_{5,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{M}_{5,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_{5,2}^{(a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z345}}$ $$\begin{aligned}\mathbb{Q}_5^{(c)}(A_{2u},2) = {} & \frac{3\sqrt{10}\mathbb{M}_{5,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{11\sqrt{14}\mathbb{M}_{5,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{112} - \frac{\sqrt{42}\mathbb{M}_{5,1}^{(a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{56} \\ & + \frac{3\sqrt{10}\mathbb{M}_{5,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{11\sqrt{14}\mathbb{M}_{5,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{112} - \frac{\sqrt{42}\mathbb{M}_{5,2}^{(a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{56}\end{aligned}$$

$\boxed{\text{z346}}$ $$\mathbb{Q}_1^{(1,-1;c)}(A_{2u}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z347}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_{2u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z348}}$ $$\begin{aligned}\mathbb{Q}_5^{(1,-1;c)}(A_{2u},1) = {} & \frac{\sqrt{30}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} \\ & + \frac{\sqrt{30}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}\end{aligned}$$

$\boxed{\text{z349}}$ $$\begin{aligned}\mathbb{Q}_5^{(1,-1;c)}(A_{2u},2) = {} & \frac{3\sqrt{10}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{11\sqrt{14}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{112} - \frac{\sqrt{42}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{56} \\ & + \frac{3\sqrt{10}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{11\sqrt{14}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{112} - \frac{\sqrt{42}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{56}\end{aligned}$$

$\boxed{\text{z350}}$ $$\mathbb{Q}_7^{(1,-1;c)}(A_{2u},1) = -\frac{5\sqrt{14}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} - \frac{3\sqrt{66}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} - \frac{\sqrt{429}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} - \frac{15\sqrt{3}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{64}$$
$$- \frac{5\sqrt{14}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} - \frac{3\sqrt{66}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} - \frac{\sqrt{429}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} - \frac{15\sqrt{3}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{64}$$

$\boxed{\text{z351}}$ $$\mathbb{Q}_7^{(1,-1;c)}(A_{2u},2) = -\frac{3\sqrt{2310}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} + \frac{43\sqrt{10}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} + \frac{9\sqrt{65}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} - \frac{11\sqrt{55}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{320}$$
$$- \frac{3\sqrt{2310}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{320} + \frac{43\sqrt{10}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{320} + \frac{9\sqrt{65}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{320} - \frac{11\sqrt{55}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{320}$$

$\boxed{\text{z352}}$ $$\mathbb{Q}_1^{(1,0;c)}(A_{2u}) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z353}}$ $$\mathbb{Q}_3^{(1,0;c)}(A_{2u}) = -\frac{\sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z354}}$ $$\mathbb{Q}_5^{(1,0;c)}(A_{2u},1) = \frac{\sqrt{6}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{3\sqrt{11}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{16}$$
$$+ \frac{\sqrt{6}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{3\sqrt{11}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{16}$$

$\boxed{\text{z355}}$ $$\mathbb{Q}_5^{(1,0;c)}(A_{2u},2a) = \frac{\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{7}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z356}}$ $$\mathbb{Q}_5^{(1,0;c)}(A_{2u},2b) = -\frac{7\sqrt{290}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{232} + \frac{\sqrt{4785}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{464} + \frac{23\sqrt{87}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{464}$$
$$- \frac{7\sqrt{290}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{232} + \frac{\sqrt{4785}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{464} + \frac{23\sqrt{87}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{464}$$

$\boxed{\text{z357}}$ $$\mathbb{Q}_7^{(1,0;c)}(A_{2u},2) = \frac{\sqrt{1914}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{116} - \frac{13\sqrt{29}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{232} + \frac{3\sqrt{1595}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{232}$$
$$+ \frac{\sqrt{1914}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{116} - \frac{13\sqrt{29}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{232} + \frac{3\sqrt{1595}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{232}$$

$\boxed{\text{z358}}$ $$\mathbb{Q}_1^{(1,1;c)}(A_{2u}) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z359}}$ $$\mathbb{Q}_3^{(1,1;c)}(A_{2u}) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z360}}$ $$\mathbb{Q}_5^{(1,1;c)}(A_{2u},1) = \frac{\sqrt{30}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{30}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z361}}$ $$\mathbb{Q}_5^{(1,1;c)}(A_{2u},2) = \frac{3\sqrt{10}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{11\sqrt{14}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{112} - \frac{\sqrt{42}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{56} + \frac{3\sqrt{10}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{11\sqrt{14}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{112} - \frac{\sqrt{42}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{56}$$

$\boxed{\text{z362}}$ $$\mathbb{G}_4^{(c)}(A_{2u},a) = -\frac{\sqrt{2}\mathbb{M}_{5,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{3\sqrt{70}\mathbb{M}_{5,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{56} + \frac{\sqrt{210}\mathbb{M}_{5,1}^{(a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{28} - \frac{\sqrt{2}\mathbb{M}_{5,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{3\sqrt{70}\mathbb{M}_{5,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{56} + \frac{\sqrt{210}\mathbb{M}_{5,2}^{(a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{28}$$

$\boxed{\text{z605}}$ $$\mathbb{G}_4^{(c)}(A_{2u},b) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z606}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_{2u},a) = -\frac{\sqrt{2}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{3\sqrt{70}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{56} + \frac{\sqrt{210}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{28} - \frac{\sqrt{2}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{3\sqrt{70}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{56} + \frac{\sqrt{210}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{28}$$

$\boxed{\text{z607}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_{2u},b) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z608}}$ $$\mathbb{G}_6^{(1,-1;c)}(A_{2u}) = \frac{\sqrt{210}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} - \frac{11\sqrt{110}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} + \frac{7\sqrt{715}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} - \frac{23\sqrt{5}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} + \frac{\sqrt{210}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{320} - \frac{11\sqrt{110}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{320} + \frac{7\sqrt{715}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{320} - \frac{23\sqrt{5}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{320}$$

$\boxed{\text{z609}}$ $$\mathbb{G}_8^{(1,-1;c)}(A_{2u},1) = \frac{\sqrt{858}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} + \frac{\sqrt{182}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} - \frac{\sqrt{7}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} - \frac{\sqrt{1001}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} + \frac{\sqrt{858}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} + \frac{\sqrt{182}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} - \frac{\sqrt{7}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} - \frac{\sqrt{1001}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{64}$$

$\boxed{\text{z610}}$ $$\mathbb{G}_4^{(1,1;c)}(A_{2u},a) = -\frac{\sqrt{2}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{3\sqrt{70}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{56} + \frac{\sqrt{210}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{28} - \frac{\sqrt{2}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{3\sqrt{70}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{56} + \frac{\sqrt{210}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{28}$$

$\boxed{\text{z611}}$ $\mathbb{G}_4^{(1,1;c)}(A_{2u},b) = \frac{\sqrt{5}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{5}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$

$\boxed{\text{z612}}$ $\mathbb{Q}_2^{(c)}(B_{1g},a) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z613}}$ $\mathbb{Q}_2^{(c)}(B_{1g},b) = \mathbb{Q}_2^{(a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z614}}$ $\mathbb{Q}_2^{(c)}(B_{1g},c) = -\mathbb{Q}_2^{(a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z615}}$ $\mathbb{Q}_2^{(c)}(B_{1g},d) = \frac{\sqrt{21}\mathbb{Q}_4^{(a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{6} - \frac{\sqrt{15}\mathbb{Q}_4^{(a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{6}$

$\boxed{\text{z616}}$ $\mathbb{Q}_4^{(c)}(B_{1g},a) = \mathbb{Q}_4^{(a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z617}}$ $\mathbb{Q}_4^{(c)}(B_{1g},b) = \frac{\sqrt{15}\mathbb{Q}_4^{(a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{6} + \frac{\sqrt{21}\mathbb{Q}_4^{(a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{6}$

$\boxed{\text{z618}}$ $\mathbb{Q}_4^{(c)}(B_{1g},c) = \frac{3\mathbb{Q}_6^{(a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{4} - \frac{\sqrt{7}\mathbb{Q}_6^{(a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{4}$

$\boxed{\text{z619}}$ $\mathbb{Q}_6^{(c)}(B_{1g},1a) = \mathbb{Q}_6^{(a)}(B_{1g},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z620}}$ $\mathbb{Q}_6^{(c)}(B_{1g},1b) = \frac{\sqrt{7}\mathbb{Q}_6^{(a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{4} + \frac{3\mathbb{Q}_6^{(a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{4}$

$\boxed{\text{z621}}$ $\mathbb{Q}_6^{(c)}(B_{1g},2) = \mathbb{Q}_6^{(a)}(B_{1g},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z622}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{1g},a) = \mathbb{Q}_2^{(1,-1;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z623}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{1g},b) = -\mathbb{Q}_2^{(1,-1;a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z624}}$ $\mathbb{Q}_2^{(1,-1;c)}(B_{1g},c) = \frac{\sqrt{21}\mathbb{Q}_4^{(1,-1;a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{6} - \frac{\sqrt{15}\mathbb{Q}_4^{(1,-1;a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{6}$

$\boxed{\text{z625}}$ $\mathbb{Q}_4^{(1,-1;c)}(B_{1g},a) = \mathbb{Q}_4^{(1,-1;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z626}}$ $\mathbb{Q}_4^{(1,-1;c)}(B_{1g},b) = \frac{\sqrt{15}\mathbb{Q}_4^{(1,-1;a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{6} + \frac{\sqrt{21}\mathbb{Q}_4^{(1,-1;a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{6}$

$\boxed{\text{z627}}$ $\mathbb{Q}_4^{(1,-1;c)}(B_{1g},c) = \frac{3\mathbb{Q}_6^{(1,-1;a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{4} - \frac{\sqrt{7}\mathbb{Q}_6^{(1,-1;a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{4}$

$\boxed{\text{z628}}$ $\mathbb{Q}_6^{(1,-1;c)}(B_{1g},1a) = \mathbb{Q}_6^{(1,-1;a)}(B_{1g},1)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z629}}$ $\mathbb{Q}_6^{(1,-1;c)}(B_{1g},1b) = \frac{\sqrt{7}\mathbb{Q}_6^{(1,-1;a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{4} + \frac{3\mathbb{Q}_6^{(1,-1;a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{4}$

$\boxed{\text{z630}}$ $\mathbb{Q}_6^{(1,-1;c)}(B_{1g},2) = \mathbb{Q}_6^{(1,-1;a)}(B_{1g},2)\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z631}}$ $\mathbb{Q}_4^{(1,0;c)}(B_{1g}) = \mathbb{G}_5^{(1,0;a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z632}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{1g},a) = \mathbb{Q}_0^{(1,1;a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z633}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{1g},b) = \mathbb{Q}_2^{(1,1;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z634}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{1g},c) = -\mathbb{Q}_2^{(1,1;a)}(A_{1g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z635}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{1g},d) = \frac{\sqrt{21}\mathbb{Q}_4^{(1,1;a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{6} - \frac{\sqrt{15}\mathbb{Q}_4^{(1,1;a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{6}$

$\boxed{\text{z636}}$ $\mathbb{Q}_4^{(1,1;c)}(B_{1g},a) = \mathbb{Q}_4^{(1,1;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z637}}$ $\mathbb{Q}_4^{(1,1;c)}(B_{1g},b) = \frac{\sqrt{15}\mathbb{Q}_4^{(1,1;a)}(A_{1g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{6} + \frac{\sqrt{21}\mathbb{Q}_4^{(1,1;a)}(A_{1g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{6}$

$\boxed{\text{z638}}$ $\mathbb{G}_3^{(1,0;c)}(B_{1g}) = \mathbb{G}_3^{(1,0;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z639}}$ $\mathbb{G}_5^{(1,0;c)}(B_{1g}) = \mathbb{G}_5^{(1,0;a)}(B_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z640}}$ $\mathbb{Q}_3^{(c)}(B_{1u}) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z641}}$ $\mathbb{Q}_5^{(c)}(B_{1u}) = -\frac{\sqrt{78}\mathbb{M}_{5,1}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{13} - \frac{\sqrt{26}\mathbb{M}_{5,1}^{(a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{26} - \frac{\sqrt{78}\mathbb{M}_{5,2}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{13} - \frac{\sqrt{26}\mathbb{M}_{5,2}^{(a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{26}$

$\boxed{\text{z642}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{1u}) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$

$\boxed{\text{z643}}$ $$\mathbb{Q}_5^{(1,-1;c)}(B_{1u}) = -\frac{\sqrt{78}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{13} - \frac{\sqrt{26}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{26} - \frac{\sqrt{78}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{13} - \frac{\sqrt{26}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{26}$$

$\boxed{\text{z644}}$ $$\mathbb{Q}_7^{(1,-1;c)}(B_{1u},1) = -\frac{3\sqrt{154}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{56} - \frac{\sqrt{182}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{56} - \frac{3\sqrt{154}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{56} - \frac{\sqrt{182}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{56}$$

$\boxed{\text{z645}}$ $$\begin{aligned}\mathbb{Q}_7^{(1,-1;c)}(B_{1u},2) = &-\frac{2\sqrt{455}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{65} - \frac{3\sqrt{70}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{280} + \frac{9\sqrt{10010}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{3640}\\ &- \frac{2\sqrt{455}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{65} - \frac{3\sqrt{70}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{280} + \frac{9\sqrt{10010}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{3640}\end{aligned}$$

$\boxed{\text{z646}}$ $$\mathbb{Q}_3^{(1,0;c)}(B_{1u},a) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z647}}$ $$\mathbb{Q}_3^{(1,0;c)}(B_{1u},b) = \frac{\sqrt{2}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z648}}$ $$\mathbb{Q}_5^{(1,0;c)}(B_{1u},a) = \frac{\sqrt{2}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z649}}$ $$\mathbb{Q}_5^{(1,0;c)}(B_{1u},b) = \frac{\sqrt{690}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{46} + \frac{2\sqrt{23}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{23} + \frac{\sqrt{690}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{46} + \frac{2\sqrt{23}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{23}$$

$\boxed{\text{z650}}$ $$\begin{aligned}\mathbb{Q}_7^{(1,0;c)}(B_{1u},1) = &-\frac{15\sqrt{4002}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2668} + \frac{\sqrt{7337}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{232} + \frac{45\sqrt{3335}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{5336}\\ &- \frac{15\sqrt{4002}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2668} + \frac{\sqrt{7337}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{232} + \frac{45\sqrt{3335}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{5336}\end{aligned}$$

$\boxed{\text{z651}}$ $$\begin{aligned}\mathbb{Q}_7^{(1,0;c)}(B_{1u},2) = &\frac{\sqrt{638}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{116} + \frac{15\sqrt{87}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{232} - \frac{\sqrt{4785}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{232}\\ &+ \frac{\sqrt{638}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{116} + \frac{15\sqrt{87}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{232} - \frac{\sqrt{4785}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{232}\end{aligned}$$

$\boxed{\text{z652}}$ $$\mathbb{Q}_3^{(1,1;c)}(B_{1u}) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z653}}$ $$\mathbb{Q}_5^{(1,1;c)}(B_{1u}) = -\frac{\sqrt{78}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{13} - \frac{\sqrt{26}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{26} - \frac{\sqrt{78}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{13} - \frac{\sqrt{26}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{26}$$

$\boxed{\text{z654}}$ $$\mathbb{G}_2^{(c)}(B_{1u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z655}}$ $$\mathbb{G}_2^{(c)}(B_{1u},b) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z656}}$ $$\mathbb{G}_4^{(c)}(B_{1u}) = \frac{\sqrt{65}\mathbb{M}_{5,1}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{3\sqrt{91}\mathbb{M}_{5,1}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{3\sqrt{273}\mathbb{M}_{5,1}^{(a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{104} + \frac{\sqrt{65}\mathbb{M}_{5,2}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{3\sqrt{91}\mathbb{M}_{5,2}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} + \frac{3\sqrt{273}\mathbb{M}_{5,2}^{(a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{104}$$

$\boxed{\text{z657}}$ $$\mathbb{G}_6^{(c)}(B_{1u},1) = \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{M}_{5,1}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{M}_{5,1}^{(a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{3\sqrt{7}\mathbb{M}_{5,2}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{M}_{5,2}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{M}_{5,2}^{(a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$\boxed{\text{z658}}$ $$\mathbb{G}_2^{(1,-1;c)}(B_{1u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z659}}$ $$\mathbb{G}_2^{(1,-1;c)}(B_{1u},b) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z660}}$ $$\mathbb{G}_4^{(1,-1;c)}(B_{1u}) = \frac{\sqrt{65}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{3\sqrt{91}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{3\sqrt{273}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{104} + \frac{\sqrt{65}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{3\sqrt{91}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} + \frac{3\sqrt{273}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{104}$$

$\boxed{\text{z661}}$ $$\mathbb{G}_6^{(1,-1;c)}(B_{1u},1a) = \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{3\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$\boxed{\text{z662}}$ $$\mathbb{G}_6^{(1,-1;c)}(B_{1u},1b) = -\frac{3\sqrt{130}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{130} + \frac{\sqrt{5}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{10} - \frac{3\sqrt{715}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{130} - \frac{3\sqrt{130}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{130} + \frac{\sqrt{5}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{10} - \frac{3\sqrt{715}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{130}$$

$\boxed{\text{z663}}$ $$\mathbb{G}_6^{(1,-1;c)}(B_{1u},2) = \frac{\sqrt{2}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z664}}$ $$\mathbb{G}_2^{(1,1;c)}(B_{1u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z665}}$ $$\mathbb{G}_2^{(1,1;c)}(B_{1u},b) = \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z666}}$ $$\mathbb{G}_4^{(1,1;c)}(B_{1u}) = \frac{\sqrt{65}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{3\sqrt{91}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{3\sqrt{273}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{104}$$
$$+ \frac{\sqrt{65}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{3\sqrt{91}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} + \frac{3\sqrt{273}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{104}$$

$\boxed{\text{z667}}$ $$\mathbb{G}_6^{(1,1;c)}(B_{1u},1) = \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$
$$+ \frac{3\sqrt{7}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$\boxed{\text{z668}}$ $$\mathbb{Q}_2^{(c)}(B_{2g},a) = \mathbb{Q}_2^{(a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$\boxed{\text{z669}}$ $$\mathbb{Q}_2^{(c)}(B_{2g},b) = \mathbb{Q}_4^{(a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$\boxed{\text{z670}}$ $$\mathbb{Q}_4^{(c)}(B_{2g},a) = \mathbb{Q}_4^{(a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$\boxed{\text{z671}}$ $$\mathbb{Q}_4^{(c)}(B_{2g},b) = \mathbb{Q}_6^{(a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$\boxed{\text{z672}}$ $$\mathbb{Q}_6^{(c)}(B_{2g},1) = \mathbb{Q}_6^{(a)}(B_{2g},1)\mathbb{Q}_0^{(b)}(A_{1g})$$

$\boxed{\text{z673}}$ $$\mathbb{Q}_6^{(c)}(B_{2g},2) = \mathbb{Q}_6^{(a)}(B_{2g},2)\mathbb{Q}_0^{(b)}(A_{1g})$$

$\boxed{\text{z674}}$ $$\mathbb{Q}_2^{(1,-1;c)}(B_{2g},a) = \mathbb{Q}_2^{(1,-1;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$\boxed{\text{z675}}$ $$\mathbb{Q}_2^{(1,-1;c)}(B_{2g},b) = \mathbb{Q}_4^{(1,-1;a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$\boxed{\text{z676}}$ $$\mathbb{Q}_4^{(1,-1;c)}(B_{2g},a) = \mathbb{Q}_4^{(1,-1;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$\boxed{\text{z677}}$ $$\mathbb{Q}_4^{(1,-1;c)}(B_{2g},b) = \mathbb{Q}_6^{(1,-1;a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$$

$\boxed{\text{z678}}$ $$\mathbb{Q}_6^{(1,-1;c)}(B_{2g},1) = \mathbb{Q}_6^{(1,-1;a)}(B_{2g},1)\mathbb{Q}_0^{(b)}(A_{1g})$$

$\boxed{\text{z679}}$ $$\mathbb{Q}_6^{(1,-1;c)}(B_{2g},2) = \mathbb{Q}_6^{(1,-1;a)}(B_{2g},2)\mathbb{Q}_0^{(b)}(A_{1g})$$

$\boxed{\text{z680}}$ $\mathbb{Q}_2^{(1,0;c)}(B_{2g},a) = \mathbb{G}_1^{(1,0;a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z681}}$ $\mathbb{Q}_2^{(1,0;c)}(B_{2g},b) = -\mathbb{G}_3^{(1,0;a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z682}}$ $\mathbb{Q}_4^{(1,0;c)}(B_{2g}) = -\frac{\sqrt{15}\mathbb{G}_5^{(1,0;a)}(A_{2g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{6} + \frac{\sqrt{21}\mathbb{G}_5^{(1,0;a)}(A_{2g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{6}$

$\boxed{\text{z683}}$ $\mathbb{Q}_6^{(1,0;c)}(B_{2g},1) = \frac{\sqrt{21}\mathbb{G}_5^{(1,0;a)}(A_{2g},1)\mathbb{Q}_2^{(b)}(B_{1g})}{6} + \frac{\sqrt{15}\mathbb{G}_5^{(1,0;a)}(A_{2g},2)\mathbb{Q}_2^{(b)}(B_{1g})}{6}$

$\boxed{\text{z684}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{2g},a) = \mathbb{Q}_2^{(1,1;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z685}}$ $\mathbb{Q}_2^{(1,1;c)}(B_{2g},b) = \mathbb{Q}_4^{(1,1;a)}(A_{2g})\mathbb{Q}_2^{(b)}(B_{1g})$

$\boxed{\text{z686}}$ $\mathbb{Q}_4^{(1,1;c)}(B_{2g}) = \mathbb{Q}_4^{(1,1;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z687}}$ $\mathbb{G}_3^{(1,0;c)}(B_{2g}) = \mathbb{G}_3^{(1,0;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z688}}$ $\mathbb{G}_5^{(1,0;c)}(B_{2g}) = \mathbb{G}_5^{(1,0;a)}(B_{2g})\mathbb{Q}_0^{(b)}(A_{1g})$

$\boxed{\text{z757}}$ $\mathbb{Q}_3^{(c)}(B_{2u}) = \frac{\sqrt{30}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$

$\boxed{\text{z758}}$ $\mathbb{Q}_5^{(c)}(B_{2u}) = -\frac{\sqrt{2730}\mathbb{M}_{5,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} + \frac{\sqrt{78}\mathbb{M}_{5,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} - \frac{5\sqrt{26}\mathbb{M}_{5,1}^{(a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{52}$
$+ \frac{\sqrt{2730}\mathbb{M}_{5,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{104} - \frac{\sqrt{78}\mathbb{M}_{5,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{104} + \frac{5\sqrt{26}\mathbb{M}_{5,2}^{(a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{52}$

$\boxed{\text{z759}}$ $\mathbb{Q}_3^{(1,-1;c)}(B_{2u}) = \frac{\sqrt{30}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$

$\boxed{\text{z760}}$ $\mathbb{Q}_5^{(1,-1;c)}(B_{2u}) = -\frac{\sqrt{2730}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} + \frac{\sqrt{78}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} - \frac{5\sqrt{26}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{52}$
$+ \frac{\sqrt{2730}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{104} - \frac{\sqrt{78}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{104} + \frac{5\sqrt{26}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{52}$

$\boxed{\text{z761}}$ $\mathbb{Q}_7^{(1,-1;c)}(B_{2u},1) = \frac{\sqrt{30030}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} - \frac{9\sqrt{130}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} + \frac{29\sqrt{5}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} - \frac{3\sqrt{715}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{320}$
$- \frac{\sqrt{30030}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{320} + \frac{9\sqrt{130}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{320} - \frac{29\sqrt{5}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{320} + \frac{3\sqrt{715}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{320}$

$\boxed{\text{z762}}$ $$\mathbb{Q}_7^{(1,-1;c)}(B_{2u},2) = \frac{15\sqrt{546}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{832} + \frac{11\sqrt{286}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{832} - \frac{3\sqrt{11}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} + \frac{115\sqrt{13}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{832}$$
$$- \frac{15\sqrt{546}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{832} - \frac{11\sqrt{286}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{832} + \frac{3\sqrt{11}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} - \frac{115\sqrt{13}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{832}$$

$\boxed{\text{z763}}$ $$\mathbb{Q}_3^{(1,0;c)}(B_{2u},a) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z764}}$ $$\mathbb{Q}_3^{(1,0;c)}(B_{2u},b) = -\frac{\sqrt{14}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{3\sqrt{2}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{14}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{3\sqrt{2}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z765}}$ $$\mathbb{Q}_5^{(1,0;c)}(B_{2u},a) = \frac{3\sqrt{2}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{14}\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{3\sqrt{2}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{14}\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z766}}$ $$\mathbb{Q}_5^{(1,0;c)}(B_{2u},b) = -\frac{\sqrt{690}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{92} + \frac{3\sqrt{1265}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{184} - \frac{11\sqrt{23}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{184}$$
$$+ \frac{\sqrt{690}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{92} - \frac{3\sqrt{1265}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{184} + \frac{11\sqrt{23}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{184}$$

$\boxed{\text{z767}}$ $$\mathbb{Q}_7^{(1,0;c)}(B_{2u},1) = \frac{\sqrt{11}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{\sqrt{6}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{32} - \frac{\sqrt{330}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{32}$$
$$- \frac{\sqrt{11}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} + \frac{\sqrt{6}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{32} + \frac{\sqrt{330}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{32}$$

$\boxed{\text{z768}}$ $$\mathbb{Q}_7^{(1,0;c)}(B_{2u},2) = -\frac{11\sqrt{69}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{184} - \frac{13\sqrt{506}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{736} - \frac{15\sqrt{230}\mathbb{T}_{6,1}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{736}$$
$$+ \frac{11\sqrt{69}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{184} + \frac{13\sqrt{506}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{736} + \frac{15\sqrt{230}\mathbb{T}_{6,2}^{(1,0;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{736}$$

$\boxed{\text{z769}}$ $$\mathbb{Q}_3^{(1,1;c)}(B_{2u}) = \frac{\sqrt{30}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z770}}$ $$\mathbb{Q}_5^{(1,1;c)}(B_{2u}) = -\frac{\sqrt{2730}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} + \frac{\sqrt{78}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} - \frac{5\sqrt{26}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{52}$$
$$+ \frac{\sqrt{2730}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{104} - \frac{\sqrt{78}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{104} + \frac{5\sqrt{26}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{52}$$

$\boxed{\text{z771}}$ $$\mathbb{G}_2^{(c)}(B_{2u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z772}}$ $$\mathbb{G}_2^{(c)}(B_{2u},b) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{30}\mathbb{M}_{3,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{M}_{3,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z773}}$ $$\mathbb{G}_4^{(c)}(B_{2u}) = \frac{\sqrt{65}\mathbb{M}_{5,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} - \frac{15\sqrt{91}\mathbb{M}_{5,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} - \frac{\sqrt{273}\mathbb{M}_{5,1}^{(a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} - \frac{\sqrt{65}\mathbb{M}_{5,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{15\sqrt{91}\mathbb{M}_{5,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{\sqrt{273}\mathbb{M}_{5,2}^{(a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{104}$$

$\boxed{\text{z774}}$ $$\mathbb{G}_6^{(c)}(B_{2u},1) = \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{M}_{5,1}^{(a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{M}_{5,1}^{(a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{3\sqrt{7}\mathbb{M}_{5,2}^{(a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{5}\mathbb{M}_{5,2}^{(a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{15}\mathbb{M}_{5,2}^{(a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z775}}$ $$\mathbb{G}_2^{(1,-1;c)}(B_{2u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,-1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,-1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z776}}$ $$\mathbb{G}_2^{(1,-1;c)}(B_{2u},b) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{30}\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z777}}$ $$\mathbb{G}_4^{(1,-1;c)}(B_{2u}) = \frac{\sqrt{65}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} - \frac{15\sqrt{91}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} - \frac{\sqrt{273}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} - \frac{\sqrt{65}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{15\sqrt{91}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{\sqrt{273}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{104}$$

$\boxed{\text{z778}}$ $$\mathbb{G}_6^{(1,-1;c)}(B_{2u},1a) = \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{3\sqrt{7}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{5}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{15}\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z779}}$ $$\mathbb{G}_6^{(1,-1;c)}(B_{2u},1b) = -\frac{3\sqrt{330}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} - \frac{19\sqrt{70}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} + \frac{3\sqrt{455}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} + \frac{7\sqrt{385}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{320} + \frac{3\sqrt{330}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{320} + \frac{19\sqrt{70}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{320} - \frac{3\sqrt{455}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{320} - \frac{7\sqrt{385}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{320}$$

$\boxed{\text{z780}}$ $$\mathbb{G}_6^{(1,-1;c)}(B_{2u},2) = -\frac{\sqrt{182}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{832} + \frac{9\sqrt{858}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{832} + \frac{7\sqrt{33}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} + \frac{9\sqrt{39}\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,2}^{(b)}(E_u)}{832} + \frac{\sqrt{182}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{832} - \frac{9\sqrt{858}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{832} - \frac{7\sqrt{33}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} - \frac{9\sqrt{39}\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4)\mathbb{T}_{1,1}^{(b)}(E_u)}{832}$$

$\boxed{\text{z781}}$ $$\mathbb{G}_2^{(1,1;c)}(B_{2u},a) = \frac{\sqrt{2}\mathbb{M}_{1,1}^{(1,1;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(E_u)}{2} - \frac{\sqrt{2}\mathbb{M}_{1,2}^{(1,1;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z782}}$ $$\mathbb{G}_2^{(1,1;c)}(B_{2u},b) = -\frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{30}\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{30}\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z783}}$ $$\mathbb{G}_4^{(1,1;c)}(B_{2u}) = \frac{\sqrt{65}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} - \frac{15\sqrt{91}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{208} - \frac{\sqrt{273}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{104} - \frac{\sqrt{65}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{15\sqrt{91}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{208} + \frac{\sqrt{273}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{104}$$

$\boxed{\text{z784}}$ $$\mathbb{G}_6^{(1,1;c)}(B_{2u},1) = \frac{3\sqrt{7}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{5}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{15}\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,2}^{(b)}(E_u)}{8} - \frac{3\sqrt{7}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} - \frac{\sqrt{5}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{15}\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z853}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z854}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z855}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,b) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$\boxed{\text{z856}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$$

$\boxed{\text{z857}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,c) = -\frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_g,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} + \frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$\boxed{\text{z858}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,c) = \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_g,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} - \frac{3\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$\boxed{\text{z859}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z860}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z861}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_g,1b) = -\frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4,1}^{(a)}(E_g,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$$

$\boxed{\text{z862}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,1b) = \frac{3\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z863}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,1c) = \frac{3\sqrt{11490}\mathbb{Q}_{6,1}^{(a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{766} - \frac{3\sqrt{21065}\mathbb{Q}_{6,1}^{(a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{766} - \frac{\sqrt{383}\mathbb{Q}_{6,1}^{(a)}(E_g,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{766}$

$\boxed{\text{z864}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,1c) = -\frac{3\sqrt{11490}\mathbb{Q}_{6,2}^{(a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{766} + \frac{3\sqrt{21065}\mathbb{Q}_{6,2}^{(a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{766} + \frac{\sqrt{383}\mathbb{Q}_{6,2}^{(a)}(E_g,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{766}$

$\boxed{\text{z865}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,2a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z866}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,2a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z867}}$ $\mathbb{Q}_{4,1}^{(c)}(E_g,2b) = -\frac{115\sqrt{383}\mathbb{Q}_{6,1}^{(a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{12256} - \frac{51\sqrt{25278}\mathbb{Q}_{6,1}^{(a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{49024} + \frac{303\sqrt{11490}\mathbb{Q}_{6,1}^{(a)}(E_g,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{49024}$

$\boxed{\text{z868}}$ $\mathbb{Q}_{4,2}^{(c)}(E_g,2b) = \frac{115\sqrt{383}\mathbb{Q}_{6,2}^{(a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{12256} + \frac{51\sqrt{25278}\mathbb{Q}_{6,2}^{(a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{49024} - \frac{303\sqrt{11490}\mathbb{Q}_{6,2}^{(a)}(E_g,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{49024}$

$\boxed{\text{z869}}$ $\mathbb{Q}_{6,1}^{(c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z870}}$ $\mathbb{Q}_{6,2}^{(c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z871}}$ $\mathbb{Q}_{6,1}^{(c)}(E_g,1b) = \frac{3\sqrt{33}\mathbb{Q}_{6,1}^{(a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{32} + \frac{35\sqrt{2}\mathbb{Q}_{6,1}^{(a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{128} + \frac{3\sqrt{110}\mathbb{Q}_{6,1}^{(a)}(E_g,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{128}$

$\boxed{\text{z872}}$ $\mathbb{Q}_{6,2}^{(c)}(E_g,1b) = -\frac{3\sqrt{33}\mathbb{Q}_{6,2}^{(a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{32} - \frac{35\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{128} - \frac{3\sqrt{110}\mathbb{Q}_{6,2}^{(a)}(E_g,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{128}$

$\boxed{\text{z873}}$ $\mathbb{Q}_{6,1}^{(c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z874}}$ $\mathbb{Q}_{6,2}^{(c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z875}}$ $\mathbb{Q}_{6,1}^{(c)}(E_g,3) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(a)}(E_g,3)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z876}}$ $\mathbb{Q}_{6,2}^{(c)}(E_g,3) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(a)}(E_g,3)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z877}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z878}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z879}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,b) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z880}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z948}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,c) = -\frac{\sqrt{14}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} + \frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z949}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,c) = \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{3\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z950}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z951}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z952}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,1b) = -\frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z953}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,1b) = \frac{3\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z954}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,1c) = \frac{3\sqrt{11490}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{766} - \frac{3\sqrt{21065}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{766} - \frac{\sqrt{383}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{766}$

$\boxed{\text{z955}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,1c) = -\frac{3\sqrt{11490}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{766} + \frac{3\sqrt{21065}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{766} + \frac{\sqrt{383}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{766}$

$\boxed{\text{z956}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,2a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z957}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,2a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z958}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,2b) = -\frac{115\sqrt{383}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{12256} - \frac{51\sqrt{25278}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{49024} + \frac{303\sqrt{11490}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{49024}$

$\boxed{\text{z959}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,2b) = \frac{115\sqrt{383}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{12256} + \frac{51\sqrt{25278}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{49024} - \frac{303\sqrt{11490}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{49024}$

$\boxed{\text{z960}}$ $\mathbb{Q}_{6,1}^{(1,-1;c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z961}}$ $\mathbb{Q}_{6,2}^{(1,-1;c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z962}}$ $\mathbb{Q}_{6,1}^{(1,-1;c)}(E_g,1b) = \frac{3\sqrt{33}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{32} + \frac{35\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{128} + \frac{3\sqrt{110}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{128}$

$\boxed{\text{z963}}$ $\mathbb{Q}_{6,2}^{(1,-1;c)}(E_g,1b) = -\frac{3\sqrt{33}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{32} - \frac{35\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{128} - \frac{3\sqrt{110}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,3)\mathbb{Q}_{2}^{(b)}(B_{1g})}{128}$

$\boxed{\text{z964}}$ $\mathbb{Q}_{6,1}^{(1,-1;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z965}}$ $\mathbb{Q}_{6,2}^{(1,-1;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z966}}$ $\mathbb{Q}_{6,1}^{(1,-1;c)}(E_g,3) = \frac{\sqrt{2}\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,3)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z967}}$ $\mathbb{Q}_{6,2}^{(1,-1;c)}(E_g,3) = \frac{\sqrt{2}\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,3)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z968}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,a) = -\frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z969}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_{2}^{(b)}(B_{1g})}{2}$

$\boxed{\text{z970}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,b) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{\sqrt{30}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z971}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,b) = -\frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} + \frac{\sqrt{30}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$

$\boxed{\text{z972}}$ $\mathbb{Q}_{4,1}^{(1,0;c)}(E_g,1a) = -\frac{3\sqrt{654}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,2)\mathbb{Q}_2^{(b)}(B_{1g})}{109} + \frac{\sqrt{218}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,3)\mathbb{Q}_2^{(b)}(B_{1g})}{218}$

$\boxed{\text{z973}}$ $\mathbb{Q}_{4,2}^{(1,0;c)}(E_g,1a) = \frac{3\sqrt{654}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,2)\mathbb{Q}_2^{(b)}(B_{1g})}{109} - \frac{\sqrt{218}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,3)\mathbb{Q}_2^{(b)}(B_{1g})}{218}$

$\boxed{\text{z974}}$ $\mathbb{Q}_{4,1}^{(1,0;c)}(E_g,1b) = -\frac{\sqrt{30}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} - \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$

$\boxed{\text{z975}}$ $\mathbb{Q}_{4,2}^{(1,0;c)}(E_g,1b) = \frac{\sqrt{30}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} + \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1043}}$ $\mathbb{Q}_{4,1}^{(1,0;c)}(E_g,2) = \frac{\sqrt{3270}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,1)\mathbb{Q}_2^{(b)}(B_{1g})}{192} - \frac{19\sqrt{4578}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,2)\mathbb{Q}_2^{(b)}(B_{1g})}{20928} - \frac{57\sqrt{1526}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,3)\mathbb{Q}_2^{(b)}(B_{1g})}{3488}$

$\boxed{\text{z1044}}$ $\mathbb{Q}_{4,2}^{(1,0;c)}(E_g,2) = -\frac{\sqrt{3270}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,1)\mathbb{Q}_2^{(b)}(B_{1g})}{192} + \frac{19\sqrt{4578}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,2)\mathbb{Q}_2^{(b)}(B_{1g})}{20928} + \frac{57\sqrt{1526}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,3)\mathbb{Q}_2^{(b)}(B_{1g})}{3488}$

$\boxed{\text{z1045}}$ $\mathbb{Q}_{6,1}^{(1,0;c)}(E_g,1) = -\frac{19\sqrt{42}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,1)\mathbb{Q}_2^{(b)}(B_{1g})}{192} - \frac{\sqrt{30}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,2)\mathbb{Q}_2^{(b)}(B_{1g})}{192} - \frac{3\sqrt{10}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,3)\mathbb{Q}_2^{(b)}(B_{1g})}{32}$

$\boxed{\text{z1046}}$ $\mathbb{Q}_{6,2}^{(1,0;c)}(E_g,1) = \frac{19\sqrt{42}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,1)\mathbb{Q}_2^{(b)}(B_{1g})}{192} + \frac{\sqrt{30}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,2)\mathbb{Q}_2^{(b)}(B_{1g})}{192} + \frac{3\sqrt{10}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,3)\mathbb{Q}_2^{(b)}(B_{1g})}{32}$

$\boxed{\text{z1047}}$ $\mathbb{Q}_{2,1}^{(1,1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1048}}$ $\mathbb{Q}_{2,2}^{(1,1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1049}}$ $\mathbb{Q}_{2,1}^{(1,1;c)}(E_g,b) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,1;a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1050}}$ $\mathbb{Q}_{2,2}^{(1,1;c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,1;a)}(E_g)\mathbb{Q}_2^{(b)}(B_{1g})}{2}$

$\boxed{\text{z1051}}$ $\mathbb{Q}_{2,1}^{(1,1;c)}(E_g,c) = -\frac{\sqrt{14}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,1)\mathbb{Q}_2^{(b)}(B_{1g})}{8} + \frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,2)\mathbb{Q}_2^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1052}}$ $\mathbb{Q}_{2,2}^{(1,1;c)}(E_g,c) = \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{3\sqrt{2}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1053}}$ $\mathbb{Q}_{4,1}^{(1,1;c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1054}}$ $\mathbb{Q}_{4,2}^{(1,1;c)}(E_g,1a) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1055}}$ $\mathbb{Q}_{4,1}^{(1,1;c)}(E_g,1b) = -\frac{3\sqrt{2}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} - \frac{\sqrt{14}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1056}}$ $\mathbb{Q}_{4,2}^{(1,1;c)}(E_g,1b) = \frac{3\sqrt{2}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,1)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8} + \frac{\sqrt{14}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,2)\mathbb{Q}_{2}^{(b)}(B_{1g})}{8}$

$\boxed{\text{z1057}}$ $\mathbb{Q}_{4,1}^{(1,1;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1058}}$ $\mathbb{Q}_{4,2}^{(1,1;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1059}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1060}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(E_g) = \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1061}}$ $\mathbb{G}_{3,1}^{(1,0;c)}(E_g,1) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1062}}$ $\mathbb{G}_{3,2}^{(1,0;c)}(E_g,1) = \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1063}}$ $\mathbb{G}_{3,1}^{(1,0;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1064}}$ $\mathbb{G}_{3,2}^{(1,0;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1065}}$ $\mathbb{G}_{5,1}^{(1,0;c)}(E_g,1) = \frac{\sqrt{2}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,1)\mathbb{Q}_{0}^{(b)}(A_{1g})}{2}$

$\boxed{\text{z1066}}$ $$\mathbb{G}_{5,2}^{(1,0;c)}(E_g,1) = \frac{\sqrt{2}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,1)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z1067}}$ $$\mathbb{G}_{5,1}^{(1,0;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z1068}}$ $$\mathbb{G}_{5,2}^{(1,0;c)}(E_g,2) = \frac{\sqrt{2}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,2)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z1069}}$ $$\mathbb{G}_{5,1}^{(1,0;c)}(E_g,3) = \frac{\sqrt{2}\mathbb{G}_{5,1}^{(1,0;a)}(E_g,3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z1070}}$ $$\mathbb{G}_{5,2}^{(1,0;c)}(E_g,3) = \frac{\sqrt{2}\mathbb{G}_{5,2}^{(1,0;a)}(E_g,3)\mathbb{Q}_0^{(b)}(A_{1g})}{2}$$

$\boxed{\text{z1331}}$ $$\mathbb{Q}_{1,1}^{(c)}(E_u) = -\frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1332}}$ $$\mathbb{Q}_{1,2}^{(c)}(E_u) = \frac{\sqrt{2}\mathbb{M}_1^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1333}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_u,1) = \frac{\sqrt{3}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z1334}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_u,1) = -\frac{\sqrt{3}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z1335}}$ $$\mathbb{Q}_{3,1}^{(c)}(E_u,2) = \frac{\sqrt{165}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{44} + \frac{2\sqrt{11}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{11} + \frac{3\sqrt{11}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{44}$$

$\boxed{\text{z1336}}$ $$\mathbb{Q}_{3,2}^{(c)}(E_u,2) = -\frac{\sqrt{165}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{44} - \frac{2\sqrt{11}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{11} + \frac{3\sqrt{11}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{44}$$

$\boxed{\text{z1337}}$ $$\mathbb{Q}_{5,1}^{(c)}(E_u,1) = -\frac{\sqrt{30}\mathbb{M}_5^{(a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{42}\mathbb{M}_5^{(a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$\boxed{\text{z1338}}$ $$\mathbb{Q}_{5,2}^{(c)}(E_u,1) = \frac{\sqrt{30}\mathbb{M}_5^{(a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{M}_5^{(a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z1339}}$ $$\mathbb{Q}_{5,1}^{(c)}(E_u,2) = -\frac{\sqrt{46}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{23} - \frac{3\sqrt{1610}\mathbb{M}_5^{(a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{368} + \frac{11\sqrt{46}\mathbb{M}_5^{(a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{368} + \frac{\sqrt{138}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{23} - \frac{\sqrt{138}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{184}$$

$\boxed{\text{z1340}}$ $$\mathbb{Q}_{5,2}^{(c)}(E_u,2) = -\frac{\sqrt{46}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{23} + \frac{3\sqrt{1610}\mathbb{M}_5^{(a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{368} - \frac{11\sqrt{46}\mathbb{M}_5^{(a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{368} - \frac{\sqrt{138}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{23} - \frac{\sqrt{138}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{184}$$

$\boxed{\text{z1341}}$ $$\mathbb{Q}_{5,1}^{(c)}(E_u,3) = \frac{\sqrt{28014}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{322} - \frac{7\sqrt{20010}\mathbb{M}_5^{(a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{5336} + \frac{3\sqrt{28014}\mathbb{M}_5^{(a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{1288} + \frac{5\sqrt{9338}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{9338} + \frac{39\sqrt{9338}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{18676}$$

$\boxed{\text{z1342}}$ $$\mathbb{Q}_{5,2}^{(c)}(E_u,3) = \frac{\sqrt{28014}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{322} + \frac{7\sqrt{20010}\mathbb{M}_5^{(a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{5336} - \frac{3\sqrt{28014}\mathbb{M}_5^{(a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{1288} - \frac{5\sqrt{9338}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{9338} + \frac{39\sqrt{9338}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{18676}$$

$\boxed{\text{z1343}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(E_u) = -\frac{\sqrt{2}\mathbb{M}_1^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1344}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{2}\mathbb{M}_1^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1345}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E_u,1) = \frac{\sqrt{3}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z1346}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E_u,1) = -\frac{\sqrt{3}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{5}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z1347}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(E_u,2) = \frac{\sqrt{165}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{44} + \frac{2\sqrt{11}\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{11} + \frac{3\sqrt{11}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{44}$$

$\boxed{\text{z1348}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(E_u,2) = -\frac{\sqrt{165}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{44} - \frac{2\sqrt{11}\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{11} + \frac{3\sqrt{11}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{44}$$

$\boxed{\text{z1349}}$ $$\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,1) = -\frac{\sqrt{30}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} - \frac{\sqrt{42}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$\boxed{\text{z1350}}$ $$\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,1) = \frac{\sqrt{30}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{42}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16} + \frac{\sqrt{14}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z1351}}$ $$\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,2) = -\frac{\sqrt{46}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{23} - \frac{3\sqrt{1610}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{368} + \frac{11\sqrt{46}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{368} + \frac{\sqrt{138}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{23} - \frac{\sqrt{138}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{184}$$

$\boxed{\text{z1352}}$ $$\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,2) = -\frac{\sqrt{46}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{23} + \frac{3\sqrt{1610}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{368} - \frac{11\sqrt{46}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{368} - \frac{\sqrt{138}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{23} - \frac{\sqrt{138}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{184}$$

$\boxed{\text{z1353}}$ $$\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u,3) = \frac{\sqrt{28014}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{322} - \frac{7\sqrt{20010}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{5336} + \frac{3\sqrt{28014}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{1288} + \frac{5\sqrt{9338}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{9338} + \frac{39\sqrt{9338}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{18676}$$

$\boxed{\text{z1354}}$ $$\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u,3) = \frac{\sqrt{28014}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{322} + \frac{7\sqrt{20010}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{5336} - \frac{3\sqrt{28014}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{1288} - \frac{5\sqrt{9338}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{9338} + \frac{39\sqrt{9338}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{18676}$$

$\boxed{\text{z1355}}$ $$\mathbb{Q}_{7,1}^{(1,-1;c)}(E_u,1) = \frac{5\sqrt{14}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} + \frac{3\sqrt{66}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} - \frac{\sqrt{429}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} - \frac{15\sqrt{3}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{64}$$

$\boxed{\text{z1356}}$ $$\mathbb{Q}_{7,2}^{(1,-1;c)}(E_u,1) = -\frac{5\sqrt{14}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} - \frac{3\sqrt{66}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} - \frac{\sqrt{429}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} - \frac{15\sqrt{3}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{64}$$

$\boxed{\text{z1357}}$ $$\mathbb{Q}_{7,1}^{(1,-1;c)}(E_u,2) = -\frac{\sqrt{2}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{6} + \frac{\sqrt{462}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} - \frac{43\sqrt{2}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{192} + \frac{\sqrt{6}\mathbb{M}_7^{(1,-1;a)}(B_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{6} + \frac{3\sqrt{13}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{64} - \frac{11\sqrt{11}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{192}$$

$\boxed{\text{z1358}}$ $$\mathbb{Q}_{7,2}^{(1,-1;c)}(E_u,2) = -\frac{\sqrt{2}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{6} - \frac{\sqrt{462}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} + \frac{43\sqrt{2}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{192} - \frac{\sqrt{6}\mathbb{M}_7^{(1,-1;a)}(B_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{6} + \frac{3\sqrt{13}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{64} - \frac{11\sqrt{11}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{192}$$

$\boxed{\text{z1359}}$ $$\mathbb{Q}_{7,1}^{(1,-1;c)}(E_u,3) = \frac{\sqrt{9230}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{213} + \frac{3\sqrt{2132130}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{22720} - \frac{13\sqrt{9230}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{13632} + \frac{\sqrt{23430}\mathbb{M}_7^{(1,-1;a)}(B_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{355} + \frac{\sqrt{27690}\mathbb{M}_7^{(1,-1;a)}(B_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{1065} - \frac{31\sqrt{355}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{4544} + \frac{47\sqrt{50765}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{68160}$$

$\boxed{\text{z1360}}$ $$\mathbb{Q}_{7,2}^{(1,-1;c)}(E_u,3) = \frac{\sqrt{9230}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{213} - \frac{3\sqrt{2132130}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{22720} + \frac{13\sqrt{9230}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{13632} - \frac{\sqrt{23430}\mathbb{M}_7^{(1,-1;a)}(B_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{355} - \frac{\sqrt{27690}\mathbb{M}_7^{(1,-1;a)}(B_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{1065} - \frac{31\sqrt{355}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{4544} + \frac{47\sqrt{50765}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{68160}$$

$\boxed{\text{z1361}}$ $$\mathbb{Q}_{7,1}^{(1,-1;c)}(E_u,4) = -\frac{77\sqrt{206894}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{620682} + \frac{\sqrt{47792514}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{49984} + \frac{4015\sqrt{206894}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{19861824} + \frac{41\sqrt{88757526}\mathbb{M}_7^{(1,-1;a)}(B_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{1137917} - \frac{271\sqrt{620682}\mathbb{M}_7^{(1,-1;a)}(B_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{620682} + \frac{55\sqrt{1344811}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{140864} - \frac{1157\sqrt{1137917}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{7047744}$$

$\boxed{\text{z1362}}$ $$\mathbb{Q}_{7,2}^{(1,-1;c)}(E_u,4) = -\frac{77\sqrt{206894}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{620682} - \frac{\sqrt{47792514}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{49984} - \frac{4015\sqrt{206894}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{19861824} - \frac{41\sqrt{88757526}\mathbb{M}_7^{(1,-1;a)}(B_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{1137917} + \frac{271\sqrt{620682}\mathbb{M}_7^{(1,-1;a)}(B_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{620682} + \frac{55\sqrt{1344811}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{140864} - \frac{1157\sqrt{1137917}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{7047744}$$

$\boxed{\text{z1363}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(E_u) = -\frac{\sqrt{14}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{14} + \frac{\sqrt{42}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{14} + \frac{\sqrt{42}\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{14}$$

$\boxed{\text{z1364}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(E_u) = -\frac{\sqrt{14}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{14} - \frac{\sqrt{42}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{14} + \frac{\sqrt{42}\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{14}$$

$\boxed{\text{z1365}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,1a) = -\frac{\sqrt{42}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{28} + \frac{3\sqrt{14}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{28} - \frac{\sqrt{14}\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{7}$$

$\boxed{\text{z1366}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,1a) = -\frac{\sqrt{42}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{28} - \frac{3\sqrt{14}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{28} - \frac{\sqrt{14}\mathbb{T}_2^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{7}$$

$\boxed{\text{z1367}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,1b) = \frac{\sqrt{273}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{39} - \frac{\sqrt{195}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{78} + \frac{\sqrt{455}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{52} + \frac{\sqrt{65}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{26} - \frac{\sqrt{65}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{52}$$

$\boxed{\text{z1368}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,1b) = \frac{\sqrt{273}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{39} - \frac{\sqrt{195}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{78} - \frac{\sqrt{455}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{52} - \frac{\sqrt{65}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{26} - \frac{\sqrt{65}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{52}$$

$\boxed{\text{z1369}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,2a) = -\frac{\sqrt{6}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4} - \frac{\sqrt{2}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z1370}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,2a) = -\frac{\sqrt{6}\mathbb{T}_2^{(1,0;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4} + \frac{\sqrt{2}\mathbb{T}_2^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z1371}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(E_u,2b) = \frac{\sqrt{5655}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{1131} + \frac{73\sqrt{7917}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{15834} - \frac{21\sqrt{377}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{1508} + \frac{31\sqrt{2639}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{5278} - \frac{83\sqrt{2639}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{10556}$$

$\boxed{\text{z1372}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(E_u,2b) = \frac{\sqrt{5655}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{1131} + \frac{73\sqrt{7917}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{15834} + \frac{21\sqrt{377}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{1508} - \frac{31\sqrt{2639}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{5278} - \frac{83\sqrt{2639}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{10556}$$

$\boxed{\text{z1373}}$ $$\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,1a) = \frac{5\sqrt{58}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{116} - \frac{41\sqrt{2030}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{8120} - \frac{7\sqrt{870}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{435} + \frac{59\sqrt{6090}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{24360} + \frac{11\sqrt{6090}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{3045}$$

$\boxed{\text{z1374}}$ $$\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,1a) = \frac{5\sqrt{58}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{116} - \frac{41\sqrt{2030}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{8120} + \frac{7\sqrt{870}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{435} - \frac{59\sqrt{6090}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{24360} + \frac{11\sqrt{6090}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{3045}$$

$\boxed{\text{z1375}}$ $$\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,1b) = \frac{\sqrt{57}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{38} - \frac{\sqrt{399}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{76} - \frac{\sqrt{798}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{152} + \frac{3\sqrt{133}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{76} + \frac{3\sqrt{1463}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{304} + \frac{\sqrt{665}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{304}$$

$\boxed{\text{z1376}}$ $$\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,1b) = \frac{\sqrt{57}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{38} - \frac{\sqrt{399}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{76} + \frac{\sqrt{798}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{152} - \frac{3\sqrt{133}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{76} + \frac{3\sqrt{1463}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{304} + \frac{\sqrt{665}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{304}$$

$\boxed{\text{z1377}}$ $$\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,2a) = \frac{5\sqrt{62}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{124} - \frac{3\sqrt{2170}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{1240} - \frac{2\sqrt{930}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{465} - \frac{23\sqrt{6510}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{3720} - \frac{2\sqrt{6510}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{465}$$

$\boxed{\text{z1378}}$ $$\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,2a) = \frac{5\sqrt{62}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{124} - \frac{3\sqrt{2170}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{1240} + \frac{2\sqrt{930}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{465} + \frac{23\sqrt{6510}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{3720} - \frac{2\sqrt{6510}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{465}$$

$\boxed{\text{z1379}}$ $$\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,2b) = -\frac{\sqrt{35}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{14} - \frac{\sqrt{5}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{28} + \frac{\sqrt{10}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{8} + \frac{\sqrt{15}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{28} + \frac{\sqrt{165}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{112} + \frac{23\sqrt{3}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{112}$$

$\boxed{\text{z1380}}$ $$\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,2b) = -\frac{\sqrt{35}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{14} - \frac{\sqrt{5}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{28} - \frac{\sqrt{10}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{8} - \frac{\sqrt{15}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{28} + \frac{\sqrt{165}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{112} + \frac{23\sqrt{3}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{112}$$

$\boxed{\text{z1381}}$ $$\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,3a) = \frac{\sqrt{930}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{93} + \frac{5\sqrt{1302}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{372} + \frac{\sqrt{62}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{62} - \frac{\sqrt{434}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{124} + \frac{\sqrt{434}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{62}$$

$\boxed{\text{z1382}}$ $$\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,3a) = \frac{\sqrt{930}\mathbb{T}_4^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{93} + \frac{5\sqrt{1302}\mathbb{T}_4^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{372} - \frac{\sqrt{62}\mathbb{T}_4^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{62} + \frac{\sqrt{434}\mathbb{T}_4^{(1,0;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{124} + \frac{\sqrt{434}\mathbb{T}_4^{(1,0;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{62}$$

$\boxed{\text{z1383}}$ $$\mathbb{Q}_{5,1}^{(1,0;c)}(E_u,3b) = -\frac{4\sqrt{97709115}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{6513941} - \frac{1131\sqrt{13958445}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{13027882} - \frac{3\sqrt{27916890}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{86564} - \frac{7\sqrt{10236193}\mathbb{T}_6^{(1,0;a)}(B_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{48977} - \frac{331\sqrt{4652815}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{13027882} - \frac{3\sqrt{51180965}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{71288} + \frac{195\sqrt{930563}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{777784}$$

$\boxed{\text{z1384}}$ $$\mathbb{Q}_{5,2}^{(1,0;c)}(E_u,3b) = -\frac{4\sqrt{97709115}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{6513941} - \frac{1131\sqrt{13958445}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{13027882} + \frac{3\sqrt{27916890}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{86564} + \frac{7\sqrt{10236193}\mathbb{T}_6^{(1,0;a)}(B_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{48977} + \frac{331\sqrt{4652815}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{13027882} - \frac{3\sqrt{51180965}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{71288} + \frac{195\sqrt{930563}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{777784}$$

$\boxed{\text{z1385}}$ $$\mathbb{Q}_{7,1}^{(1,0;c)}(E_u,1) = \frac{2603\sqrt{830698897}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{474685084} - \frac{10901\sqrt{118671271}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{949370168} + \frac{213\sqrt{237342542}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{11039188} + \frac{98\sqrt{19580759715}\mathbb{T}_6^{(1,0;a)}(B_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{118671271} + \frac{18741\sqrt{356013813}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{949370168} - \frac{9\sqrt{3916151943}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{1298728} - \frac{59\sqrt{1780069065}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{14169704}$$

$\boxed{\text{z1386}}$ $$\mathbb{Q}_{7,2}^{(1,0;c)}(E_u,1) = \frac{2603\sqrt{830698897}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{474685084} - \frac{10901\sqrt{118671271}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{949370168} - \frac{213\sqrt{237342542}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{11039188} - \frac{98\sqrt{19580759715}\mathbb{T}_6^{(1,0;a)}(B_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{118671271} - \frac{18741\sqrt{356013813}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{949370168} - \frac{9\sqrt{3916151943}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{1298728} - \frac{59\sqrt{1780069065}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{14169704}$$

$\boxed{\text{z1387}}$ $$\mathbb{Q}_{7,1}^{(1,0;c)}(E_u,2) = -\frac{141\sqrt{7756023}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{940124} + \frac{99\sqrt{54292161}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{13161736} - \frac{269\sqrt{108584322}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{6580868} + \frac{132\sqrt{8226085}\mathbb{T}_6^{(1,0;a)}(B_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{1645217} + \frac{663\sqrt{18097387}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{13161736} - \frac{2087\sqrt{1645217}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{13161736} - \frac{39\sqrt{90486935}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{13161736}$$

$\boxed{\text{z1388}}$ $$\mathbb{Q}_{7,2}^{(1,0;c)}(E_u,2) = -\frac{141\sqrt{7756023}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{940124} + \frac{99\sqrt{54292161}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{13161736} + \frac{269\sqrt{108584322}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{6580868} - \frac{132\sqrt{8226085}\mathbb{T}_6^{(1,0;a)}(B_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{1645217} - \frac{663\sqrt{18097387}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{13161736} - \frac{2087\sqrt{1645217}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{13161736} - \frac{39\sqrt{90486935}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{13161736}$$

$\boxed{\text{z1389}}$ $$\mathbb{Q}_{7,1}^{(1,0;c)}(E_u,3) = \frac{8\sqrt{7727214}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{117079} - \frac{435\sqrt{54090498}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{13112848} - \frac{8\sqrt{27045249}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{819553} + \frac{1075\sqrt{8195530}\mathbb{T}_6^{(1,0;a)}(B_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{6556424} - \frac{691\sqrt{18030166}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{13112848} - \frac{66\sqrt{1639106}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{819553} + \frac{30\sqrt{90150830}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{819553}$$

$\boxed{\text{z1390}}$ $$\mathbb{Q}_{7,2}^{(1,0;c)}(E_u,3) = \frac{8\sqrt{7727214}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{117079} - \frac{435\sqrt{54090498}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{13112848} + \frac{8\sqrt{27045249}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{819553} - \frac{1075\sqrt{8195530}\mathbb{T}_6^{(1,0;a)}(B_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{6556424} + \frac{691\sqrt{18030166}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{13112848} - \frac{66\sqrt{1639106}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{819553} + \frac{30\sqrt{90150830}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{819553}$$

$\boxed{\text{z1391}}$ $$\mathbb{Q}_{7,1}^{(1,0;c)}(E_u,4) = -\frac{2\sqrt{16898}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{1207} - \frac{193\sqrt{2414}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{19312} + \frac{2\sqrt{1207}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{1207} + \frac{\sqrt{398310}\mathbb{T}_6^{(1,0;a)}(B_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{9656} - \frac{43\sqrt{7242}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{19312} + \frac{\sqrt{79662}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{2414} - \frac{5\sqrt{36210}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{2414}$$

$\boxed{\text{z1392}}$ $$\mathbb{Q}_{7,2}^{(1,0;c)}(E_u,4) = -\frac{2\sqrt{16898}\mathbb{T}_6^{(1,0;a)}(A_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{1207} - \frac{193\sqrt{2414}\mathbb{T}_6^{(1,0;a)}(A_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{19312} - \frac{2\sqrt{1207}\mathbb{T}_6^{(1,0;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{1207} - \frac{\sqrt{398310}\mathbb{T}_6^{(1,0;a)}(B_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{9656} + \frac{43\sqrt{7242}\mathbb{T}_6^{(1,0;a)}(B_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{19312} + \frac{\sqrt{79662}\mathbb{T}_6^{(1,0;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{2414} - \frac{5\sqrt{36210}\mathbb{T}_6^{(1,0;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{2414}$$

$\boxed{\text{z1393}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(E_u) = -\frac{\sqrt{2}\mathbb{M}_1^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1394}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{M}_1^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{2}$$

$\boxed{\text{z1395}}$ $$\mathbb{Q}_{3,1}^{(1,1;c)}(E_u,1)=\frac{\sqrt{3}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}-\frac{\sqrt{5}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4}$$

$\boxed{\text{z1396}}$ $$\mathbb{Q}_{3,2}^{(1,1;c)}(E_u,1)=-\frac{\sqrt{3}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}-\frac{\sqrt{5}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4}$$

$\boxed{\text{z1397}}$ $$\mathbb{Q}_{3,1}^{(1,1;c)}(E_u,2)=\frac{\sqrt{165}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{44}+\frac{2\sqrt{11}\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{11}+\frac{3\sqrt{11}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{44}$$

$\boxed{\text{z1398}}$ $$\mathbb{Q}_{3,2}^{(1,1;c)}(E_u,2)=-\frac{\sqrt{165}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{44}-\frac{2\sqrt{11}\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{11}+\frac{3\sqrt{11}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{44}$$

$\boxed{\text{z1399}}$ $$\mathbb{Q}_{5,1}^{(1,1;c)}(E_u,1)=-\frac{\sqrt{30}\mathbb{M}_5^{(1,1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{16}-\frac{\sqrt{42}\mathbb{M}_5^{(1,1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16}+\frac{\sqrt{14}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{8}$$

$\boxed{\text{z1400}}$ $$\mathbb{Q}_{5,2}^{(1,1;c)}(E_u,1)=\frac{\sqrt{30}\mathbb{M}_5^{(1,1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{16}+\frac{\sqrt{42}\mathbb{M}_5^{(1,1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16}+\frac{\sqrt{14}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{8}$$

$\boxed{\text{z1401}}$ $$\mathbb{Q}_{5,1}^{(1,1;c)}(E_u,2)=-\frac{\sqrt{46}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{23}-\frac{3\sqrt{1610}\mathbb{M}_5^{(1,1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{368}+\frac{11\sqrt{46}\mathbb{M}_5^{(1,1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{368}+\frac{\sqrt{138}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{23}-\frac{\sqrt{138}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{184}$$

$\boxed{\text{z1402}}$ $$\mathbb{Q}_{5,2}^{(1,1;c)}(E_u,2)=-\frac{\sqrt{46}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{23}+\frac{3\sqrt{1610}\mathbb{M}_5^{(1,1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{368}-\frac{11\sqrt{46}\mathbb{M}_5^{(1,1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{368}-\frac{\sqrt{138}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{23}-\frac{\sqrt{138}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{184}$$

$\boxed{\text{z1403}}$ $$\mathbb{Q}_{5,1}^{(1,1;c)}(E_u,3)=\frac{\sqrt{28014}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{322}-\frac{7\sqrt{20010}\mathbb{M}_5^{(1,1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{5336}+\frac{3\sqrt{28014}\mathbb{M}_5^{(1,1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{1288}$$
$$+\frac{5\sqrt{9338}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{9338}+\frac{39\sqrt{9338}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{18676}$$

$\boxed{\text{z1404}}$ $$\mathbb{Q}_{5,2}^{(1,1;c)}(E_u,3)=\frac{\sqrt{28014}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{322}+\frac{7\sqrt{20010}\mathbb{M}_5^{(1,1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{5336}-\frac{3\sqrt{28014}\mathbb{M}_5^{(1,1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{1288}$$
$$-\frac{5\sqrt{9338}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{9338}+\frac{39\sqrt{9338}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{18676}$$

$\boxed{\text{z1405}}$ $$\mathbb{G}_{2,1}^{(c)}(E_u)=-\frac{\sqrt{110}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22}+\frac{\sqrt{66}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22}-\frac{\sqrt{66}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22}$$

$\boxed{\text{z1406}}$ $$\mathbb{G}_{2,2}^{(c)}(E_u)=\frac{\sqrt{110}\mathbb{M}_3^{(a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22}-\frac{\sqrt{66}\mathbb{M}_3^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22}-\frac{\sqrt{66}\mathbb{M}_3^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22}$$

$\boxed{\text{z1407}}$ $$\mathbb{G}_{4,1}^{(c)}(E_u,1) = -\frac{3\sqrt{100282}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{6916} + \frac{49\sqrt{71630}\mathbb{M}_5^{(a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{28652} + \frac{3\sqrt{100282}\mathbb{M}_5^{(a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{6916} + \frac{79\sqrt{300846}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{200564} + \frac{83\sqrt{300846}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{100282}$$

$\boxed{\text{z1408}}$ $$\mathbb{G}_{4,2}^{(c)}(E_u,1) = -\frac{3\sqrt{100282}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{6916} - \frac{49\sqrt{71630}\mathbb{M}_5^{(a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{28652} - \frac{3\sqrt{100282}\mathbb{M}_5^{(a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{6916} - \frac{79\sqrt{300846}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{200564} + \frac{83\sqrt{300846}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{100282}$$

$\boxed{\text{z1409}}$ $$\mathbb{G}_{4,1}^{(c)}(E_u,2) = \frac{7\sqrt{2470}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{988} + \frac{3\sqrt{3458}\mathbb{M}_5^{(a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{988} - \frac{7\sqrt{2470}\mathbb{M}_5^{(a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{988} + \frac{5\sqrt{7410}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{988} - \frac{\sqrt{7410}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{494}$$

$\boxed{\text{z1410}}$ $$\mathbb{G}_{4,2}^{(c)}(E_u,2) = \frac{7\sqrt{2470}\mathbb{M}_5^{(a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{988} - \frac{3\sqrt{3458}\mathbb{M}_5^{(a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{988} + \frac{7\sqrt{2470}\mathbb{M}_5^{(a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{988} - \frac{5\sqrt{7410}\mathbb{M}_5^{(a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{988} - \frac{\sqrt{7410}\mathbb{M}_5^{(a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{494}$$

$\boxed{\text{z1411}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = -\frac{\sqrt{110}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} + \frac{\sqrt{66}\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22}$$

$\boxed{\text{z1412}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{110}\mathbb{M}_3^{(1,-1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22}$$

$\boxed{\text{z1413}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(E_u,1) = -\frac{3\sqrt{100282}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{6916} + \frac{49\sqrt{71630}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{28652} + \frac{3\sqrt{100282}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{6916} + \frac{79\sqrt{300846}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{200564} + \frac{83\sqrt{300846}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{100282}$$

$\boxed{\text{z1414}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(E_u,1) = -\frac{3\sqrt{100282}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{6916} - \frac{49\sqrt{71630}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{28652} - \frac{3\sqrt{100282}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{6916} - \frac{79\sqrt{300846}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{200564} + \frac{83\sqrt{300846}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{100282}$$

$\boxed{\text{z1415}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(E_u,2) = \frac{7\sqrt{2470}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{988} + \frac{3\sqrt{3458}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{988} - \frac{7\sqrt{2470}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{988} + \frac{5\sqrt{7410}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{988} - \frac{\sqrt{7410}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{494}$$

$\boxed{\text{z1416}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(E_u,2) = \frac{7\sqrt{2470}\mathbb{M}_5^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{988} - \frac{3\sqrt{3458}\mathbb{M}_5^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{988} + \frac{7\sqrt{2470}\mathbb{M}_5^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{988} - \frac{5\sqrt{7410}\mathbb{M}_5^{(1,-1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{988} - \frac{\sqrt{7410}\mathbb{M}_5^{(1,-1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{494}$$

$\boxed{\text{z1417}}$ $$\mathbb{G}_{6,1}^{(1,-1;c)}(E_u,1) = -\frac{451\sqrt{305970}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{4773132} - \frac{3\sqrt{160270}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{5720} + \frac{121\sqrt{305970}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{151528} + \frac{311\sqrt{14584570}\mathbb{M}_7^{(1,-1;a)}(B_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{9375795} + \frac{31973\sqrt{101990}\mathbb{M}_7^{(1,-1;a)}(B_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{23865660} + \frac{11\sqrt{1988805}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{67704} + \frac{1051\sqrt{1682835}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{16936920}$$

$\boxed{\text{z1418}}$ $$\mathbb{G}_{6,2}^{(1,-1;c)}(E_u,1) = -\frac{451\sqrt{305970}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{4773132} + \frac{3\sqrt{160270}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{5720} - \frac{121\sqrt{305970}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{151528} - \frac{311\sqrt{14584570}\mathbb{M}_7^{(1,-1;a)}(B_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{9375795} - \frac{31973\sqrt{101990}\mathbb{M}_7^{(1,-1;a)}(B_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{23865660} + \frac{11\sqrt{1988805}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{67704} + \frac{1051\sqrt{1682835}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{16936920}$$

$\boxed{\text{z1419}}$ $$\mathbb{G}_{6,1}^{(1,-1;c)}(E_u,2) = \frac{36493\sqrt{772653}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{241067736} - \frac{5259\sqrt{404723}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{7652944} - \frac{1753\sqrt{772653}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{7652944} + \frac{407\sqrt{36829793}\mathbb{M}_7^{(1,-1;a)}(B_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{17219124} + \frac{36749\sqrt{257551}\mathbb{M}_7^{(1,-1;a)}(B_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{241067736} - \frac{2881\sqrt{20088978}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{321423648} - \frac{110311\sqrt{16998366}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{964270944}$$

$\boxed{\text{z1420}}$ $$\mathbb{G}_{6,2}^{(1,-1;c)}(E_u,2) = \frac{36493\sqrt{772653}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{241067736} + \frac{5259\sqrt{404723}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{7652944} + \frac{1753\sqrt{772653}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{7652944} - \frac{407\sqrt{36829793}\mathbb{M}_7^{(1,-1;a)}(B_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{17219124} - \frac{36749\sqrt{257551}\mathbb{M}_7^{(1,-1;a)}(B_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{241067736} - \frac{2881\sqrt{20088978}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{321423648} - \frac{110311\sqrt{16998366}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{964270944}$$

$\boxed{\text{z1421}}$ $$\mathbb{G}_{6,1}^{(1,-1;c)}(E_u,3) = -\frac{75\sqrt{3348163}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{294344} - \frac{15\sqrt{15784197}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{588688} - \frac{15\sqrt{3348163}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{588688} + \frac{61\sqrt{8499183}\mathbb{M}_7^{(1,-1;a)}(B_{1g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{441516} - \frac{11\sqrt{10044489}\mathbb{M}_7^{(1,-1;a)}(B_{1g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{883032} - \frac{495\sqrt{515102}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{1177376} + \frac{17\sqrt{73659586}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{1177376}$$

$\boxed{\text{z1422}}$ $$\mathbb{G}_{6,2}^{(1,-1;c)}(E_u,3) = -\frac{75\sqrt{3348163}\mathbb{M}_7^{(1,-1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{294344} + \frac{15\sqrt{15784197}\mathbb{M}_7^{(1,-1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{588688} + \frac{15\sqrt{3348163}\mathbb{M}_7^{(1,-1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{588688} - \frac{61\sqrt{8499183}\mathbb{M}_7^{(1,-1;a)}(B_{1g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{441516} + \frac{11\sqrt{10044489}\mathbb{M}_7^{(1,-1;a)}(B_{1g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{883032} - \frac{495\sqrt{515102}\mathbb{M}_7^{(1,-1;a)}(B_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{1177376} + \frac{17\sqrt{73659586}\mathbb{M}_7^{(1,-1;a)}(B_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{1177376}$$

$\boxed{\text{z1423}}$ $$\mathbb{G}_{2,1}^{(1,1;c)}(E_u) = -\frac{\sqrt{110}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} + \frac{\sqrt{66}\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22}$$

$\boxed{\text{z1424}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{110}\mathbb{M}_3^{(1,1;a)}(A_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{22} - \frac{\sqrt{66}\mathbb{M}_3^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{22}$$

$\boxed{\text{z1425}}$ $$\mathbb{G}_{4,1}^{(1,1;c)}(E_u,1) = -\frac{3\sqrt{100282}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{6916} + \frac{49\sqrt{71630}\mathbb{M}_5^{(1,1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{28652} + \frac{3\sqrt{100282}\mathbb{M}_5^{(1,1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{6916} + \frac{79\sqrt{300846}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{200564} + \frac{83\sqrt{300846}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{100282}$$

$\boxed{\text{z1426}}$ $$\mathbb{G}_{4,2}^{(1,1;c)}(E_u,1) = -\frac{3\sqrt{100282}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{6916} - \frac{49\sqrt{71630}\mathbb{M}_5^{(1,1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{28652} - \frac{3\sqrt{100282}\mathbb{M}_5^{(1,1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{6916} - \frac{79\sqrt{300846}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{200564} + \frac{83\sqrt{300846}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{100282}$$

$\boxed{\text{z1427}}$ $$\mathbb{G}_{4,1}^{(1,1;c)}(E_u,2) = \frac{7\sqrt{2470}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{988} + \frac{3\sqrt{3458}\mathbb{M}_5^{(1,1;a)}(A_{2g},1)\mathbb{T}_{1,2}^{(b)}(E_u)}{988} - \frac{7\sqrt{2470}\mathbb{M}_5^{(1,1;a)}(A_{2g},2)\mathbb{T}_{1,2}^{(b)}(E_u)}{988} + \frac{5\sqrt{7410}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,1}^{(b)}(E_u)}{988} - \frac{\sqrt{7410}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,2}^{(b)}(E_u)}{494}$$

$\boxed{\text{z1428}}$ $$\mathbb{G}_{4,2}^{(1,1;c)}(E_u,2) = \frac{7\sqrt{2470}\mathbb{M}_5^{(1,1;a)}(A_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{988} - \frac{3\sqrt{3458}\mathbb{M}_5^{(1,1;a)}(A_{2g},1)\mathbb{T}_{1,1}^{(b)}(E_u)}{988} + \frac{7\sqrt{2470}\mathbb{M}_5^{(1,1;a)}(A_{2g},2)\mathbb{T}_{1,1}^{(b)}(E_u)}{988} - \frac{5\sqrt{7410}\mathbb{M}_5^{(1,1;a)}(B_{1g})\mathbb{T}_{1,2}^{(b)}(E_u)}{988} - \frac{\sqrt{7410}\mathbb{M}_5^{(1,1;a)}(B_{2g})\mathbb{T}_{1,1}^{(b)}(E_u)}{494}$$

Atomic SAMB

- bra: $\langle\frac{3}{2},\frac{3}{2};d|$, $\langle\frac{3}{2},\frac{1}{2};d|$, $\langle\frac{3}{2},-\frac{1}{2};d|$, $\langle\frac{3}{2},-\frac{3}{2};d|$, $\langle\frac{5}{2},\frac{5}{2};d|$, $\langle\frac{5}{2},\frac{3}{2};d|$, $\langle\frac{5}{2},\frac{1}{2};d|$, $\langle\frac{5}{2},-\frac{1}{2};d|$, $\langle\frac{5}{2},-\frac{3}{2};d|$, $\langle\frac{5}{2},-\frac{5}{2};d|$

- ket: $|\frac{3}{2},\frac{3}{2};d\rangle$, $|\frac{3}{2},\frac{1}{2};d\rangle$, $|\frac{3}{2},-\frac{1}{2};d\rangle$, $|\frac{3}{2},-\frac{3}{2};d\rangle$, $|\frac{5}{2},\frac{5}{2};d\rangle$, $|\frac{5}{2},\frac{3}{2};d\rangle$, $|\frac{5}{2},\frac{1}{2};d\rangle$, $|\frac{5}{2},-\frac{1}{2};d\rangle$, $|\frac{5}{2},-\frac{3}{2};d\rangle$, $|\frac{5}{2},-\frac{5}{2};d\rangle$

$\boxed{\text{x1}}$ $$\mathbb{Q}_0^{(a)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} \end{bmatrix}$$

$\boxed{\text{x2}}$ $$\mathbb{Q}_2^{(a)}(A_{1g}) = \begin{bmatrix} -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{35} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} \end{bmatrix}$$

$\boxed{\text{x3}}$ $\mathbb{Q}_4^{(a)}(A_{1g},1) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 \\
0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{15}}{60} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 \\
\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\
0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}}{30} & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 \\
\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & \frac{\sqrt{15}}{60}
\end{bmatrix}$

$\boxed{\text{x4}}$ $\mathbb{Q}_4^{(a)}(A_{1g},2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & -\frac{\sqrt{105}}{30} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}}{30} & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} & 0 \\
\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} \\
0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{21}}{42} & -\frac{\sqrt{105}}{60} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 \\
-\frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} & 0 & 0 & 0 & \frac{\sqrt{21}}{84}
\end{bmatrix}$

$\boxed{\text{x5}}$ $\mathbb{Q}_4^{(a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x6}}$ $\mathbb{Q}_2^{(a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{70} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 \\ -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} \\ 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{70} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{70} & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 & -\frac{3\sqrt{42}}{70} & 0 \\ -\frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{70} & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} \\ 0 & -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x7}}$ $\mathbb{Q}_4^{(a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\
0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{14}}{14} & -\frac{3\sqrt{70}}{140} & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\
-\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} \\
0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x8}}$ $\mathbb{Q}_2^{(a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{70} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{7}i}{10} & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 \\
-\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} \\
0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{70} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{70} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}i}{70} & -\frac{\sqrt{210}i}{70} & 0 & 0 & 0 & \frac{3\sqrt{42}i}{70} & 0 \\
-\frac{\sqrt{42}i}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{70} & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} \\
0 & -\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{70} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x9}}$ $\mathbb{Q}_4^{(a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\
0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{140} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{14}i}{14} & \frac{3\sqrt{70}i}{140} & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 \\
\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{140} \\
0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{140} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x10}}$ $\mathbb{Q}_{2,1}^{(a)}(E_g) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{7}i}{10} & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & \frac{3\sqrt{42}i}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{7}i}{70} & 0 \\
0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 & -\frac{\sqrt{105}i}{70} \\
-\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{7}i}{70} & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & \frac{\sqrt{42}i}{35} & 0 & 0 & 0 \\
-\frac{3\sqrt{42}i}{140} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{35} & 0 \\
0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{35} & 0 & -\frac{\sqrt{105}i}{35} \\
0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0
\end{bmatrix}$$

$\boxed{\text{x11}}$ $\mathbb{Q}^{(a)}_{2,2}(E_g) = \begin{bmatrix}
0 & -\frac{\sqrt{7}}{10} & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & -\frac{3\sqrt{42}}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{70} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{7}}{10} & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{7}}{70} & 0 \\
0 & 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 & \frac{\sqrt{105}}{70} \\
\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{7}}{70} & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & -\frac{\sqrt{42}}{35} & 0 & 0 & 0 \\
-\frac{3\sqrt{42}}{140} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{14}}{28} & 0 & -\frac{3\sqrt{42}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{35} & 0 \\
0 & 0 & -\frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{35} & 0 & \frac{\sqrt{105}}{35} \\
0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0
\end{bmatrix}$

$\boxed{\text{x12}}$ $\mathbb{Q}^{(a)}_{4,1}(E_g, 1) = \begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{40} & 0 & \frac{\sqrt{2}i}{8} & 0 & \frac{i}{8} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & -\frac{\sqrt{6}i}{8} & 0 & -\frac{\sqrt{15}i}{40} \\
0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{40} & 0 & \frac{\sqrt{6}i}{8} & 0 & \frac{\sqrt{3}i}{8} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & -\frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{5}i}{40} \\
-\frac{\sqrt{5}i}{40} & 0 & -\frac{\sqrt{15}i}{40} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & \frac{\sqrt{10}i}{40} & 0 & 0 \\
0 & \frac{\sqrt{3}i}{8} & 0 & \frac{i}{8} & -\frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 \\
-\frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{40} \\
0 & \frac{\sqrt{6}i}{8} & 0 & \frac{\sqrt{2}i}{8} & -\frac{\sqrt{10}i}{40} & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 \\
-\frac{i}{8} & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{5}i}{20} \\
0 & \frac{\sqrt{15}i}{40} & 0 & \frac{\sqrt{5}i}{40} & 0 & 0 & \frac{\sqrt{10}i}{40} & 0 & \frac{\sqrt{5}i}{20} & 0
\end{bmatrix}$

$\boxed{\text{x13}}$ $\mathbb{Q}^{(a)}_{4,2}(E_g,1) = \begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{5}}{40} & 0 & -\frac{\sqrt{2}}{8} & 0 & \frac{1}{8} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & \frac{\sqrt{6}}{8} & 0 & -\frac{\sqrt{15}}{40} \\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{40} & 0 & \frac{\sqrt{6}}{8} & 0 & -\frac{\sqrt{3}}{8} & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & -\frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{5}}{40} \\
\frac{\sqrt{5}}{40} & 0 & -\frac{\sqrt{15}}{40} & 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{10}}{40} & 0 & 0 \\
0 & -\frac{\sqrt{3}}{8} & 0 & \frac{1}{8} & -\frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 \\
-\frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & -\frac{\sqrt{10}}{40} \\
0 & \frac{\sqrt{6}}{8} & 0 & -\frac{\sqrt{2}}{8} & \frac{\sqrt{10}}{40} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 \\
\frac{1}{8} & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{5}}{20} \\
0 & -\frac{\sqrt{15}}{40} & 0 & \frac{\sqrt{5}}{40} & 0 & 0 & -\frac{\sqrt{10}}{40} & 0 & \frac{\sqrt{5}}{20} & 0
\end{bmatrix}$

$\boxed{\text{x14}}$ $\mathbb{Q}^{(a)}_{4,1}(E_g,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{280} & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{7}i}{8} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{105}i}{40} \\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{40} & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{21}i}{56} & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{8} & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{35}i}{280} \\
-\frac{\sqrt{35}i}{280} & 0 & \frac{\sqrt{105}i}{40} & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & -\frac{\sqrt{70}i}{40} & 0 & 0 \\
0 & \frac{\sqrt{21}i}{56} & 0 & -\frac{\sqrt{7}i}{8} & -\frac{\sqrt{35}i}{140} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 \\
-\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{70}i}{40} \\
0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{14}i}{56} & \frac{\sqrt{70}i}{40} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 \\
\frac{\sqrt{7}i}{8} & 0 & -\frac{\sqrt{21}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{35}i}{140} \\
0 & -\frac{\sqrt{105}i}{40} & 0 & \frac{\sqrt{35}i}{280} & 0 & 0 & -\frac{\sqrt{70}i}{40} & 0 & \frac{\sqrt{35}i}{140} & 0
\end{bmatrix}$

$\boxed{\text{x15}}$ $\mathbb{Q}_{4,2}^{(a)}(E_g,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{35}}{280} & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{7}}{8} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{105}}{40} \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}}{40} & 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{21}}{56} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{8} & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{35}}{280} \\
\frac{\sqrt{35}}{280} & 0 & \frac{\sqrt{105}}{40} & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & -\frac{\sqrt{70}}{40} & 0 & 0 \\
0 & -\frac{\sqrt{21}}{56} & 0 & -\frac{\sqrt{7}}{8} & -\frac{\sqrt{35}}{140} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 \\
-\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{70}}{40} \\
0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{14}}{56} & -\frac{\sqrt{70}}{40} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 \\
-\frac{\sqrt{7}}{8} & 0 & -\frac{\sqrt{21}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{35}}{140} \\
0 & \frac{\sqrt{105}}{40} & 0 & \frac{\sqrt{35}}{280} & 0 & 0 & \frac{\sqrt{70}}{40} & 0 & \frac{\sqrt{35}}{140} & 0
\end{bmatrix}$

$\boxed{\text{x16}}$ $\mathbb{Q}_{2}^{(1,-1;a)}(A_{1g}) = \begin{bmatrix}
-\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{50} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{75} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{30}}{75} & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{30}}{75} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{75} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15}
\end{bmatrix}$

$\boxed{\text{x17}}$ $\mathbb{Q}_4^{(1,-1;a)}(A_{1g},1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}}{60} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{60} & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 \\ \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} \end{bmatrix}$

$\boxed{\text{x18}}$ $\mathbb{Q}_4^{(1,-1;a)}(A_{1g},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{60} & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{60} & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 \\ \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & \frac{\sqrt{105}}{30} \\ 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 \\ -\frac{\sqrt{105}}{60} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} \end{bmatrix}$

$\boxed{\text{x19}}$ $\mathbb{Q}_4^{(1,-1;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x20}}$ $\mathbb{Q}_2^{(1,-1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{50} & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & -\frac{\sqrt{30}}{25} & 0 \\ -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} \\ 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}}{50} & \frac{1}{5} & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 \\ -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 & \frac{1}{5} \\ 0 & -\frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x21}}$ $\mathbb{Q}_4^{(1,-1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{28} & \frac{3\sqrt{70}}{70} & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 \\ -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} \\ 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x22}}$ $\mathbb{Q}_2^{(1,-1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{50} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{50} & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & \frac{\sqrt{30}i}{25} & 0 \\ -\frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{25} & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} \\ 0 & -\frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{50} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{25} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{5}i}{50} & \frac{i}{5} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{25} & 0 \\ -\frac{3\sqrt{5}i}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{25} & 0 & 0 & 0 & -\frac{i}{5} \\ 0 & -\frac{\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & \frac{i}{5} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x23}}$ $\mathbb{Q}_4^{(1,-1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & -\frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 \\ \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} \\ 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{70} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x24}}$ $\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{30}i}{50} & 0 & 0 & \frac{3\sqrt{2}i}{20} & 0 & \frac{9\sqrt{5}i}{100} & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{100} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{50} & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{\sqrt{30}i}{100} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}i}{100} & 0 & -\frac{3\sqrt{2}i}{20} \\ -\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{100} & 0 & 0 & \frac{\sqrt{2}i}{5} & 0 & -\frac{2\sqrt{5}i}{25} & 0 & 0 & 0 \\ -\frac{9\sqrt{5}i}{100} & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & \frac{2\sqrt{5}i}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{20} & 0 & \frac{9\sqrt{5}i}{100} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}i}{25} & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{100} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{25} & 0 & \frac{\sqrt{2}i}{5} \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 \end{bmatrix}$

$\boxed{\text{x25}}$ $\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & -\frac{9\sqrt{5}}{100} & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{100} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{30}}{100} & 0 \\ 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}}{100} & 0 & \frac{3\sqrt{2}}{20} \\ \frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{100} & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & \frac{2\sqrt{5}}{25} & 0 & 0 & 0 \\ -\frac{9\sqrt{5}}{100} & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{20} & 0 & -\frac{9\sqrt{5}}{100} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{25} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{100} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{25} & 0 & -\frac{\sqrt{2}}{5} \\ 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 \end{bmatrix}$

$\boxed{\text{x26}}$ $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{80} & 0 & \frac{\sqrt{2}i}{16} & 0 & \frac{i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{15}i}{80} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{80} & 0 & \frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{3}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{16} & 0 & -\frac{\sqrt{2}i}{16} & 0 & -\frac{\sqrt{5}i}{80} \\ -\frac{\sqrt{5}i}{80} & 0 & -\frac{\sqrt{15}i}{80} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{16} & 0 & \frac{i}{16} & \frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{16} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} \\ 0 & \frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{2}i}{16} & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ -\frac{i}{16} & 0 & -\frac{\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{5}i}{10} \\ 0 & \frac{\sqrt{15}i}{80} & 0 & \frac{\sqrt{5}i}{80} & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{5}i}{10} & 0 \end{bmatrix}$

$\boxed{\text{x27}}$ $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1) = \begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{5}}{80} & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{1}{16} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{16} & 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{15}}{80} \\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{80} & 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{3}}{16} & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{16} & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{5}}{80} \\
\frac{\sqrt{5}}{80} & 0 & -\frac{\sqrt{15}}{80} & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 \\
0 & -\frac{\sqrt{3}}{16} & 0 & \frac{1}{16} & \frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \\
-\frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{6}}{16} & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & \frac{\sqrt{10}}{20} \\
0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{2}}{16} & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\
\frac{1}{16} & 0 & -\frac{\sqrt{3}}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{5}}{10} \\
0 & -\frac{\sqrt{15}}{80} & 0 & \frac{\sqrt{5}}{80} & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{5}}{10} & 0
\end{bmatrix}$

$\boxed{\text{x28}}$ $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{560} & 0 & \frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{7}i}{16} & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{112} & 0 & -\frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{105}i}{80} \\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{80} & 0 & \frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{21}i}{112} & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{16} & 0 & -\frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{35}i}{560} \\
-\frac{\sqrt{35}i}{560} & 0 & \frac{\sqrt{105}i}{80} & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{70}i}{20} & 0 & 0 \\
0 & \frac{\sqrt{21}i}{112} & 0 & -\frac{\sqrt{7}i}{16} & \frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\
-\frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{20} \\
0 & \frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{14}i}{112} & -\frac{\sqrt{70}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 \\
\frac{\sqrt{7}i}{16} & 0 & -\frac{\sqrt{21}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{35}i}{70} \\
0 & -\frac{\sqrt{105}i}{80} & 0 & \frac{\sqrt{35}i}{560} & 0 & 0 & \frac{\sqrt{70}i}{20} & 0 & -\frac{\sqrt{35}i}{70} & 0
\end{bmatrix}$

$\boxed{\text{x29}}$ $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{560} & 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{7}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{112} & 0 & \frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{105}}{80} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{80} & 0 & \frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{21}}{112} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{16} & 0 & -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{35}}{560} \\ \frac{\sqrt{35}}{560} & 0 & \frac{\sqrt{105}}{80} & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{70}}{20} & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{112} & 0 & -\frac{\sqrt{7}}{16} & \frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}}{20} \\ 0 & \frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{14}}{112} & \frac{\sqrt{70}}{20} & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\ -\frac{\sqrt{7}}{16} & 0 & -\frac{\sqrt{21}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{35}}{70} \\ 0 & \frac{\sqrt{105}}{80} & 0 & \frac{\sqrt{35}}{560} & 0 & 0 & -\frac{\sqrt{70}}{20} & 0 & -\frac{\sqrt{35}}{70} & 0 \end{bmatrix}$

$\boxed{\text{x30}}$ $\mathbb{Q}_{0}^{(1,1;a)}(A_{1g}) = \begin{bmatrix} -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} \end{bmatrix}$

$\boxed{\text{x31}}$ $\mathbb{Q}_2^{(1,1;a)}(A_{1g}) = \begin{bmatrix}
\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{70}}{175} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}}{175} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 \\
-\frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{350} & 0 & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 \\
0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{350} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70}
\end{bmatrix}$

$\boxed{\text{x32}}$ $\mathbb{Q}_2^{(1,1;a)}(B_{1g}) = \begin{bmatrix}
0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{70}}{175} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}}{25} & \frac{4\sqrt{21}}{105} & 0 & 0 & 0 & -\frac{8\sqrt{105}}{525} & 0 \\
\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{8\sqrt{105}}{525} & 0 & 0 & 0 & -\frac{4\sqrt{21}}{105} \\
0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}}{175} & 0 & 0 & 0 \\
0 & \frac{4\sqrt{21}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{140} & 0 & 0 & 0 \\
0 & 0 & \frac{8\sqrt{105}}{525} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{700} & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{70}}{175} & -\frac{3\sqrt{14}}{140} & 0 & 0 & 0 & -\frac{9\sqrt{70}}{700} & 0 \\
-\frac{2\sqrt{70}}{175} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{700} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{140} \\
0 & -\frac{8\sqrt{105}}{525} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{700} & 0 & 0 & 0 \\
0 & 0 & -\frac{4\sqrt{21}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{140} & 0 & 0
\end{bmatrix}$

$\boxed{\text{x33}}$ $\mathbb{Q}_2^{(1,1;a)}(B_{2g}) = \begin{bmatrix}
0 & 0 & -\frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}i}{175} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}i}{25} & \frac{4\sqrt{21}i}{105} & 0 & 0 & 0 & \frac{8\sqrt{105}i}{525} & 0 \\
\frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & \frac{8\sqrt{105}i}{525} & 0 & 0 & 0 & \frac{4\sqrt{21}i}{105} \\
0 & \frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}i}{175} & 0 & 0 & 0 \\
0 & -\frac{4\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{140} & 0 & 0 & 0 \\
0 & 0 & -\frac{8\sqrt{105}i}{525} & 0 & 0 & 0 & 0 & \frac{9\sqrt{70}i}{700} & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{70}i}{175} & -\frac{3\sqrt{14}i}{140} & 0 & 0 & 0 & \frac{9\sqrt{70}i}{700} & 0 \\
-\frac{2\sqrt{70}i}{175} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}i}{700} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{140} \\
0 & -\frac{8\sqrt{105}i}{525} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}i}{700} & 0 & 0 & 0 \\
0 & 0 & -\frac{4\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{140} & 0 & 0
\end{bmatrix}$

$\boxed{\text{x34}}$ $\mathbb{Q}_{2,1}^{(1,1;a)}(E_g) = \begin{bmatrix}
0 & -\frac{\sqrt{105}i}{25} & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 & \frac{3\sqrt{70}i}{175} & 0 & 0 & 0 \\
\frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}i}{525} & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}i}{25} & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & \frac{2\sqrt{105}i}{525} & 0 \\
0 & 0 & -\frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{175} & 0 & -\frac{2\sqrt{7}i}{35} \\
-\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & 0 & 0 \\
0 & \frac{2\sqrt{105}i}{525} & 0 & 0 & -\frac{3\sqrt{7}i}{70} & 0 & \frac{3\sqrt{70}i}{350} & 0 & 0 & 0 \\
-\frac{3\sqrt{70}i}{175} & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 & -\frac{3\sqrt{70}i}{350} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{210}i}{105} & 0 & \frac{3\sqrt{70}i}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{350} & 0 \\
0 & 0 & -\frac{2\sqrt{105}i}{525} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{350} & 0 & -\frac{3\sqrt{7}i}{70} \\
0 & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0
\end{bmatrix}$

$\boxed{\text{x35}}$ $\mathbb{Q}_{2,2}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{105}}{25} & 0 & 0 & \frac{2\sqrt{7}}{35} & 0 & -\frac{3\sqrt{70}}{175} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{525} & 0 & -\frac{\sqrt{210}}{105} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{25} & 0 & 0 & -\frac{\sqrt{210}}{105} & 0 & -\frac{2\sqrt{105}}{525} & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{175} & 0 & \frac{2\sqrt{7}}{35} \\ \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{105}}{525} & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 & -\frac{3\sqrt{70}}{350} & 0 & 0 & 0 \\ -\frac{3\sqrt{70}}{175} & 0 & -\frac{\sqrt{210}}{105} & 0 & 0 & -\frac{3\sqrt{70}}{350} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{105} & 0 & -\frac{3\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{350} & 0 \\ 0 & 0 & -\frac{2\sqrt{105}}{525} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{350} & 0 & \frac{3\sqrt{7}}{70} \\ 0 & 0 & 0 & \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{70} & 0 \end{bmatrix}$

$\boxed{\text{x36}}$ $\mathbb{G}_{1}^{(1,0;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x37}}$ $\mathbb{G}_3^{(1,0;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x38}}$ $\mathbb{G}_3^{(1,0;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x39}}$ $\mathbb{G}_3^{(1,0;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x40}}$ $\mathbb{G}_{1,1}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{30}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{2}i}{4} \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{20} & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x41}}$ $\mathbb{G}_{1,2}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{2}}{4} \\ \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{20} & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x42}}$ $\mathbb{G}_{3,1}^{(1,0;a)}(E_g, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{16} & 0 & -\frac{3\sqrt{30}i}{80} & 0 & \frac{\sqrt{15}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{5}i}{80} & 0 & \frac{\sqrt{10}i}{80} & 0 & \frac{5i}{16} \\ 0 & 0 & 0 & 0 & -\frac{5i}{16} & 0 & -\frac{\sqrt{10}i}{80} & 0 & \frac{7\sqrt{5}i}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{16} & 0 & \frac{3\sqrt{30}i}{80} & 0 & -\frac{\sqrt{3}i}{16} \\ -\frac{\sqrt{3}i}{16} & 0 & \frac{5i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{7\sqrt{5}i}{80} & 0 & \frac{\sqrt{15}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{30}i}{80} & 0 & \frac{\sqrt{10}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{80} & 0 & -\frac{3\sqrt{30}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{16} & 0 & -\frac{7\sqrt{5}i}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5i}{16} & 0 & \frac{\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x43}}$ $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{16} & 0 & \frac{3\sqrt{30}}{80} & 0 & \frac{\sqrt{15}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{5}}{80} & 0 & -\frac{\sqrt{10}}{80} & 0 & \frac{5}{16} \\ 0 & 0 & 0 & 0 & \frac{5}{16} & 0 & -\frac{\sqrt{10}}{80} & 0 & -\frac{7\sqrt{5}}{80} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{16} & 0 & \frac{3\sqrt{30}}{80} & 0 & \frac{\sqrt{3}}{16} \\ \frac{\sqrt{3}}{16} & 0 & \frac{5}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{5}}{80} & 0 & \frac{\sqrt{15}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{30}}{80} & 0 & -\frac{\sqrt{10}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{80} & 0 & \frac{3\sqrt{30}}{80} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{16} & 0 & -\frac{7\sqrt{5}}{80} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5}{16} & 0 & \frac{\sqrt{3}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x44}}$ $\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{16} & 0 & -\frac{3\sqrt{2}i}{16} & 0 & -\frac{3i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{3}i}{48} & 0 & \frac{\sqrt{6}i}{48} & 0 & -\frac{\sqrt{15}i}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{16} & 0 & -\frac{\sqrt{6}i}{48} & 0 & \frac{7\sqrt{3}i}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3i}{16} & 0 & \frac{3\sqrt{2}i}{16} & 0 & -\frac{\sqrt{5}i}{16} \\ -\frac{\sqrt{5}i}{16} & 0 & -\frac{\sqrt{15}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{7\sqrt{3}i}{48} & 0 & -\frac{3i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{2}i}{16} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{48} & 0 & -\frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3i}{16} & 0 & -\frac{7\sqrt{3}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{16} & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x45}}$ $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{16} & 0 & \frac{3\sqrt{2}}{16} & 0 & -\frac{3}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{3}}{48} & 0 & -\frac{\sqrt{6}}{48} & 0 & -\frac{\sqrt{15}}{16} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{16} & 0 & -\frac{\sqrt{6}}{48} & 0 & -\frac{7\sqrt{3}}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3}{16} & 0 & \frac{3\sqrt{2}}{16} & 0 & \frac{\sqrt{5}}{16} \\ \frac{\sqrt{5}}{16} & 0 & -\frac{\sqrt{15}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{3}}{48} & 0 & -\frac{3}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{2}}{16} & 0 & -\frac{\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{48} & 0 & \frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3}{16} & 0 & -\frac{7\sqrt{3}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{16} & 0 & \frac{\sqrt{5}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x46}}$ $\mathbb{M}_1^{(a)}(A_{2g}) = \begin{bmatrix} \frac{9\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} \end{bmatrix}$

$\boxed{\text{x47}}$ $\mathbb{M}_3^{(a)}(A_{2g}) = \begin{bmatrix} -\frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{7\sqrt{5}}{50} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{5}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \end{bmatrix}$

$\boxed{\text{x48}}$ $\mathbb{M}_3^{(a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{5} & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & \frac{i}{10} & 0 \\ -\frac{i}{5} & 0 & 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 \\ -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x49}}$ $\mathbb{M}_3^{(a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{5} & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & -\frac{1}{10} & 0 \\ -\frac{1}{5} & 0 & 0 & 0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 \\ -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & \frac{\sqrt{30}}{20} \\ 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x50}}$ $\mathbb{M}_{1,1}^{(a)}(E_g) = \begin{bmatrix} 0 & \frac{3\sqrt{15}}{50} & 0 & 0 & -\frac{1}{10} & 0 & \frac{\sqrt{10}}{100} & 0 & 0 & 0 \\ \frac{3\sqrt{15}}{50} & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & -\frac{\sqrt{15}}{50} & 0 & \frac{\sqrt{30}}{100} & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{25} & 0 & \frac{3\sqrt{15}}{50} & 0 & 0 & -\frac{\sqrt{30}}{100} & 0 & \frac{\sqrt{15}}{50} & 0 \\ 0 & 0 & \frac{3\sqrt{15}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{100} & 0 & \frac{1}{10} \\ -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{50} & 0 & 0 & \frac{1}{5} & 0 & \frac{2\sqrt{10}}{25} & 0 & 0 & 0 \\ \frac{\sqrt{10}}{100} & 0 & -\frac{\sqrt{30}}{100} & 0 & 0 & \frac{2\sqrt{10}}{25} & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{100} & 0 & -\frac{\sqrt{10}}{100} & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & \frac{2\sqrt{10}}{25} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{50} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{25} & 0 & \frac{1}{5} \\ 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 \end{bmatrix}$

$\boxed{\text{x51}}$ $\mathbb{M}^{(a)}_{1,2}(E_g) = \begin{bmatrix} 0 & \frac{3\sqrt{15}i}{50} & 0 & 0 & \frac{i}{10} & 0 & \frac{\sqrt{10}i}{100} & 0 & 0 & 0 \\ -\frac{3\sqrt{15}i}{50} & 0 & \frac{3\sqrt{5}i}{25} & 0 & 0 & \frac{\sqrt{15}i}{50} & 0 & \frac{\sqrt{30}i}{100} & 0 & 0 \\ 0 & -\frac{3\sqrt{5}i}{25} & 0 & \frac{3\sqrt{15}i}{50} & 0 & 0 & \frac{\sqrt{30}i}{100} & 0 & \frac{\sqrt{15}i}{50} & 0 \\ 0 & 0 & -\frac{3\sqrt{15}i}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{100} & 0 & \frac{i}{10} \\ -\frac{i}{10} & 0 & 0 & 0 & 0 & \frac{i}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{50} & 0 & 0 & -\frac{i}{5} & 0 & \frac{2\sqrt{10}i}{25} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{100} & 0 & -\frac{\sqrt{30}i}{100} & 0 & 0 & -\frac{2\sqrt{10}i}{25} & 0 & \frac{3\sqrt{5}i}{25} & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{100} & 0 & -\frac{\sqrt{10}i}{100} & 0 & 0 & -\frac{3\sqrt{5}i}{25} & 0 & \frac{2\sqrt{10}i}{25} & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}i}{25} & 0 & \frac{i}{5} \\ 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 \end{bmatrix}$

$\boxed{\text{x52}}$ $\mathbb{M}^{(a)}_{3,1}(E_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{15}}{50} & 0 & -\frac{\sqrt{5}}{10} & -\frac{3}{40} & 0 & \frac{9\sqrt{10}}{200} & 0 & -\frac{3\sqrt{5}}{40} & 0 \\ \frac{\sqrt{15}}{50} & 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & \frac{7\sqrt{15}}{200} & 0 & -\frac{\sqrt{30}}{200} & 0 & -\frac{\sqrt{3}}{8} \\ 0 & -\frac{3\sqrt{5}}{50} & 0 & \frac{\sqrt{15}}{50} & \frac{\sqrt{3}}{8} & 0 & \frac{\sqrt{30}}{200} & 0 & -\frac{7\sqrt{15}}{200} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{15}}{50} & 0 & 0 & \frac{3\sqrt{5}}{40} & 0 & -\frac{9\sqrt{10}}{200} & 0 & \frac{3}{40} \\ -\frac{3}{40} & 0 & \frac{\sqrt{3}}{8} & 0 & 0 & \frac{3}{20} & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 \\ 0 & \frac{7\sqrt{15}}{200} & 0 & \frac{3\sqrt{5}}{40} & \frac{3}{20} & 0 & -\frac{3\sqrt{10}}{200} & 0 & -\frac{\sqrt{5}}{10} & 0 \\ \frac{9\sqrt{10}}{200} & 0 & \frac{\sqrt{30}}{200} & 0 & 0 & -\frac{3\sqrt{10}}{200} & 0 & -\frac{3\sqrt{5}}{50} & 0 & -\frac{\sqrt{2}}{8} \\ 0 & -\frac{\sqrt{30}}{200} & 0 & -\frac{9\sqrt{10}}{200} & -\frac{\sqrt{2}}{8} & 0 & -\frac{3\sqrt{5}}{50} & 0 & -\frac{3\sqrt{10}}{200} & 0 \\ -\frac{3\sqrt{5}}{40} & 0 & -\frac{7\sqrt{15}}{200} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & -\frac{3\sqrt{10}}{200} & 0 & \frac{3}{20} \\ 0 & -\frac{\sqrt{3}}{8} & 0 & \frac{3}{40} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & \frac{3}{20} & 0 \end{bmatrix}$

$\boxed{\text{x53}}$ $\mathbb{M}^{(a)}_{3,2}(E_g,1) = \begin{bmatrix} 0 & \frac{\sqrt{15}i}{50} & 0 & \frac{\sqrt{5}i}{10} & \frac{3i}{40} & 0 & \frac{9\sqrt{10}i}{200} & 0 & \frac{3\sqrt{5}i}{40} & 0 \\ -\frac{\sqrt{15}i}{50} & 0 & -\frac{3\sqrt{5}i}{50} & 0 & 0 & -\frac{7\sqrt{15}i}{200} & 0 & -\frac{\sqrt{30}i}{200} & 0 & \frac{\sqrt{3}i}{8} \\ 0 & \frac{3\sqrt{5}i}{50} & 0 & \frac{\sqrt{15}i}{50} & \frac{\sqrt{3}i}{8} & 0 & -\frac{\sqrt{30}i}{200} & 0 & -\frac{7\sqrt{15}i}{200} & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{15}i}{50} & 0 & 0 & \frac{3\sqrt{5}i}{40} & 0 & \frac{9\sqrt{10}i}{200} & 0 & \frac{3i}{40} \\ -\frac{3i}{40} & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & \frac{3i}{20} & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 \\ 0 & \frac{7\sqrt{15}i}{200} & 0 & -\frac{3\sqrt{5}i}{40} & -\frac{3i}{20} & 0 & -\frac{3\sqrt{10}i}{200} & 0 & \frac{\sqrt{5}i}{10} & 0 \\ -\frac{9\sqrt{10}i}{200} & 0 & \frac{\sqrt{30}i}{200} & 0 & 0 & \frac{3\sqrt{10}i}{200} & 0 & -\frac{3\sqrt{5}i}{50} & 0 & \frac{\sqrt{2}i}{8} \\ 0 & \frac{\sqrt{30}i}{200} & 0 & -\frac{9\sqrt{10}i}{200} & -\frac{\sqrt{2}i}{8} & 0 & \frac{3\sqrt{5}i}{50} & 0 & -\frac{3\sqrt{10}i}{200} & 0 \\ -\frac{3\sqrt{5}i}{40} & 0 & \frac{7\sqrt{15}i}{200} & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & \frac{3\sqrt{10}i}{200} & 0 & \frac{3i}{20} \\ 0 & -\frac{\sqrt{3}i}{8} & 0 & -\frac{3i}{40} & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & -\frac{3i}{20} & 0 \end{bmatrix}$

$\boxed{\text{x54}}$ $\mathbb{M}^{(a)}_{3,1}(E_g,2) = \begin{bmatrix} 0 & \frac{1}{10} & 0 & \frac{\sqrt{3}}{10} & -\frac{\sqrt{15}}{40} & 0 & \frac{3\sqrt{6}}{40} & 0 & \frac{3\sqrt{3}}{40} & 0 \\ \frac{1}{10} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & \frac{7}{40} & 0 & -\frac{\sqrt{2}}{40} & 0 & \frac{3\sqrt{5}}{40} \\ 0 & -\frac{\sqrt{3}}{10} & 0 & \frac{1}{10} & -\frac{3\sqrt{5}}{40} & 0 & \frac{\sqrt{2}}{40} & 0 & -\frac{7}{40} & 0 \\ \frac{\sqrt{3}}{10} & 0 & \frac{1}{10} & 0 & 0 & -\frac{3\sqrt{3}}{40} & 0 & -\frac{3\sqrt{6}}{40} & 0 & \frac{\sqrt{15}}{40} \\ -\frac{\sqrt{15}}{40} & 0 & -\frac{3\sqrt{5}}{40} & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & \frac{\sqrt{30}}{40} & 0 & 0 \\ 0 & \frac{7}{40} & 0 & -\frac{3\sqrt{3}}{40} & \frac{\sqrt{15}}{20} & 0 & -\frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{3}}{10} & 0 \\ \frac{3\sqrt{6}}{40} & 0 & \frac{\sqrt{2}}{40} & 0 & 0 & -\frac{\sqrt{6}}{40} & 0 & -\frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{30}}{40} \\ 0 & -\frac{\sqrt{2}}{40} & 0 & -\frac{3\sqrt{6}}{40} & \frac{\sqrt{30}}{40} & 0 & -\frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{6}}{40} & 0 \\ \frac{3\sqrt{3}}{40} & 0 & -\frac{7}{40} & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{6}}{40} & 0 & \frac{\sqrt{15}}{20} \\ 0 & \frac{3\sqrt{5}}{40} & 0 & \frac{\sqrt{15}}{40} & 0 & 0 & \frac{\sqrt{30}}{40} & 0 & \frac{\sqrt{15}}{20} & 0 \end{bmatrix}$

$\boxed{\text{x55}}$ $\mathbb{M}_{3,2}^{(a)}(E_g,2) = \begin{bmatrix} 0 & \frac{i}{10} & 0 & -\frac{\sqrt{3}i}{10} & \frac{\sqrt{15}i}{40} & 0 & \frac{3\sqrt{6}i}{40} & 0 & -\frac{3\sqrt{3}i}{40} & 0 \\ -\frac{i}{10} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & -\frac{7i}{40} & 0 & -\frac{\sqrt{2}i}{40} & 0 & -\frac{3\sqrt{5}i}{40} \\ 0 & \frac{\sqrt{3}i}{10} & 0 & \frac{i}{10} & -\frac{3\sqrt{5}i}{40} & 0 & -\frac{\sqrt{2}i}{40} & 0 & -\frac{7i}{40} & 0 \\ \frac{\sqrt{3}i}{10} & 0 & -\frac{i}{10} & 0 & 0 & -\frac{3\sqrt{3}i}{40} & 0 & \frac{3\sqrt{6}i}{40} & 0 & \frac{\sqrt{15}i}{40} \\ -\frac{\sqrt{15}i}{40} & 0 & \frac{3\sqrt{5}i}{40} & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 \\ 0 & \frac{7i}{40} & 0 & \frac{3\sqrt{3}i}{40} & -\frac{\sqrt{15}i}{20} & 0 & -\frac{\sqrt{6}i}{40} & 0 & -\frac{\sqrt{3}i}{10} & 0 \\ -\frac{3\sqrt{6}i}{40} & 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & \frac{\sqrt{6}i}{40} & 0 & -\frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{30}i}{40} \\ 0 & \frac{\sqrt{2}i}{40} & 0 & -\frac{3\sqrt{6}i}{40} & \frac{\sqrt{30}i}{40} & 0 & \frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{6}i}{40} & 0 \\ \frac{3\sqrt{3}i}{40} & 0 & \frac{7i}{40} & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{6}i}{40} & 0 & \frac{\sqrt{15}i}{20} \\ 0 & \frac{3\sqrt{5}i}{40} & 0 & -\frac{\sqrt{15}i}{40} & 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & -\frac{\sqrt{15}i}{20} & 0 \end{bmatrix}$

$\boxed{\text{x56}}$ $\mathbb{M}_{5}^{(1,-1;a)}(A_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x57}}$ $\mathbb{M}_1^{(1,-1;a)}(A_{2g}) =$

$$\begin{bmatrix}
-\frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\
-\frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{50} & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{50} & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{50} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10}
\end{bmatrix}$$

$\boxed{\text{x58}}$ $\mathbb{M}_3^{(1,-1;a)}(A_{2g}) =$

$$\begin{bmatrix}
\frac{\sqrt{105}}{350} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{105}}{350} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{105}}{350} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}}{350} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 \\
\frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 \\
0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 \\
0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{175} & 0 & 0 \\
0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{25} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35}
\end{bmatrix}$$

$\boxed{\text{x59}}$ $\mathbb{M}_5^{(1,-1;a)}(A_{2g},1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} \end{bmatrix}$

$\boxed{\text{x60}}$ $\mathbb{M}_5^{(1,-1;a)}(A_{2g},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x61}}$ $\mathbb{M}_3^{(1,-1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}i}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{70} & -\frac{2\sqrt{105}i}{105} & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{105} & 0 \\ \frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & \frac{2\sqrt{105}i}{105} \\ 0 & -\frac{\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}i}{35} & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{105}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{14}i}{35} & -\frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{70} & 0 \\ \frac{2\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{70} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{70} \\ 0 & \frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{105}i}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x62}}$ $\mathbb{M}_5^{(1,-1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & \frac{\sqrt{30}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x63}}$ $\mathbb{M}_3^{(1,-1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{70} & -\frac{2\sqrt{105}}{105} & 0 & 0 & 0 & \frac{2\sqrt{21}}{105} & 0 \\ \frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{105} & 0 & 0 & 0 & -\frac{2\sqrt{105}}{105} \\ 0 & -\frac{\sqrt{21}}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{105}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{70} & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & -\frac{3\sqrt{70}}{70} & 0 & 0 & 0 & \frac{3\sqrt{14}}{70} & 0 \\ \frac{2\sqrt{14}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{70} & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} \\ 0 & \frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{105}}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x64}}$ $\mathbb{M}_5^{(1,-1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & \frac{\sqrt{30}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{12} & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x65}}$ $\mathbb{M}_{1,1}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & -\frac{\sqrt{5}}{25} & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{50} & 0 & -\frac{\sqrt{10}}{25} & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & -\frac{\sqrt{15}}{25} & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{25} & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & \frac{\sqrt{15}}{25} & 0 & -\frac{\sqrt{30}}{25} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & -\frac{\sqrt{2}}{5} \\ \frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{25} & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & \frac{2\sqrt{5}}{25} & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{25} & 0 & \frac{\sqrt{15}}{25} & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{25} & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & \frac{2\sqrt{5}}{25} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & \frac{\sqrt{2}}{10} \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 \end{bmatrix}$

$\boxed{\text{x66}}$ $\mathbb{M}_{1,2}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & -\frac{\sqrt{30}i}{50} & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & -\frac{\sqrt{5}i}{25} & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{50} & 0 & -\frac{\sqrt{10}i}{25} & 0 & 0 & -\frac{\sqrt{30}i}{25} & 0 & -\frac{\sqrt{15}i}{25} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{25} & 0 & -\frac{\sqrt{30}i}{50} & 0 & 0 & -\frac{\sqrt{15}i}{25} & 0 & -\frac{\sqrt{30}i}{25} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{25} & 0 & -\frac{\sqrt{2}i}{5} \\ \frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{25} & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & \frac{2\sqrt{5}i}{25} & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{25} & 0 & \frac{\sqrt{15}i}{25} & 0 & 0 & -\frac{2\sqrt{5}i}{25} & 0 & \frac{3\sqrt{10}i}{50} & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{25} & 0 & \frac{\sqrt{5}i}{25} & 0 & 0 & -\frac{3\sqrt{10}i}{50} & 0 & \frac{2\sqrt{5}i}{25} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{25} & 0 & \frac{\sqrt{2}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 \end{bmatrix}$

$\boxed{\text{x67}}$ $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1) = \begin{bmatrix}
0 & -\frac{3\sqrt{35}}{700} & 0 & \frac{\sqrt{105}}{140} & \frac{\sqrt{21}}{70} & 0 & -\frac{3\sqrt{210}}{350} & 0 & \frac{\sqrt{105}}{70} & 0 \\
-\frac{3\sqrt{35}}{700} & 0 & \frac{3\sqrt{105}}{700} & 0 & 0 & -\frac{\sqrt{35}}{50} & 0 & \frac{\sqrt{70}}{350} & 0 & \frac{\sqrt{7}}{14} \\
0 & \frac{3\sqrt{105}}{700} & 0 & -\frac{3\sqrt{35}}{700} & -\frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{70}}{350} & 0 & \frac{\sqrt{35}}{50} & 0 \\
\frac{\sqrt{105}}{140} & 0 & -\frac{3\sqrt{35}}{700} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & \frac{3\sqrt{210}}{350} & 0 & -\frac{\sqrt{21}}{70} \\
\frac{\sqrt{21}}{70} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{3\sqrt{21}}{70} & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 \\
0 & -\frac{\sqrt{35}}{50} & 0 & -\frac{\sqrt{105}}{70} & \frac{3\sqrt{21}}{70} & 0 & -\frac{3\sqrt{210}}{700} & 0 & -\frac{\sqrt{105}}{35} & 0 \\
-\frac{3\sqrt{210}}{350} & 0 & -\frac{\sqrt{70}}{350} & 0 & 0 & -\frac{3\sqrt{210}}{700} & 0 & -\frac{3\sqrt{105}}{175} & 0 & -\frac{\sqrt{42}}{28} \\
0 & \frac{\sqrt{70}}{350} & 0 & \frac{3\sqrt{210}}{350} & -\frac{\sqrt{42}}{28} & 0 & -\frac{3\sqrt{105}}{175} & 0 & -\frac{3\sqrt{210}}{700} & 0 \\
\frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{35}}{50} & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & -\frac{3\sqrt{210}}{700} & 0 & \frac{3\sqrt{21}}{70} \\
0 & \frac{\sqrt{7}}{14} & 0 & -\frac{\sqrt{21}}{70} & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & \frac{3\sqrt{21}}{70} & 0
\end{bmatrix}$

$\boxed{\text{x68}}$ $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1) = \begin{bmatrix}
0 & -\frac{3\sqrt{35}i}{700} & 0 & -\frac{\sqrt{105}i}{140} & -\frac{\sqrt{21}i}{70} & 0 & -\frac{3\sqrt{210}i}{350} & 0 & -\frac{\sqrt{105}i}{70} & 0 \\
\frac{3\sqrt{35}i}{700} & 0 & \frac{3\sqrt{105}i}{700} & 0 & 0 & \frac{\sqrt{35}i}{50} & 0 & \frac{\sqrt{70}i}{350} & 0 & -\frac{\sqrt{7}i}{14} \\
0 & -\frac{3\sqrt{105}i}{700} & 0 & -\frac{3\sqrt{35}i}{700} & -\frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{70}i}{350} & 0 & \frac{\sqrt{35}i}{50} & 0 \\
\frac{\sqrt{105}i}{140} & 0 & \frac{3\sqrt{35}i}{700} & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & -\frac{3\sqrt{210}i}{350} & 0 & -\frac{\sqrt{21}i}{70} \\
\frac{\sqrt{21}i}{70} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & \frac{3\sqrt{21}i}{70} & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{50} & 0 & \frac{\sqrt{105}i}{70} & -\frac{3\sqrt{21}i}{70} & 0 & -\frac{3\sqrt{210}i}{700} & 0 & \frac{\sqrt{105}i}{35} & 0 \\
\frac{3\sqrt{210}i}{350} & 0 & -\frac{\sqrt{70}i}{350} & 0 & 0 & \frac{3\sqrt{210}i}{700} & 0 & -\frac{3\sqrt{105}i}{175} & 0 & \frac{\sqrt{42}i}{28} \\
0 & -\frac{\sqrt{70}i}{350} & 0 & \frac{3\sqrt{210}i}{350} & -\frac{\sqrt{42}i}{28} & 0 & \frac{3\sqrt{105}i}{175} & 0 & -\frac{3\sqrt{210}i}{700} & 0 \\
\frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{35}i}{50} & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & \frac{3\sqrt{210}i}{700} & 0 & \frac{3\sqrt{21}i}{70} \\
0 & \frac{\sqrt{7}i}{14} & 0 & \frac{\sqrt{21}i}{70} & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & -\frac{3\sqrt{21}i}{70} & 0
\end{bmatrix}$

$\boxed{\text{x69}}$ $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2) = \begin{bmatrix}
0 & -\frac{\sqrt{21}}{140} & 0 & -\frac{3\sqrt{7}}{140} & \frac{\sqrt{35}}{70} & 0 & -\frac{3\sqrt{14}}{70} & 0 & -\frac{3\sqrt{7}}{70} & 0 \\
-\frac{\sqrt{21}}{140} & 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & -\frac{\sqrt{21}}{30} & 0 & \frac{\sqrt{42}}{210} & 0 & -\frac{\sqrt{105}}{70} \\
0 & \frac{3\sqrt{7}}{140} & 0 & -\frac{\sqrt{21}}{140} & \frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{42}}{210} & 0 & \frac{\sqrt{21}}{30} & 0 \\
-\frac{3\sqrt{7}}{140} & 0 & -\frac{\sqrt{21}}{140} & 0 & 0 & \frac{3\sqrt{7}}{70} & 0 & \frac{3\sqrt{14}}{70} & 0 & -\frac{\sqrt{35}}{70} \\
\frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 \\
0 & -\frac{\sqrt{21}}{30} & 0 & \frac{3\sqrt{7}}{70} & \frac{3\sqrt{35}}{70} & 0 & -\frac{3\sqrt{14}}{140} & 0 & \frac{3\sqrt{7}}{35} & 0 \\
-\frac{3\sqrt{14}}{70} & 0 & -\frac{\sqrt{42}}{210} & 0 & 0 & -\frac{3\sqrt{14}}{140} & 0 & -\frac{3\sqrt{7}}{35} & 0 & \frac{3\sqrt{70}}{140} \\
0 & \frac{\sqrt{42}}{210} & 0 & \frac{3\sqrt{14}}{70} & \frac{3\sqrt{70}}{140} & 0 & -\frac{3\sqrt{7}}{35} & 0 & -\frac{3\sqrt{14}}{140} & 0 \\
-\frac{3\sqrt{7}}{70} & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & \frac{3\sqrt{7}}{35} & 0 & -\frac{3\sqrt{14}}{140} & 0 & \frac{3\sqrt{35}}{70} \\
0 & -\frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & \frac{3\sqrt{35}}{70} & 0
\end{bmatrix}$

$\boxed{\text{x70}}$ $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2) = \begin{bmatrix}
0 & -\frac{\sqrt{21}i}{140} & 0 & \frac{3\sqrt{7}i}{140} & -\frac{\sqrt{35}i}{70} & 0 & -\frac{3\sqrt{14}i}{70} & 0 & \frac{3\sqrt{7}i}{70} & 0 \\
\frac{\sqrt{21}i}{140} & 0 & \frac{3\sqrt{7}i}{140} & 0 & 0 & \frac{\sqrt{21}i}{30} & 0 & \frac{\sqrt{42}i}{210} & 0 & \frac{\sqrt{105}i}{70} \\
0 & -\frac{3\sqrt{7}i}{140} & 0 & -\frac{\sqrt{21}i}{140} & \frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{42}i}{210} & 0 & \frac{\sqrt{21}i}{30} & 0 \\
-\frac{3\sqrt{7}i}{140} & 0 & \frac{\sqrt{21}i}{140} & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 & -\frac{3\sqrt{14}i}{70} & 0 & -\frac{\sqrt{35}i}{70} \\
\frac{\sqrt{35}i}{70} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & -\frac{3\sqrt{70}i}{140} & 0 & 0 \\
0 & -\frac{\sqrt{21}i}{30} & 0 & -\frac{3\sqrt{7}i}{70} & -\frac{3\sqrt{35}i}{70} & 0 & -\frac{3\sqrt{14}i}{140} & 0 & -\frac{3\sqrt{7}i}{35} & 0 \\
\frac{3\sqrt{14}i}{70} & 0 & -\frac{\sqrt{42}i}{210} & 0 & 0 & \frac{3\sqrt{14}i}{140} & 0 & -\frac{3\sqrt{7}i}{35} & 0 & -\frac{3\sqrt{70}i}{140} \\
0 & -\frac{\sqrt{42}i}{210} & 0 & \frac{3\sqrt{14}i}{70} & \frac{3\sqrt{70}i}{140} & 0 & \frac{3\sqrt{7}i}{35} & 0 & -\frac{3\sqrt{14}i}{140} & 0 \\
-\frac{3\sqrt{7}i}{70} & 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & \frac{3\sqrt{7}i}{35} & 0 & \frac{3\sqrt{14}i}{140} & 0 & \frac{3\sqrt{35}i}{70} \\
0 & -\frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & \frac{3\sqrt{70}i}{140} & 0 & -\frac{3\sqrt{35}i}{70} & 0
\end{bmatrix}$

$\boxed{\text{x71}}$ $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{112} & 0 & -\frac{\sqrt{70}}{48} & 0 & \frac{3\sqrt{7}}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{112} & 0 & -\frac{5\sqrt{14}}{112} & 0 & \frac{5\sqrt{7}}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{112} & 0 & \frac{5\sqrt{7}}{56} & 0 & -\frac{\sqrt{70}}{48} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{48} & 0 & \frac{5\sqrt{7}}{56} & 0 & -\frac{5\sqrt{14}}{112} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{48} & 0 & -\frac{5\sqrt{14}}{112} & 0 & \frac{\sqrt{35}}{112} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{16} & 0 & -\frac{\sqrt{70}}{48} & 0 & \frac{\sqrt{35}}{112} & 0 \end{bmatrix}$

$\boxed{\text{x72}}$ $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{112} & 0 & \frac{\sqrt{70}i}{48} & 0 & \frac{3\sqrt{7}i}{16} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{112} & 0 & -\frac{5\sqrt{14}i}{112} & 0 & -\frac{5\sqrt{7}i}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{112} & 0 & \frac{5\sqrt{7}i}{56} & 0 & \frac{\sqrt{70}i}{48} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{48} & 0 & -\frac{5\sqrt{7}i}{56} & 0 & -\frac{5\sqrt{14}i}{112} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{48} & 0 & \frac{5\sqrt{14}i}{112} & 0 & \frac{\sqrt{35}i}{112} \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{16} & 0 & -\frac{\sqrt{70}i}{48} & 0 & -\frac{\sqrt{35}i}{112} & 0 \end{bmatrix}$

$\boxed{\text{x73}}$ $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{16} & 0 & \frac{3\sqrt{2}}{16} & 0 & \frac{\sqrt{5}}{16} \\ 0 & 0 & 0 & 0 & \frac{1}{16} & 0 & -\frac{\sqrt{10}}{16} & 0 & -\frac{3\sqrt{5}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{5}}{8} & 0 & \frac{3\sqrt{2}}{16} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{16} & 0 & \frac{\sqrt{5}}{8} & 0 & -\frac{\sqrt{10}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{16} & 0 & -\frac{\sqrt{10}}{16} & 0 & \frac{1}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{16} & 0 & \frac{3\sqrt{2}}{16} & 0 & \frac{1}{16} & 0 \end{bmatrix}$

$\boxed{\text{x74}}$ $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{16} & 0 & -\frac{3\sqrt{2}i}{16} & 0 & \frac{\sqrt{5}i}{16} \\ 0 & 0 & 0 & 0 & -\frac{i}{16} & 0 & -\frac{\sqrt{10}i}{16} & 0 & \frac{3\sqrt{5}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & \frac{\sqrt{5}i}{8} & 0 & -\frac{3\sqrt{2}i}{16} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{16} & 0 & -\frac{\sqrt{5}i}{8} & 0 & -\frac{\sqrt{10}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & \frac{i}{16} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{16} & 0 & \frac{3\sqrt{2}i}{16} & 0 & -\frac{i}{16} & 0 \end{bmatrix}$

$\boxed{\text{x75}}$ $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & -\frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{15}}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & -\frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{15}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{15}}{12} & 0 & -\frac{\sqrt{6}}{24} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{15}}{12} & 0 & -\frac{\sqrt{30}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & -\frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{3}}{24} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{8} & 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{3}}{24} & 0 \end{bmatrix}$

$\boxed{\text{x76}}$ $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{24} & 0 & \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{15}i}{8} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{24} & 0 & -\frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{15}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & \frac{\sqrt{15}i}{12} & 0 & \frac{\sqrt{6}i}{24} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{15}i}{12} & 0 & -\frac{\sqrt{30}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & \frac{\sqrt{30}i}{24} & 0 & \frac{\sqrt{3}i}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{8} & 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{3}i}{24} & 0 \end{bmatrix}$

$\boxed{\text{x77}}$ $\mathbb{M}_1^{(1,1;a)}(A_{2g}) = \begin{bmatrix} \frac{3\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{50} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{175} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{175} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{175} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} \end{bmatrix}$

$\boxed{\text{x78}}$ $\mathbb{M}_3^{(1,1;a)}(A_{2g}) = \begin{bmatrix} -\frac{6\sqrt{35}}{175} & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 \\ 0 & \frac{18\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 & 0 \\ 0 & 0 & -\frac{18\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 \\ 0 & 0 & 0 & \frac{6\sqrt{35}}{175} & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{75} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{525} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}}{525} & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{75} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{105} \end{bmatrix}$

$\boxed{\text{x79}}$ $\mathbb{M}_3^{(1,1;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & \frac{6\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 & 0 \\ 0 & 0 & 0 & -\frac{6\sqrt{7}i}{35} & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{140} & 0 \\ -\frac{6\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{35}i}{140} \\ 0 & \frac{6\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{140} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{210} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{210} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & \frac{\sqrt{210}i}{210} & 0 & 0 & 0 & \frac{\sqrt{42}i}{210} & 0 \\ \frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{210} & 0 & 0 & 0 & \frac{\sqrt{210}i}{210} \\ 0 & \frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{210} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{210} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x80}}$ $\mathbb{M}_3^{(1,1;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & -\frac{6\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{140} & 0 & 0 \\ 0 & 0 & 0 & \frac{6\sqrt{7}}{35} & -\frac{3\sqrt{35}}{140} & 0 & 0 & 0 & \frac{3\sqrt{7}}{140} & 0 \\ -\frac{6\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & -\frac{3\sqrt{35}}{140} \\ 0 & \frac{6\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{140} & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{210} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{210} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{42}}{140} & \frac{\sqrt{210}}{210} & 0 & 0 & 0 & -\frac{\sqrt{42}}{210} & 0 \\ \frac{3\sqrt{42}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{210} & 0 & 0 & 0 & -\frac{\sqrt{210}}{210} \\ 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{210} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{210} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x81}}$ $\mathbb{M}_{1,1}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{210}}{50} & 0 & 0 & \frac{\sqrt{14}}{20} & 0 & -\frac{\sqrt{35}}{100} & 0 & 0 & 0 \\ \frac{\sqrt{210}}{50} & 0 & \frac{\sqrt{70}}{25} & 0 & 0 & \frac{\sqrt{210}}{100} & 0 & -\frac{\sqrt{105}}{100} & 0 & 0 \\ 0 & \frac{\sqrt{70}}{25} & 0 & \frac{\sqrt{210}}{50} & 0 & 0 & \frac{\sqrt{105}}{100} & 0 & -\frac{\sqrt{210}}{100} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{100} & 0 & -\frac{\sqrt{14}}{20} \\ \frac{\sqrt{14}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{100} & 0 & 0 & -\frac{\sqrt{14}}{35} & 0 & -\frac{4\sqrt{35}}{175} & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{100} & 0 & \frac{\sqrt{105}}{100} & 0 & 0 & -\frac{4\sqrt{35}}{175} & 0 & -\frac{3\sqrt{70}}{175} & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{100} & 0 & \frac{\sqrt{35}}{100} & 0 & 0 & -\frac{3\sqrt{70}}{175} & 0 & -\frac{4\sqrt{35}}{175} & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{100} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{175} & 0 & -\frac{\sqrt{14}}{35} \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{35} & 0 \end{bmatrix}$

$\boxed{\text{x82}}$ $\mathbb{M}_{1,2}^{(1,1;a)}(E_g) = \begin{bmatrix} 0 & \frac{\sqrt{210}i}{50} & 0 & 0 & -\frac{\sqrt{14}i}{20} & 0 & -\frac{\sqrt{35}i}{100} & 0 & 0 & 0 \\ -\frac{\sqrt{210}i}{50} & 0 & \frac{\sqrt{70}i}{25} & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 & -\frac{\sqrt{105}i}{100} & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{25} & 0 & \frac{\sqrt{210}i}{50} & 0 & 0 & -\frac{\sqrt{105}i}{100} & 0 & -\frac{\sqrt{210}i}{100} & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{100} & 0 & -\frac{\sqrt{14}i}{20} \\ \frac{\sqrt{14}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{100} & 0 & 0 & \frac{\sqrt{14}i}{35} & 0 & -\frac{4\sqrt{35}i}{175} & 0 & 0 & 0 \\ \frac{\sqrt{35}i}{100} & 0 & \frac{\sqrt{105}i}{100} & 0 & 0 & \frac{4\sqrt{35}i}{175} & 0 & -\frac{3\sqrt{70}i}{175} & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{100} & 0 & \frac{\sqrt{35}i}{100} & 0 & 0 & \frac{3\sqrt{70}i}{175} & 0 & -\frac{4\sqrt{35}i}{175} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{100} & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}i}{175} & 0 & -\frac{\sqrt{14}i}{35} \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{35} & 0 \end{bmatrix}$

$\boxed{\text{x83}}$ $\mathbb{M}_{3,1}^{(1,1;a)}(E_g,1) = \begin{bmatrix}
0 & \frac{3\sqrt{105}}{175} & 0 & -\frac{3\sqrt{35}}{35} & \frac{9\sqrt{7}}{560} & 0 & -\frac{27\sqrt{70}}{2800} & 0 & \frac{9\sqrt{35}}{560} & 0 \\
\frac{3\sqrt{105}}{175} & 0 & -\frac{9\sqrt{35}}{175} & 0 & 0 & -\frac{3\sqrt{105}}{400} & 0 & \frac{3\sqrt{210}}{2800} & 0 & \frac{3\sqrt{21}}{112} \\
0 & -\frac{9\sqrt{35}}{175} & 0 & \frac{3\sqrt{105}}{175} & -\frac{3\sqrt{21}}{112} & 0 & -\frac{3\sqrt{210}}{2800} & 0 & \frac{3\sqrt{105}}{400} & 0 \\
-\frac{3\sqrt{35}}{35} & 0 & \frac{3\sqrt{105}}{175} & 0 & 0 & -\frac{9\sqrt{35}}{560} & 0 & \frac{27\sqrt{70}}{2800} & 0 & -\frac{9\sqrt{7}}{560} \\
\frac{9\sqrt{7}}{560} & 0 & -\frac{3\sqrt{21}}{112} & 0 & 0 & -\frac{\sqrt{7}}{70} & 0 & \frac{\sqrt{14}}{84} & 0 & 0 \\
0 & -\frac{3\sqrt{105}}{400} & 0 & -\frac{9\sqrt{35}}{560} & -\frac{\sqrt{7}}{70} & 0 & \frac{\sqrt{70}}{700} & 0 & \frac{\sqrt{35}}{105} & 0 \\
-\frac{27\sqrt{70}}{2800} & 0 & -\frac{3\sqrt{210}}{2800} & 0 & 0 & \frac{\sqrt{70}}{700} & 0 & \frac{\sqrt{35}}{175} & 0 & \frac{\sqrt{14}}{84} \\
0 & \frac{3\sqrt{210}}{2800} & 0 & \frac{27\sqrt{70}}{2800} & \frac{\sqrt{14}}{84} & 0 & \frac{\sqrt{35}}{175} & 0 & \frac{\sqrt{70}}{700} & 0 \\
\frac{9\sqrt{35}}{560} & 0 & \frac{3\sqrt{105}}{400} & 0 & 0 & \frac{\sqrt{35}}{105} & 0 & \frac{\sqrt{70}}{700} & 0 & -\frac{\sqrt{7}}{70} \\
0 & \frac{3\sqrt{21}}{112} & 0 & -\frac{9\sqrt{7}}{560} & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & -\frac{\sqrt{7}}{70} & 0
\end{bmatrix}$

$\boxed{\text{x84}}$ $\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1) = \begin{bmatrix}
0 & \frac{3\sqrt{105}i}{175} & 0 & \frac{3\sqrt{35}i}{35} & -\frac{9\sqrt{7}i}{560} & 0 & -\frac{27\sqrt{70}i}{2800} & 0 & -\frac{9\sqrt{35}i}{560} & 0 \\
-\frac{3\sqrt{105}i}{175} & 0 & -\frac{9\sqrt{35}i}{175} & 0 & 0 & \frac{3\sqrt{105}i}{400} & 0 & \frac{3\sqrt{210}i}{2800} & 0 & -\frac{3\sqrt{21}i}{112} \\
0 & \frac{9\sqrt{35}i}{175} & 0 & \frac{3\sqrt{105}i}{175} & -\frac{3\sqrt{21}i}{112} & 0 & \frac{3\sqrt{210}i}{2800} & 0 & \frac{3\sqrt{105}i}{400} & 0 \\
-\frac{3\sqrt{35}i}{35} & 0 & -\frac{3\sqrt{105}i}{175} & 0 & 0 & -\frac{9\sqrt{35}i}{560} & 0 & -\frac{27\sqrt{70}i}{2800} & 0 & -\frac{9\sqrt{7}i}{560} \\
\frac{9\sqrt{7}i}{560} & 0 & \frac{3\sqrt{21}i}{112} & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 \\
0 & -\frac{3\sqrt{105}i}{400} & 0 & \frac{9\sqrt{35}i}{560} & \frac{\sqrt{7}i}{70} & 0 & \frac{\sqrt{70}i}{700} & 0 & -\frac{\sqrt{35}i}{105} & 0 \\
\frac{27\sqrt{70}i}{2800} & 0 & -\frac{3\sqrt{210}i}{2800} & 0 & 0 & -\frac{\sqrt{70}i}{700} & 0 & \frac{\sqrt{35}i}{175} & 0 & -\frac{\sqrt{14}i}{84} \\
0 & -\frac{3\sqrt{210}i}{2800} & 0 & \frac{27\sqrt{70}i}{2800} & \frac{\sqrt{14}i}{84} & 0 & -\frac{\sqrt{35}i}{175} & 0 & \frac{\sqrt{70}i}{700} & 0 \\
\frac{9\sqrt{35}i}{560} & 0 & -\frac{3\sqrt{105}i}{400} & 0 & 0 & \frac{\sqrt{35}i}{105} & 0 & -\frac{\sqrt{70}i}{700} & 0 & -\frac{\sqrt{7}i}{70} \\
0 & \frac{3\sqrt{21}i}{112} & 0 & \frac{9\sqrt{7}i}{560} & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & \frac{\sqrt{7}i}{70} & 0
\end{bmatrix}$

$\boxed{\text{x85}}$ $\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & \frac{3\sqrt{7}}{35} & 0 & \frac{3\sqrt{21}}{35} & \frac{3\sqrt{105}}{560} & 0 & -\frac{9\sqrt{42}}{560} & 0 & -\frac{9\sqrt{21}}{560} & 0 \\
\frac{3\sqrt{7}}{35} & 0 & -\frac{3\sqrt{21}}{35} & 0 & 0 & -\frac{3\sqrt{7}}{80} & 0 & \frac{3\sqrt{14}}{560} & 0 & -\frac{9\sqrt{35}}{560} \\
0 & -\frac{3\sqrt{21}}{35} & 0 & \frac{3\sqrt{7}}{35} & \frac{9\sqrt{35}}{560} & 0 & -\frac{3\sqrt{14}}{560} & 0 & \frac{3\sqrt{7}}{80} & 0 \\
\frac{3\sqrt{21}}{35} & 0 & \frac{3\sqrt{7}}{35} & 0 & 0 & \frac{9\sqrt{21}}{560} & 0 & \frac{9\sqrt{42}}{560} & 0 & -\frac{3\sqrt{105}}{560} \\
\frac{3\sqrt{105}}{560} & 0 & \frac{9\sqrt{35}}{560} & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 \\
0 & -\frac{3\sqrt{7}}{80} & 0 & \frac{9\sqrt{21}}{560} & -\frac{\sqrt{105}}{210} & 0 & \frac{\sqrt{42}}{420} & 0 & -\frac{\sqrt{21}}{105} & 0 \\
-\frac{9\sqrt{42}}{560} & 0 & -\frac{3\sqrt{14}}{560} & 0 & 0 & \frac{\sqrt{42}}{420} & 0 & \frac{\sqrt{21}}{105} & 0 & -\frac{\sqrt{210}}{420} \\
0 & \frac{3\sqrt{14}}{560} & 0 & \frac{9\sqrt{42}}{560} & -\frac{\sqrt{210}}{420} & 0 & \frac{\sqrt{21}}{105} & 0 & \frac{\sqrt{42}}{420} & 0 \\
-\frac{9\sqrt{21}}{560} & 0 & \frac{3\sqrt{7}}{80} & 0 & 0 & -\frac{\sqrt{21}}{105} & 0 & \frac{\sqrt{42}}{420} & 0 & -\frac{\sqrt{105}}{210} \\
0 & -\frac{9\sqrt{35}}{560} & 0 & -\frac{3\sqrt{105}}{560} & 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & -\frac{\sqrt{105}}{210} & 0
\end{bmatrix}$$

$\boxed{\text{x86}}$ $\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & \frac{3\sqrt{7}i}{35} & 0 & -\frac{3\sqrt{21}i}{35} & -\frac{3\sqrt{105}i}{560} & 0 & -\frac{9\sqrt{42}i}{560} & 0 & \frac{9\sqrt{21}i}{560} & 0 \\
-\frac{3\sqrt{7}i}{35} & 0 & -\frac{3\sqrt{21}i}{35} & 0 & 0 & \frac{3\sqrt{7}i}{80} & 0 & \frac{3\sqrt{14}i}{560} & 0 & \frac{9\sqrt{35}i}{560} \\
0 & \frac{3\sqrt{21}i}{35} & 0 & \frac{3\sqrt{7}i}{35} & \frac{9\sqrt{35}i}{560} & 0 & \frac{3\sqrt{14}i}{560} & 0 & \frac{3\sqrt{7}i}{80} & 0 \\
\frac{3\sqrt{21}i}{35} & 0 & -\frac{3\sqrt{7}i}{35} & 0 & 0 & \frac{9\sqrt{21}i}{560} & 0 & -\frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{105}i}{560} \\
\frac{3\sqrt{105}i}{560} & 0 & -\frac{9\sqrt{35}i}{560} & 0 & 0 & -\frac{\sqrt{105}i}{210} & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 \\
0 & -\frac{3\sqrt{7}i}{80} & 0 & -\frac{9\sqrt{21}i}{560} & \frac{\sqrt{105}i}{210} & 0 & \frac{\sqrt{42}i}{420} & 0 & \frac{\sqrt{21}i}{105} & 0 \\
\frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{14}i}{560} & 0 & 0 & -\frac{\sqrt{42}i}{420} & 0 & \frac{\sqrt{21}i}{105} & 0 & \frac{\sqrt{210}i}{420} \\
0 & -\frac{3\sqrt{14}i}{560} & 0 & \frac{9\sqrt{42}i}{560} & -\frac{\sqrt{210}i}{420} & 0 & -\frac{\sqrt{21}i}{105} & 0 & \frac{\sqrt{42}i}{420} & 0 \\
-\frac{9\sqrt{21}i}{560} & 0 & -\frac{3\sqrt{7}i}{80} & 0 & 0 & -\frac{\sqrt{21}i}{105} & 0 & -\frac{\sqrt{42}i}{420} & 0 & -\frac{\sqrt{105}i}{210} \\
0 & -\frac{9\sqrt{35}i}{560} & 0 & \frac{3\sqrt{105}i}{560} & 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & \frac{\sqrt{105}i}{210} & 0
\end{bmatrix}$$

$\boxed{\text{x87}}$ $\mathbb{T}_2^{(1,0;a)}(A_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x88}}$ $\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x89}}$ $\mathbb{T}_4^{(1,0;a)}(A_{1g},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & \frac{\sqrt{21}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{12} & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x90}}$ $\mathbb{T}_4^{(1,0;a)}(A_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x91}}$ $\mathbb{T}_2^{(1,0;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{21} & 0 & 0 & 0 & \frac{\sqrt{210}i}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x92}}$ $\mathbb{T}_4^{(1,0;a)}(B_{1g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x93}}$ $\mathbb{T}_2^{(1,0;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 & \frac{\sqrt{210}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x94}}$ $\mathbb{T}_4^{(1,0;a)}(B_{2g}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & \frac{\sqrt{105}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x95}}$ $\mathbb{T}_{2,1}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{84} & 0 & \frac{\sqrt{42}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & 0 & -\frac{\sqrt{70}}{28} \\ \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{7}}{28} & 0 & -\frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{21}}{84} & 0 & -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x96}}$ $\mathbb{T}_{2,2}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & \frac{3\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & \frac{5\sqrt{21}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{84} & 0 & \frac{\sqrt{42}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{28} & 0 & -\frac{\sqrt{70}i}{28} \\ \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{7}i}{28} & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{21}i}{84} & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x97}}$ $\mathbb{T}_{4,1}^{(1,0;a)}(E_g,1) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{16} & 0 & \frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{5}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{16} & 0 & -\frac{\sqrt{30}}{16} & 0 & -\frac{\sqrt{3}}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{16} & 0 & \frac{\sqrt{30}}{16} & 0 & \frac{\sqrt{15}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{16} & 0 & -\frac{\sqrt{10}}{16} & 0 & -\frac{1}{16} \\ \frac{1}{16} & 0 & \frac{\sqrt{3}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{16} & 0 & -\frac{\sqrt{5}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{30}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{16} & 0 & -\frac{\sqrt{10}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{15}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{16} & 0 & -\frac{1}{16} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x98}}$ $\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{i}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & -\frac{\sqrt{5}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{16} & 0 & -\frac{\sqrt{30}i}{16} & 0 & \frac{\sqrt{3}i}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{16} & 0 & -\frac{\sqrt{30}i}{16} & 0 & \frac{\sqrt{15}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & -\frac{i}{16} \\ \frac{i}{16} & 0 & -\frac{\sqrt{3}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{16} & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{16} & 0 & \frac{\sqrt{30}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{16} & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{16} & 0 & -\frac{\sqrt{15}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{16} & 0 & \frac{i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x99}}$ $\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{35}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{21}}{16} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{16} & 0 & \frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{105}}{112} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{16} & 0 & -\frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{7}}{112} \\ \frac{\sqrt{7}}{112} & 0 & -\frac{\sqrt{21}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{112} & 0 & \frac{\sqrt{35}}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{210}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{16} & 0 & \frac{\sqrt{105}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{16} & 0 & -\frac{\sqrt{7}}{112} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x100}}$ $\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2) = \begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{112} & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{35}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{21}i}{16} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{16} & 0 & -\frac{\sqrt{210}i}{112} & 0 & \frac{\sqrt{105}i}{112} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{16} & 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{7}i}{112} \\ \frac{\sqrt{7}i}{112} & 0 & \frac{\sqrt{21}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{35}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{210}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{16} & 0 & -\frac{\sqrt{105}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{16} & 0 & \frac{\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

- bra: $\langle \frac{3}{2}, \frac{3}{2}; d|$, $\langle \frac{3}{2}, \frac{1}{2}; d|$, $\langle \frac{3}{2}, -\frac{1}{2}; d|$, $\langle \frac{3}{2}, -\frac{3}{2}; d|$, $\langle \frac{5}{2}, \frac{5}{2}; d|$, $\langle \frac{5}{2}, \frac{3}{2}; d|$, $\langle \frac{5}{2}, \frac{1}{2}; d|$, $\langle \frac{5}{2}, -\frac{1}{2}; d|$, $\langle \frac{5}{2}, -\frac{3}{2}; d|$, $\langle \frac{5}{2}, -\frac{5}{2}; d|$
- ket: $|\frac{5}{2}, \frac{5}{2}; f\rangle$, $|\frac{5}{2}, \frac{3}{2}; f\rangle$, $|\frac{5}{2}, \frac{1}{2}; f\rangle$, $|\frac{5}{2}, -\frac{1}{2}; f\rangle$, $|\frac{5}{2}, -\frac{3}{2}; f\rangle$, $|\frac{5}{2}, -\frac{5}{2}; f\rangle$, $|\frac{7}{2}, \frac{7}{2}; f\rangle$, $|\frac{7}{2}, \frac{5}{2}; f\rangle$, $|\frac{7}{2}, \frac{3}{2}; f\rangle$, $|\frac{7}{2}, \frac{1}{2}; f\rangle$, $|\frac{7}{2}, -\frac{1}{2}; f\rangle$, $|\frac{7}{2}, -\frac{3}{2}; f\rangle$, $|\frac{7}{2}, -\frac{5}{2}; f\rangle$, $|\frac{7}{2}, -\frac{7}{2}; f\rangle$

$\boxed{\text{x101}}$ $\mathbb{Q}_5^{(a)}(A_{1u}) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{10} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x102}}$ $\mathbb{Q}_1^{(a)}(A_{2u}) = \begin{bmatrix}
0 & \frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{1}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{1}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0
\end{bmatrix}$

$\boxed{\text{x103}}$ $\mathbb{Q}_3^{(a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 \\ \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 \end{bmatrix}$

$\boxed{\text{x104}}$ $\mathbb{Q}_5^{(a)}(A_{2u}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 \\ -\frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 \end{bmatrix}$

$\boxed{\text{x105}}$ $\mathbb{Q}_5^{(a)}(A_{2u},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x106}}$ $\mathbb{Q}_3^{(a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{3\sqrt{70}i}{140} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 \\ 0 & 0 & -\frac{3\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} \\ 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 \\ \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x107}}$ $\mathbb{Q}_5^{(a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{60} & 0 & 0 & 0 & \frac{\sqrt{105}i}{60} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{20} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{20} & 0 & 0 & 0 & \frac{\sqrt{7}i}{20} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{60} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{60} \\
0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{105} & 0 & 0 & 0 \\
\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & -\frac{2\sqrt{42}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{70} & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{70} & 0 & 0 & 0 & \frac{2\sqrt{42}i}{105} & 0 \\
0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{30} \\
0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x108}}$ $\mathbb{Q}_3^{(a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 \\
\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 \\
0 & -\frac{\sqrt{105}}{140} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 \\
0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & \frac{\sqrt{6}}{24} \\
0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 \\
\frac{\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\
0 & \frac{\sqrt{70}}{140} & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 \\
0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\
0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x109}}$ $\mathbb{Q}_5^{(a)}(B_{2u}) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{60} & 0 & 0 & 0 & \frac{\sqrt{105}}{60} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 & 0 & -\frac{\sqrt{21}}{20} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{20} & 0 & 0 & 0 & \frac{\sqrt{7}}{20} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} & 0 & 0 & 0 & -\frac{\sqrt{3}}{60} \\
0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{3}}{30} & 0 & 0 & 0 & -\frac{\sqrt{105}}{105} & 0 & 0 & 0 \\
-\frac{\sqrt{7}}{28} & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & \frac{2\sqrt{42}}{105} & 0 & 0 & 0 & -\frac{\sqrt{14}}{70} & 0 & 0 \\
0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{14}}{70} & 0 & 0 & 0 & \frac{2\sqrt{42}}{105} & 0 \\
0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{105} & 0 & 0 & 0 & -\frac{\sqrt{3}}{30} \\
0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x110}}$ $\mathbb{Q}_{1,1}^{(a)}(E_u) = \begin{bmatrix}
-\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}}{20} & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{5}}{70} & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2}}{35} & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\
0 & 0 & -\frac{3}{70} & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2}}{35} & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{30}}{28} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & \frac{\sqrt{42}}{28}
\end{bmatrix}$

$\boxed{\text{x111}}$ $\mathbb{Q}^{(a)}_{1,2}(E_u) = \begin{bmatrix}
-\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{5}i}{70} & 0 & \frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2}i}{35} & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\
0 & 0 & -\frac{3i}{70} & 0 & \frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2}i}{35} & 0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{30}i}{28} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{42}i}{28}
\end{bmatrix}$

$\boxed{\text{x112}}$ $\mathbb{Q}^{(a)}_{3,1}(E_u, 1) = \begin{bmatrix}
-\frac{3\sqrt{105}}{560} & 0 & \frac{9\sqrt{42}}{560} & 0 & -\frac{3\sqrt{21}}{112} & 0 & 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{210}}{336} & 0 & 0 \\
0 & \frac{3\sqrt{7}}{80} & 0 & -\frac{3\sqrt{14}}{560} & 0 & -\frac{3\sqrt{35}}{112} & -\frac{\sqrt{30}}{48} & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{210}}{168} & 0 \\
\frac{3\sqrt{35}}{112} & 0 & \frac{3\sqrt{14}}{560} & 0 & -\frac{3\sqrt{7}}{80} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & -\frac{\sqrt{30}}{48} \\
0 & \frac{3\sqrt{21}}{112} & 0 & -\frac{9\sqrt{42}}{560} & 0 & \frac{3\sqrt{105}}{560} & 0 & 0 & -\frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 \\
0 & -\frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 \\
-\frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{42}}{280} & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{210}}{112} & 0 & 0 \\
0 & \frac{\sqrt{42}}{280} & 0 & \frac{\sqrt{21}}{70} & 0 & \frac{\sqrt{210}}{168} & \frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{7}}{112} & 0 & -\frac{3\sqrt{35}}{112} & 0 \\
\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{21}}{70} & 0 & \frac{\sqrt{42}}{280} & 0 & 0 & \frac{3\sqrt{35}}{112} & 0 & \frac{\sqrt{7}}{112} & 0 & -\frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{5}}{16} \\
0 & \frac{\sqrt{21}}{42} & 0 & \frac{\sqrt{42}}{280} & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 \\
0 & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{2}}{16}
\end{bmatrix}$

$\boxed{\text{x113}}$ $\mathbb{Q}_{3,2}^{(a)}(E_u,1) =$

$$\begin{bmatrix}
-\frac{3\sqrt{105}i}{560} & 0 & -\frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{21}i}{112} & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{210}i}{336} & 0 & 0 \\
0 & \frac{3\sqrt{7}i}{80} & 0 & \frac{3\sqrt{14}i}{560} & 0 & -\frac{3\sqrt{35}i}{112} & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{210}i}{168} & 0 \\
-\frac{3\sqrt{35}i}{112} & 0 & \frac{3\sqrt{14}i}{560} & 0 & \frac{3\sqrt{7}i}{80} & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} \\
0 & -\frac{3\sqrt{21}i}{112} & 0 & -\frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{105}i}{560} & 0 & 0 & \frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{70}i}{112} & 0 \\
0 & \frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 \\
-\frac{\sqrt{105}i}{140} & 0 & -\frac{\sqrt{42}i}{280} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 \\
0 & \frac{\sqrt{42}i}{280} & 0 & -\frac{\sqrt{21}i}{70} & 0 & \frac{\sqrt{210}i}{168} & -\frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{105}i}{112} & 0 & \frac{\sqrt{7}i}{112} & 0 & -\frac{3\sqrt{35}i}{112} & 0 \\
-\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{21}i}{70} & 0 & -\frac{\sqrt{42}i}{280} & 0 & 0 & -\frac{3\sqrt{35}i}{112} & 0 & \frac{\sqrt{7}i}{112} & 0 & \frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{5}i}{16} \\
0 & -\frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{42}i}{280} & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{70}i}{112} & 0 \\
0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{2}i}{16}
\end{bmatrix}$$

$\boxed{\text{x114}}$ $\mathbb{Q}_{3,1}^{(a)}(E_u,2) =$

$$\begin{bmatrix}
-\frac{3\sqrt{7}}{112} & 0 & \frac{9\sqrt{70}}{560} & 0 & \frac{9\sqrt{35}}{560} & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 & \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 \\
0 & \frac{\sqrt{105}}{80} & 0 & -\frac{\sqrt{210}}{560} & 0 & \frac{3\sqrt{21}}{112} & \frac{\sqrt{2}}{16} & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{14}}{56} & 0 \\
-\frac{3\sqrt{21}}{112} & 0 & \frac{\sqrt{210}}{560} & 0 & -\frac{\sqrt{105}}{80} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & \frac{\sqrt{2}}{16} \\
0 & -\frac{9\sqrt{35}}{560} & 0 & -\frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{7}}{112} & 0 & 0 & \frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{5\sqrt{42}}{336} & 0 \\
0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{30}}{48} & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & 0 \\
-\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{70}}{280} & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & \frac{5\sqrt{42}}{336} & 0 & \frac{\sqrt{210}}{168} & 0 & \frac{3\sqrt{14}}{112} & 0 & 0 \\
0 & \frac{\sqrt{70}}{280} & 0 & \frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{14}}{56} & -\frac{\sqrt{3}}{16} & 0 & \frac{5\sqrt{7}}{112} & 0 & -\frac{\sqrt{105}}{336} & 0 & \frac{3\sqrt{21}}{112} & 0 \\
-\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & -\frac{3\sqrt{21}}{112} & 0 & \frac{\sqrt{105}}{336} & 0 & -\frac{5\sqrt{7}}{112} & 0 & \frac{\sqrt{3}}{16} \\
0 & -\frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{70}}{280} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{5\sqrt{42}}{336} & 0 \\
0 & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{30}}{48}
\end{bmatrix}$$

$\boxed{\text{x115}}$ $\mathbb{Q}^{(a)}_{3,2}(E_u,2) =$

$$\begin{bmatrix}
-\frac{3\sqrt{7}i}{112} & 0 & -\frac{9\sqrt{70}i}{560} & 0 & \frac{9\sqrt{35}i}{560} & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 \\
0 & \frac{\sqrt{105}i}{80} & 0 & \frac{\sqrt{210}i}{560} & 0 & \frac{3\sqrt{21}i}{112} & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{14}i}{56} & 0 \\
\frac{3\sqrt{21}i}{112} & 0 & \frac{\sqrt{210}i}{560} & 0 & \frac{\sqrt{105}i}{80} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} \\
0 & \frac{9\sqrt{35}i}{560} & 0 & -\frac{9\sqrt{70}i}{560} & 0 & -\frac{3\sqrt{7}i}{112} & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{5\sqrt{42}i}{336} & 0 \\
0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 \\
-\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{70}i}{280} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & \frac{5\sqrt{42}i}{336} & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{3\sqrt{14}i}{112} & 0 & 0 \\
0 & \frac{\sqrt{70}i}{280} & 0 & -\frac{\sqrt{35}i}{70} & 0 & -\frac{\sqrt{14}i}{56} & \frac{\sqrt{3}i}{16} & 0 & \frac{5\sqrt{7}i}{112} & 0 & \frac{\sqrt{105}i}{336} & 0 & \frac{3\sqrt{21}i}{112} & 0 \\
\frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{35}i}{70} & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & \frac{3\sqrt{21}i}{112} & 0 & \frac{\sqrt{105}i}{336} & 0 & \frac{5\sqrt{7}i}{112} & 0 & \frac{\sqrt{3}i}{16} \\
0 & \frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{70}i}{280} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{3\sqrt{14}i}{112} & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{5\sqrt{42}i}{336} & 0 \\
0 & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{30}i}{48}
\end{bmatrix}$$

$\boxed{\text{x116}}$ $\mathbb{Q}^{(a)}_{5,1}(E_u,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{80} & 0 & \frac{1}{16} & 0 & -\frac{7\sqrt{15}}{240} & 0 & \frac{3\sqrt{35}}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{240} & 0 & \frac{3\sqrt{5}}{80} & 0 & -\frac{\sqrt{3}}{16} & 0 & \frac{7\sqrt{15}}{240} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{15}}{240} & 0 & -\frac{\sqrt{3}}{16} & 0 & \frac{3\sqrt{5}}{80} & 0 & -\frac{\sqrt{105}}{240} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{80} & 0 & -\frac{7\sqrt{15}}{240} & 0 & \frac{1}{16} & 0 & -\frac{\sqrt{5}}{80} & 0 \\
0 & -\frac{\sqrt{30}}{224} & 0 & \frac{\sqrt{15}}{48} & 0 & -\frac{3\sqrt{6}}{32} & -\frac{\sqrt{7}}{224} & 0 & \frac{5\sqrt{3}}{224} & 0 & -\frac{\sqrt{5}}{32} & 0 & \frac{3}{32} & 0 \\
-\frac{\sqrt{30}}{224} & 0 & \frac{5\sqrt{3}}{112} & 0 & -\frac{5\sqrt{6}}{96} & 0 & 0 & \frac{23\sqrt{5}}{1120} & 0 & -\frac{13}{224} & 0 & \frac{\sqrt{15}}{160} & 0 & \frac{3\sqrt{35}}{160} \\
0 & \frac{5\sqrt{3}}{112} & 0 & -\frac{5\sqrt{6}}{112} & 0 & \frac{\sqrt{15}}{48} & \frac{\sqrt{70}}{160} & 0 & -\frac{11\sqrt{30}}{1120} & 0 & \frac{\sqrt{2}}{224} & 0 & \frac{3\sqrt{10}}{160} & 0 \\
\frac{\sqrt{15}}{48} & 0 & -\frac{5\sqrt{6}}{112} & 0 & \frac{5\sqrt{3}}{112} & 0 & 0 & -\frac{3\sqrt{10}}{160} & 0 & -\frac{\sqrt{2}}{224} & 0 & \frac{11\sqrt{30}}{1120} & 0 & -\frac{\sqrt{70}}{160} \\
0 & -\frac{5\sqrt{6}}{96} & 0 & \frac{5\sqrt{3}}{112} & 0 & -\frac{\sqrt{30}}{224} & -\frac{3\sqrt{35}}{160} & 0 & -\frac{\sqrt{15}}{160} & 0 & \frac{13}{224} & 0 & -\frac{23\sqrt{5}}{1120} & 0 \\
-\frac{3\sqrt{6}}{32} & 0 & \frac{\sqrt{15}}{48} & 0 & -\frac{\sqrt{30}}{224} & 0 & 0 & -\frac{3}{32} & 0 & \frac{\sqrt{5}}{32} & 0 & -\frac{5\sqrt{3}}{224} & 0 & \frac{\sqrt{7}}{224}
\end{bmatrix}$$

$\boxed{\text{x117}}$ $\mathbb{Q}^{(a)}_{5,2}(E_u,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{80} & 0 & -\frac{i}{16} & 0 & -\frac{7\sqrt{15}i}{240} & 0 & -\frac{3\sqrt{35}i}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{240} & 0 & \frac{3\sqrt{5}i}{80} & 0 & \frac{\sqrt{3}i}{16} & 0 & \frac{7\sqrt{15}i}{240} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{15}i}{240} & 0 & -\frac{\sqrt{3}i}{16} & 0 & -\frac{3\sqrt{5}i}{80} & 0 & -\frac{\sqrt{105}i}{240} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{80} & 0 & \frac{7\sqrt{15}i}{240} & 0 & \frac{i}{16} & 0 & \frac{\sqrt{5}i}{80} & 0 \\
0 & \frac{\sqrt{30}i}{224} & 0 & \frac{\sqrt{15}i}{48} & 0 & \frac{3\sqrt{6}i}{32} & -\frac{\sqrt{7}i}{224} & 0 & -\frac{5\sqrt{3}i}{224} & 0 & -\frac{\sqrt{5}i}{32} & 0 & -\frac{3i}{32} & 0 \\
-\frac{\sqrt{30}i}{224} & 0 & -\frac{5\sqrt{3}i}{112} & 0 & -\frac{5\sqrt{6}i}{96} & 0 & 0 & \frac{23\sqrt{5}i}{1120} & 0 & \frac{13i}{224} & 0 & \frac{\sqrt{15}i}{160} & 0 & -\frac{3\sqrt{35}i}{160} \\
0 & \frac{5\sqrt{3}i}{112} & 0 & \frac{5\sqrt{6}i}{112} & 0 & \frac{\sqrt{15}i}{48} & -\frac{\sqrt{70}i}{160} & 0 & -\frac{11\sqrt{30}i}{1120} & 0 & -\frac{\sqrt{2}i}{224} & 0 & \frac{3\sqrt{10}i}{160} & 0 \\
-\frac{\sqrt{15}i}{48} & 0 & -\frac{5\sqrt{6}i}{112} & 0 & -\frac{5\sqrt{3}i}{112} & 0 & 0 & \frac{3\sqrt{10}i}{160} & 0 & -\frac{\sqrt{2}i}{224} & 0 & -\frac{11\sqrt{30}i}{1120} & 0 & -\frac{\sqrt{70}i}{160} \\
0 & \frac{5\sqrt{6}i}{96} & 0 & \frac{5\sqrt{3}i}{112} & 0 & \frac{\sqrt{30}i}{224} & -\frac{3\sqrt{35}i}{160} & 0 & \frac{\sqrt{15}i}{160} & 0 & \frac{13i}{224} & 0 & \frac{23\sqrt{5}i}{1120} & 0 \\
-\frac{3\sqrt{6}i}{32} & 0 & -\frac{\sqrt{15}i}{48} & 0 & -\frac{\sqrt{30}i}{224} & 0 & 0 & -\frac{3i}{32} & 0 & -\frac{\sqrt{5}i}{32} & 0 & -\frac{5\sqrt{3}i}{224} & 0 & -\frac{\sqrt{7}i}{224}
\end{bmatrix}$$

$\boxed{\text{x118}}$ $\mathbb{Q}^{(a)}_{5,1}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{80} & 0 & \frac{\sqrt{35}}{80} & 0 & \frac{3\sqrt{21}}{80} & 0 & \frac{1}{16} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}}{80} & 0 & \frac{3\sqrt{7}}{80} & 0 & -\frac{\sqrt{105}}{80} & 0 & -\frac{3\sqrt{21}}{80} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{80} & 0 & -\frac{\sqrt{105}}{80} & 0 & \frac{3\sqrt{7}}{80} & 0 & \frac{3\sqrt{3}}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{16} & 0 & \frac{3\sqrt{21}}{80} & 0 & \frac{\sqrt{35}}{80} & 0 & -\frac{\sqrt{7}}{80} & 0 \\
0 & -\frac{\sqrt{42}}{224} & 0 & -\frac{3\sqrt{21}}{112} & 0 & -\frac{\sqrt{210}}{224} & -\frac{\sqrt{5}}{160} & 0 & \frac{\sqrt{105}}{224} & 0 & \frac{9\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{224} & 0 \\
-\frac{\sqrt{42}}{224} & 0 & \frac{\sqrt{105}}{112} & 0 & \frac{3\sqrt{210}}{224} & 0 & 0 & \frac{23\sqrt{7}}{1120} & 0 & -\frac{13\sqrt{35}}{1120} & 0 & -\frac{9\sqrt{21}}{1120} & 0 & \frac{1}{32} \\
0 & \frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{210}}{112} & 0 & -\frac{3\sqrt{21}}{112} & -\frac{9\sqrt{2}}{160} & 0 & -\frac{11\sqrt{42}}{1120} & 0 & \frac{\sqrt{70}}{1120} & 0 & -\frac{27\sqrt{14}}{1120} & 0 \\
-\frac{3\sqrt{21}}{112} & 0 & -\frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{105}}{112} & 0 & 0 & \frac{27\sqrt{14}}{1120} & 0 & -\frac{\sqrt{70}}{1120} & 0 & \frac{11\sqrt{42}}{1120} & 0 & \frac{9\sqrt{2}}{160} \\
0 & \frac{3\sqrt{210}}{224} & 0 & \frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{42}}{224} & -\frac{1}{32} & 0 & \frac{9\sqrt{21}}{1120} & 0 & \frac{13\sqrt{35}}{1120} & 0 & -\frac{23\sqrt{7}}{1120} & 0 \\
-\frac{\sqrt{210}}{224} & 0 & -\frac{3\sqrt{21}}{112} & 0 & -\frac{\sqrt{42}}{224} & 0 & 0 & -\frac{\sqrt{35}}{224} & 0 & -\frac{9\sqrt{7}}{224} & 0 & -\frac{\sqrt{105}}{224} & 0 & \frac{\sqrt{5}}{160}
\end{bmatrix}$$

$\boxed{\text{x119}}$ $\mathbb{Q}_{5,2}^{(a)}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{80} & 0 & -\frac{\sqrt{35}i}{80} & 0 & \frac{3\sqrt{21}i}{80} & 0 & -\frac{i}{16} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{80} & 0 & \frac{3\sqrt{7}i}{80} & 0 & \frac{\sqrt{105}i}{80} & 0 & -\frac{3\sqrt{21}i}{80} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}i}{80} & 0 & -\frac{\sqrt{105}i}{80} & 0 & -\frac{3\sqrt{7}i}{80} & 0 & \frac{3\sqrt{3}i}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{16} & 0 & -\frac{3\sqrt{21}i}{80} & 0 & \frac{\sqrt{35}i}{80} & 0 & \frac{\sqrt{7}i}{80} & 0 \\
0 & \frac{\sqrt{42}i}{224} & 0 & -\frac{3\sqrt{21}i}{112} & 0 & \frac{\sqrt{210}i}{224} & -\frac{\sqrt{5}i}{160} & 0 & -\frac{\sqrt{105}i}{224} & 0 & \frac{9\sqrt{7}i}{224} & 0 & -\frac{\sqrt{35}i}{224} & 0 \\
-\frac{\sqrt{42}i}{224} & 0 & -\frac{\sqrt{105}i}{112} & 0 & \frac{3\sqrt{210}i}{224} & 0 & 0 & \frac{23\sqrt{7}i}{1120} & 0 & \frac{13\sqrt{35}i}{1120} & 0 & -\frac{9\sqrt{21}i}{1120} & 0 & -\frac{i}{32} \\
0 & \frac{\sqrt{105}i}{112} & 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{3\sqrt{21}i}{112} & \frac{9\sqrt{2}i}{160} & 0 & -\frac{11\sqrt{42}i}{1120} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & -\frac{27\sqrt{14}i}{1120} & 0 \\
\frac{3\sqrt{21}i}{112} & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{105}i}{112} & 0 & 0 & -\frac{27\sqrt{14}i}{1120} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & -\frac{11\sqrt{42}i}{1120} & 0 & \frac{9\sqrt{2}i}{160} \\
0 & -\frac{3\sqrt{210}i}{224} & 0 & \frac{\sqrt{105}i}{112} & 0 & \frac{\sqrt{42}i}{224} & -\frac{i}{32} & 0 & -\frac{9\sqrt{21}i}{1120} & 0 & \frac{13\sqrt{35}i}{1120} & 0 & \frac{23\sqrt{7}i}{1120} & 0 \\
-\frac{\sqrt{210}i}{224} & 0 & \frac{3\sqrt{21}i}{112} & 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 & -\frac{\sqrt{35}i}{224} & 0 & \frac{9\sqrt{7}i}{224} & 0 & -\frac{\sqrt{105}i}{224} & 0 & -\frac{\sqrt{5}i}{160}
\end{bmatrix}$$

$\boxed{\text{x120}}$ $\mathbb{Q}_{5,1}^{(a)}(E_u,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{120} & 0 & \frac{\sqrt{105}}{120} & 0 & -\frac{\sqrt{7}}{40} & 0 & -\frac{\sqrt{3}}{8} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{40} & 0 & \frac{\sqrt{21}}{40} & 0 & -\frac{\sqrt{35}}{40} & 0 & \frac{\sqrt{7}}{40} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{40} & 0 & -\frac{\sqrt{35}}{40} & 0 & \frac{\sqrt{21}}{40} & 0 & -\frac{1}{40} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & -\frac{\sqrt{7}}{40} & 0 & \frac{\sqrt{105}}{120} & 0 & -\frac{\sqrt{21}}{120} & 0 \\
0 & -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{7}}{56} & 0 & \frac{3\sqrt{70}}{112} & -\frac{\sqrt{15}}{240} & 0 & \frac{\sqrt{35}}{112} & 0 & -\frac{\sqrt{21}}{112} & 0 & -\frac{\sqrt{105}}{112} & 0 \\
-\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{35}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & \frac{23\sqrt{21}}{1680} & 0 & -\frac{13\sqrt{105}}{1680} & 0 & \frac{3\sqrt{7}}{560} & 0 & -\frac{\sqrt{3}}{16} \\
0 & \frac{\sqrt{35}}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{7}}{56} & \frac{\sqrt{6}}{80} & 0 & -\frac{11\sqrt{14}}{560} & 0 & \frac{\sqrt{210}}{1680} & 0 & \frac{3\sqrt{42}}{560} & 0 \\
\frac{\sqrt{7}}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{35}}{56} & 0 & 0 & -\frac{3\sqrt{42}}{560} & 0 & -\frac{\sqrt{210}}{1680} & 0 & \frac{11\sqrt{14}}{560} & 0 & -\frac{\sqrt{6}}{80} \\
0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{35}}{56} & 0 & -\frac{\sqrt{14}}{112} & \frac{\sqrt{3}}{16} & 0 & -\frac{3\sqrt{7}}{560} & 0 & \frac{13\sqrt{105}}{1680} & 0 & -\frac{23\sqrt{21}}{1680} & 0 \\
\frac{3\sqrt{70}}{112} & 0 & \frac{\sqrt{7}}{56} & 0 & -\frac{\sqrt{14}}{112} & 0 & 0 & \frac{\sqrt{105}}{112} & 0 & \frac{\sqrt{21}}{112} & 0 & -\frac{\sqrt{35}}{112} & 0 & \frac{\sqrt{15}}{240}
\end{bmatrix}$$

$\boxed{\text{x121}}$ $\mathbb{Q}_{5,2}^{(a)}(E_u,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{120} & 0 & -\frac{\sqrt{105}i}{120} & 0 & -\frac{\sqrt{7}i}{40} & 0 & \frac{\sqrt{3}i}{8} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{40} & 0 & \frac{\sqrt{21}i}{40} & 0 & \frac{\sqrt{35}i}{40} & 0 & \frac{\sqrt{7}i}{40} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{40} & 0 & -\frac{\sqrt{35}i}{40} & 0 & -\frac{\sqrt{21}i}{40} & 0 & -\frac{i}{40} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & \frac{\sqrt{7}i}{40} & 0 & \frac{\sqrt{105}i}{120} & 0 & \frac{\sqrt{21}i}{120} & 0 \\
0 & \frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{7}i}{56} & 0 & -\frac{3\sqrt{70}i}{112} & -\frac{\sqrt{15}i}{240} & 0 & -\frac{\sqrt{35}i}{112} & 0 & -\frac{\sqrt{21}i}{112} & 0 & \frac{\sqrt{105}i}{112} & 0 \\
-\frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{35}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & \frac{23\sqrt{21}i}{1680} & 0 & \frac{13\sqrt{105}i}{1680} & 0 & \frac{3\sqrt{7}i}{560} & 0 & \frac{\sqrt{3}i}{16} \\
0 & \frac{\sqrt{35}i}{56} & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{7}i}{56} & -\frac{\sqrt{6}i}{80} & 0 & -\frac{11\sqrt{14}i}{560} & 0 & -\frac{\sqrt{210}i}{1680} & 0 & \frac{3\sqrt{42}i}{560} & 0 \\
-\frac{\sqrt{7}i}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{35}i}{56} & 0 & 0 & \frac{3\sqrt{42}i}{560} & 0 & -\frac{\sqrt{210}i}{1680} & 0 & -\frac{11\sqrt{14}i}{560} & 0 & -\frac{\sqrt{6}i}{80} \\
0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{35}i}{56} & 0 & \frac{\sqrt{14}i}{112} & \frac{\sqrt{3}i}{16} & 0 & \frac{3\sqrt{7}i}{560} & 0 & \frac{13\sqrt{105}i}{1680} & 0 & \frac{23\sqrt{21}i}{1680} & 0 \\
\frac{3\sqrt{70}i}{112} & 0 & -\frac{\sqrt{7}i}{56} & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & \frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{21}i}{112} & 0 & -\frac{\sqrt{35}i}{112} & 0 & -\frac{\sqrt{15}i}{240}
\end{bmatrix}$$

$\boxed{\text{x122}}$ $\mathbb{Q}_{5}^{(1,-1;a)}(A_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{100} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{35} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{175} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{25} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x123}}$ $\mathbb{Q}_3^{(1,-1;a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 \\ \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 \end{bmatrix}$

$\boxed{\text{x124}}$ $\mathbb{Q}_5^{(1,-1;a)}(A_{2u}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 \\ -\frac{\sqrt{5}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{35} & 0 \end{bmatrix}$

$\boxed{\text{x125}}$ $\mathbb{Q}_5^{(1,-1;a)}(A_{2u},2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{100} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{35} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{175} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{25} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{35} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x126}}$ $\mathbb{Q}_3^{(1,-1;a)}(B_{1u}) = \begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 \\
\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & \frac{\sqrt{30}i}{210} & 0 & 0 & \frac{3i}{28} & 0 & 0 & 0 & \frac{3\sqrt{3}i}{28} & 0 & 0 \\
0 & -\frac{\sqrt{30}i}{210} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & \frac{3\sqrt{3}i}{28} & 0 & 0 & 0 & \frac{3i}{28} & 0 \\
0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} \\
0 & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 \\
\frac{3i}{28} & 0 & 0 & 0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 \\
0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & \frac{3i}{28} & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 \\
0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} \\
0 & 0 & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{21} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x127}}$ $\mathbb{Q}_5^{(1,-1;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{200} & 0 & 0 & 0 & \frac{\sqrt{14}i}{40} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{200} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{200} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{200} & 0 & 0 & 0 & \frac{\sqrt{210}i}{200} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{40} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{200} \\
0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & -\frac{\sqrt{10}i}{25} & 0 & 0 & 0 & \frac{2\sqrt{14}i}{35} & 0 & 0 & 0 \\
\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & \frac{8\sqrt{35}i}{175} & 0 & 0 & 0 & \frac{2\sqrt{105}i}{175} & 0 & 0 \\
0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 & -\frac{2\sqrt{105}i}{175} & 0 & 0 & 0 & -\frac{8\sqrt{35}i}{175} & 0 \\
0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}i}{35} & 0 & 0 & 0 & \frac{\sqrt{10}i}{25} \\
0 & 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x128}}$ $\mathbb{Q}_3^{(1,-1;a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{5}}{35} & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 \\
\frac{\sqrt{6}}{42} & 0 & 0 & 0 & -\frac{\sqrt{30}}{210} & 0 & 0 & \frac{3}{28} & 0 & 0 & 0 & -\frac{3\sqrt{3}}{28} & 0 & 0 \\
0 & -\frac{\sqrt{30}}{210} & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & \frac{3\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{3}{28} & 0 \\
0 & 0 & -\frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} \\
0 & 0 & \frac{3}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 \\
\frac{3}{28} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 \\
0 & \frac{3\sqrt{5}}{140} & 0 & 0 & 0 & -\frac{3}{28} & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 \\
0 & 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} \\
0 & 0 & 0 & -\frac{3}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x129}}$ $\mathbb{Q}_5^{(1,-1;a)}(B_{2u}) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{200} & 0 & 0 & 0 & \frac{\sqrt{14}}{40} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{200} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{200} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{200} & 0 & 0 & 0 & \frac{\sqrt{210}}{200} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{40} & 0 & 0 & 0 & -\frac{\sqrt{10}}{200} \\
0 & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & \frac{\sqrt{10}}{25} & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 & 0 \\
-\frac{\sqrt{210}}{420} & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & -\frac{8\sqrt{35}}{175} & 0 & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 \\
0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & \frac{\sqrt{210}}{420} & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 & 0 & -\frac{8\sqrt{35}}{175} & 0 \\
0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 & 0 & \frac{\sqrt{10}}{25} \\
0 & 0 & 0 & \frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x130}}$ $\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,1) = \begin{bmatrix}
-\frac{\sqrt{30}}{280} & 0 & \frac{3\sqrt{3}}{140} & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & -\frac{3\sqrt{5}}{56} & 0 & \frac{3}{28} & 0 & -\frac{\sqrt{15}}{56} & 0 & 0 \\
0 & \frac{\sqrt{2}}{40} & 0 & -\frac{1}{140} & 0 & -\frac{\sqrt{10}}{56} & -\frac{\sqrt{105}}{56} & 0 & 0 & 0 & \frac{3\sqrt{3}}{56} & 0 & -\frac{\sqrt{15}}{28} & 0 \\
\frac{\sqrt{10}}{56} & 0 & \frac{1}{140} & 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & \frac{3\sqrt{3}}{56} & 0 & 0 & 0 & -\frac{\sqrt{105}}{56} \\
0 & \frac{\sqrt{6}}{56} & 0 & -\frac{3\sqrt{3}}{140} & 0 & \frac{\sqrt{30}}{280} & 0 & 0 & -\frac{\sqrt{15}}{56} & 0 & \frac{3}{28} & 0 & -\frac{3\sqrt{5}}{56} & 0 \\
0 & -\frac{3\sqrt{30}}{280} & 0 & \frac{\sqrt{15}}{56} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{3}}{14} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 \\
-\frac{3\sqrt{30}}{280} & 0 & \frac{3\sqrt{3}}{280} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{1}{14} & 0 & \frac{\sqrt{15}}{28} & 0 & 0 \\
0 & \frac{3\sqrt{3}}{280} & 0 & \frac{3\sqrt{6}}{140} & 0 & \frac{\sqrt{15}}{56} & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{2}}{56} & 0 & \frac{3\sqrt{10}}{56} & 0 \\
\frac{\sqrt{15}}{56} & 0 & \frac{3\sqrt{6}}{140} & 0 & \frac{3\sqrt{3}}{280} & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & -\frac{\sqrt{2}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{70}}{56} \\
0 & \frac{\sqrt{6}}{28} & 0 & \frac{3\sqrt{3}}{280} & 0 & -\frac{3\sqrt{30}}{280} & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & \frac{1}{14} & 0 & \frac{\sqrt{5}}{28} & 0 \\
0 & 0 & \frac{\sqrt{15}}{56} & 0 & -\frac{3\sqrt{30}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{3}}{14} & 0 & -\frac{\sqrt{7}}{28}
\end{bmatrix}$

$\boxed{\text{x131}}$ $\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,1) = \begin{bmatrix}
-\frac{\sqrt{30}i}{280} & 0 & -\frac{3\sqrt{3}i}{140} & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & -\frac{3i}{28} & 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 \\
0 & \frac{\sqrt{2}i}{40} & 0 & \frac{i}{140} & 0 & -\frac{\sqrt{10}i}{56} & \frac{\sqrt{105}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{56} & 0 & -\frac{\sqrt{15}i}{28} & 0 \\
-\frac{\sqrt{10}i}{56} & 0 & \frac{i}{140} & 0 & \frac{\sqrt{2}i}{40} & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{3\sqrt{3}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{56} \\
0 & -\frac{\sqrt{6}i}{56} & 0 & -\frac{3\sqrt{3}i}{140} & 0 & -\frac{\sqrt{30}i}{280} & 0 & 0 & \frac{\sqrt{15}i}{56} & 0 & \frac{3i}{28} & 0 & \frac{3\sqrt{5}i}{56} & 0 \\
0 & \frac{3\sqrt{30}i}{280} & 0 & \frac{\sqrt{15}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{3}i}{14} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 \\
-\frac{3\sqrt{30}i}{280} & 0 & -\frac{3\sqrt{3}i}{280} & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{i}{14} & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 \\
0 & \frac{3\sqrt{3}i}{280} & 0 & -\frac{3\sqrt{6}i}{140} & 0 & \frac{\sqrt{15}i}{56} & \frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 & -\frac{\sqrt{2}i}{56} & 0 & \frac{3\sqrt{10}i}{56} & 0 \\
-\frac{\sqrt{15}i}{56} & 0 & \frac{3\sqrt{6}i}{140} & 0 & -\frac{3\sqrt{3}i}{280} & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & -\frac{\sqrt{2}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{70}i}{56} \\
0 & -\frac{\sqrt{6}i}{28} & 0 & \frac{3\sqrt{3}i}{280} & 0 & \frac{3\sqrt{30}i}{280} & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{i}{14} & 0 & -\frac{\sqrt{5}i}{28} & 0 \\
0 & 0 & -\frac{\sqrt{15}i}{56} & 0 & -\frac{3\sqrt{30}i}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{3}i}{14} & 0 & \frac{\sqrt{7}i}{28}
\end{bmatrix}$

$\boxed{\text{x132}}$ $\mathbb{Q}_{3,1}^{(1,-1;a)}(E_u,2) = \begin{bmatrix}
-\frac{\sqrt{2}}{56} & 0 & \frac{3\sqrt{5}}{140} & 0 & \frac{3\sqrt{10}}{280} & 0 & 0 & -\frac{5\sqrt{3}}{56} & 0 & \frac{\sqrt{15}}{28} & 0 & \frac{3}{56} & 0 & 0 \\
0 & \frac{\sqrt{30}}{120} & 0 & -\frac{\sqrt{15}}{420} & 0 & \frac{\sqrt{6}}{56} & \frac{3\sqrt{7}}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & \frac{3}{28} & 0 \\
-\frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{15}}{420} & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{3}{28} & 0 & \frac{3\sqrt{5}}{56} & 0 & 0 & 0 & \frac{3\sqrt{7}}{56} \\
0 & -\frac{3\sqrt{10}}{280} & 0 & -\frac{3\sqrt{5}}{140} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & \frac{3}{56} & 0 & \frac{\sqrt{15}}{28} & 0 & -\frac{5\sqrt{3}}{56} & 0 \\
0 & -\frac{3\sqrt{2}}{56} & 0 & -\frac{3}{56} & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 \\
-\frac{3\sqrt{2}}{56} & 0 & \frac{3\sqrt{5}}{280} & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 & -\frac{5\sqrt{3}}{84} & 0 & -\frac{\sqrt{15}}{42} & 0 & -\frac{3}{28} & 0 & 0 \\
0 & \frac{3\sqrt{5}}{280} & 0 & \frac{3\sqrt{10}}{140} & 0 & -\frac{3}{56} & \frac{\sqrt{42}}{56} & 0 & -\frac{5\sqrt{2}}{56} & 0 & \frac{\sqrt{30}}{168} & 0 & -\frac{3\sqrt{6}}{56} & 0 \\
-\frac{3}{56} & 0 & \frac{3\sqrt{10}}{140} & 0 & \frac{3\sqrt{5}}{280} & 0 & 0 & \frac{3\sqrt{6}}{56} & 0 & -\frac{\sqrt{30}}{168} & 0 & \frac{5\sqrt{2}}{56} & 0 & -\frac{\sqrt{42}}{56} \\
0 & -\frac{3\sqrt{10}}{140} & 0 & \frac{3\sqrt{5}}{280} & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & \frac{3}{28} & 0 & \frac{\sqrt{15}}{42} & 0 & \frac{5\sqrt{3}}{84} & 0 \\
0 & 0 & -\frac{3}{56} & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{105}}{84}
\end{bmatrix}$

$\boxed{\text{x133}}$ $\mathbb{Q}_{3,2}^{(1,-1;a)}(E_u,2) = \begin{bmatrix}
-\frac{\sqrt{2}i}{56} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & \frac{3\sqrt{10}i}{280} & 0 & 0 & -\frac{5\sqrt{3}i}{56} & 0 & -\frac{\sqrt{15}i}{28} & 0 & \frac{3i}{56} & 0 & 0 \\
0 & \frac{\sqrt{30}i}{120} & 0 & \frac{\sqrt{15}i}{420} & 0 & \frac{\sqrt{6}i}{56} & -\frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & \frac{3i}{28} & 0 \\
\frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{15}i}{420} & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & -\frac{3i}{28} & 0 & \frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{7}i}{56} \\
0 & \frac{3\sqrt{10}i}{280} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & -\frac{3i}{56} & 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{5\sqrt{3}i}{56} & 0 \\
0 & \frac{3\sqrt{2}i}{56} & 0 & -\frac{3i}{56} & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\
-\frac{3\sqrt{2}i}{56} & 0 & -\frac{3\sqrt{5}i}{280} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & -\frac{5\sqrt{3}i}{84} & 0 & \frac{\sqrt{15}i}{42} & 0 & -\frac{3i}{28} & 0 & 0 \\
0 & \frac{3\sqrt{5}i}{280} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & -\frac{3i}{56} & -\frac{\sqrt{42}i}{56} & 0 & -\frac{5\sqrt{2}i}{56} & 0 & -\frac{\sqrt{30}i}{168} & 0 & -\frac{3\sqrt{6}i}{56} & 0 \\
\frac{3i}{56} & 0 & \frac{3\sqrt{10}i}{140} & 0 & -\frac{3\sqrt{5}i}{280} & 0 & 0 & -\frac{3\sqrt{6}i}{56} & 0 & -\frac{\sqrt{30}i}{168} & 0 & -\frac{5\sqrt{2}i}{56} & 0 & -\frac{\sqrt{42}i}{56} \\
0 & \frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{5}i}{280} & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & -\frac{3i}{28} & 0 & \frac{\sqrt{15}i}{42} & 0 & -\frac{5\sqrt{3}i}{84} & 0 \\
0 & 0 & \frac{3i}{56} & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{105}i}{84}
\end{bmatrix}$

$\boxed{\text{x134}}$ $\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,1) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{160} & 0 & \frac{\sqrt{30}}{160} & 0 & -\frac{7\sqrt{2}}{160} & 0 & \frac{3\sqrt{42}}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{160} & 0 & \frac{3\sqrt{6}}{160} & 0 & -\frac{3\sqrt{10}}{160} & 0 & \frac{7\sqrt{2}}{160} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{2}}{160} & 0 & -\frac{3\sqrt{10}}{160} & 0 & \frac{3\sqrt{6}}{160} & 0 & -\frac{\sqrt{14}}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{160} & 0 & -\frac{7\sqrt{2}}{160} & 0 & \frac{\sqrt{30}}{160} & 0 & -\frac{\sqrt{6}}{160} & 0 \\
0 & -\frac{1}{112} & 0 & \frac{\sqrt{2}}{48} & 0 & -\frac{3\sqrt{5}}{80} & \frac{\sqrt{210}}{560} & 0 & -\frac{3\sqrt{10}}{112} & 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{3\sqrt{30}}{80} & 0 \\
-\frac{1}{112} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{\sqrt{5}}{48} & 0 & 0 & -\frac{23\sqrt{6}}{560} & 0 & \frac{13\sqrt{30}}{560} & 0 & -\frac{3\sqrt{2}}{80} & 0 & -\frac{3\sqrt{42}}{80} \\
0 & \frac{\sqrt{10}}{112} & 0 & -\frac{\sqrt{5}}{56} & 0 & \frac{\sqrt{2}}{48} & -\frac{\sqrt{21}}{40} & 0 & \frac{33}{280} & 0 & -\frac{\sqrt{15}}{280} & 0 & -\frac{3\sqrt{3}}{40} & 0 \\
\frac{\sqrt{2}}{48} & 0 & -\frac{\sqrt{5}}{56} & 0 & \frac{\sqrt{10}}{112} & 0 & 0 & \frac{3\sqrt{3}}{40} & 0 & \frac{\sqrt{15}}{280} & 0 & -\frac{33}{280} & 0 & \frac{\sqrt{21}}{40} \\
0 & -\frac{\sqrt{5}}{48} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{1}{112} & \frac{3\sqrt{42}}{80} & 0 & \frac{3\sqrt{2}}{80} & 0 & -\frac{13\sqrt{30}}{560} & 0 & \frac{23\sqrt{6}}{560} & 0 \\
-\frac{3\sqrt{5}}{80} & 0 & \frac{\sqrt{2}}{48} & 0 & -\frac{1}{112} & 0 & 0 & \frac{3\sqrt{30}}{80} & 0 & -\frac{\sqrt{6}}{16} & 0 & \frac{3\sqrt{10}}{112} & 0 & -\frac{\sqrt{210}}{560}
\end{bmatrix}$

$\boxed{\text{x135}}$ $\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{160} & 0 & -\frac{\sqrt{30}i}{160} & 0 & -\frac{7\sqrt{2}i}{160} & 0 & -\frac{3\sqrt{42}i}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{160} & 0 & \frac{3\sqrt{6}i}{160} & 0 & \frac{3\sqrt{10}i}{160} & 0 & \frac{7\sqrt{2}i}{160} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{2}i}{160} & 0 & -\frac{3\sqrt{10}i}{160} & 0 & -\frac{3\sqrt{6}i}{160} & 0 & -\frac{\sqrt{14}i}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{160} & 0 & \frac{7\sqrt{2}i}{160} & 0 & \frac{\sqrt{30}i}{160} & 0 & \frac{\sqrt{6}i}{160} & 0 \\
0 & \frac{i}{112} & 0 & \frac{\sqrt{2}i}{48} & 0 & \frac{3\sqrt{5}i}{80} & \frac{\sqrt{210}i}{560} & 0 & \frac{3\sqrt{10}i}{112} & 0 & \frac{\sqrt{6}i}{16} & 0 & \frac{3\sqrt{30}i}{80} & 0 \\
-\frac{i}{112} & 0 & -\frac{\sqrt{10}i}{112} & 0 & -\frac{\sqrt{5}i}{48} & 0 & 0 & -\frac{23\sqrt{6}i}{560} & 0 & -\frac{13\sqrt{30}i}{560} & 0 & -\frac{3\sqrt{2}i}{80} & 0 & \frac{3\sqrt{42}i}{80} \\
0 & \frac{\sqrt{10}i}{112} & 0 & \frac{\sqrt{5}i}{56} & 0 & \frac{\sqrt{2}i}{48} & \frac{\sqrt{21}i}{40} & 0 & \frac{33i}{280} & 0 & \frac{\sqrt{15}i}{280} & 0 & -\frac{3\sqrt{3}i}{40} & 0 \\
-\frac{\sqrt{2}i}{48} & 0 & -\frac{\sqrt{5}i}{56} & 0 & -\frac{\sqrt{10}i}{112} & 0 & 0 & -\frac{3\sqrt{3}i}{40} & 0 & \frac{\sqrt{15}i}{280} & 0 & \frac{33i}{280} & 0 & \frac{\sqrt{21}i}{40} \\
0 & \frac{\sqrt{5}i}{48} & 0 & \frac{\sqrt{10}i}{112} & 0 & \frac{i}{112} & \frac{3\sqrt{42}i}{80} & 0 & -\frac{3\sqrt{2}i}{80} & 0 & -\frac{13\sqrt{30}i}{560} & 0 & -\frac{23\sqrt{6}i}{560} & 0 \\
-\frac{3\sqrt{5}i}{80} & 0 & -\frac{\sqrt{2}i}{48} & 0 & -\frac{i}{112} & 0 & 0 & \frac{3\sqrt{30}i}{80} & 0 & \frac{\sqrt{6}i}{16} & 0 & \frac{3\sqrt{10}i}{112} & 0 & \frac{\sqrt{210}i}{560}
\end{bmatrix}$$

$\boxed{\text{x136}}$ $\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{800} & 0 & \frac{\sqrt{42}}{160} & 0 & \frac{9\sqrt{70}}{800} & 0 & \frac{\sqrt{30}}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{800} & 0 & \frac{3\sqrt{210}}{800} & 0 & -\frac{3\sqrt{14}}{160} & 0 & -\frac{9\sqrt{70}}{800} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{800} & 0 & -\frac{3\sqrt{14}}{160} & 0 & \frac{3\sqrt{210}}{800} & 0 & \frac{9\sqrt{10}}{800} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{160} & 0 & \frac{9\sqrt{70}}{800} & 0 & \frac{\sqrt{42}}{160} & 0 & -\frac{\sqrt{210}}{800} & 0 \\
0 & -\frac{\sqrt{35}}{560} & 0 & -\frac{3\sqrt{70}}{560} & 0 & -\frac{\sqrt{7}}{112} & \frac{\sqrt{6}}{80} & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{9\sqrt{210}}{560} & 0 & -\frac{\sqrt{42}}{112} & 0 \\
-\frac{\sqrt{35}}{560} & 0 & \frac{\sqrt{14}}{112} & 0 & \frac{3\sqrt{7}}{112} & 0 & 0 & -\frac{23\sqrt{210}}{2800} & 0 & \frac{13\sqrt{42}}{560} & 0 & \frac{27\sqrt{70}}{2800} & 0 & -\frac{\sqrt{30}}{80} \\
0 & \frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{7}}{56} & 0 & -\frac{3\sqrt{70}}{560} & \frac{9\sqrt{15}}{200} & 0 & \frac{33\sqrt{35}}{1400} & 0 & -\frac{\sqrt{21}}{280} & 0 & \frac{27\sqrt{105}}{1400} & 0 \\
-\frac{3\sqrt{70}}{560} & 0 & -\frac{\sqrt{7}}{56} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & -\frac{27\sqrt{105}}{1400} & 0 & \frac{\sqrt{21}}{280} & 0 & -\frac{33\sqrt{35}}{1400} & 0 & -\frac{9\sqrt{15}}{200} \\
0 & \frac{3\sqrt{7}}{112} & 0 & \frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{35}}{560} & \frac{\sqrt{30}}{80} & 0 & -\frac{27\sqrt{70}}{2800} & 0 & -\frac{13\sqrt{42}}{560} & 0 & \frac{23\sqrt{210}}{2800} & 0 \\
-\frac{\sqrt{7}}{112} & 0 & -\frac{3\sqrt{70}}{560} & 0 & -\frac{\sqrt{35}}{560} & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & \frac{9\sqrt{210}}{560} & 0 & \frac{3\sqrt{14}}{112} & 0 & -\frac{\sqrt{6}}{80}
\end{bmatrix}$$

$\boxed{\text{x137}}$ $\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{800} & 0 & -\frac{\sqrt{42}i}{160} & 0 & \frac{9\sqrt{70}i}{800} & 0 & -\frac{\sqrt{30}i}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{800} & 0 & \frac{3\sqrt{210}i}{800} & 0 & \frac{3\sqrt{14}i}{160} & 0 & -\frac{9\sqrt{70}i}{800} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{70}i}{800} & 0 & -\frac{3\sqrt{14}i}{160} & 0 & -\frac{3\sqrt{210}i}{800} & 0 & \frac{9\sqrt{10}i}{800} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{160} & 0 & -\frac{9\sqrt{70}i}{800} & 0 & \frac{\sqrt{42}i}{160} & 0 & \frac{\sqrt{210}i}{800} & 0 \\
0 & \frac{\sqrt{35}i}{560} & 0 & -\frac{3\sqrt{70}i}{560} & 0 & \frac{\sqrt{7}i}{112} & \frac{\sqrt{6}i}{80} & 0 & \frac{3\sqrt{14}i}{112} & 0 & -\frac{9\sqrt{210}i}{560} & 0 & \frac{\sqrt{42}i}{112} & 0 \\
-\frac{\sqrt{35}i}{560} & 0 & -\frac{\sqrt{14}i}{112} & 0 & \frac{3\sqrt{7}i}{112} & 0 & 0 & -\frac{23\sqrt{210}i}{2800} & 0 & -\frac{13\sqrt{42}i}{560} & 0 & \frac{27\sqrt{70}i}{2800} & 0 & \frac{\sqrt{30}i}{80} \\
0 & \frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{7}i}{56} & 0 & -\frac{3\sqrt{70}i}{560} & -\frac{9\sqrt{15}i}{200} & 0 & \frac{33\sqrt{35}i}{1400} & 0 & \frac{\sqrt{21}i}{280} & 0 & \frac{27\sqrt{105}i}{1400} & 0 \\
\frac{3\sqrt{70}i}{560} & 0 & -\frac{\sqrt{7}i}{56} & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & \frac{27\sqrt{105}i}{1400} & 0 & \frac{\sqrt{21}i}{280} & 0 & \frac{33\sqrt{35}i}{1400} & 0 & -\frac{9\sqrt{15}i}{200} \\
0 & -\frac{3\sqrt{7}i}{112} & 0 & \frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{35}i}{560} & \frac{\sqrt{30}i}{80} & 0 & \frac{27\sqrt{70}i}{2800} & 0 & -\frac{13\sqrt{42}i}{560} & 0 & -\frac{23\sqrt{210}i}{2800} & 0 \\
-\frac{\sqrt{7}i}{112} & 0 & \frac{3\sqrt{70}i}{560} & 0 & -\frac{\sqrt{35}i}{560} & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & -\frac{9\sqrt{210}i}{560} & 0 & \frac{3\sqrt{14}i}{112} & 0 & \frac{\sqrt{6}i}{80}
\end{bmatrix}$$

$\boxed{\text{x138}}$ $\mathbb{Q}_{5,1}^{(1,-1;a)}(E_u,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{400} & 0 & \frac{\sqrt{14}}{80} & 0 & -\frac{\sqrt{210}}{400} & 0 & -\frac{3\sqrt{10}}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{400} & 0 & \frac{3\sqrt{70}}{400} & 0 & -\frac{\sqrt{42}}{80} & 0 & \frac{\sqrt{210}}{400} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{400} & 0 & -\frac{\sqrt{42}}{80} & 0 & \frac{3\sqrt{70}}{400} & 0 & -\frac{\sqrt{30}}{400} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{80} & 0 & -\frac{\sqrt{210}}{400} & 0 & \frac{\sqrt{14}}{80} & 0 & -\frac{\sqrt{70}}{400} & 0 \\
0 & -\frac{\sqrt{105}}{840} & 0 & \frac{\sqrt{210}}{840} & 0 & \frac{\sqrt{21}}{56} & \frac{\sqrt{2}}{40} & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{3\sqrt{70}}{280} & 0 & \frac{3\sqrt{14}}{56} & 0 \\
-\frac{\sqrt{105}}{840} & 0 & \frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{21}}{168} & 0 & 0 & -\frac{23\sqrt{70}}{1400} & 0 & \frac{13\sqrt{14}}{280} & 0 & -\frac{3\sqrt{210}}{1400} & 0 & \frac{3\sqrt{10}}{40} \\
0 & \frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{210}}{840} & -\frac{3\sqrt{5}}{100} & 0 & \frac{11\sqrt{105}}{700} & 0 & -\frac{\sqrt{7}}{140} & 0 & -\frac{9\sqrt{35}}{700} & 0 \\
\frac{\sqrt{210}}{840} & 0 & -\frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{9\sqrt{35}}{700} & 0 & \frac{\sqrt{7}}{140} & 0 & -\frac{11\sqrt{105}}{700} & 0 & \frac{3\sqrt{5}}{100} \\
0 & -\frac{\sqrt{21}}{168} & 0 & \frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{105}}{840} & -\frac{3\sqrt{10}}{40} & 0 & \frac{3\sqrt{210}}{1400} & 0 & -\frac{13\sqrt{14}}{280} & 0 & \frac{23\sqrt{70}}{1400} & 0 \\
\frac{\sqrt{21}}{56} & 0 & \frac{\sqrt{210}}{840} & 0 & -\frac{\sqrt{105}}{840} & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & -\frac{3\sqrt{70}}{280} & 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{2}}{40}
\end{bmatrix}$$

$\boxed{\text{x139}}$ $\mathbb{Q}_{5,2}^{(1,-1;a)}(E_u,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{400} & 0 & -\frac{\sqrt{14}i}{80} & 0 & -\frac{\sqrt{210}i}{400} & 0 & \frac{3\sqrt{10}i}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{400} & 0 & \frac{3\sqrt{70}i}{400} & 0 & \frac{\sqrt{42}i}{80} & 0 & \frac{\sqrt{210}i}{400} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{400} & 0 & -\frac{\sqrt{42}i}{80} & 0 & -\frac{3\sqrt{70}i}{400} & 0 & -\frac{\sqrt{30}i}{400} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & \frac{\sqrt{210}i}{400} & 0 & \frac{\sqrt{14}i}{80} & 0 & \frac{\sqrt{70}i}{400} & 0 \\
0 & \frac{\sqrt{105}i}{840} & 0 & \frac{\sqrt{210}i}{840} & 0 & -\frac{\sqrt{21}i}{56} & \frac{\sqrt{2}i}{40} & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{3\sqrt{70}i}{280} & 0 & -\frac{3\sqrt{14}i}{56} & 0 \\
-\frac{\sqrt{105}i}{840} & 0 & -\frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{21}i}{168} & 0 & 0 & -\frac{23\sqrt{70}i}{1400} & 0 & -\frac{13\sqrt{14}i}{280} & 0 & -\frac{3\sqrt{210}i}{1400} & 0 & -\frac{3\sqrt{10}i}{40} \\
0 & \frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{210}i}{840} & \frac{3\sqrt{5}i}{100} & 0 & \frac{11\sqrt{105}i}{700} & 0 & \frac{\sqrt{7}i}{140} & 0 & -\frac{9\sqrt{35}i}{700} & 0 \\
-\frac{\sqrt{210}i}{840} & 0 & -\frac{\sqrt{21}i}{84} & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & -\frac{9\sqrt{35}i}{700} & 0 & \frac{\sqrt{7}i}{140} & 0 & \frac{11\sqrt{105}i}{700} & 0 & \frac{3\sqrt{5}i}{100} \\
0 & \frac{\sqrt{21}i}{168} & 0 & \frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{105}i}{840} & -\frac{3\sqrt{10}i}{40} & 0 & -\frac{3\sqrt{210}i}{1400} & 0 & -\frac{13\sqrt{14}i}{280} & 0 & -\frac{23\sqrt{70}i}{1400} & 0 \\
\frac{\sqrt{21}i}{56} & 0 & -\frac{\sqrt{210}i}{840} & 0 & -\frac{\sqrt{105}i}{840} & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & \frac{3\sqrt{70}i}{280} & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{2}i}{40}
\end{bmatrix}$$

$\boxed{\text{x140}}$ $\mathbb{Q}_{5}^{(1,0;a)}(A_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{50} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{50} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{140} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{2100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{150} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{2100} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{140} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x141}}$ $\mathbb{Q}_1^{(1,0;a)}(A_{2u}) = \begin{bmatrix}
0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0
\end{bmatrix}$

$\boxed{\text{x142}}$ $\mathbb{Q}_3^{(1,0;a)}(A_{2u}) = \begin{bmatrix}
0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 \\
-\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0
\end{bmatrix}$

$\boxed{\text{x143}}$ $\mathbb{Q}_5^{(1,0;a)}(A_{2u},1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 \\ \frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{420} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{420} & 0 \end{bmatrix}$

$\boxed{\text{x144}}$ $\mathbb{Q}_5^{(1,0;a)}(A_{2u},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{2100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{2100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x145}}$ $\mathbb{Q}_3^{(1,0;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{210}i}{280} & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\
-\frac{\sqrt{7}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{280} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 \\
0 & \frac{\sqrt{35}i}{280} & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 \\
0 & 0 & \frac{\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} \\
0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & 0 & \frac{\sqrt{70}i}{168} & 0 & 0 & 0 \\
-\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & \frac{\sqrt{7}i}{168} & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & 0 \\
0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{\sqrt{21}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{168} & 0 \\
0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{24} \\
0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x146}}$ $\mathbb{Q}_5^{(1,0;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{100} & 0 & 0 & 0 & \frac{\sqrt{14}i}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{100} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{100} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{100} & 0 & 0 & 0 & \frac{\sqrt{210}i}{100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{100} \\
0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{420} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & \frac{\sqrt{10}i}{300} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{210} & 0 & 0 & 0 \\
-\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{2\sqrt{35}i}{525} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{1050} & 0 & 0 \\
0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & \frac{\sqrt{105}i}{1050} & 0 & 0 & 0 & \frac{2\sqrt{35}i}{525} & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{210} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{300} \\
0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{420} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x147}}$ $\mathbb{Q}_3^{(1,0;a)}(B_{2u}) = \begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{210}}{280} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 \\
-\frac{\sqrt{7}}{56} & 0 & 0 & 0 & \frac{\sqrt{35}}{280} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 \\
0 & \frac{\sqrt{35}}{280} & 0 & 0 & 0 & -\frac{\sqrt{7}}{56} & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 \\
0 & 0 & \frac{\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & \frac{\sqrt{2}}{8} \\
0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & -\frac{\sqrt{70}}{168} & 0 & 0 & 0 \\
-\frac{\sqrt{42}}{28} & 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{7}}{168} & 0 & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 & 0 \\
0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 & 0 & 0 & \frac{\sqrt{7}}{168} & 0 \\
0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{168} & 0 & 0 & 0 & \frac{\sqrt{2}}{24} \\
0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x148}}$ $\mathbb{Q}_5^{(1,0;a)}(B_{2u}) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{100} & 0 & 0 & 0 & \frac{\sqrt{14}}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{100} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{100} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{100} & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{20} & 0 & 0 & 0 & -\frac{\sqrt{10}}{100} \\
0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{420} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{10}}{300} & 0 & 0 & 0 & -\frac{\sqrt{14}}{210} & 0 & 0 & 0 \\
\frac{\sqrt{210}}{140} & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & \frac{2\sqrt{35}}{525} & 0 & 0 & 0 & -\frac{\sqrt{105}}{1050} & 0 & 0 \\
0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & -\frac{\sqrt{105}}{1050} & 0 & 0 & 0 & \frac{2\sqrt{35}}{525} & 0 \\
0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{210} & 0 & 0 & 0 & -\frac{\sqrt{10}}{300} \\
0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{420} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x149}}$ $\mathbb{Q}_{1,1}^{(1,0;a)}(E_u) = \begin{bmatrix}
\frac{\sqrt{10}}{20} & 0 & -\frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{6}}{20} & 0 & -\frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}}{20} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{1}{20} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{10}}{70} & 0 & \frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\
0 & \frac{6}{35} & 0 & \frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 \\
0 & 0 & \frac{9\sqrt{2}}{70} & 0 & \frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{6}{35} & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{15}}{28} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & \frac{\sqrt{21}}{28}
\end{bmatrix}$

$\boxed{\text{x150}}$ $\mathbb{Q}_{1,2}^{(1,0;a)}(E_u) = \begin{bmatrix}
\frac{\sqrt{10}i}{20} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{i}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{10}i}{70} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\
0 & \frac{6i}{35} & 0 & -\frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 \\
0 & 0 & \frac{9\sqrt{2}i}{70} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 \\
0 & 0 & 0 & \frac{6i}{35} & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & -\frac{\sqrt{15}i}{28} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & -\frac{\sqrt{21}i}{28}
\end{bmatrix}$

$\boxed{\text{x151}}$ $\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,1) = \begin{bmatrix}
\frac{3\sqrt{35}}{1120} & 0 & -\frac{9\sqrt{14}}{1120} & 0 & \frac{3\sqrt{7}}{224} & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 \\
0 & -\frac{\sqrt{21}}{160} & 0 & \frac{\sqrt{42}}{1120} & 0 & \frac{\sqrt{105}}{224} & -\frac{\sqrt{10}}{16} & 0 & 0 & 0 & \frac{3\sqrt{14}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 \\
-\frac{\sqrt{105}}{224} & 0 & -\frac{\sqrt{42}}{1120} & 0 & \frac{\sqrt{21}}{160} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{3\sqrt{14}}{112} & 0 & 0 & 0 & -\frac{\sqrt{10}}{16} \\
0 & -\frac{3\sqrt{7}}{224} & 0 & \frac{9\sqrt{14}}{1120} & 0 & -\frac{3\sqrt{35}}{1120} & 0 & 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{210}}{112} & 0 \\
0 & \frac{3\sqrt{35}}{140} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & \frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{210}}{672} & 0 & 0 & 0 \\
\frac{3\sqrt{35}}{140} & 0 & -\frac{3\sqrt{14}}{280} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{210}}{672} & 0 & \frac{\sqrt{42}}{336} & 0 & -\frac{\sqrt{70}}{224} & 0 & 0 \\
0 & -\frac{3\sqrt{14}}{280} & 0 & -\frac{3\sqrt{7}}{70} & 0 & -\frac{\sqrt{70}}{56} & \frac{\sqrt{15}}{96} & 0 & \frac{\sqrt{35}}{224} & 0 & -\frac{\sqrt{21}}{672} & 0 & -\frac{\sqrt{105}}{224} & 0 \\
-\frac{\sqrt{70}}{56} & 0 & -\frac{3\sqrt{7}}{70} & 0 & -\frac{3\sqrt{14}}{280} & 0 & 0 & \frac{\sqrt{105}}{224} & 0 & \frac{\sqrt{21}}{672} & 0 & -\frac{\sqrt{35}}{224} & 0 & -\frac{\sqrt{15}}{96} \\
0 & -\frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{14}}{280} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & -\frac{\sqrt{42}}{336} & 0 & -\frac{\sqrt{210}}{672} & 0 \\
0 & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{672} & 0 & -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{6}}{96}
\end{bmatrix}$

$\boxed{\text{x152}}$ $\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,1) = \begin{bmatrix}
\frac{3\sqrt{35}i}{1120} & 0 & \frac{9\sqrt{14}i}{1120} & 0 & \frac{3\sqrt{7}i}{224} & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 \\
0 & -\frac{\sqrt{21}i}{160} & 0 & -\frac{\sqrt{42}i}{1120} & 0 & \frac{\sqrt{105}i}{224} & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 \\
\frac{\sqrt{105}i}{224} & 0 & -\frac{\sqrt{42}i}{1120} & 0 & -\frac{\sqrt{21}i}{160} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{3\sqrt{14}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} \\
0 & \frac{3\sqrt{7}i}{224} & 0 & \frac{9\sqrt{14}i}{1120} & 0 & \frac{3\sqrt{35}i}{1120} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{210}i}{112} & 0 \\
0 & -\frac{3\sqrt{35}i}{140} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{6}i}{96} & 0 & -\frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{210}i}{672} & 0 & 0 & 0 \\
\frac{3\sqrt{35}i}{140} & 0 & \frac{3\sqrt{14}i}{280} & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & \frac{\sqrt{210}i}{672} & 0 & -\frac{\sqrt{42}i}{336} & 0 & -\frac{\sqrt{70}i}{224} & 0 & 0 \\
0 & -\frac{3\sqrt{14}i}{280} & 0 & \frac{3\sqrt{7}i}{70} & 0 & -\frac{\sqrt{70}i}{56} & -\frac{\sqrt{15}i}{96} & 0 & \frac{\sqrt{35}i}{224} & 0 & \frac{\sqrt{21}i}{672} & 0 & -\frac{\sqrt{105}i}{224} & 0 \\
\frac{\sqrt{70}i}{56} & 0 & -\frac{3\sqrt{7}i}{70} & 0 & \frac{3\sqrt{14}i}{280} & 0 & 0 & -\frac{\sqrt{105}i}{224} & 0 & \frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{35}i}{224} & 0 & -\frac{\sqrt{15}i}{96} \\
0 & \frac{\sqrt{7}i}{14} & 0 & -\frac{3\sqrt{14}i}{280} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & -\frac{\sqrt{42}i}{336} & 0 & \frac{\sqrt{210}i}{672} & 0 \\
0 & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{672} & 0 & -\frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{6}i}{96}
\end{bmatrix}$

$\boxed{\text{x153}}$ $\mathbb{Q}_{3,1}^{(1,0;a)}(E_u,2) = \begin{bmatrix}
\frac{\sqrt{21}}{224} & 0 & -\frac{3\sqrt{210}}{1120} & 0 & -\frac{3\sqrt{105}}{1120} & 0 & 0 & -\frac{5\sqrt{14}}{112} & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 \\
0 & -\frac{\sqrt{35}}{160} & 0 & \frac{\sqrt{70}}{1120} & 0 & -\frac{3\sqrt{7}}{224} & \frac{\sqrt{6}}{16} & 0 & 0 & 0 & \frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{42}}{56} & 0 \\
\frac{3\sqrt{7}}{224} & 0 & -\frac{\sqrt{70}}{1120} & 0 & \frac{\sqrt{35}}{160} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{210}}{112} & 0 & 0 & 0 & \frac{\sqrt{6}}{16} \\
0 & \frac{3\sqrt{105}}{1120} & 0 & \frac{3\sqrt{210}}{1120} & 0 & -\frac{\sqrt{21}}{224} & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{5\sqrt{14}}{112} & 0 \\
0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{10}}{96} & 0 & \frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{14}}{224} & 0 & 0 & 0 \\
\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & \frac{5\sqrt{14}}{672} & 0 & \frac{\sqrt{70}}{336} & 0 & \frac{\sqrt{42}}{224} & 0 & 0 \\
0 & -\frac{\sqrt{210}}{280} & 0 & -\frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{42}}{56} & -\frac{1}{32} & 0 & \frac{5\sqrt{21}}{672} & 0 & -\frac{\sqrt{35}}{672} & 0 & \frac{3\sqrt{7}}{224} & 0 \\
\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{210}}{280} & 0 & 0 & -\frac{3\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{672} & 0 & -\frac{5\sqrt{21}}{672} & 0 & \frac{1}{32} \\
0 & \frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{42}}{224} & 0 & -\frac{\sqrt{70}}{336} & 0 & -\frac{5\sqrt{14}}{672} & 0 \\
0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{224} & 0 & -\frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{10}}{96}
\end{bmatrix}$

$\boxed{\text{x154}}$ $\mathbb{Q}_{3,2}^{(1,0;a)}(E_u,2) = \begin{bmatrix}
\frac{\sqrt{21}i}{224} & 0 & \frac{3\sqrt{210}i}{1120} & 0 & -\frac{3\sqrt{105}i}{1120} & 0 & 0 & -\frac{5\sqrt{14}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{160} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & -\frac{3\sqrt{7}i}{224} & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & \frac{\sqrt{42}i}{56} & 0 \\
-\frac{3\sqrt{7}i}{224} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & -\frac{\sqrt{35}i}{160} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{210}i}{112} & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} \\
0 & -\frac{3\sqrt{105}i}{1120} & 0 & \frac{3\sqrt{210}i}{1120} & 0 & \frac{\sqrt{21}i}{224} & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{5\sqrt{14}i}{112} & 0 \\
0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & -\frac{\sqrt{10}i}{96} & 0 & -\frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{14}i}{224} & 0 & 0 & 0 \\
\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & \frac{5\sqrt{14}i}{672} & 0 & -\frac{\sqrt{70}i}{336} & 0 & \frac{\sqrt{42}i}{224} & 0 & 0 \\
0 & -\frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{42}i}{56} & \frac{i}{32} & 0 & \frac{5\sqrt{21}i}{672} & 0 & \frac{\sqrt{35}i}{672} & 0 & \frac{3\sqrt{7}i}{224} & 0 \\
-\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{210}i}{280} & 0 & 0 & \frac{3\sqrt{7}i}{224} & 0 & \frac{\sqrt{35}i}{672} & 0 & \frac{5\sqrt{21}i}{672} & 0 & \frac{i}{32} \\
0 & -\frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{224} & 0 & -\frac{\sqrt{70}i}{336} & 0 & \frac{5\sqrt{14}i}{672} & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{224} & 0 & -\frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{10}i}{96}
\end{bmatrix}$

$\boxed{\text{x155}}$ $\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{80} & 0 & \frac{\sqrt{30}}{80} & 0 & -\frac{7\sqrt{2}}{80} & 0 & \frac{3\sqrt{42}}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{80} & 0 & \frac{3\sqrt{6}}{80} & 0 & -\frac{3\sqrt{10}}{80} & 0 & \frac{7\sqrt{2}}{80} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{2}}{80} & 0 & -\frac{3\sqrt{10}}{80} & 0 & \frac{3\sqrt{6}}{80} & 0 & -\frac{\sqrt{14}}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{80} & 0 & -\frac{7\sqrt{2}}{80} & 0 & \frac{\sqrt{30}}{80} & 0 & -\frac{\sqrt{6}}{80} & 0 \\
0 & \frac{3}{112} & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{9\sqrt{5}}{80} & -\frac{\sqrt{210}}{6720} & 0 & \frac{\sqrt{10}}{448} & 0 & -\frac{\sqrt{6}}{192} & 0 & \frac{\sqrt{30}}{320} & 0 \\
\frac{3}{112} & 0 & -\frac{3\sqrt{10}}{112} & 0 & \frac{\sqrt{5}}{16} & 0 & 0 & \frac{23\sqrt{6}}{6720} & 0 & -\frac{13\sqrt{30}}{6720} & 0 & \frac{\sqrt{2}}{320} & 0 & \frac{\sqrt{42}}{320} \\
0 & -\frac{3\sqrt{10}}{112} & 0 & \frac{3\sqrt{5}}{56} & 0 & -\frac{\sqrt{2}}{16} & \frac{\sqrt{21}}{480} & 0 & -\frac{11}{1120} & 0 & \frac{\sqrt{15}}{3360} & 0 & \frac{\sqrt{3}}{160} & 0 \\
-\frac{\sqrt{2}}{16} & 0 & \frac{3\sqrt{5}}{56} & 0 & -\frac{3\sqrt{10}}{112} & 0 & 0 & -\frac{\sqrt{3}}{160} & 0 & -\frac{\sqrt{15}}{3360} & 0 & \frac{11}{1120} & 0 & -\frac{\sqrt{21}}{480} \\
0 & \frac{\sqrt{5}}{16} & 0 & -\frac{3\sqrt{10}}{112} & 0 & \frac{3}{112} & -\frac{\sqrt{42}}{320} & 0 & -\frac{\sqrt{2}}{320} & 0 & \frac{13\sqrt{30}}{6720} & 0 & -\frac{23\sqrt{6}}{6720} & 0 \\
\frac{9\sqrt{5}}{80} & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{3}{112} & 0 & 0 & -\frac{\sqrt{30}}{320} & 0 & \frac{\sqrt{6}}{192} & 0 & -\frac{\sqrt{10}}{448} & 0 & \frac{\sqrt{210}}{6720}
\end{bmatrix}$$

$\boxed{\text{x156}}$ $\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{80} & 0 & -\frac{\sqrt{30}i}{80} & 0 & -\frac{7\sqrt{2}i}{80} & 0 & -\frac{3\sqrt{42}i}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{80} & 0 & \frac{3\sqrt{6}i}{80} & 0 & \frac{3\sqrt{10}i}{80} & 0 & \frac{7\sqrt{2}i}{80} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{2}i}{80} & 0 & -\frac{3\sqrt{10}i}{80} & 0 & -\frac{3\sqrt{6}i}{80} & 0 & -\frac{\sqrt{14}i}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{80} & 0 & \frac{7\sqrt{2}i}{80} & 0 & \frac{\sqrt{30}i}{80} & 0 & \frac{\sqrt{6}i}{80} & 0 \\
0 & -\frac{3i}{112} & 0 & -\frac{\sqrt{2}i}{16} & 0 & -\frac{9\sqrt{5}i}{80} & -\frac{\sqrt{210}i}{6720} & 0 & -\frac{\sqrt{10}i}{448} & 0 & -\frac{\sqrt{6}i}{192} & 0 & -\frac{\sqrt{30}i}{320} & 0 \\
\frac{3i}{112} & 0 & \frac{3\sqrt{10}i}{112} & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 & \frac{23\sqrt{6}i}{6720} & 0 & \frac{13\sqrt{30}i}{6720} & 0 & \frac{\sqrt{2}i}{320} & 0 & -\frac{\sqrt{42}i}{320} \\
0 & -\frac{3\sqrt{10}i}{112} & 0 & -\frac{3\sqrt{5}i}{56} & 0 & -\frac{\sqrt{2}i}{16} & -\frac{\sqrt{21}i}{480} & 0 & -\frac{11i}{1120} & 0 & -\frac{\sqrt{15}i}{3360} & 0 & \frac{\sqrt{3}i}{160} & 0 \\
\frac{\sqrt{2}i}{16} & 0 & \frac{3\sqrt{5}i}{56} & 0 & \frac{3\sqrt{10}i}{112} & 0 & 0 & \frac{\sqrt{3}i}{160} & 0 & -\frac{\sqrt{15}i}{3360} & 0 & -\frac{11i}{1120} & 0 & -\frac{\sqrt{21}i}{480} \\
0 & -\frac{\sqrt{5}i}{16} & 0 & -\frac{3\sqrt{10}i}{112} & 0 & -\frac{3i}{112} & -\frac{\sqrt{42}i}{320} & 0 & \frac{\sqrt{2}i}{320} & 0 & \frac{13\sqrt{30}i}{6720} & 0 & \frac{23\sqrt{6}i}{6720} & 0 \\
\frac{9\sqrt{5}i}{80} & 0 & \frac{\sqrt{2}i}{16} & 0 & \frac{3i}{112} & 0 & 0 & -\frac{\sqrt{30}i}{320} & 0 & -\frac{\sqrt{6}i}{192} & 0 & -\frac{\sqrt{10}i}{448} & 0 & -\frac{\sqrt{210}i}{6720}
\end{bmatrix}$$

$\boxed{\text{x157}}$ $\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{400} & 0 & \frac{\sqrt{42}}{80} & 0 & \frac{9\sqrt{70}}{400} & 0 & \frac{\sqrt{30}}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{400} & 0 & \frac{3\sqrt{210}}{400} & 0 & -\frac{3\sqrt{14}}{80} & 0 & -\frac{9\sqrt{70}}{400} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{400} & 0 & -\frac{3\sqrt{14}}{80} & 0 & \frac{3\sqrt{210}}{400} & 0 & \frac{9\sqrt{10}}{400} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{80} & 0 & \frac{9\sqrt{70}}{400} & 0 & \frac{\sqrt{42}}{80} & 0 & -\frac{\sqrt{210}}{400} & 0 \\
0 & \frac{3\sqrt{35}}{560} & 0 & \frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{7}}{112} & -\frac{\sqrt{6}}{960} & 0 & \frac{\sqrt{14}}{448} & 0 & \frac{3\sqrt{210}}{2240} & 0 & \frac{\sqrt{42}}{1344} & 0 \\
\frac{3\sqrt{35}}{560} & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{9\sqrt{7}}{112} & 0 & 0 & \frac{23\sqrt{210}}{33600} & 0 & -\frac{13\sqrt{42}}{6720} & 0 & -\frac{9\sqrt{70}}{11200} & 0 & \frac{\sqrt{30}}{960} \\
0 & -\frac{3\sqrt{14}}{112} & 0 & \frac{3\sqrt{7}}{56} & 0 & \frac{9\sqrt{70}}{560} & -\frac{3\sqrt{15}}{800} & 0 & -\frac{11\sqrt{35}}{5600} & 0 & \frac{\sqrt{21}}{3360} & 0 & -\frac{9\sqrt{105}}{5600} & 0 \\
\frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{7}}{56} & 0 & -\frac{3\sqrt{14}}{112} & 0 & 0 & \frac{9\sqrt{105}}{5600} & 0 & -\frac{\sqrt{21}}{3360} & 0 & \frac{11\sqrt{35}}{5600} & 0 & \frac{3\sqrt{15}}{800} \\
0 & -\frac{9\sqrt{7}}{112} & 0 & -\frac{3\sqrt{14}}{112} & 0 & \frac{3\sqrt{35}}{560} & -\frac{\sqrt{30}}{960} & 0 & \frac{9\sqrt{70}}{11200} & 0 & \frac{13\sqrt{42}}{6720} & 0 & -\frac{23\sqrt{210}}{33600} & 0 \\
\frac{3\sqrt{7}}{112} & 0 & \frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{35}}{560} & 0 & 0 & -\frac{\sqrt{42}}{1344} & 0 & -\frac{3\sqrt{210}}{2240} & 0 & -\frac{\sqrt{14}}{448} & 0 & \frac{\sqrt{6}}{960}
\end{bmatrix}$$

$\boxed{\text{x158}}$ $\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{400} & 0 & -\frac{\sqrt{42}i}{80} & 0 & \frac{9\sqrt{70}i}{400} & 0 & -\frac{\sqrt{30}i}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{400} & 0 & \frac{3\sqrt{210}i}{400} & 0 & \frac{3\sqrt{14}i}{80} & 0 & -\frac{9\sqrt{70}i}{400} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{70}i}{400} & 0 & -\frac{3\sqrt{14}i}{80} & 0 & -\frac{3\sqrt{210}i}{400} & 0 & \frac{9\sqrt{10}i}{400} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{80} & 0 & -\frac{9\sqrt{70}i}{400} & 0 & \frac{\sqrt{42}i}{80} & 0 & \frac{\sqrt{210}i}{400} & 0 \\
0 & -\frac{3\sqrt{35}i}{560} & 0 & \frac{9\sqrt{70}i}{560} & 0 & -\frac{3\sqrt{7}i}{112} & -\frac{\sqrt{6}i}{960} & 0 & -\frac{\sqrt{14}i}{448} & 0 & \frac{3\sqrt{210}i}{2240} & 0 & -\frac{\sqrt{42}i}{1344} & 0 \\
\frac{3\sqrt{35}i}{560} & 0 & \frac{3\sqrt{14}i}{112} & 0 & -\frac{9\sqrt{7}i}{112} & 0 & 0 & \frac{23\sqrt{210}i}{33600} & 0 & \frac{13\sqrt{42}i}{6720} & 0 & -\frac{9\sqrt{70}i}{11200} & 0 & -\frac{\sqrt{30}i}{960} \\
0 & -\frac{3\sqrt{14}i}{112} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & \frac{9\sqrt{70}i}{560} & \frac{3\sqrt{15}i}{800} & 0 & -\frac{11\sqrt{35}i}{5600} & 0 & -\frac{\sqrt{21}i}{3360} & 0 & -\frac{9\sqrt{105}i}{5600} & 0 \\
-\frac{9\sqrt{70}i}{560} & 0 & \frac{3\sqrt{7}i}{56} & 0 & \frac{3\sqrt{14}i}{112} & 0 & 0 & -\frac{9\sqrt{105}i}{5600} & 0 & -\frac{\sqrt{21}i}{3360} & 0 & -\frac{11\sqrt{35}i}{5600} & 0 & \frac{3\sqrt{15}i}{800} \\
0 & \frac{9\sqrt{7}i}{112} & 0 & -\frac{3\sqrt{14}i}{112} & 0 & -\frac{3\sqrt{35}i}{560} & -\frac{\sqrt{30}i}{960} & 0 & -\frac{9\sqrt{70}i}{11200} & 0 & \frac{13\sqrt{42}i}{6720} & 0 & \frac{23\sqrt{210}i}{33600} & 0 \\
\frac{3\sqrt{7}i}{112} & 0 & -\frac{9\sqrt{70}i}{560} & 0 & \frac{3\sqrt{35}i}{560} & 0 & 0 & -\frac{\sqrt{42}i}{1344} & 0 & \frac{3\sqrt{210}i}{2240} & 0 & -\frac{\sqrt{14}i}{448} & 0 & -\frac{\sqrt{6}i}{960}
\end{bmatrix}$$

$\boxed{\text{x159}}$ $\mathbb{Q}_{5,1}^{(1,0;a)}(E_u,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{200} & 0 & \frac{\sqrt{14}}{40} & 0 & -\frac{\sqrt{210}}{200} & 0 & -\frac{3\sqrt{10}}{40} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{200} & 0 & \frac{3\sqrt{70}}{200} & 0 & -\frac{\sqrt{42}}{40} & 0 & \frac{\sqrt{210}}{200} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{200} & 0 & -\frac{\sqrt{42}}{40} & 0 & \frac{3\sqrt{70}}{200} & 0 & -\frac{\sqrt{30}}{200} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{40} & 0 & -\frac{\sqrt{210}}{200} & 0 & \frac{\sqrt{14}}{40} & 0 & -\frac{\sqrt{70}}{200} & 0 \\
0 & \frac{\sqrt{105}}{280} & 0 & -\frac{\sqrt{210}}{280} & 0 & -\frac{3\sqrt{21}}{56} & -\frac{\sqrt{2}}{480} & 0 & \frac{\sqrt{42}}{672} & 0 & -\frac{\sqrt{70}}{1120} & 0 & -\frac{\sqrt{14}}{224} & 0 \\
\frac{\sqrt{105}}{280} & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{21}}{56} & 0 & 0 & \frac{23\sqrt{70}}{16800} & 0 & -\frac{13\sqrt{14}}{3360} & 0 & \frac{\sqrt{210}}{5600} & 0 & -\frac{\sqrt{10}}{160} \\
0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{210}}{280} & \frac{\sqrt{5}}{400} & 0 & -\frac{11\sqrt{105}}{8400} & 0 & \frac{\sqrt{7}}{1680} & 0 & \frac{3\sqrt{35}}{2800} & 0 \\
-\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & -\frac{3\sqrt{35}}{2800} & 0 & -\frac{\sqrt{7}}{1680} & 0 & \frac{11\sqrt{105}}{8400} & 0 & -\frac{\sqrt{5}}{400} \\
0 & \frac{\sqrt{21}}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{105}}{280} & \frac{\sqrt{10}}{160} & 0 & -\frac{\sqrt{210}}{5600} & 0 & \frac{13\sqrt{14}}{3360} & 0 & -\frac{23\sqrt{70}}{16800} & 0 \\
-\frac{3\sqrt{21}}{56} & 0 & -\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{105}}{280} & 0 & 0 & \frac{\sqrt{14}}{224} & 0 & \frac{\sqrt{70}}{1120} & 0 & -\frac{\sqrt{42}}{672} & 0 & \frac{\sqrt{2}}{480}
\end{bmatrix}$$

$\boxed{\text{x160}}$ $\mathbb{Q}_{5,2}^{(1,0;a)}(E_u,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{200} & 0 & -\frac{\sqrt{14}i}{40} & 0 & -\frac{\sqrt{210}i}{200} & 0 & \frac{3\sqrt{10}i}{40} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{200} & 0 & \frac{3\sqrt{70}i}{200} & 0 & \frac{\sqrt{42}i}{40} & 0 & \frac{\sqrt{210}i}{200} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{200} & 0 & -\frac{\sqrt{42}i}{40} & 0 & -\frac{3\sqrt{70}i}{200} & 0 & -\frac{\sqrt{30}i}{200} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{40} & 0 & \frac{\sqrt{210}i}{200} & 0 & \frac{\sqrt{14}i}{40} & 0 & \frac{\sqrt{70}i}{200} & 0 \\
0 & -\frac{\sqrt{105}i}{280} & 0 & -\frac{\sqrt{210}i}{280} & 0 & \frac{3\sqrt{21}i}{56} & -\frac{\sqrt{2}i}{480} & 0 & -\frac{\sqrt{42}i}{672} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & \frac{\sqrt{14}i}{224} & 0 \\
\frac{\sqrt{105}i}{280} & 0 & \frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{21}i}{56} & 0 & 0 & \frac{23\sqrt{70}i}{16800} & 0 & \frac{13\sqrt{14}i}{3360} & 0 & \frac{\sqrt{210}i}{5600} & 0 & \frac{\sqrt{10}i}{160} \\
0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{210}i}{280} & -\frac{\sqrt{5}i}{400} & 0 & -\frac{11\sqrt{105}i}{8400} & 0 & -\frac{\sqrt{7}i}{1680} & 0 & \frac{3\sqrt{35}i}{2800} & 0 \\
\frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & \frac{3\sqrt{35}i}{2800} & 0 & -\frac{\sqrt{7}i}{1680} & 0 & -\frac{11\sqrt{105}i}{8400} & 0 & -\frac{\sqrt{5}i}{400} \\
0 & -\frac{\sqrt{21}i}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{105}i}{280} & \frac{\sqrt{10}i}{160} & 0 & \frac{\sqrt{210}i}{5600} & 0 & \frac{13\sqrt{14}i}{3360} & 0 & \frac{23\sqrt{70}i}{16800} & 0 \\
-\frac{3\sqrt{21}i}{56} & 0 & \frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{105}i}{280} & 0 & 0 & \frac{\sqrt{14}i}{224} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & -\frac{\sqrt{42}i}{672} & 0 & -\frac{\sqrt{2}i}{480}
\end{bmatrix}$$

$\boxed{\text{x161}}$ $\mathbb{Q}_1^{(1,1;a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 \end{bmatrix}$

$\boxed{\text{x162}}$ $\mathbb{Q}_3^{(1,1;a)}(A_{2u}) = \begin{bmatrix} 0 & \frac{9}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\ \frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 \end{bmatrix}$

$\boxed{\text{x163}}$ $\mathbb{Q}_3^{(1,1;a)}(B_{1u}) = \begin{bmatrix}
0 & 0 & 0 & -\frac{3\sqrt{30}i}{56} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\
-\frac{15i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 \\
0 & \frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & \frac{15i}{56} & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 \\
0 & 0 & \frac{3\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \\
0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\
\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & \frac{i}{56} & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 \\
0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & 0 & -\frac{i}{56} & 0 \\
0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \\
0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x164}}$ $\mathbb{Q}_3^{(1,1;a)}(B_{2u}) = \begin{bmatrix}
0 & 0 & 0 & \frac{3\sqrt{30}}{56} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\
-\frac{15}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 \\
0 & \frac{3\sqrt{5}}{56} & 0 & 0 & 0 & -\frac{15}{56} & 0 & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 \\
0 & 0 & \frac{3\sqrt{30}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} \\
0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\
\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & \frac{1}{56} & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 \\
0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 & \frac{1}{56} & 0 \\
0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} \\
0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x165}}$ $\mathbb{Q}_{1,1}^{(1,1;a)}(E_u) = \begin{bmatrix}
\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}}{20} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 \\
-\frac{2\sqrt{5}}{35} & 0 & -\frac{4\sqrt{2}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 \\
0 & -\frac{4\sqrt{2}}{35} & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 \\
0 & 0 & -\frac{6}{35} & 0 & -\frac{4\sqrt{2}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{4\sqrt{2}}{35} & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 \\
0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & \frac{\sqrt{42}}{56}
\end{bmatrix}$

$\boxed{\text{x166}}$ $\mathbb{Q}_{1,2}^{(1,1;a)}(E_u) = \begin{bmatrix}
\frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 \\
-\frac{2\sqrt{5}i}{35} & 0 & \frac{4\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{56} & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 \\
0 & -\frac{4\sqrt{2}i}{35} & 0 & \frac{6i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\
0 & 0 & -\frac{6i}{35} & 0 & \frac{4\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{4\sqrt{2}i}{35} & 0 & \frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 \\
0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & -\frac{\sqrt{42}i}{56}
\end{bmatrix}$

$\boxed{\text{x167}}$ $\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,1) =$

$$\begin{bmatrix}
\frac{9\sqrt{5}}{224} & 0 & -\frac{27\sqrt{2}}{224} & 0 & \frac{45}{224} & 0 & 0 & -\frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{10}}{112} & 0 & 0 \\
0 & -\frac{3\sqrt{3}}{32} & 0 & \frac{3\sqrt{6}}{224} & 0 & \frac{15\sqrt{15}}{224} & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & \frac{3\sqrt{2}}{112} & 0 & -\frac{\sqrt{10}}{56} & 0 \\
-\frac{15\sqrt{15}}{224} & 0 & -\frac{3\sqrt{6}}{224} & 0 & \frac{3\sqrt{3}}{32} & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{3\sqrt{2}}{112} & 0 & 0 & 0 & -\frac{\sqrt{70}}{112} \\
0 & -\frac{45}{224} & 0 & \frac{27\sqrt{2}}{224} & 0 & -\frac{9\sqrt{5}}{224} & 0 & 0 & -\frac{\sqrt{10}}{112} & 0 & \frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{30}}{112} & 0 \\
0 & -\frac{\sqrt{5}}{28} & 0 & \frac{5\sqrt{10}}{168} & 0 & 0 & -\frac{\sqrt{42}}{224} & 0 & \frac{3\sqrt{2}}{112} & 0 & -\frac{\sqrt{30}}{224} & 0 & 0 & 0 \\
-\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{2}}{56} & 0 & \frac{5}{42} & 0 & 0 & \frac{\sqrt{30}}{224} & 0 & \frac{\sqrt{6}}{112} & 0 & -\frac{3\sqrt{10}}{224} & 0 & 0 \\
0 & \frac{\sqrt{2}}{56} & 0 & \frac{1}{14} & 0 & \frac{5\sqrt{10}}{168} & \frac{\sqrt{105}}{224} & 0 & \frac{3\sqrt{5}}{224} & 0 & -\frac{\sqrt{3}}{224} & 0 & -\frac{3\sqrt{15}}{224} & 0 \\
\frac{5\sqrt{10}}{168} & 0 & \frac{1}{14} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & \frac{3\sqrt{15}}{224} & 0 & \frac{\sqrt{3}}{224} & 0 & -\frac{3\sqrt{5}}{224} & 0 & -\frac{\sqrt{105}}{224} \\
0 & \frac{5}{42} & 0 & \frac{\sqrt{2}}{56} & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & \frac{3\sqrt{10}}{224} & 0 & -\frac{\sqrt{6}}{112} & 0 & -\frac{\sqrt{30}}{224} & 0 \\
0 & 0 & \frac{5\sqrt{10}}{168} & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{224} & 0 & -\frac{3\sqrt{2}}{112} & 0 & \frac{\sqrt{42}}{224}
\end{bmatrix}$$

$\boxed{\text{x168}}$ $\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,1) =$

$$\begin{bmatrix}
\frac{9\sqrt{5}i}{224} & 0 & \frac{27\sqrt{2}i}{224} & 0 & \frac{45i}{224} & 0 & 0 & -\frac{\sqrt{30}i}{112} & 0 & -\frac{\sqrt{6}i}{56} & 0 & -\frac{\sqrt{10}i}{112} & 0 & 0 \\
0 & -\frac{3\sqrt{3}i}{32} & 0 & -\frac{3\sqrt{6}i}{224} & 0 & \frac{15\sqrt{15}i}{224} & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{112} & 0 & -\frac{\sqrt{10}i}{56} & 0 \\
\frac{15\sqrt{15}i}{224} & 0 & -\frac{3\sqrt{6}i}{224} & 0 & -\frac{3\sqrt{3}i}{32} & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{3\sqrt{2}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} \\
0 & \frac{45i}{224} & 0 & \frac{27\sqrt{2}i}{224} & 0 & \frac{9\sqrt{5}i}{224} & 0 & 0 & \frac{\sqrt{10}i}{112} & 0 & \frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{30}i}{112} & 0 \\
0 & \frac{\sqrt{5}i}{28} & 0 & \frac{5\sqrt{10}i}{168} & 0 & 0 & -\frac{\sqrt{42}i}{224} & 0 & -\frac{3\sqrt{2}i}{112} & 0 & -\frac{\sqrt{30}i}{224} & 0 & 0 & 0 \\
-\frac{\sqrt{5}i}{28} & 0 & -\frac{\sqrt{2}i}{56} & 0 & \frac{5i}{42} & 0 & 0 & \frac{\sqrt{30}i}{224} & 0 & -\frac{\sqrt{6}i}{112} & 0 & -\frac{3\sqrt{10}i}{224} & 0 & 0 \\
0 & \frac{\sqrt{2}i}{56} & 0 & -\frac{i}{14} & 0 & \frac{5\sqrt{10}i}{168} & -\frac{\sqrt{105}i}{224} & 0 & \frac{3\sqrt{5}i}{224} & 0 & \frac{\sqrt{3}i}{224} & 0 & -\frac{3\sqrt{15}i}{224} & 0 \\
-\frac{5\sqrt{10}i}{168} & 0 & \frac{i}{14} & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & -\frac{3\sqrt{15}i}{224} & 0 & \frac{\sqrt{3}i}{224} & 0 & \frac{3\sqrt{5}i}{224} & 0 & -\frac{\sqrt{105}i}{224} \\
0 & -\frac{5i}{42} & 0 & \frac{\sqrt{2}i}{56} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & -\frac{3\sqrt{10}i}{224} & 0 & -\frac{\sqrt{6}i}{112} & 0 & \frac{\sqrt{30}i}{224} & 0 \\
0 & 0 & -\frac{5\sqrt{10}i}{168} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{224} & 0 & -\frac{3\sqrt{2}i}{112} & 0 & -\frac{\sqrt{42}i}{224}
\end{bmatrix}$$

$\boxed{\text{x169}}$ $\mathbb{Q}_{3,1}^{(1,1;a)}(E_u,2) = \begin{bmatrix}
\frac{15\sqrt{3}}{224} & 0 & -\frac{9\sqrt{30}}{224} & 0 & -\frac{9\sqrt{15}}{224} & 0 & 0 & -\frac{5\sqrt{2}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{6}}{112} & 0 & 0 \\
0 & -\frac{3\sqrt{5}}{32} & 0 & \frac{3\sqrt{10}}{224} & 0 & -\frac{45}{224} & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & \frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{6}}{56} & 0 \\
\frac{45}{224} & 0 & -\frac{3\sqrt{10}}{224} & 0 & \frac{3\sqrt{5}}{32} & 0 & 0 & \frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{30}}{112} & 0 & 0 & 0 & \frac{\sqrt{42}}{112} \\
0 & \frac{9\sqrt{15}}{224} & 0 & \frac{9\sqrt{30}}{224} & 0 & -\frac{15\sqrt{3}}{224} & 0 & 0 & \frac{\sqrt{6}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{5\sqrt{2}}{112} & 0 \\
0 & -\frac{5\sqrt{3}}{84} & 0 & -\frac{5\sqrt{6}}{168} & 0 & 0 & -\frac{\sqrt{70}}{224} & 0 & \frac{\sqrt{30}}{112} & 0 & \frac{3\sqrt{2}}{224} & 0 & 0 & 0 \\
-\frac{5\sqrt{3}}{84} & 0 & \frac{\sqrt{30}}{168} & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & \frac{5\sqrt{2}}{224} & 0 & \frac{\sqrt{10}}{112} & 0 & \frac{3\sqrt{6}}{224} & 0 & 0 \\
0 & \frac{\sqrt{30}}{168} & 0 & \frac{\sqrt{15}}{42} & 0 & -\frac{5\sqrt{6}}{168} & -\frac{3\sqrt{7}}{224} & 0 & \frac{5\sqrt{3}}{224} & 0 & -\frac{\sqrt{5}}{224} & 0 & \frac{9}{224} & 0 \\
-\frac{5\sqrt{6}}{168} & 0 & \frac{\sqrt{15}}{42} & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & -\frac{9}{224} & 0 & \frac{\sqrt{5}}{224} & 0 & -\frac{5\sqrt{3}}{224} & 0 & \frac{3\sqrt{7}}{224} \\
0 & -\frac{\sqrt{15}}{42} & 0 & \frac{\sqrt{30}}{168} & 0 & -\frac{5\sqrt{3}}{84} & 0 & 0 & -\frac{3\sqrt{6}}{224} & 0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{5\sqrt{2}}{224} & 0 \\
0 & 0 & -\frac{5\sqrt{6}}{168} & 0 & -\frac{5\sqrt{3}}{84} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{224} & 0 & -\frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{70}}{224}
\end{bmatrix}$

$\boxed{\text{x170}}$ $\mathbb{Q}_{3,2}^{(1,1;a)}(E_u,2) = \begin{bmatrix}
\frac{15\sqrt{3}i}{224} & 0 & \frac{9\sqrt{30}i}{224} & 0 & -\frac{9\sqrt{15}i}{224} & 0 & 0 & -\frac{5\sqrt{2}i}{112} & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{6}i}{112} & 0 & 0 \\
0 & -\frac{3\sqrt{5}i}{32} & 0 & -\frac{3\sqrt{10}i}{224} & 0 & -\frac{45i}{224} & -\frac{\sqrt{42}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{112} & 0 & \frac{\sqrt{6}i}{56} & 0 \\
-\frac{45i}{224} & 0 & -\frac{3\sqrt{10}i}{224} & 0 & -\frac{3\sqrt{5}i}{32} & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{30}i}{112} & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} \\
0 & -\frac{9\sqrt{15}i}{224} & 0 & \frac{9\sqrt{30}i}{224} & 0 & \frac{15\sqrt{3}i}{224} & 0 & 0 & -\frac{\sqrt{6}i}{112} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{5\sqrt{2}i}{112} & 0 \\
0 & \frac{5\sqrt{3}i}{84} & 0 & -\frac{5\sqrt{6}i}{168} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & -\frac{\sqrt{30}i}{112} & 0 & \frac{3\sqrt{2}i}{224} & 0 & 0 & 0 \\
-\frac{5\sqrt{3}i}{84} & 0 & -\frac{\sqrt{30}i}{168} & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & \frac{5\sqrt{2}i}{224} & 0 & -\frac{\sqrt{10}i}{112} & 0 & \frac{3\sqrt{6}i}{224} & 0 & 0 \\
0 & \frac{\sqrt{30}i}{168} & 0 & -\frac{\sqrt{15}i}{42} & 0 & -\frac{5\sqrt{6}i}{168} & \frac{3\sqrt{7}i}{224} & 0 & \frac{5\sqrt{3}i}{224} & 0 & \frac{\sqrt{5}i}{224} & 0 & \frac{9i}{224} & 0 \\
\frac{5\sqrt{6}i}{168} & 0 & \frac{\sqrt{15}i}{42} & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & \frac{9i}{224} & 0 & \frac{\sqrt{5}i}{224} & 0 & \frac{5\sqrt{3}i}{224} & 0 & \frac{3\sqrt{7}i}{224} \\
0 & \frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{30}i}{168} & 0 & \frac{5\sqrt{3}i}{84} & 0 & 0 & \frac{3\sqrt{6}i}{224} & 0 & -\frac{\sqrt{10}i}{112} & 0 & \frac{5\sqrt{2}i}{224} & 0 \\
0 & 0 & \frac{5\sqrt{6}i}{168} & 0 & -\frac{5\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{224} & 0 & -\frac{\sqrt{30}i}{112} & 0 & -\frac{\sqrt{70}i}{224}
\end{bmatrix}$

$\boxed{\text{x171}}$ $\mathbb{G}_2^{(a)}(A_{1u}) =$

$$\begin{bmatrix}
0 & \frac{3\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 \\
-\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0
\end{bmatrix}$$

$\boxed{\text{x172}}$ $\mathbb{G}_4^{(a)}(A_{1u}, 1) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 \\
0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{40} \\
0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 \\
\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 \\
\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 \\
0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0 \\
0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} \\
0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\
\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & 0 \\
0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0
\end{bmatrix}$$

$\boxed{\text{x173}}$ $\mathbb{G}_4^{(a)}(A_{1u},2) = \begin{bmatrix} 0 & -\frac{\sqrt{15}i}{84} & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & -\frac{3\sqrt{6}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{40} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{56} & 0 & 0 & 0 & \frac{\sqrt{42}i}{40} \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{40} & 0 & 0 & 0 & \frac{\sqrt{30}i}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & \frac{\sqrt{15}i}{84} & 0 & 0 & \frac{3\sqrt{2}i}{40} & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{56} & 0 & 0 \\ \frac{\sqrt{15}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 & \frac{\sqrt{7}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & \frac{\sqrt{15}i}{84} & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 \end{bmatrix}$

$\boxed{\text{x174}}$ $\mathbb{G}_4^{(a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x175}}$ $\mathbb{G}_2^{(a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & \frac{\sqrt{5}i}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & \frac{2\sqrt{10}i}{35} & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & \frac{i}{28} & 0 & 0 \\
0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 & 0 & \frac{i}{28} & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 \\
0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{140} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \\
0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & \frac{2\sqrt{5}i}{35} & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 \\
0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 \\
0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\
0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x176}}$ $\mathbb{G}_4^{(a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & \frac{3\sqrt{42}i}{280} & 0 & 0 & 0 & \frac{9\sqrt{30}i}{280} & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & -\frac{33\sqrt{2}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{6}i}{280} & 0 & 0 \\
0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & \frac{3\sqrt{6}i}{280} & 0 & 0 & 0 & -\frac{33\sqrt{2}i}{280} & 0 \\
0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{30}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{42}i}{280} \\
0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 \\
-\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & \frac{9\sqrt{3}i}{140} & 0 & 0 & 0 & -\frac{17i}{140} & 0 & 0 \\
0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & \frac{17i}{140} & 0 & 0 & 0 & -\frac{9\sqrt{3}i}{140} & 0 \\
0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{42}i}{140} \\
0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x177}}$ $\mathbb{G}_2^{(a)}(B_{2u}) = \begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{5}}{140} & 0 & 0 & 0 \\
\frac{\sqrt{2}}{7} & 0 & 0 & 0 & \frac{2\sqrt{10}}{35} & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & \frac{1}{28} & 0 & 0 \\
0 & \frac{2\sqrt{10}}{35} & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & -\frac{1}{28} & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 \\
0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{140} & 0 & 0 & 0 & \frac{\sqrt{7}}{28} \\
0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & \frac{2\sqrt{5}}{35} & 0 & 0 & 0 \\
\frac{\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 \\
0 & \frac{3\sqrt{15}}{140} & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 \\
0 & 0 & \frac{3\sqrt{15}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{35} & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\
0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x178}}$ $\mathbb{G}_4^{(a)}(B_{2u}) = \begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & \frac{3\sqrt{42}}{280} & 0 & 0 & 0 & -\frac{9\sqrt{30}}{280} & 0 & 0 & 0 \\
-\frac{\sqrt{3}}{28} & 0 & 0 & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & -\frac{33\sqrt{2}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{6}}{280} & 0 & 0 \\
0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & \frac{3\sqrt{6}}{280} & 0 & 0 & 0 & \frac{33\sqrt{2}}{280} & 0 \\
0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{30}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{42}}{280} \\
0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 \\
-\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & \frac{9\sqrt{3}}{140} & 0 & 0 & 0 & \frac{17}{140} & 0 & 0 \\
0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & \frac{17}{140} & 0 & 0 & 0 & \frac{9\sqrt{3}}{140} & 0 \\
0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 & -\frac{3\sqrt{42}}{140} \\
0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x179}}$ $\mathbb{G}^{(a)}_{2,1}(E_u) = \begin{bmatrix}
\frac{\sqrt{6}}{14} & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & \frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{10}}{70} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & \frac{1}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{70} & 0 & \frac{1}{14} & 0 \\
0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{6}}{28} & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & \frac{11\sqrt{5}}{140} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{5}}{140} & 0 & -\frac{1}{28} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{\sqrt{35}}{28}
\end{bmatrix}$

$\boxed{\text{x180}}$ $\mathbb{G}^{(a)}_{2,2}(E_u) = \begin{bmatrix}
\frac{\sqrt{6}i}{14} & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{10}i}{70} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & -\frac{i}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{70} & 0 & -\frac{i}{14} & 0 \\
0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & -\frac{11\sqrt{5}i}{140} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{5}i}{140} & 0 & \frac{i}{28} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & \frac{\sqrt{35}i}{28}
\end{bmatrix}$

$\boxed{\text{x181}}$ $\mathbb{G}^{(a)}_{4,1}(E_u,1) =$

$$\begin{bmatrix}
\frac{\sqrt{7}}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{35}}{112} & 0 & 0 & \frac{9\sqrt{42}}{560} & 0 & \frac{3\sqrt{210}}{280} & 0 & \frac{9\sqrt{14}}{560} & 0 & 0 \\
0 & -\frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{21}}{112} & -\frac{3\sqrt{2}}{80} & 0 & -\frac{3\sqrt{42}}{140} & 0 & -\frac{3\sqrt{70}}{560} & 0 & \frac{3\sqrt{14}}{280} & 0 \\
\frac{\sqrt{21}}{112} & 0 & \frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{105}}{112} & 0 & 0 & \frac{3\sqrt{14}}{280} & 0 & -\frac{3\sqrt{70}}{560} & 0 & -\frac{3\sqrt{42}}{140} & 0 & -\frac{3\sqrt{2}}{80} \\
0 & -\frac{\sqrt{35}}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{7}}{112} & 0 & 0 & \frac{9\sqrt{14}}{560} & 0 & \frac{3\sqrt{210}}{280} & 0 & \frac{9\sqrt{42}}{560} & 0 \\
0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{30}}{80} & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & 0 \\
\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{42}}{560} & 0 & -\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{14}}{80} & 0 & 0 \\
0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & \frac{3\sqrt{3}}{80} & 0 & \frac{\sqrt{7}}{560} & 0 & -\frac{\sqrt{105}}{80} & 0 & -\frac{\sqrt{21}}{560} & 0 \\
\frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & \frac{\sqrt{21}}{560} & 0 & \frac{\sqrt{105}}{80} & 0 & -\frac{\sqrt{7}}{560} & 0 & -\frac{3\sqrt{3}}{80} \\
0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{14}}{80} & 0 & \frac{\sqrt{210}}{280} & 0 & \frac{13\sqrt{42}}{560} & 0 \\
0 & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{30}}{80}
\end{bmatrix}$$

$\boxed{\text{x182}}$ $\mathbb{G}^{(a)}_{4,2}(E_u,1) =$

$$\begin{bmatrix}
\frac{\sqrt{7}i}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{35}i}{112} & 0 & 0 & \frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{210}i}{280} & 0 & \frac{9\sqrt{14}i}{560} & 0 & 0 \\
0 & -\frac{\sqrt{105}i}{112} & 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{21}i}{112} & \frac{3\sqrt{2}i}{80} & 0 & -\frac{3\sqrt{42}i}{140} & 0 & \frac{3\sqrt{70}i}{560} & 0 & \frac{3\sqrt{14}i}{280} & 0 \\
-\frac{\sqrt{21}i}{112} & 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{105}i}{112} & 0 & 0 & -\frac{3\sqrt{14}i}{280} & 0 & -\frac{3\sqrt{70}i}{560} & 0 & \frac{3\sqrt{42}i}{140} & 0 & -\frac{3\sqrt{2}i}{80} \\
0 & \frac{\sqrt{35}i}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{7}i}{112} & 0 & 0 & -\frac{9\sqrt{14}i}{560} & 0 & \frac{3\sqrt{210}i}{280} & 0 & -\frac{9\sqrt{42}i}{560} & 0 \\
0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & \frac{\sqrt{30}i}{80} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 \\
\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{42}i}{560} & 0 & \frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{14}i}{80} & 0 & 0 \\
0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & -\frac{3\sqrt{3}i}{80} & 0 & \frac{\sqrt{7}i}{560} & 0 & \frac{\sqrt{105}i}{80} & 0 & -\frac{\sqrt{21}i}{560} & 0 \\
-\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{21}i}{560} & 0 & \frac{\sqrt{105}i}{80} & 0 & \frac{\sqrt{7}i}{560} & 0 & -\frac{3\sqrt{3}i}{80} \\
0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{80} & 0 & \frac{\sqrt{210}i}{280} & 0 & -\frac{13\sqrt{42}i}{560} & 0 \\
0 & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{30}i}{80}
\end{bmatrix}$$

$\boxed{\text{x183}}$ $\mathbb{G}^{(a)}_{4,1}(E_u,2) =$

$$\begin{bmatrix}
\frac{1}{112} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{\sqrt{5}}{16} & 0 & 0 & \frac{9\sqrt{6}}{560} & 0 & \frac{3\sqrt{30}}{280} & 0 & -\frac{9\sqrt{2}}{80} & 0 & 0 \\
0 & -\frac{\sqrt{15}}{112} & 0 & -\frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{3}}{16} & \frac{3\sqrt{14}}{80} & 0 & -\frac{3\sqrt{6}}{140} & 0 & -\frac{3\sqrt{10}}{560} & 0 & -\frac{3\sqrt{2}}{40} & 0 \\
-\frac{\sqrt{3}}{16} & 0 & \frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{15}}{112} & 0 & 0 & -\frac{3\sqrt{2}}{40} & 0 & -\frac{3\sqrt{10}}{560} & 0 & -\frac{3\sqrt{6}}{140} & 0 & \frac{3\sqrt{14}}{80} \\
0 & \frac{\sqrt{5}}{16} & 0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{1}{112} & 0 & 0 & -\frac{9\sqrt{2}}{80} & 0 & \frac{3\sqrt{30}}{280} & 0 & \frac{9\sqrt{6}}{560} & 0 \\
0 & -\frac{1}{28} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{210}}{560} & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 \\
\frac{1}{28} & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{6}}{560} & 0 & -\frac{\sqrt{30}}{280} & 0 & -\frac{7\sqrt{2}}{80} & 0 & 0 \\
0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & -\frac{3\sqrt{21}}{80} & 0 & \frac{1}{560} & 0 & -\frac{\sqrt{15}}{80} & 0 & \frac{\sqrt{3}}{80} & 0 \\
-\frac{\sqrt{2}}{8} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & -\frac{\sqrt{3}}{80} & 0 & \frac{\sqrt{15}}{80} & 0 & -\frac{1}{560} & 0 & \frac{3\sqrt{21}}{80} \\
0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{1}{28} & 0 & 0 & \frac{7\sqrt{2}}{80} & 0 & \frac{\sqrt{30}}{280} & 0 & \frac{13\sqrt{6}}{560} & 0 \\
0 & 0 & \frac{\sqrt{2}}{8} & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{210}}{560}
\end{bmatrix}$$

$\boxed{\text{x184}}$ $\mathbb{G}^{(a)}_{4,2}(E_u,2) =$

$$\begin{bmatrix}
\frac{i}{112} & 0 & -\frac{\sqrt{10}i}{112} & 0 & -\frac{\sqrt{5}i}{16} & 0 & 0 & \frac{9\sqrt{6}i}{560} & 0 & -\frac{3\sqrt{30}i}{280} & 0 & -\frac{9\sqrt{2}i}{80} & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{112} & 0 & \frac{\sqrt{30}i}{112} & 0 & \frac{\sqrt{3}i}{16} & -\frac{3\sqrt{14}i}{80} & 0 & -\frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{10}i}{560} & 0 & -\frac{3\sqrt{2}i}{40} & 0 \\
\frac{\sqrt{3}i}{16} & 0 & \frac{\sqrt{30}i}{112} & 0 & -\frac{\sqrt{15}i}{112} & 0 & 0 & \frac{3\sqrt{2}i}{40} & 0 & -\frac{3\sqrt{10}i}{560} & 0 & \frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{14}i}{80} \\
0 & -\frac{\sqrt{5}i}{16} & 0 & -\frac{\sqrt{10}i}{112} & 0 & \frac{i}{112} & 0 & 0 & \frac{9\sqrt{2}i}{80} & 0 & \frac{3\sqrt{30}i}{280} & 0 & -\frac{9\sqrt{6}i}{560} & 0 \\
0 & \frac{i}{28} & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & \frac{\sqrt{210}i}{560} & 0 & -\frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 \\
\frac{i}{28} & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{6}i}{560} & 0 & \frac{\sqrt{30}i}{280} & 0 & -\frac{7\sqrt{2}i}{80} & 0 & 0 \\
0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & \frac{3\sqrt{21}i}{80} & 0 & \frac{i}{560} & 0 & \frac{\sqrt{15}i}{80} & 0 & \frac{\sqrt{3}i}{80} & 0 \\
\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & \frac{\sqrt{3}i}{80} & 0 & \frac{\sqrt{15}i}{80} & 0 & \frac{i}{560} & 0 & \frac{3\sqrt{21}i}{80} \\
0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{i}{28} & 0 & 0 & -\frac{7\sqrt{2}i}{80} & 0 & \frac{\sqrt{30}i}{280} & 0 & -\frac{13\sqrt{6}i}{560} & 0 \\
0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{210}i}{560}
\end{bmatrix}$$

$\boxed{\text{x185}}$ $\mathbb{G}_2^{(1,-1;a)}(A_{1u}) =$

$$\begin{bmatrix}
0 & -\frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}i}{35} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}i}{35} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 \\
\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{4\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{4\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{70} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0
\end{bmatrix}$$

$\boxed{\text{x186}}$ $\mathbb{G}_4^{(1,-1;a)}(A_{1u}, 1) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{7}i}{168} & 0 & 0 & 0 & \frac{\sqrt{35}i}{168} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & \frac{\sqrt{10}i}{24} \\
0 & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 & 0 \\
-\frac{\sqrt{35}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{168} & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 \\
-\frac{\sqrt{7}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{168} & 0 & 0 \\
0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & \frac{\sqrt{70}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 \\
0 & 0 & -\frac{\sqrt{7}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} \\
0 & 0 & 0 & -\frac{\sqrt{7}i}{42} & 0 & 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{168} & 0 & 0 & 0 \\
-\frac{\sqrt{35}i}{84} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{42} & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{84} & 0 & 0 & \frac{5\sqrt{14}i}{168} & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & 0
\end{bmatrix}$$

$\boxed{\text{x187}}$ $\mathbb{G}_4^{(1,-1;a)}(A_{1u},2) = \begin{bmatrix}
0 & \frac{\sqrt{5}i}{168} & 0 & 0 & 0 & -\frac{i}{24} & 0 & 0 & \frac{5\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 \\
0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{10}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{24} \\
0 & 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & -\frac{\sqrt{14}i}{24} & 0 & 0 & 0 & -\frac{5\sqrt{10}i}{168} & 0 & 0 & 0 \\
\frac{i}{24} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{168} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & \frac{5\sqrt{2}i}{56} & 0 & 0 \\
-\frac{\sqrt{5}i}{84} & 0 & 0 & 0 & \frac{i}{12} & 0 & 0 & -\frac{5\sqrt{30}i}{168} & 0 & 0 & 0 & \frac{\sqrt{10}i}{24} & 0 & 0 \\
0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 & \frac{i}{12} & 0 & 0 & \frac{5\sqrt{2}i}{42} & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\
0 & 0 & -\frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{15}i}{168} & 0 & 0 & 0 & \frac{\sqrt{21}i}{24} \\
0 & 0 & 0 & -\frac{\sqrt{5}i}{42} & 0 & 0 & -\frac{\sqrt{21}i}{24} & 0 & 0 & 0 & -\frac{5\sqrt{15}i}{168} & 0 & 0 & 0 \\
\frac{i}{12} & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{42} & 0 & 0 \\
0 & \frac{i}{12} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{84} & 0 & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 & \frac{5\sqrt{30}i}{168} & 0
\end{bmatrix}$

$\boxed{\text{x188}}$ $\mathbb{G}_6^{(1,-1;a)}(A_{1u},1) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{264} & 0 & 0 & 0 & -\frac{7\sqrt{11}i}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{88} & 0 & 0 & 0 & \frac{7\sqrt{165}i}{264} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}i}{264} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{66}i}{264} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{165}i}{264} & 0 & 0 & 0 & \frac{\sqrt{55}i}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{11}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{264} & 0
\end{bmatrix}$

$\boxed{\text{x189}}$ $\mathbb{G}_6^{(1,-1;a)}(A_{1u},2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{264} & 0 & 0 & 0 & \frac{\sqrt{77}i}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{264} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{264} & 0 & 0 & 0 & \frac{\sqrt{330}i}{264} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{462}i}{264} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{264} & 0 & 0 & 0 & \frac{\sqrt{385}i}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{231}i}{264} & 0
\end{bmatrix}$

$\boxed{\text{x190}}$ $\mathbb{G}_4^{(1,-1;a)}(A_{2u}) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x191}}$ $\mathbb{G}_6^{(1,-1;a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x192}}$ $\mathbb{G}_2^{(1,-1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{3i}{70} & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{35} & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 \\ 0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} \\ 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3i}{35} & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & \frac{2\sqrt{3}i}{35} & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & \frac{3i}{35} & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 & 0 & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 \\ 0 & \frac{3i}{35} & 0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & \frac{\sqrt{30}i}{35} & 0 \\ 0 & 0 & \frac{3i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x193}}$ $\mathbb{G}_4^{(1,-1;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 \\
\frac{i}{56} & 0 & 0 & 0 & \frac{\sqrt{5}i}{56} & 0 & 0 & \frac{11\sqrt{6}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 \\
0 & -\frac{\sqrt{5}i}{56} & 0 & 0 & 0 & -\frac{i}{56} & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & \frac{11\sqrt{6}i}{168} & 0 \\
0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \\
0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\
\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & \frac{9i}{56} & 0 & 0 & 0 & -\frac{17\sqrt{3}i}{168} & 0 & 0 \\
0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & \frac{17\sqrt{3}i}{168} & 0 & 0 & 0 & -\frac{9i}{56} & 0 \\
0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} \\
0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x194}}$ $\mathbb{G}_6^{(1,-1;a)}(B_{1u}, 1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{24} & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{24} & 0 & 0 & 0 & \frac{\sqrt{35}i}{24} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 & \frac{\sqrt{14}i}{24} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{24} & 0 & 0 & 0 & -\frac{i}{24} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & \frac{\sqrt{7}i}{24} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x195}}$ $\mathbb{G}_6^{(1,-1;a)}(B_{1u},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{264} & 0 & 0 & 0 & \frac{\sqrt{11}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{264} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{264} & 0 & 0 & 0 & \frac{\sqrt{55}i}{264} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{264} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x196}}$ $\mathbb{G}_2^{(1,-1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{3}{70} & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & 0 & 0 & -\frac{\sqrt{3}}{35} & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{70} & 0 & 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & \frac{3\sqrt{5}}{35} & 0 & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 \\ 0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{35} & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} \\ 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{35} & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & \frac{2\sqrt{3}}{35} & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{35} & 0 & 0 & 0 & \frac{3}{35} & 0 & 0 & \frac{\sqrt{30}}{35} & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 \\ 0 & -\frac{3}{35} & 0 & 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & \frac{\sqrt{30}}{35} & 0 \\ 0 & 0 & -\frac{3}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{35} & 0 & 0 & 0 & \frac{\sqrt{105}}{70} \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x197}}$ $\mathbb{G}_4^{(1,-1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & 0 & 0 \\ \frac{1}{56} & 0 & 0 & 0 & -\frac{\sqrt{5}}{56} & 0 & 0 & \frac{11\sqrt{6}}{168} & 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{56} & 0 & 0 & 0 & \frac{1}{56} & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{11\sqrt{6}}{168} & 0 \\ 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} \\ 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{28} & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & \frac{9}{56} & 0 & 0 & 0 & \frac{17\sqrt{3}}{168} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & \frac{17\sqrt{3}}{168} & 0 & 0 & 0 & \frac{9}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x198}}$ $\mathbb{G}_6^{(1,-1;a)}(B_{2u}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x199}}$ $\mathbb{G}_6^{(1,-1;a)}(B_{2u},2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{66} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{66} & 0 & 0 & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{66} & 0 & 0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 & 0 & -\frac{\sqrt{154}}{66} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 & \frac{\sqrt{11}}{66} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{66} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x200}}$ $\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) = \begin{bmatrix}
-\frac{3\sqrt{10}}{140} & 0 & -\frac{9}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & -\frac{2\sqrt{3}}{35} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{6}}{140} & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & -\frac{6}{35} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}}{28} & 0 & -\frac{\sqrt{6}}{140} & 0 & 0 & 0 & 0 & -\frac{6}{35} & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 \\
0 & 0 & 0 & \frac{9}{140} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{35} & 0 & -\frac{2\sqrt{15}}{35} & 0 \\
0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{3}{28} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{10}}{35} & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{140} & 0 & \frac{11\sqrt{3}}{140} & 0 & 0 & 0 & 0 \\
0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 \\
0 & 0 & 0 & \frac{2}{35} & 0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{3}}{140} & 0 & -\frac{\sqrt{15}}{140} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{28} & 0 & -\frac{\sqrt{21}}{28}
\end{bmatrix}$

$\boxed{\text{x201}}$ $\mathbb{G}_{2,2}^{(1,-1;a)}(E_u) = \begin{bmatrix} -\frac{3\sqrt{10}i}{140} & 0 & \frac{9i}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & \frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{140} & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & \frac{6i}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & -\frac{6i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{9i}{140} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 \\ 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{35} & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{140} & 0 & -\frac{11\sqrt{3}i}{140} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 \\ 0 & 0 & 0 & \frac{2i}{35} & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{3}i}{140} & 0 & \frac{\sqrt{15}i}{140} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{28} & 0 & \frac{\sqrt{21}i}{28} \end{bmatrix}$

$\boxed{\text{x202}}$ $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} -\frac{\sqrt{21}}{672} & 0 & -\frac{\sqrt{210}}{672} & 0 & -\frac{\sqrt{105}}{672} & 0 & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 \\ 0 & \frac{\sqrt{35}}{224} & 0 & \frac{\sqrt{70}}{224} & 0 & \frac{\sqrt{7}}{224} & \frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{42}}{168} & 0 \\ -\frac{\sqrt{7}}{224} & 0 & -\frac{\sqrt{70}}{224} & 0 & -\frac{\sqrt{35}}{224} & 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{6}}{48} \\ 0 & \frac{\sqrt{105}}{672} & 0 & \frac{\sqrt{210}}{672} & 0 & \frac{\sqrt{21}}{672} & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{3\sqrt{14}}{112} & 0 \\ 0 & \frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{\sqrt{10}}{32} & 0 & \frac{5\sqrt{210}}{336} & 0 & \frac{5\sqrt{14}}{224} & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{84} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{14}}{224} & 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{42}}{96} & 0 & 0 \\ 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} & \frac{3}{32} & 0 & \frac{\sqrt{21}}{672} & 0 & -\frac{\sqrt{35}}{32} & 0 & -\frac{\sqrt{7}}{224} & 0 \\ -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{32} & 0 & -\frac{\sqrt{21}}{672} & 0 & -\frac{3}{32} \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & -\frac{\sqrt{42}}{96} & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{13\sqrt{14}}{224} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{224} & 0 & -\frac{5\sqrt{210}}{336} & 0 & -\frac{\sqrt{10}}{32} \end{bmatrix}$

$\boxed{\text{x203}}$ $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,1) =$

$$\begin{bmatrix}
-\frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{210}i}{672} & 0 & -\frac{\sqrt{105}i}{672} & 0 & 0 & -\frac{3\sqrt{14}i}{112} & 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 \\
0 & \frac{\sqrt{35}i}{224} & 0 & -\frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{7}i}{224} & -\frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{42}i}{168} & 0 \\
\frac{\sqrt{7}i}{224} & 0 & -\frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{35}i}{224} & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{14}i}{28} & 0 & \frac{\sqrt{6}i}{48} \\
0 & -\frac{\sqrt{105}i}{672} & 0 & \frac{\sqrt{210}i}{672} & 0 & -\frac{\sqrt{21}i}{672} & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{3\sqrt{14}i}{112} & 0 \\
0 & -\frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & \frac{\sqrt{10}i}{32} & 0 & -\frac{5\sqrt{210}i}{336} & 0 & \frac{5\sqrt{14}i}{224} & 0 & 0 & 0 \\
-\frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{14}i}{224} & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{42}i}{96} & 0 & 0 \\
0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{168} & -\frac{3i}{32} & 0 & \frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{35}i}{32} & 0 & -\frac{\sqrt{7}i}{224} & 0 \\
\frac{\sqrt{42}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{7}i}{224} & 0 & \frac{\sqrt{35}i}{32} & 0 & \frac{\sqrt{21}i}{672} & 0 & -\frac{3i}{32} \\
0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & \frac{\sqrt{42}i}{96} & 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{13\sqrt{14}i}{224} & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{224} & 0 & -\frac{5\sqrt{210}i}{336} & 0 & \frac{\sqrt{10}i}{32}
\end{bmatrix}$$

$\boxed{\text{x204}}$ $\mathbb{G}_{4,1}^{(1,-1;a)}(E_u,2) =$

$$\begin{bmatrix}
-\frac{\sqrt{3}}{672} & 0 & -\frac{\sqrt{30}}{672} & 0 & \frac{\sqrt{15}}{96} & 0 & 0 & -\frac{3\sqrt{2}}{112} & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{6}}{16} & 0 & 0 \\
0 & \frac{\sqrt{5}}{224} & 0 & \frac{\sqrt{10}}{224} & 0 & -\frac{1}{32} & -\frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{2}}{28} & 0 & \frac{\sqrt{30}}{336} & 0 & \frac{\sqrt{6}}{24} & 0 \\
\frac{1}{32} & 0 & -\frac{\sqrt{10}}{224} & 0 & -\frac{\sqrt{5}}{224} & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{30}}{336} & 0 & \frac{\sqrt{2}}{28} & 0 & -\frac{\sqrt{42}}{48} \\
0 & -\frac{\sqrt{15}}{96} & 0 & \frac{\sqrt{30}}{672} & 0 & \frac{\sqrt{3}}{672} & 0 & 0 & \frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{10}}{56} & 0 & -\frac{3\sqrt{2}}{112} & 0 \\
0 & \frac{\sqrt{3}}{84} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & \frac{5\sqrt{30}}{336} & 0 & -\frac{5\sqrt{2}}{32} & 0 & 0 & 0 \\
-\frac{\sqrt{3}}{84} & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}}{224} & 0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{7\sqrt{6}}{96} & 0 & 0 \\
0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & -\frac{3\sqrt{7}}{32} & 0 & \frac{\sqrt{3}}{672} & 0 & -\frac{\sqrt{5}}{32} & 0 & \frac{1}{32} & 0 \\
\frac{\sqrt{6}}{24} & 0 & 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & -\frac{1}{32} & 0 & \frac{\sqrt{5}}{32} & 0 & -\frac{\sqrt{3}}{672} & 0 & \frac{3\sqrt{7}}{32} \\
0 & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 & \frac{7\sqrt{6}}{96} & 0 & \frac{\sqrt{10}}{112} & 0 & \frac{13\sqrt{2}}{224} & 0 \\
0 & 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{32} & 0 & -\frac{5\sqrt{30}}{336} & 0 & -\frac{\sqrt{70}}{224}
\end{bmatrix}$$

$\boxed{\text{x205}}$ $\mathbb{G}_{4,2}^{(1,-1;a)}(E_u,2) =$

$$\begin{bmatrix}
-\frac{\sqrt{3}i}{672} & 0 & \frac{\sqrt{30}i}{672} & 0 & \frac{\sqrt{15}i}{96} & 0 & 0 & -\frac{3\sqrt{2}i}{112} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 \\
0 & \frac{\sqrt{5}i}{224} & 0 & -\frac{\sqrt{10}i}{224} & 0 & -\frac{i}{32} & \frac{\sqrt{42}i}{48} & 0 & \frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{30}i}{336} & 0 & \frac{\sqrt{6}i}{24} & 0 \\
-\frac{i}{32} & 0 & -\frac{\sqrt{10}i}{224} & 0 & \frac{\sqrt{5}i}{224} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{30}i}{336} & 0 & -\frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{42}i}{48} \\
0 & \frac{\sqrt{15}i}{96} & 0 & \frac{\sqrt{30}i}{672} & 0 & -\frac{\sqrt{3}i}{672} & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{3\sqrt{2}i}{112} & 0 \\
0 & -\frac{\sqrt{3}i}{84} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & \frac{\sqrt{70}i}{224} & 0 & -\frac{5\sqrt{30}i}{336} & 0 & -\frac{5\sqrt{2}i}{32} & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{84} & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}i}{224} & 0 & \frac{\sqrt{10}i}{112} & 0 & -\frac{7\sqrt{6}i}{96} & 0 & 0 \\
0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & \frac{3\sqrt{7}i}{32} & 0 & \frac{\sqrt{3}i}{672} & 0 & \frac{\sqrt{5}i}{32} & 0 & \frac{i}{32} & 0 \\
-\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & \frac{i}{32} & 0 & \frac{\sqrt{5}i}{32} & 0 & \frac{\sqrt{3}i}{672} & 0 & \frac{3\sqrt{7}i}{32} \\
0 & 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & \frac{\sqrt{3}i}{84} & 0 & 0 & -\frac{7\sqrt{6}i}{96} & 0 & \frac{\sqrt{10}i}{112} & 0 & -\frac{13\sqrt{2}i}{224} & 0 \\
0 & 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{32} & 0 & -\frac{5\sqrt{30}i}{336} & 0 & \frac{\sqrt{70}i}{224}
\end{bmatrix}$$

$\boxed{\text{x206}}$ $\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{1056} & 0 & \frac{\sqrt{154}}{352} & 0 & -\frac{\sqrt{2310}}{352} & 0 & -\frac{\sqrt{462}}{96} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1056} & 0 & -\frac{5\sqrt{462}}{1056} & 0 & \frac{3\sqrt{770}}{352} & 0 & \frac{\sqrt{330}}{96} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{176} & 0 & \frac{\sqrt{385}}{176} & 0 & \frac{5\sqrt{231}}{528} & 0 & -\frac{\sqrt{1155}}{176} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{176} & 0 & -\frac{5\sqrt{231}}{528} & 0 & -\frac{\sqrt{385}}{176} & 0 & \frac{\sqrt{165}}{176} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{96} & 0 & -\frac{3\sqrt{770}}{352} & 0 & \frac{5\sqrt{462}}{1056} & 0 & \frac{\sqrt{2310}}{1056} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{96} & 0 & \frac{\sqrt{2310}}{352} & 0 & -\frac{\sqrt{154}}{352} & 0 & -\frac{\sqrt{66}}{1056}
\end{bmatrix}$$

$\boxed{\text{x207}}$ $\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,1) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{1056} & 0 & -\frac{\sqrt{154}i}{352} & 0 & -\frac{\sqrt{2310}i}{352} & 0 & \frac{\sqrt{462}i}{96} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1056} & 0 & \frac{5\sqrt{462}i}{1056} & 0 & \frac{3\sqrt{770}i}{352} & 0 & -\frac{\sqrt{330}i}{96} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{176} & 0 & \frac{\sqrt{385}i}{176} & 0 & -\frac{5\sqrt{231}i}{528} & 0 & -\frac{\sqrt{1155}i}{176} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{176} & 0 & -\frac{5\sqrt{231}i}{528} & 0 & \frac{\sqrt{385}i}{176} & 0 & \frac{\sqrt{165}i}{176} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{96} & 0 & \frac{3\sqrt{770}i}{352} & 0 & \frac{5\sqrt{462}i}{1056} & 0 & -\frac{\sqrt{2310}i}{1056} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{96} & 0 & -\frac{\sqrt{2310}i}{352} & 0 & -\frac{\sqrt{154}i}{352} & 0 & \frac{\sqrt{66}i}{1056}
\end{bmatrix}$

$\boxed{\text{x208}}$ $\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{64} & 0 & \frac{\sqrt{21}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 & \frac{\sqrt{7}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{64} & 0 & -\frac{5\sqrt{7}}{64} & 0 & -\frac{\sqrt{105}}{64} & 0 & -\frac{\sqrt{5}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{64} & 0 & \frac{\sqrt{210}}{64} & 0 & \frac{5\sqrt{14}}{64} & 0 & \frac{\sqrt{70}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{64} & 0 & -\frac{5\sqrt{14}}{64} & 0 & -\frac{\sqrt{210}}{64} & 0 & -\frac{\sqrt{10}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{64} & 0 & \frac{\sqrt{105}}{64} & 0 & \frac{5\sqrt{7}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{64} & 0 & -\frac{\sqrt{35}}{64} & 0 & -\frac{\sqrt{21}}{64} & 0 & -\frac{1}{64}
\end{bmatrix}$

$\boxed{\text{x209}}$ $\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{64} & 0 & -\frac{\sqrt{21}i}{64} & 0 & \frac{\sqrt{35}i}{64} & 0 & -\frac{\sqrt{7}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{64} & 0 & \frac{5\sqrt{7}i}{64} & 0 & -\frac{\sqrt{105}i}{64} & 0 & \frac{\sqrt{5}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{64} & 0 & \frac{\sqrt{210}i}{64} & 0 & -\frac{5\sqrt{14}i}{64} & 0 & \frac{\sqrt{70}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{64} & 0 & -\frac{5\sqrt{14}i}{64} & 0 & \frac{\sqrt{210}i}{64} & 0 & -\frac{\sqrt{10}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{64} & 0 & -\frac{\sqrt{105}i}{64} & 0 & \frac{5\sqrt{7}i}{64} & 0 & -\frac{\sqrt{35}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{64} & 0 & \frac{\sqrt{35}i}{64} & 0 & -\frac{\sqrt{21}i}{64} & 0 & \frac{i}{64}
\end{bmatrix}$$

$\boxed{\text{x210}}$ $\mathbb{G}_{6,1}^{(1,-1;a)}(E_u,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{2112} & 0 & \frac{\sqrt{1155}}{2112} & 0 & -\frac{9\sqrt{77}}{704} & 0 & \frac{\sqrt{385}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{2112} & 0 & -\frac{5\sqrt{385}}{2112} & 0 & \frac{9\sqrt{231}}{704} & 0 & -\frac{5\sqrt{11}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{22}}{704} & 0 & \frac{5\sqrt{462}}{2112} & 0 & \frac{5\sqrt{770}}{2112} & 0 & -\frac{9\sqrt{154}}{704} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{704} & 0 & -\frac{5\sqrt{770}}{2112} & 0 & -\frac{5\sqrt{462}}{2112} & 0 & \frac{9\sqrt{22}}{704} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{64} & 0 & -\frac{9\sqrt{231}}{704} & 0 & \frac{5\sqrt{385}}{2112} & 0 & \frac{5\sqrt{77}}{2112} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{64} & 0 & \frac{9\sqrt{77}}{704} & 0 & -\frac{\sqrt{1155}}{2112} & 0 & -\frac{\sqrt{55}}{2112}
\end{bmatrix}$$

$\boxed{\text{x211}}$ $\mathbb{G}_{6,2}^{(1,-1;a)}(E_u,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{2112} & 0 & -\frac{\sqrt{1155}i}{2112} & 0 & -\frac{9\sqrt{77}i}{704} & 0 & -\frac{\sqrt{385}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{2112} & 0 & \frac{5\sqrt{385}i}{2112} & 0 & \frac{9\sqrt{231}i}{704} & 0 & \frac{5\sqrt{11}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{22}i}{704} & 0 & \frac{5\sqrt{462}i}{2112} & 0 & -\frac{5\sqrt{770}i}{2112} & 0 & -\frac{9\sqrt{154}i}{704} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{154}i}{704} & 0 & -\frac{5\sqrt{770}i}{2112} & 0 & \frac{5\sqrt{462}i}{2112} & 0 & \frac{9\sqrt{22}i}{704} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}i}{64} & 0 & \frac{9\sqrt{231}i}{704} & 0 & \frac{5\sqrt{385}i}{2112} & 0 & -\frac{5\sqrt{77}i}{2112} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{64} & 0 & -\frac{9\sqrt{77}i}{704} & 0 & -\frac{\sqrt{1155}i}{2112} & 0 & \frac{\sqrt{55}i}{2112}
\end{bmatrix}$$

$\boxed{\text{x212}}$ $\mathbb{G}_{2}^{(1,0;a)}(A_{1u}) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{70} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 \\
-\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0
\end{bmatrix}$$

$\boxed{\text{x213}}$ $\mathbb{G}_4^{(1,0;a)}(A_{1u},1) = \begin{bmatrix}
0 & -\frac{\sqrt{105}i}{840} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{168} & 0 & 0 & \frac{9\sqrt{42}i}{280} & 0 & 0 & 0 & \frac{9\sqrt{14}i}{280} & 0 \\
0 & 0 & \frac{\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{6}i}{40} \\
0 & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & \frac{3\sqrt{6}i}{40} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 & 0 \\
\frac{\sqrt{21}i}{168} & 0 & 0 & 0 & \frac{\sqrt{105}i}{840} & 0 & 0 & \frac{9\sqrt{14}i}{280} & 0 & 0 & 0 & \frac{9\sqrt{42}i}{280} & 0 & 0 \\
\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{70}i}{280} & 0 & 0 & 0 & \frac{\sqrt{210}i}{840} & 0 & 0 \\
0 & -\frac{3\sqrt{105}i}{140} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & -\frac{\sqrt{42}i}{210} & 0 & 0 & 0 & \frac{\sqrt{14}i}{140} & 0 \\
0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{280} & 0 & 0 & 0 & \frac{i}{40} \\
0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & -\frac{i}{40} & 0 & 0 & 0 & \frac{\sqrt{35}i}{280} & 0 & 0 & 0 \\
\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{105}i}{140} & 0 & 0 & -\frac{\sqrt{14}i}{140} & 0 & 0 & 0 & \frac{\sqrt{42}i}{210} & 0 & 0 \\
0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & -\frac{\sqrt{210}i}{840} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0
\end{bmatrix}$

$\boxed{\text{x214}}$ $\mathbb{G}_4^{(1,0;a)}(A_{1u},2) = \begin{bmatrix}
0 & -\frac{\sqrt{3}i}{168} & 0 & 0 & 0 & \frac{\sqrt{15}i}{120} & 0 & 0 & \frac{9\sqrt{30}i}{280} & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{200} & 0 \\
0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{200} \\
0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & -\frac{3\sqrt{210}i}{200} & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{56} & 0 & 0 & 0 \\
-\frac{\sqrt{15}i}{120} & 0 & 0 & 0 & \frac{\sqrt{3}i}{168} & 0 & 0 & -\frac{9\sqrt{10}i}{200} & 0 & 0 & 0 & \frac{9\sqrt{30}i}{280} & 0 & 0 \\
\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{120} & 0 & 0 \\
0 & -\frac{3\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & -\frac{\sqrt{30}i}{210} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{100} & 0 \\
0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{56} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{200} \\
0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & \frac{\sqrt{35}i}{200} & 0 & 0 & 0 & \frac{i}{56} & 0 & 0 & 0 \\
-\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{28} & 0 & 0 & \frac{\sqrt{10}i}{100} & 0 & 0 & 0 & \frac{\sqrt{30}i}{210} & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & \frac{\sqrt{6}i}{120} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0
\end{bmatrix}$

$\boxed{\text{x215}}$ $\mathbb{G}_4^{(1,0;a)}(A_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{210}}{700} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{100} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{210}}{700} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{350} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{100} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x216}}$ $\mathbb{G}_2^{(1,0;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{10}i}{70} & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\
0 & -\frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 \\
0 & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \\
0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{105} & 0 & 0 & 0 \\
-\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 \\
0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 \\
0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} \\
0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{84} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x217}}$ $\mathbb{G}_4^{(1,0;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & -\frac{9\sqrt{210}i}{1400} & 0 & 0 & 0 & -\frac{27\sqrt{6}i}{280} & 0 & 0 & 0 \\
-\frac{\sqrt{15}i}{280} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & \frac{99\sqrt{10}i}{1400} & 0 & 0 & 0 & -\frac{9\sqrt{30}i}{1400} & 0 & 0 \\
0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & \frac{\sqrt{15}i}{280} & 0 & 0 & -\frac{9\sqrt{30}i}{1400} & 0 & 0 & 0 & \frac{99\sqrt{10}i}{1400} & 0 \\
0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{6}i}{280} & 0 & 0 & 0 & -\frac{9\sqrt{210}i}{1400} \\
0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & \frac{3\sqrt{210}i}{1400} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{280} & 0 & 0 & 0 \\
-\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & -\frac{9\sqrt{15}i}{1400} & 0 & 0 & 0 & \frac{17\sqrt{5}i}{1400} & 0 & 0 \\
0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & -\frac{17\sqrt{5}i}{1400} & 0 & 0 & 0 & \frac{9\sqrt{15}i}{1400} & 0 \\
0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{1400} \\
0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x218}}$ $\mathbb{G}_2^{(1,0;a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 \\
\frac{\sqrt{3}}{21} & 0 & 0 & 0 & \frac{2\sqrt{15}}{105} & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\
0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 \\
0 & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\
0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}}{105} & 0 & 0 & 0 \\
\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 \\
0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 \\
0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{105} & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} \\
0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x219}}$ $\mathbb{G}_4^{(1,0;a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & -\frac{9\sqrt{210}}{1400} & 0 & 0 & 0 & \frac{27\sqrt{6}}{280} & 0 & 0 & 0 \\
-\frac{\sqrt{15}}{280} & 0 & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & \frac{99\sqrt{10}}{1400} & 0 & 0 & 0 & \frac{9\sqrt{30}}{1400} & 0 & 0 \\
0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & -\frac{\sqrt{15}}{280} & 0 & 0 & -\frac{9\sqrt{30}}{1400} & 0 & 0 & 0 & -\frac{99\sqrt{10}}{1400} & 0 \\
0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{6}}{280} & 0 & 0 & 0 & \frac{9\sqrt{210}}{1400} \\
0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & \frac{3\sqrt{210}}{1400} & 0 & 0 & 0 & \frac{\sqrt{6}}{280} & 0 & 0 & 0 \\
-\frac{9\sqrt{10}}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & -\frac{9\sqrt{15}}{1400} & 0 & 0 & 0 & -\frac{17\sqrt{5}}{1400} & 0 & 0 \\
0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{17\sqrt{5}}{1400} & 0 & 0 & 0 & -\frac{9\sqrt{15}}{1400} & 0 \\
0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{280} & 0 & 0 & 0 & \frac{3\sqrt{210}}{1400} \\
0 & 0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x220}}$ $\mathbb{G}_{2,1}^{(1,0;a)}(E_u) =$

$$\begin{bmatrix}
\frac{1}{14} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{15}}{210} & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{30}}{84} & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 \\
0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 \\
\frac{3}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{168} & 0 & -\frac{11\sqrt{30}}{840} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} & 0 & -\frac{1}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{10}}{70} & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}}{840} & 0 & \frac{\sqrt{6}}{168} & 0 \\
0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{210}}{168}
\end{bmatrix}$$

$\boxed{\text{x221}}$ $\mathbb{G}_{2,2}^{(1,0;a)}(E_u) =$

$$\begin{bmatrix}
\frac{i}{14} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{210} & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 \\
0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 \\
\frac{3i}{14} & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{168} & 0 & \frac{11\sqrt{30}i}{840} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & \frac{\sqrt{15}i}{60} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 & \frac{i}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}i}{840} & 0 & -\frac{\sqrt{6}i}{168} & 0 \\
0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{210}i}{168}
\end{bmatrix}$$

$\boxed{\text{x222}}$ $\mathbb{G}_{4,1}^{(1,0;a)}(E_u, 1) =$

$$\begin{bmatrix}
\frac{\sqrt{35}}{1120} & 0 & \frac{\sqrt{14}}{224} & 0 & \frac{\sqrt{7}}{224} & 0 & 0 & -\frac{27\sqrt{210}}{2800} & 0 & -\frac{9\sqrt{42}}{280} & 0 & -\frac{27\sqrt{70}}{2800} & 0 & 0 \\
0 & -\frac{\sqrt{21}}{224} & 0 & -\frac{\sqrt{42}}{224} & 0 & -\frac{\sqrt{105}}{1120} & \frac{9\sqrt{10}}{400} & 0 & \frac{9\sqrt{210}}{700} & 0 & \frac{9\sqrt{14}}{560} & 0 & -\frac{9\sqrt{70}}{1400} & 0 \\
\frac{\sqrt{105}}{1120} & 0 & \frac{\sqrt{42}}{224} & 0 & \frac{\sqrt{21}}{224} & 0 & 0 & -\frac{9\sqrt{70}}{1400} & 0 & \frac{9\sqrt{14}}{560} & 0 & \frac{9\sqrt{210}}{700} & 0 & \frac{9\sqrt{10}}{400} \\
0 & -\frac{\sqrt{7}}{224} & 0 & -\frac{\sqrt{14}}{224} & 0 & -\frac{\sqrt{35}}{1120} & 0 & 0 & -\frac{27\sqrt{70}}{2800} & 0 & -\frac{9\sqrt{42}}{280} & 0 & -\frac{27\sqrt{210}}{2800} & 0 \\
0 & -\frac{3\sqrt{35}}{140} & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & -\frac{\sqrt{6}}{160} & 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{210}}{1120} & 0 & 0 & 0 \\
\frac{3\sqrt{35}}{140} & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{210}}{5600} & 0 & \frac{\sqrt{42}}{560} & 0 & -\frac{\sqrt{70}}{800} & 0 & 0 \\
0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & \frac{3\sqrt{70}}{280} & -\frac{3\sqrt{15}}{800} & 0 & -\frac{\sqrt{35}}{5600} & 0 & \frac{\sqrt{21}}{160} & 0 & \frac{\sqrt{105}}{5600} & 0 \\
\frac{3\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{105}}{5600} & 0 & -\frac{\sqrt{21}}{160} & 0 & \frac{\sqrt{35}}{5600} & 0 & \frac{3\sqrt{15}}{800} \\
0 & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & \frac{\sqrt{70}}{800} & 0 & -\frac{\sqrt{42}}{560} & 0 & -\frac{13\sqrt{210}}{5600} & 0 \\
0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & -\frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{1120} & 0 & \frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{6}}{160}
\end{bmatrix}$$

$\boxed{\text{x223}}$ $\mathbb{G}_{4,2}^{(1,0;a)}(E_u,1) =$

$$\begin{bmatrix}
\frac{\sqrt{35}i}{1120} & 0 & -\frac{\sqrt{14}i}{224} & 0 & \frac{\sqrt{7}i}{224} & 0 & 0 & -\frac{27\sqrt{210}i}{2800} & 0 & \frac{9\sqrt{42}i}{280} & 0 & -\frac{27\sqrt{70}i}{2800} & 0 & 0 \\
0 & -\frac{\sqrt{21}i}{224} & 0 & \frac{\sqrt{42}i}{224} & 0 & -\frac{\sqrt{105}i}{1120} & -\frac{9\sqrt{10}i}{400} & 0 & \frac{9\sqrt{210}i}{700} & 0 & -\frac{9\sqrt{14}i}{560} & 0 & -\frac{9\sqrt{70}i}{1400} & 0 \\
-\frac{\sqrt{105}i}{1120} & 0 & \frac{\sqrt{42}i}{224} & 0 & -\frac{\sqrt{21}i}{224} & 0 & 0 & \frac{9\sqrt{70}i}{1400} & 0 & \frac{9\sqrt{14}i}{560} & 0 & -\frac{9\sqrt{210}i}{700} & 0 & \frac{9\sqrt{10}i}{400} \\
0 & \frac{\sqrt{7}i}{224} & 0 & -\frac{\sqrt{14}i}{224} & 0 & \frac{\sqrt{35}i}{1120} & 0 & 0 & \frac{27\sqrt{70}i}{2800} & 0 & -\frac{9\sqrt{42}i}{280} & 0 & \frac{27\sqrt{210}i}{2800} & 0 \\
0 & \frac{3\sqrt{35}i}{140} & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & -\frac{\sqrt{6}i}{160} & 0 & \frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{210}i}{1120} & 0 & 0 & 0 \\
\frac{3\sqrt{35}i}{140} & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{210}i}{5600} & 0 & -\frac{\sqrt{42}i}{560} & 0 & -\frac{\sqrt{70}i}{800} & 0 & 0 \\
0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} & \frac{3\sqrt{15}i}{800} & 0 & -\frac{\sqrt{35}i}{5600} & 0 & -\frac{\sqrt{21}i}{160} & 0 & \frac{\sqrt{105}i}{5600} & 0 \\
-\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & \frac{\sqrt{105}i}{5600} & 0 & -\frac{\sqrt{21}i}{160} & 0 & -\frac{\sqrt{35}i}{5600} & 0 & \frac{3\sqrt{15}i}{800} \\
0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & -\frac{\sqrt{70}i}{800} & 0 & -\frac{\sqrt{42}i}{560} & 0 & \frac{13\sqrt{210}i}{5600} & 0 \\
0 & 0 & \frac{3\sqrt{70}i}{280} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{1120} & 0 & \frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{6}i}{160}
\end{bmatrix}$$

$\boxed{\text{x224}}$ $\mathbb{G}_{4,1}^{(1,0;a)}(E_u,2) =$

$$\begin{bmatrix}
\frac{\sqrt{5}}{1120} & 0 & \frac{\sqrt{2}}{224} & 0 & -\frac{1}{32} & 0 & 0 & -\frac{27\sqrt{30}}{2800} & 0 & -\frac{9\sqrt{6}}{280} & 0 & \frac{27\sqrt{10}}{400} & 0 & 0 \\
0 & -\frac{\sqrt{3}}{224} & 0 & -\frac{\sqrt{6}}{224} & 0 & \frac{\sqrt{15}}{160} & -\frac{9\sqrt{70}}{400} & 0 & \frac{9\sqrt{30}}{700} & 0 & \frac{9\sqrt{2}}{560} & 0 & \frac{9\sqrt{10}}{200} & 0 \\
-\frac{\sqrt{15}}{160} & 0 & \frac{\sqrt{6}}{224} & 0 & \frac{\sqrt{3}}{224} & 0 & 0 & \frac{9\sqrt{10}}{200} & 0 & \frac{9\sqrt{2}}{560} & 0 & \frac{9\sqrt{30}}{700} & 0 & -\frac{9\sqrt{70}}{400} \\
0 & \frac{1}{32} & 0 & -\frac{\sqrt{2}}{224} & 0 & -\frac{\sqrt{5}}{1120} & 0 & 0 & \frac{27\sqrt{10}}{400} & 0 & -\frac{9\sqrt{6}}{280} & 0 & -\frac{27\sqrt{30}}{2800} & 0 \\
0 & -\frac{3\sqrt{5}}{140} & 0 & \frac{3\sqrt{10}}{40} & 0 & 0 & -\frac{\sqrt{42}}{1120} & 0 & -\frac{\sqrt{2}}{112} & 0 & \frac{\sqrt{30}}{160} & 0 & 0 & 0 \\
\frac{3\sqrt{5}}{140} & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{30}}{5600} & 0 & \frac{\sqrt{6}}{560} & 0 & \frac{7\sqrt{10}}{800} & 0 & 0 \\
0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{40} & \frac{3\sqrt{105}}{800} & 0 & -\frac{\sqrt{5}}{5600} & 0 & \frac{\sqrt{3}}{160} & 0 & -\frac{\sqrt{15}}{800} & 0 \\
-\frac{3\sqrt{10}}{40} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & \frac{\sqrt{15}}{800} & 0 & -\frac{\sqrt{3}}{160} & 0 & \frac{\sqrt{5}}{5600} & 0 & -\frac{3\sqrt{105}}{800} \\
0 & 0 & 0 & \frac{3\sqrt{2}}{56} & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & -\frac{7\sqrt{10}}{800} & 0 & -\frac{\sqrt{6}}{560} & 0 & -\frac{13\sqrt{30}}{5600} & 0 \\
0 & 0 & \frac{3\sqrt{10}}{40} & 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{160} & 0 & \frac{\sqrt{2}}{112} & 0 & \frac{\sqrt{42}}{1120}
\end{bmatrix}$$

$\boxed{\text{x225}}$ $\mathbb{G}_{4,2}^{(1,0;a)}(E_u,2) =$

$$\begin{bmatrix}
\frac{\sqrt{5}i}{1120} & 0 & -\frac{\sqrt{2}i}{224} & 0 & -\frac{i}{32} & 0 & 0 & -\frac{27\sqrt{30}i}{2800} & 0 & \frac{9\sqrt{6}i}{280} & 0 & \frac{27\sqrt{10}i}{400} & 0 & 0 \\
0 & -\frac{\sqrt{3}i}{224} & 0 & \frac{\sqrt{6}i}{224} & 0 & \frac{\sqrt{15}i}{160} & \frac{9\sqrt{70}i}{400} & 0 & \frac{9\sqrt{30}i}{700} & 0 & -\frac{9\sqrt{2}i}{560} & 0 & \frac{9\sqrt{10}i}{200} & 0 \\
\frac{\sqrt{15}i}{160} & 0 & \frac{\sqrt{6}i}{224} & 0 & -\frac{\sqrt{3}i}{224} & 0 & 0 & -\frac{9\sqrt{10}i}{200} & 0 & \frac{9\sqrt{2}i}{560} & 0 & -\frac{9\sqrt{30}i}{700} & 0 & -\frac{9\sqrt{70}i}{400} \\
0 & -\frac{i}{32} & 0 & -\frac{\sqrt{2}i}{224} & 0 & \frac{\sqrt{5}i}{1120} & 0 & 0 & -\frac{27\sqrt{10}i}{400} & 0 & -\frac{9\sqrt{6}i}{280} & 0 & \frac{27\sqrt{30}i}{2800} & 0 \\
0 & \frac{3\sqrt{5}i}{140} & 0 & \frac{3\sqrt{10}i}{40} & 0 & 0 & -\frac{\sqrt{42}i}{1120} & 0 & \frac{\sqrt{2}i}{112} & 0 & \frac{\sqrt{30}i}{160} & 0 & 0 & 0 \\
\frac{3\sqrt{5}i}{140} & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{30}i}{5600} & 0 & -\frac{\sqrt{6}i}{560} & 0 & \frac{7\sqrt{10}i}{800} & 0 & 0 \\
0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{40} & -\frac{3\sqrt{105}i}{800} & 0 & -\frac{\sqrt{5}i}{5600} & 0 & -\frac{\sqrt{3}i}{160} & 0 & -\frac{\sqrt{15}i}{800} & 0 \\
\frac{3\sqrt{10}i}{40} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & -\frac{\sqrt{15}i}{800} & 0 & -\frac{\sqrt{3}i}{160} & 0 & -\frac{\sqrt{5}i}{5600} & 0 & -\frac{3\sqrt{105}i}{800} \\
0 & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & \frac{7\sqrt{10}i}{800} & 0 & -\frac{\sqrt{6}i}{560} & 0 & \frac{13\sqrt{30}i}{5600} & 0 \\
0 & 0 & -\frac{3\sqrt{10}i}{40} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{160} & 0 & \frac{\sqrt{2}i}{112} & 0 & -\frac{\sqrt{42}i}{1120}
\end{bmatrix}$$

$\boxed{\text{x226}}$ $\mathbb{G}_{0}^{(1,1;a)}(A_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x227}}$ $\mathbb{G}_2^{(1,1;a)}(A_{1u}) = \begin{bmatrix}
0 & \frac{12i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}i}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}i}{140} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{12i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 \\
\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{105} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{105} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0
\end{bmatrix}$

$\boxed{\text{x228}}$ $\mathbb{G}_4^{(1,1;a)}(A_{1u},1) = \begin{bmatrix}
0 & -\frac{\sqrt{770}i}{210} & 0 & 0 & 0 & -\frac{\sqrt{154}i}{42} & 0 & 0 & \frac{\sqrt{77}i}{140} & 0 & 0 & 0 & \frac{\sqrt{231}i}{420} & 0 \\
0 & 0 & \frac{\sqrt{1155}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{420} & 0 & 0 & 0 & \frac{\sqrt{11}i}{60} \\
0 & 0 & 0 & -\frac{\sqrt{1155}i}{105} & 0 & 0 & \frac{\sqrt{11}i}{60} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{420} & 0 & 0 & 0 \\
\frac{\sqrt{154}i}{42} & 0 & 0 & 0 & \frac{\sqrt{770}i}{210} & 0 & 0 & \frac{\sqrt{231}i}{420} & 0 & 0 & 0 & \frac{\sqrt{77}i}{140} & 0 & 0 \\
-\frac{\sqrt{770}i}{840} & 0 & 0 & 0 & -\frac{\sqrt{154}i}{168} & 0 & 0 & \frac{\sqrt{1155}i}{1155} & 0 & 0 & 0 & \frac{\sqrt{385}i}{1155} & 0 & 0 \\
0 & \frac{\sqrt{770}i}{280} & 0 & 0 & 0 & -\frac{\sqrt{154}i}{168} & 0 & 0 & -\frac{4\sqrt{77}i}{1155} & 0 & 0 & 0 & \frac{2\sqrt{231}i}{1155} & 0 \\
0 & 0 & -\frac{\sqrt{770}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{2310} & 0 & 0 & 0 & \frac{\sqrt{66}i}{330} \\
0 & 0 & 0 & -\frac{\sqrt{770}i}{420} & 0 & 0 & -\frac{\sqrt{66}i}{330} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{2310} & 0 & 0 & 0 \\
-\frac{\sqrt{154}i}{168} & 0 & 0 & 0 & \frac{\sqrt{770}i}{280} & 0 & 0 & -\frac{2\sqrt{231}i}{1155} & 0 & 0 & 0 & \frac{4\sqrt{77}i}{1155} & 0 & 0 \\
0 & -\frac{\sqrt{154}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{840} & 0 & 0 & -\frac{\sqrt{385}i}{1155} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{1155} & 0
\end{bmatrix}$

$\boxed{\text{x229}}$ $\mathbb{G}_4^{(1,1;a)}(A_{1u},2) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{22}i}{42} & 0 & 0 & 0 & \frac{\sqrt{110}i}{30} & 0 & 0 & \frac{\sqrt{55}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{165}i}{300} & 0 \\
0 & 0 & \frac{\sqrt{33}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{300} \\
0 & 0 & 0 & -\frac{\sqrt{33}i}{21} & 0 & 0 & -\frac{\sqrt{385}i}{300} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{84} & 0 & 0 & 0 \\
-\frac{\sqrt{110}i}{30} & 0 & 0 & 0 & \frac{\sqrt{22}i}{42} & 0 & 0 & -\frac{\sqrt{165}i}{300} & 0 & 0 & 0 & \frac{\sqrt{55}i}{140} & 0 & 0 \\
-\frac{\sqrt{22}i}{168} & 0 & 0 & 0 & \frac{\sqrt{110}i}{120} & 0 & 0 & \frac{\sqrt{33}i}{231} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{165} & 0 & 0 \\
0 & \frac{\sqrt{22}i}{56} & 0 & 0 & 0 & \frac{\sqrt{110}i}{120} & 0 & 0 & -\frac{4\sqrt{55}i}{1155} & 0 & 0 & 0 & -\frac{2\sqrt{165}i}{825} & 0 \\
0 & 0 & -\frac{\sqrt{22}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{462} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1650} \\
0 & 0 & 0 & -\frac{\sqrt{22}i}{84} & 0 & 0 & \frac{\sqrt{2310}i}{1650} & 0 & 0 & 0 & \frac{\sqrt{66}i}{462} & 0 & 0 & 0 \\
\frac{\sqrt{110}i}{120} & 0 & 0 & 0 & \frac{\sqrt{22}i}{56} & 0 & 0 & \frac{2\sqrt{165}i}{825} & 0 & 0 & 0 & \frac{4\sqrt{55}i}{1155} & 0 & 0 \\
0 & \frac{\sqrt{110}i}{120} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{168} & 0 & 0 & \frac{\sqrt{11}i}{165} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{231} & 0
\end{bmatrix}$$

$\boxed{\text{x230}}$ $\mathbb{G}_4^{(1,1;a)}(A_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{350} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{150} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2310}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{231}}{1155} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}}{1925} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{275} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{275} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{231}}{1155} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x231}}$ $\mathbb{G}_2^{(1,1;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{2\sqrt{6}i}{35} & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{280} & 0 & 0 & 0 \\
-\frac{4\sqrt{5}i}{35} & 0 & 0 & 0 & \frac{8i}{35} & 0 & 0 & -\frac{3\sqrt{30}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{280} & 0 & 0 \\
0 & -\frac{8i}{35} & 0 & 0 & 0 & \frac{4\sqrt{5}i}{35} & 0 & 0 & -\frac{3\sqrt{10}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{280} & 0 \\
0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{280} \\
0 & 0 & \frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{105} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & \frac{\sqrt{70}i}{105} & 0 & 0 & 0 & -\frac{4\sqrt{2}i}{105} & 0 & 0 & 0 \\
\frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & \frac{4\sqrt{5}i}{105} & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{105} & 0 & 0 \\
0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{30}i}{140} & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & -\frac{4\sqrt{5}i}{105} & 0 \\
0 & 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{2}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{105} \\
0 & 0 & 0 & \frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{105} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x232}}$ $\mathbb{G}_4^{(1,1;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{33}i}{21} & 0 & 0 & -\frac{\sqrt{385}i}{700} & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{110}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{14} & 0 & 0 & \frac{11\sqrt{165}i}{2100} & 0 & 0 & 0 & -\frac{\sqrt{55}i}{700} & 0 & 0 \\
0 & \frac{\sqrt{22}i}{14} & 0 & 0 & 0 & \frac{\sqrt{110}i}{70} & 0 & 0 & -\frac{\sqrt{55}i}{700} & 0 & 0 & 0 & \frac{11\sqrt{165}i}{2100} & 0 \\
0 & 0 & -\frac{\sqrt{33}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{700} \\
0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{385} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & \frac{3\sqrt{385}i}{1925} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{385} & 0 & 0 & 0 \\
\frac{\sqrt{165}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & -\frac{9\sqrt{110}i}{3850} & 0 & 0 & 0 & \frac{17\sqrt{330}i}{11550} & 0 & 0 \\
0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & -\frac{17\sqrt{330}i}{11550} & 0 & 0 & 0 & \frac{9\sqrt{110}i}{3850} & 0 \\
0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{385} & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{1925} \\
0 & 0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}i}{385} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x233}}$ $\mathbb{G}_2^{(1,1;a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & \frac{3\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{280} & 0 & 0 & 0 \\
\frac{4\sqrt{5}}{35} & 0 & 0 & 0 & \frac{8}{35} & 0 & 0 & \frac{3\sqrt{30}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{280} & 0 & 0 \\
0 & \frac{8}{35} & 0 & 0 & 0 & \frac{4\sqrt{5}}{35} & 0 & 0 & \frac{3\sqrt{10}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{30}}{280} & 0 \\
0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} \\
0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{105} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & -\frac{\sqrt{70}}{105} & 0 & 0 & 0 & -\frac{4\sqrt{2}}{105} & 0 & 0 & 0 \\
-\frac{3\sqrt{30}}{140} & 0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & -\frac{4\sqrt{5}}{105} & 0 & 0 & 0 & -\frac{2\sqrt{15}}{105} & 0 & 0 \\
0 & -\frac{9\sqrt{6}}{140} & 0 & 0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & -\frac{2\sqrt{15}}{105} & 0 & 0 & 0 & -\frac{4\sqrt{5}}{105} & 0 \\
0 & 0 & -\frac{9\sqrt{6}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{2}}{105} & 0 & 0 & 0 & -\frac{\sqrt{70}}{105} \\
0 & 0 & 0 & -\frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{105} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x234}}$ $\mathbb{G}_4^{(1,1;a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{33}}{21} & 0 & 0 & -\frac{\sqrt{385}}{700} & 0 & 0 & 0 & \frac{3\sqrt{11}}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{110}}{70} & 0 & 0 & 0 & \frac{\sqrt{22}}{14} & 0 & 0 & \frac{11\sqrt{165}}{2100} & 0 & 0 & 0 & \frac{\sqrt{55}}{700} & 0 & 0 \\
0 & \frac{\sqrt{22}}{14} & 0 & 0 & 0 & -\frac{\sqrt{110}}{70} & 0 & 0 & -\frac{\sqrt{55}}{700} & 0 & 0 & 0 & -\frac{11\sqrt{165}}{2100} & 0 \\
0 & 0 & -\frac{\sqrt{33}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{140} & 0 & 0 & 0 & \frac{\sqrt{385}}{700} \\
0 & 0 & -\frac{\sqrt{165}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{385} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{33}}{84} & 0 & 0 & \frac{3\sqrt{385}}{1925} & 0 & 0 & 0 & \frac{\sqrt{11}}{385} & 0 & 0 & 0 \\
\frac{\sqrt{165}}{140} & 0 & 0 & 0 & \frac{\sqrt{33}}{84} & 0 & 0 & -\frac{9\sqrt{110}}{3850} & 0 & 0 & 0 & -\frac{17\sqrt{330}}{11550} & 0 & 0 \\
0 & -\frac{\sqrt{33}}{84} & 0 & 0 & 0 & -\frac{\sqrt{165}}{140} & 0 & 0 & -\frac{17\sqrt{330}}{11550} & 0 & 0 & 0 & -\frac{9\sqrt{110}}{3850} & 0 \\
0 & 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{385} & 0 & 0 & 0 & \frac{3\sqrt{385}}{1925} \\
0 & 0 & 0 & \frac{\sqrt{165}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{385} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x235}}$ $\mathbb{G}_{2,1}^{(1,1;a)}(E_u) = \begin{bmatrix}
\frac{2\sqrt{15}}{35} & 0 & \frac{3\sqrt{6}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 \\
0 & -\frac{2}{35} & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{3\sqrt{6}}{140} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{2}}{7} & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{140} & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{6}}{35} & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & -\frac{3\sqrt{10}}{140} & 0 \\
0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{42} & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{15}}{70} & 0 & \frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{210} & 0 & -\frac{11\sqrt{2}}{210} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{105} & 0 & -\frac{1}{15} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & \frac{1}{15} & 0 & -\frac{\sqrt{15}}{105} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{6}}{70} & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{11\sqrt{2}}{210} & 0 & \frac{\sqrt{10}}{210} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & \frac{\sqrt{14}}{42}
\end{bmatrix}$

$\boxed{\text{x236}}$ $\mathbb{G}_{2,2}^{(1,1;a)}(E_u) = \begin{bmatrix}
\frac{2\sqrt{15}i}{35} & 0 & -\frac{3\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 \\
0 & -\frac{2i}{35} & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{6}i}{140} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{10}i}{140} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{6}i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{140} & 0 & \frac{3\sqrt{10}i}{140} & 0 \\
0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{42} & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{15}i}{70} & 0 & -\frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{210} & 0 & \frac{11\sqrt{2}i}{210} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{105} & 0 & \frac{i}{15} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & \frac{i}{15} & 0 & \frac{\sqrt{15}i}{105} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{6}i}{70} & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{11\sqrt{2}i}{210} & 0 & -\frac{\sqrt{10}i}{210} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & -\frac{\sqrt{14}i}{42}
\end{bmatrix}$

$\boxed{\text{x237}}$ $\mathbb{G}_{4,1}^{(1,1;a)}(E_u,1) =$

$$\begin{bmatrix}
\frac{\sqrt{2310}}{840} & 0 & \frac{\sqrt{231}}{84} & 0 & \frac{\sqrt{462}}{168} & 0 & 0 & -\frac{3\sqrt{385}}{1400} & 0 & -\frac{\sqrt{77}}{140} & 0 & -\frac{\sqrt{1155}}{1400} & 0 & 0 \\
0 & -\frac{\sqrt{154}}{56} & 0 & -\frac{\sqrt{77}}{28} & 0 & -\frac{\sqrt{770}}{280} & \frac{\sqrt{165}}{600} & 0 & \frac{\sqrt{385}}{350} & 0 & \frac{\sqrt{231}}{840} & 0 & -\frac{\sqrt{1155}}{2100} & 0 \\
\frac{\sqrt{770}}{280} & 0 & \frac{\sqrt{77}}{28} & 0 & \frac{\sqrt{154}}{56} & 0 & 0 & -\frac{\sqrt{1155}}{2100} & 0 & \frac{\sqrt{231}}{840} & 0 & \frac{\sqrt{385}}{350} & 0 & \frac{\sqrt{165}}{600} \\
0 & -\frac{\sqrt{462}}{168} & 0 & -\frac{\sqrt{231}}{84} & 0 & -\frac{\sqrt{2310}}{840} & 0 & 0 & -\frac{\sqrt{1155}}{1400} & 0 & -\frac{\sqrt{77}}{140} & 0 & -\frac{3\sqrt{385}}{1400} & 0 \\
0 & \frac{\sqrt{2310}}{840} & 0 & \frac{\sqrt{1155}}{840} & 0 & 0 & -\frac{\sqrt{11}}{220} & 0 & -\frac{\sqrt{231}}{462} & 0 & -\frac{\sqrt{385}}{1540} & 0 & 0 & 0 \\
-\frac{\sqrt{2310}}{840} & 0 & -\frac{\sqrt{231}}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{385}}{7700} & 0 & \frac{\sqrt{77}}{770} & 0 & -\frac{\sqrt{1155}}{3300} & 0 & 0 \\
0 & \frac{\sqrt{231}}{168} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{840} & -\frac{3\sqrt{110}}{2200} & 0 & -\frac{\sqrt{2310}}{46200} & 0 & \frac{\sqrt{154}}{440} & 0 & \frac{\sqrt{770}}{15400} & 0 \\
-\frac{\sqrt{1155}}{840} & 0 & 0 & 0 & \frac{\sqrt{231}}{168} & 0 & 0 & -\frac{\sqrt{770}}{15400} & 0 & -\frac{\sqrt{154}}{440} & 0 & \frac{\sqrt{2310}}{46200} & 0 & \frac{3\sqrt{110}}{2200} \\
0 & 0 & 0 & -\frac{\sqrt{231}}{168} & 0 & -\frac{\sqrt{2310}}{840} & 0 & 0 & \frac{\sqrt{1155}}{3300} & 0 & -\frac{\sqrt{77}}{770} & 0 & -\frac{13\sqrt{385}}{7700} & 0 \\
0 & 0 & \frac{\sqrt{1155}}{840} & 0 & \frac{\sqrt{2310}}{840} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{1540} & 0 & \frac{\sqrt{231}}{462} & 0 & \frac{\sqrt{11}}{220}
\end{bmatrix}$$

$\boxed{\text{x238}}$ $\mathbb{G}_{4,2}^{(1,1;a)}(E_u,1) =$

$$\begin{bmatrix}
\frac{\sqrt{2310}i}{840} & 0 & -\frac{\sqrt{231}i}{84} & 0 & \frac{\sqrt{462}i}{168} & 0 & 0 & -\frac{3\sqrt{385}i}{1400} & 0 & \frac{\sqrt{77}i}{140} & 0 & -\frac{\sqrt{1155}i}{1400} & 0 & 0 \\
0 & -\frac{\sqrt{154}i}{56} & 0 & \frac{\sqrt{77}i}{28} & 0 & -\frac{\sqrt{770}i}{280} & -\frac{\sqrt{165}i}{600} & 0 & \frac{\sqrt{385}i}{350} & 0 & -\frac{\sqrt{231}i}{840} & 0 & -\frac{\sqrt{1155}i}{2100} & 0 \\
-\frac{\sqrt{770}i}{280} & 0 & \frac{\sqrt{77}i}{28} & 0 & -\frac{\sqrt{154}i}{56} & 0 & 0 & \frac{\sqrt{1155}i}{2100} & 0 & \frac{\sqrt{231}i}{840} & 0 & -\frac{\sqrt{385}i}{350} & 0 & \frac{\sqrt{165}i}{600} \\
0 & \frac{\sqrt{462}i}{168} & 0 & -\frac{\sqrt{231}i}{84} & 0 & \frac{\sqrt{2310}i}{840} & 0 & 0 & \frac{\sqrt{1155}i}{1400} & 0 & -\frac{\sqrt{77}i}{140} & 0 & \frac{3\sqrt{385}i}{1400} & 0 \\
0 & -\frac{\sqrt{2310}i}{840} & 0 & \frac{\sqrt{1155}i}{840} & 0 & 0 & -\frac{\sqrt{11}i}{220} & 0 & \frac{\sqrt{231}i}{462} & 0 & -\frac{\sqrt{385}i}{1540} & 0 & 0 & 0 \\
-\frac{\sqrt{2310}i}{840} & 0 & \frac{\sqrt{231}i}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{385}i}{7700} & 0 & -\frac{\sqrt{77}i}{770} & 0 & -\frac{\sqrt{1155}i}{3300} & 0 & 0 \\
0 & \frac{\sqrt{231}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{840} & \frac{3\sqrt{110}i}{2200} & 0 & -\frac{\sqrt{2310}i}{46200} & 0 & -\frac{\sqrt{154}i}{440} & 0 & \frac{\sqrt{770}i}{15400} & 0 \\
\frac{\sqrt{1155}i}{840} & 0 & 0 & 0 & -\frac{\sqrt{231}i}{168} & 0 & 0 & \frac{\sqrt{770}i}{15400} & 0 & -\frac{\sqrt{154}i}{440} & 0 & -\frac{\sqrt{2310}i}{46200} & 0 & \frac{3\sqrt{110}i}{2200} \\
0 & 0 & 0 & -\frac{\sqrt{231}i}{168} & 0 & \frac{\sqrt{2310}i}{840} & 0 & 0 & -\frac{\sqrt{1155}i}{3300} & 0 & -\frac{\sqrt{77}i}{770} & 0 & \frac{13\sqrt{385}i}{7700} & 0 \\
0 & 0 & -\frac{\sqrt{1155}i}{840} & 0 & \frac{\sqrt{2310}i}{840} & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{1540} & 0 & \frac{\sqrt{231}i}{462} & 0 & -\frac{\sqrt{11}i}{220}
\end{bmatrix}$$

$\boxed{\text{x239}}$ $\mathbb{G}_{4,1}^{(1,1;a)}(E_u,2) =$

$$\begin{bmatrix}
\frac{\sqrt{330}}{840} & 0 & \frac{\sqrt{33}}{84} & 0 & -\frac{\sqrt{66}}{24} & 0 & 0 & -\frac{3\sqrt{55}}{1400} & 0 & -\frac{\sqrt{11}}{140} & 0 & \frac{\sqrt{165}}{200} & 0 & 0 \\
0 & -\frac{\sqrt{22}}{56} & 0 & -\frac{\sqrt{11}}{28} & 0 & \frac{\sqrt{110}}{40} & -\frac{\sqrt{1155}}{600} & 0 & \frac{\sqrt{55}}{350} & 0 & \frac{\sqrt{33}}{840} & 0 & \frac{\sqrt{165}}{300} & 0 \\
-\frac{\sqrt{110}}{40} & 0 & \frac{\sqrt{11}}{28} & 0 & \frac{\sqrt{22}}{56} & 0 & 0 & \frac{\sqrt{165}}{300} & 0 & \frac{\sqrt{33}}{840} & 0 & \frac{\sqrt{55}}{350} & 0 & -\frac{\sqrt{1155}}{600} \\
0 & \frac{\sqrt{66}}{24} & 0 & -\frac{\sqrt{33}}{84} & 0 & -\frac{\sqrt{330}}{840} & 0 & 0 & \frac{\sqrt{165}}{200} & 0 & -\frac{\sqrt{11}}{140} & 0 & -\frac{3\sqrt{55}}{1400} & 0 \\
0 & \frac{\sqrt{330}}{840} & 0 & -\frac{\sqrt{165}}{120} & 0 & 0 & -\frac{\sqrt{77}}{1540} & 0 & -\frac{\sqrt{33}}{462} & 0 & \frac{\sqrt{55}}{220} & 0 & 0 & 0 \\
-\frac{\sqrt{330}}{840} & 0 & -\frac{\sqrt{33}}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{55}}{7700} & 0 & \frac{\sqrt{11}}{770} & 0 & \frac{7\sqrt{165}}{3300} & 0 & 0 \\
0 & \frac{\sqrt{33}}{168} & 0 & 0 & 0 & \frac{\sqrt{165}}{120} & \frac{3\sqrt{770}}{2200} & 0 & -\frac{\sqrt{330}}{46200} & 0 & \frac{\sqrt{22}}{440} & 0 & -\frac{\sqrt{110}}{2200} & 0 \\
\frac{\sqrt{165}}{120} & 0 & 0 & 0 & \frac{\sqrt{33}}{168} & 0 & 0 & \frac{\sqrt{110}}{2200} & 0 & -\frac{\sqrt{22}}{440} & 0 & \frac{\sqrt{330}}{46200} & 0 & -\frac{3\sqrt{770}}{2200} \\
0 & 0 & 0 & -\frac{\sqrt{33}}{168} & 0 & -\frac{\sqrt{330}}{840} & 0 & 0 & -\frac{7\sqrt{165}}{3300} & 0 & -\frac{\sqrt{11}}{770} & 0 & -\frac{13\sqrt{55}}{7700} & 0 \\
0 & 0 & -\frac{\sqrt{165}}{120} & 0 & \frac{\sqrt{330}}{840} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{220} & 0 & \frac{\sqrt{33}}{462} & 0 & \frac{\sqrt{77}}{1540}
\end{bmatrix}$$

$\boxed{\text{x240}}$ $\mathbb{G}_{4,2}^{(1,1;a)}(E_u,2) =$

$$\begin{bmatrix}
\frac{\sqrt{330}i}{840} & 0 & -\frac{\sqrt{33}i}{84} & 0 & -\frac{\sqrt{66}i}{24} & 0 & 0 & -\frac{3\sqrt{55}i}{1400} & 0 & \frac{\sqrt{11}i}{140} & 0 & \frac{\sqrt{165}i}{200} & 0 & 0 \\
0 & -\frac{\sqrt{22}i}{56} & 0 & \frac{\sqrt{11}i}{28} & 0 & \frac{\sqrt{110}i}{40} & \frac{\sqrt{1155}i}{600} & 0 & \frac{\sqrt{55}i}{350} & 0 & -\frac{\sqrt{33}i}{840} & 0 & \frac{\sqrt{165}i}{300} & 0 \\
\frac{\sqrt{110}i}{40} & 0 & \frac{\sqrt{11}i}{28} & 0 & -\frac{\sqrt{22}i}{56} & 0 & 0 & -\frac{\sqrt{165}i}{300} & 0 & \frac{\sqrt{33}i}{840} & 0 & -\frac{\sqrt{55}i}{350} & 0 & -\frac{\sqrt{1155}i}{600} \\
0 & -\frac{\sqrt{66}i}{24} & 0 & -\frac{\sqrt{33}i}{84} & 0 & \frac{\sqrt{330}i}{840} & 0 & 0 & -\frac{\sqrt{165}i}{200} & 0 & -\frac{\sqrt{11}i}{140} & 0 & \frac{3\sqrt{55}i}{1400} & 0 \\
0 & -\frac{\sqrt{330}i}{840} & 0 & -\frac{\sqrt{165}i}{120} & 0 & 0 & -\frac{\sqrt{77}i}{1540} & 0 & \frac{\sqrt{33}i}{462} & 0 & \frac{\sqrt{55}i}{220} & 0 & 0 & 0 \\
-\frac{\sqrt{330}i}{840} & 0 & \frac{\sqrt{33}i}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{55}i}{7700} & 0 & -\frac{\sqrt{11}i}{770} & 0 & \frac{7\sqrt{165}i}{3300} & 0 & 0 \\
0 & \frac{\sqrt{33}i}{168} & 0 & 0 & 0 & \frac{\sqrt{165}i}{120} & -\frac{3\sqrt{770}i}{2200} & 0 & -\frac{\sqrt{330}i}{46200} & 0 & -\frac{\sqrt{22}i}{440} & 0 & -\frac{\sqrt{110}i}{2200} & 0 \\
-\frac{\sqrt{165}i}{120} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{168} & 0 & 0 & -\frac{\sqrt{110}i}{2200} & 0 & -\frac{\sqrt{22}i}{440} & 0 & -\frac{\sqrt{330}i}{46200} & 0 & -\frac{3\sqrt{770}i}{2200} \\
0 & 0 & 0 & -\frac{\sqrt{33}i}{168} & 0 & \frac{\sqrt{330}i}{840} & 0 & 0 & \frac{7\sqrt{165}i}{3300} & 0 & -\frac{\sqrt{11}i}{770} & 0 & \frac{13\sqrt{55}i}{7700} & 0 \\
0 & 0 & \frac{\sqrt{165}i}{120} & 0 & \frac{\sqrt{330}i}{840} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{220} & 0 & \frac{\sqrt{33}i}{462} & 0 & -\frac{\sqrt{77}i}{1540}
\end{bmatrix}$$

$\boxed{\text{x241}}$ $\mathbb{M}_2^{(a)}(A_{1u}) = \begin{bmatrix} 0 & \frac{3\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 \\ -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 \end{bmatrix}$

$\boxed{\text{x242}}$ $\mathbb{M}_4^{(a)}(A_{1u}, 1) = \begin{bmatrix} 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 \\ 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{84} & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 \\ \frac{\sqrt{21}}{84} & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & \frac{\sqrt{210}}{105} & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 \\ 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & -\frac{\sqrt{5}}{20} \\ 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{84} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & -\frac{\sqrt{210}}{105} & 0 & 0 \\ 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \end{bmatrix}$

$\boxed{\text{x243}}$ $\mathbb{M}_4^{(a)}(A_{1u},2) = \begin{bmatrix} 0 & -\frac{\sqrt{15}}{84} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & -\frac{3\sqrt{6}}{56} & 0 & 0 & 0 & \frac{3\sqrt{2}}{40} & 0 \\ 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{56} & 0 & 0 & 0 & \frac{\sqrt{42}}{40} \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & \frac{\sqrt{42}}{40} & 0 & 0 & 0 & \frac{\sqrt{30}}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & 0 & \frac{3\sqrt{2}}{40} & 0 & 0 & 0 & -\frac{3\sqrt{6}}{56} & 0 & 0 \\ \frac{\sqrt{15}}{84} & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & \frac{\sqrt{7}}{20} \\ 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 \end{bmatrix}$

$\boxed{\text{x244}}$ $\mathbb{M}_4^{(a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{20} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x245}}$ $\mathbb{M}_2^{(a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{5}}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{2}}{7} & 0 & 0 & 0 & \frac{2\sqrt{10}}{35} & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & \frac{1}{28} & 0 & 0 \\
0 & -\frac{2\sqrt{10}}{35} & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & \frac{1}{28} & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 \\
0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{140} & 0 & 0 & 0 & \frac{\sqrt{7}}{28} \\
0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & \frac{2\sqrt{5}}{35} & 0 & 0 & 0 \\
-\frac{\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 \\
0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 \\
0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\
0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x246}}$ $\mathbb{M}_4^{(a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & \frac{3\sqrt{42}}{280} & 0 & 0 & 0 & \frac{9\sqrt{30}}{280} & 0 & 0 & 0 \\
-\frac{\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & -\frac{33\sqrt{2}}{280} & 0 & 0 & 0 & \frac{3\sqrt{6}}{280} & 0 & 0 \\
0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & \frac{3\sqrt{6}}{280} & 0 & 0 & 0 & -\frac{33\sqrt{2}}{280} & 0 \\
0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{30}}{280} & 0 & 0 & 0 & \frac{3\sqrt{42}}{280} \\
0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & -\frac{3\sqrt{42}}{140} & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 \\
-\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & \frac{9\sqrt{3}}{140} & 0 & 0 & 0 & -\frac{17}{140} & 0 & 0 \\
0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & \frac{17}{140} & 0 & 0 & 0 & -\frac{9\sqrt{3}}{140} & 0 \\
0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 & \frac{3\sqrt{42}}{140} \\
0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x247}}$ $\mathbb{M}_2^{(a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 \\
0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & \frac{i}{28} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 \\
0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} \\
0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{15}i}{140} & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & \frac{3\sqrt{15}i}{140} & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 \\
0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 \\
0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\
0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x248}}$ $\mathbb{M}_4^{(a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & -\frac{3\sqrt{42}i}{280} & 0 & 0 & 0 & \frac{9\sqrt{30}i}{280} & 0 & 0 & 0 \\
\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & \frac{33\sqrt{2}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{6}i}{280} & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & -\frac{3\sqrt{6}i}{280} & 0 & 0 & 0 & -\frac{33\sqrt{2}i}{280} & 0 \\
0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{30}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{42}i}{280} \\
0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & \frac{3\sqrt{42}i}{140} & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 \\
\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & -\frac{9\sqrt{3}i}{140} & 0 & 0 & 0 & -\frac{17i}{140} & 0 & 0 \\
0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & -\frac{17i}{140} & 0 & 0 & 0 & -\frac{9\sqrt{3}i}{140} & 0 \\
0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 & \frac{3\sqrt{42}i}{140} \\
0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x249}}$ $\mathbb{M}_{2,1}^{(a)}(E_u) = \begin{bmatrix} -\frac{\sqrt{6}i}{14} & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{70} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & -\frac{i}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{70} & 0 & -\frac{i}{14} & 0 \\ 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & -\frac{11\sqrt{5}i}{140} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & \frac{11\sqrt{5}i}{140} & 0 & \frac{i}{28} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{\sqrt{35}i}{28} \end{bmatrix}$

$\boxed{\text{x250}}$ $\mathbb{M}_{2,2}^{(a)}(E_u) = \begin{bmatrix} \frac{\sqrt{6}}{14} & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{70} & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & -\frac{1}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{70} & 0 & -\frac{1}{14} & 0 \\ 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & -\frac{11\sqrt{5}}{140} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{5}}{140} & 0 & \frac{1}{28} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & \frac{\sqrt{35}}{28} \end{bmatrix}$

$\boxed{\text{x251}}$ $\mathbb{M}_{4,1}^{(a)}(E_u,1) = \begin{bmatrix}
-\frac{\sqrt{7}i}{112} & 0 & -\frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{35}i}{112} & 0 & 0 & -\frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{210}i}{280} & 0 & -\frac{9\sqrt{14}i}{560} & 0 & 0 \\
0 & \frac{\sqrt{105}i}{112} & 0 & \frac{\sqrt{210}i}{112} & 0 & \frac{\sqrt{21}i}{112} & \frac{3\sqrt{2}i}{80} & 0 & \frac{3\sqrt{42}i}{140} & 0 & \frac{3\sqrt{70}i}{560} & 0 & -\frac{3\sqrt{14}i}{280} & 0 \\
-\frac{\sqrt{21}i}{112} & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{105}i}{112} & 0 & 0 & -\frac{3\sqrt{14}i}{280} & 0 & \frac{3\sqrt{70}i}{560} & 0 & \frac{3\sqrt{42}i}{140} & 0 & \frac{3\sqrt{2}i}{80} \\
0 & \frac{\sqrt{35}i}{112} & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{7}i}{112} & 0 & 0 & -\frac{9\sqrt{14}i}{560} & 0 & -\frac{3\sqrt{210}i}{280} & 0 & -\frac{9\sqrt{42}i}{560} & 0 \\
0 & \frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{30}i}{80} & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 & 0 \\
-\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{42}i}{560} & 0 & \frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{14}i}{80} & 0 & 0 \\
0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & -\frac{3\sqrt{3}i}{80} & 0 & -\frac{\sqrt{7}i}{560} & 0 & \frac{\sqrt{105}i}{80} & 0 & \frac{\sqrt{21}i}{560} & 0 \\
-\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & -\frac{\sqrt{21}i}{560} & 0 & -\frac{\sqrt{105}i}{80} & 0 & \frac{\sqrt{7}i}{560} & 0 & \frac{3\sqrt{3}i}{80} \\
0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & \frac{\sqrt{14}i}{80} & 0 & -\frac{\sqrt{210}i}{280} & 0 & -\frac{13\sqrt{42}i}{560} & 0 \\
0 & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{30}i}{80}
\end{bmatrix}$

$\boxed{\text{x252}}$ $\mathbb{M}_{4,2}^{(a)}(E_u,1) = \begin{bmatrix}
\frac{\sqrt{7}}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{35}}{112} & 0 & 0 & \frac{9\sqrt{42}}{560} & 0 & -\frac{3\sqrt{210}}{280} & 0 & \frac{9\sqrt{14}}{560} & 0 & 0 \\
0 & -\frac{\sqrt{105}}{112} & 0 & \frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{21}}{112} & \frac{3\sqrt{2}}{80} & 0 & -\frac{3\sqrt{42}}{140} & 0 & \frac{3\sqrt{70}}{560} & 0 & \frac{3\sqrt{14}}{280} & 0 \\
-\frac{\sqrt{21}}{112} & 0 & \frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{105}}{112} & 0 & 0 & -\frac{3\sqrt{14}}{280} & 0 & -\frac{3\sqrt{70}}{560} & 0 & \frac{3\sqrt{42}}{140} & 0 & -\frac{3\sqrt{2}}{80} \\
0 & \frac{\sqrt{35}}{112} & 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{7}}{112} & 0 & 0 & -\frac{9\sqrt{14}}{560} & 0 & \frac{3\sqrt{210}}{280} & 0 & -\frac{9\sqrt{42}}{560} & 0 \\
0 & \frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{30}}{80} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & 0 \\
\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{42}}{560} & 0 & \frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{14}}{80} & 0 & 0 \\
0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & -\frac{3\sqrt{3}}{80} & 0 & \frac{\sqrt{7}}{560} & 0 & \frac{\sqrt{105}}{80} & 0 & -\frac{\sqrt{21}}{560} & 0 \\
-\frac{\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{21}}{560} & 0 & \frac{\sqrt{105}}{80} & 0 & \frac{\sqrt{7}}{560} & 0 & -\frac{3\sqrt{3}}{80} \\
0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{14}}{80} & 0 & \frac{\sqrt{210}}{280} & 0 & -\frac{13\sqrt{42}}{560} & 0 \\
0 & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{30}}{80}
\end{bmatrix}$

$\boxed{\text{x253}}$ $\mathbb{M}_{4,1}^{(a)}(E_u,2) = \begin{bmatrix} -\frac{i}{112} & 0 & -\frac{\sqrt{10}i}{112} & 0 & \frac{\sqrt{5}i}{16} & 0 & 0 & -\frac{9\sqrt{6}i}{560} & 0 & -\frac{3\sqrt{30}i}{280} & 0 & \frac{9\sqrt{2}i}{80} & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{112} & 0 & \frac{\sqrt{30}i}{112} & 0 & -\frac{\sqrt{3}i}{16} & -\frac{3\sqrt{14}i}{80} & 0 & \frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{10}i}{560} & 0 & \frac{3\sqrt{2}i}{40} & 0 \\ \frac{\sqrt{3}i}{16} & 0 & -\frac{\sqrt{30}i}{112} & 0 & -\frac{\sqrt{15}i}{112} & 0 & 0 & \frac{3\sqrt{2}i}{40} & 0 & \frac{3\sqrt{10}i}{560} & 0 & \frac{3\sqrt{6}i}{140} & 0 & -\frac{3\sqrt{14}i}{80} \\ 0 & -\frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{10}i}{112} & 0 & \frac{i}{112} & 0 & 0 & \frac{9\sqrt{2}i}{80} & 0 & -\frac{3\sqrt{30}i}{280} & 0 & -\frac{9\sqrt{6}i}{560} & 0 \\ 0 & \frac{i}{28} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & -\frac{\sqrt{210}i}{560} & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 & 0 \\ -\frac{i}{28} & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{6}i}{560} & 0 & \frac{\sqrt{30}i}{280} & 0 & \frac{7\sqrt{2}i}{80} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & \frac{3\sqrt{21}i}{80} & 0 & -\frac{i}{560} & 0 & \frac{\sqrt{15}i}{80} & 0 & -\frac{\sqrt{3}i}{80} & 0 \\ \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & \frac{\sqrt{3}i}{80} & 0 & -\frac{\sqrt{15}i}{80} & 0 & \frac{i}{560} & 0 & -\frac{3\sqrt{21}i}{80} \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & -\frac{i}{28} & 0 & 0 & -\frac{7\sqrt{2}i}{80} & 0 & -\frac{\sqrt{30}i}{280} & 0 & -\frac{13\sqrt{6}i}{560} & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & \frac{i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{210}i}{560} \end{bmatrix}$

$\boxed{\text{x254}}$ $\mathbb{M}_{4,2}^{(a)}(E_u,2) = \begin{bmatrix} \frac{1}{112} & 0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{\sqrt{5}}{16} & 0 & 0 & \frac{9\sqrt{6}}{560} & 0 & -\frac{3\sqrt{30}}{280} & 0 & -\frac{9\sqrt{2}}{80} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{112} & 0 & \frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{3}}{16} & -\frac{3\sqrt{14}}{80} & 0 & -\frac{3\sqrt{6}}{140} & 0 & \frac{3\sqrt{10}}{560} & 0 & -\frac{3\sqrt{2}}{40} & 0 \\ \frac{\sqrt{3}}{16} & 0 & \frac{\sqrt{30}}{112} & 0 & -\frac{\sqrt{15}}{112} & 0 & 0 & \frac{3\sqrt{2}}{40} & 0 & -\frac{3\sqrt{10}}{560} & 0 & \frac{3\sqrt{6}}{140} & 0 & \frac{3\sqrt{14}}{80} \\ 0 & -\frac{\sqrt{5}}{16} & 0 & -\frac{\sqrt{10}}{112} & 0 & \frac{1}{112} & 0 & 0 & \frac{9\sqrt{2}}{80} & 0 & \frac{3\sqrt{30}}{280} & 0 & -\frac{9\sqrt{6}}{560} & 0 \\ 0 & \frac{1}{28} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & \frac{\sqrt{210}}{560} & 0 & -\frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 \\ \frac{1}{28} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{6}}{560} & 0 & \frac{\sqrt{30}}{280} & 0 & -\frac{7\sqrt{2}}{80} & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & \frac{3\sqrt{21}}{80} & 0 & \frac{1}{560} & 0 & \frac{\sqrt{15}}{80} & 0 & \frac{\sqrt{3}}{80} & 0 \\ \frac{\sqrt{2}}{8} & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & \frac{\sqrt{3}}{80} & 0 & \frac{\sqrt{15}}{80} & 0 & \frac{1}{560} & 0 & \frac{3\sqrt{21}}{80} \\ 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{1}{28} & 0 & 0 & -\frac{7\sqrt{2}}{80} & 0 & \frac{\sqrt{30}}{280} & 0 & -\frac{13\sqrt{6}}{560} & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{210}}{560} \end{bmatrix}$

$\boxed{\text{x255}}$ $\mathbb{M}_2^{(1,-1;a)}(A_{1u}) = \begin{bmatrix} 0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 \\ \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 \end{bmatrix}$

$\boxed{\text{x256}}$ $\mathbb{M}_4^{(1,-1;a)}(A_{1u}, 1) = \begin{bmatrix} 0 & \frac{\sqrt{7}}{168} & 0 & 0 & 0 & \frac{\sqrt{35}}{168} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 & \frac{\sqrt{10}}{24} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{\sqrt{10}}{24} & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{168} & 0 & 0 & 0 & -\frac{\sqrt{7}}{168} & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 \\ -\frac{\sqrt{7}}{84} & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & -\frac{5\sqrt{14}}{168} & 0 & 0 \\ 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & \frac{\sqrt{70}}{42} & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{168} & 0 & 0 & 0 & -\frac{\sqrt{15}}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & -\frac{5\sqrt{21}}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{84} & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & -\frac{\sqrt{70}}{42} & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & 0 & -\frac{\sqrt{7}}{84} & 0 & 0 & \frac{5\sqrt{14}}{168} & 0 & 0 & 0 & \frac{5\sqrt{42}}{168} & 0 \end{bmatrix}$

$\boxed{\text{x257}}$ $\mathbb{M}_4^{(1,-1;a)}(A_{1u},2) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{5}}{168} & 0 & 0 & 0 & -\frac{1}{24} & 0 & 0 & \frac{5\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 \\
0 & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{10}}{168} & 0 & 0 & 0 & -\frac{\sqrt{14}}{24} \\
0 & 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & -\frac{\sqrt{14}}{24} & 0 & 0 & 0 & -\frac{5\sqrt{10}}{168} & 0 & 0 & 0 \\
\frac{1}{24} & 0 & 0 & 0 & -\frac{\sqrt{5}}{168} & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & \frac{5\sqrt{2}}{56} & 0 & 0 \\
-\frac{\sqrt{5}}{84} & 0 & 0 & 0 & \frac{1}{12} & 0 & 0 & -\frac{5\sqrt{30}}{168} & 0 & 0 & 0 & \frac{\sqrt{10}}{24} & 0 & 0 \\
0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & \frac{1}{12} & 0 & 0 & \frac{5\sqrt{2}}{42} & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\
0 & 0 & -\frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{15}}{168} & 0 & 0 & 0 & \frac{\sqrt{21}}{24} \\
0 & 0 & 0 & -\frac{\sqrt{5}}{42} & 0 & 0 & -\frac{\sqrt{21}}{24} & 0 & 0 & 0 & -\frac{5\sqrt{15}}{168} & 0 & 0 & 0 \\
\frac{1}{12} & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & -\frac{5\sqrt{2}}{42} & 0 & 0 & 0 \\
0 & \frac{1}{12} & 0 & 0 & 0 & -\frac{\sqrt{5}}{84} & 0 & 0 & -\frac{\sqrt{10}}{24} & 0 & 0 & 0 & \frac{5\sqrt{30}}{168} & 0
\end{bmatrix}$$

$\boxed{\text{x258}}$ $\mathbb{M}_6^{(1,-1;a)}(A_{1u},1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{264} & 0 & 0 & 0 & -\frac{7\sqrt{11}}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 & 0 & \frac{7\sqrt{165}}{264} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{264} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{264} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{264} & 0 & 0 & 0 & -\frac{5\sqrt{66}}{264} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{165}}{264} & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{11}}{88} & 0 & 0 & 0 & -\frac{\sqrt{33}}{264} & 0
\end{bmatrix}$$

$\boxed{\text{x259}}$ $\mathbb{M}_6^{(1,-1;a)}(A_{1u},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{264} & 0 & 0 & 0 & \frac{\sqrt{77}}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{88} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{264} & 0 & 0 & 0 & \frac{\sqrt{330}}{264} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 & 0 & -\frac{5\sqrt{462}}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{264} & 0 & 0 & 0 & \frac{\sqrt{385}}{88} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & 0 & -\frac{\sqrt{231}}{264} & 0 \end{bmatrix}$

$\boxed{\text{x260}}$ $\mathbb{M}_4^{(1,-1;a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x261}}$ $\mathbb{M}_6^{(1,-1;a)}(A_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x262}}$ $\mathbb{M}_2^{(1,-1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{3}{70} & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & -\frac{\sqrt{3}}{35} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{70} & 0 & 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{35} & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 \\ 0 & 0 & \frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{35} & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} \\ 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{35} & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & \frac{2\sqrt{3}}{35} & 0 & 0 & 0 \\ \frac{\sqrt{5}}{35} & 0 & 0 & 0 & \frac{3}{35} & 0 & 0 & -\frac{\sqrt{30}}{35} & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 \\ 0 & \frac{3}{35} & 0 & 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & \frac{\sqrt{30}}{35} & 0 \\ 0 & 0 & \frac{3}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{35} & 0 & 0 & 0 & \frac{\sqrt{105}}{70} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x263}}$ $\mathbb{M}_4^{(1,-1;a)}(B_{1u}) = \begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 \\
\frac{1}{56} & 0 & 0 & 0 & \frac{\sqrt{5}}{56} & 0 & 0 & \frac{11\sqrt{6}}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 \\
0 & -\frac{\sqrt{5}}{56} & 0 & 0 & 0 & -\frac{1}{56} & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & \frac{11\sqrt{6}}{168} & 0 \\
0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} \\
0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 \\
\frac{\sqrt{6}}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & \frac{9}{56} & 0 & 0 & 0 & -\frac{17\sqrt{3}}{168} & 0 & 0 \\
0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & \frac{17\sqrt{3}}{168} & 0 & 0 & 0 & -\frac{9}{56} & 0 \\
0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} \\
0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x264}}$ $\mathbb{M}_6^{(1,-1;a)}(B_{1u},1) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{24} & 0 & 0 & 0 & \frac{\sqrt{5}}{8} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{24} & 0 & 0 & 0 & \frac{\sqrt{35}}{24} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{24} & 0 & 0 & 0 & -\frac{\sqrt{42}}{24} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 & \frac{\sqrt{14}}{24} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{24} & 0 & 0 & 0 & -\frac{1}{24} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & \frac{\sqrt{7}}{24} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x265}}$ $\mathbb{M}_6^{(1,-1;a)}(B_{1u},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{264} & 0 & 0 & 0 & \frac{\sqrt{11}}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{264} & 0 & 0 & 0 & -\frac{5\sqrt{77}}{264} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{264} & 0 & 0 & 0 & \frac{\sqrt{2310}}{264} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{264} & 0 & 0 & 0 & -\frac{\sqrt{770}}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{264} & 0 & 0 & 0 & \frac{\sqrt{55}}{264} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{8} & 0 & 0 & 0 & -\frac{\sqrt{385}}{264} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x266}}$ $\mathbb{M}_2^{(1,-1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{3i}{70} & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 & \frac{\sqrt{3}i}{35} & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & \frac{\sqrt{6}i}{35} & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & \frac{\sqrt{15}i}{35} & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{35} & 0 & 0 & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & \frac{3\sqrt{5}i}{35} & 0 \\ 0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{35} & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} \\ 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3i}{35} & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{35} & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & -\frac{3i}{35} & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 \\ 0 & \frac{3i}{35} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 \\ 0 & 0 & \frac{3i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x267}}$ $\mathbb{M}_4^{(1,-1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 \\ -\frac{i}{56} & 0 & 0 & 0 & \frac{\sqrt{5}i}{56} & 0 & 0 & -\frac{11\sqrt{6}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{56} & 0 & 0 & 0 & -\frac{i}{56} & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & \frac{11\sqrt{6}i}{168} & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \\ 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & -\frac{9i}{56} & 0 & 0 & 0 & -\frac{17\sqrt{3}i}{168} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & -\frac{17\sqrt{3}i}{168} & 0 & 0 & 0 & -\frac{9i}{56} & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x268}}$ $\mathbb{M}_6^{(1,-1;a)}(B_{2u},1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x269}}$ $\mathbb{M}_6^{(1,-1;a)}(B_{2u},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{66} & 0 & 0 & 0 & \frac{\sqrt{385}i}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{462}i}{66} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{66} & 0 & 0 & 0 & \frac{\sqrt{154}i}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{66} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x270}}$ $\mathbb{M}_{2,1}^{(1,-1;a)}(E_u) = \begin{bmatrix} \frac{3\sqrt{10}i}{140} & 0 & \frac{9i}{140} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{35} & 0 & \frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{140} & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{35} & 0 & \frac{6i}{35} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & \frac{6i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{9i}{140} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 \\ 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{35} & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{140} & 0 & -\frac{11\sqrt{3}i}{140} & 0 & 0 & 0 & 0 \\ 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{140} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2i}{35} & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & \frac{11\sqrt{3}i}{140} & 0 & \frac{\sqrt{15}i}{140} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{28} & 0 & \frac{\sqrt{21}i}{28} \end{bmatrix}$

$\boxed{\text{x271}}$ $\mathbb{M}_{2,2}^{(1,-1;a)}(E_u) = \begin{bmatrix} -\frac{3\sqrt{10}}{140} & 0 & \frac{9}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & \frac{2\sqrt{3}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{140} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & \frac{6}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{6}}{140} & 0 & 0 & 0 & 0 & -\frac{6}{35} & 0 & \frac{2\sqrt{15}}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{9}{140} & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{35} & 0 & \frac{2\sqrt{15}}{35} & 0 \\ 0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{3}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{35} & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{140} & 0 & -\frac{11\sqrt{3}}{140} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{35} & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{3}}{140} & 0 & \frac{\sqrt{15}}{140} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{28} & 0 & \frac{\sqrt{21}}{28} \end{bmatrix}$

$\boxed{\text{x272}}$ $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u, 1) = \begin{bmatrix} \frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{210}i}{672} & 0 & \frac{\sqrt{105}i}{672} & 0 & 0 & \frac{3\sqrt{14}i}{112} & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{224} & 0 & -\frac{\sqrt{70}i}{224} & 0 & -\frac{\sqrt{7}i}{224} & -\frac{\sqrt{6}i}{48} & 0 & -\frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{42}i}{168} & 0 \\ \frac{\sqrt{7}i}{224} & 0 & \frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{35}i}{224} & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{14}i}{28} & 0 & -\frac{\sqrt{6}i}{48} \\ 0 & -\frac{\sqrt{105}i}{672} & 0 & -\frac{\sqrt{210}i}{672} & 0 & -\frac{\sqrt{21}i}{672} & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{3\sqrt{14}i}{112} & 0 \\ 0 & -\frac{\sqrt{21}i}{84} & 0 & -\frac{\sqrt{42}i}{168} & 0 & 0 & -\frac{\sqrt{10}i}{32} & 0 & -\frac{5\sqrt{210}i}{336} & 0 & -\frac{5\sqrt{14}i}{224} & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{14}i}{224} & 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{42}i}{96} & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & \frac{\sqrt{42}i}{168} & -\frac{3i}{32} & 0 & -\frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{35}i}{32} & 0 & \frac{\sqrt{7}i}{224} & 0 \\ \frac{\sqrt{42}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{7}i}{224} & 0 & -\frac{\sqrt{35}i}{32} & 0 & \frac{\sqrt{21}i}{672} & 0 & \frac{3i}{32} \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & \frac{\sqrt{42}i}{96} & 0 & -\frac{\sqrt{70}i}{112} & 0 & -\frac{13\sqrt{14}i}{224} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{224} & 0 & \frac{5\sqrt{210}i}{336} & 0 & \frac{\sqrt{10}i}{32} \end{bmatrix}$

$\boxed{\text{x273}}$ $\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,1) =$

$$\begin{bmatrix}
-\frac{\sqrt{21}}{672} & 0 & \frac{\sqrt{210}}{672} & 0 & -\frac{\sqrt{105}}{672} & 0 & 0 & -\frac{3\sqrt{14}}{112} & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 \\
0 & \frac{\sqrt{35}}{224} & 0 & -\frac{\sqrt{70}}{224} & 0 & \frac{\sqrt{7}}{224} & -\frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{14}}{28} & 0 & -\frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{42}}{168} & 0 \\
\frac{\sqrt{7}}{224} & 0 & -\frac{\sqrt{70}}{224} & 0 & \frac{\sqrt{35}}{224} & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{14}}{28} & 0 & \frac{\sqrt{6}}{48} \\
0 & -\frac{\sqrt{105}}{672} & 0 & \frac{\sqrt{210}}{672} & 0 & -\frac{\sqrt{21}}{672} & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{3\sqrt{14}}{112} & 0 \\
0 & -\frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{\sqrt{10}}{32} & 0 & -\frac{5\sqrt{210}}{336} & 0 & \frac{5\sqrt{14}}{224} & 0 & 0 & 0 \\
-\frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{14}}{224} & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{42}}{96} & 0 & 0 \\
0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} & -\frac{3}{32} & 0 & \frac{\sqrt{21}}{672} & 0 & \frac{\sqrt{35}}{32} & 0 & -\frac{\sqrt{7}}{224} & 0 \\
\frac{\sqrt{42}}{168} & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & -\frac{\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{32} & 0 & \frac{\sqrt{21}}{672} & 0 & -\frac{3}{32} \\
0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & \frac{\sqrt{42}}{96} & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{13\sqrt{14}}{224} & 0 \\
0 & 0 & -\frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{224} & 0 & -\frac{5\sqrt{210}}{336} & 0 & \frac{\sqrt{10}}{32}
\end{bmatrix}$$

$\boxed{\text{x274}}$ $\mathbb{M}_{4,1}^{(1,-1;a)}(E_u,2) =$

$$\begin{bmatrix}
\frac{\sqrt{3}i}{672} & 0 & \frac{\sqrt{30}i}{672} & 0 & -\frac{\sqrt{15}i}{96} & 0 & 0 & \frac{3\sqrt{2}i}{112} & 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 \\
0 & -\frac{\sqrt{5}i}{224} & 0 & -\frac{\sqrt{10}i}{224} & 0 & \frac{i}{32} & \frac{\sqrt{42}i}{48} & 0 & -\frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{30}i}{336} & 0 & -\frac{\sqrt{6}i}{24} & 0 \\
-\frac{i}{32} & 0 & \frac{\sqrt{10}i}{224} & 0 & \frac{\sqrt{5}i}{224} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{30}i}{336} & 0 & -\frac{\sqrt{2}i}{28} & 0 & \frac{\sqrt{42}i}{48} \\
0 & \frac{\sqrt{15}i}{96} & 0 & -\frac{\sqrt{30}i}{672} & 0 & -\frac{\sqrt{3}i}{672} & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{3\sqrt{2}i}{112} & 0 \\
0 & -\frac{\sqrt{3}i}{84} & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & -\frac{5\sqrt{30}i}{336} & 0 & \frac{5\sqrt{2}i}{32} & 0 & 0 & 0 \\
\frac{\sqrt{3}i}{84} & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{2}i}{224} & 0 & \frac{\sqrt{10}i}{112} & 0 & \frac{7\sqrt{6}i}{96} & 0 & 0 \\
0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} & \frac{3\sqrt{7}i}{32} & 0 & -\frac{\sqrt{3}i}{672} & 0 & \frac{\sqrt{5}i}{32} & 0 & -\frac{i}{32} & 0 \\
-\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{168} & 0 & 0 & \frac{i}{32} & 0 & -\frac{\sqrt{5}i}{32} & 0 & \frac{\sqrt{3}i}{672} & 0 & -\frac{3\sqrt{7}i}{32} \\
0 & 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & \frac{\sqrt{3}i}{84} & 0 & 0 & -\frac{7\sqrt{6}i}{96} & 0 & -\frac{\sqrt{10}i}{112} & 0 & -\frac{13\sqrt{2}i}{224} & 0 \\
0 & 0 & \frac{\sqrt{6}i}{24} & 0 & -\frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{32} & 0 & \frac{5\sqrt{30}i}{336} & 0 & \frac{\sqrt{70}i}{224}
\end{bmatrix}$$

$\boxed{\text{x275}}$ $\mathbb{M}_{4,2}^{(1,-1;a)}(E_u,2) =$

$$\begin{bmatrix}
-\frac{\sqrt{3}}{672} & 0 & \frac{\sqrt{30}}{672} & 0 & \frac{\sqrt{15}}{96} & 0 & 0 & -\frac{3\sqrt{2}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{6}}{16} & 0 & 0 \\
0 & \frac{\sqrt{5}}{224} & 0 & -\frac{\sqrt{10}}{224} & 0 & -\frac{1}{32} & \frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{2}}{28} & 0 & -\frac{\sqrt{30}}{336} & 0 & \frac{\sqrt{6}}{24} & 0 \\
-\frac{1}{32} & 0 & -\frac{\sqrt{10}}{224} & 0 & \frac{\sqrt{5}}{224} & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{30}}{336} & 0 & -\frac{\sqrt{2}}{28} & 0 & -\frac{\sqrt{42}}{48} \\
0 & \frac{\sqrt{15}}{96} & 0 & \frac{\sqrt{30}}{672} & 0 & -\frac{\sqrt{3}}{672} & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{3\sqrt{2}}{112} & 0 \\
0 & -\frac{\sqrt{3}}{84} & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & -\frac{5\sqrt{30}}{336} & 0 & -\frac{5\sqrt{2}}{32} & 0 & 0 & 0 \\
-\frac{\sqrt{3}}{84} & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}}{224} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{7\sqrt{6}}{96} & 0 & 0 \\
0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & \frac{3\sqrt{7}}{32} & 0 & \frac{\sqrt{3}}{672} & 0 & \frac{\sqrt{5}}{32} & 0 & \frac{1}{32} & 0 \\
-\frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & 0 & \frac{1}{32} & 0 & \frac{\sqrt{5}}{32} & 0 & \frac{\sqrt{3}}{672} & 0 & \frac{3\sqrt{7}}{32} \\
0 & 0 & 0 & -\frac{\sqrt{30}}{168} & 0 & \frac{\sqrt{3}}{84} & 0 & 0 & -\frac{7\sqrt{6}}{96} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{13\sqrt{2}}{224} & 0 \\
0 & 0 & \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{32} & 0 & -\frac{5\sqrt{30}}{336} & 0 & \frac{\sqrt{70}}{224}
\end{bmatrix}$$

$\boxed{\text{x276}}$ $\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{1056} & 0 & -\frac{\sqrt{154}i}{352} & 0 & \frac{\sqrt{2310}i}{352} & 0 & \frac{\sqrt{462}i}{96} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1056} & 0 & \frac{5\sqrt{462}i}{1056} & 0 & -\frac{3\sqrt{770}i}{352} & 0 & -\frac{\sqrt{330}i}{96} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{176} & 0 & -\frac{\sqrt{385}i}{176} & 0 & -\frac{5\sqrt{231}i}{528} & 0 & \frac{\sqrt{1155}i}{176} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{176} & 0 & \frac{5\sqrt{231}i}{528} & 0 & \frac{\sqrt{385}i}{176} & 0 & -\frac{\sqrt{165}i}{176} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{96} & 0 & \frac{3\sqrt{770}i}{352} & 0 & -\frac{5\sqrt{462}i}{1056} & 0 & -\frac{\sqrt{2310}i}{1056} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{96} & 0 & -\frac{\sqrt{2310}i}{352} & 0 & \frac{\sqrt{154}i}{352} & 0 & \frac{\sqrt{66}i}{1056}
\end{bmatrix}$$

$\boxed{\text{x277}}$ $\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{1056} & 0 & -\frac{\sqrt{154}}{352} & 0 & -\frac{\sqrt{2310}}{352} & 0 & \frac{\sqrt{462}}{96} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1056} & 0 & \frac{5\sqrt{462}}{1056} & 0 & \frac{3\sqrt{770}}{352} & 0 & -\frac{\sqrt{330}}{96} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{176} & 0 & \frac{\sqrt{385}}{176} & 0 & -\frac{5\sqrt{231}}{528} & 0 & -\frac{\sqrt{1155}}{176} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{176} & 0 & -\frac{5\sqrt{231}}{528} & 0 & \frac{\sqrt{385}}{176} & 0 & \frac{\sqrt{165}}{176} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{96} & 0 & \frac{3\sqrt{770}}{352} & 0 & \frac{5\sqrt{462}}{1056} & 0 & -\frac{\sqrt{2310}}{1056} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{96} & 0 & -\frac{\sqrt{2310}}{352} & 0 & -\frac{\sqrt{154}}{352} & 0 & \frac{\sqrt{66}}{1056}
\end{bmatrix}$$

$\boxed{\text{x278}}$ $\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{64} & 0 & -\frac{\sqrt{21}i}{64} & 0 & -\frac{\sqrt{35}i}{64} & 0 & -\frac{\sqrt{7}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{64} & 0 & \frac{5\sqrt{7}i}{64} & 0 & \frac{\sqrt{105}i}{64} & 0 & \frac{\sqrt{5}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{64} & 0 & -\frac{\sqrt{210}i}{64} & 0 & -\frac{5\sqrt{14}i}{64} & 0 & -\frac{\sqrt{70}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{64} & 0 & \frac{5\sqrt{14}i}{64} & 0 & \frac{\sqrt{210}i}{64} & 0 & \frac{\sqrt{10}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{64} & 0 & -\frac{\sqrt{105}i}{64} & 0 & -\frac{5\sqrt{7}i}{64} & 0 & -\frac{\sqrt{35}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{64} & 0 & \frac{\sqrt{35}i}{64} & 0 & \frac{\sqrt{21}i}{64} & 0 & \frac{i}{64}
\end{bmatrix}$$

$\boxed{\text{x279}}$ $\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{64} & 0 & -\frac{\sqrt{21}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 & -\frac{\sqrt{7}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{64} & 0 & \frac{5\sqrt{7}}{64} & 0 & -\frac{\sqrt{105}}{64} & 0 & \frac{\sqrt{5}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{64} & 0 & \frac{\sqrt{210}}{64} & 0 & -\frac{5\sqrt{14}}{64} & 0 & \frac{\sqrt{70}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{64} & 0 & -\frac{5\sqrt{14}}{64} & 0 & \frac{\sqrt{210}}{64} & 0 & -\frac{\sqrt{10}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{64} & 0 & -\frac{\sqrt{105}}{64} & 0 & \frac{5\sqrt{7}}{64} & 0 & -\frac{\sqrt{35}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 & -\frac{\sqrt{21}}{64} & 0 & \frac{1}{64}
\end{bmatrix}$$

$\boxed{\text{x280}}$ $\mathbb{M}_{6,1}^{(1,-1;a)}(E_u,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{2112} & 0 & -\frac{\sqrt{1155}i}{2112} & 0 & \frac{9\sqrt{77}i}{704} & 0 & -\frac{\sqrt{385}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{2112} & 0 & \frac{5\sqrt{385}i}{2112} & 0 & -\frac{9\sqrt{231}i}{704} & 0 & \frac{5\sqrt{11}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{22}i}{704} & 0 & -\frac{5\sqrt{462}i}{2112} & 0 & -\frac{5\sqrt{770}i}{2112} & 0 & \frac{9\sqrt{154}i}{704} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{154}i}{704} & 0 & \frac{5\sqrt{770}i}{2112} & 0 & \frac{5\sqrt{462}i}{2112} & 0 & -\frac{9\sqrt{22}i}{704} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}i}{64} & 0 & \frac{9\sqrt{231}i}{704} & 0 & -\frac{5\sqrt{385}i}{2112} & 0 & -\frac{5\sqrt{77}i}{2112} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{64} & 0 & -\frac{9\sqrt{77}i}{704} & 0 & \frac{\sqrt{1155}i}{2112} & 0 & \frac{\sqrt{55}i}{2112}
\end{bmatrix}$$

$\boxed{\text{x281}}$ $\mathbb{M}_{6,2}^{(1,-1;a)}(E_u,3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{2112} & 0 & -\frac{\sqrt{1155}}{2112} & 0 & -\frac{9\sqrt{77}}{704} & 0 & -\frac{\sqrt{385}}{64} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{2112} & 0 & \frac{5\sqrt{385}}{2112} & 0 & \frac{9\sqrt{231}}{704} & 0 & \frac{5\sqrt{11}}{64} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{22}}{704} & 0 & \frac{5\sqrt{462}}{2112} & 0 & -\frac{5\sqrt{770}}{2112} & 0 & -\frac{9\sqrt{154}}{704} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{154}}{704} & 0 & -\frac{5\sqrt{770}}{2112} & 0 & \frac{5\sqrt{462}}{2112} & 0 & \frac{9\sqrt{22}}{704} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{64} & 0 & \frac{9\sqrt{231}}{704} & 0 & \frac{5\sqrt{385}}{2112} & 0 & -\frac{5\sqrt{77}}{2112} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{64} & 0 & -\frac{9\sqrt{77}}{704} & 0 & -\frac{\sqrt{1155}}{2112} & 0 & \frac{\sqrt{55}}{2112} \end{bmatrix}$

$\boxed{\text{x282}}$ $\mathbb{M}_{2}^{(1,0;a)}(A_{1u}) = \begin{bmatrix} 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 \\ \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 \end{bmatrix}$

$\boxed{\text{x283}}$ $\mathbb{M}_4^{(1,0;a)}(A_{1u},1) = \begin{bmatrix}
0 & \frac{\sqrt{105}}{840} & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0 & -\frac{9\sqrt{42}}{280} & 0 & 0 & 0 & -\frac{9\sqrt{14}}{280} & 0 \\
0 & 0 & -\frac{\sqrt{70}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{6}}{40} \\
0 & 0 & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & -\frac{3\sqrt{6}}{40} & 0 & 0 & 0 & \frac{3\sqrt{210}}{280} & 0 & 0 & 0 \\
-\frac{\sqrt{21}}{168} & 0 & 0 & 0 & -\frac{\sqrt{105}}{840} & 0 & 0 & -\frac{9\sqrt{14}}{280} & 0 & 0 & 0 & -\frac{9\sqrt{42}}{280} & 0 & 0 \\
-\frac{\sqrt{105}}{140} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{\sqrt{210}}{840} & 0 & 0 \\
0 & \frac{3\sqrt{105}}{140} & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{42}}{210} & 0 & 0 & 0 & -\frac{\sqrt{14}}{140} & 0 \\
0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{280} & 0 & 0 & 0 & -\frac{1}{40} \\
0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & \frac{1}{40} & 0 & 0 & 0 & -\frac{\sqrt{35}}{280} & 0 & 0 & 0 \\
-\frac{\sqrt{21}}{28} & 0 & 0 & 0 & \frac{3\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}}{210} & 0 & 0 \\
0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{210}}{840} & 0 & 0 & 0 & \frac{\sqrt{70}}{280} & 0
\end{bmatrix}$

$\boxed{\text{x284}}$ $\mathbb{M}_4^{(1,0;a)}(A_{1u},2) = \begin{bmatrix}
0 & \frac{\sqrt{3}}{168} & 0 & 0 & 0 & -\frac{\sqrt{15}}{120} & 0 & 0 & -\frac{9\sqrt{30}}{280} & 0 & 0 & 0 & \frac{9\sqrt{10}}{200} & 0 \\
0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{56} & 0 & 0 & 0 & \frac{3\sqrt{210}}{200} \\
0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & \frac{3\sqrt{210}}{200} & 0 & 0 & 0 & \frac{3\sqrt{6}}{56} & 0 & 0 & 0 \\
\frac{\sqrt{15}}{120} & 0 & 0 & 0 & -\frac{\sqrt{3}}{168} & 0 & 0 & \frac{9\sqrt{10}}{200} & 0 & 0 & 0 & -\frac{9\sqrt{30}}{280} & 0 & 0 \\
-\frac{\sqrt{3}}{28} & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & \frac{\sqrt{6}}{120} & 0 & 0 \\
0 & \frac{3\sqrt{3}}{28} & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & \frac{\sqrt{30}}{210} & 0 & 0 & 0 & \frac{\sqrt{10}}{100} & 0 \\
0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{56} & 0 & 0 & 0 & \frac{\sqrt{35}}{200} \\
0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & -\frac{\sqrt{35}}{200} & 0 & 0 & 0 & -\frac{1}{56} & 0 & 0 & 0 \\
\frac{\sqrt{15}}{20} & 0 & 0 & 0 & \frac{3\sqrt{3}}{28} & 0 & 0 & -\frac{\sqrt{10}}{100} & 0 & 0 & 0 & -\frac{\sqrt{30}}{210} & 0 & 0 \\
0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & -\frac{\sqrt{6}}{120} & 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0
\end{bmatrix}$

$\boxed{\text{x285}}$ $\mathbb{M}_4^{(1,0;a)}(A_{2u}) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{210}i}{700} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}i}{100} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{210}i}{700} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{140} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{350} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{100} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{140} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x286}}$ $\mathbb{M}_2^{(1,0;a)}(B_{1u}) = \begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 \\
\frac{\sqrt{3}}{21} & 0 & 0 & 0 & -\frac{2\sqrt{15}}{105} & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 \\
0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 \\
0 & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & \frac{\sqrt{42}}{28} \\
0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & \frac{\sqrt{30}}{105} & 0 & 0 & 0 \\
\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 \\
0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 \\
0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{105} & 0 & 0 & 0 & \frac{\sqrt{42}}{84} \\
0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x287}}$ $\mathbb{M}_4^{(1,0;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & \frac{9\sqrt{210}}{1400} & 0 & 0 & 0 & \frac{27\sqrt{6}}{280} & 0 & 0 & 0 \\
\frac{\sqrt{15}}{280} & 0 & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & -\frac{99\sqrt{10}}{1400} & 0 & 0 & 0 & \frac{9\sqrt{30}}{1400} & 0 & 0 \\
0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 & -\frac{\sqrt{15}}{280} & 0 & 0 & \frac{9\sqrt{30}}{1400} & 0 & 0 & 0 & -\frac{99\sqrt{10}}{1400} & 0 \\
0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{6}}{280} & 0 & 0 & 0 & \frac{9\sqrt{210}}{1400} \\
0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & -\frac{3\sqrt{210}}{1400} & 0 & 0 & 0 & \frac{\sqrt{6}}{280} & 0 & 0 & 0 \\
\frac{9\sqrt{10}}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & \frac{9\sqrt{15}}{1400} & 0 & 0 & 0 & -\frac{17\sqrt{5}}{1400} & 0 & 0 \\
0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & \frac{17\sqrt{5}}{1400} & 0 & 0 & 0 & -\frac{9\sqrt{15}}{1400} & 0 \\
0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{280} & 0 & 0 & 0 & \frac{3\sqrt{210}}{1400} \\
0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x288}}$ $\mathbb{M}_2^{(1,0;a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{10}i}{70} & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 \\
\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\
0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 \\
0 & 0 & \frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \\
0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{105} & 0 & 0 & 0 \\
\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 \\
0 & \frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 \\
0 & 0 & \frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} \\
0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x289}}$ $\mathbb{M}_4^{(1,0;a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & -\frac{9\sqrt{210}i}{1400} & 0 & 0 & 0 & \frac{27\sqrt{6}i}{280} & 0 & 0 & 0 \\
-\frac{\sqrt{15}i}{280} & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & \frac{99\sqrt{10}i}{1400} & 0 & 0 & 0 & \frac{9\sqrt{30}i}{1400} & 0 & 0 \\
0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{280} & 0 & 0 & -\frac{9\sqrt{30}i}{1400} & 0 & 0 & 0 & -\frac{99\sqrt{10}i}{1400} & 0 \\
0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{6}i}{280} & 0 & 0 & 0 & \frac{9\sqrt{210}i}{1400} \\
0 & 0 & \frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & \frac{3\sqrt{210}i}{1400} & 0 & 0 & 0 & \frac{\sqrt{6}i}{280} & 0 & 0 & 0 \\
-\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & -\frac{9\sqrt{15}i}{1400} & 0 & 0 & 0 & -\frac{17\sqrt{5}i}{1400} & 0 & 0 \\
0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{9\sqrt{10}i}{140} & 0 & 0 & -\frac{17\sqrt{5}i}{1400} & 0 & 0 & 0 & -\frac{9\sqrt{15}i}{1400} & 0 \\
0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{210}i}{1400} \\
0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x290}}$ $\mathbb{M}_{2,1}^{(1,0;a)}(E_u) =$

$$\begin{bmatrix}
\frac{i}{14} & 0 & \frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{210} & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & \frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & -\frac{\sqrt{6}i}{14} & 0 \\
0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 \\
\frac{3i}{14} & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{168} & 0 & -\frac{11\sqrt{30}i}{840} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} & 0 & -\frac{i}{28} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}i}{840} & 0 & \frac{\sqrt{6}i}{168} & 0 \\
0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{210}i}{168}
\end{bmatrix}$$

$\boxed{\text{x291}}$ $\mathbb{M}_{2,2}^{(1,0;a)}(E_u) = \begin{bmatrix} -\frac{1}{14} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{210} & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{84} & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}}{140} & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & -\frac{\sqrt{6}}{14} & 0 \\ 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3}{14} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{168} & 0 & -\frac{11\sqrt{30}}{840} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{60} & 0 & -\frac{1}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{30}}{840} & 0 & \frac{\sqrt{6}}{168} & 0 \\ 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{210}}{168} \end{bmatrix}$

$\boxed{\text{x292}}$ $\mathbb{M}_{4,1}^{(1,0;a)}(E_u, 1) = \begin{bmatrix} \frac{\sqrt{35}i}{1120} & 0 & \frac{\sqrt{14}i}{224} & 0 & \frac{\sqrt{7}i}{224} & 0 & 0 & -\frac{27\sqrt{210}i}{2800} & 0 & -\frac{9\sqrt{42}i}{280} & 0 & -\frac{27\sqrt{70}i}{2800} & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{224} & 0 & -\frac{\sqrt{42}i}{224} & 0 & -\frac{\sqrt{105}i}{1120} & \frac{9\sqrt{10}i}{400} & 0 & \frac{9\sqrt{210}i}{700} & 0 & \frac{9\sqrt{14}i}{560} & 0 & -\frac{9\sqrt{70}i}{1400} & 0 \\ \frac{\sqrt{105}i}{1120} & 0 & \frac{\sqrt{42}i}{224} & 0 & \frac{\sqrt{21}i}{224} & 0 & 0 & -\frac{9\sqrt{70}i}{1400} & 0 & \frac{9\sqrt{14}i}{560} & 0 & \frac{9\sqrt{210}i}{700} & 0 & \frac{9\sqrt{10}i}{400} \\ 0 & -\frac{\sqrt{7}i}{224} & 0 & -\frac{\sqrt{14}i}{224} & 0 & -\frac{\sqrt{35}i}{1120} & 0 & 0 & -\frac{27\sqrt{70}i}{2800} & 0 & -\frac{9\sqrt{42}i}{280} & 0 & -\frac{27\sqrt{210}i}{2800} & 0 \\ 0 & -\frac{3\sqrt{35}i}{140} & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & -\frac{\sqrt{6}i}{160} & 0 & -\frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{210}i}{1120} & 0 & 0 & 0 \\ \frac{3\sqrt{35}i}{140} & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{210}i}{5600} & 0 & \frac{\sqrt{42}i}{560} & 0 & -\frac{\sqrt{70}i}{800} & 0 & 0 \\ 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} & -\frac{3\sqrt{15}i}{800} & 0 & -\frac{\sqrt{35}i}{5600} & 0 & \frac{\sqrt{21}i}{160} & 0 & \frac{\sqrt{105}i}{5600} & 0 \\ \frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{105}i}{5600} & 0 & -\frac{\sqrt{21}i}{160} & 0 & \frac{\sqrt{35}i}{5600} & 0 & \frac{3\sqrt{15}i}{800} \\ 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & \frac{3\sqrt{35}i}{140} & 0 & 0 & \frac{\sqrt{70}i}{800} & 0 & -\frac{\sqrt{42}i}{560} & 0 & -\frac{13\sqrt{210}i}{5600} & 0 \\ 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{1120} & 0 & \frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{6}i}{160} \end{bmatrix}$

$\boxed{\text{x293}}$ $\mathbb{M}_{4,2}^{(1,0;a)}(E_u,1) =$

$$\begin{bmatrix}
-\frac{\sqrt{35}}{1120} & 0 & \frac{\sqrt{14}}{224} & 0 & -\frac{\sqrt{7}}{224} & 0 & 0 & \frac{27\sqrt{210}}{2800} & 0 & -\frac{9\sqrt{42}}{280} & 0 & \frac{27\sqrt{70}}{2800} & 0 & 0 \\
0 & \frac{\sqrt{21}}{224} & 0 & -\frac{\sqrt{42}}{224} & 0 & \frac{\sqrt{105}}{1120} & \frac{9\sqrt{10}}{400} & 0 & -\frac{9\sqrt{210}}{700} & 0 & \frac{9\sqrt{14}}{560} & 0 & \frac{9\sqrt{70}}{1400} & 0 \\
\frac{\sqrt{105}}{1120} & 0 & -\frac{\sqrt{42}}{224} & 0 & \frac{\sqrt{21}}{224} & 0 & 0 & -\frac{9\sqrt{70}}{1400} & 0 & -\frac{9\sqrt{14}}{560} & 0 & \frac{9\sqrt{210}}{700} & 0 & -\frac{9\sqrt{10}}{400} \\
0 & -\frac{\sqrt{7}}{224} & 0 & \frac{\sqrt{14}}{224} & 0 & -\frac{\sqrt{35}}{1120} & 0 & 0 & -\frac{27\sqrt{70}}{2800} & 0 & \frac{9\sqrt{42}}{280} & 0 & -\frac{27\sqrt{210}}{2800} & 0 \\
0 & -\frac{3\sqrt{35}}{140} & 0 & \frac{3\sqrt{70}}{280} & 0 & 0 & \frac{\sqrt{6}}{160} & 0 & -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{210}}{1120} & 0 & 0 & 0 \\
-\frac{3\sqrt{35}}{140} & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{210}}{5600} & 0 & \frac{\sqrt{42}}{560} & 0 & \frac{\sqrt{70}}{800} & 0 & 0 \\
0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} & -\frac{3\sqrt{15}}{800} & 0 & \frac{\sqrt{35}}{5600} & 0 & \frac{\sqrt{21}}{160} & 0 & -\frac{\sqrt{105}}{5600} & 0 \\
\frac{3\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & -\frac{\sqrt{105}}{5600} & 0 & \frac{\sqrt{21}}{160} & 0 & \frac{\sqrt{35}}{5600} & 0 & -\frac{3\sqrt{15}}{800} \\
0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & \frac{\sqrt{70}}{800} & 0 & \frac{\sqrt{42}}{560} & 0 & -\frac{13\sqrt{210}}{5600} & 0 \\
0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{1120} & 0 & -\frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{6}}{160}
\end{bmatrix}$$

$\boxed{\text{x294}}$ $\mathbb{M}_{4,1}^{(1,0;a)}(E_u,2) =$

$$\begin{bmatrix}
\frac{\sqrt{5}i}{1120} & 0 & \frac{\sqrt{2}i}{224} & 0 & -\frac{i}{32} & 0 & 0 & -\frac{27\sqrt{30}i}{2800} & 0 & -\frac{9\sqrt{6}i}{280} & 0 & \frac{27\sqrt{10}i}{400} & 0 & 0 \\
0 & -\frac{\sqrt{3}i}{224} & 0 & -\frac{\sqrt{6}i}{224} & 0 & \frac{\sqrt{15}i}{160} & -\frac{9\sqrt{70}i}{400} & 0 & \frac{9\sqrt{30}i}{700} & 0 & \frac{9\sqrt{2}i}{560} & 0 & \frac{9\sqrt{10}i}{200} & 0 \\
-\frac{\sqrt{15}i}{160} & 0 & \frac{\sqrt{6}i}{224} & 0 & \frac{\sqrt{3}i}{224} & 0 & 0 & \frac{9\sqrt{10}i}{200} & 0 & \frac{9\sqrt{2}i}{560} & 0 & \frac{9\sqrt{30}i}{700} & 0 & -\frac{9\sqrt{70}i}{400} \\
0 & \frac{i}{32} & 0 & -\frac{\sqrt{2}i}{224} & 0 & -\frac{\sqrt{5}i}{1120} & 0 & 0 & \frac{27\sqrt{10}i}{400} & 0 & -\frac{9\sqrt{6}i}{280} & 0 & -\frac{27\sqrt{30}i}{2800} & 0 \\
0 & -\frac{3\sqrt{5}i}{140} & 0 & \frac{3\sqrt{10}i}{40} & 0 & 0 & -\frac{\sqrt{42}i}{1120} & 0 & -\frac{\sqrt{2}i}{112} & 0 & \frac{\sqrt{30}i}{160} & 0 & 0 & 0 \\
\frac{3\sqrt{5}i}{140} & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & \frac{13\sqrt{30}i}{5600} & 0 & \frac{\sqrt{6}i}{560} & 0 & \frac{7\sqrt{10}i}{800} & 0 & 0 \\
0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{40} & \frac{3\sqrt{105}i}{800} & 0 & -\frac{\sqrt{5}i}{5600} & 0 & \frac{\sqrt{3}i}{160} & 0 & -\frac{\sqrt{15}i}{800} & 0 \\
-\frac{3\sqrt{10}i}{40} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & \frac{\sqrt{15}i}{800} & 0 & -\frac{\sqrt{3}i}{160} & 0 & \frac{\sqrt{5}i}{5600} & 0 & -\frac{3\sqrt{105}i}{800} \\
0 & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & -\frac{7\sqrt{10}i}{800} & 0 & -\frac{\sqrt{6}i}{560} & 0 & -\frac{13\sqrt{30}i}{5600} & 0 \\
0 & 0 & \frac{3\sqrt{10}i}{40} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{160} & 0 & \frac{\sqrt{2}i}{112} & 0 & \frac{\sqrt{42}i}{1120}
\end{bmatrix}$$

$\boxed{\text{x295}}$ $\mathbb{M}_{4,2}^{(1,0;a)}(E_u,2) = \begin{bmatrix} -\frac{\sqrt{5}}{1120} & 0 & \frac{\sqrt{2}}{224} & 0 & \frac{1}{32} & 0 & 0 & \frac{27\sqrt{30}}{2800} & 0 & -\frac{9\sqrt{6}}{280} & 0 & -\frac{27\sqrt{10}}{400} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{224} & 0 & -\frac{\sqrt{6}}{224} & 0 & -\frac{\sqrt{15}}{160} & -\frac{9\sqrt{70}}{400} & 0 & -\frac{9\sqrt{30}}{700} & 0 & \frac{9\sqrt{2}}{560} & 0 & -\frac{9\sqrt{10}}{200} & 0 \\ -\frac{\sqrt{15}}{160} & 0 & -\frac{\sqrt{6}}{224} & 0 & \frac{\sqrt{3}}{224} & 0 & 0 & \frac{9\sqrt{10}}{200} & 0 & -\frac{9\sqrt{2}}{560} & 0 & \frac{9\sqrt{30}}{700} & 0 & \frac{9\sqrt{70}}{400} \\ 0 & \frac{1}{32} & 0 & \frac{\sqrt{2}}{224} & 0 & -\frac{\sqrt{5}}{1120} & 0 & 0 & \frac{27\sqrt{10}}{400} & 0 & \frac{9\sqrt{6}}{280} & 0 & -\frac{27\sqrt{30}}{2800} & 0 \\ 0 & -\frac{3\sqrt{5}}{140} & 0 & -\frac{3\sqrt{10}}{40} & 0 & 0 & \frac{\sqrt{42}}{1120} & 0 & -\frac{\sqrt{2}}{112} & 0 & -\frac{\sqrt{30}}{160} & 0 & 0 & 0 \\ -\frac{3\sqrt{5}}{140} & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{30}}{5600} & 0 & \frac{\sqrt{6}}{560} & 0 & -\frac{7\sqrt{10}}{800} & 0 & 0 \\ 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 & \frac{3\sqrt{10}}{40} & \frac{3\sqrt{105}}{800} & 0 & \frac{\sqrt{5}}{5600} & 0 & \frac{\sqrt{3}}{160} & 0 & \frac{\sqrt{15}}{800} & 0 \\ -\frac{3\sqrt{10}}{40} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & \frac{\sqrt{15}}{800} & 0 & \frac{\sqrt{3}}{160} & 0 & \frac{\sqrt{5}}{5600} & 0 & \frac{3\sqrt{105}}{800} \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & -\frac{7\sqrt{10}}{800} & 0 & \frac{\sqrt{6}}{560} & 0 & -\frac{13\sqrt{30}}{5600} & 0 \\ 0 & 0 & \frac{3\sqrt{10}}{40} & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{160} & 0 & -\frac{\sqrt{2}}{112} & 0 & \frac{\sqrt{42}}{1120} \end{bmatrix}$

$\boxed{\text{x296}}$ $\mathbb{M}_{0}^{(1,1;a)}(A_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x297}}$ $\mathbb{M}_2^{(1,1;a)}(A_{1u}) = \begin{bmatrix}
0 & \frac{12}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{140} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{140} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{12}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 \\
\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{105} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{105} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0
\end{bmatrix}$

$\boxed{\text{x298}}$ $\mathbb{M}_4^{(1,1;a)}(A_{1u}, 1) = \begin{bmatrix}
0 & -\frac{\sqrt{770}}{210} & 0 & 0 & 0 & -\frac{\sqrt{154}}{42} & 0 & 0 & \frac{\sqrt{77}}{140} & 0 & 0 & 0 & \frac{\sqrt{231}}{420} & 0 \\
0 & 0 & \frac{\sqrt{1155}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{420} & 0 & 0 & 0 & \frac{\sqrt{11}}{60} \\
0 & 0 & 0 & -\frac{\sqrt{1155}}{105} & 0 & 0 & \frac{\sqrt{11}}{60} & 0 & 0 & 0 & -\frac{\sqrt{385}}{420} & 0 & 0 & 0 \\
\frac{\sqrt{154}}{42} & 0 & 0 & 0 & \frac{\sqrt{770}}{210} & 0 & 0 & \frac{\sqrt{231}}{420} & 0 & 0 & 0 & \frac{\sqrt{77}}{140} & 0 & 0 \\
-\frac{\sqrt{770}}{840} & 0 & 0 & 0 & -\frac{\sqrt{154}}{168} & 0 & 0 & \frac{\sqrt{1155}}{1155} & 0 & 0 & 0 & \frac{\sqrt{385}}{1155} & 0 & 0 \\
0 & \frac{\sqrt{770}}{280} & 0 & 0 & 0 & -\frac{\sqrt{154}}{168} & 0 & 0 & -\frac{4\sqrt{77}}{1155} & 0 & 0 & 0 & \frac{2\sqrt{231}}{1155} & 0 \\
0 & 0 & -\frac{\sqrt{770}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{2310} & 0 & 0 & 0 & \frac{\sqrt{66}}{330} \\
0 & 0 & 0 & -\frac{\sqrt{770}}{420} & 0 & 0 & -\frac{\sqrt{66}}{330} & 0 & 0 & 0 & \frac{\sqrt{2310}}{2310} & 0 & 0 & 0 \\
-\frac{\sqrt{154}}{168} & 0 & 0 & 0 & \frac{\sqrt{770}}{280} & 0 & 0 & -\frac{2\sqrt{231}}{1155} & 0 & 0 & 0 & \frac{4\sqrt{77}}{1155} & 0 & 0 \\
0 & -\frac{\sqrt{154}}{168} & 0 & 0 & 0 & -\frac{\sqrt{770}}{840} & 0 & 0 & -\frac{\sqrt{385}}{1155} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{1155} & 0
\end{bmatrix}$

$\boxed{\text{x299}}$ $\mathbb{M}_4^{(1,1;a)}(A_{1u},2) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{22}}{42} & 0 & 0 & 0 & \frac{\sqrt{110}}{30} & 0 & 0 & \frac{\sqrt{55}}{140} & 0 & 0 & 0 & -\frac{\sqrt{165}}{300} & 0 \\
0 & 0 & \frac{\sqrt{33}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{84} & 0 & 0 & 0 & -\frac{\sqrt{385}}{300} \\
0 & 0 & 0 & -\frac{\sqrt{33}}{21} & 0 & 0 & -\frac{\sqrt{385}}{300} & 0 & 0 & 0 & -\frac{\sqrt{11}}{84} & 0 & 0 & 0 \\
-\frac{\sqrt{110}}{30} & 0 & 0 & 0 & \frac{\sqrt{22}}{42} & 0 & 0 & -\frac{\sqrt{165}}{300} & 0 & 0 & 0 & \frac{\sqrt{55}}{140} & 0 & 0 \\
-\frac{\sqrt{22}}{168} & 0 & 0 & 0 & \frac{\sqrt{110}}{120} & 0 & 0 & \frac{\sqrt{33}}{231} & 0 & 0 & 0 & -\frac{\sqrt{11}}{165} & 0 & 0 \\
0 & \frac{\sqrt{22}}{56} & 0 & 0 & 0 & \frac{\sqrt{110}}{120} & 0 & 0 & -\frac{4\sqrt{55}}{1155} & 0 & 0 & 0 & -\frac{2\sqrt{165}}{825} & 0 \\
0 & 0 & -\frac{\sqrt{22}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{462} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1650} \\
0 & 0 & 0 & -\frac{\sqrt{22}}{84} & 0 & 0 & \frac{\sqrt{2310}}{1650} & 0 & 0 & 0 & \frac{\sqrt{66}}{462} & 0 & 0 & 0 \\
\frac{\sqrt{110}}{120} & 0 & 0 & 0 & \frac{\sqrt{22}}{56} & 0 & 0 & \frac{2\sqrt{165}}{825} & 0 & 0 & 0 & \frac{4\sqrt{55}}{1155} & 0 & 0 \\
0 & \frac{\sqrt{110}}{120} & 0 & 0 & 0 & -\frac{\sqrt{22}}{168} & 0 & 0 & \frac{\sqrt{11}}{165} & 0 & 0 & 0 & -\frac{\sqrt{33}}{231} & 0
\end{bmatrix}$$

$\boxed{\text{x300}}$ $\mathbb{M}_4^{(1,1;a)}(A_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{350} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{150} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2310}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{231}i}{1155} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}i}{1925} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{275} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{275} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}i}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{231}i}{1155} & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x301}}$ $\mathbb{M}_2^{(1,1;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & -\frac{3\sqrt{70}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{280} & 0 & 0 & 0 \\
-\frac{4\sqrt{5}}{35} & 0 & 0 & 0 & \frac{8}{35} & 0 & 0 & -\frac{3\sqrt{30}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{280} & 0 & 0 \\
0 & -\frac{8}{35} & 0 & 0 & 0 & \frac{4\sqrt{5}}{35} & 0 & 0 & -\frac{3\sqrt{10}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{30}}{280} & 0 \\
0 & 0 & -\frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{280} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{280} \\
0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{105} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & \frac{\sqrt{70}}{105} & 0 & 0 & 0 & -\frac{4\sqrt{2}}{105} & 0 & 0 & 0 \\
\frac{3\sqrt{30}}{140} & 0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & \frac{4\sqrt{5}}{105} & 0 & 0 & 0 & -\frac{2\sqrt{15}}{105} & 0 & 0 \\
0 & \frac{9\sqrt{6}}{140} & 0 & 0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 & -\frac{4\sqrt{5}}{105} & 0 \\
0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{2}}{105} & 0 & 0 & 0 & -\frac{\sqrt{70}}{105} \\
0 & 0 & 0 & \frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{105} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x302}}$ $\mathbb{M}_4^{(1,1;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{33}}{21} & 0 & 0 & -\frac{\sqrt{385}}{700} & 0 & 0 & 0 & -\frac{3\sqrt{11}}{140} & 0 & 0 & 0 \\
-\frac{\sqrt{110}}{70} & 0 & 0 & 0 & -\frac{\sqrt{22}}{14} & 0 & 0 & \frac{11\sqrt{165}}{2100} & 0 & 0 & 0 & -\frac{\sqrt{55}}{700} & 0 & 0 \\
0 & \frac{\sqrt{22}}{14} & 0 & 0 & 0 & \frac{\sqrt{110}}{70} & 0 & 0 & -\frac{\sqrt{55}}{700} & 0 & 0 & 0 & \frac{11\sqrt{165}}{2100} & 0 \\
0 & 0 & -\frac{\sqrt{33}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{140} & 0 & 0 & 0 & -\frac{\sqrt{385}}{700} \\
0 & 0 & \frac{\sqrt{165}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{385} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & \frac{3\sqrt{385}}{1925} & 0 & 0 & 0 & -\frac{\sqrt{11}}{385} & 0 & 0 & 0 \\
\frac{\sqrt{165}}{140} & 0 & 0 & 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & -\frac{9\sqrt{110}}{3850} & 0 & 0 & 0 & \frac{17\sqrt{330}}{11550} & 0 & 0 \\
0 & -\frac{\sqrt{33}}{84} & 0 & 0 & 0 & \frac{\sqrt{165}}{140} & 0 & 0 & -\frac{17\sqrt{330}}{11550} & 0 & 0 & 0 & \frac{9\sqrt{110}}{3850} & 0 \\
0 & 0 & -\frac{\sqrt{33}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{385} & 0 & 0 & 0 & -\frac{3\sqrt{385}}{1925} \\
0 & 0 & 0 & \frac{\sqrt{165}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{385} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x303}}$ $\mathbb{M}_2^{(1,1;a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & -\frac{3\sqrt{70}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{280} & 0 & 0 & 0 \\
-\frac{4\sqrt{5}i}{35} & 0 & 0 & 0 & -\frac{8i}{35} & 0 & 0 & -\frac{3\sqrt{30}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{10}i}{280} & 0 & 0 \\
0 & -\frac{8i}{35} & 0 & 0 & 0 & -\frac{4\sqrt{5}i}{35} & 0 & 0 & -\frac{3\sqrt{10}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{30}i}{280} & 0 \\
0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{280} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{280} \\
0 & 0 & -\frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{105} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{9\sqrt{6}i}{140} & 0 & 0 & \frac{\sqrt{70}i}{105} & 0 & 0 & 0 & \frac{4\sqrt{2}i}{105} & 0 & 0 & 0 \\
\frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & -\frac{9\sqrt{6}i}{140} & 0 & 0 & \frac{4\sqrt{5}i}{105} & 0 & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 \\
0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{140} & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & \frac{4\sqrt{5}i}{105} & 0 \\
0 & 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{2}i}{105} & 0 & 0 & 0 & \frac{\sqrt{70}i}{105} \\
0 & 0 & 0 & \frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{105} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x304}}$ $\mathbb{M}_4^{(1,1;a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & \frac{\sqrt{33}i}{21} & 0 & 0 & \frac{\sqrt{385}i}{700} & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{140} & 0 & 0 & 0 \\
\frac{\sqrt{110}i}{70} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{14} & 0 & 0 & -\frac{11\sqrt{165}i}{2100} & 0 & 0 & 0 & -\frac{\sqrt{55}i}{700} & 0 & 0 \\
0 & -\frac{\sqrt{22}i}{14} & 0 & 0 & 0 & \frac{\sqrt{110}i}{70} & 0 & 0 & \frac{\sqrt{55}i}{700} & 0 & 0 & 0 & \frac{11\sqrt{165}i}{2100} & 0 \\
0 & 0 & \frac{\sqrt{33}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{700} \\
0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{385} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & -\frac{3\sqrt{385}i}{1925} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{385} & 0 & 0 & 0 \\
-\frac{\sqrt{165}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{84} & 0 & 0 & \frac{9\sqrt{110}i}{3850} & 0 & 0 & 0 & \frac{17\sqrt{330}i}{11550} & 0 & 0 \\
0 & \frac{\sqrt{33}i}{84} & 0 & 0 & 0 & \frac{\sqrt{165}i}{140} & 0 & 0 & \frac{17\sqrt{330}i}{11550} & 0 & 0 & 0 & \frac{9\sqrt{110}i}{3850} & 0 \\
0 & 0 & \frac{\sqrt{33}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{385} & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{1925} \\
0 & 0 & 0 & -\frac{\sqrt{165}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{385} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x305}}$ $\mathbb{M}_{2,1}^{(1,1;a)}(E_u) = \begin{bmatrix} -\frac{2\sqrt{15}i}{35} & 0 & -\frac{3\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 \\ 0 & \frac{2i}{35} & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{6}i}{140} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{7} & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{10}i}{140} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{6}i}{35} & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{140} & 0 & \frac{3\sqrt{10}i}{140} & 0 \\ 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{42} & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{15}i}{70} & 0 & -\frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{210} & 0 & \frac{11\sqrt{2}i}{210} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{105} & 0 & \frac{i}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & -\frac{i}{15} & 0 & \frac{\sqrt{15}i}{105} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{6}i}{70} & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{2}i}{210} & 0 & -\frac{\sqrt{10}i}{210} & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & -\frac{\sqrt{14}i}{42} \end{bmatrix}$

$\boxed{\text{x306}}$ $\mathbb{M}_{2,2}^{(1,1;a)}(E_u) = \begin{bmatrix} \frac{2\sqrt{15}}{35} & 0 & -\frac{3\sqrt{6}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & \frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2}{35} & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & \frac{3\sqrt{6}}{140} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{140} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{6}}{35} & 0 & \frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & \frac{3\sqrt{10}}{140} & 0 \\ 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{42} & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{15}}{70} & 0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{210} & 0 & \frac{11\sqrt{2}}{210} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{105} & 0 & \frac{1}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & \frac{1}{15} & 0 & \frac{\sqrt{15}}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{6}}{70} & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & \frac{11\sqrt{2}}{210} & 0 & -\frac{\sqrt{10}}{210} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & -\frac{\sqrt{14}}{42} \end{bmatrix}$

$\boxed{\text{x307}}$ $\mathbb{M}_{4,1}^{(1,1;a)}(E_u,1) =$

$$\begin{bmatrix}
-\frac{\sqrt{2310}i}{840} & 0 & -\frac{\sqrt{231}i}{84} & 0 & -\frac{\sqrt{462}i}{168} & 0 & 0 & \frac{3\sqrt{385}i}{1400} & 0 & \frac{\sqrt{77}i}{140} & 0 & \frac{\sqrt{1155}i}{1400} & 0 & 0 \\
0 & \frac{\sqrt{154}i}{56} & 0 & \frac{\sqrt{77}i}{28} & 0 & \frac{\sqrt{770}i}{280} & -\frac{\sqrt{165}i}{600} & 0 & -\frac{\sqrt{385}i}{350} & 0 & -\frac{\sqrt{231}i}{840} & 0 & \frac{\sqrt{1155}i}{2100} & 0 \\
-\frac{\sqrt{770}i}{280} & 0 & -\frac{\sqrt{77}i}{28} & 0 & -\frac{\sqrt{154}i}{56} & 0 & 0 & \frac{\sqrt{1155}i}{2100} & 0 & -\frac{\sqrt{231}i}{840} & 0 & -\frac{\sqrt{385}i}{350} & 0 & -\frac{\sqrt{165}i}{600} \\
0 & \frac{\sqrt{462}i}{168} & 0 & \frac{\sqrt{231}i}{84} & 0 & \frac{\sqrt{2310}i}{840} & 0 & 0 & \frac{\sqrt{1155}i}{1400} & 0 & \frac{\sqrt{77}i}{140} & 0 & \frac{3\sqrt{385}i}{1400} & 0 \\
0 & -\frac{\sqrt{2310}i}{840} & 0 & -\frac{\sqrt{1155}i}{840} & 0 & 0 & \frac{\sqrt{11}i}{220} & 0 & \frac{\sqrt{231}i}{462} & 0 & \frac{\sqrt{385}i}{1540} & 0 & 0 & 0 \\
\frac{\sqrt{2310}i}{840} & 0 & \frac{\sqrt{231}i}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{385}i}{7700} & 0 & -\frac{\sqrt{77}i}{770} & 0 & \frac{\sqrt{1155}i}{3300} & 0 & 0 \\
0 & -\frac{\sqrt{231}i}{168} & 0 & 0 & 0 & \frac{\sqrt{1155}i}{840} & \frac{3\sqrt{110}i}{2200} & 0 & \frac{\sqrt{2310}i}{46200} & 0 & -\frac{\sqrt{154}i}{440} & 0 & -\frac{\sqrt{770}i}{15400} & 0 \\
\frac{\sqrt{1155}i}{840} & 0 & 0 & 0 & -\frac{\sqrt{231}i}{168} & 0 & 0 & \frac{\sqrt{770}i}{15400} & 0 & \frac{\sqrt{154}i}{440} & 0 & -\frac{\sqrt{2310}i}{46200} & 0 & -\frac{3\sqrt{110}i}{2200} \\
0 & 0 & 0 & \frac{\sqrt{231}i}{168} & 0 & \frac{\sqrt{2310}i}{840} & 0 & 0 & -\frac{\sqrt{1155}i}{3300} & 0 & \frac{\sqrt{77}i}{770} & 0 & \frac{13\sqrt{385}i}{7700} & 0 \\
0 & 0 & -\frac{\sqrt{1155}i}{840} & 0 & -\frac{\sqrt{2310}i}{840} & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{1540} & 0 & -\frac{\sqrt{231}i}{462} & 0 & -\frac{\sqrt{11}i}{220}
\end{bmatrix}$$

$\boxed{\text{x308}}$ $\mathbb{M}_{4,2}^{(1,1;a)}(E_u,1) =$

$$\begin{bmatrix}
\frac{\sqrt{2310}}{840} & 0 & -\frac{\sqrt{231}}{84} & 0 & \frac{\sqrt{462}}{168} & 0 & 0 & -\frac{3\sqrt{385}}{1400} & 0 & \frac{\sqrt{77}}{140} & 0 & -\frac{\sqrt{1155}}{1400} & 0 & 0 \\
0 & -\frac{\sqrt{154}}{56} & 0 & \frac{\sqrt{77}}{28} & 0 & -\frac{\sqrt{770}}{280} & -\frac{\sqrt{165}}{600} & 0 & \frac{\sqrt{385}}{350} & 0 & -\frac{\sqrt{231}}{840} & 0 & -\frac{\sqrt{1155}}{2100} & 0 \\
-\frac{\sqrt{770}}{280} & 0 & \frac{\sqrt{77}}{28} & 0 & -\frac{\sqrt{154}}{56} & 0 & 0 & \frac{\sqrt{1155}}{2100} & 0 & \frac{\sqrt{231}}{840} & 0 & -\frac{\sqrt{385}}{350} & 0 & \frac{\sqrt{165}}{600} \\
0 & \frac{\sqrt{462}}{168} & 0 & -\frac{\sqrt{231}}{84} & 0 & \frac{\sqrt{2310}}{840} & 0 & 0 & \frac{\sqrt{1155}}{1400} & 0 & -\frac{\sqrt{77}}{140} & 0 & \frac{3\sqrt{385}}{1400} & 0 \\
0 & -\frac{\sqrt{2310}}{840} & 0 & \frac{\sqrt{1155}}{840} & 0 & 0 & -\frac{\sqrt{11}}{220} & 0 & \frac{\sqrt{231}}{462} & 0 & -\frac{\sqrt{385}}{1540} & 0 & 0 & 0 \\
-\frac{\sqrt{2310}}{840} & 0 & \frac{\sqrt{231}}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{385}}{7700} & 0 & -\frac{\sqrt{77}}{770} & 0 & -\frac{\sqrt{1155}}{3300} & 0 & 0 \\
0 & \frac{\sqrt{231}}{168} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{840} & \frac{3\sqrt{110}}{2200} & 0 & -\frac{\sqrt{2310}}{46200} & 0 & -\frac{\sqrt{154}}{440} & 0 & \frac{\sqrt{770}}{15400} & 0 \\
\frac{\sqrt{1155}}{840} & 0 & 0 & 0 & -\frac{\sqrt{231}}{168} & 0 & 0 & \frac{\sqrt{770}}{15400} & 0 & -\frac{\sqrt{154}}{440} & 0 & -\frac{\sqrt{2310}}{46200} & 0 & \frac{3\sqrt{110}}{2200} \\
0 & 0 & 0 & -\frac{\sqrt{231}}{168} & 0 & \frac{\sqrt{2310}}{840} & 0 & 0 & -\frac{\sqrt{1155}}{3300} & 0 & -\frac{\sqrt{77}}{770} & 0 & \frac{13\sqrt{385}}{7700} & 0 \\
0 & 0 & -\frac{\sqrt{1155}}{840} & 0 & \frac{\sqrt{2310}}{840} & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{1540} & 0 & \frac{\sqrt{231}}{462} & 0 & -\frac{\sqrt{11}}{220}
\end{bmatrix}$$

$\boxed{\text{x309}}$ $\mathbb{M}_{4,1}^{(1,1;a)}(E_u,2) =$

$$\begin{bmatrix}
-\frac{\sqrt{330}i}{840} & 0 & -\frac{\sqrt{33}i}{84} & 0 & \frac{\sqrt{66}i}{24} & 0 & 0 & \frac{3\sqrt{55}i}{1400} & 0 & \frac{\sqrt{11}i}{140} & 0 & -\frac{\sqrt{165}i}{200} & 0 & 0 \\
0 & \frac{\sqrt{22}i}{56} & 0 & \frac{\sqrt{11}i}{28} & 0 & -\frac{\sqrt{110}i}{40} & \frac{\sqrt{1155}i}{600} & 0 & -\frac{\sqrt{55}i}{350} & 0 & -\frac{\sqrt{33}i}{840} & 0 & -\frac{\sqrt{165}i}{300} & 0 \\
\frac{\sqrt{110}i}{40} & 0 & -\frac{\sqrt{11}i}{28} & 0 & -\frac{\sqrt{22}i}{56} & 0 & 0 & -\frac{\sqrt{165}i}{300} & 0 & -\frac{\sqrt{33}i}{840} & 0 & -\frac{\sqrt{55}i}{350} & 0 & \frac{\sqrt{1155}i}{600} \\
0 & -\frac{\sqrt{66}i}{24} & 0 & \frac{\sqrt{33}i}{84} & 0 & \frac{\sqrt{330}i}{840} & 0 & 0 & -\frac{\sqrt{165}i}{200} & 0 & \frac{\sqrt{11}i}{140} & 0 & \frac{3\sqrt{55}i}{1400} & 0 \\
0 & -\frac{\sqrt{330}i}{840} & 0 & \frac{\sqrt{165}i}{120} & 0 & 0 & \frac{\sqrt{77}i}{1540} & 0 & \frac{\sqrt{33}i}{462} & 0 & -\frac{\sqrt{55}i}{220} & 0 & 0 & 0 \\
\frac{\sqrt{330}i}{840} & 0 & \frac{\sqrt{33}i}{168} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{55}i}{7700} & 0 & -\frac{\sqrt{11}i}{770} & 0 & -\frac{7\sqrt{165}i}{3300} & 0 & 0 \\
0 & -\frac{\sqrt{33}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{165}i}{120} & -\frac{3\sqrt{770}i}{2200} & 0 & \frac{\sqrt{330}i}{46200} & 0 & -\frac{\sqrt{22}i}{440} & 0 & \frac{\sqrt{110}i}{2200} & 0 \\
-\frac{\sqrt{165}i}{120} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{168} & 0 & 0 & -\frac{\sqrt{110}i}{2200} & 0 & \frac{\sqrt{22}i}{440} & 0 & -\frac{\sqrt{330}i}{46200} & 0 & \frac{3\sqrt{770}i}{2200} \\
0 & 0 & 0 & \frac{\sqrt{33}i}{168} & 0 & \frac{\sqrt{330}i}{840} & 0 & 0 & \frac{7\sqrt{165}i}{3300} & 0 & \frac{\sqrt{11}i}{770} & 0 & \frac{13\sqrt{55}i}{7700} & 0 \\
0 & 0 & \frac{\sqrt{165}i}{120} & 0 & -\frac{\sqrt{330}i}{840} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{220} & 0 & -\frac{\sqrt{33}i}{462} & 0 & -\frac{\sqrt{77}i}{1540}
\end{bmatrix}$$

$\boxed{\text{x310}}$ $\mathbb{M}_{4,2}^{(1,1;a)}(E_u,2) =$

$$\begin{bmatrix}
\frac{\sqrt{330}}{840} & 0 & -\frac{\sqrt{33}}{84} & 0 & -\frac{\sqrt{66}}{24} & 0 & 0 & -\frac{3\sqrt{55}}{1400} & 0 & \frac{\sqrt{11}}{140} & 0 & \frac{\sqrt{165}}{200} & 0 & 0 \\
0 & -\frac{\sqrt{22}}{56} & 0 & \frac{\sqrt{11}}{28} & 0 & \frac{\sqrt{110}}{40} & \frac{\sqrt{1155}}{600} & 0 & \frac{\sqrt{55}}{350} & 0 & -\frac{\sqrt{33}}{840} & 0 & \frac{\sqrt{165}}{300} & 0 \\
\frac{\sqrt{110}}{40} & 0 & \frac{\sqrt{11}}{28} & 0 & -\frac{\sqrt{22}}{56} & 0 & 0 & -\frac{\sqrt{165}}{300} & 0 & \frac{\sqrt{33}}{840} & 0 & -\frac{\sqrt{55}}{350} & 0 & -\frac{\sqrt{1155}}{600} \\
0 & -\frac{\sqrt{66}}{24} & 0 & -\frac{\sqrt{33}}{84} & 0 & \frac{\sqrt{330}}{840} & 0 & 0 & -\frac{\sqrt{165}}{200} & 0 & -\frac{\sqrt{11}}{140} & 0 & \frac{3\sqrt{55}}{1400} & 0 \\
0 & -\frac{\sqrt{330}}{840} & 0 & -\frac{\sqrt{165}}{120} & 0 & 0 & -\frac{\sqrt{77}}{1540} & 0 & \frac{\sqrt{33}}{462} & 0 & \frac{\sqrt{55}}{220} & 0 & 0 & 0 \\
-\frac{\sqrt{330}}{840} & 0 & \frac{\sqrt{33}}{168} & 0 & 0 & 0 & 0 & \frac{13\sqrt{55}}{7700} & 0 & -\frac{\sqrt{11}}{770} & 0 & \frac{7\sqrt{165}}{3300} & 0 & 0 \\
0 & \frac{\sqrt{33}}{168} & 0 & 0 & 0 & \frac{\sqrt{165}}{120} & -\frac{3\sqrt{770}}{2200} & 0 & -\frac{\sqrt{330}}{46200} & 0 & -\frac{\sqrt{22}}{440} & 0 & -\frac{\sqrt{110}}{2200} & 0 \\
-\frac{\sqrt{165}}{120} & 0 & 0 & 0 & -\frac{\sqrt{33}}{168} & 0 & 0 & -\frac{\sqrt{110}}{2200} & 0 & -\frac{\sqrt{22}}{440} & 0 & -\frac{\sqrt{330}}{46200} & 0 & -\frac{3\sqrt{770}}{2200} \\
0 & 0 & 0 & -\frac{\sqrt{33}}{168} & 0 & \frac{\sqrt{330}}{840} & 0 & 0 & \frac{7\sqrt{165}}{3300} & 0 & -\frac{\sqrt{11}}{770} & 0 & \frac{13\sqrt{55}}{7700} & 0 \\
0 & 0 & \frac{\sqrt{165}}{120} & 0 & \frac{\sqrt{330}}{840} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{220} & 0 & \frac{\sqrt{33}}{462} & 0 & -\frac{\sqrt{77}}{1540}
\end{bmatrix}$$

$\boxed{\text{x311}}$ $\mathbb{T}_5^{(a)}(A_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x312}}$ $\mathbb{T}_1^{(a)}(A_{2u}) = \begin{bmatrix} 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 \end{bmatrix}$

$\boxed{\text{x313}}$ $\mathbb{T}_3^{(a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{3\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 \\ \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 \end{bmatrix}$

$\boxed{\text{x314}}$ $\mathbb{T}_5^{(a)}(A_{2u}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 \\ -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 \end{bmatrix}$

$\boxed{\text{x315}}$ $\mathbb{T}_5^{(a)}(A_{2u},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x316}}$ $\mathbb{T}_3^{(a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 \\ 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 \\ 0 & 0 & \frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & 0 & \frac{\sqrt{6}}{24} \\ 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{140} & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 \\ 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x317}}$ $\mathbb{T}_5^{(a)}(B_{1u}) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{60} & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 & 0 & \frac{\sqrt{21}}{20} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{20} & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{60} & 0 & 0 & 0 & \frac{\sqrt{3}}{60} \\
0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & -\frac{\sqrt{3}}{30} & 0 & 0 & 0 & \frac{\sqrt{105}}{105} & 0 & 0 & 0 \\
-\frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & \frac{2\sqrt{42}}{105} & 0 & 0 & 0 & \frac{\sqrt{14}}{70} & 0 & 0 \\
0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{\sqrt{14}}{70} & 0 & 0 & 0 & -\frac{2\sqrt{42}}{105} & 0 \\
0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{105} & 0 & 0 & 0 & \frac{\sqrt{3}}{30} \\
0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x318}}$ $\mathbb{T}_3^{(a)}(B_{2u}) = \begin{bmatrix}
0 & 0 & 0 & -\frac{3\sqrt{70}i}{140} & 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 \\
\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 \\
0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 \\
0 & 0 & -\frac{3\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} \\
0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 \\
\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 \\
0 & \frac{\sqrt{70}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 \\
0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\
0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x319}}$ $\mathbb{T}_5^{(a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{60} & 0 & 0 & 0 & \frac{\sqrt{105}i}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{20} & 0 & 0 & 0 & \frac{\sqrt{7}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{60} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{60} \\ 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & -\frac{\sqrt{3}i}{30} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{105} & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & \frac{2\sqrt{42}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{70} & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{\sqrt{14}i}{70} & 0 & 0 & 0 & \frac{2\sqrt{42}i}{105} & 0 \\ 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{105} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{30} \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x320}}$ $\mathbb{T}_{1,1}^{(a)}(E_u) = \begin{bmatrix} -\frac{\sqrt{5}i}{10} & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{10} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{70} & 0 & -\frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{35} & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3i}{70} & 0 & -\frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{35} & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{30}i}{28} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & \frac{\sqrt{42}i}{28} \end{bmatrix}$

$\boxed{\text{x321}}$ $\mathbb{T}^{(a)}_{1,2}(E_u) = \begin{bmatrix} \frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{10} & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{20} & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{70} & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{35} & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{3}{70} & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{35} & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{30}}{28} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & \frac{\sqrt{42}}{28} \end{bmatrix}$

$\boxed{\text{x322}}$ $\mathbb{T}^{(a)}_{3,1}(E_u, 1) = \begin{bmatrix} -\frac{3\sqrt{105}i}{560} & 0 & \frac{9\sqrt{42}i}{560} & 0 & -\frac{3\sqrt{21}i}{112} & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{210}i}{336} & 0 & 0 \\ 0 & \frac{3\sqrt{7}i}{80} & 0 & -\frac{3\sqrt{14}i}{560} & 0 & -\frac{3\sqrt{35}i}{112} & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{210}i}{168} & 0 \\ \frac{3\sqrt{35}i}{112} & 0 & \frac{3\sqrt{14}i}{560} & 0 & -\frac{3\sqrt{7}i}{80} & 0 & 0 & -\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} \\ 0 & \frac{3\sqrt{21}i}{112} & 0 & -\frac{9\sqrt{42}i}{560} & 0 & \frac{3\sqrt{105}i}{560} & 0 & 0 & -\frac{\sqrt{210}i}{336} & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 \\ 0 & -\frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 & -\frac{\sqrt{2}i}{16} & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 \\ -\frac{\sqrt{105}i}{140} & 0 & \frac{\sqrt{42}i}{280} & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{280} & 0 & \frac{\sqrt{21}i}{70} & 0 & \frac{\sqrt{210}i}{168} & \frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{7}i}{112} & 0 & -\frac{3\sqrt{35}i}{112} & 0 \\ \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{21}i}{70} & 0 & \frac{\sqrt{42}i}{280} & 0 & 0 & \frac{3\sqrt{35}i}{112} & 0 & \frac{\sqrt{7}i}{112} & 0 & -\frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{5}i}{16} \\ 0 & \frac{\sqrt{21}i}{42} & 0 & \frac{\sqrt{42}i}{280} & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{2}i}{16} \end{bmatrix}$

$\boxed{\text{x323}}$ $\mathbb{T}^{(a)}_{3,2}(E_u,1) = \begin{bmatrix} \frac{3\sqrt{105}}{560} & 0 & \frac{9\sqrt{42}}{560} & 0 & \frac{3\sqrt{21}}{112} & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{210}}{336} & 0 & 0 \\ 0 & -\frac{3\sqrt{7}}{80} & 0 & -\frac{3\sqrt{14}}{560} & 0 & \frac{3\sqrt{35}}{112} & -\frac{\sqrt{30}}{48} & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{210}}{168} & 0 \\ \frac{3\sqrt{35}}{112} & 0 & -\frac{3\sqrt{14}}{560} & 0 & -\frac{3\sqrt{7}}{80} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & \frac{\sqrt{30}}{48} \\ 0 & \frac{3\sqrt{21}}{112} & 0 & \frac{9\sqrt{42}}{560} & 0 & \frac{3\sqrt{105}}{560} & 0 & 0 & -\frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 \\ 0 & -\frac{\sqrt{105}}{140} & 0 & -\frac{\sqrt{210}}{168} & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 \\ \frac{\sqrt{105}}{140} & 0 & \frac{\sqrt{42}}{280} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & -\frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{210}}{112} & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{280} & 0 & \frac{\sqrt{21}}{70} & 0 & -\frac{\sqrt{210}}{168} & \frac{\sqrt{5}}{16} & 0 & -\frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{7}}{112} & 0 & \frac{3\sqrt{35}}{112} & 0 \\ \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{21}}{70} & 0 & \frac{\sqrt{42}}{280} & 0 & 0 & \frac{3\sqrt{35}}{112} & 0 & -\frac{\sqrt{7}}{112} & 0 & -\frac{\sqrt{105}}{112} & 0 & \frac{\sqrt{5}}{16} \\ 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{42}}{280} & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 & \frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{2}}{16} \end{bmatrix}$

$\boxed{\text{x324}}$ $\mathbb{T}^{(a)}_{3,1}(E_u,2) = \begin{bmatrix} -\frac{3\sqrt{7}i}{112} & 0 & \frac{9\sqrt{70}i}{560} & 0 & \frac{9\sqrt{35}i}{560} & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{80} & 0 & -\frac{\sqrt{210}i}{560} & 0 & \frac{3\sqrt{21}i}{112} & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{14}i}{56} & 0 \\ -\frac{3\sqrt{21}i}{112} & 0 & \frac{\sqrt{210}i}{560} & 0 & -\frac{\sqrt{105}i}{80} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} \\ 0 & -\frac{9\sqrt{35}i}{560} & 0 & -\frac{9\sqrt{70}i}{560} & 0 & \frac{3\sqrt{7}i}{112} & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{210}i}{168} & 0 & -\frac{5\sqrt{42}i}{336} & 0 \\ 0 & -\frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & \frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{28} & 0 & \frac{\sqrt{70}i}{280} & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & \frac{5\sqrt{42}i}{336} & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{3\sqrt{14}i}{112} & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{280} & 0 & \frac{\sqrt{35}i}{70} & 0 & -\frac{\sqrt{14}i}{56} & -\frac{\sqrt{3}i}{16} & 0 & \frac{5\sqrt{7}i}{112} & 0 & -\frac{\sqrt{105}i}{336} & 0 & \frac{3\sqrt{21}i}{112} & 0 \\ -\frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{70}i}{280} & 0 & 0 & -\frac{3\sqrt{21}i}{112} & 0 & \frac{\sqrt{105}i}{336} & 0 & -\frac{5\sqrt{7}i}{112} & 0 & \frac{\sqrt{3}i}{16} \\ 0 & -\frac{\sqrt{35}i}{70} & 0 & \frac{\sqrt{70}i}{280} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & -\frac{3\sqrt{14}i}{112} & 0 & -\frac{\sqrt{210}i}{168} & 0 & -\frac{5\sqrt{42}i}{336} & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{30}i}{48} \end{bmatrix}$

$\boxed{\text{x325}}$ $\mathbb{T}^{(a)}_{3,2}(E_u,2) =$

$$\begin{bmatrix}
\frac{3\sqrt{7}}{112} & 0 & \frac{9\sqrt{70}}{560} & 0 & -\frac{9\sqrt{35}}{560} & 0 & 0 & \frac{5\sqrt{42}}{336} & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{14}}{112} & 0 & 0 \\
0 & -\frac{\sqrt{105}}{80} & 0 & -\frac{\sqrt{210}}{560} & 0 & -\frac{3\sqrt{21}}{112} & \frac{\sqrt{2}}{16} & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{14}}{56} & 0 \\
-\frac{3\sqrt{21}}{112} & 0 & -\frac{\sqrt{210}}{560} & 0 & -\frac{\sqrt{105}}{80} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} \\
0 & -\frac{9\sqrt{35}}{560} & 0 & \frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{7}}{112} & 0 & 0 & \frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{210}}{168} & 0 & -\frac{5\sqrt{42}}{336} & 0 \\
0 & -\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & \frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 \\
\frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{70}}{280} & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{3\sqrt{14}}{112} & 0 & 0 \\
0 & -\frac{\sqrt{70}}{280} & 0 & \frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{14}}{56} & -\frac{\sqrt{3}}{16} & 0 & -\frac{5\sqrt{7}}{112} & 0 & -\frac{\sqrt{105}}{336} & 0 & -\frac{3\sqrt{21}}{112} & 0 \\
-\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{35}}{70} & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & -\frac{3\sqrt{21}}{112} & 0 & -\frac{\sqrt{105}}{336} & 0 & -\frac{5\sqrt{7}}{112} & 0 & -\frac{\sqrt{3}}{16} \\
0 & -\frac{\sqrt{35}}{70} & 0 & -\frac{\sqrt{70}}{280} & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & -\frac{3\sqrt{14}}{112} & 0 & \frac{\sqrt{210}}{168} & 0 & -\frac{5\sqrt{42}}{336} & 0 \\
0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{30}}{48}
\end{bmatrix}$$

$\boxed{\text{x326}}$ $\mathbb{T}^{(a)}_{5,1}(E_u,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{80} & 0 & \frac{i}{16} & 0 & -\frac{7\sqrt{15}i}{240} & 0 & \frac{3\sqrt{35}i}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{240} & 0 & \frac{3\sqrt{5}i}{80} & 0 & -\frac{\sqrt{3}i}{16} & 0 & \frac{7\sqrt{15}i}{240} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{15}i}{240} & 0 & -\frac{\sqrt{3}i}{16} & 0 & \frac{3\sqrt{5}i}{80} & 0 & -\frac{\sqrt{105}i}{240} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{80} & 0 & -\frac{7\sqrt{15}i}{240} & 0 & \frac{i}{16} & 0 & -\frac{\sqrt{5}i}{80} & 0 \\
0 & -\frac{\sqrt{30}i}{224} & 0 & \frac{\sqrt{15}i}{48} & 0 & -\frac{3\sqrt{6}i}{32} & -\frac{\sqrt{7}i}{224} & 0 & \frac{5\sqrt{3}i}{224} & 0 & -\frac{\sqrt{5}i}{32} & 0 & \frac{3i}{32} & 0 \\
-\frac{\sqrt{30}i}{224} & 0 & \frac{5\sqrt{3}i}{112} & 0 & -\frac{5\sqrt{6}i}{96} & 0 & 0 & \frac{23\sqrt{5}i}{1120} & 0 & -\frac{13i}{224} & 0 & \frac{\sqrt{15}i}{160} & 0 & \frac{3\sqrt{35}i}{160} \\
0 & \frac{5\sqrt{3}i}{112} & 0 & -\frac{5\sqrt{6}i}{112} & 0 & \frac{\sqrt{15}i}{48} & \frac{\sqrt{70}i}{160} & 0 & -\frac{11\sqrt{30}i}{1120} & 0 & \frac{\sqrt{2}i}{224} & 0 & \frac{3\sqrt{10}i}{160} & 0 \\
\frac{\sqrt{15}i}{48} & 0 & -\frac{5\sqrt{6}i}{112} & 0 & \frac{5\sqrt{3}i}{112} & 0 & 0 & -\frac{3\sqrt{10}i}{160} & 0 & -\frac{\sqrt{2}i}{224} & 0 & \frac{11\sqrt{30}i}{1120} & 0 & -\frac{\sqrt{70}i}{160} \\
0 & -\frac{5\sqrt{6}i}{96} & 0 & \frac{5\sqrt{3}i}{112} & 0 & -\frac{\sqrt{30}i}{224} & -\frac{3\sqrt{35}i}{160} & 0 & -\frac{\sqrt{15}i}{160} & 0 & \frac{13i}{224} & 0 & -\frac{23\sqrt{5}i}{1120} & 0 \\
-\frac{3\sqrt{6}i}{32} & 0 & \frac{\sqrt{15}i}{48} & 0 & -\frac{\sqrt{30}i}{224} & 0 & 0 & -\frac{3i}{32} & 0 & \frac{\sqrt{5}i}{32} & 0 & -\frac{5\sqrt{3}i}{224} & 0 & \frac{\sqrt{7}i}{224}
\end{bmatrix}$$

$\boxed{\text{x327}}$ $\mathbb{T}^{(a)}_{5,2}(E_u,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{80} & 0 & \frac{1}{16} & 0 & \frac{7\sqrt{15}}{240} & 0 & \frac{3\sqrt{35}}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{240} & 0 & -\frac{3\sqrt{5}}{80} & 0 & -\frac{\sqrt{3}}{16} & 0 & -\frac{7\sqrt{15}}{240} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{15}}{240} & 0 & \frac{\sqrt{3}}{16} & 0 & \frac{3\sqrt{5}}{80} & 0 & \frac{\sqrt{105}}{240} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{80} & 0 & -\frac{7\sqrt{15}}{240} & 0 & -\frac{1}{16} & 0 & -\frac{\sqrt{5}}{80} & 0 \\
0 & -\frac{\sqrt{30}}{224} & 0 & -\frac{\sqrt{15}}{48} & 0 & -\frac{3\sqrt{6}}{32} & \frac{\sqrt{7}}{224} & 0 & \frac{5\sqrt{3}}{224} & 0 & \frac{\sqrt{5}}{32} & 0 & \frac{3}{32} & 0 \\
\frac{\sqrt{30}}{224} & 0 & \frac{5\sqrt{3}}{112} & 0 & \frac{5\sqrt{6}}{96} & 0 & 0 & -\frac{23\sqrt{5}}{1120} & 0 & -\frac{13}{224} & 0 & -\frac{\sqrt{15}}{160} & 0 & \frac{3\sqrt{35}}{160} \\
0 & -\frac{5\sqrt{3}}{112} & 0 & -\frac{5\sqrt{6}}{112} & 0 & -\frac{\sqrt{15}}{48} & \frac{\sqrt{70}}{160} & 0 & \frac{11\sqrt{30}}{1120} & 0 & \frac{\sqrt{2}}{224} & 0 & -\frac{3\sqrt{10}}{160} & 0 \\
\frac{\sqrt{15}}{48} & 0 & \frac{5\sqrt{6}}{112} & 0 & \frac{5\sqrt{3}}{112} & 0 & 0 & -\frac{3\sqrt{10}}{160} & 0 & \frac{\sqrt{2}}{224} & 0 & \frac{11\sqrt{30}}{1120} & 0 & \frac{\sqrt{70}}{160} \\
0 & -\frac{5\sqrt{6}}{96} & 0 & -\frac{5\sqrt{3}}{112} & 0 & -\frac{\sqrt{30}}{224} & \frac{3\sqrt{35}}{160} & 0 & -\frac{\sqrt{15}}{160} & 0 & -\frac{13}{224} & 0 & -\frac{23\sqrt{5}}{1120} & 0 \\
\frac{3\sqrt{6}}{32} & 0 & \frac{\sqrt{15}}{48} & 0 & \frac{\sqrt{30}}{224} & 0 & 0 & \frac{3}{32} & 0 & \frac{\sqrt{5}}{32} & 0 & \frac{5\sqrt{3}}{224} & 0 & \frac{\sqrt{7}}{224}
\end{bmatrix}$$

$\boxed{\text{x328}}$ $\mathbb{T}^{(a)}_{5,1}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{80} & 0 & \frac{\sqrt{35}i}{80} & 0 & \frac{3\sqrt{21}i}{80} & 0 & \frac{i}{16} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}i}{80} & 0 & \frac{3\sqrt{7}i}{80} & 0 & -\frac{\sqrt{105}i}{80} & 0 & -\frac{3\sqrt{21}i}{80} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{80} & 0 & -\frac{\sqrt{105}i}{80} & 0 & \frac{3\sqrt{7}i}{80} & 0 & \frac{3\sqrt{3}i}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{16} & 0 & \frac{3\sqrt{21}i}{80} & 0 & \frac{\sqrt{35}i}{80} & 0 & -\frac{\sqrt{7}i}{80} & 0 \\
0 & -\frac{\sqrt{42}i}{224} & 0 & -\frac{3\sqrt{21}i}{112} & 0 & -\frac{\sqrt{210}i}{224} & -\frac{\sqrt{5}i}{160} & 0 & \frac{\sqrt{105}i}{224} & 0 & \frac{9\sqrt{7}i}{224} & 0 & \frac{\sqrt{35}i}{224} & 0 \\
-\frac{\sqrt{42}i}{224} & 0 & \frac{\sqrt{105}i}{112} & 0 & \frac{3\sqrt{210}i}{224} & 0 & 0 & \frac{23\sqrt{7}i}{1120} & 0 & -\frac{13\sqrt{35}i}{1120} & 0 & -\frac{9\sqrt{21}i}{1120} & 0 & \frac{i}{32} \\
0 & \frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{3\sqrt{21}i}{112} & -\frac{9\sqrt{2}i}{160} & 0 & -\frac{11\sqrt{42}i}{1120} & 0 & \frac{\sqrt{70}i}{1120} & 0 & -\frac{27\sqrt{14}i}{1120} & 0 \\
-\frac{3\sqrt{21}i}{112} & 0 & -\frac{\sqrt{210}i}{112} & 0 & \frac{\sqrt{105}i}{112} & 0 & 0 & \frac{27\sqrt{14}i}{1120} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & \frac{11\sqrt{42}i}{1120} & 0 & \frac{9\sqrt{2}i}{160} \\
0 & \frac{3\sqrt{210}i}{224} & 0 & \frac{\sqrt{105}i}{112} & 0 & -\frac{\sqrt{42}i}{224} & -\frac{i}{32} & 0 & \frac{9\sqrt{21}i}{1120} & 0 & \frac{13\sqrt{35}i}{1120} & 0 & -\frac{23\sqrt{7}i}{1120} & 0 \\
-\frac{\sqrt{210}i}{224} & 0 & -\frac{3\sqrt{21}i}{112} & 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 & -\frac{\sqrt{35}i}{224} & 0 & -\frac{9\sqrt{7}i}{224} & 0 & -\frac{\sqrt{105}i}{224} & 0 & \frac{\sqrt{5}i}{160}
\end{bmatrix}$$

$\boxed{\text{x329}}$ $\mathbb{T}^{(a)}_{5,2}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{80} & 0 & \frac{\sqrt{35}}{80} & 0 & -\frac{3\sqrt{21}}{80} & 0 & \frac{1}{16} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}}{80} & 0 & -\frac{3\sqrt{7}}{80} & 0 & -\frac{\sqrt{105}}{80} & 0 & \frac{3\sqrt{21}}{80} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{80} & 0 & \frac{\sqrt{105}}{80} & 0 & \frac{3\sqrt{7}}{80} & 0 & -\frac{3\sqrt{3}}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{16} & 0 & \frac{3\sqrt{21}}{80} & 0 & -\frac{\sqrt{35}}{80} & 0 & -\frac{\sqrt{7}}{80} & 0 \\
0 & -\frac{\sqrt{42}}{224} & 0 & \frac{3\sqrt{21}}{112} & 0 & -\frac{\sqrt{210}}{224} & \frac{\sqrt{5}}{160} & 0 & \frac{\sqrt{105}}{224} & 0 & -\frac{9\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{224} & 0 \\
\frac{\sqrt{42}}{224} & 0 & \frac{\sqrt{105}}{112} & 0 & -\frac{3\sqrt{210}}{224} & 0 & 0 & -\frac{23\sqrt{7}}{1120} & 0 & -\frac{13\sqrt{35}}{1120} & 0 & \frac{9\sqrt{21}}{1120} & 0 & \frac{1}{32} \\
0 & -\frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{210}}{112} & 0 & \frac{3\sqrt{21}}{112} & -\frac{9\sqrt{2}}{160} & 0 & \frac{11\sqrt{42}}{1120} & 0 & \frac{\sqrt{70}}{1120} & 0 & \frac{27\sqrt{14}}{1120} & 0 \\
-\frac{3\sqrt{21}}{112} & 0 & \frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{105}}{112} & 0 & 0 & \frac{27\sqrt{14}}{1120} & 0 & \frac{\sqrt{70}}{1120} & 0 & \frac{11\sqrt{42}}{1120} & 0 & -\frac{9\sqrt{2}}{160} \\
0 & \frac{3\sqrt{210}}{224} & 0 & -\frac{\sqrt{105}}{112} & 0 & -\frac{\sqrt{42}}{224} & \frac{1}{32} & 0 & \frac{9\sqrt{21}}{1120} & 0 & -\frac{13\sqrt{35}}{1120} & 0 & -\frac{23\sqrt{7}}{1120} & 0 \\
\frac{\sqrt{210}}{224} & 0 & -\frac{3\sqrt{21}}{112} & 0 & \frac{\sqrt{42}}{224} & 0 & 0 & \frac{\sqrt{35}}{224} & 0 & -\frac{9\sqrt{7}}{224} & 0 & \frac{\sqrt{105}}{224} & 0 & \frac{\sqrt{5}}{160}
\end{bmatrix}$$

$\boxed{\text{x330}}$ $\mathbb{T}^{(a)}_{5,1}(E_u,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{120} & 0 & \frac{\sqrt{105}i}{120} & 0 & -\frac{\sqrt{7}i}{40} & 0 & -\frac{\sqrt{3}i}{8} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{40} & 0 & \frac{\sqrt{21}i}{40} & 0 & -\frac{\sqrt{35}i}{40} & 0 & \frac{\sqrt{7}i}{40} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{40} & 0 & -\frac{\sqrt{35}i}{40} & 0 & \frac{\sqrt{21}i}{40} & 0 & -\frac{i}{40} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & -\frac{\sqrt{7}i}{40} & 0 & \frac{\sqrt{105}i}{120} & 0 & -\frac{\sqrt{21}i}{120} & 0 \\
0 & -\frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{7}i}{56} & 0 & \frac{3\sqrt{70}i}{112} & -\frac{\sqrt{15}i}{240} & 0 & \frac{\sqrt{35}i}{112} & 0 & -\frac{\sqrt{21}i}{112} & 0 & -\frac{\sqrt{105}i}{112} & 0 \\
-\frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{35}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & \frac{23\sqrt{21}i}{1680} & 0 & -\frac{13\sqrt{105}i}{1680} & 0 & \frac{3\sqrt{7}i}{560} & 0 & -\frac{\sqrt{3}i}{16} \\
0 & \frac{\sqrt{35}i}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{7}i}{56} & \frac{\sqrt{6}i}{80} & 0 & -\frac{11\sqrt{14}i}{560} & 0 & \frac{\sqrt{210}i}{1680} & 0 & \frac{3\sqrt{42}i}{560} & 0 \\
\frac{\sqrt{7}i}{56} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{\sqrt{35}i}{56} & 0 & 0 & -\frac{3\sqrt{42}i}{560} & 0 & -\frac{\sqrt{210}i}{1680} & 0 & \frac{11\sqrt{14}i}{560} & 0 & -\frac{\sqrt{6}i}{80} \\
0 & -\frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{35}i}{56} & 0 & -\frac{\sqrt{14}i}{112} & \frac{\sqrt{3}i}{16} & 0 & -\frac{3\sqrt{7}i}{560} & 0 & \frac{13\sqrt{105}i}{1680} & 0 & -\frac{23\sqrt{21}i}{1680} & 0 \\
\frac{3\sqrt{70}i}{112} & 0 & \frac{\sqrt{7}i}{56} & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & \frac{\sqrt{105}i}{112} & 0 & \frac{\sqrt{21}i}{112} & 0 & -\frac{\sqrt{35}i}{112} & 0 & \frac{\sqrt{15}i}{240}
\end{bmatrix}$$

$\boxed{\text{x331}}$ $\mathbb{T}^{(a)}_{5,2}(E_u,3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{120} & 0 & \frac{\sqrt{105}}{120} & 0 & \frac{\sqrt{7}}{40} & 0 & -\frac{\sqrt{3}}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{40} & 0 & -\frac{\sqrt{21}}{40} & 0 & -\frac{\sqrt{35}}{40} & 0 & -\frac{\sqrt{7}}{40} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{40} & 0 & \frac{\sqrt{35}}{40} & 0 & \frac{\sqrt{21}}{40} & 0 & \frac{1}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} & 0 & -\frac{\sqrt{7}}{40} & 0 & -\frac{\sqrt{105}}{120} & 0 & -\frac{\sqrt{21}}{120} & 0 \\ 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{7}}{56} & 0 & \frac{3\sqrt{70}}{112} & \frac{\sqrt{15}}{240} & 0 & \frac{\sqrt{35}}{112} & 0 & \frac{\sqrt{21}}{112} & 0 & -\frac{\sqrt{105}}{112} & 0 \\ \frac{\sqrt{14}}{112} & 0 & \frac{\sqrt{35}}{56} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & -\frac{23\sqrt{21}}{1680} & 0 & -\frac{13\sqrt{105}}{1680} & 0 & -\frac{3\sqrt{7}}{560} & 0 & -\frac{\sqrt{3}}{16} \\ 0 & -\frac{\sqrt{35}}{56} & 0 & -\frac{\sqrt{70}}{56} & 0 & -\frac{\sqrt{7}}{56} & \frac{\sqrt{6}}{80} & 0 & \frac{11\sqrt{14}}{560} & 0 & \frac{\sqrt{210}}{1680} & 0 & -\frac{3\sqrt{42}}{560} & 0 \\ \frac{\sqrt{7}}{56} & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{35}}{56} & 0 & 0 & -\frac{3\sqrt{42}}{560} & 0 & \frac{\sqrt{210}}{1680} & 0 & \frac{11\sqrt{14}}{560} & 0 & \frac{\sqrt{6}}{80} \\ 0 & -\frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{35}}{56} & 0 & -\frac{\sqrt{14}}{112} & -\frac{\sqrt{3}}{16} & 0 & -\frac{3\sqrt{7}}{560} & 0 & -\frac{13\sqrt{105}}{1680} & 0 & -\frac{23\sqrt{21}}{1680} & 0 \\ -\frac{3\sqrt{70}}{112} & 0 & \frac{\sqrt{7}}{56} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & -\frac{\sqrt{105}}{112} & 0 & \frac{\sqrt{21}}{112} & 0 & \frac{\sqrt{35}}{112} & 0 & \frac{\sqrt{15}}{240} \end{bmatrix}$

$\boxed{\text{x332}}$ $\mathbb{T}^{(1,-1;a)}_{5}(A_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{100} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{35} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x333}}$ $\mathbb{T}_3^{(1,-1;a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 \\ \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 \end{bmatrix}$

$\boxed{\text{x334}}$ $\mathbb{T}_5^{(1,-1;a)}(A_{2u}, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 \\ -\frac{\sqrt{5}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 \end{bmatrix}$

$\boxed{\text{x335}}$ $\mathbb{T}_5^{(1,-1;a)}(A_{2u},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{100} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x336}}$ $\mathbb{T}_3^{(1,-1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{5}}{35} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & -\frac{\sqrt{30}}{210} & 0 & 0 & -\frac{3}{28} & 0 & 0 & 0 & -\frac{3\sqrt{3}}{28} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{210} & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & -\frac{3\sqrt{3}}{28} & 0 & 0 & 0 & -\frac{3}{28} & 0 \\ 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & \frac{\sqrt{21}}{28} \\ 0 & 0 & \frac{3}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 \\ -\frac{3}{28} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 \\ 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & 0 & -\frac{3}{28} & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 \\ 0 & 0 & \frac{3\sqrt{5}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} \\ 0 & 0 & 0 & \frac{3}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x337}}$ $\mathbb{T}_5^{(1,-1;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{200} & 0 & 0 & 0 & -\frac{\sqrt{14}}{40} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{200} & 0 & 0 & 0 & \frac{3\sqrt{70}}{200} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{200} & 0 & 0 & 0 & -\frac{\sqrt{210}}{200} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{40} & 0 & 0 & 0 & \frac{\sqrt{10}}{200} \\
0 & 0 & \frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & \frac{\sqrt{10}}{25} & 0 & 0 & 0 & -\frac{2\sqrt{14}}{35} & 0 & 0 & 0 \\
-\frac{\sqrt{210}}{420} & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & -\frac{8\sqrt{35}}{175} & 0 & 0 & 0 & -\frac{2\sqrt{105}}{175} & 0 & 0 \\
0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 & 0 & \frac{8\sqrt{35}}{175} & 0 \\
0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 & 0 & -\frac{\sqrt{10}}{25} \\
0 & 0 & 0 & \frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x338}}$ $\mathbb{T}_3^{(1,-1;a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 \\
\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{210} & 0 & 0 & \frac{3i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{28} & 0 & 0 \\
0 & -\frac{\sqrt{30}i}{210} & 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & \frac{3\sqrt{3}i}{28} & 0 & 0 & 0 & -\frac{3i}{28} & 0 \\
0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} \\
0 & 0 & \frac{3i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 \\
\frac{3i}{28} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 \\
0 & \frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & -\frac{3i}{28} & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 \\
0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} \\
0 & 0 & 0 & -\frac{3i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{21} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x339}}$ $\mathbb{T}_5^{(1,-1;a)}(B_{2u}) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{200} & 0 & 0 & 0 & \frac{\sqrt{14}i}{40} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{200} & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{200} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{200} & 0 & 0 & 0 & \frac{\sqrt{210}i}{200} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{40} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{200} \\
0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & \frac{\sqrt{10}i}{25} & 0 & 0 & 0 & \frac{2\sqrt{14}i}{35} & 0 & 0 & 0 \\
-\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & -\frac{8\sqrt{35}i}{175} & 0 & 0 & 0 & \frac{2\sqrt{105}i}{175} & 0 & 0 \\
0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 & \frac{2\sqrt{105}i}{175} & 0 & 0 & 0 & -\frac{8\sqrt{35}i}{175} & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}i}{35} & 0 & 0 & 0 & \frac{\sqrt{10}i}{25} \\
0 & 0 & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{35} & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x340}}$ $\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,1) = \begin{bmatrix}
-\frac{\sqrt{30}i}{280} & 0 & \frac{3\sqrt{3}i}{140} & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & \frac{3i}{28} & 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 \\
0 & \frac{\sqrt{2}i}{40} & 0 & -\frac{i}{140} & 0 & -\frac{\sqrt{10}i}{56} & -\frac{\sqrt{105}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{3}i}{56} & 0 & -\frac{\sqrt{15}i}{28} & 0 \\
\frac{\sqrt{10}i}{56} & 0 & \frac{i}{140} & 0 & -\frac{\sqrt{2}i}{40} & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & \frac{3\sqrt{3}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{56} \\
0 & \frac{\sqrt{6}i}{56} & 0 & -\frac{3\sqrt{3}i}{140} & 0 & \frac{\sqrt{30}i}{280} & 0 & 0 & -\frac{\sqrt{15}i}{56} & 0 & \frac{3i}{28} & 0 & -\frac{3\sqrt{5}i}{56} & 0 \\
0 & -\frac{3\sqrt{30}i}{280} & 0 & \frac{\sqrt{15}i}{56} & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{3}i}{14} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 \\
-\frac{3\sqrt{30}i}{280} & 0 & \frac{3\sqrt{3}i}{280} & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & -\frac{i}{14} & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 \\
0 & \frac{3\sqrt{3}i}{280} & 0 & \frac{3\sqrt{6}i}{140} & 0 & \frac{\sqrt{15}i}{56} & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{2}i}{56} & 0 & \frac{3\sqrt{10}i}{56} & 0 \\
\frac{\sqrt{15}i}{56} & 0 & \frac{3\sqrt{6}i}{140} & 0 & \frac{3\sqrt{3}i}{280} & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & -\frac{\sqrt{2}i}{56} & 0 & \frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{70}i}{56} \\
0 & \frac{\sqrt{6}i}{28} & 0 & \frac{3\sqrt{3}i}{280} & 0 & -\frac{3\sqrt{30}i}{280} & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & \frac{i}{14} & 0 & \frac{\sqrt{5}i}{28} & 0 \\
0 & 0 & \frac{\sqrt{15}i}{56} & 0 & -\frac{3\sqrt{30}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{3}i}{14} & 0 & -\frac{\sqrt{7}i}{28}
\end{bmatrix}$

$\boxed{\text{x341}}$ $\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,1) =$

$$\begin{bmatrix}
\frac{\sqrt{30}}{280} & 0 & \frac{3\sqrt{3}}{140} & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & \frac{3}{28} & 0 & \frac{\sqrt{15}}{56} & 0 & 0 \\
0 & -\frac{\sqrt{2}}{40} & 0 & -\frac{1}{140} & 0 & \frac{\sqrt{10}}{56} & -\frac{\sqrt{105}}{56} & 0 & 0 & 0 & \frac{3\sqrt{3}}{56} & 0 & \frac{\sqrt{15}}{28} & 0 \\
\frac{\sqrt{10}}{56} & 0 & -\frac{1}{140} & 0 & -\frac{\sqrt{2}}{40} & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{3\sqrt{3}}{56} & 0 & 0 & 0 & \frac{\sqrt{105}}{56} \\
0 & \frac{\sqrt{6}}{56} & 0 & \frac{3\sqrt{3}}{140} & 0 & \frac{\sqrt{30}}{280} & 0 & 0 & -\frac{\sqrt{15}}{56} & 0 & -\frac{3}{28} & 0 & -\frac{3\sqrt{5}}{56} & 0 \\
0 & -\frac{3\sqrt{30}}{280} & 0 & -\frac{\sqrt{15}}{56} & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 \\
\frac{3\sqrt{30}}{280} & 0 & \frac{3\sqrt{3}}{280} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & -\frac{1}{14} & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 \\
0 & -\frac{3\sqrt{3}}{280} & 0 & \frac{3\sqrt{6}}{140} & 0 & -\frac{\sqrt{15}}{56} & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{2}}{56} & 0 & -\frac{3\sqrt{10}}{56} & 0 \\
\frac{\sqrt{15}}{56} & 0 & -\frac{3\sqrt{6}}{140} & 0 & \frac{3\sqrt{3}}{280} & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & \frac{\sqrt{2}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & -\frac{\sqrt{70}}{56} \\
0 & \frac{\sqrt{6}}{28} & 0 & -\frac{3\sqrt{3}}{280} & 0 & -\frac{3\sqrt{30}}{280} & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{1}{14} & 0 & \frac{\sqrt{5}}{28} & 0 \\
0 & 0 & \frac{\sqrt{15}}{56} & 0 & \frac{3\sqrt{30}}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{\sqrt{7}}{28}
\end{bmatrix}$$

$\boxed{\text{x342}}$ $\mathbb{T}_{3,1}^{(1,-1;a)}(E_u,2) =$

$$\begin{bmatrix}
-\frac{\sqrt{2}i}{56} & 0 & \frac{3\sqrt{5}i}{140} & 0 & \frac{3\sqrt{10}i}{280} & 0 & 0 & -\frac{5\sqrt{3}i}{56} & 0 & \frac{\sqrt{15}i}{28} & 0 & \frac{3i}{56} & 0 & 0 \\
0 & \frac{\sqrt{30}i}{120} & 0 & -\frac{\sqrt{15}i}{420} & 0 & \frac{\sqrt{6}i}{56} & \frac{3\sqrt{7}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}i}{56} & 0 & \frac{3i}{28} & 0 \\
-\frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{15}i}{420} & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & \frac{3i}{28} & 0 & \frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{7}i}{56} \\
0 & -\frac{3\sqrt{10}i}{280} & 0 & -\frac{3\sqrt{5}i}{140} & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & \frac{3i}{56} & 0 & \frac{\sqrt{15}i}{28} & 0 & -\frac{5\sqrt{3}i}{56} & 0 \\
0 & -\frac{3\sqrt{2}i}{56} & 0 & -\frac{3i}{56} & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\
-\frac{3\sqrt{2}i}{56} & 0 & \frac{3\sqrt{5}i}{280} & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & -\frac{5\sqrt{3}i}{84} & 0 & -\frac{\sqrt{15}i}{42} & 0 & -\frac{3i}{28} & 0 & 0 \\
0 & \frac{3\sqrt{5}i}{280} & 0 & \frac{3\sqrt{10}i}{140} & 0 & -\frac{3i}{56} & \frac{\sqrt{42}i}{56} & 0 & -\frac{5\sqrt{2}i}{56} & 0 & \frac{\sqrt{30}i}{168} & 0 & -\frac{3\sqrt{6}i}{56} & 0 \\
-\frac{3i}{56} & 0 & \frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{5}i}{280} & 0 & 0 & \frac{3\sqrt{6}i}{56} & 0 & -\frac{\sqrt{30}i}{168} & 0 & \frac{5\sqrt{2}i}{56} & 0 & -\frac{\sqrt{42}i}{56} \\
0 & -\frac{3\sqrt{10}i}{140} & 0 & \frac{3\sqrt{5}i}{280} & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & \frac{3i}{28} & 0 & \frac{\sqrt{15}i}{42} & 0 & \frac{5\sqrt{3}i}{84} & 0 \\
0 & 0 & -\frac{3i}{56} & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{105}i}{84}
\end{bmatrix}$$

$\boxed{\text{x343}}$ $\mathbb{T}_{3,2}^{(1,-1;a)}(E_u,2) = \begin{bmatrix}
\frac{\sqrt{2}}{56} & 0 & \frac{3\sqrt{5}}{140} & 0 & -\frac{3\sqrt{10}}{280} & 0 & 0 & \frac{5\sqrt{3}}{56} & 0 & \frac{\sqrt{15}}{28} & 0 & -\frac{3}{56} & 0 & 0 \\
0 & -\frac{\sqrt{30}}{120} & 0 & -\frac{\sqrt{15}}{420} & 0 & -\frac{\sqrt{6}}{56} & \frac{3\sqrt{7}}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & -\frac{3}{28} & 0 \\
-\frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{15}}{420} & 0 & -\frac{\sqrt{30}}{120} & 0 & 0 & \frac{3}{28} & 0 & -\frac{3\sqrt{5}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{7}}{56} \\
0 & -\frac{3\sqrt{10}}{280} & 0 & \frac{3\sqrt{5}}{140} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & \frac{3}{56} & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{5\sqrt{3}}{56} & 0 \\
0 & -\frac{3\sqrt{2}}{56} & 0 & \frac{3}{56} & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 \\
\frac{3\sqrt{2}}{56} & 0 & \frac{3\sqrt{5}}{280} & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & \frac{5\sqrt{3}}{84} & 0 & -\frac{\sqrt{15}}{42} & 0 & \frac{3}{28} & 0 & 0 \\
0 & -\frac{3\sqrt{5}}{280} & 0 & \frac{3\sqrt{10}}{140} & 0 & \frac{3}{56} & \frac{\sqrt{42}}{56} & 0 & \frac{5\sqrt{2}}{56} & 0 & \frac{\sqrt{30}}{168} & 0 & \frac{3\sqrt{6}}{56} & 0 \\
-\frac{3}{56} & 0 & -\frac{3\sqrt{10}}{140} & 0 & \frac{3\sqrt{5}}{280} & 0 & 0 & \frac{3\sqrt{6}}{56} & 0 & \frac{\sqrt{30}}{168} & 0 & \frac{5\sqrt{2}}{56} & 0 & \frac{\sqrt{42}}{56} \\
0 & -\frac{3\sqrt{10}}{140} & 0 & -\frac{3\sqrt{5}}{280} & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & \frac{3}{28} & 0 & -\frac{\sqrt{15}}{42} & 0 & \frac{5\sqrt{3}}{84} & 0 \\
0 & 0 & -\frac{3}{56} & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{105}}{84}
\end{bmatrix}$

$\boxed{\text{x344}}$ $\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,1) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{160} & 0 & \frac{\sqrt{30}i}{160} & 0 & -\frac{7\sqrt{2}i}{160} & 0 & \frac{3\sqrt{42}i}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{160} & 0 & \frac{3\sqrt{6}i}{160} & 0 & -\frac{3\sqrt{10}i}{160} & 0 & \frac{7\sqrt{2}i}{160} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{2}i}{160} & 0 & -\frac{3\sqrt{10}i}{160} & 0 & \frac{3\sqrt{6}i}{160} & 0 & -\frac{\sqrt{14}i}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{160} & 0 & -\frac{7\sqrt{2}i}{160} & 0 & \frac{\sqrt{30}i}{160} & 0 & -\frac{\sqrt{6}i}{160} & 0 \\
0 & -\frac{i}{112} & 0 & \frac{\sqrt{2}i}{48} & 0 & -\frac{3\sqrt{5}i}{80} & \frac{\sqrt{210}i}{560} & 0 & -\frac{3\sqrt{10}i}{112} & 0 & \frac{\sqrt{6}i}{16} & 0 & -\frac{3\sqrt{30}i}{80} & 0 \\
-\frac{i}{112} & 0 & \frac{\sqrt{10}i}{112} & 0 & -\frac{\sqrt{5}i}{48} & 0 & 0 & -\frac{23\sqrt{6}i}{560} & 0 & \frac{13\sqrt{30}i}{560} & 0 & -\frac{3\sqrt{2}i}{80} & 0 & -\frac{3\sqrt{42}i}{80} \\
0 & \frac{\sqrt{10}i}{112} & 0 & -\frac{\sqrt{5}i}{56} & 0 & \frac{\sqrt{2}i}{48} & -\frac{\sqrt{21}i}{40} & 0 & \frac{33i}{280} & 0 & -\frac{\sqrt{15}i}{280} & 0 & -\frac{3\sqrt{3}i}{40} & 0 \\
\frac{\sqrt{2}i}{48} & 0 & -\frac{\sqrt{5}i}{56} & 0 & \frac{\sqrt{10}i}{112} & 0 & 0 & \frac{3\sqrt{3}i}{40} & 0 & \frac{\sqrt{15}i}{280} & 0 & -\frac{33i}{280} & 0 & \frac{\sqrt{21}i}{40} \\
0 & -\frac{\sqrt{5}i}{48} & 0 & \frac{\sqrt{10}i}{112} & 0 & -\frac{i}{112} & \frac{3\sqrt{42}i}{80} & 0 & \frac{3\sqrt{2}i}{80} & 0 & -\frac{13\sqrt{30}i}{560} & 0 & \frac{23\sqrt{6}i}{560} & 0 \\
-\frac{3\sqrt{5}i}{80} & 0 & \frac{\sqrt{2}i}{48} & 0 & -\frac{i}{112} & 0 & 0 & \frac{3\sqrt{30}i}{80} & 0 & -\frac{\sqrt{6}i}{16} & 0 & \frac{3\sqrt{10}i}{112} & 0 & -\frac{\sqrt{210}i}{560}
\end{bmatrix}$

$\boxed{\text{x345}}$ $\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{160} & 0 & \frac{\sqrt{30}}{160} & 0 & \frac{7\sqrt{2}}{160} & 0 & \frac{3\sqrt{42}}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{160} & 0 & -\frac{3\sqrt{6}}{160} & 0 & -\frac{3\sqrt{10}}{160} & 0 & -\frac{7\sqrt{2}}{160} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{2}}{160} & 0 & \frac{3\sqrt{10}}{160} & 0 & \frac{3\sqrt{6}}{160} & 0 & \frac{\sqrt{14}}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{160} & 0 & -\frac{7\sqrt{2}}{160} & 0 & -\frac{\sqrt{30}}{160} & 0 & -\frac{\sqrt{6}}{160} & 0 \\
0 & -\frac{1}{112} & 0 & -\frac{\sqrt{2}}{48} & 0 & -\frac{3\sqrt{5}}{80} & -\frac{\sqrt{210}}{560} & 0 & -\frac{3\sqrt{10}}{112} & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{3\sqrt{30}}{80} & 0 \\
\frac{1}{112} & 0 & \frac{\sqrt{10}}{112} & 0 & \frac{\sqrt{5}}{48} & 0 & 0 & \frac{23\sqrt{6}}{560} & 0 & \frac{13\sqrt{30}}{560} & 0 & \frac{3\sqrt{2}}{80} & 0 & -\frac{3\sqrt{42}}{80} \\
0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{\sqrt{5}}{56} & 0 & -\frac{\sqrt{2}}{48} & -\frac{\sqrt{21}}{40} & 0 & -\frac{33}{280} & 0 & -\frac{\sqrt{15}}{280} & 0 & \frac{3\sqrt{3}}{40} & 0 \\
\frac{\sqrt{2}}{48} & 0 & \frac{\sqrt{5}}{56} & 0 & \frac{\sqrt{10}}{112} & 0 & 0 & \frac{3\sqrt{3}}{40} & 0 & -\frac{\sqrt{15}}{280} & 0 & -\frac{33}{280} & 0 & -\frac{\sqrt{21}}{40} \\
0 & -\frac{\sqrt{5}}{48} & 0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{1}{112} & -\frac{3\sqrt{42}}{80} & 0 & \frac{3\sqrt{2}}{80} & 0 & \frac{13\sqrt{30}}{560} & 0 & \frac{23\sqrt{6}}{560} & 0 \\
\frac{3\sqrt{5}}{80} & 0 & \frac{\sqrt{2}}{48} & 0 & \frac{1}{112} & 0 & 0 & -\frac{3\sqrt{30}}{80} & 0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{3\sqrt{10}}{112} & 0 & -\frac{\sqrt{210}}{560}
\end{bmatrix}$$

$\boxed{\text{x346}}$ $\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{800} & 0 & \frac{\sqrt{42}i}{160} & 0 & \frac{9\sqrt{70}i}{800} & 0 & \frac{\sqrt{30}i}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}i}{800} & 0 & \frac{3\sqrt{210}i}{800} & 0 & -\frac{3\sqrt{14}i}{160} & 0 & -\frac{9\sqrt{70}i}{800} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}i}{800} & 0 & -\frac{3\sqrt{14}i}{160} & 0 & \frac{3\sqrt{210}i}{800} & 0 & \frac{9\sqrt{10}i}{800} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{160} & 0 & \frac{9\sqrt{70}i}{800} & 0 & \frac{\sqrt{42}i}{160} & 0 & -\frac{\sqrt{210}i}{800} & 0 \\
0 & -\frac{\sqrt{35}i}{560} & 0 & -\frac{3\sqrt{70}i}{560} & 0 & -\frac{\sqrt{7}i}{112} & \frac{\sqrt{6}i}{80} & 0 & -\frac{3\sqrt{14}i}{112} & 0 & -\frac{9\sqrt{210}i}{560} & 0 & -\frac{\sqrt{42}i}{112} & 0 \\
-\frac{\sqrt{35}i}{560} & 0 & \frac{\sqrt{14}i}{112} & 0 & \frac{3\sqrt{7}i}{112} & 0 & 0 & -\frac{23\sqrt{210}i}{2800} & 0 & \frac{13\sqrt{42}i}{560} & 0 & \frac{27\sqrt{70}i}{2800} & 0 & -\frac{\sqrt{30}i}{80} \\
0 & \frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{7}i}{56} & 0 & -\frac{3\sqrt{70}i}{560} & \frac{9\sqrt{15}i}{200} & 0 & \frac{33\sqrt{35}i}{1400} & 0 & -\frac{\sqrt{21}i}{280} & 0 & \frac{27\sqrt{105}i}{1400} & 0 \\
-\frac{3\sqrt{70}i}{560} & 0 & -\frac{\sqrt{7}i}{56} & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & -\frac{27\sqrt{105}i}{1400} & 0 & \frac{\sqrt{21}i}{280} & 0 & -\frac{33\sqrt{35}i}{1400} & 0 & -\frac{9\sqrt{15}i}{200} \\
0 & \frac{3\sqrt{7}i}{112} & 0 & \frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{35}i}{560} & \frac{\sqrt{30}i}{80} & 0 & -\frac{27\sqrt{70}i}{2800} & 0 & -\frac{13\sqrt{42}i}{560} & 0 & \frac{23\sqrt{210}i}{2800} & 0 \\
-\frac{\sqrt{7}i}{112} & 0 & -\frac{3\sqrt{70}i}{560} & 0 & -\frac{\sqrt{35}i}{560} & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & \frac{9\sqrt{210}i}{560} & 0 & \frac{3\sqrt{14}i}{112} & 0 & -\frac{\sqrt{6}i}{80}
\end{bmatrix}$$

$\boxed{\text{x347}}$ $\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{800} & 0 & \frac{\sqrt{42}}{160} & 0 & -\frac{9\sqrt{70}}{800} & 0 & \frac{\sqrt{30}}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{800} & 0 & -\frac{3\sqrt{210}}{800} & 0 & -\frac{3\sqrt{14}}{160} & 0 & \frac{9\sqrt{70}}{800} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{70}}{800} & 0 & \frac{3\sqrt{14}}{160} & 0 & \frac{3\sqrt{210}}{800} & 0 & -\frac{9\sqrt{10}}{800} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{160} & 0 & \frac{9\sqrt{70}}{800} & 0 & -\frac{\sqrt{42}}{160} & 0 & -\frac{\sqrt{210}}{800} & 0 \\
0 & -\frac{\sqrt{35}}{560} & 0 & \frac{3\sqrt{70}}{560} & 0 & -\frac{\sqrt{7}}{112} & -\frac{\sqrt{6}}{80} & 0 & -\frac{3\sqrt{14}}{112} & 0 & \frac{9\sqrt{210}}{560} & 0 & -\frac{\sqrt{42}}{112} & 0 \\
\frac{\sqrt{35}}{560} & 0 & \frac{\sqrt{14}}{112} & 0 & -\frac{3\sqrt{7}}{112} & 0 & 0 & \frac{23\sqrt{210}}{2800} & 0 & \frac{13\sqrt{42}}{560} & 0 & -\frac{27\sqrt{70}}{2800} & 0 & -\frac{\sqrt{30}}{80} \\
0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{7}}{56} & 0 & \frac{3\sqrt{70}}{560} & \frac{9\sqrt{15}}{200} & 0 & -\frac{33\sqrt{35}}{1400} & 0 & -\frac{\sqrt{21}}{280} & 0 & -\frac{27\sqrt{105}}{1400} & 0 \\
-\frac{3\sqrt{70}}{560} & 0 & \frac{\sqrt{7}}{56} & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & -\frac{27\sqrt{105}}{1400} & 0 & -\frac{\sqrt{21}}{280} & 0 & -\frac{33\sqrt{35}}{1400} & 0 & \frac{9\sqrt{15}}{200} \\
0 & \frac{3\sqrt{7}}{112} & 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{35}}{560} & -\frac{\sqrt{30}}{80} & 0 & -\frac{27\sqrt{70}}{2800} & 0 & \frac{13\sqrt{42}}{560} & 0 & \frac{23\sqrt{210}}{2800} & 0 \\
\frac{\sqrt{7}}{112} & 0 & -\frac{3\sqrt{70}}{560} & 0 & \frac{\sqrt{35}}{560} & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & \frac{9\sqrt{210}}{560} & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{\sqrt{6}}{80}
\end{bmatrix}$

$\boxed{\text{x348}}$ $\mathbb{T}_{5,1}^{(1,-1;a)}(E_u,3) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{400} & 0 & \frac{\sqrt{14}i}{80} & 0 & -\frac{\sqrt{210}i}{400} & 0 & -\frac{3\sqrt{10}i}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{400} & 0 & \frac{3\sqrt{70}i}{400} & 0 & -\frac{\sqrt{42}i}{80} & 0 & \frac{\sqrt{210}i}{400} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{400} & 0 & -\frac{\sqrt{42}i}{80} & 0 & \frac{3\sqrt{70}i}{400} & 0 & -\frac{\sqrt{30}i}{400} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{80} & 0 & -\frac{\sqrt{210}i}{400} & 0 & \frac{\sqrt{14}i}{80} & 0 & -\frac{\sqrt{70}i}{400} & 0 \\
0 & -\frac{\sqrt{105}i}{840} & 0 & \frac{\sqrt{210}i}{840} & 0 & \frac{\sqrt{21}i}{56} & \frac{\sqrt{2}i}{40} & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{3\sqrt{70}i}{280} & 0 & \frac{3\sqrt{14}i}{56} & 0 \\
-\frac{\sqrt{105}i}{840} & 0 & \frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{21}i}{168} & 0 & 0 & -\frac{23\sqrt{70}i}{1400} & 0 & \frac{13\sqrt{14}i}{280} & 0 & -\frac{3\sqrt{210}i}{1400} & 0 & \frac{3\sqrt{10}i}{40} \\
0 & \frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{210}i}{840} & -\frac{3\sqrt{5}i}{100} & 0 & \frac{11\sqrt{105}i}{700} & 0 & -\frac{\sqrt{7}i}{140} & 0 & -\frac{9\sqrt{35}i}{700} & 0 \\
\frac{\sqrt{210}i}{840} & 0 & -\frac{\sqrt{21}i}{84} & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & \frac{9\sqrt{35}i}{700} & 0 & \frac{\sqrt{7}i}{140} & 0 & -\frac{11\sqrt{105}i}{700} & 0 & \frac{3\sqrt{5}i}{100} \\
0 & -\frac{\sqrt{21}i}{168} & 0 & \frac{\sqrt{42}i}{168} & 0 & -\frac{\sqrt{105}i}{840} & -\frac{3\sqrt{10}i}{40} & 0 & \frac{3\sqrt{210}i}{1400} & 0 & -\frac{13\sqrt{14}i}{280} & 0 & \frac{23\sqrt{70}i}{1400} & 0 \\
\frac{\sqrt{21}i}{56} & 0 & \frac{\sqrt{210}i}{840} & 0 & -\frac{\sqrt{105}i}{840} & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & -\frac{3\sqrt{70}i}{280} & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{2}i}{40}
\end{bmatrix}$

$\boxed{\text{x349}}$ $\mathbb{T}_{5,2}^{(1,-1;a)}(E_u,3) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{400} & 0 & \frac{\sqrt{14}}{80} & 0 & \frac{\sqrt{210}}{400} & 0 & -\frac{3\sqrt{10}}{80} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{400} & 0 & -\frac{3\sqrt{70}}{400} & 0 & -\frac{\sqrt{42}}{80} & 0 & -\frac{\sqrt{210}}{400} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{400} & 0 & \frac{\sqrt{42}}{80} & 0 & \frac{3\sqrt{70}}{400} & 0 & \frac{\sqrt{30}}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{80} & 0 & -\frac{\sqrt{210}}{400} & 0 & -\frac{\sqrt{14}}{80} & 0 & -\frac{\sqrt{70}}{400} & 0 \\ 0 & -\frac{\sqrt{105}}{840} & 0 & -\frac{\sqrt{210}}{840} & 0 & \frac{\sqrt{21}}{56} & -\frac{\sqrt{2}}{40} & 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{3\sqrt{70}}{280} & 0 & \frac{3\sqrt{14}}{56} & 0 \\ \frac{\sqrt{105}}{840} & 0 & \frac{\sqrt{42}}{168} & 0 & \frac{\sqrt{21}}{168} & 0 & 0 & \frac{23\sqrt{70}}{1400} & 0 & \frac{13\sqrt{14}}{280} & 0 & \frac{3\sqrt{210}}{1400} & 0 & \frac{3\sqrt{10}}{40} \\ 0 & -\frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{21}}{84} & 0 & -\frac{\sqrt{210}}{840} & -\frac{3\sqrt{5}}{100} & 0 & -\frac{11\sqrt{105}}{700} & 0 & -\frac{\sqrt{7}}{140} & 0 & \frac{9\sqrt{35}}{700} & 0 \\ \frac{\sqrt{210}}{840} & 0 & \frac{\sqrt{21}}{84} & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & \frac{9\sqrt{35}}{700} & 0 & -\frac{\sqrt{7}}{140} & 0 & -\frac{11\sqrt{105}}{700} & 0 & -\frac{3\sqrt{5}}{100} \\ 0 & -\frac{\sqrt{21}}{168} & 0 & -\frac{\sqrt{42}}{168} & 0 & -\frac{\sqrt{105}}{840} & \frac{3\sqrt{10}}{40} & 0 & \frac{3\sqrt{210}}{1400} & 0 & \frac{13\sqrt{14}}{280} & 0 & \frac{23\sqrt{70}}{1400} & 0 \\ -\frac{\sqrt{21}}{56} & 0 & \frac{\sqrt{210}}{840} & 0 & \frac{\sqrt{105}}{840} & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & -\frac{3\sqrt{70}}{280} & 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{2}}{40} \end{bmatrix}$

$\boxed{\text{x350}}$ $\mathbb{T}_{5}^{(1,0;a)}(A_{1u}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{2100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{2100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x351}}$ $\mathbb{T}_1^{(1,0;a)}(A_{2u}) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 \end{bmatrix}$

$\boxed{\text{x352}}$ $\mathbb{T}_3^{(1,0;a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 \end{bmatrix}$

$\boxed{\text{x353}}$ $\mathbb{T}_5^{(1,0;a)}(A_{2u},1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{420} & 0 \end{bmatrix}$

$\boxed{\text{x354}}$ $\mathbb{T}_5^{(1,0;a)}(A_{2u},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{2100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{2100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x355}}$ $\mathbb{T}_3^{(1,0;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{210}}{280} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 \\
-\frac{\sqrt{7}}{56} & 0 & 0 & 0 & -\frac{\sqrt{35}}{280} & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 \\
0 & \frac{\sqrt{35}}{280} & 0 & 0 & 0 & \frac{\sqrt{7}}{56} & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 \\
0 & 0 & \frac{\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} \\
0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & \frac{\sqrt{70}}{168} & 0 & 0 & 0 \\
-\frac{\sqrt{42}}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{7}}{168} & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0 \\
0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 & 0 & 0 & -\frac{\sqrt{7}}{168} & 0 \\
0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}}{24} \\
0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x356}}$ $\mathbb{T}_5^{(1,0;a)}(B_{1u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{100} & 0 & 0 & 0 & \frac{\sqrt{14}}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 & 0 & 0 & -\frac{3\sqrt{70}}{100} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{100} & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{20} & 0 & 0 & 0 & -\frac{\sqrt{10}}{100} \\
0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{420} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & \frac{\sqrt{10}}{300} & 0 & 0 & 0 & -\frac{\sqrt{14}}{210} & 0 & 0 & 0 \\
-\frac{\sqrt{210}}{140} & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & -\frac{2\sqrt{35}}{525} & 0 & 0 & 0 & -\frac{\sqrt{105}}{1050} & 0 & 0 \\
0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & \frac{\sqrt{105}}{1050} & 0 & 0 & 0 & \frac{2\sqrt{35}}{525} & 0 \\
0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{210} & 0 & 0 & 0 & -\frac{\sqrt{10}}{300} \\
0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{420} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x357}}$ $\mathbb{T}_3^{(1,0;a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & -\frac{\sqrt{210}i}{280} & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 \\
\frac{\sqrt{7}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{280} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{280} & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 \\
0 & 0 & -\frac{\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} \\
0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & -\frac{\sqrt{2}i}{24} & 0 & 0 & 0 & \frac{\sqrt{70}i}{168} & 0 & 0 & 0 \\
\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & -\frac{\sqrt{7}i}{168} & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & 0 \\
0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{168} & 0 \\
0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{168} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{24} \\
0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x358}}$ $\mathbb{T}_5^{(1,0;a)}(B_{2u}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{100} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{20} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{100} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{100} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{100} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{100} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{20} & 0 & 0 & 0 & \frac{\sqrt{10}i}{100} \\
0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{420} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & \frac{\sqrt{10}i}{300} & 0 & 0 & 0 & \frac{\sqrt{14}i}{210} & 0 & 0 & 0 \\
-\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & -\frac{2\sqrt{35}i}{525} & 0 & 0 & 0 & \frac{\sqrt{105}i}{1050} & 0 & 0 \\
0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & \frac{\sqrt{105}i}{1050} & 0 & 0 & 0 & -\frac{2\sqrt{35}i}{525} & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{210} & 0 & 0 & 0 & \frac{\sqrt{10}i}{300} \\
0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{420} & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x359}}$ $\mathbb{T}_{1,1}^{(1,0;a)}(E_u) = \begin{bmatrix} -\frac{\sqrt{10}i}{20} & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{20} & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{20} & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{10}i}{70} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\ 0 & -\frac{6i}{35} & 0 & -\frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{9\sqrt{2}i}{70} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{6i}{35} & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & -\frac{\sqrt{15}i}{28} & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & -\frac{\sqrt{21}i}{28} \end{bmatrix}$

$\boxed{\text{x360}}$ $\mathbb{T}_{1,2}^{(1,0;a)}(E_u) = \begin{bmatrix} \frac{\sqrt{10}}{20} & 0 & \frac{1}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{20} & 0 & \frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{20} & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{20} & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{10}}{70} & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\ 0 & \frac{6}{35} & 0 & -\frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{2}}{70} & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{6}{35} & 0 & -\frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & -\frac{\sqrt{15}}{28} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & -\frac{\sqrt{21}}{28} \end{bmatrix}$

$\boxed{\text{x361}}$ $\mathbb{T}_{3,1}^{(1,0;a)}(E_u,1) =$

$$\begin{bmatrix}
-\frac{3\sqrt{35}i}{1120} & 0 & \frac{9\sqrt{14}i}{1120} & 0 & -\frac{3\sqrt{7}i}{224} & 0 & 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 \\
0 & \frac{\sqrt{21}i}{160} & 0 & -\frac{\sqrt{42}i}{1120} & 0 & -\frac{\sqrt{105}i}{224} & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{112} & 0 & \frac{\sqrt{70}i}{56} & 0 \\
\frac{\sqrt{105}i}{224} & 0 & \frac{\sqrt{42}i}{1120} & 0 & -\frac{\sqrt{21}i}{160} & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{3\sqrt{14}i}{112} & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} \\
0 & \frac{3\sqrt{7}i}{224} & 0 & -\frac{9\sqrt{14}i}{1120} & 0 & \frac{3\sqrt{35}i}{1120} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{210}i}{112} & 0 \\
0 & -\frac{3\sqrt{35}i}{140} & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & \frac{\sqrt{6}i}{96} & 0 & -\frac{\sqrt{14}i}{112} & 0 & \frac{\sqrt{210}i}{672} & 0 & 0 & 0 \\
-\frac{3\sqrt{35}i}{140} & 0 & \frac{3\sqrt{14}i}{280} & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & -\frac{\sqrt{210}i}{672} & 0 & -\frac{\sqrt{42}i}{336} & 0 & \frac{\sqrt{70}i}{224} & 0 & 0 \\
0 & \frac{3\sqrt{14}i}{280} & 0 & \frac{3\sqrt{7}i}{70} & 0 & \frac{\sqrt{70}i}{56} & -\frac{\sqrt{15}i}{96} & 0 & -\frac{\sqrt{35}i}{224} & 0 & \frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{105}i}{224} & 0 \\
\frac{\sqrt{70}i}{56} & 0 & \frac{3\sqrt{7}i}{70} & 0 & \frac{3\sqrt{14}i}{280} & 0 & 0 & -\frac{\sqrt{105}i}{224} & 0 & -\frac{\sqrt{21}i}{672} & 0 & \frac{\sqrt{35}i}{224} & 0 & \frac{\sqrt{15}i}{96} \\
0 & \frac{\sqrt{7}i}{14} & 0 & \frac{3\sqrt{14}i}{280} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{42}i}{336} & 0 & \frac{\sqrt{210}i}{672} & 0 \\
0 & 0 & \frac{\sqrt{70}i}{56} & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{672} & 0 & \frac{\sqrt{14}i}{112} & 0 & -\frac{\sqrt{6}i}{96}
\end{bmatrix}$$

$\boxed{\text{x362}}$ $\mathbb{T}_{3,2}^{(1,0;a)}(E_u,1) =$

$$\begin{bmatrix}
\frac{3\sqrt{35}}{1120} & 0 & \frac{9\sqrt{14}}{1120} & 0 & \frac{3\sqrt{7}}{224} & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{70}}{112} & 0 & 0 \\
0 & -\frac{\sqrt{21}}{160} & 0 & -\frac{\sqrt{42}}{1120} & 0 & \frac{\sqrt{105}}{224} & \frac{\sqrt{10}}{16} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 \\
\frac{\sqrt{105}}{224} & 0 & -\frac{\sqrt{42}}{1120} & 0 & -\frac{\sqrt{21}}{160} & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{3\sqrt{14}}{112} & 0 & 0 & 0 & -\frac{\sqrt{10}}{16} \\
0 & \frac{3\sqrt{7}}{224} & 0 & \frac{9\sqrt{14}}{1120} & 0 & \frac{3\sqrt{35}}{1120} & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{210}}{112} & 0 \\
0 & -\frac{3\sqrt{35}}{140} & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & -\frac{\sqrt{6}}{96} & 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{210}}{672} & 0 & 0 & 0 \\
\frac{3\sqrt{35}}{140} & 0 & \frac{3\sqrt{14}}{280} & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & \frac{\sqrt{210}}{672} & 0 & -\frac{\sqrt{42}}{336} & 0 & -\frac{\sqrt{70}}{224} & 0 & 0 \\
0 & -\frac{3\sqrt{14}}{280} & 0 & \frac{3\sqrt{7}}{70} & 0 & -\frac{\sqrt{70}}{56} & -\frac{\sqrt{15}}{96} & 0 & \frac{\sqrt{35}}{224} & 0 & \frac{\sqrt{21}}{672} & 0 & -\frac{\sqrt{105}}{224} & 0 \\
\frac{\sqrt{70}}{56} & 0 & -\frac{3\sqrt{7}}{70} & 0 & \frac{3\sqrt{14}}{280} & 0 & 0 & -\frac{\sqrt{105}}{224} & 0 & \frac{\sqrt{21}}{672} & 0 & \frac{\sqrt{35}}{224} & 0 & -\frac{\sqrt{15}}{96} \\
0 & \frac{\sqrt{7}}{14} & 0 & -\frac{3\sqrt{14}}{280} & 0 & -\frac{3\sqrt{35}}{140} & 0 & 0 & -\frac{\sqrt{70}}{224} & 0 & -\frac{\sqrt{42}}{336} & 0 & \frac{\sqrt{210}}{672} & 0 \\
0 & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{672} & 0 & -\frac{\sqrt{14}}{112} & 0 & -\frac{\sqrt{6}}{96}
\end{bmatrix}$$

$\boxed{\text{x363}}$ $\mathbb{T}_{3,1}^{(1,0;a)}(E_u,2) =$

$$\begin{bmatrix}
-\frac{\sqrt{21}i}{224} & 0 & \frac{3\sqrt{210}i}{1120} & 0 & \frac{3\sqrt{105}i}{1120} & 0 & 0 & \frac{5\sqrt{14}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 \\
0 & \frac{\sqrt{35}i}{160} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & \frac{3\sqrt{7}i}{224} & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{42}i}{56} & 0 \\
-\frac{3\sqrt{7}i}{224} & 0 & \frac{\sqrt{70}i}{1120} & 0 & -\frac{\sqrt{35}i}{160} & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} \\
0 & -\frac{3\sqrt{105}i}{1120} & 0 & -\frac{3\sqrt{210}i}{1120} & 0 & \frac{\sqrt{21}i}{224} & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{70}i}{56} & 0 & \frac{5\sqrt{14}i}{112} & 0 \\
0 & -\frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & \frac{\sqrt{10}i}{96} & 0 & -\frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{14}i}{224} & 0 & 0 & 0 \\
-\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & -\frac{5\sqrt{14}i}{672} & 0 & -\frac{\sqrt{70}i}{336} & 0 & -\frac{\sqrt{42}i}{224} & 0 & 0 \\
0 & \frac{\sqrt{210}i}{280} & 0 & \frac{\sqrt{105}i}{70} & 0 & -\frac{\sqrt{42}i}{56} & \frac{i}{32} & 0 & -\frac{5\sqrt{21}i}{672} & 0 & \frac{\sqrt{35}i}{672} & 0 & -\frac{3\sqrt{7}i}{224} & 0 \\
-\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{210}i}{280} & 0 & 0 & \frac{3\sqrt{7}i}{224} & 0 & -\frac{\sqrt{35}i}{672} & 0 & \frac{5\sqrt{21}i}{672} & 0 & -\frac{i}{32} \\
0 & -\frac{\sqrt{105}i}{70} & 0 & \frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{224} & 0 & \frac{\sqrt{70}i}{336} & 0 & \frac{5\sqrt{14}i}{672} & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{224} & 0 & \frac{\sqrt{210}i}{336} & 0 & -\frac{\sqrt{10}i}{96}
\end{bmatrix}$$

$\boxed{\text{x364}}$ $\mathbb{T}_{3,2}^{(1,0;a)}(E_u,2) =$

$$\begin{bmatrix}
\frac{\sqrt{21}}{224} & 0 & \frac{3\sqrt{210}}{1120} & 0 & -\frac{3\sqrt{105}}{1120} & 0 & 0 & -\frac{5\sqrt{14}}{112} & 0 & -\frac{\sqrt{70}}{56} & 0 & \frac{\sqrt{42}}{112} & 0 & 0 \\
0 & -\frac{\sqrt{35}}{160} & 0 & -\frac{\sqrt{70}}{1120} & 0 & -\frac{3\sqrt{7}}{224} & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{42}}{56} & 0 \\
-\frac{3\sqrt{7}}{224} & 0 & -\frac{\sqrt{70}}{1120} & 0 & -\frac{\sqrt{35}}{160} & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{210}}{112} & 0 & 0 & 0 & \frac{\sqrt{6}}{16} \\
0 & -\frac{3\sqrt{105}}{1120} & 0 & \frac{3\sqrt{210}}{1120} & 0 & \frac{\sqrt{21}}{224} & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{70}}{56} & 0 & \frac{5\sqrt{14}}{112} & 0 \\
0 & -\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & -\frac{\sqrt{10}}{96} & 0 & -\frac{\sqrt{210}}{336} & 0 & \frac{\sqrt{14}}{224} & 0 & 0 & 0 \\
\frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & \frac{5\sqrt{14}}{672} & 0 & -\frac{\sqrt{70}}{336} & 0 & \frac{\sqrt{42}}{224} & 0 & 0 \\
0 & -\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{42}}{56} & \frac{1}{32} & 0 & \frac{5\sqrt{21}}{672} & 0 & \frac{\sqrt{35}}{672} & 0 & \frac{3\sqrt{7}}{224} & 0 \\
-\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{105}}{70} & 0 & \frac{\sqrt{210}}{280} & 0 & 0 & \frac{3\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{672} & 0 & \frac{5\sqrt{21}}{672} & 0 & \frac{1}{32} \\
0 & -\frac{\sqrt{105}}{70} & 0 & -\frac{\sqrt{210}}{280} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & \frac{\sqrt{42}}{224} & 0 & -\frac{\sqrt{70}}{336} & 0 & \frac{5\sqrt{14}}{672} & 0 \\
0 & 0 & -\frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{224} & 0 & -\frac{\sqrt{210}}{336} & 0 & -\frac{\sqrt{10}}{96}
\end{bmatrix}$$

$\boxed{\text{x365}}$ $\mathbb{T}_{5,1}^{(1,0;a)}(E_u,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{80} & 0 & -\frac{\sqrt{30}i}{80} & 0 & \frac{7\sqrt{2}i}{80} & 0 & -\frac{3\sqrt{42}i}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{80} & 0 & -\frac{3\sqrt{6}i}{80} & 0 & \frac{3\sqrt{10}i}{80} & 0 & -\frac{7\sqrt{2}i}{80} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{2}i}{80} & 0 & \frac{3\sqrt{10}i}{80} & 0 & -\frac{3\sqrt{6}i}{80} & 0 & \frac{\sqrt{14}i}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{80} & 0 & \frac{7\sqrt{2}i}{80} & 0 & -\frac{\sqrt{30}i}{80} & 0 & \frac{\sqrt{6}i}{80} & 0 \\
0 & -\frac{3i}{112} & 0 & \frac{\sqrt{2}i}{16} & 0 & -\frac{9\sqrt{5}i}{80} & \frac{\sqrt{210}i}{6720} & 0 & -\frac{\sqrt{10}i}{448} & 0 & \frac{\sqrt{6}i}{192} & 0 & -\frac{\sqrt{30}i}{320} & 0 \\
-\frac{3i}{112} & 0 & \frac{3\sqrt{10}i}{112} & 0 & -\frac{\sqrt{5}i}{16} & 0 & 0 & -\frac{23\sqrt{6}i}{6720} & 0 & \frac{13\sqrt{30}i}{6720} & 0 & -\frac{\sqrt{2}i}{320} & 0 & -\frac{\sqrt{42}i}{320} \\
0 & \frac{3\sqrt{10}i}{112} & 0 & -\frac{3\sqrt{5}i}{56} & 0 & \frac{\sqrt{2}i}{16} & -\frac{\sqrt{21}i}{480} & 0 & \frac{11i}{1120} & 0 & -\frac{\sqrt{15}i}{3360} & 0 & -\frac{\sqrt{3}i}{160} & 0 \\
\frac{\sqrt{2}i}{16} & 0 & -\frac{3\sqrt{5}i}{56} & 0 & \frac{3\sqrt{10}i}{112} & 0 & 0 & \frac{\sqrt{3}i}{160} & 0 & \frac{\sqrt{15}i}{3360} & 0 & -\frac{11i}{1120} & 0 & \frac{\sqrt{21}i}{480} \\
0 & -\frac{\sqrt{5}i}{16} & 0 & \frac{3\sqrt{10}i}{112} & 0 & -\frac{3i}{112} & \frac{\sqrt{42}i}{320} & 0 & \frac{\sqrt{2}i}{320} & 0 & -\frac{13\sqrt{30}i}{6720} & 0 & \frac{23\sqrt{6}i}{6720} & 0 \\
-\frac{9\sqrt{5}i}{80} & 0 & \frac{\sqrt{2}i}{16} & 0 & -\frac{3i}{112} & 0 & 0 & \frac{\sqrt{30}i}{320} & 0 & -\frac{\sqrt{6}i}{192} & 0 & \frac{\sqrt{10}i}{448} & 0 & -\frac{\sqrt{210}i}{6720}
\end{bmatrix}$$

$\boxed{\text{x366}}$ $\mathbb{T}_{5,2}^{(1,0;a)}(E_u,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{80} & 0 & -\frac{\sqrt{30}}{80} & 0 & -\frac{7\sqrt{2}}{80} & 0 & -\frac{3\sqrt{42}}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{80} & 0 & \frac{3\sqrt{6}}{80} & 0 & \frac{3\sqrt{10}}{80} & 0 & \frac{7\sqrt{2}}{80} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{2}}{80} & 0 & -\frac{3\sqrt{10}}{80} & 0 & -\frac{3\sqrt{6}}{80} & 0 & -\frac{\sqrt{14}}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{80} & 0 & \frac{7\sqrt{2}}{80} & 0 & \frac{\sqrt{30}}{80} & 0 & \frac{\sqrt{6}}{80} & 0 \\
0 & -\frac{3}{112} & 0 & -\frac{\sqrt{2}}{16} & 0 & -\frac{9\sqrt{5}}{80} & -\frac{\sqrt{210}}{6720} & 0 & -\frac{\sqrt{10}}{448} & 0 & -\frac{\sqrt{6}}{192} & 0 & -\frac{\sqrt{30}}{320} & 0 \\
\frac{3}{112} & 0 & \frac{3\sqrt{10}}{112} & 0 & \frac{\sqrt{5}}{16} & 0 & 0 & \frac{23\sqrt{6}}{6720} & 0 & \frac{13\sqrt{30}}{6720} & 0 & \frac{\sqrt{2}}{320} & 0 & -\frac{\sqrt{42}}{320} \\
0 & -\frac{3\sqrt{10}}{112} & 0 & -\frac{3\sqrt{5}}{56} & 0 & -\frac{\sqrt{2}}{16} & -\frac{\sqrt{21}}{480} & 0 & -\frac{11}{1120} & 0 & -\frac{\sqrt{15}}{3360} & 0 & \frac{\sqrt{3}}{160} & 0 \\
\frac{\sqrt{2}}{16} & 0 & \frac{3\sqrt{5}}{56} & 0 & \frac{3\sqrt{10}}{112} & 0 & 0 & \frac{\sqrt{3}}{160} & 0 & -\frac{\sqrt{15}}{3360} & 0 & -\frac{11}{1120} & 0 & -\frac{\sqrt{21}}{480} \\
0 & -\frac{\sqrt{5}}{16} & 0 & -\frac{3\sqrt{10}}{112} & 0 & -\frac{3}{112} & -\frac{\sqrt{42}}{320} & 0 & \frac{\sqrt{2}}{320} & 0 & \frac{13\sqrt{30}}{6720} & 0 & \frac{23\sqrt{6}}{6720} & 0 \\
\frac{9\sqrt{5}}{80} & 0 & \frac{\sqrt{2}}{16} & 0 & \frac{3}{112} & 0 & 0 & -\frac{\sqrt{30}}{320} & 0 & -\frac{\sqrt{6}}{192} & 0 & -\frac{\sqrt{10}}{448} & 0 & -\frac{\sqrt{210}}{6720}
\end{bmatrix}$$

$\boxed{\text{x367}}$ $\mathbb{T}_{5,1}^{(1,0;a)}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{400} & 0 & -\frac{\sqrt{42}i}{80} & 0 & -\frac{9\sqrt{70}i}{400} & 0 & -\frac{\sqrt{30}i}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{400} & 0 & -\frac{3\sqrt{210}i}{400} & 0 & \frac{3\sqrt{14}i}{80} & 0 & \frac{9\sqrt{70}i}{400} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{70}i}{400} & 0 & \frac{3\sqrt{14}i}{80} & 0 & -\frac{3\sqrt{210}i}{400} & 0 & -\frac{9\sqrt{10}i}{400} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{80} & 0 & -\frac{9\sqrt{70}i}{400} & 0 & -\frac{\sqrt{42}i}{80} & 0 & \frac{\sqrt{210}i}{400} & 0 \\
0 & -\frac{3\sqrt{35}i}{560} & 0 & -\frac{9\sqrt{70}i}{560} & 0 & -\frac{3\sqrt{7}i}{112} & \frac{\sqrt{6}i}{960} & 0 & -\frac{\sqrt{14}i}{448} & 0 & -\frac{3\sqrt{210}i}{2240} & 0 & -\frac{\sqrt{42}i}{1344} & 0 \\
-\frac{3\sqrt{35}i}{560} & 0 & \frac{3\sqrt{14}i}{112} & 0 & \frac{9\sqrt{7}i}{112} & 0 & 0 & -\frac{23\sqrt{210}i}{33600} & 0 & \frac{13\sqrt{42}i}{6720} & 0 & \frac{9\sqrt{70}i}{11200} & 0 & -\frac{\sqrt{30}i}{960} \\
0 & \frac{3\sqrt{14}i}{112} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & -\frac{9\sqrt{70}i}{560} & \frac{3\sqrt{15}i}{800} & 0 & \frac{11\sqrt{35}i}{5600} & 0 & -\frac{\sqrt{21}i}{3360} & 0 & \frac{9\sqrt{105}i}{5600} & 0 \\
-\frac{9\sqrt{70}i}{560} & 0 & -\frac{3\sqrt{7}i}{56} & 0 & \frac{3\sqrt{14}i}{112} & 0 & 0 & -\frac{9\sqrt{105}i}{5600} & 0 & \frac{\sqrt{21}i}{3360} & 0 & -\frac{11\sqrt{35}i}{5600} & 0 & -\frac{3\sqrt{15}i}{800} \\
0 & \frac{9\sqrt{7}i}{112} & 0 & \frac{3\sqrt{14}i}{112} & 0 & -\frac{3\sqrt{35}i}{560} & \frac{\sqrt{30}i}{960} & 0 & -\frac{9\sqrt{70}i}{11200} & 0 & -\frac{13\sqrt{42}i}{6720} & 0 & \frac{23\sqrt{210}i}{33600} & 0 \\
-\frac{3\sqrt{7}i}{112} & 0 & -\frac{9\sqrt{70}i}{560} & 0 & -\frac{3\sqrt{35}i}{560} & 0 & 0 & \frac{\sqrt{42}i}{1344} & 0 & \frac{3\sqrt{210}i}{2240} & 0 & \frac{\sqrt{14}i}{448} & 0 & -\frac{\sqrt{6}i}{960}
\end{bmatrix}$$

$\boxed{\text{x368}}$ $\mathbb{T}_{5,2}^{(1,0;a)}(E_u,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{400} & 0 & -\frac{\sqrt{42}}{80} & 0 & \frac{9\sqrt{70}}{400} & 0 & -\frac{\sqrt{30}}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}}{400} & 0 & \frac{3\sqrt{210}}{400} & 0 & \frac{3\sqrt{14}}{80} & 0 & -\frac{9\sqrt{70}}{400} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{70}}{400} & 0 & -\frac{3\sqrt{14}}{80} & 0 & -\frac{3\sqrt{210}}{400} & 0 & \frac{9\sqrt{10}}{400} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{80} & 0 & -\frac{9\sqrt{70}}{400} & 0 & \frac{\sqrt{42}}{80} & 0 & \frac{\sqrt{210}}{400} & 0 \\
0 & -\frac{3\sqrt{35}}{560} & 0 & \frac{9\sqrt{70}}{560} & 0 & -\frac{3\sqrt{7}}{112} & -\frac{\sqrt{6}}{960} & 0 & -\frac{\sqrt{14}}{448} & 0 & \frac{3\sqrt{210}}{2240} & 0 & -\frac{\sqrt{42}}{1344} & 0 \\
\frac{3\sqrt{35}}{560} & 0 & \frac{3\sqrt{14}}{112} & 0 & -\frac{9\sqrt{7}}{112} & 0 & 0 & \frac{23\sqrt{210}}{33600} & 0 & \frac{13\sqrt{42}}{6720} & 0 & -\frac{9\sqrt{70}}{11200} & 0 & -\frac{\sqrt{30}}{960} \\
0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{3\sqrt{7}}{56} & 0 & \frac{9\sqrt{70}}{560} & \frac{3\sqrt{15}}{800} & 0 & -\frac{11\sqrt{35}}{5600} & 0 & -\frac{\sqrt{21}}{3360} & 0 & -\frac{9\sqrt{105}}{5600} & 0 \\
-\frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{7}}{56} & 0 & \frac{3\sqrt{14}}{112} & 0 & 0 & -\frac{9\sqrt{105}}{5600} & 0 & -\frac{\sqrt{21}}{3360} & 0 & -\frac{11\sqrt{35}}{5600} & 0 & \frac{3\sqrt{15}}{800} \\
0 & \frac{9\sqrt{7}}{112} & 0 & -\frac{3\sqrt{14}}{112} & 0 & -\frac{3\sqrt{35}}{560} & -\frac{\sqrt{30}}{960} & 0 & -\frac{9\sqrt{70}}{11200} & 0 & \frac{13\sqrt{42}}{6720} & 0 & \frac{23\sqrt{210}}{33600} & 0 \\
\frac{3\sqrt{7}}{112} & 0 & -\frac{9\sqrt{70}}{560} & 0 & \frac{3\sqrt{35}}{560} & 0 & 0 & -\frac{\sqrt{42}}{1344} & 0 & \frac{3\sqrt{210}}{2240} & 0 & -\frac{\sqrt{14}}{448} & 0 & -\frac{\sqrt{6}}{960}
\end{bmatrix}$$

$\boxed{\text{x369}}$ $\mathbb{T}_{5,1}^{(1,0;a)}(E_u,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{200} & 0 & -\frac{\sqrt{14}i}{40} & 0 & \frac{\sqrt{210}i}{200} & 0 & \frac{3\sqrt{10}i}{40} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{200} & 0 & -\frac{3\sqrt{70}i}{200} & 0 & \frac{\sqrt{42}i}{40} & 0 & -\frac{\sqrt{210}i}{200} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{200} & 0 & \frac{\sqrt{42}i}{40} & 0 & -\frac{3\sqrt{70}i}{200} & 0 & \frac{\sqrt{30}i}{200} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{40} & 0 & \frac{\sqrt{210}i}{200} & 0 & -\frac{\sqrt{14}i}{40} & 0 & \frac{\sqrt{70}i}{200} & 0 \\
0 & -\frac{\sqrt{105}i}{280} & 0 & \frac{\sqrt{210}i}{280} & 0 & \frac{3\sqrt{21}i}{56} & \frac{\sqrt{2}i}{480} & 0 & -\frac{\sqrt{42}i}{672} & 0 & \frac{\sqrt{70}i}{1120} & 0 & \frac{\sqrt{14}i}{224} & 0 \\
-\frac{\sqrt{105}i}{280} & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{21}i}{56} & 0 & 0 & -\frac{23\sqrt{70}i}{16800} & 0 & \frac{13\sqrt{14}i}{3360} & 0 & -\frac{\sqrt{210}i}{5600} & 0 & \frac{\sqrt{10}i}{160} \\
0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{210}i}{280} & -\frac{\sqrt{5}i}{400} & 0 & \frac{11\sqrt{105}i}{8400} & 0 & -\frac{\sqrt{7}i}{1680} & 0 & -\frac{3\sqrt{35}i}{2800} & 0 \\
\frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & \frac{3\sqrt{35}i}{2800} & 0 & \frac{\sqrt{7}i}{1680} & 0 & -\frac{11\sqrt{105}i}{8400} & 0 & \frac{\sqrt{5}i}{400} \\
0 & -\frac{\sqrt{21}i}{56} & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{105}i}{280} & -\frac{\sqrt{10}i}{160} & 0 & \frac{\sqrt{210}i}{5600} & 0 & -\frac{13\sqrt{14}i}{3360} & 0 & \frac{23\sqrt{70}i}{16800} & 0 \\
\frac{3\sqrt{21}i}{56} & 0 & \frac{\sqrt{210}i}{280} & 0 & -\frac{\sqrt{105}i}{280} & 0 & 0 & -\frac{\sqrt{14}i}{224} & 0 & -\frac{\sqrt{70}i}{1120} & 0 & \frac{\sqrt{42}i}{672} & 0 & -\frac{\sqrt{2}i}{480}
\end{bmatrix}$$

$\boxed{\text{x370}}$ $\mathbb{T}_{5,2}^{(1,0;a)}(E_u,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{200} & 0 & -\frac{\sqrt{14}}{40} & 0 & -\frac{\sqrt{210}}{200} & 0 & \frac{3\sqrt{10}}{40} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{200} & 0 & \frac{3\sqrt{70}}{200} & 0 & \frac{\sqrt{42}}{40} & 0 & \frac{\sqrt{210}}{200} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{200} & 0 & -\frac{\sqrt{42}}{40} & 0 & -\frac{3\sqrt{70}}{200} & 0 & -\frac{\sqrt{30}}{200} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{40} & 0 & \frac{\sqrt{210}}{200} & 0 & \frac{\sqrt{14}}{40} & 0 & \frac{\sqrt{70}}{200} & 0 \\
0 & -\frac{\sqrt{105}}{280} & 0 & -\frac{\sqrt{210}}{280} & 0 & \frac{3\sqrt{21}}{56} & -\frac{\sqrt{2}}{480} & 0 & -\frac{\sqrt{42}}{672} & 0 & -\frac{\sqrt{70}}{1120} & 0 & \frac{\sqrt{14}}{224} & 0 \\
\frac{\sqrt{105}}{280} & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{21}}{56} & 0 & 0 & \frac{23\sqrt{70}}{16800} & 0 & \frac{13\sqrt{14}}{3360} & 0 & \frac{\sqrt{210}}{5600} & 0 & \frac{\sqrt{10}}{160} \\
0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{210}}{280} & -\frac{\sqrt{5}}{400} & 0 & -\frac{11\sqrt{105}}{8400} & 0 & -\frac{\sqrt{7}}{1680} & 0 & \frac{3\sqrt{35}}{2800} & 0 \\
\frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & \frac{3\sqrt{35}}{2800} & 0 & -\frac{\sqrt{7}}{1680} & 0 & -\frac{11\sqrt{105}}{8400} & 0 & -\frac{\sqrt{5}}{400} \\
0 & -\frac{\sqrt{21}}{56} & 0 & -\frac{\sqrt{42}}{56} & 0 & -\frac{\sqrt{105}}{280} & \frac{\sqrt{10}}{160} & 0 & \frac{\sqrt{210}}{5600} & 0 & \frac{13\sqrt{14}}{3360} & 0 & \frac{23\sqrt{70}}{16800} & 0 \\
-\frac{3\sqrt{21}}{56} & 0 & \frac{\sqrt{210}}{280} & 0 & \frac{\sqrt{105}}{280} & 0 & 0 & \frac{\sqrt{14}}{224} & 0 & -\frac{\sqrt{70}}{1120} & 0 & -\frac{\sqrt{42}}{672} & 0 & -\frac{\sqrt{2}}{480}
\end{bmatrix}$$

$\boxed{\text{x371}}$ $\mathbb{T}_1^{(1,1;a)}(A_{2u}) = \begin{bmatrix} 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 \end{bmatrix}$

$\boxed{\text{x372}}$ $\mathbb{T}_3^{(1,1;a)}(A_{2u}) = \begin{bmatrix} 0 & \frac{9i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 \\ \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 \end{bmatrix}$

$\boxed{\text{x373}}$ $\mathbb{T}_3^{(1,1;a)}(B_{1u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{3\sqrt{30}}{56} & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\ \frac{15}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 \\ 0 & -\frac{3\sqrt{5}}{56} & 0 & 0 & 0 & -\frac{15}{56} & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 \\ 0 & 0 & -\frac{3\sqrt{30}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} \\ 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\ -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & -\frac{1}{56} & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & \frac{1}{56} & 0 \\ 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & \frac{\sqrt{14}}{56} \\ 0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x374}}$ $\mathbb{T}_3^{(1,1;a)}(B_{2u}) = \begin{bmatrix} 0 & 0 & 0 & \frac{3\sqrt{30}i}{56} & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\ -\frac{15i}{56} & 0 & 0 & 0 & \frac{3\sqrt{5}i}{56} & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 \\ 0 & \frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & -\frac{15i}{56} & 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 \\ 0 & 0 & \frac{3\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} \\ 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\ \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & \frac{i}{56} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & 0 & \frac{i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} \\ 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x375}}$ $\mathbb{T}_{1,1}^{(1,1;a)}(E_u) = \begin{bmatrix} \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{10} & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{5}i}{35} & 0 & -\frac{4\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{2}i}{35} & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{6i}{35} & 0 & -\frac{4\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{2}i}{35} & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{30}i}{56} & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & \frac{\sqrt{42}i}{56} \end{bmatrix}$

$\boxed{\text{x376}}$ $\mathbb{T}_{1,2}^{(1,1;a)}(E_u) = \begin{bmatrix} -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{10} & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{5}}{35} & 0 & -\frac{4\sqrt{2}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 \\ 0 & \frac{4\sqrt{2}}{35} & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 \\ 0 & 0 & \frac{6}{35} & 0 & -\frac{4\sqrt{2}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{2}}{35} & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & \frac{\sqrt{42}}{56} \end{bmatrix}$

$\boxed{\text{x377}}$ $\mathbb{T}_{3,1}^{(1,1;a)}(E_u,1) =$

$$\begin{bmatrix}
\frac{9\sqrt{5}i}{224} & 0 & -\frac{27\sqrt{2}i}{224} & 0 & \frac{45i}{224} & 0 & 0 & -\frac{\sqrt{30}i}{112} & 0 & \frac{\sqrt{6}i}{56} & 0 & -\frac{\sqrt{10}i}{112} & 0 & 0 \\
0 & -\frac{3\sqrt{3}i}{32} & 0 & \frac{3\sqrt{6}i}{224} & 0 & \frac{15\sqrt{15}i}{224} & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{112} & 0 & -\frac{\sqrt{10}i}{56} & 0 \\
-\frac{15\sqrt{15}i}{224} & 0 & -\frac{3\sqrt{6}i}{224} & 0 & \frac{3\sqrt{3}i}{32} & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{3\sqrt{2}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} \\
0 & -\frac{45i}{224} & 0 & \frac{27\sqrt{2}i}{224} & 0 & -\frac{9\sqrt{5}i}{224} & 0 & 0 & -\frac{\sqrt{10}i}{112} & 0 & \frac{\sqrt{6}i}{56} & 0 & -\frac{\sqrt{30}i}{112} & 0 \\
0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{5\sqrt{10}i}{168} & 0 & 0 & -\frac{\sqrt{42}i}{224} & 0 & \frac{3\sqrt{2}i}{112} & 0 & -\frac{\sqrt{30}i}{224} & 0 & 0 & 0 \\
-\frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{2}i}{56} & 0 & \frac{5i}{42} & 0 & 0 & \frac{\sqrt{30}i}{224} & 0 & \frac{\sqrt{6}i}{112} & 0 & -\frac{3\sqrt{10}i}{224} & 0 & 0 \\
0 & \frac{\sqrt{2}i}{56} & 0 & \frac{i}{14} & 0 & \frac{5\sqrt{10}i}{168} & \frac{\sqrt{105}i}{224} & 0 & \frac{3\sqrt{5}i}{224} & 0 & -\frac{\sqrt{3}i}{224} & 0 & -\frac{3\sqrt{15}i}{224} & 0 \\
\frac{5\sqrt{10}i}{168} & 0 & \frac{i}{14} & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & \frac{3\sqrt{15}i}{224} & 0 & \frac{\sqrt{3}i}{224} & 0 & -\frac{3\sqrt{5}i}{224} & 0 & -\frac{\sqrt{105}i}{224} \\
0 & \frac{5i}{42} & 0 & \frac{\sqrt{2}i}{56} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & \frac{3\sqrt{10}i}{224} & 0 & -\frac{\sqrt{6}i}{112} & 0 & -\frac{\sqrt{30}i}{224} & 0 \\
0 & 0 & \frac{5\sqrt{10}i}{168} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{224} & 0 & -\frac{3\sqrt{2}i}{112} & 0 & \frac{\sqrt{42}i}{224}
\end{bmatrix}$$

$\boxed{\text{x378}}$ $\mathbb{T}_{3,2}^{(1,1;a)}(E_u,1) =$

$$\begin{bmatrix}
-\frac{9\sqrt{5}}{224} & 0 & -\frac{27\sqrt{2}}{224} & 0 & -\frac{45}{224} & 0 & 0 & \frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{10}}{112} & 0 & 0 \\
0 & \frac{3\sqrt{3}}{32} & 0 & \frac{3\sqrt{6}}{224} & 0 & -\frac{15\sqrt{15}}{224} & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & \frac{3\sqrt{2}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 \\
-\frac{15\sqrt{15}}{224} & 0 & \frac{3\sqrt{6}}{224} & 0 & \frac{3\sqrt{3}}{32} & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & -\frac{3\sqrt{2}}{112} & 0 & 0 & 0 & \frac{\sqrt{70}}{112} \\
0 & -\frac{45}{224} & 0 & -\frac{27\sqrt{2}}{224} & 0 & -\frac{9\sqrt{5}}{224} & 0 & 0 & -\frac{\sqrt{10}}{112} & 0 & -\frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{30}}{112} & 0 \\
0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{5\sqrt{10}}{168} & 0 & 0 & \frac{\sqrt{42}}{224} & 0 & \frac{3\sqrt{2}}{112} & 0 & \frac{\sqrt{30}}{224} & 0 & 0 & 0 \\
\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{2}}{56} & 0 & -\frac{5}{42} & 0 & 0 & -\frac{\sqrt{30}}{224} & 0 & \frac{\sqrt{6}}{112} & 0 & \frac{3\sqrt{10}}{224} & 0 & 0 \\
0 & -\frac{\sqrt{2}}{56} & 0 & \frac{1}{14} & 0 & -\frac{5\sqrt{10}}{168} & \frac{\sqrt{105}}{224} & 0 & -\frac{3\sqrt{5}}{224} & 0 & -\frac{\sqrt{3}}{224} & 0 & \frac{3\sqrt{15}}{224} & 0 \\
\frac{5\sqrt{10}}{168} & 0 & -\frac{1}{14} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & \frac{3\sqrt{15}}{224} & 0 & -\frac{\sqrt{3}}{224} & 0 & -\frac{3\sqrt{5}}{224} & 0 & \frac{\sqrt{105}}{224} \\
0 & \frac{5}{42} & 0 & -\frac{\sqrt{2}}{56} & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & \frac{3\sqrt{10}}{224} & 0 & \frac{\sqrt{6}}{112} & 0 & -\frac{\sqrt{30}}{224} & 0 \\
0 & 0 & \frac{5\sqrt{10}}{168} & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{224} & 0 & \frac{3\sqrt{2}}{112} & 0 & \frac{\sqrt{42}}{224}
\end{bmatrix}$$

$\boxed{\text{x379}}$ $\mathbb{T}_{3,1}^{(1,1;a)}(E_u,2) =$

$$\begin{bmatrix}
\frac{15\sqrt{3}i}{224} & 0 & -\frac{9\sqrt{30}i}{224} & 0 & -\frac{9\sqrt{15}i}{224} & 0 & 0 & -\frac{5\sqrt{2}i}{112} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{6}i}{112} & 0 & 0 \\
0 & -\frac{3\sqrt{5}i}{32} & 0 & \frac{3\sqrt{10}i}{224} & 0 & -\frac{45i}{224} & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & \frac{\sqrt{30}i}{112} & 0 & \frac{\sqrt{6}i}{56} & 0 \\
\frac{45i}{224} & 0 & -\frac{3\sqrt{10}i}{224} & 0 & \frac{3\sqrt{5}i}{32} & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{30}i}{112} & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} \\
0 & \frac{9\sqrt{15}i}{224} & 0 & \frac{9\sqrt{30}i}{224} & 0 & -\frac{15\sqrt{3}i}{224} & 0 & 0 & \frac{\sqrt{6}i}{112} & 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{5\sqrt{2}i}{112} & 0 \\
0 & -\frac{5\sqrt{3}i}{84} & 0 & -\frac{5\sqrt{6}i}{168} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & \frac{\sqrt{30}i}{112} & 0 & \frac{3\sqrt{2}i}{224} & 0 & 0 & 0 \\
-\frac{5\sqrt{3}i}{84} & 0 & \frac{\sqrt{30}i}{168} & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & \frac{5\sqrt{2}i}{224} & 0 & \frac{\sqrt{10}i}{112} & 0 & \frac{3\sqrt{6}i}{224} & 0 & 0 \\
0 & \frac{\sqrt{30}i}{168} & 0 & \frac{\sqrt{15}i}{42} & 0 & -\frac{5\sqrt{6}i}{168} & -\frac{3\sqrt{7}i}{224} & 0 & \frac{5\sqrt{3}i}{224} & 0 & -\frac{\sqrt{5}i}{224} & 0 & \frac{9i}{224} & 0 \\
-\frac{5\sqrt{6}i}{168} & 0 & \frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & -\frac{9i}{224} & 0 & \frac{\sqrt{5}i}{224} & 0 & -\frac{5\sqrt{3}i}{224} & 0 & \frac{3\sqrt{7}i}{224} \\
0 & -\frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{30}i}{168} & 0 & -\frac{5\sqrt{3}i}{84} & 0 & 0 & -\frac{3\sqrt{6}i}{224} & 0 & -\frac{\sqrt{10}i}{112} & 0 & -\frac{5\sqrt{2}i}{224} & 0 \\
0 & 0 & -\frac{5\sqrt{6}i}{168} & 0 & -\frac{5\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{224} & 0 & -\frac{\sqrt{30}i}{112} & 0 & \frac{\sqrt{70}i}{224}
\end{bmatrix}$$

$\boxed{\text{x380}}$ $\mathbb{T}_{3,2}^{(1,1;a)}(E_u,2) =$

$$\begin{bmatrix}
-\frac{15\sqrt{3}}{224} & 0 & -\frac{9\sqrt{30}}{224} & 0 & \frac{9\sqrt{15}}{224} & 0 & 0 & \frac{5\sqrt{2}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{6}}{112} & 0 & 0 \\
0 & \frac{3\sqrt{5}}{32} & 0 & \frac{3\sqrt{10}}{224} & 0 & \frac{45}{224} & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & \frac{\sqrt{30}}{112} & 0 & -\frac{\sqrt{6}}{56} & 0 \\
\frac{45}{224} & 0 & \frac{3\sqrt{10}}{224} & 0 & \frac{3\sqrt{5}}{32} & 0 & 0 & \frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{30}}{112} & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} \\
0 & \frac{9\sqrt{15}}{224} & 0 & -\frac{9\sqrt{30}}{224} & 0 & -\frac{15\sqrt{3}}{224} & 0 & 0 & \frac{\sqrt{6}}{112} & 0 & -\frac{\sqrt{10}}{56} & 0 & -\frac{5\sqrt{2}}{112} & 0 \\
0 & -\frac{5\sqrt{3}}{84} & 0 & \frac{5\sqrt{6}}{168} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & \frac{\sqrt{30}}{112} & 0 & -\frac{3\sqrt{2}}{224} & 0 & 0 & 0 \\
\frac{5\sqrt{3}}{84} & 0 & \frac{\sqrt{30}}{168} & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & -\frac{5\sqrt{2}}{224} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{3\sqrt{6}}{224} & 0 & 0 \\
0 & -\frac{\sqrt{30}}{168} & 0 & \frac{\sqrt{15}}{42} & 0 & \frac{5\sqrt{6}}{168} & -\frac{3\sqrt{7}}{224} & 0 & -\frac{5\sqrt{3}}{224} & 0 & -\frac{\sqrt{5}}{224} & 0 & -\frac{9}{224} & 0 \\
-\frac{5\sqrt{6}}{168} & 0 & -\frac{\sqrt{15}}{42} & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & -\frac{9}{224} & 0 & -\frac{\sqrt{5}}{224} & 0 & -\frac{5\sqrt{3}}{224} & 0 & -\frac{3\sqrt{7}}{224} \\
0 & -\frac{\sqrt{15}}{42} & 0 & -\frac{\sqrt{30}}{168} & 0 & -\frac{5\sqrt{3}}{84} & 0 & 0 & -\frac{3\sqrt{6}}{224} & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{5\sqrt{2}}{224} & 0 \\
0 & 0 & -\frac{5\sqrt{6}}{168} & 0 & \frac{5\sqrt{3}}{84} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{224} & 0 & \frac{\sqrt{30}}{112} & 0 & \frac{\sqrt{70}}{224}
\end{bmatrix}$$

- bra: $\langle\frac{5}{2},\frac{5}{2};f|$, $\langle\frac{5}{2},\frac{3}{2};f|$, $\langle\frac{5}{2},\frac{1}{2};f|$, $\langle\frac{5}{2},-\frac{1}{2};f|$, $\langle\frac{5}{2},-\frac{3}{2};f|$, $\langle\frac{5}{2},-\frac{5}{2};f|$, $\langle\frac{7}{2},\frac{7}{2};f|$, $\langle\frac{7}{2},\frac{5}{2};f|$, $\langle\frac{7}{2},\frac{3}{2};f|$, $\langle\frac{7}{2},\frac{1}{2};f|$, $\langle\frac{7}{2},-\frac{1}{2};f|$, $\langle\frac{7}{2},-\frac{3}{2};f|$, $\langle\frac{7}{2},-\frac{5}{2};f|$, $\langle\frac{7}{2},-\frac{7}{2};f|$
- ket: $|\frac{5}{2},\frac{5}{2};f\rangle$, $|\frac{5}{2},\frac{3}{2};f\rangle$, $|\frac{5}{2},\frac{1}{2};f\rangle$, $|\frac{5}{2},-\frac{1}{2};f\rangle$, $|\frac{5}{2},-\frac{3}{2};f\rangle$, $|\frac{5}{2},-\frac{5}{2};f\rangle$, $|\frac{7}{2},\frac{7}{2};f\rangle$, $|\frac{7}{2},\frac{5}{2};f\rangle$, $|\frac{7}{2},\frac{3}{2};f\rangle$, $|\frac{7}{2},\frac{1}{2};f\rangle$, $|\frac{7}{2},-\frac{1}{2};f\rangle$, $|\frac{7}{2},-\frac{3}{2};f\rangle$, $|\frac{7}{2},-\frac{5}{2};f\rangle$, $|\frac{7}{2},-\frac{7}{2};f\rangle$

$\boxed{\text{x381}}$ $\mathbb{Q}_0^{(a)}(A_{1g}) =$

$$\begin{bmatrix}
\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14}
\end{bmatrix}$$

$\boxed{\text{x382}}$ $\mathbb{Q}_2^{(a)}(A_{1g}) =$

$$\begin{bmatrix}
-\frac{5\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{25\sqrt{42}}{588} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{25\sqrt{42}}{588} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{588} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{84}
\end{bmatrix}$$

$\boxed{\text{x383}}$ $\mathbb{Q}_4^{(a)}(A_{1g},1) =$

$$\begin{bmatrix}
\frac{\sqrt{33}}{84} & 0 & 0 & 0 & \frac{\sqrt{165}}{84} & 0 & 0 & \frac{5\sqrt{22}}{154} & 0 & 0 & 0 & \frac{5\sqrt{66}}{462} & 0 & 0 \\
0 & -\frac{\sqrt{33}}{28} & 0 & 0 & 0 & \frac{\sqrt{165}}{84} & 0 & 0 & -\frac{2\sqrt{330}}{231} & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & 0 \\
0 & 0 & \frac{\sqrt{33}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{154} & 0 & 0 & 0 & \frac{\sqrt{385}}{154} \\
0 & 0 & 0 & \frac{\sqrt{33}}{42} & 0 & 0 & -\frac{\sqrt{385}}{154} & 0 & 0 & 0 & \frac{5\sqrt{11}}{154} & 0 & 0 & 0 \\
\frac{\sqrt{165}}{84} & 0 & 0 & 0 & -\frac{\sqrt{33}}{28} & 0 & 0 & -\frac{\sqrt{110}}{77} & 0 & 0 & 0 & \frac{2\sqrt{330}}{231} & 0 & 0 \\
0 & \frac{\sqrt{165}}{84} & 0 & 0 & 0 & \frac{\sqrt{33}}{84} & 0 & 0 & -\frac{5\sqrt{66}}{462} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{154} & 0 \\
0 & 0 & 0 & -\frac{\sqrt{385}}{154} & 0 & 0 & \frac{\sqrt{33}}{44} & 0 & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 \\
\frac{5\sqrt{22}}{154} & 0 & 0 & 0 & -\frac{\sqrt{110}}{77} & 0 & 0 & -\frac{13\sqrt{33}}{308} & 0 & 0 & 0 & \frac{15\sqrt{11}}{308} & 0 & 0 \\
0 & -\frac{2\sqrt{330}}{231} & 0 & 0 & 0 & -\frac{5\sqrt{66}}{462} & 0 & 0 & -\frac{3\sqrt{33}}{308} & 0 & 0 & 0 & \frac{15\sqrt{11}}{308} & 0 \\
0 & 0 & -\frac{5\sqrt{11}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{33}}{308} & 0 & 0 & 0 & \frac{\sqrt{1155}}{308} \\
0 & 0 & 0 & \frac{5\sqrt{11}}{154} & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & \frac{9\sqrt{33}}{308} & 0 & 0 & 0 \\
\frac{5\sqrt{66}}{462} & 0 & 0 & 0 & \frac{2\sqrt{330}}{231} & 0 & 0 & \frac{15\sqrt{11}}{308} & 0 & 0 & 0 & -\frac{3\sqrt{33}}{308} & 0 & 0 \\
0 & \frac{\sqrt{110}}{77} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{154} & 0 & 0 & \frac{15\sqrt{11}}{308} & 0 & 0 & 0 & -\frac{13\sqrt{33}}{308} & 0 \\
0 & 0 & \frac{\sqrt{385}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & \frac{\sqrt{33}}{44}
\end{bmatrix}$$

$\boxed{\text{x384}}$ $\mathbb{Q}_4^{(a)}(A_{1g},2) =$

$$\begin{bmatrix}
\frac{\sqrt{1155}}{588} & 0 & 0 & 0 & -\frac{\sqrt{231}}{84} & 0 & 0 & \frac{5\sqrt{770}}{1078} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{462} & 0 & 0 \\
0 & -\frac{\sqrt{1155}}{196} & 0 & 0 & 0 & -\frac{\sqrt{231}}{84} & 0 & 0 & -\frac{10\sqrt{462}}{1617} & 0 & 0 & 0 & -\frac{\sqrt{154}}{77} & 0 \\
0 & 0 & \frac{\sqrt{1155}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{385}}{1078} & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} \\
0 & 0 & 0 & \frac{\sqrt{1155}}{294} & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & \frac{5\sqrt{385}}{1078} & 0 & 0 & 0 \\
-\frac{\sqrt{231}}{84} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{196} & 0 & 0 & \frac{\sqrt{154}}{77} & 0 & 0 & 0 & \frac{10\sqrt{462}}{1617} & 0 & 0 \\
0 & -\frac{\sqrt{231}}{84} & 0 & 0 & 0 & \frac{\sqrt{1155}}{588} & 0 & 0 & \frac{\sqrt{2310}}{462} & 0 & 0 & 0 & -\frac{5\sqrt{770}}{1078} & 0 \\
0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & -\frac{\sqrt{33}}{44} & 0 & 0 & 0 \\
\frac{5\sqrt{770}}{1078} & 0 & 0 & 0 & \frac{\sqrt{154}}{77} & 0 & 0 & -\frac{13\sqrt{1155}}{2156} & 0 & 0 & 0 & -\frac{3\sqrt{385}}{308} & 0 & 0 \\
0 & -\frac{10\sqrt{462}}{1617} & 0 & 0 & 0 & \frac{\sqrt{2310}}{462} & 0 & 0 & -\frac{3\sqrt{1155}}{2156} & 0 & 0 & 0 & -\frac{3\sqrt{385}}{308} & 0 \\
0 & 0 & -\frac{5\sqrt{385}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{1155}}{2156} & 0 & 0 & 0 & -\frac{\sqrt{33}}{44} \\
0 & 0 & 0 & \frac{5\sqrt{385}}{1078} & 0 & 0 & -\frac{\sqrt{33}}{44} & 0 & 0 & 0 & \frac{9\sqrt{1155}}{2156} & 0 & 0 & 0 \\
-\frac{\sqrt{2310}}{462} & 0 & 0 & 0 & \frac{10\sqrt{462}}{1617} & 0 & 0 & -\frac{3\sqrt{385}}{308} & 0 & 0 & 0 & -\frac{3\sqrt{1155}}{2156} & 0 & 0 \\
0 & -\frac{\sqrt{154}}{77} & 0 & 0 & 0 & -\frac{5\sqrt{770}}{1078} & 0 & 0 & -\frac{3\sqrt{385}}{308} & 0 & 0 & 0 & -\frac{13\sqrt{1155}}{2156} & 0 \\
0 & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{44} & 0 & 0 & 0 & \frac{\sqrt{1155}}{308}
\end{bmatrix}$$

$\boxed{\text{x385}}$ $\mathbb{Q}_6^{(a)}(A_{1g},1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{616} & 0 & 0 & 0 & \frac{\sqrt{462}}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & -\frac{\sqrt{770}}{88} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{308} & 0 & 0 & 0 & \frac{\sqrt{55}}{44} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & 0 & \frac{5\sqrt{77}}{308} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{88} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{88} & 0 & 0 & 0 & \frac{\sqrt{154}}{616} & 0 \\
0 & 0 & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & -\frac{\sqrt{231}}{1848} & 0 & 0 & 0 & \frac{\sqrt{165}}{88} & 0 & 0 & 0 \\
-\frac{\sqrt{154}}{616} & 0 & 0 & 0 & \frac{\sqrt{770}}{88} & 0 & 0 & \frac{5\sqrt{231}}{1848} & 0 & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 \\
0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & -\frac{\sqrt{462}}{88} & 0 & 0 & -\frac{3\sqrt{231}}{616} & 0 & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 \\
0 & 0 & -\frac{5\sqrt{77}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{1848} & 0 & 0 & 0 & \frac{\sqrt{165}}{88} \\
0 & 0 & 0 & \frac{5\sqrt{77}}{308} & 0 & 0 & \frac{\sqrt{165}}{88} & 0 & 0 & 0 & \frac{5\sqrt{231}}{1848} & 0 & 0 & 0 \\
\frac{\sqrt{462}}{88} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{616} & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & 0 & -\frac{3\sqrt{231}}{616} & 0 & 0 \\
0 & -\frac{\sqrt{770}}{88} & 0 & 0 & 0 & \frac{\sqrt{154}}{616} & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & 0 & \frac{5\sqrt{231}}{1848} & 0 \\
0 & 0 & \frac{\sqrt{55}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{88} & 0 & 0 & 0 & -\frac{\sqrt{231}}{1848}
\end{bmatrix}$$

$\boxed{\text{x386}}$ $\mathbb{Q}_6^{(a)}(A_{1g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{88} & 0 & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{88} & 0 & 0 & 0 & \frac{\sqrt{110}}{88} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{44} & 0 & 0 & 0 & -\frac{\sqrt{385}}{308} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{308} & 0 & 0 & 0 & \frac{5\sqrt{11}}{44} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{88} & 0 & 0 & 0 & \frac{\sqrt{22}}{88} & 0 \\
0 & 0 & 0 & \frac{\sqrt{385}}{308} & 0 & 0 & -\frac{\sqrt{33}}{264} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{616} & 0 & 0 & 0 \\
-\frac{\sqrt{22}}{88} & 0 & 0 & 0 & -\frac{\sqrt{110}}{88} & 0 & 0 & \frac{5\sqrt{33}}{264} & 0 & 0 & 0 & \frac{\sqrt{11}}{88} & 0 & 0 \\
0 & \frac{\sqrt{330}}{88} & 0 & 0 & 0 & \frac{\sqrt{66}}{88} & 0 & 0 & -\frac{3\sqrt{33}}{88} & 0 & 0 & 0 & \frac{\sqrt{11}}{88} & 0 \\
0 & 0 & -\frac{5\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{264} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{616} \\
0 & 0 & 0 & \frac{5\sqrt{11}}{44} & 0 & 0 & -\frac{\sqrt{1155}}{616} & 0 & 0 & 0 & \frac{5\sqrt{33}}{264} & 0 & 0 & 0 \\
-\frac{\sqrt{66}}{88} & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & \frac{\sqrt{11}}{88} & 0 & 0 & 0 & -\frac{3\sqrt{33}}{88} & 0 & 0 \\
0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 & \frac{\sqrt{22}}{88} & 0 & 0 & \frac{\sqrt{11}}{88} & 0 & 0 & 0 & \frac{5\sqrt{33}}{264} & 0 \\
0 & 0 & -\frac{\sqrt{385}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{616} & 0 & 0 & 0 & -\frac{\sqrt{33}}{264}
\end{bmatrix}$$

$\boxed{\text{x387}}$ $\mathbb{Q}_4^{(a)}(A_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{66}i}{77} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{77} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{66}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{231}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{2\sqrt{66}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{154} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{2\sqrt{66}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{231}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}i}{154} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x388}}$ $\mathbb{Q}_6^{(a)}(A_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{22} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{154} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{770}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{308} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{308} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{55}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{770}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{308} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x389}}$ $\mathbb{Q}_2^{(a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{294} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{9\sqrt{7}}{98} & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 \\
-\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & -\frac{9\sqrt{7}}{98} & 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 \\
0 & -\frac{9\sqrt{7}}{98} & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 \\
0 & 0 & -\frac{9\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} \\
0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{294} & 0 & 0 & 0 \\
0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{210}}{294} & 0 & 0 & 0 \\
-\frac{\sqrt{105}}{294} & 0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 & 0 & 0 & -\frac{5\sqrt{210}}{294} & 0 & 0 \\
0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 & \frac{\sqrt{105}}{294} & 0 & 0 & -\frac{5\sqrt{210}}{294} & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 \\
0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}}{294} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} \\
0 & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x390}}$ $\mathbb{Q}_4^{(a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & -\frac{3\sqrt{154}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & \frac{3\sqrt{66}}{154} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1078} & 0 & 0 & 0 \\
-\frac{3\sqrt{154}}{196} & 0 & 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & -\frac{9\sqrt{231}}{1078} & 0 & 0 & 0 & \frac{17\sqrt{77}}{1078} & 0 & 0 \\
0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 & -\frac{3\sqrt{154}}{196} & 0 & 0 & -\frac{17\sqrt{77}}{1078} & 0 & 0 & 0 & \frac{9\sqrt{231}}{1078} & 0 \\
0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1078} & 0 & 0 & 0 & -\frac{3\sqrt{66}}{154} \\
0 & 0 & 0 & -\frac{3\sqrt{154}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{539} & 0 & 0 & 0 \\
0 & \frac{3\sqrt{66}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{9\sqrt{231}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{1078} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{17\sqrt{77}}{1078} & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0 & 0 & \frac{6\sqrt{231}}{539} & 0 & 0 & 0 \\
-\frac{5\sqrt{462}}{539} & 0 & 0 & 0 & \frac{\sqrt{2310}}{1078} & 0 & 0 & -\frac{3\sqrt{77}}{1078} & 0 & 0 & 0 & \frac{6\sqrt{231}}{539} & 0 & 0 \\
0 & -\frac{\sqrt{2310}}{1078} & 0 & 0 & 0 & \frac{5\sqrt{462}}{539} & 0 & 0 & \frac{6\sqrt{231}}{539} & 0 & 0 & 0 & -\frac{3\sqrt{77}}{1078} & 0 \\
0 & 0 & \frac{17\sqrt{77}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{231}}{539} & 0 & 0 & 0 & -\frac{3\sqrt{165}}{154} \\
0 & 0 & 0 & \frac{9\sqrt{231}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{1078} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{66}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x391}}$ $\mathbb{Q}_6^{(a)}(B_{1g},1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{210}}{56} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & \frac{\sqrt{42}}{168} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 \\
0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} \\
0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 \\
\frac{\sqrt{6}}{24} & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 \\
0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 & -\frac{1}{8} & 0 \\
0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{24} & 0 & 0 & 0 & \frac{\sqrt{105}}{168} \\
0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 & 0 \\
-\frac{\sqrt{210}}{56} & 0 & 0 & 0 & \frac{\sqrt{42}}{168} & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x392}}$ $\mathbb{Q}_6^{(a)}(B_{1g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 & 0 & -\frac{\sqrt{462}}{56} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1848} & 0 & 0 & 0 & \frac{5\sqrt{66}}{264} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{132} & 0 & 0 & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{44} & 0 & 0 & 0 & \frac{\sqrt{165}}{132} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}}{264} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1848} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{56} & 0 & 0 & 0 & \frac{\sqrt{330}}{264} & 0 & 0 & 0 \\
0 & \frac{\sqrt{2310}}{1848} & 0 & 0 & 0 & \frac{\sqrt{462}}{56} & 0 & 0 & -\frac{5\sqrt{231}}{1848} & 0 & 0 & 0 & -\frac{\sqrt{77}}{56} & 0 \\
0 & 0 & -\frac{\sqrt{165}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 & -\frac{\sqrt{77}}{56} \\
0 & 0 & 0 & \frac{\sqrt{55}}{44} & 0 & 0 & -\frac{5\sqrt{231}}{1848} & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 \\
-\frac{\sqrt{330}}{264} & 0 & 0 & 0 & -\frac{5\sqrt{66}}{264} & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 \\
0 & \frac{5\sqrt{66}}{264} & 0 & 0 & 0 & \frac{\sqrt{330}}{264} & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 \\
0 & 0 & -\frac{\sqrt{55}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 & -\frac{5\sqrt{231}}{1848} \\
0 & 0 & 0 & \frac{\sqrt{165}}{132} & 0 & 0 & -\frac{\sqrt{77}}{56} & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 \\
-\frac{\sqrt{462}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1848} & 0 & 0 & -\frac{\sqrt{77}}{56} & 0 & 0 & 0 & -\frac{5\sqrt{231}}{1848} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x393}}$ $\mathbb{Q}_2^{(a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{294} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{7}i}{98} & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & \frac{2\sqrt{21}i}{147} & 0 & 0 & 0 \\
-\frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & \frac{9\sqrt{7}i}{98} & 0 & 0 & \frac{\sqrt{210}i}{147} & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 \\
0 & -\frac{9\sqrt{7}i}{98} & 0 & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & \frac{\sqrt{210}i}{147} & 0 \\
0 & 0 & -\frac{9\sqrt{7}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{147} & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} \\
0 & 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{294} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{70}i}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & \frac{5\sqrt{210}i}{294} & 0 & 0 & 0 \\
-\frac{\sqrt{105}i}{294} & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{147} & 0 & 0 & -\frac{5\sqrt{70}i}{196} & 0 & 0 & 0 & \frac{5\sqrt{210}i}{294} & 0 & 0 \\
0 & -\frac{2\sqrt{21}i}{147} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{294} & 0 & 0 & -\frac{5\sqrt{210}i}{294} & 0 & 0 & 0 & \frac{5\sqrt{70}i}{196} & 0 \\
0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{294} & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} \\
0 & 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}i}{196} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x394}}$ $\mathbb{Q}_4^{(a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & -\frac{3\sqrt{154}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}i}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{770}i}{196} & 0 & 0 & -\frac{3\sqrt{66}i}{154} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1078} & 0 & 0 & 0 \\
\frac{3\sqrt{154}i}{196} & 0 & 0 & 0 & \frac{\sqrt{770}i}{196} & 0 & 0 & \frac{9\sqrt{231}i}{1078} & 0 & 0 & 0 & \frac{17\sqrt{77}i}{1078} & 0 & 0 \\
0 & -\frac{\sqrt{770}i}{196} & 0 & 0 & 0 & -\frac{3\sqrt{154}i}{196} & 0 & 0 & \frac{17\sqrt{77}i}{1078} & 0 & 0 & 0 & \frac{9\sqrt{231}i}{1078} & 0 \\
0 & 0 & -\frac{\sqrt{770}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1078} & 0 & 0 & 0 & -\frac{3\sqrt{66}i}{154} \\
0 & 0 & 0 & \frac{3\sqrt{154}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}i}{539} & 0 & 0 & 0 \\
0 & \frac{3\sqrt{66}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{9\sqrt{231}i}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{1078} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{17\sqrt{77}i}{1078} & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & \frac{6\sqrt{231}i}{539} & 0 & 0 & 0 \\
\frac{5\sqrt{462}i}{539} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1078} & 0 & 0 & \frac{3\sqrt{77}i}{1078} & 0 & 0 & 0 & \frac{6\sqrt{231}i}{539} & 0 & 0 \\
0 & \frac{\sqrt{2310}i}{1078} & 0 & 0 & 0 & \frac{5\sqrt{462}i}{539} & 0 & 0 & -\frac{6\sqrt{231}i}{539} & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{1078} & 0 \\
0 & 0 & -\frac{17\sqrt{77}i}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{231}i}{539} & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} \\
0 & 0 & 0 & -\frac{9\sqrt{231}i}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}i}{1078} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{66}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x395}}$ $\mathbb{Q}_6^{(a)}(B_{2g},1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x396}}$ $\mathbb{Q}_6^{(a)}(B_{2g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{462} & 0 & 0 & 0 & -\frac{\sqrt{330}i}{66} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{33} & 0 & 0 & 0 & \frac{\sqrt{11}i}{11} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{11} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{33} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{66} & 0 & 0 & 0 & \frac{\sqrt{462}i}{462} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{462}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{33}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{11}i}{11} & 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & \frac{\sqrt{33}i}{66} & 0 & 0 & 0 \\
-\frac{\sqrt{66}i}{66} & 0 & 0 & 0 & \frac{\sqrt{330}i}{66} & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & \frac{\sqrt{33}i}{66} & 0 & 0 \\
0 & \frac{\sqrt{330}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{66}i}{66} & 0 & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 \\
0 & 0 & -\frac{\sqrt{11}i}{11} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{66} & 0 & 0 & 0 & \frac{\sqrt{1155}i}{462} \\
0 & 0 & 0 & \frac{\sqrt{33}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x397}}$ $\mathbb{Q}_{2,1}^{(a)}(E_g) = \begin{bmatrix}
0 & \frac{3\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{84} & 0 & \frac{5\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{70}i}{98} & 0 & \frac{3\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{588} & 0 & \frac{11\sqrt{21}i}{588} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{70}i}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{7}i}{49} & 0 & -\frac{3\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{21}i}{588} & 0 & -\frac{\sqrt{105}i}{588} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{196} & 0 & -\frac{5\sqrt{3}i}{84} \\
-\frac{5\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}i}{588} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & \frac{5\sqrt{42}i}{147} & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{7}i}{196} & 0 & \frac{\sqrt{70}i}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{147} & 0 & \frac{5\sqrt{210}i}{588} & 0 & 0 & 0 & 0 \\
0 & -\frac{11\sqrt{21}i}{588} & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{588} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{11\sqrt{21}i}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{588} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & \frac{5\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}i}{588} & 0 & -\frac{5\sqrt{42}i}{147} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{147} & 0 & -\frac{5\sqrt{2}i}{28} \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0
\end{bmatrix}$

$\boxed{\text{x398}}$ $\mathbb{Q}_{2,2}^{(a)}(E_g) = \begin{bmatrix}
0 & -\frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{84} & 0 & -\frac{5\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{70}}{98} & 0 & -\frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{588} & 0 & -\frac{11\sqrt{21}}{588} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{196} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & -\frac{\sqrt{70}}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{7}}{49} & 0 & \frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{21}}{588} & 0 & \frac{\sqrt{105}}{588} & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{196} & 0 & \frac{5\sqrt{3}}{84} \\
\frac{5\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{105}}{588} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & -\frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{7}}{196} & 0 & -\frac{\sqrt{70}}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{147} & 0 & -\frac{5\sqrt{210}}{588} & 0 & 0 & 0 & 0 \\
0 & -\frac{11\sqrt{21}}{588} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}}{588} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}}{84} & 0 & -\frac{11\sqrt{21}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}}{588} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}}{196} & 0 & -\frac{5\sqrt{7}}{196} & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}}{588} & 0 & \frac{5\sqrt{42}}{147} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{147} & 0 & \frac{5\sqrt{2}}{28} \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0
\end{bmatrix}$

$\boxed{\text{x399}}$ $\mathbb{Q}_{4,1}^{(a)}(E_g,1) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{11}i}{28} & 0 & \frac{\sqrt{22}i}{56} & 0 & 0 & \frac{\sqrt{2310}i}{616} & 0 & \frac{5\sqrt{110}i}{308} & 0 & \frac{5\sqrt{66}i}{616} & 0 & 0 & 0 \\
-\frac{\sqrt{11}i}{28} & 0 & -\frac{\sqrt{110}i}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{66}i}{616} & 0 & -\frac{\sqrt{330}i}{308} & 0 & \frac{\sqrt{22}i}{88} & 0 & 0 \\
0 & \frac{\sqrt{110}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{56} & \frac{3\sqrt{231}i}{616} & 0 & \frac{\sqrt{11}i}{616} & 0 & -\frac{\sqrt{165}i}{88} & 0 & -\frac{\sqrt{33}i}{616} & 0 \\
-\frac{\sqrt{22}i}{56} & 0 & 0 & 0 & \frac{\sqrt{110}i}{56} & 0 & 0 & \frac{\sqrt{33}i}{616} & 0 & \frac{\sqrt{165}i}{88} & 0 & -\frac{\sqrt{11}i}{616} & 0 & -\frac{3\sqrt{231}i}{616} \\
0 & 0 & 0 & -\frac{\sqrt{110}i}{56} & 0 & -\frac{\sqrt{11}i}{28} & 0 & 0 & -\frac{\sqrt{22}i}{88} & 0 & \frac{\sqrt{330}i}{308} & 0 & \frac{13\sqrt{66}i}{616} & 0 \\
0 & 0 & \frac{\sqrt{22}i}{56} & 0 & \frac{\sqrt{11}i}{28} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}i}{616} & 0 & -\frac{5\sqrt{110}i}{308} & 0 & -\frac{\sqrt{2310}i}{616} \\
-\frac{\sqrt{2310}i}{616} & 0 & -\frac{3\sqrt{231}i}{616} & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{308} & 0 & \frac{3\sqrt{77}i}{308} & 0 & 0 & 0 & 0 \\
0 & \frac{13\sqrt{66}i}{616} & 0 & -\frac{\sqrt{33}i}{616} & 0 & 0 & -\frac{3\sqrt{385}i}{308} & 0 & -\frac{3\sqrt{165}i}{308} & 0 & \frac{3\sqrt{11}i}{154} & 0 & 0 & 0 \\
-\frac{5\sqrt{110}i}{308} & 0 & -\frac{\sqrt{11}i}{616} & 0 & \frac{\sqrt{22}i}{88} & 0 & 0 & \frac{3\sqrt{165}i}{308} & 0 & -\frac{9\sqrt{33}i}{308} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{330}i}{308} & 0 & -\frac{\sqrt{165}i}{88} & 0 & \frac{5\sqrt{66}i}{616} & -\frac{3\sqrt{77}i}{308} & 0 & \frac{9\sqrt{33}i}{308} & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{154} & 0 \\
-\frac{5\sqrt{66}i}{616} & 0 & \frac{\sqrt{165}i}{88} & 0 & -\frac{\sqrt{330}i}{308} & 0 & 0 & -\frac{3\sqrt{11}i}{154} & 0 & 0 & 0 & \frac{9\sqrt{33}i}{308} & 0 & -\frac{3\sqrt{77}i}{308} \\
0 & -\frac{\sqrt{22}i}{88} & 0 & \frac{\sqrt{11}i}{616} & 0 & \frac{5\sqrt{110}i}{308} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{33}i}{308} & 0 & \frac{3\sqrt{165}i}{308} & 0 \\
0 & 0 & \frac{\sqrt{33}i}{616} & 0 & -\frac{13\sqrt{66}i}{616} & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}i}{154} & 0 & -\frac{3\sqrt{165}i}{308} & 0 & -\frac{3\sqrt{385}i}{308} \\
0 & 0 & 0 & \frac{3\sqrt{231}i}{616} & 0 & \frac{\sqrt{2310}i}{616} & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}i}{308} & 0 & \frac{3\sqrt{385}i}{308} & 0
\end{bmatrix}$$

$\boxed{\text{x400}}$ $\mathbb{Q}_{4,2}^{(a)}(E_g,1) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{11}}{28} & 0 & \frac{\sqrt{22}}{56} & 0 & 0 & \frac{\sqrt{2310}}{616} & 0 & -\frac{5\sqrt{110}}{308} & 0 & \frac{5\sqrt{66}}{616} & 0 & 0 & 0 \\
-\frac{\sqrt{11}}{28} & 0 & \frac{\sqrt{110}}{56} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{66}}{616} & 0 & \frac{\sqrt{330}}{308} & 0 & \frac{\sqrt{22}}{88} & 0 & 0 \\
0 & \frac{\sqrt{110}}{56} & 0 & 0 & 0 & -\frac{\sqrt{22}}{56} & -\frac{3\sqrt{231}}{616} & 0 & \frac{\sqrt{11}}{616} & 0 & \frac{\sqrt{165}}{88} & 0 & -\frac{\sqrt{33}}{616} & 0 \\
\frac{\sqrt{22}}{56} & 0 & 0 & 0 & -\frac{\sqrt{110}}{56} & 0 & 0 & -\frac{\sqrt{33}}{616} & 0 & \frac{\sqrt{165}}{88} & 0 & \frac{\sqrt{11}}{616} & 0 & -\frac{3\sqrt{231}}{616} \\
0 & 0 & 0 & -\frac{\sqrt{110}}{56} & 0 & \frac{\sqrt{11}}{28} & 0 & 0 & \frac{\sqrt{22}}{88} & 0 & \frac{\sqrt{330}}{308} & 0 & -\frac{13\sqrt{66}}{616} & 0 \\
0 & 0 & -\frac{\sqrt{22}}{56} & 0 & \frac{\sqrt{11}}{28} & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{616} & 0 & -\frac{5\sqrt{110}}{308} & 0 & \frac{\sqrt{2310}}{616} \\
\frac{\sqrt{2310}}{616} & 0 & -\frac{3\sqrt{231}}{616} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}}{308} & 0 & \frac{3\sqrt{77}}{308} & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{66}}{616} & 0 & -\frac{\sqrt{33}}{616} & 0 & 0 & -\frac{3\sqrt{385}}{308} & 0 & \frac{3\sqrt{165}}{308} & 0 & \frac{3\sqrt{11}}{154} & 0 & 0 & 0 \\
-\frac{5\sqrt{110}}{308} & 0 & \frac{\sqrt{11}}{616} & 0 & \frac{\sqrt{22}}{88} & 0 & 0 & \frac{3\sqrt{165}}{308} & 0 & \frac{9\sqrt{33}}{308} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{330}}{308} & 0 & \frac{\sqrt{165}}{88} & 0 & \frac{5\sqrt{66}}{616} & \frac{3\sqrt{77}}{308} & 0 & \frac{9\sqrt{33}}{308} & 0 & 0 & 0 & -\frac{3\sqrt{11}}{154} & 0 \\
\frac{5\sqrt{66}}{616} & 0 & \frac{\sqrt{165}}{88} & 0 & \frac{\sqrt{330}}{308} & 0 & 0 & \frac{3\sqrt{11}}{154} & 0 & 0 & 0 & -\frac{9\sqrt{33}}{308} & 0 & -\frac{3\sqrt{77}}{308} \\
0 & \frac{\sqrt{22}}{88} & 0 & \frac{\sqrt{11}}{616} & 0 & -\frac{5\sqrt{110}}{308} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{33}}{308} & 0 & -\frac{3\sqrt{165}}{308} & 0 \\
0 & 0 & -\frac{\sqrt{33}}{616} & 0 & -\frac{13\sqrt{66}}{616} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{154} & 0 & -\frac{3\sqrt{165}}{308} & 0 & \frac{3\sqrt{385}}{308} \\
0 & 0 & 0 & -\frac{3\sqrt{231}}{616} & 0 & \frac{\sqrt{2310}}{616} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{308} & 0 & \frac{3\sqrt{385}}{308} & 0
\end{bmatrix}$$

$\boxed{\text{x401}}$ $\mathbb{Q}^{(a)}_{4,1}(E_g,2) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{77}i}{196} & 0 & -\frac{\sqrt{154}i}{56} & 0 & 0 & \frac{\sqrt{330}i}{616} & 0 & \frac{5\sqrt{770}i}{2156} & 0 & -\frac{5\sqrt{462}i}{616} & 0 & 0 & 0 \\
-\frac{\sqrt{77}i}{196} & 0 & -\frac{\sqrt{770}i}{392} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{462}i}{4312} & 0 & -\frac{\sqrt{2310}i}{2156} & 0 & -\frac{\sqrt{154}i}{88} & 0 & 0 \\
0 & \frac{\sqrt{770}i}{392} & 0 & 0 & 0 & \frac{\sqrt{154}i}{56} & -\frac{3\sqrt{33}i}{88} & 0 & \frac{\sqrt{77}i}{4312} & 0 & -\frac{\sqrt{1155}i}{616} & 0 & \frac{\sqrt{231}i}{616} & 0 \\
\frac{\sqrt{154}i}{56} & 0 & 0 & 0 & \frac{\sqrt{770}i}{392} & 0 & 0 & -\frac{\sqrt{231}i}{616} & 0 & \frac{\sqrt{1155}i}{616} & 0 & -\frac{\sqrt{77}i}{4312} & 0 & \frac{3\sqrt{33}i}{88} \\
0 & 0 & 0 & -\frac{\sqrt{770}i}{392} & 0 & -\frac{\sqrt{77}i}{196} & 0 & 0 & \frac{\sqrt{154}i}{88} & 0 & \frac{\sqrt{2310}i}{2156} & 0 & \frac{13\sqrt{462}i}{4312} & 0 \\
0 & 0 & -\frac{\sqrt{154}i}{56} & 0 & \frac{\sqrt{77}i}{196} & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{616} & 0 & -\frac{5\sqrt{770}i}{2156} & 0 & -\frac{\sqrt{330}i}{616} \\
-\frac{\sqrt{330}i}{616} & 0 & \frac{3\sqrt{33}i}{88} & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}i}{308} & 0 & -\frac{3\sqrt{11}i}{44} & 0 & 0 & 0 & 0 \\
0 & \frac{13\sqrt{462}i}{4312} & 0 & \frac{\sqrt{231}i}{616} & 0 & 0 & -\frac{3\sqrt{55}i}{308} & 0 & -\frac{3\sqrt{1155}i}{2156} & 0 & -\frac{3\sqrt{77}i}{154} & 0 & 0 & 0 \\
-\frac{5\sqrt{770}i}{2156} & 0 & -\frac{\sqrt{77}i}{4312} & 0 & -\frac{\sqrt{154}i}{88} & 0 & 0 & \frac{3\sqrt{1155}i}{2156} & 0 & -\frac{9\sqrt{231}i}{2156} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{2310}i}{2156} & 0 & -\frac{\sqrt{1155}i}{616} & 0 & -\frac{5\sqrt{462}i}{616} & \frac{3\sqrt{11}i}{44} & 0 & \frac{9\sqrt{231}i}{2156} & 0 & 0 & 0 & \frac{3\sqrt{77}i}{154} & 0 \\
\frac{5\sqrt{462}i}{616} & 0 & \frac{\sqrt{1155}i}{616} & 0 & -\frac{\sqrt{2310}i}{2156} & 0 & 0 & \frac{3\sqrt{77}i}{154} & 0 & 0 & 0 & \frac{9\sqrt{231}i}{2156} & 0 & \frac{3\sqrt{11}i}{44} \\
0 & \frac{\sqrt{154}i}{88} & 0 & \frac{\sqrt{77}i}{4312} & 0 & \frac{5\sqrt{770}i}{2156} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{231}i}{2156} & 0 & \frac{3\sqrt{1155}i}{2156} & 0 \\
0 & 0 & -\frac{\sqrt{231}i}{616} & 0 & -\frac{13\sqrt{462}i}{4312} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{154} & 0 & -\frac{3\sqrt{1155}i}{2156} & 0 & -\frac{3\sqrt{55}i}{308} \\
0 & 0 & 0 & -\frac{3\sqrt{33}i}{88} & 0 & \frac{\sqrt{330}i}{616} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{44} & 0 & \frac{3\sqrt{55}i}{308} & 0
\end{bmatrix}$$

$\boxed{\text{x402}}$ $\mathbb{Q}^{(a)}_{4,2}(E_g,2) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{77}}{196} & 0 & -\frac{\sqrt{154}}{56} & 0 & 0 & \frac{\sqrt{330}}{616} & 0 & -\frac{5\sqrt{770}}{2156} & 0 & -\frac{5\sqrt{462}}{616} & 0 & 0 & 0 \\
-\frac{\sqrt{77}}{196} & 0 & \frac{\sqrt{770}}{392} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{462}}{4312} & 0 & \frac{\sqrt{2310}}{2156} & 0 & -\frac{\sqrt{154}}{88} & 0 & 0 \\
0 & \frac{\sqrt{770}}{392} & 0 & 0 & 0 & \frac{\sqrt{154}}{56} & \frac{3\sqrt{33}}{88} & 0 & \frac{\sqrt{77}}{4312} & 0 & \frac{\sqrt{1155}}{616} & 0 & \frac{\sqrt{231}}{616} & 0 \\
-\frac{\sqrt{154}}{56} & 0 & 0 & 0 & -\frac{\sqrt{770}}{392} & 0 & 0 & \frac{\sqrt{231}}{616} & 0 & \frac{\sqrt{1155}}{616} & 0 & \frac{\sqrt{77}}{4312} & 0 & \frac{3\sqrt{33}}{88} \\
0 & 0 & 0 & -\frac{\sqrt{770}}{392} & 0 & \frac{\sqrt{77}}{196} & 0 & 0 & -\frac{\sqrt{154}}{88} & 0 & \frac{\sqrt{2310}}{2156} & 0 & -\frac{13\sqrt{462}}{4312} & 0 \\
0 & 0 & \frac{\sqrt{154}}{56} & 0 & \frac{\sqrt{77}}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{616} & 0 & -\frac{5\sqrt{770}}{2156} & 0 & \frac{\sqrt{330}}{616} \\
\frac{\sqrt{330}}{616} & 0 & \frac{3\sqrt{33}}{88} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{308} & 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{462}}{4312} & 0 & \frac{\sqrt{231}}{616} & 0 & 0 & -\frac{3\sqrt{55}}{308} & 0 & \frac{3\sqrt{1155}}{2156} & 0 & -\frac{3\sqrt{77}}{154} & 0 & 0 & 0 \\
-\frac{5\sqrt{770}}{2156} & 0 & \frac{\sqrt{77}}{4312} & 0 & -\frac{\sqrt{154}}{88} & 0 & 0 & \frac{3\sqrt{1155}}{2156} & 0 & \frac{9\sqrt{231}}{2156} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{2310}}{2156} & 0 & \frac{\sqrt{1155}}{616} & 0 & -\frac{5\sqrt{462}}{616} & -\frac{3\sqrt{11}}{44} & 0 & \frac{9\sqrt{231}}{2156} & 0 & 0 & 0 & \frac{3\sqrt{77}}{154} & 0 \\
-\frac{5\sqrt{462}}{616} & 0 & \frac{\sqrt{1155}}{616} & 0 & \frac{\sqrt{2310}}{2156} & 0 & 0 & -\frac{3\sqrt{77}}{154} & 0 & 0 & 0 & -\frac{9\sqrt{231}}{2156} & 0 & \frac{3\sqrt{11}}{44} \\
0 & -\frac{\sqrt{154}}{88} & 0 & \frac{\sqrt{77}}{4312} & 0 & -\frac{5\sqrt{770}}{2156} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{231}}{2156} & 0 & -\frac{3\sqrt{1155}}{2156} & 0 \\
0 & 0 & \frac{\sqrt{231}}{616} & 0 & -\frac{13\sqrt{462}}{4312} & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{154} & 0 & -\frac{3\sqrt{1155}}{2156} & 0 & \frac{3\sqrt{55}}{308} \\
0 & 0 & 0 & \frac{3\sqrt{33}}{88} & 0 & \frac{\sqrt{330}}{616} & 0 & 0 & 0 & 0 & \frac{3\sqrt{11}}{44} & 0 & \frac{3\sqrt{55}}{308} & 0
\end{bmatrix}$$

$\boxed{\text{x403}}$ $\mathbb{Q}_{6,1}^{(a)}(E_g,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{1232} & 0 & \frac{\sqrt{33}i}{176} & 0 & -\frac{3\sqrt{55}i}{176} & 0 & -\frac{\sqrt{11}i}{16} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{176} & 0 & -\frac{5\sqrt{11}i}{176} & 0 & \frac{3\sqrt{165}i}{176} & 0 & \frac{\sqrt{385}i}{112} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}i}{1232} & 0 & \frac{\sqrt{330}i}{176} & 0 & \frac{5\sqrt{22}i}{176} & 0 & -\frac{3\sqrt{110}i}{176} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}i}{176} & 0 & -\frac{5\sqrt{22}i}{176} & 0 & -\frac{\sqrt{330}i}{176} & 0 & \frac{3\sqrt{770}i}{1232} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{112} & 0 & -\frac{3\sqrt{165}i}{176} & 0 & \frac{5\sqrt{11}i}{176} & 0 & \frac{\sqrt{55}i}{176} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{16} & 0 & \frac{3\sqrt{55}i}{176} & 0 & -\frac{\sqrt{33}i}{176} & 0 & -\frac{\sqrt{77}i}{1232} \\
-\frac{\sqrt{77}i}{1232} & 0 & \frac{3\sqrt{770}i}{1232} & 0 & \frac{\sqrt{385}i}{112} & 0 & 0 & \frac{\sqrt{462}i}{1232} & 0 & -\frac{\sqrt{2310}i}{616} & 0 & -\frac{\sqrt{154}i}{112} & 0 & 0 \\
0 & \frac{\sqrt{55}i}{176} & 0 & -\frac{3\sqrt{110}i}{176} & 0 & -\frac{\sqrt{11}i}{16} & -\frac{\sqrt{462}i}{1232} & 0 & -\frac{\sqrt{22}i}{88} & 0 & \frac{\sqrt{330}i}{176} & 0 & 0 & 0 \\
-\frac{\sqrt{33}i}{176} & 0 & -\frac{\sqrt{330}i}{176} & 0 & \frac{3\sqrt{165}i}{176} & 0 & 0 & \frac{\sqrt{22}i}{88} & 0 & \frac{\sqrt{110}i}{176} & 0 & 0 & 0 & \frac{\sqrt{154}i}{112} \\
0 & \frac{5\sqrt{11}i}{176} & 0 & \frac{5\sqrt{22}i}{176} & 0 & -\frac{3\sqrt{55}i}{176} & \frac{\sqrt{2310}i}{616} & 0 & -\frac{\sqrt{110}i}{176} & 0 & 0 & 0 & -\frac{\sqrt{330}i}{176} & 0 \\
\frac{3\sqrt{55}i}{176} & 0 & -\frac{5\sqrt{22}i}{176} & 0 & -\frac{5\sqrt{11}i}{176} & 0 & 0 & -\frac{\sqrt{330}i}{176} & 0 & 0 & 0 & -\frac{\sqrt{110}i}{176} & 0 & \frac{\sqrt{2310}i}{616} \\
0 & -\frac{3\sqrt{165}i}{176} & 0 & \frac{\sqrt{330}i}{176} & 0 & \frac{\sqrt{33}i}{176} & \frac{\sqrt{154}i}{112} & 0 & 0 & 0 & \frac{\sqrt{110}i}{176} & 0 & \frac{\sqrt{22}i}{88} & 0 \\
\frac{\sqrt{11}i}{16} & 0 & \frac{3\sqrt{110}i}{176} & 0 & -\frac{\sqrt{55}i}{176} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{176} & 0 & -\frac{\sqrt{22}i}{88} & 0 & -\frac{\sqrt{462}i}{1232} \\
0 & -\frac{\sqrt{385}i}{112} & 0 & -\frac{3\sqrt{770}i}{1232} & 0 & \frac{\sqrt{77}i}{1232} & 0 & 0 & -\frac{\sqrt{154}i}{112} & 0 & -\frac{\sqrt{2310}i}{616} & 0 & \frac{\sqrt{462}i}{1232} & 0
\end{bmatrix}$$

$\boxed{\text{x404}}$ $\mathbb{Q}_{6,2}^{(a)}(E_g,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{1232} & 0 & -\frac{\sqrt{33}}{176} & 0 & -\frac{3\sqrt{55}}{176} & 0 & \frac{\sqrt{11}}{16} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{176} & 0 & \frac{5\sqrt{11}}{176} & 0 & \frac{3\sqrt{165}}{176} & 0 & -\frac{\sqrt{385}}{112} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{1232} & 0 & \frac{\sqrt{330}}{176} & 0 & -\frac{5\sqrt{22}}{176} & 0 & -\frac{3\sqrt{110}}{176} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}}{176} & 0 & -\frac{5\sqrt{22}}{176} & 0 & \frac{\sqrt{330}}{176} & 0 & \frac{3\sqrt{770}}{1232} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{112} & 0 & \frac{3\sqrt{165}}{176} & 0 & \frac{5\sqrt{11}}{176} & 0 & -\frac{\sqrt{55}}{176} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{16} & 0 & -\frac{3\sqrt{55}}{176} & 0 & -\frac{\sqrt{33}}{176} & 0 & \frac{\sqrt{77}}{1232} \\
\frac{\sqrt{77}}{1232} & 0 & \frac{3\sqrt{770}}{1232} & 0 & -\frac{\sqrt{385}}{112} & 0 & 0 & -\frac{\sqrt{462}}{1232} & 0 & -\frac{\sqrt{2310}}{616} & 0 & \frac{\sqrt{154}}{112} & 0 & 0 \\
0 & -\frac{\sqrt{55}}{176} & 0 & -\frac{3\sqrt{110}}{176} & 0 & \frac{\sqrt{11}}{16} & -\frac{\sqrt{462}}{1232} & 0 & \frac{\sqrt{22}}{88} & 0 & \frac{\sqrt{330}}{176} & 0 & 0 & 0 \\
-\frac{\sqrt{33}}{176} & 0 & \frac{\sqrt{330}}{176} & 0 & \frac{3\sqrt{165}}{176} & 0 & 0 & \frac{\sqrt{22}}{88} & 0 & -\frac{\sqrt{110}}{176} & 0 & 0 & 0 & -\frac{\sqrt{154}}{112} \\
0 & \frac{5\sqrt{11}}{176} & 0 & -\frac{5\sqrt{22}}{176} & 0 & -\frac{3\sqrt{55}}{176} & -\frac{\sqrt{2310}}{616} & 0 & -\frac{\sqrt{110}}{176} & 0 & 0 & 0 & -\frac{\sqrt{330}}{176} & 0 \\
-\frac{3\sqrt{55}}{176} & 0 & -\frac{5\sqrt{22}}{176} & 0 & \frac{5\sqrt{11}}{176} & 0 & 0 & \frac{\sqrt{330}}{176} & 0 & 0 & 0 & \frac{\sqrt{110}}{176} & 0 & \frac{\sqrt{2310}}{616} \\
0 & \frac{3\sqrt{165}}{176} & 0 & \frac{\sqrt{330}}{176} & 0 & -\frac{\sqrt{33}}{176} & \frac{\sqrt{154}}{112} & 0 & 0 & 0 & \frac{\sqrt{110}}{176} & 0 & -\frac{\sqrt{22}}{88} & 0 \\
\frac{\sqrt{11}}{16} & 0 & -\frac{3\sqrt{110}}{176} & 0 & -\frac{\sqrt{55}}{176} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{176} & 0 & -\frac{\sqrt{22}}{88} & 0 & \frac{\sqrt{462}}{1232} \\
0 & -\frac{\sqrt{385}}{112} & 0 & \frac{3\sqrt{770}}{1232} & 0 & \frac{\sqrt{77}}{1232} & 0 & 0 & -\frac{\sqrt{154}}{112} & 0 & \frac{\sqrt{2310}}{616} & 0 & \frac{\sqrt{462}}{1232} & 0
\end{bmatrix}$$

$\boxed{\text{x405}}$ $\mathbb{Q}_{6,1}^{(a)}(E_g,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{448} & 0 & \frac{3\sqrt{2}i}{64} & 0 & \frac{\sqrt{30}i}{64} & 0 & \frac{\sqrt{6}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{64} & 0 & -\frac{5\sqrt{6}i}{64} & 0 & -\frac{3\sqrt{10}i}{64} & 0 & -\frac{\sqrt{210}i}{448} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{224} & 0 & \frac{3\sqrt{5}i}{32} & 0 & \frac{5\sqrt{3}i}{32} & 0 & \frac{\sqrt{15}i}{32} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{32} & 0 & -\frac{5\sqrt{3}i}{32} & 0 & -\frac{3\sqrt{5}i}{32} & 0 & -\frac{\sqrt{105}i}{224} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{448} & 0 & \frac{3\sqrt{10}i}{64} & 0 & \frac{5\sqrt{6}i}{64} & 0 & \frac{\sqrt{30}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{64} & 0 & -\frac{\sqrt{30}i}{64} & 0 & -\frac{3\sqrt{2}i}{64} & 0 & -\frac{\sqrt{42}i}{448} \\
-\frac{\sqrt{42}i}{448} & 0 & -\frac{\sqrt{105}i}{224} & 0 & -\frac{\sqrt{210}i}{448} & 0 & 0 & \frac{3\sqrt{7}i}{224} & 0 & \frac{\sqrt{35}i}{112} & 0 & \frac{\sqrt{21}i}{224} & 0 & 0 \\
0 & \frac{\sqrt{30}i}{64} & 0 & \frac{\sqrt{15}i}{32} & 0 & \frac{\sqrt{6}i}{64} & -\frac{3\sqrt{7}i}{224} & 0 & -\frac{\sqrt{3}i}{16} & 0 & -\frac{\sqrt{5}i}{32} & 0 & 0 & 0 \\
-\frac{3\sqrt{2}i}{64} & 0 & -\frac{3\sqrt{5}i}{32} & 0 & -\frac{3\sqrt{10}i}{64} & 0 & 0 & \frac{\sqrt{3}i}{16} & 0 & \frac{\sqrt{15}i}{32} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{224} \\
0 & \frac{5\sqrt{6}i}{64} & 0 & \frac{5\sqrt{3}i}{32} & 0 & \frac{\sqrt{30}i}{64} & -\frac{\sqrt{35}i}{112} & 0 & -\frac{\sqrt{15}i}{32} & 0 & 0 & 0 & \frac{\sqrt{5}i}{32} & 0 \\
-\frac{\sqrt{30}i}{64} & 0 & -\frac{5\sqrt{3}i}{32} & 0 & -\frac{5\sqrt{6}i}{64} & 0 & 0 & \frac{\sqrt{5}i}{32} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{32} & 0 & -\frac{\sqrt{35}i}{112} \\
0 & \frac{3\sqrt{10}i}{64} & 0 & \frac{3\sqrt{5}i}{32} & 0 & \frac{3\sqrt{2}i}{64} & -\frac{\sqrt{21}i}{224} & 0 & 0 & 0 & \frac{\sqrt{15}i}{32} & 0 & \frac{\sqrt{3}i}{16} & 0 \\
-\frac{\sqrt{6}i}{64} & 0 & -\frac{\sqrt{15}i}{32} & 0 & -\frac{\sqrt{30}i}{64} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{32} & 0 & -\frac{\sqrt{3}i}{16} & 0 & -\frac{3\sqrt{7}i}{224} \\
0 & \frac{\sqrt{210}i}{448} & 0 & \frac{\sqrt{105}i}{224} & 0 & \frac{\sqrt{42}i}{448} & 0 & 0 & \frac{\sqrt{21}i}{224} & 0 & \frac{\sqrt{35}i}{112} & 0 & \frac{3\sqrt{7}i}{224} & 0
\end{bmatrix}$

$\boxed{\text{x406}}$ $\mathbb{Q}_{6,2}^{(a)}(E_g,2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{448} & 0 & -\frac{3\sqrt{2}}{64} & 0 & \frac{\sqrt{30}}{64} & 0 & -\frac{\sqrt{6}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{64} & 0 & \frac{5\sqrt{6}}{64} & 0 & -\frac{3\sqrt{10}}{64} & 0 & \frac{\sqrt{210}}{448} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{224} & 0 & \frac{3\sqrt{5}}{32} & 0 & -\frac{5\sqrt{3}}{32} & 0 & \frac{\sqrt{15}}{32} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{32} & 0 & -\frac{5\sqrt{3}}{32} & 0 & \frac{3\sqrt{5}}{32} & 0 & -\frac{\sqrt{105}}{224} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{448} & 0 & -\frac{3\sqrt{10}}{64} & 0 & \frac{5\sqrt{6}}{64} & 0 & -\frac{\sqrt{30}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{64} & 0 & \frac{\sqrt{30}}{64} & 0 & -\frac{3\sqrt{2}}{64} & 0 & \frac{\sqrt{42}}{448} \\
\frac{\sqrt{42}}{448} & 0 & -\frac{\sqrt{105}}{224} & 0 & \frac{\sqrt{210}}{448} & 0 & 0 & -\frac{3\sqrt{7}}{224} & 0 & \frac{\sqrt{35}}{112} & 0 & -\frac{\sqrt{21}}{224} & 0 & 0 \\
0 & -\frac{\sqrt{30}}{64} & 0 & \frac{\sqrt{15}}{32} & 0 & -\frac{\sqrt{6}}{64} & -\frac{3\sqrt{7}}{224} & 0 & \frac{\sqrt{3}}{16} & 0 & -\frac{\sqrt{5}}{32} & 0 & 0 & 0 \\
-\frac{3\sqrt{2}}{64} & 0 & \frac{3\sqrt{5}}{32} & 0 & -\frac{3\sqrt{10}}{64} & 0 & 0 & \frac{\sqrt{3}}{16} & 0 & -\frac{\sqrt{15}}{32} & 0 & 0 & 0 & \frac{\sqrt{21}}{224} \\
0 & \frac{5\sqrt{6}}{64} & 0 & -\frac{5\sqrt{3}}{32} & 0 & \frac{\sqrt{30}}{64} & \frac{\sqrt{35}}{112} & 0 & -\frac{\sqrt{15}}{32} & 0 & 0 & 0 & \frac{\sqrt{5}}{32} & 0 \\
\frac{\sqrt{30}}{64} & 0 & -\frac{5\sqrt{3}}{32} & 0 & \frac{5\sqrt{6}}{64} & 0 & 0 & -\frac{\sqrt{5}}{32} & 0 & 0 & 0 & \frac{\sqrt{15}}{32} & 0 & -\frac{\sqrt{35}}{112} \\
0 & -\frac{3\sqrt{10}}{64} & 0 & \frac{3\sqrt{5}}{32} & 0 & -\frac{3\sqrt{2}}{64} & -\frac{\sqrt{21}}{224} & 0 & 0 & 0 & \frac{\sqrt{15}}{32} & 0 & -\frac{\sqrt{3}}{16} & 0 \\
-\frac{\sqrt{6}}{64} & 0 & \frac{\sqrt{15}}{32} & 0 & -\frac{\sqrt{30}}{64} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{32} & 0 & -\frac{\sqrt{3}}{16} & 0 & \frac{3\sqrt{7}}{224} \\
0 & \frac{\sqrt{210}}{448} & 0 & -\frac{\sqrt{105}}{224} & 0 & \frac{\sqrt{42}}{448} & 0 & 0 & \frac{\sqrt{21}}{224} & 0 & -\frac{\sqrt{35}}{112} & 0 & \frac{3\sqrt{7}}{224} & 0
\end{bmatrix}$

$\boxed{\text{x407}}$ $\mathbb{Q}_{6,1}^{(a)}(E_g,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{14784} & 0 & \frac{\sqrt{110}i}{704} & 0 & -\frac{9\sqrt{66}i}{704} & 0 & \frac{\sqrt{330}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}i}{2112} & 0 & -\frac{5\sqrt{330}i}{2112} & 0 & \frac{27\sqrt{22}i}{704} & 0 & -\frac{5\sqrt{462}i}{448} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{231}i}{2464} & 0 & \frac{5\sqrt{11}i}{352} & 0 & \frac{5\sqrt{165}i}{1056} & 0 & -\frac{9\sqrt{33}i}{352} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{33}i}{352} & 0 & -\frac{5\sqrt{165}i}{1056} & 0 & -\frac{5\sqrt{11}i}{352} & 0 & \frac{9\sqrt{231}i}{2464} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{448} & 0 & -\frac{27\sqrt{22}i}{704} & 0 & \frac{5\sqrt{330}i}{2112} & 0 & \frac{5\sqrt{66}i}{2112} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{64} & 0 & \frac{9\sqrt{66}i}{704} & 0 & -\frac{\sqrt{110}i}{704} & 0 & -\frac{\sqrt{2310}i}{14784} \\
-\frac{\sqrt{2310}i}{14784} & 0 & \frac{9\sqrt{231}i}{2464} & 0 & -\frac{5\sqrt{462}i}{448} & 0 & 0 & \frac{\sqrt{385}i}{2464} & 0 & -\frac{9\sqrt{77}i}{1232} & 0 & \frac{\sqrt{1155}i}{224} & 0 & 0 \\
0 & \frac{5\sqrt{66}i}{2112} & 0 & -\frac{9\sqrt{33}i}{352} & 0 & \frac{\sqrt{330}i}{64} & -\frac{\sqrt{385}i}{2464} & 0 & -\frac{\sqrt{165}i}{528} & 0 & \frac{9\sqrt{11}i}{352} & 0 & 0 & 0 \\
-\frac{\sqrt{110}i}{704} & 0 & -\frac{5\sqrt{11}i}{352} & 0 & \frac{27\sqrt{22}i}{704} & 0 & 0 & \frac{\sqrt{165}i}{528} & 0 & \frac{5\sqrt{33}i}{1056} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{224} \\
0 & \frac{5\sqrt{330}i}{2112} & 0 & \frac{5\sqrt{165}i}{1056} & 0 & -\frac{9\sqrt{66}i}{704} & \frac{9\sqrt{77}i}{1232} & 0 & -\frac{5\sqrt{33}i}{1056} & 0 & 0 & 0 & -\frac{9\sqrt{11}i}{352} & 0 \\
\frac{9\sqrt{66}i}{704} & 0 & -\frac{5\sqrt{165}i}{1056} & 0 & -\frac{5\sqrt{330}i}{2112} & 0 & 0 & -\frac{9\sqrt{11}i}{352} & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{1056} & 0 & \frac{9\sqrt{77}i}{1232} \\
0 & -\frac{27\sqrt{22}i}{704} & 0 & \frac{5\sqrt{11}i}{352} & 0 & \frac{\sqrt{110}i}{704} & -\frac{\sqrt{1155}i}{224} & 0 & 0 & 0 & \frac{5\sqrt{33}i}{1056} & 0 & \frac{\sqrt{165}i}{528} & 0 \\
-\frac{\sqrt{330}i}{64} & 0 & \frac{9\sqrt{33}i}{352} & 0 & -\frac{5\sqrt{66}i}{2112} & 0 & 0 & 0 & 0 & \frac{9\sqrt{11}i}{352} & 0 & -\frac{\sqrt{165}i}{528} & 0 & -\frac{\sqrt{385}i}{2464} \\
0 & \frac{5\sqrt{462}i}{448} & 0 & -\frac{9\sqrt{231}i}{2464} & 0 & \frac{\sqrt{2310}i}{14784} & 0 & 0 & \frac{\sqrt{1155}i}{224} & 0 & -\frac{9\sqrt{77}i}{1232} & 0 & \frac{\sqrt{385}i}{2464} & 0
\end{bmatrix}$$

$\boxed{\text{x408}}$ $\mathbb{Q}_{6,2}^{(a)}(E_g,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{14784} & 0 & -\frac{\sqrt{110}}{704} & 0 & -\frac{9\sqrt{66}}{704} & 0 & -\frac{\sqrt{330}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}}{2112} & 0 & \frac{5\sqrt{330}}{2112} & 0 & \frac{27\sqrt{22}}{704} & 0 & \frac{5\sqrt{462}}{448} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{231}}{2464} & 0 & \frac{5\sqrt{11}}{352} & 0 & -\frac{5\sqrt{165}}{1056} & 0 & -\frac{9\sqrt{33}}{352} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{33}}{352} & 0 & -\frac{5\sqrt{165}}{1056} & 0 & \frac{5\sqrt{11}}{352} & 0 & \frac{9\sqrt{231}}{2464} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{448} & 0 & \frac{27\sqrt{22}}{704} & 0 & \frac{5\sqrt{330}}{2112} & 0 & -\frac{5\sqrt{66}}{2112} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{64} & 0 & -\frac{9\sqrt{66}}{704} & 0 & -\frac{\sqrt{110}}{704} & 0 & \frac{\sqrt{2310}}{14784} \\
\frac{\sqrt{2310}}{14784} & 0 & \frac{9\sqrt{231}}{2464} & 0 & \frac{5\sqrt{462}}{448} & 0 & 0 & -\frac{\sqrt{385}}{2464} & 0 & -\frac{9\sqrt{77}}{1232} & 0 & -\frac{\sqrt{1155}}{224} & 0 & 0 \\
0 & -\frac{5\sqrt{66}}{2112} & 0 & -\frac{9\sqrt{33}}{352} & 0 & -\frac{\sqrt{330}}{64} & -\frac{\sqrt{385}}{2464} & 0 & \frac{\sqrt{165}}{528} & 0 & \frac{9\sqrt{11}}{352} & 0 & 0 & 0 \\
-\frac{\sqrt{110}}{704} & 0 & \frac{5\sqrt{11}}{352} & 0 & \frac{27\sqrt{22}}{704} & 0 & 0 & \frac{\sqrt{165}}{528} & 0 & -\frac{5\sqrt{33}}{1056} & 0 & 0 & 0 & \frac{\sqrt{1155}}{224} \\
0 & \frac{5\sqrt{330}}{2112} & 0 & -\frac{5\sqrt{165}}{1056} & 0 & -\frac{9\sqrt{66}}{704} & -\frac{9\sqrt{77}}{1232} & 0 & -\frac{5\sqrt{33}}{1056} & 0 & 0 & 0 & -\frac{9\sqrt{11}}{352} & 0 \\
-\frac{9\sqrt{66}}{704} & 0 & -\frac{5\sqrt{165}}{1056} & 0 & \frac{5\sqrt{330}}{2112} & 0 & 0 & \frac{9\sqrt{11}}{352} & 0 & 0 & 0 & \frac{5\sqrt{33}}{1056} & 0 & \frac{9\sqrt{77}}{1232} \\
0 & \frac{27\sqrt{22}}{704} & 0 & \frac{5\sqrt{11}}{352} & 0 & -\frac{\sqrt{110}}{704} & -\frac{\sqrt{1155}}{224} & 0 & 0 & 0 & \frac{5\sqrt{33}}{1056} & 0 & -\frac{\sqrt{165}}{528} & 0 \\
-\frac{\sqrt{330}}{64} & 0 & -\frac{9\sqrt{33}}{352} & 0 & -\frac{5\sqrt{66}}{2112} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{11}}{352} & 0 & -\frac{\sqrt{165}}{528} & 0 & \frac{\sqrt{385}}{2464} \\
0 & \frac{5\sqrt{462}}{448} & 0 & \frac{9\sqrt{231}}{2464} & 0 & \frac{\sqrt{2310}}{14784} & 0 & 0 & \frac{\sqrt{1155}}{224} & 0 & \frac{9\sqrt{77}}{1232} & 0 & \frac{\sqrt{385}}{2464} & 0
\end{bmatrix}$$

$\boxed{\text{x409}}$ $\mathbb{Q}_2^{(1,-1;a)}(A_{1g}) =$

$$\begin{bmatrix}
-\frac{5\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{4\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{4\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{196} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{98} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14}
\end{bmatrix}$$

$\boxed{\text{x410}}$ $\mathbb{Q}_4^{(1,-1;a)}(A_{1g}, 1) =$

$$\begin{bmatrix}
\frac{1}{42} & 0 & 0 & 0 & \frac{\sqrt{5}}{42} & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & \frac{5\sqrt{2}}{84} & 0 & 0 \\
0 & -\frac{1}{14} & 0 & 0 & 0 & \frac{\sqrt{5}}{42} & 0 & 0 & -\frac{\sqrt{10}}{21} & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 \\
0 & 0 & \frac{1}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{84} & 0 & 0 & 0 & \frac{\sqrt{105}}{84} \\
0 & 0 & 0 & \frac{1}{21} & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & \frac{5\sqrt{3}}{84} & 0 & 0 & 0 \\
\frac{\sqrt{5}}{42} & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & \frac{\sqrt{10}}{21} & 0 & 0 \\
0 & \frac{\sqrt{5}}{42} & 0 & 0 & 0 & \frac{1}{42} & 0 & 0 & -\frac{5\sqrt{2}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & -\frac{1}{6} & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 \\
\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & \frac{13}{42} & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 \\
0 & -\frac{\sqrt{10}}{21} & 0 & 0 & 0 & -\frac{5\sqrt{2}}{84} & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 \\
0 & 0 & -\frac{5\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} \\
0 & 0 & 0 & \frac{5\sqrt{3}}{84} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 \\
\frac{5\sqrt{2}}{84} & 0 & 0 & 0 & \frac{\sqrt{10}}{21} & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 \\
0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & \frac{13}{42} & 0 \\
0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & -\frac{1}{6}
\end{bmatrix}$$

$\boxed{\text{x411}}$ $\mathbb{Q}_4^{(1,-1;a)}(A_{1g},2) =$

$$\begin{bmatrix}
\frac{\sqrt{35}}{294} & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & \frac{5\sqrt{210}}{588} & 0 & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 \\
0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & -\frac{5\sqrt{14}}{147} & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 \\
0 & 0 & \frac{\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{588} & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\
0 & 0 & 0 & \frac{\sqrt{35}}{147} & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & \frac{5\sqrt{105}}{588} & 0 & 0 & 0 \\
-\frac{\sqrt{7}}{42} & 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & \frac{5\sqrt{14}}{147} & 0 & 0 \\
0 & -\frac{\sqrt{7}}{42} & 0 & 0 & 0 & \frac{\sqrt{35}}{294} & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{210}}{588} & 0 \\
0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & -\frac{\sqrt{35}}{42} & 0 & 0 & 0 & \frac{1}{6} & 0 & 0 & 0 \\
\frac{5\sqrt{210}}{588} & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & \frac{13\sqrt{35}}{294} & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 \\
0 & -\frac{5\sqrt{14}}{147} & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 \\
0 & 0 & -\frac{5\sqrt{105}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & \frac{1}{6} \\
0 & 0 & 0 & \frac{5\sqrt{105}}{588} & 0 & 0 & \frac{1}{6} & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 \\
-\frac{\sqrt{70}}{84} & 0 & 0 & 0 & \frac{5\sqrt{14}}{147} & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 \\
0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & -\frac{5\sqrt{210}}{588} & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & \frac{13\sqrt{35}}{294} & 0 \\
0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{6} & 0 & 0 & 0 & -\frac{\sqrt{35}}{42}
\end{bmatrix}$$

$\boxed{\text{x412}}$ $\mathbb{Q}_6^{(1,-1;a)}(A_{1g},1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{1848} & 0 & 0 & 0 & \frac{\sqrt{77}}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{616} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{264} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & 0 & 0 & \frac{\sqrt{330}}{264} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 & 0 & \frac{5\sqrt{462}}{1848} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{264} & 0 & 0 & 0 & -\frac{\sqrt{385}}{616} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & 0 & \frac{\sqrt{231}}{1848} & 0 \\
0 & 0 & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 & \frac{\sqrt{154}}{616} & 0 & 0 & 0 & -\frac{3\sqrt{110}}{88} & 0 & 0 & 0 \\
-\frac{\sqrt{231}}{1848} & 0 & 0 & 0 & \frac{\sqrt{1155}}{264} & 0 & 0 & -\frac{5\sqrt{154}}{616} & 0 & 0 & 0 & \frac{\sqrt{462}}{88} & 0 & 0 \\
0 & \frac{\sqrt{385}}{616} & 0 & 0 & 0 & -\frac{\sqrt{77}}{88} & 0 & 0 & \frac{9\sqrt{154}}{616} & 0 & 0 & 0 & \frac{\sqrt{462}}{88} & 0 \\
0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}}{616} & 0 & 0 & 0 & -\frac{3\sqrt{110}}{88} \\
0 & 0 & 0 & \frac{5\sqrt{462}}{1848} & 0 & 0 & -\frac{3\sqrt{110}}{88} & 0 & 0 & 0 & -\frac{5\sqrt{154}}{616} & 0 & 0 & 0 \\
\frac{\sqrt{77}}{88} & 0 & 0 & 0 & -\frac{\sqrt{385}}{616} & 0 & 0 & \frac{\sqrt{462}}{88} & 0 & 0 & 0 & \frac{9\sqrt{154}}{616} & 0 & 0 \\
0 & -\frac{\sqrt{1155}}{264} & 0 & 0 & 0 & \frac{\sqrt{231}}{1848} & 0 & 0 & \frac{\sqrt{462}}{88} & 0 & 0 & 0 & -\frac{5\sqrt{154}}{616} & 0 \\
0 & 0 & \frac{\sqrt{330}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}}{88} & 0 & 0 & 0 & \frac{\sqrt{154}}{616}
\end{bmatrix}$$

$\boxed{\text{x413}}$ $\mathbb{Q}_6^{(1,-1;a)}(A_{1g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{264} & 0 & 0 & 0 & -\frac{\sqrt{11}}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{88} & 0 & 0 & 0 & \frac{\sqrt{165}}{264} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}}{264} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1848} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1848} & 0 & 0 & 0 & \frac{5\sqrt{66}}{264} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{88} & 0 & 0 & 0 & \frac{\sqrt{33}}{264} & 0 \\
0 & 0 & 0 & \frac{\sqrt{2310}}{1848} & 0 & 0 & \frac{\sqrt{22}}{88} & 0 & 0 & 0 & \frac{3\sqrt{770}}{616} & 0 & 0 & 0 \\
-\frac{\sqrt{33}}{264} & 0 & 0 & 0 & -\frac{\sqrt{165}}{264} & 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 \\
0 & \frac{\sqrt{55}}{88} & 0 & 0 & \frac{\sqrt{11}}{88} & 0 & 0 & \frac{9\sqrt{22}}{88} & 0 & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 \\
0 & 0 & -\frac{5\sqrt{66}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & \frac{3\sqrt{770}}{616} \\
0 & 0 & 0 & \frac{5\sqrt{66}}{264} & 0 & 0 & \frac{3\sqrt{770}}{616} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 \\
-\frac{\sqrt{11}}{88} & 0 & 0 & 0 & -\frac{\sqrt{55}}{88} & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & 0 & \frac{9\sqrt{22}}{88} & 0 & 0 \\
0 & \frac{\sqrt{165}}{264} & 0 & 0 & 0 & \frac{\sqrt{33}}{264} & 0 & 0 & -\frac{\sqrt{66}}{88} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 \\
0 & 0 & -\frac{\sqrt{2310}}{1848} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{616} & 0 & 0 & 0 & \frac{\sqrt{22}}{88}
\end{bmatrix}$$

$\boxed{\text{x414}}$ $\mathbb{Q}_4^{(1,-1;a)}(A_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x415}}$ $\mathbb{Q}_6^{(1,-1;a)}(A_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{132} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{462} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{154} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{330}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{154} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x416}}$ $\mathbb{Q}_2^{(1,-1;a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}}{49} & 0 & 0 & 0 \\
-\frac{\sqrt{70}}{98} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 \\
0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & 0 & -\frac{\sqrt{105}}{49} & 0 \\
0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} \\
0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{196} & 0 & 0 & 0 \\
0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 \\
-\frac{\sqrt{210}}{196} & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 \\
0 & -\frac{\sqrt{42}}{49} & 0 & 0 & 0 & \frac{\sqrt{210}}{196} & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 \\
0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & \frac{\sqrt{3}}{14} \\
0 & 0 & 0 & -\frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x417}}$ $\mathbb{Q}_4^{(1,-1;a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{210}}{294} & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}}{196} & 0 & 0 & 0 \\
-\frac{\sqrt{42}}{98} & 0 & 0 & 0 & \frac{\sqrt{210}}{294} & 0 & 0 & -\frac{9\sqrt{7}}{196} & 0 & 0 & 0 & \frac{17\sqrt{21}}{588} & 0 & 0 \\
0 & \frac{\sqrt{210}}{294} & 0 & 0 & 0 & -\frac{\sqrt{42}}{98} & 0 & 0 & -\frac{17\sqrt{21}}{588} & 0 & 0 & 0 & \frac{9\sqrt{7}}{196} & 0 \\
0 & 0 & \frac{\sqrt{210}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{196} & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} \\
0 & 0 & 0 & -\frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 \\
0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{9\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{17\sqrt{21}}{588} & 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 & 0 & -\frac{4\sqrt{7}}{49} & 0 & 0 & 0 \\
-\frac{5\sqrt{14}}{98} & 0 & 0 & 0 & \frac{\sqrt{70}}{196} & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & -\frac{4\sqrt{7}}{49} & 0 & 0 \\
0 & -\frac{\sqrt{70}}{196} & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & -\frac{4\sqrt{7}}{49} & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 \\
0 & 0 & \frac{17\sqrt{21}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{7}}{49} & 0 & 0 & 0 & \frac{\sqrt{5}}{7} \\
0 & 0 & 0 & \frac{9\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x418}}$ $\mathbb{Q}_6^{(1,-1;a)}(B_{1g}, 1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{24} & 0 & 0 & 0 & -\frac{\sqrt{35}}{56} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{168} & 0 & 0 & 0 & -\frac{\sqrt{5}}{24} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{2}}{24} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{24} & 0 & 0 & 0 & \frac{\sqrt{7}}{168} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{56} & 0 & 0 & 0 & -\frac{1}{24} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{7}}{168} & 0 & 0 & 0 & \frac{\sqrt{35}}{56} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 \\
0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & \frac{\sqrt{210}}{56} \\
0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 \\
\frac{1}{24} & 0 & 0 & 0 & \frac{\sqrt{5}}{24} & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 \\
0 & -\frac{\sqrt{5}}{24} & 0 & 0 & 0 & -\frac{1}{24} & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 \\
0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} \\
0 & 0 & 0 & -\frac{\sqrt{2}}{24} & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 \\
-\frac{\sqrt{35}}{56} & 0 & 0 & 0 & \frac{\sqrt{7}}{168} & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x419}}$ $\mathbb{Q}_6^{(1,-1;a)}(B_{1g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{264} & 0 & 0 & 0 & -\frac{\sqrt{77}}{56} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{1848} & 0 & 0 & 0 & \frac{5\sqrt{11}}{264} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{264} & 0 & 0 & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{264} & 0 & 0 & 0 & \frac{\sqrt{110}}{264} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{264} & 0 & 0 & 0 & -\frac{\sqrt{385}}{1848} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{56} & 0 & 0 & 0 & \frac{\sqrt{55}}{264} & 0 & 0 & 0 \\
0 & \frac{\sqrt{385}}{1848} & 0 & 0 & 0 & \frac{\sqrt{77}}{56} & 0 & 0 & \frac{5\sqrt{154}}{616} & 0 & 0 & 0 & \frac{\sqrt{462}}{56} & 0 \\
0 & 0 & -\frac{\sqrt{110}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & \frac{\sqrt{462}}{56} \\
0 & 0 & 0 & \frac{\sqrt{330}}{264} & 0 & 0 & \frac{5\sqrt{154}}{616} & 0 & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 \\
-\frac{\sqrt{55}}{264} & 0 & 0 & 0 & -\frac{5\sqrt{11}}{264} & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & 0 \\
0 & \frac{5\sqrt{11}}{264} & 0 & 0 & 0 & \frac{\sqrt{55}}{264} & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 \\
0 & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & 0 & 0 & \frac{5\sqrt{154}}{616} \\
0 & 0 & 0 & \frac{\sqrt{110}}{264} & 0 & 0 & \frac{\sqrt{462}}{56} & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 \\
-\frac{\sqrt{77}}{56} & 0 & 0 & 0 & -\frac{\sqrt{385}}{1848} & 0 & 0 & \frac{\sqrt{462}}{56} & 0 & 0 & 0 & \frac{5\sqrt{154}}{616} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x420}}$ $\mathbb{Q}_2^{(1,-1;a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{3\sqrt{14}i}{98} & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & \frac{\sqrt{42}i}{49} & 0 & 0 & 0 \\
-\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{98} & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 \\
0 & -\frac{3\sqrt{14}i}{98} & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 \\
0 & 0 & -\frac{3\sqrt{14}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{49} & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} \\
0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{196} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 \\
-\frac{\sqrt{210}i}{196} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{49} & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 \\
0 & -\frac{\sqrt{42}i}{49} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{196} & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 \\
0 & 0 & -\frac{3\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} \\
0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x421}}$ $\mathbb{Q}_4^{(1,-1;a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{210}i}{294} & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & 0 & 0 \\
\frac{\sqrt{42}i}{98} & 0 & 0 & 0 & \frac{\sqrt{210}i}{294} & 0 & 0 & \frac{9\sqrt{7}i}{196} & 0 & 0 & 0 & \frac{17\sqrt{21}i}{588} & 0 & 0 \\
0 & -\frac{\sqrt{210}i}{294} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{98} & 0 & 0 & \frac{17\sqrt{21}i}{588} & 0 & 0 & 0 & \frac{9\sqrt{7}i}{196} & 0 \\
0 & 0 & -\frac{\sqrt{210}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} \\
0 & 0 & 0 & \frac{\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{98} & 0 & 0 & 0 \\
0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{9\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{147} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{17\sqrt{21}i}{588} & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & -\frac{4\sqrt{7}i}{49} & 0 & 0 & 0 \\
\frac{5\sqrt{14}i}{98} & 0 & 0 & 0 & \frac{\sqrt{70}i}{196} & 0 & 0 & -\frac{\sqrt{21}i}{147} & 0 & 0 & 0 & -\frac{4\sqrt{7}i}{49} & 0 & 0 \\
0 & \frac{\sqrt{70}i}{196} & 0 & 0 & 0 & \frac{5\sqrt{14}i}{98} & 0 & 0 & \frac{4\sqrt{7}i}{49} & 0 & 0 & 0 & \frac{\sqrt{21}i}{147} & 0 \\
0 & 0 & -\frac{17\sqrt{21}i}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}i}{49} & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} \\
0 & 0 & 0 & -\frac{9\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{147} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x422}}$ $\mathbb{Q}_6^{(1,-1;a)}(B_{2g}, 1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x423}}$ $\mathbb{Q}_6^{(1,-1;a)}(B_{2g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{66} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{462} & 0 & 0 & 0 & -\frac{\sqrt{55}i}{66} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{66} & 0 & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{66} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{66} & 0 & 0 & 0 & \frac{\sqrt{77}i}{462} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{66} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{77}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{22}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{22} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{66}i}{66} & 0 & 0 & \frac{\sqrt{770}i}{154} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{22} & 0 & 0 & 0 \\
-\frac{\sqrt{11}i}{66} & 0 & 0 & 0 & \frac{\sqrt{55}i}{66} & 0 & 0 & -\frac{\sqrt{66}i}{22} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{22} & 0 & 0 \\
0 & \frac{\sqrt{55}i}{66} & 0 & 0 & 0 & -\frac{\sqrt{11}i}{66} & 0 & 0 & \frac{\sqrt{22}i}{22} & 0 & 0 & 0 & \frac{\sqrt{66}i}{22} & 0 \\
0 & 0 & -\frac{\sqrt{66}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{22} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{154} \\
0 & 0 & 0 & \frac{\sqrt{22}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{22} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{154} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x424}}$ $\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{56} & 0 & \frac{15\sqrt{14}i}{392} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{35}i}{49} & 0 & \frac{\sqrt{14}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{392} & 0 & \frac{11\sqrt{42}i}{392} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{14}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{196} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{3\sqrt{35}i}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{14}i}{49} & 0 & -\frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{42}i}{392} & 0 & -\frac{\sqrt{210}i}{392} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}i}{392} & 0 & -\frac{5\sqrt{6}i}{56} \\
-\frac{5\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{210}i}{392} & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & -\frac{2\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{15\sqrt{14}i}{392} & 0 & \frac{3\sqrt{35}i}{196} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{49} & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 \\
0 & -\frac{11\sqrt{42}i}{392} & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{11\sqrt{42}i}{392} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{35}i}{196} & 0 & \frac{15\sqrt{14}i}{392} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & \frac{2\sqrt{21}i}{49} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{392} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{49} & 0 & \frac{3i}{14} \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0
\end{bmatrix}$$

$\boxed{\text{x425}}$ $\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{56} & 0 & -\frac{15\sqrt{14}}{392} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{35}}{49} & 0 & -\frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{392} & 0 & -\frac{11\sqrt{42}}{392} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{196} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{3\sqrt{35}}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{14}}{49} & 0 & \frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{42}}{392} & 0 & \frac{\sqrt{210}}{392} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}}{392} & 0 & \frac{5\sqrt{6}}{56} \\
\frac{5\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{210}}{392} & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{15\sqrt{14}}{392} & 0 & -\frac{3\sqrt{35}}{196} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 \\
0 & -\frac{11\sqrt{42}}{392} & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{21}}{28} & 0 & -\frac{11\sqrt{42}}{392} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{35}}{196} & 0 & -\frac{15\sqrt{14}}{392} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & -\frac{2\sqrt{21}}{49} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{210}}{392} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 & -\frac{3}{14} \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0
\end{bmatrix}$$

$\boxed{\text{x426}}$ $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g, 1) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{3}i}{42} & 0 & \frac{\sqrt{6}i}{84} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{5\sqrt{30}i}{168} & 0 & \frac{5\sqrt{2}i}{112} & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{42} & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}i}{112} & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 \\
0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & \frac{3\sqrt{7}i}{112} & 0 & \frac{\sqrt{3}i}{336} & 0 & -\frac{\sqrt{5}i}{16} & 0 & -\frac{i}{112} & 0 \\
-\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & \frac{i}{112} & 0 & \frac{\sqrt{5}i}{16} & 0 & -\frac{\sqrt{3}i}{336} & 0 & -\frac{3\sqrt{7}i}{112} \\
0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & -\frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{13\sqrt{2}i}{112} & 0 \\
0 & 0 & \frac{\sqrt{6}i}{84} & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{112} & 0 & -\frac{5\sqrt{30}i}{168} & 0 & -\frac{\sqrt{70}i}{112} \\
-\frac{\sqrt{70}i}{112} & 0 & -\frac{3\sqrt{7}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\
0 & \frac{13\sqrt{2}i}{112} & 0 & -\frac{i}{112} & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 \\
-\frac{5\sqrt{30}i}{168} & 0 & -\frac{\sqrt{3}i}{336} & 0 & \frac{\sqrt{6}i}{48} & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{5}i}{16} & 0 & \frac{5\sqrt{2}i}{112} & \frac{\sqrt{21}i}{42} & 0 & -\frac{3i}{14} & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 \\
-\frac{5\sqrt{2}i}{112} & 0 & \frac{\sqrt{5}i}{16} & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{3i}{14} & 0 & \frac{\sqrt{21}i}{42} \\
0 & -\frac{\sqrt{6}i}{48} & 0 & \frac{\sqrt{3}i}{336} & 0 & \frac{5\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & -\frac{\sqrt{5}i}{14} & 0 \\
0 & 0 & \frac{i}{112} & 0 & -\frac{13\sqrt{2}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & \frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{105}i}{42} \\
0 & 0 & 0 & \frac{3\sqrt{7}i}{112} & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & -\frac{\sqrt{105}i}{42} & 0
\end{bmatrix}$$

$\boxed{\text{x427}}$ $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{3}}{42} & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{5\sqrt{30}}{168} & 0 & \frac{5\sqrt{2}}{112} & 0 & 0 & 0 \\
-\frac{\sqrt{3}}{42} & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{6}}{48} & 0 & 0 \\
0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & -\frac{3\sqrt{7}}{112} & 0 & \frac{\sqrt{3}}{336} & 0 & \frac{\sqrt{5}}{16} & 0 & -\frac{1}{112} & 0 \\
\frac{\sqrt{6}}{84} & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & -\frac{1}{112} & 0 & \frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{3}}{336} & 0 & -\frac{3\sqrt{7}}{112} \\
0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & \frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{13\sqrt{2}}{112} & 0 \\
0 & 0 & -\frac{\sqrt{6}}{84} & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{112} & 0 & -\frac{5\sqrt{30}}{168} & 0 & \frac{\sqrt{70}}{112} \\
\frac{\sqrt{70}}{112} & 0 & -\frac{3\sqrt{7}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{2}}{112} & 0 & -\frac{1}{112} & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 \\
-\frac{5\sqrt{30}}{168} & 0 & \frac{\sqrt{3}}{336} & 0 & \frac{\sqrt{6}}{48} & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{5}}{16} & 0 & \frac{5\sqrt{2}}{112} & -\frac{\sqrt{21}}{42} & 0 & -\frac{3}{14} & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 \\
\frac{5\sqrt{2}}{112} & 0 & \frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & \frac{3}{14} & 0 & \frac{\sqrt{21}}{42} \\
0 & \frac{\sqrt{6}}{48} & 0 & \frac{\sqrt{3}}{336} & 0 & -\frac{5\sqrt{30}}{168} & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & \frac{\sqrt{5}}{14} & 0 \\
0 & 0 & -\frac{1}{112} & 0 & -\frac{13\sqrt{2}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & \frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{105}}{42} \\
0 & 0 & 0 & -\frac{3\sqrt{7}}{112} & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & -\frac{\sqrt{105}}{42} & 0
\end{bmatrix}$$

$\boxed{\text{x428}}$ $\mathbb{Q}_{4,1}^{(1,-1;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{21}i}{294} & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & \frac{\sqrt{10}i}{112} & 0 & \frac{5\sqrt{210}i}{1176} & 0 & -\frac{5\sqrt{14}i}{112} & 0 & 0 & 0 \\
-\frac{\sqrt{21}i}{294} & 0 & -\frac{\sqrt{210}i}{588} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{14}i}{784} & 0 & -\frac{\sqrt{70}i}{392} & 0 & -\frac{\sqrt{42}i}{48} & 0 & 0 \\
0 & \frac{\sqrt{210}i}{588} & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & -\frac{3i}{16} & 0 & \frac{\sqrt{21}i}{2352} & 0 & -\frac{\sqrt{35}i}{112} & 0 & \frac{\sqrt{7}i}{112} & 0 \\
\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & \frac{\sqrt{210}i}{588} & 0 & 0 & -\frac{\sqrt{7}i}{112} & 0 & \frac{\sqrt{35}i}{112} & 0 & -\frac{\sqrt{21}i}{2352} & 0 & \frac{3i}{16} \\
0 & 0 & 0 & -\frac{\sqrt{210}i}{588} & 0 & -\frac{\sqrt{21}i}{294} & 0 & 0 & \frac{\sqrt{42}i}{48} & 0 & \frac{\sqrt{70}i}{392} & 0 & \frac{13\sqrt{14}i}{784} & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & \frac{\sqrt{21}i}{294} & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{112} & 0 & -\frac{5\sqrt{210}i}{1176} & 0 & -\frac{\sqrt{10}i}{112} \\
-\frac{\sqrt{10}i}{112} & 0 & \frac{3i}{16} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\
0 & \frac{13\sqrt{14}i}{784} & 0 & \frac{\sqrt{7}i}{112} & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & \frac{\sqrt{35}i}{98} & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 \\
-\frac{5\sqrt{210}i}{1176} & 0 & -\frac{\sqrt{21}i}{2352} & 0 & -\frac{\sqrt{42}i}{48} & 0 & 0 & -\frac{\sqrt{35}i}{98} & 0 & \frac{3\sqrt{7}i}{98} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{70}i}{392} & 0 & -\frac{\sqrt{35}i}{112} & 0 & -\frac{5\sqrt{14}i}{112} & -\frac{\sqrt{3}i}{6} & 0 & -\frac{3\sqrt{7}i}{98} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 \\
\frac{5\sqrt{14}i}{112} & 0 & \frac{\sqrt{35}i}{112} & 0 & -\frac{\sqrt{70}i}{392} & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{98} & 0 & -\frac{\sqrt{3}i}{6} \\
0 & \frac{\sqrt{42}i}{48} & 0 & \frac{\sqrt{21}i}{2352} & 0 & \frac{5\sqrt{210}i}{1176} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{98} & 0 & -\frac{\sqrt{35}i}{98} & 0 \\
0 & 0 & -\frac{\sqrt{7}i}{112} & 0 & -\frac{13\sqrt{14}i}{784} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & 0 & \frac{\sqrt{35}i}{98} & 0 & \frac{\sqrt{15}i}{42} \\
0 & 0 & 0 & -\frac{3i}{16} & 0 & \frac{\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{15}i}{42} & 0
\end{bmatrix}$$

$\boxed{\text{x429}}$ $\mathbb{Q}_{4,2}^{(1,-1;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{21}}{294} & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & \frac{\sqrt{10}}{112} & 0 & -\frac{5\sqrt{210}}{1176} & 0 & -\frac{5\sqrt{14}}{112} & 0 & 0 & 0 \\
-\frac{\sqrt{21}}{294} & 0 & \frac{\sqrt{210}}{588} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{14}}{784} & 0 & \frac{\sqrt{70}}{392} & 0 & -\frac{\sqrt{42}}{48} & 0 & 0 \\
0 & \frac{\sqrt{210}}{588} & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & \frac{3}{16} & 0 & \frac{\sqrt{21}}{2352} & 0 & \frac{\sqrt{35}}{112} & 0 & \frac{\sqrt{7}}{112} & 0 \\
-\frac{\sqrt{42}}{84} & 0 & 0 & 0 & -\frac{\sqrt{210}}{588} & 0 & 0 & \frac{\sqrt{7}}{112} & 0 & \frac{\sqrt{35}}{112} & 0 & \frac{\sqrt{21}}{2352} & 0 & \frac{3}{16} \\
0 & 0 & 0 & -\frac{\sqrt{210}}{588} & 0 & \frac{\sqrt{21}}{294} & 0 & 0 & -\frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{70}}{392} & 0 & -\frac{13\sqrt{14}}{784} & 0 \\
0 & 0 & \frac{\sqrt{42}}{84} & 0 & \frac{\sqrt{21}}{294} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{112} & 0 & -\frac{5\sqrt{210}}{1176} & 0 & \frac{\sqrt{10}}{112} \\
\frac{\sqrt{10}}{112} & 0 & \frac{3}{16} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{14}}{784} & 0 & \frac{\sqrt{7}}{112} & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & -\frac{\sqrt{35}}{98} & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 \\
-\frac{5\sqrt{210}}{1176} & 0 & \frac{\sqrt{21}}{2352} & 0 & -\frac{\sqrt{42}}{48} & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{70}}{392} & 0 & \frac{\sqrt{35}}{112} & 0 & -\frac{5\sqrt{14}}{112} & \frac{\sqrt{3}}{6} & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 \\
-\frac{5\sqrt{14}}{112} & 0 & \frac{\sqrt{35}}{112} & 0 & \frac{\sqrt{70}}{392} & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & -\frac{\sqrt{3}}{6} \\
0 & -\frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{21}}{2352} & 0 & -\frac{5\sqrt{210}}{1176} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & \frac{\sqrt{35}}{98} & 0 \\
0 & 0 & \frac{\sqrt{7}}{112} & 0 & -\frac{13\sqrt{14}}{784} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & \frac{\sqrt{35}}{98} & 0 & -\frac{\sqrt{15}}{42} \\
0 & 0 & 0 & \frac{3}{16} & 0 & \frac{\sqrt{10}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{15}}{42} & 0
\end{bmatrix}$$

$\boxed{\text{x430}}$ $\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{7392} & 0 & \frac{\sqrt{22}i}{352} & 0 & -\frac{\sqrt{330}i}{352} & 0 & -\frac{\sqrt{66}i}{96} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{1056} & 0 & -\frac{5\sqrt{66}i}{1056} & 0 & \frac{3\sqrt{110}i}{352} & 0 & \frac{\sqrt{2310}i}{672} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{1232} & 0 & \frac{\sqrt{55}i}{176} & 0 & \frac{5\sqrt{33}i}{528} & 0 & -\frac{\sqrt{165}i}{176} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{176} & 0 & -\frac{5\sqrt{33}i}{528} & 0 & -\frac{\sqrt{55}i}{176} & 0 & \frac{\sqrt{1155}i}{1232} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{672} & 0 & -\frac{3\sqrt{110}i}{352} & 0 & \frac{5\sqrt{66}i}{1056} & 0 & \frac{\sqrt{330}i}{1056} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{96} & 0 & \frac{\sqrt{330}i}{352} & 0 & -\frac{\sqrt{22}i}{352} & 0 & -\frac{\sqrt{462}i}{7392} \\
-\frac{\sqrt{462}i}{7392} & 0 & \frac{\sqrt{1155}i}{1232} & 0 & \frac{\sqrt{2310}i}{672} & 0 & 0 & -\frac{3\sqrt{77}i}{616} & 0 & \frac{3\sqrt{385}i}{308} & 0 & \frac{\sqrt{231}i}{56} & 0 & 0 \\
0 & \frac{\sqrt{330}i}{1056} & 0 & -\frac{\sqrt{165}i}{176} & 0 & -\frac{\sqrt{66}i}{96} & \frac{3\sqrt{77}i}{616} & 0 & \frac{\sqrt{33}i}{44} & 0 & -\frac{3\sqrt{55}i}{88} & 0 & 0 & 0 \\
-\frac{\sqrt{22}i}{352} & 0 & -\frac{\sqrt{55}i}{176} & 0 & \frac{3\sqrt{110}i}{352} & 0 & 0 & -\frac{\sqrt{33}i}{44} & 0 & -\frac{\sqrt{165}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{231}i}{56} \\
0 & \frac{5\sqrt{66}i}{1056} & 0 & \frac{5\sqrt{33}i}{528} & 0 & -\frac{\sqrt{330}i}{352} & -\frac{3\sqrt{385}i}{308} & 0 & \frac{\sqrt{165}i}{88} & 0 & 0 & 0 & \frac{3\sqrt{55}i}{88} & 0 \\
\frac{\sqrt{330}i}{352} & 0 & -\frac{5\sqrt{33}i}{528} & 0 & -\frac{5\sqrt{66}i}{1056} & 0 & 0 & \frac{3\sqrt{55}i}{88} & 0 & 0 & 0 & \frac{\sqrt{165}i}{88} & 0 & -\frac{3\sqrt{385}i}{308} \\
0 & -\frac{3\sqrt{110}i}{352} & 0 & \frac{\sqrt{55}i}{176} & 0 & \frac{\sqrt{22}i}{352} & -\frac{\sqrt{231}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{165}i}{88} & 0 & -\frac{\sqrt{33}i}{44} & 0 \\
\frac{\sqrt{66}i}{96} & 0 & \frac{\sqrt{165}i}{176} & 0 & -\frac{\sqrt{330}i}{1056} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}i}{88} & 0 & \frac{\sqrt{33}i}{44} & 0 & \frac{3\sqrt{77}i}{616} \\
0 & -\frac{\sqrt{2310}i}{672} & 0 & -\frac{\sqrt{1155}i}{1232} & 0 & \frac{\sqrt{462}i}{7392} & 0 & 0 & \frac{\sqrt{231}i}{56} & 0 & \frac{3\sqrt{385}i}{308} & 0 & -\frac{3\sqrt{77}i}{616} & 0
\end{bmatrix}$$

$\boxed{\text{x431}}$ $\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{7392} & 0 & -\frac{\sqrt{22}}{352} & 0 & -\frac{\sqrt{330}}{352} & 0 & \frac{\sqrt{66}}{96} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{1056} & 0 & \frac{5\sqrt{66}}{1056} & 0 & \frac{3\sqrt{110}}{352} & 0 & -\frac{\sqrt{2310}}{672} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{1232} & 0 & \frac{\sqrt{55}}{176} & 0 & -\frac{5\sqrt{33}}{528} & 0 & -\frac{\sqrt{165}}{176} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{176} & 0 & -\frac{5\sqrt{33}}{528} & 0 & \frac{\sqrt{55}}{176} & 0 & \frac{\sqrt{1155}}{1232} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{672} & 0 & \frac{3\sqrt{110}}{352} & 0 & \frac{5\sqrt{66}}{1056} & 0 & -\frac{\sqrt{330}}{1056} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{96} & 0 & -\frac{\sqrt{330}}{352} & 0 & -\frac{\sqrt{22}}{352} & 0 & \frac{\sqrt{462}}{7392} \\
\frac{\sqrt{462}}{7392} & 0 & \frac{\sqrt{1155}}{1232} & 0 & -\frac{\sqrt{2310}}{672} & 0 & 0 & \frac{3\sqrt{77}}{616} & 0 & \frac{3\sqrt{385}}{308} & 0 & -\frac{\sqrt{231}}{56} & 0 & 0 \\
0 & -\frac{\sqrt{330}}{1056} & 0 & -\frac{\sqrt{165}}{176} & 0 & \frac{\sqrt{66}}{96} & \frac{3\sqrt{77}}{616} & 0 & -\frac{\sqrt{33}}{44} & 0 & -\frac{3\sqrt{55}}{88} & 0 & 0 & 0 \\
-\frac{\sqrt{22}}{352} & 0 & \frac{\sqrt{55}}{176} & 0 & \frac{3\sqrt{110}}{352} & 0 & 0 & -\frac{\sqrt{33}}{44} & 0 & \frac{\sqrt{165}}{88} & 0 & 0 & 0 & \frac{\sqrt{231}}{56} \\
0 & \frac{5\sqrt{66}}{1056} & 0 & -\frac{5\sqrt{33}}{528} & 0 & -\frac{\sqrt{330}}{352} & \frac{3\sqrt{385}}{308} & 0 & \frac{\sqrt{165}}{88} & 0 & 0 & 0 & \frac{3\sqrt{55}}{88} & 0 \\
-\frac{\sqrt{330}}{352} & 0 & -\frac{5\sqrt{33}}{528} & 0 & \frac{5\sqrt{66}}{1056} & 0 & 0 & -\frac{3\sqrt{55}}{88} & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & -\frac{3\sqrt{385}}{308} \\
0 & \frac{3\sqrt{110}}{352} & 0 & \frac{\sqrt{55}}{176} & 0 & -\frac{\sqrt{22}}{352} & -\frac{\sqrt{231}}{56} & 0 & 0 & 0 & -\frac{\sqrt{165}}{88} & 0 & \frac{\sqrt{33}}{44} & 0 \\
\frac{\sqrt{66}}{96} & 0 & -\frac{\sqrt{165}}{176} & 0 & -\frac{\sqrt{330}}{1056} & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}}{88} & 0 & \frac{\sqrt{33}}{44} & 0 & -\frac{3\sqrt{77}}{616} \\
0 & -\frac{\sqrt{2310}}{672} & 0 & \frac{\sqrt{1155}}{1232} & 0 & \frac{\sqrt{462}}{7392} & 0 & 0 & \frac{\sqrt{231}}{56} & 0 & -\frac{3\sqrt{385}}{308} & 0 & -\frac{3\sqrt{77}}{616} & 0
\end{bmatrix}$$

$\boxed{\text{x432}}$ $\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{448} & 0 & \frac{\sqrt{3}i}{64} & 0 & \frac{\sqrt{5}i}{64} & 0 & \frac{i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{64} & 0 & -\frac{5i}{64} & 0 & -\frac{\sqrt{15}i}{64} & 0 & -\frac{\sqrt{35}i}{448} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{448} & 0 & \frac{\sqrt{30}i}{64} & 0 & \frac{5\sqrt{2}i}{64} & 0 & \frac{\sqrt{10}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{64} & 0 & -\frac{5\sqrt{2}i}{64} & 0 & -\frac{\sqrt{30}i}{64} & 0 & -\frac{\sqrt{70}i}{448} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{448} & 0 & \frac{\sqrt{15}i}{64} & 0 & \frac{5i}{64} & 0 & \frac{\sqrt{5}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{64} & 0 & -\frac{\sqrt{5}i}{64} & 0 & -\frac{\sqrt{3}i}{64} & 0 & -\frac{\sqrt{7}i}{448} \\
-\frac{\sqrt{7}i}{448} & 0 & -\frac{\sqrt{70}i}{448} & 0 & -\frac{\sqrt{35}i}{448} & 0 & 0 & -\frac{3\sqrt{42}i}{224} & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{3\sqrt{14}i}{224} & 0 & 0 \\
0 & \frac{\sqrt{5}i}{64} & 0 & \frac{\sqrt{10}i}{64} & 0 & \frac{i}{64} & \frac{3\sqrt{42}i}{224} & 0 & \frac{3\sqrt{2}i}{16} & 0 & \frac{\sqrt{30}i}{32} & 0 & 0 & 0 \\
-\frac{\sqrt{3}i}{64} & 0 & -\frac{\sqrt{30}i}{64} & 0 & -\frac{\sqrt{15}i}{64} & 0 & 0 & -\frac{3\sqrt{2}i}{16} & 0 & -\frac{3\sqrt{10}i}{32} & 0 & 0 & 0 & \frac{3\sqrt{14}i}{224} \\
0 & \frac{5i}{64} & 0 & \frac{5\sqrt{2}i}{64} & 0 & \frac{\sqrt{5}i}{64} & \frac{\sqrt{210}i}{112} & 0 & \frac{3\sqrt{10}i}{32} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{32} & 0 \\
-\frac{\sqrt{5}i}{64} & 0 & -\frac{5\sqrt{2}i}{64} & 0 & -\frac{5i}{64} & 0 & 0 & -\frac{\sqrt{30}i}{32} & 0 & 0 & 0 & \frac{3\sqrt{10}i}{32} & 0 & \frac{\sqrt{210}i}{112} \\
0 & \frac{\sqrt{15}i}{64} & 0 & \frac{\sqrt{30}i}{64} & 0 & \frac{\sqrt{3}i}{64} & \frac{3\sqrt{14}i}{224} & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{32} & 0 & -\frac{3\sqrt{2}i}{16} & 0 \\
-\frac{i}{64} & 0 & -\frac{\sqrt{10}i}{64} & 0 & -\frac{\sqrt{5}i}{64} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{32} & 0 & \frac{3\sqrt{2}i}{16} & 0 & \frac{3\sqrt{42}i}{224} \\
0 & \frac{\sqrt{35}i}{448} & 0 & \frac{\sqrt{70}i}{448} & 0 & \frac{\sqrt{7}i}{448} & 0 & 0 & -\frac{3\sqrt{14}i}{224} & 0 & -\frac{\sqrt{210}i}{112} & 0 & -\frac{3\sqrt{42}i}{224} & 0
\end{bmatrix}$$

$\boxed{\text{x433}}$ $\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{448} & 0 & -\frac{\sqrt{3}}{64} & 0 & \frac{\sqrt{5}}{64} & 0 & -\frac{1}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{64} & 0 & \frac{5}{64} & 0 & -\frac{\sqrt{15}}{64} & 0 & \frac{\sqrt{35}}{448} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{448} & 0 & \frac{\sqrt{30}}{64} & 0 & -\frac{5\sqrt{2}}{64} & 0 & \frac{\sqrt{10}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{64} & 0 & -\frac{5\sqrt{2}}{64} & 0 & \frac{\sqrt{30}}{64} & 0 & -\frac{\sqrt{70}}{448} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{448} & 0 & -\frac{\sqrt{15}}{64} & 0 & \frac{5}{64} & 0 & -\frac{\sqrt{5}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{64} & 0 & \frac{\sqrt{5}}{64} & 0 & -\frac{\sqrt{3}}{64} & 0 & \frac{\sqrt{7}}{448} \\
\frac{\sqrt{7}}{448} & 0 & -\frac{\sqrt{70}}{448} & 0 & \frac{\sqrt{35}}{448} & 0 & 0 & \frac{3\sqrt{42}}{224} & 0 & -\frac{\sqrt{210}}{112} & 0 & \frac{3\sqrt{14}}{224} & 0 & 0 \\
0 & -\frac{\sqrt{5}}{64} & 0 & \frac{\sqrt{10}}{64} & 0 & -\frac{1}{64} & \frac{3\sqrt{42}}{224} & 0 & -\frac{3\sqrt{2}}{16} & 0 & \frac{\sqrt{30}}{32} & 0 & 0 & 0 \\
-\frac{\sqrt{3}}{64} & 0 & \frac{\sqrt{30}}{64} & 0 & -\frac{\sqrt{15}}{64} & 0 & 0 & -\frac{3\sqrt{2}}{16} & 0 & \frac{3\sqrt{10}}{32} & 0 & 0 & 0 & -\frac{3\sqrt{14}}{224} \\
0 & \frac{5}{64} & 0 & -\frac{5\sqrt{2}}{64} & 0 & \frac{\sqrt{5}}{64} & -\frac{\sqrt{210}}{112} & 0 & \frac{3\sqrt{10}}{32} & 0 & 0 & 0 & -\frac{\sqrt{30}}{32} & 0 \\
\frac{\sqrt{5}}{64} & 0 & -\frac{5\sqrt{2}}{64} & 0 & \frac{5}{64} & 0 & 0 & \frac{\sqrt{30}}{32} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{32} & 0 & \frac{\sqrt{210}}{112} \\
0 & -\frac{\sqrt{15}}{64} & 0 & \frac{\sqrt{30}}{64} & 0 & -\frac{\sqrt{3}}{64} & \frac{3\sqrt{14}}{224} & 0 & 0 & 0 & -\frac{3\sqrt{10}}{32} & 0 & \frac{3\sqrt{2}}{16} & 0 \\
-\frac{1}{64} & 0 & \frac{\sqrt{10}}{64} & 0 & -\frac{\sqrt{5}}{64} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{32} & 0 & \frac{3\sqrt{2}}{16} & 0 & -\frac{3\sqrt{42}}{224} \\
0 & \frac{\sqrt{35}}{448} & 0 & -\frac{\sqrt{70}}{448} & 0 & \frac{\sqrt{7}}{448} & 0 & 0 & -\frac{3\sqrt{14}}{224} & 0 & \frac{\sqrt{210}}{112} & 0 & -\frac{3\sqrt{42}}{224} & 0
\end{bmatrix}$$

$\boxed{\text{x434}}$ $\mathbb{Q}_{6,1}^{(1,-1;a)}(E_g,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{14784} & 0 & \frac{\sqrt{165}i}{2112} & 0 & -\frac{9\sqrt{11}i}{704} & 0 & \frac{\sqrt{55}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}i}{2112} & 0 & -\frac{5\sqrt{55}i}{2112} & 0 & \frac{9\sqrt{33}i}{704} & 0 & -\frac{5\sqrt{77}i}{448} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{154}i}{4928} & 0 & \frac{5\sqrt{66}i}{2112} & 0 & \frac{5\sqrt{110}i}{2112} & 0 & -\frac{9\sqrt{22}i}{704} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{22}i}{704} & 0 & -\frac{5\sqrt{110}i}{2112} & 0 & -\frac{5\sqrt{66}i}{2112} & 0 & \frac{9\sqrt{154}i}{4928} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{448} & 0 & -\frac{9\sqrt{33}i}{704} & 0 & \frac{5\sqrt{55}i}{2112} & 0 & \frac{5\sqrt{11}i}{2112} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{64} & 0 & \frac{9\sqrt{11}i}{704} & 0 & -\frac{\sqrt{165}i}{2112} & 0 & -\frac{\sqrt{385}i}{14784} \\
-\frac{\sqrt{385}i}{14784} & 0 & \frac{9\sqrt{154}i}{4928} & 0 & -\frac{5\sqrt{77}i}{448} & 0 & 0 & -\frac{\sqrt{2310}i}{2464} & 0 & \frac{9\sqrt{462}i}{1232} & 0 & -\frac{3\sqrt{770}i}{224} & 0 & 0 \\
0 & \frac{5\sqrt{11}i}{2112} & 0 & -\frac{9\sqrt{22}i}{704} & 0 & \frac{\sqrt{55}i}{64} & \frac{\sqrt{2310}i}{2464} & 0 & \frac{\sqrt{110}i}{176} & 0 & -\frac{9\sqrt{66}i}{352} & 0 & 0 & 0 \\
-\frac{\sqrt{165}i}{2112} & 0 & -\frac{5\sqrt{66}i}{2112} & 0 & \frac{9\sqrt{33}i}{704} & 0 & 0 & -\frac{\sqrt{110}i}{176} & 0 & -\frac{5\sqrt{22}i}{352} & 0 & 0 & 0 & \frac{3\sqrt{770}i}{224} \\
0 & \frac{5\sqrt{55}i}{2112} & 0 & \frac{5\sqrt{110}i}{2112} & 0 & -\frac{9\sqrt{11}i}{704} & -\frac{9\sqrt{462}i}{1232} & 0 & \frac{5\sqrt{22}i}{352} & 0 & 0 & 0 & \frac{9\sqrt{66}i}{352} & 0 \\
\frac{9\sqrt{11}i}{704} & 0 & -\frac{5\sqrt{110}i}{2112} & 0 & -\frac{5\sqrt{55}i}{2112} & 0 & 0 & \frac{9\sqrt{66}i}{352} & 0 & 0 & 0 & \frac{5\sqrt{22}i}{352} & 0 & -\frac{9\sqrt{462}i}{1232} \\
0 & -\frac{9\sqrt{33}i}{704} & 0 & \frac{5\sqrt{66}i}{2112} & 0 & \frac{\sqrt{165}i}{2112} & \frac{3\sqrt{770}i}{224} & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{352} & 0 & -\frac{\sqrt{110}i}{176} & 0 \\
-\frac{\sqrt{55}i}{64} & 0 & \frac{9\sqrt{22}i}{704} & 0 & -\frac{5\sqrt{11}i}{2112} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{66}i}{352} & 0 & \frac{\sqrt{110}i}{176} & 0 & \frac{\sqrt{2310}i}{2464} \\
0 & \frac{5\sqrt{77}i}{448} & 0 & -\frac{9\sqrt{154}i}{4928} & 0 & \frac{\sqrt{385}i}{14784} & 0 & 0 & -\frac{3\sqrt{770}i}{224} & 0 & \frac{9\sqrt{462}i}{1232} & 0 & -\frac{\sqrt{2310}i}{2464} & 0
\end{bmatrix}$$

$\boxed{\text{x435}}$ $\mathbb{Q}_{6,2}^{(1,-1;a)}(E_g,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{14784} & 0 & -\frac{\sqrt{165}}{2112} & 0 & -\frac{9\sqrt{11}}{704} & 0 & -\frac{\sqrt{55}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}}{2112} & 0 & \frac{5\sqrt{55}}{2112} & 0 & \frac{9\sqrt{33}}{704} & 0 & \frac{5\sqrt{77}}{448} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{4928} & 0 & \frac{5\sqrt{66}}{2112} & 0 & -\frac{5\sqrt{110}}{2112} & 0 & -\frac{9\sqrt{22}}{704} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{22}}{704} & 0 & -\frac{5\sqrt{110}}{2112} & 0 & \frac{5\sqrt{66}}{2112} & 0 & \frac{9\sqrt{154}}{4928} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{448} & 0 & \frac{9\sqrt{33}}{704} & 0 & \frac{5\sqrt{55}}{2112} & 0 & -\frac{5\sqrt{11}}{2112} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{64} & 0 & -\frac{9\sqrt{11}}{704} & 0 & -\frac{\sqrt{165}}{2112} & 0 & \frac{\sqrt{385}}{14784} \\
\frac{\sqrt{385}}{14784} & 0 & \frac{9\sqrt{154}}{4928} & 0 & \frac{5\sqrt{77}}{448} & 0 & 0 & \frac{\sqrt{2310}}{2464} & 0 & \frac{9\sqrt{462}}{1232} & 0 & \frac{3\sqrt{770}}{224} & 0 & 0 \\
0 & -\frac{5\sqrt{11}}{2112} & 0 & -\frac{9\sqrt{22}}{704} & 0 & -\frac{\sqrt{55}}{64} & \frac{\sqrt{2310}}{2464} & 0 & -\frac{\sqrt{110}}{176} & 0 & -\frac{9\sqrt{66}}{352} & 0 & 0 & 0 \\
-\frac{\sqrt{165}}{2112} & 0 & \frac{5\sqrt{66}}{2112} & 0 & \frac{9\sqrt{33}}{704} & 0 & 0 & -\frac{\sqrt{110}}{176} & 0 & \frac{5\sqrt{22}}{352} & 0 & 0 & 0 & -\frac{3\sqrt{770}}{224} \\
0 & \frac{5\sqrt{55}}{2112} & 0 & -\frac{5\sqrt{110}}{2112} & 0 & -\frac{9\sqrt{11}}{704} & \frac{9\sqrt{462}}{1232} & 0 & \frac{5\sqrt{22}}{352} & 0 & 0 & 0 & \frac{9\sqrt{66}}{352} & 0 \\
-\frac{9\sqrt{11}}{704} & 0 & -\frac{5\sqrt{110}}{2112} & 0 & \frac{5\sqrt{55}}{2112} & 0 & 0 & -\frac{9\sqrt{66}}{352} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{352} & 0 & -\frac{9\sqrt{462}}{1232} \\
0 & \frac{9\sqrt{33}}{704} & 0 & \frac{5\sqrt{66}}{2112} & 0 & -\frac{\sqrt{165}}{2112} & \frac{3\sqrt{770}}{224} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{352} & 0 & \frac{\sqrt{110}}{176} & 0 \\
-\frac{\sqrt{55}}{64} & 0 & -\frac{9\sqrt{22}}{704} & 0 & -\frac{5\sqrt{11}}{2112} & 0 & 0 & 0 & 0 & \frac{9\sqrt{66}}{352} & 0 & \frac{\sqrt{110}}{176} & 0 & -\frac{\sqrt{2310}}{2464} \\
0 & \frac{5\sqrt{77}}{448} & 0 & \frac{9\sqrt{154}}{4928} & 0 & \frac{\sqrt{385}}{14784} & 0 & 0 & -\frac{3\sqrt{770}}{224} & 0 & -\frac{9\sqrt{462}}{1232} & 0 & -\frac{\sqrt{2310}}{2464} & 0
\end{bmatrix}$$

$\boxed{\text{x436}}$ $\mathbb{Q}_{0}^{(1,1;a)}(A_{1g}) =$

$$\begin{bmatrix}
-\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28}
\end{bmatrix}$$

$\boxed{\text{x437}}$ $\mathbb{Q}_2^{(1,1;a)}(A_{1g}) =$

$$\begin{bmatrix}
\frac{15\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{6\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{6\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{147} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{98} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14}
\end{bmatrix}$$

$\boxed{\text{x438}}$ $\mathbb{Q}_4^{(1,1;a)}(A_{1g}, 1) =$

$$\begin{bmatrix}
-\frac{\sqrt{110}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{84} & 0 & 0 & \frac{2\sqrt{165}}{231} & 0 & 0 & 0 & \frac{2\sqrt{55}}{231} & 0 & 0 \\
0 & \frac{\sqrt{110}}{28} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{84} & 0 & 0 & -\frac{8\sqrt{11}}{231} & 0 & 0 & 0 & \frac{4\sqrt{33}}{231} & 0 \\
0 & 0 & -\frac{\sqrt{110}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{231} & 0 & 0 & 0 & \frac{\sqrt{462}}{231} \\
0 & 0 & 0 & -\frac{\sqrt{110}}{42} & 0 & 0 & -\frac{\sqrt{462}}{231} & 0 & 0 & 0 & \frac{\sqrt{330}}{231} & 0 & 0 & 0 \\
-\frac{5\sqrt{22}}{84} & 0 & 0 & 0 & \frac{\sqrt{110}}{28} & 0 & 0 & -\frac{4\sqrt{33}}{231} & 0 & 0 & 0 & \frac{8\sqrt{11}}{231} & 0 & 0 \\
0 & -\frac{5\sqrt{22}}{84} & 0 & 0 & 0 & -\frac{\sqrt{110}}{84} & 0 & 0 & -\frac{2\sqrt{55}}{231} & 0 & 0 & 0 & -\frac{2\sqrt{165}}{231} & 0 \\
0 & 0 & 0 & -\frac{\sqrt{462}}{231} & 0 & 0 & \frac{\sqrt{110}}{264} & 0 & 0 & 0 & \frac{5\sqrt{154}}{1848} & 0 & 0 & 0 \\
\frac{2\sqrt{165}}{231} & 0 & 0 & 0 & -\frac{4\sqrt{33}}{231} & 0 & 0 & -\frac{13\sqrt{110}}{1848} & 0 & 0 & 0 & \frac{5\sqrt{330}}{1848} & 0 & 0 \\
0 & -\frac{8\sqrt{11}}{231} & 0 & 0 & 0 & -\frac{2\sqrt{55}}{231} & 0 & 0 & -\frac{\sqrt{110}}{616} & 0 & 0 & 0 & \frac{5\sqrt{330}}{1848} & 0 \\
0 & 0 & -\frac{\sqrt{330}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{110}}{616} & 0 & 0 & 0 & \frac{5\sqrt{154}}{1848} \\
0 & 0 & 0 & \frac{\sqrt{330}}{231} & 0 & 0 & \frac{5\sqrt{154}}{1848} & 0 & 0 & 0 & \frac{3\sqrt{110}}{616} & 0 & 0 & 0 \\
\frac{2\sqrt{55}}{231} & 0 & 0 & 0 & \frac{8\sqrt{11}}{231} & 0 & 0 & \frac{5\sqrt{330}}{1848} & 0 & 0 & 0 & -\frac{\sqrt{110}}{616} & 0 & 0 \\
0 & \frac{4\sqrt{33}}{231} & 0 & 0 & 0 & -\frac{2\sqrt{165}}{231} & 0 & 0 & \frac{5\sqrt{330}}{1848} & 0 & 0 & 0 & -\frac{13\sqrt{110}}{1848} & 0 \\
0 & 0 & \frac{\sqrt{462}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{1848} & 0 & 0 & 0 & \frac{\sqrt{110}}{264}
\end{bmatrix}$$

$\boxed{\text{x439}}$ $\mathbb{Q}_4^{(1,1;a)}(A_{1g},2) =$

$$\begin{bmatrix}
-\frac{5\sqrt{154}}{588} & 0 & 0 & 0 & \frac{\sqrt{770}}{84} & 0 & 0 & \frac{10\sqrt{231}}{1617} & 0 & 0 & 0 & -\frac{2\sqrt{77}}{231} & 0 & 0 \\
0 & \frac{5\sqrt{154}}{196} & 0 & 0 & 0 & \frac{\sqrt{770}}{84} & 0 & 0 & -\frac{8\sqrt{385}}{1617} & 0 & 0 & 0 & -\frac{4\sqrt{1155}}{1155} & 0 \\
0 & 0 & -\frac{5\sqrt{154}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{1617} & 0 & 0 & 0 & -\frac{\sqrt{330}}{165} \\
0 & 0 & 0 & -\frac{5\sqrt{154}}{294} & 0 & 0 & \frac{\sqrt{330}}{165} & 0 & 0 & 0 & \frac{5\sqrt{462}}{1617} & 0 & 0 & 0 \\
\frac{\sqrt{770}}{84} & 0 & 0 & 0 & \frac{5\sqrt{154}}{196} & 0 & 0 & \frac{4\sqrt{1155}}{1155} & 0 & 0 & 0 & \frac{8\sqrt{385}}{1617} & 0 & 0 \\
0 & \frac{\sqrt{770}}{84} & 0 & 0 & 0 & -\frac{5\sqrt{154}}{588} & 0 & 0 & \frac{2\sqrt{77}}{231} & 0 & 0 & 0 & -\frac{10\sqrt{231}}{1617} & 0 \\
0 & 0 & 0 & \frac{\sqrt{330}}{165} & 0 & 0 & \frac{5\sqrt{154}}{1848} & 0 & 0 & 0 & -\frac{\sqrt{110}}{264} & 0 & 0 & 0 \\
\frac{10\sqrt{231}}{1617} & 0 & 0 & 0 & \frac{4\sqrt{1155}}{1155} & 0 & 0 & -\frac{65\sqrt{154}}{12936} & 0 & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & 0 \\
0 & -\frac{8\sqrt{385}}{1617} & 0 & 0 & 0 & \frac{2\sqrt{77}}{231} & 0 & 0 & -\frac{5\sqrt{154}}{4312} & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 \\
0 & 0 & -\frac{5\sqrt{462}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{154}}{4312} & 0 & 0 & 0 & -\frac{\sqrt{110}}{264} \\
0 & 0 & 0 & \frac{5\sqrt{462}}{1617} & 0 & 0 & -\frac{\sqrt{110}}{264} & 0 & 0 & 0 & \frac{15\sqrt{154}}{4312} & 0 & 0 & 0 \\
-\frac{2\sqrt{77}}{231} & 0 & 0 & 0 & \frac{8\sqrt{385}}{1617} & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & 0 & 0 & -\frac{5\sqrt{154}}{4312} & 0 & 0 \\
0 & -\frac{4\sqrt{1155}}{1155} & 0 & 0 & 0 & -\frac{10\sqrt{231}}{1617} & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & 0 & 0 & -\frac{65\sqrt{154}}{12936} & 0 \\
0 & 0 & -\frac{\sqrt{330}}{165} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{264} & 0 & 0 & 0 & \frac{5\sqrt{154}}{1848}
\end{bmatrix}$$

$\boxed{\text{x440}}$ $\mathbb{Q}_4^{(1,1;a)}(A_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{33}i}{231} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{55}i}{385} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{770}i}{385} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{33}i}{231} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{924} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{308} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{33}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{308} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{924} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{4\sqrt{33}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}i}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{924} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x441}}$ $\mathbb{Q}_2^{(1,1;a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{147} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{42}}{196} & 0 & 0 & \frac{\sqrt{10}}{21} & 0 & 0 & 0 & -\frac{4\sqrt{14}}{147} & 0 & 0 & 0 \\
\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & \frac{9\sqrt{42}}{196} & 0 & 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & 0 & -\frac{2\sqrt{105}}{147} & 0 & 0 \\
0 & \frac{9\sqrt{42}}{196} & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & \frac{2\sqrt{105}}{147} & 0 & 0 & 0 & -\frac{4\sqrt{35}}{147} & 0 \\
0 & 0 & \frac{9\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{14}}{147} & 0 & 0 & 0 & -\frac{\sqrt{10}}{21} \\
0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{147} & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{105}}{147} & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 \\
-\frac{\sqrt{70}}{147} & 0 & 0 & 0 & \frac{4\sqrt{14}}{147} & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 \\
0 & -\frac{4\sqrt{14}}{147} & 0 & 0 & 0 & \frac{\sqrt{70}}{147} & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 \\
0 & 0 & -\frac{2\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & -\frac{1}{14} \\
0 & 0 & 0 & -\frac{4\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x442}}$ $\mathbb{Q}_4^{(1,1;a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{1155}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{231}}{294} & 0 & 0 & \frac{6\sqrt{55}}{385} & 0 & 0 & 0 & -\frac{2\sqrt{77}}{539} & 0 & 0 & 0 \\
\frac{\sqrt{1155}}{98} & 0 & 0 & 0 & -\frac{5\sqrt{231}}{294} & 0 & 0 & -\frac{9\sqrt{770}}{2695} & 0 & 0 & 0 & \frac{17\sqrt{2310}}{8085} & 0 & 0 \\
0 & -\frac{5\sqrt{231}}{294} & 0 & 0 & 0 & \frac{\sqrt{1155}}{98} & 0 & 0 & -\frac{17\sqrt{2310}}{8085} & 0 & 0 & 0 & \frac{9\sqrt{770}}{2695} & 0 \\
0 & 0 & -\frac{5\sqrt{231}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 & 0 & -\frac{6\sqrt{55}}{385} \\
0 & 0 & 0 & \frac{\sqrt{1155}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}}{539} & 0 & 0 & 0 \\
0 & \frac{6\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{9\sqrt{770}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{6468} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{17\sqrt{2310}}{8085} & 0 & 0 & -\frac{5\sqrt{22}}{308} & 0 & 0 & 0 & \frac{\sqrt{770}}{539} & 0 & 0 & 0 \\
-\frac{4\sqrt{385}}{539} & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 & -\frac{\sqrt{2310}}{6468} & 0 & 0 & 0 & \frac{\sqrt{770}}{539} & 0 & 0 \\
0 & -\frac{2\sqrt{77}}{539} & 0 & 0 & 0 & \frac{4\sqrt{385}}{539} & 0 & 0 & \frac{\sqrt{770}}{539} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{6468} & 0 \\
0 & 0 & \frac{17\sqrt{2310}}{8085} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{539} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{308} \\
0 & 0 & 0 & \frac{9\sqrt{770}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{6468} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{6\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{308} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x443}}$ $\mathbb{Q}_2^{(1,1;a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{147} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{9\sqrt{42}i}{196} & 0 & 0 & \frac{\sqrt{10}i}{21} & 0 & 0 & 0 & \frac{4\sqrt{14}i}{147} & 0 & 0 & 0 \\
\frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & -\frac{9\sqrt{42}i}{196} & 0 & 0 & \frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & \frac{2\sqrt{105}i}{147} & 0 & 0 \\
0 & \frac{9\sqrt{42}i}{196} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & 0 & \frac{2\sqrt{105}i}{147} & 0 & 0 & 0 & \frac{4\sqrt{35}i}{147} & 0 \\
0 & 0 & \frac{9\sqrt{42}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{14}i}{147} & 0 & 0 & 0 & \frac{\sqrt{10}i}{21} \\
0 & 0 & 0 & \frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{147} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{105}i}{147} & 0 & 0 & -\frac{i}{14} & 0 & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 & 0 \\
-\frac{\sqrt{70}i}{147} & 0 & 0 & 0 & -\frac{4\sqrt{14}i}{147} & 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 \\
0 & -\frac{4\sqrt{14}i}{147} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{147} & 0 & 0 & -\frac{\sqrt{35}i}{49} & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 \\
0 & 0 & -\frac{2\sqrt{105}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{49} & 0 & 0 & 0 & \frac{i}{14} \\
0 & 0 & 0 & -\frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x444}}$ $\mathbb{Q}_4^{(1,1;a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{1155}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}i}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{231}i}{294} & 0 & 0 & -\frac{6\sqrt{55}i}{385} & 0 & 0 & 0 & -\frac{2\sqrt{77}i}{539} & 0 & 0 & 0 \\
-\frac{\sqrt{1155}i}{98} & 0 & 0 & 0 & -\frac{5\sqrt{231}i}{294} & 0 & 0 & \frac{9\sqrt{770}i}{2695} & 0 & 0 & 0 & \frac{17\sqrt{2310}i}{8085} & 0 & 0 \\
0 & \frac{5\sqrt{231}i}{294} & 0 & 0 & 0 & \frac{\sqrt{1155}i}{98} & 0 & 0 & \frac{17\sqrt{2310}i}{8085} & 0 & 0 & 0 & \frac{9\sqrt{770}i}{2695} & 0 \\
0 & 0 & \frac{5\sqrt{231}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{77}i}{539} & 0 & 0 & 0 & -\frac{6\sqrt{55}i}{385} \\
0 & 0 & 0 & -\frac{\sqrt{1155}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}i}{539} & 0 & 0 & 0 \\
0 & \frac{6\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{9\sqrt{770}i}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{6468} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{17\sqrt{2310}i}{8085} & 0 & 0 & \frac{5\sqrt{22}i}{308} & 0 & 0 & 0 & \frac{\sqrt{770}i}{539} & 0 & 0 & 0 \\
\frac{4\sqrt{385}i}{539} & 0 & 0 & 0 & \frac{2\sqrt{77}i}{539} & 0 & 0 & \frac{\sqrt{2310}i}{6468} & 0 & 0 & 0 & \frac{\sqrt{770}i}{539} & 0 & 0 \\
0 & \frac{2\sqrt{77}i}{539} & 0 & 0 & 0 & \frac{4\sqrt{385}i}{539} & 0 & 0 & -\frac{\sqrt{770}i}{539} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{6468} & 0 \\
0 & 0 & -\frac{17\sqrt{2310}i}{8085} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{539} & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{308} \\
0 & 0 & 0 & -\frac{9\sqrt{770}i}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{6468} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{6\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}i}{308} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x445}}$ $\mathbb{Q}_{2,1}^{(1,1;a)}(E_g) = \begin{bmatrix}
0 & -\frac{3\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{42} & 0 & \frac{5\sqrt{42}i}{294} & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{105}i}{98} & 0 & -\frac{3\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{294} & 0 & \frac{11\sqrt{14}i}{294} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{147} & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & \frac{\sqrt{105}i}{147} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{42}i}{98} & 0 & \frac{3\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{14}i}{294} & 0 & -\frac{\sqrt{70}i}{294} & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{294} & 0 & -\frac{5\sqrt{2}i}{42} \\
-\frac{5\sqrt{2}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{70}i}{294} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & \frac{2\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{42}i}{294} & 0 & \frac{\sqrt{105}i}{147} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}i}{49} & 0 & \frac{\sqrt{35}i}{98} & 0 & 0 & 0 & 0 \\
0 & -\frac{11\sqrt{14}i}{294} & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & \frac{11\sqrt{14}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}i}{147} & 0 & \frac{5\sqrt{42}i}{294} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{98} & 0 & -\frac{2\sqrt{7}i}{49} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}i}{49} & 0 & -\frac{\sqrt{3}i}{14} \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0
\end{bmatrix}$

$\boxed{\text{x446}}$ $\mathbb{Q}_{2,2}^{(1,1;a)}(E_g) = \begin{bmatrix}
0 & \frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{42} & 0 & -\frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{105}}{98} & 0 & \frac{3\sqrt{42}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{294} & 0 & -\frac{11\sqrt{14}}{294} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{147} & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{42}}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{21} & 0 & -\frac{\sqrt{105}}{147} & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{42}}{98} & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & -\frac{11\sqrt{14}}{294} & 0 & \frac{\sqrt{70}}{294} & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{294} & 0 & \frac{5\sqrt{2}}{42} \\
\frac{5\sqrt{2}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{70}}{294} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{42}}{294} & 0 & -\frac{\sqrt{105}}{147} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}}{49} & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 \\
0 & -\frac{11\sqrt{14}}{294} & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{7}}{21} & 0 & -\frac{11\sqrt{14}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}}{147} & 0 & -\frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & \frac{2\sqrt{7}}{49} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}}{49} & 0 & \frac{\sqrt{3}}{14} \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0
\end{bmatrix}$

$\boxed{\text{x447}}$ $\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{330}i}{84} & 0 & -\frac{\sqrt{165}i}{84} & 0 & 0 & \frac{\sqrt{77}i}{154} & 0 & \frac{5\sqrt{33}i}{231} & 0 & \frac{\sqrt{55}i}{154} & 0 & 0 & 0 \\
\frac{\sqrt{330}i}{84} & 0 & \frac{5\sqrt{33}i}{84} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{55}i}{770} & 0 & -\frac{\sqrt{11}i}{77} & 0 & \frac{\sqrt{165}i}{330} & 0 & 0 \\
0 & -\frac{5\sqrt{33}i}{84} & 0 & 0 & 0 & \frac{\sqrt{165}i}{84} & \frac{3\sqrt{770}i}{1540} & 0 & \frac{\sqrt{330}i}{4620} & 0 & -\frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{110}i}{1540} & 0 \\
\frac{\sqrt{165}i}{84} & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{84} & 0 & 0 & \frac{\sqrt{110}i}{1540} & 0 & \frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{330}i}{4620} & 0 & -\frac{3\sqrt{770}i}{1540} \\
0 & 0 & 0 & \frac{5\sqrt{33}i}{84} & 0 & \frac{\sqrt{330}i}{84} & 0 & 0 & -\frac{\sqrt{165}i}{330} & 0 & \frac{\sqrt{11}i}{77} & 0 & \frac{13\sqrt{55}i}{770} & 0 \\
0 & 0 & -\frac{\sqrt{165}i}{84} & 0 & -\frac{\sqrt{330}i}{84} & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{154} & 0 & -\frac{5\sqrt{33}i}{231} & 0 & -\frac{\sqrt{77}i}{154} \\
-\frac{\sqrt{77}i}{154} & 0 & -\frac{3\sqrt{770}i}{1540} & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{1848} & 0 & \frac{\sqrt{2310}i}{1848} & 0 & 0 & 0 & 0 \\
0 & \frac{13\sqrt{55}i}{770} & 0 & -\frac{\sqrt{110}i}{1540} & 0 & 0 & -\frac{5\sqrt{462}i}{1848} & 0 & -\frac{5\sqrt{22}i}{616} & 0 & \frac{\sqrt{330}i}{924} & 0 & 0 & 0 \\
-\frac{5\sqrt{33}i}{231} & 0 & -\frac{\sqrt{330}i}{4620} & 0 & \frac{\sqrt{165}i}{330} & 0 & 0 & \frac{5\sqrt{22}i}{616} & 0 & -\frac{3\sqrt{110}i}{616} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{11}i}{77} & 0 & -\frac{\sqrt{22}i}{44} & 0 & \frac{\sqrt{55}i}{154} & -\frac{\sqrt{2310}i}{1848} & 0 & \frac{3\sqrt{110}i}{616} & 0 & 0 & 0 & -\frac{\sqrt{330}i}{924} & 0 \\
-\frac{\sqrt{55}i}{154} & 0 & \frac{\sqrt{22}i}{44} & 0 & -\frac{\sqrt{11}i}{77} & 0 & 0 & -\frac{\sqrt{330}i}{924} & 0 & 0 & 0 & \frac{3\sqrt{110}i}{616} & 0 & -\frac{\sqrt{2310}i}{1848} \\
0 & -\frac{\sqrt{165}i}{330} & 0 & \frac{\sqrt{330}i}{4620} & 0 & \frac{5\sqrt{33}i}{231} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}i}{616} & 0 & \frac{5\sqrt{22}i}{616} & 0 \\
0 & 0 & \frac{\sqrt{110}i}{1540} & 0 & -\frac{13\sqrt{55}i}{770} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{924} & 0 & -\frac{5\sqrt{22}i}{616} & 0 & -\frac{5\sqrt{462}i}{1848} \\
0 & 0 & 0 & \frac{3\sqrt{770}i}{1540} & 0 & \frac{\sqrt{77}i}{154} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1848} & 0 & \frac{5\sqrt{462}i}{1848} & 0
\end{bmatrix}$$

$\boxed{\text{x448}}$ $\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{330}}{84} & 0 & -\frac{\sqrt{165}}{84} & 0 & 0 & \frac{\sqrt{77}}{154} & 0 & -\frac{5\sqrt{33}}{231} & 0 & \frac{\sqrt{55}}{154} & 0 & 0 & 0 \\
\frac{\sqrt{330}}{84} & 0 & -\frac{5\sqrt{33}}{84} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{55}}{770} & 0 & \frac{\sqrt{11}}{77} & 0 & \frac{\sqrt{165}}{330} & 0 & 0 \\
0 & -\frac{5\sqrt{33}}{84} & 0 & 0 & 0 & \frac{\sqrt{165}}{84} & -\frac{3\sqrt{770}}{1540} & 0 & \frac{\sqrt{330}}{4620} & 0 & \frac{\sqrt{22}}{44} & 0 & -\frac{\sqrt{110}}{1540} & 0 \\
-\frac{\sqrt{165}}{84} & 0 & 0 & 0 & \frac{5\sqrt{33}}{84} & 0 & 0 & -\frac{\sqrt{110}}{1540} & 0 & \frac{\sqrt{22}}{44} & 0 & \frac{\sqrt{330}}{4620} & 0 & -\frac{3\sqrt{770}}{1540} \\
0 & 0 & 0 & \frac{5\sqrt{33}}{84} & 0 & -\frac{\sqrt{330}}{84} & 0 & 0 & \frac{\sqrt{165}}{330} & 0 & \frac{\sqrt{11}}{77} & 0 & -\frac{13\sqrt{55}}{770} & 0 \\
0 & 0 & \frac{\sqrt{165}}{84} & 0 & -\frac{\sqrt{330}}{84} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{154} & 0 & -\frac{5\sqrt{33}}{231} & 0 & \frac{\sqrt{77}}{154} \\
\frac{\sqrt{77}}{154} & 0 & -\frac{3\sqrt{770}}{1540} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & \frac{\sqrt{2310}}{1848} & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{55}}{770} & 0 & -\frac{\sqrt{110}}{1540} & 0 & 0 & -\frac{5\sqrt{462}}{1848} & 0 & \frac{5\sqrt{22}}{616} & 0 & \frac{\sqrt{330}}{924} & 0 & 0 & 0 \\
-\frac{5\sqrt{33}}{231} & 0 & \frac{\sqrt{330}}{4620} & 0 & \frac{\sqrt{165}}{330} & 0 & 0 & \frac{5\sqrt{22}}{616} & 0 & \frac{3\sqrt{110}}{616} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{11}}{77} & 0 & \frac{\sqrt{22}}{44} & 0 & \frac{\sqrt{55}}{154} & \frac{\sqrt{2310}}{1848} & 0 & \frac{3\sqrt{110}}{616} & 0 & 0 & 0 & -\frac{\sqrt{330}}{924} & 0 \\
\frac{\sqrt{55}}{154} & 0 & \frac{\sqrt{22}}{44} & 0 & \frac{\sqrt{11}}{77} & 0 & 0 & \frac{\sqrt{330}}{924} & 0 & 0 & 0 & -\frac{3\sqrt{110}}{616} & 0 & -\frac{\sqrt{2310}}{1848} \\
0 & \frac{\sqrt{165}}{330} & 0 & \frac{\sqrt{330}}{4620} & 0 & -\frac{5\sqrt{33}}{231} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{110}}{616} & 0 & -\frac{5\sqrt{22}}{616} & 0 \\
0 & 0 & -\frac{\sqrt{110}}{1540} & 0 & -\frac{13\sqrt{55}}{770} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{924} & 0 & -\frac{5\sqrt{22}}{616} & 0 & \frac{5\sqrt{462}}{1848} \\
0 & 0 & 0 & -\frac{3\sqrt{770}}{1540} & 0 & \frac{\sqrt{77}}{154} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1848} & 0 & \frac{5\sqrt{462}}{1848} & 0
\end{bmatrix}$$

$\boxed{\text{x449}}$ $\mathbb{Q}_{4,1}^{(1,1;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{2310}i}{588} & 0 & \frac{\sqrt{1155}i}{84} & 0 & 0 & \frac{\sqrt{11}i}{154} & 0 & \frac{5\sqrt{231}i}{1617} & 0 & -\frac{\sqrt{385}i}{154} & 0 & 0 & 0 \\
\frac{\sqrt{2310}i}{588} & 0 & \frac{5\sqrt{231}i}{588} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{385}i}{5390} & 0 & -\frac{\sqrt{77}i}{539} & 0 & -\frac{\sqrt{1155}i}{330} & 0 & 0 \\
0 & -\frac{5\sqrt{231}i}{588} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{84} & -\frac{3\sqrt{110}i}{220} & 0 & \frac{\sqrt{2310}i}{32340} & 0 & -\frac{\sqrt{154}i}{308} & 0 & \frac{\sqrt{770}i}{1540} & 0 \\
-\frac{\sqrt{1155}i}{84} & 0 & 0 & 0 & -\frac{5\sqrt{231}i}{588} & 0 & 0 & -\frac{\sqrt{770}i}{1540} & 0 & \frac{\sqrt{154}i}{308} & 0 & -\frac{\sqrt{2310}i}{32340} & 0 & \frac{3\sqrt{110}i}{220} \\
0 & 0 & 0 & \frac{5\sqrt{231}i}{588} & 0 & \frac{\sqrt{2310}i}{588} & 0 & 0 & \frac{\sqrt{1155}i}{330} & 0 & \frac{\sqrt{77}i}{539} & 0 & \frac{13\sqrt{385}i}{5390} & 0 \\
0 & 0 & \frac{\sqrt{1155}i}{84} & 0 & -\frac{\sqrt{2310}i}{588} & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{154} & 0 & -\frac{5\sqrt{231}i}{1617} & 0 & -\frac{\sqrt{11}i}{154} \\
-\frac{\sqrt{11}i}{154} & 0 & \frac{3\sqrt{110}i}{220} & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}i}{1848} & 0 & -\frac{\sqrt{330}i}{264} & 0 & 0 & 0 & 0 \\
0 & \frac{13\sqrt{385}i}{5390} & 0 & \frac{\sqrt{770}i}{1540} & 0 & 0 & -\frac{5\sqrt{66}i}{1848} & 0 & -\frac{5\sqrt{154}i}{4312} & 0 & -\frac{\sqrt{2310}i}{924} & 0 & 0 & 0 \\
-\frac{5\sqrt{231}i}{1617} & 0 & -\frac{\sqrt{2310}i}{32340} & 0 & -\frac{\sqrt{1155}i}{330} & 0 & 0 & \frac{5\sqrt{154}i}{4312} & 0 & -\frac{3\sqrt{770}i}{4312} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{77}i}{539} & 0 & -\frac{\sqrt{154}i}{308} & 0 & -\frac{\sqrt{385}i}{154} & \frac{\sqrt{330}i}{264} & 0 & \frac{3\sqrt{770}i}{4312} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{924} & 0 \\
\frac{\sqrt{385}i}{154} & 0 & \frac{\sqrt{154}i}{308} & 0 & -\frac{\sqrt{77}i}{539} & 0 & 0 & \frac{\sqrt{2310}i}{924} & 0 & 0 & 0 & \frac{3\sqrt{770}i}{4312} & 0 & \frac{\sqrt{330}i}{264} \\
0 & \frac{\sqrt{1155}i}{330} & 0 & \frac{\sqrt{2310}i}{32340} & 0 & \frac{5\sqrt{231}i}{1617} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}i}{4312} & 0 & \frac{5\sqrt{154}i}{4312} & 0 \\
0 & 0 & -\frac{\sqrt{770}i}{1540} & 0 & -\frac{13\sqrt{385}i}{5390} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{924} & 0 & -\frac{5\sqrt{154}i}{4312} & 0 & -\frac{5\sqrt{66}i}{1848} \\
0 & 0 & 0 & -\frac{3\sqrt{110}i}{220} & 0 & \frac{\sqrt{11}i}{154} & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{264} & 0 & \frac{5\sqrt{66}i}{1848} & 0
\end{bmatrix}$$

$\boxed{\text{x450}}$ $\mathbb{Q}_{4,2}^{(1,1;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{2310}}{588} & 0 & \frac{\sqrt{1155}}{84} & 0 & 0 & \frac{\sqrt{11}}{154} & 0 & -\frac{5\sqrt{231}}{1617} & 0 & -\frac{\sqrt{385}}{154} & 0 & 0 & 0 \\
\frac{\sqrt{2310}}{588} & 0 & -\frac{5\sqrt{231}}{588} & 0 & 0 & 0 & 0 & -\frac{13\sqrt{385}}{5390} & 0 & \frac{\sqrt{77}}{539} & 0 & -\frac{\sqrt{1155}}{330} & 0 & 0 \\
0 & -\frac{5\sqrt{231}}{588} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{84} & \frac{3\sqrt{110}}{220} & 0 & \frac{\sqrt{2310}}{32340} & 0 & \frac{\sqrt{154}}{308} & 0 & \frac{\sqrt{770}}{1540} & 0 \\
\frac{\sqrt{1155}}{84} & 0 & 0 & 0 & \frac{5\sqrt{231}}{588} & 0 & 0 & \frac{\sqrt{770}}{1540} & 0 & \frac{\sqrt{154}}{308} & 0 & \frac{\sqrt{2310}}{32340} & 0 & \frac{3\sqrt{110}}{220} \\
0 & 0 & 0 & \frac{5\sqrt{231}}{588} & 0 & -\frac{\sqrt{2310}}{588} & 0 & 0 & -\frac{\sqrt{1155}}{330} & 0 & \frac{\sqrt{77}}{539} & 0 & -\frac{13\sqrt{385}}{5390} & 0 \\
0 & 0 & -\frac{\sqrt{1155}}{84} & 0 & -\frac{\sqrt{2310}}{588} & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{154} & 0 & -\frac{5\sqrt{231}}{1617} & 0 & \frac{\sqrt{11}}{154} \\
\frac{\sqrt{11}}{154} & 0 & \frac{3\sqrt{110}}{220} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}}{1848} & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{385}}{5390} & 0 & \frac{\sqrt{770}}{1540} & 0 & 0 & -\frac{5\sqrt{66}}{1848} & 0 & \frac{5\sqrt{154}}{4312} & 0 & -\frac{\sqrt{2310}}{924} & 0 & 0 & 0 \\
-\frac{5\sqrt{231}}{1617} & 0 & \frac{\sqrt{2310}}{32340} & 0 & -\frac{\sqrt{1155}}{330} & 0 & 0 & \frac{5\sqrt{154}}{4312} & 0 & \frac{3\sqrt{770}}{4312} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{77}}{539} & 0 & \frac{\sqrt{154}}{308} & 0 & -\frac{\sqrt{385}}{154} & -\frac{\sqrt{330}}{264} & 0 & \frac{3\sqrt{770}}{4312} & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} & 0 \\
-\frac{\sqrt{385}}{154} & 0 & \frac{\sqrt{154}}{308} & 0 & \frac{\sqrt{77}}{539} & 0 & 0 & -\frac{\sqrt{2310}}{924} & 0 & 0 & 0 & -\frac{3\sqrt{770}}{4312} & 0 & \frac{\sqrt{330}}{264} \\
0 & -\frac{\sqrt{1155}}{330} & 0 & \frac{\sqrt{2310}}{32340} & 0 & -\frac{5\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}}{4312} & 0 & -\frac{5\sqrt{154}}{4312} & 0 \\
0 & 0 & \frac{\sqrt{770}}{1540} & 0 & -\frac{13\sqrt{385}}{5390} & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} & 0 & -\frac{5\sqrt{154}}{4312} & 0 & \frac{5\sqrt{66}}{1848} \\
0 & 0 & 0 & \frac{3\sqrt{110}}{220} & 0 & \frac{\sqrt{11}}{154} & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{264} & 0 & \frac{5\sqrt{66}}{1848} & 0
\end{bmatrix}$$

$\boxed{\text{x451}}$ $\mathbb{G}_5^{(1,0;a)}(A_{1g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{60} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x452}}$ $\mathbb{G}_1^{(1,0;a)}(A_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x453}}$ $\mathbb{G}_3^{(1,0;a)}(A_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x454}}$ $\mathbb{G}_5^{(1,0;a)}(A_{2g}, 1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x455}}$ $\mathbb{G}_5^{(1,0;a)}(A_{2g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{30} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x456}}$ $\mathbb{G}_3^{(1,0;a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{12} & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{12} & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & -\frac{1}{12} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & 0 & -\frac{\sqrt{14}}{12} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 \\
0 & \frac{\sqrt{14}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2}}{6} & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{1}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x457}}$ $\mathbb{G}_5^{(1,0;a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{60} & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{15} & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & \frac{2\sqrt{5}}{15} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & -\frac{\sqrt{70}}{60} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & 0 \\
0 & \frac{\sqrt{70}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{5}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{10}}{12} & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{5}}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x458}}$ $\mathbb{G}_3^{(1,0;a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{12} & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{12} & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & -\frac{i}{12} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{12} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 \\
0 & \frac{\sqrt{14}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x459}}$ $\mathbb{G}_5^{(1,0;a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{60} & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & \frac{\sqrt{70}i}{60} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{70}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{5}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{5}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x460}}$ $\mathbb{G}_{1,1}^{(1,0;a)}(E_g) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{210}i}{56} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{6}i}{8} \\
-\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{42}i}{56} & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{21}i}{28} & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x461}}$ $\mathbb{G}_{1,2}^{(1,0;a)}(E_g) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{35}}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{210}}{56} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{6}}{8} \\
\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{42}}{56} & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x462}}$ $\mathbb{G}_{3,1}^{(1,0;a)}(E_g, 1) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{48} & 0 & -\frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & -\frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{48} & 0 & -\frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{3}i}{48} & 0 & \frac{\sqrt{15}i}{16} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{16} & 0 & -\frac{\sqrt{3}i}{48} & 0 & \frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{105}i}{48} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{30}i}{48} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & \frac{\sqrt{2}i}{8} & 0 & -\frac{\sqrt{42}i}{48} \\
-\frac{\sqrt{42}i}{48} & 0 & \frac{\sqrt{105}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{30}i}{48} & 0 & \frac{\sqrt{15}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2}i}{8} & 0 & \frac{\sqrt{5}i}{16} & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{6}i}{24} & 0 & \frac{\sqrt{3}i}{48} & 0 & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{30}i}{48} & 0 & -\frac{\sqrt{3}i}{48} & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{10}i}{16} & 0 & -\frac{\sqrt{5}i}{16} & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{15}i}{16} & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}i}{48} & 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x463}}$ $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{48} & 0 & \frac{\sqrt{6}}{24} & 0 & \frac{\sqrt{10}}{16} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{48} & 0 & -\frac{\sqrt{5}}{16} & 0 & -\frac{\sqrt{3}}{48} & 0 & \frac{\sqrt{15}}{16} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{16} & 0 & -\frac{\sqrt{3}}{48} & 0 & -\frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{105}}{48} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{16} & 0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{30}}{48} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & \frac{\sqrt{2}}{8} & 0 & \frac{\sqrt{42}}{48} \\
\frac{\sqrt{42}}{48} & 0 & \frac{\sqrt{105}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{30}}{48} & 0 & \frac{\sqrt{15}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2}}{8} & 0 & -\frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{10}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{6}}{24} & 0 & -\frac{\sqrt{3}}{48} & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{30}}{48} & 0 & -\frac{\sqrt{3}}{48} & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}}{16} & 0 & -\frac{\sqrt{5}}{16} & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{15}}{16} & 0 & -\frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}}{48} & 0 & \frac{\sqrt{42}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x464}}$ $\mathbb{G}_{3,1}^{(1,0;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{48} & 0 & -\frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{48} & 0 & -\frac{\sqrt{10}i}{24} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{16} & 0 & -\frac{5\sqrt{3}i}{48} & 0 & \frac{\sqrt{5}i}{48} & 0 & -\frac{3i}{16} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{16} & 0 & -\frac{\sqrt{5}i}{48} & 0 & \frac{5\sqrt{3}i}{48} & 0 & -\frac{\sqrt{7}i}{16} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & \frac{\sqrt{10}i}{24} & 0 & \frac{5\sqrt{2}i}{48} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & \frac{\sqrt{30}i}{24} & 0 & -\frac{\sqrt{70}i}{48} \\
-\frac{\sqrt{70}i}{48} & 0 & -\frac{\sqrt{7}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{2}i}{48} & 0 & -\frac{3i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{30}i}{24} & 0 & \frac{5\sqrt{3}i}{48} & 0 & -\frac{\sqrt{6}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}i}{24} & 0 & \frac{\sqrt{5}i}{48} & 0 & -\frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{2}i}{16} & 0 & -\frac{\sqrt{5}i}{48} & 0 & -\frac{\sqrt{10}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{6}i}{16} & 0 & -\frac{5\sqrt{3}i}{48} & 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{3i}{16} & 0 & -\frac{5\sqrt{2}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{7}i}{16} & 0 & \frac{\sqrt{70}i}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x465}}$ $\mathbb{G}_{3,2}^{(1,0;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{48} & 0 & \frac{\sqrt{30}}{24} & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{48} & 0 & \frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{16} & 0 & -\frac{5\sqrt{3}}{48} & 0 & -\frac{\sqrt{5}}{48} & 0 & -\frac{3}{16} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{16} & 0 & -\frac{\sqrt{5}}{48} & 0 & -\frac{5\sqrt{3}}{48} & 0 & -\frac{\sqrt{7}}{16} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 & \frac{\sqrt{10}}{24} & 0 & -\frac{5\sqrt{2}}{48} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & \frac{\sqrt{30}}{24} & 0 & \frac{\sqrt{70}}{48} \\
\frac{\sqrt{70}}{48} & 0 & -\frac{\sqrt{7}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{2}}{48} & 0 & -\frac{3}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{30}}{24} & 0 & -\frac{5\sqrt{3}}{48} & 0 & -\frac{\sqrt{6}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}}{24} & 0 & -\frac{\sqrt{5}}{48} & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2}}{16} & 0 & -\frac{\sqrt{5}}{48} & 0 & \frac{\sqrt{10}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{6}}{16} & 0 & -\frac{5\sqrt{3}}{48} & 0 & \frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3}{16} & 0 & -\frac{5\sqrt{2}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{7}}{16} & 0 & \frac{\sqrt{70}}{48} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x466}}$ $\mathbb{G}_{5,1}^{(1,0;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{192} & 0 & -\frac{5\sqrt{70}i}{448} & 0 & \frac{5\sqrt{42}i}{192} & 0 & -\frac{\sqrt{210}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{42}i}{1344} & 0 & \frac{13\sqrt{210}i}{1344} & 0 & -\frac{\sqrt{14}i}{64} & 0 & -\frac{7\sqrt{6}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{3}i}{96} & 0 & \frac{11\sqrt{7}i}{224} & 0 & -\frac{\sqrt{105}i}{672} & 0 & -\frac{\sqrt{21}i}{32} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{32} & 0 & \frac{\sqrt{105}i}{672} & 0 & -\frac{11\sqrt{7}i}{224} & 0 & \frac{7\sqrt{3}i}{96} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{6}i}{64} & 0 & \frac{\sqrt{14}i}{64} & 0 & -\frac{13\sqrt{210}i}{1344} & 0 & \frac{23\sqrt{42}i}{1344} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{64} & 0 & -\frac{5\sqrt{42}i}{192} & 0 & \frac{5\sqrt{70}i}{448} & 0 & -\frac{\sqrt{30}i}{192} \\
-\frac{\sqrt{30}i}{192} & 0 & \frac{7\sqrt{3}i}{96} & 0 & -\frac{7\sqrt{6}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{23\sqrt{42}i}{1344} & 0 & -\frac{\sqrt{21}i}{32} & 0 & -\frac{\sqrt{210}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{70}i}{448} & 0 & -\frac{11\sqrt{7}i}{224} & 0 & -\frac{\sqrt{14}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{210}i}{1344} & 0 & -\frac{\sqrt{105}i}{672} & 0 & \frac{5\sqrt{42}i}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{42}i}{192} & 0 & \frac{\sqrt{105}i}{672} & 0 & \frac{13\sqrt{210}i}{1344} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{14}i}{64} & 0 & \frac{11\sqrt{7}i}{224} & 0 & -\frac{5\sqrt{70}i}{448} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{210}i}{64} & 0 & \frac{\sqrt{21}i}{32} & 0 & -\frac{23\sqrt{42}i}{1344} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{7\sqrt{6}i}{64} & 0 & -\frac{7\sqrt{3}i}{96} & 0 & \frac{\sqrt{30}i}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x467}}$ $\mathbb{G}_{5,2}^{(1,0;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{192} & 0 & \frac{5\sqrt{70}}{448} & 0 & \frac{5\sqrt{42}}{192} & 0 & \frac{\sqrt{210}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{42}}{1344} & 0 & -\frac{13\sqrt{210}}{1344} & 0 & -\frac{\sqrt{14}}{64} & 0 & \frac{7\sqrt{6}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{3}}{96} & 0 & \frac{11\sqrt{7}}{224} & 0 & \frac{\sqrt{105}}{672} & 0 & -\frac{\sqrt{21}}{32} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{32} & 0 & \frac{\sqrt{105}}{672} & 0 & \frac{11\sqrt{7}}{224} & 0 & \frac{7\sqrt{3}}{96} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{6}}{64} & 0 & -\frac{\sqrt{14}}{64} & 0 & -\frac{13\sqrt{210}}{1344} & 0 & -\frac{23\sqrt{42}}{1344} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{64} & 0 & \frac{5\sqrt{42}}{192} & 0 & \frac{5\sqrt{70}}{448} & 0 & \frac{\sqrt{30}}{192} \\
\frac{\sqrt{30}}{192} & 0 & \frac{7\sqrt{3}}{96} & 0 & \frac{7\sqrt{6}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{23\sqrt{42}}{1344} & 0 & -\frac{\sqrt{21}}{32} & 0 & \frac{\sqrt{210}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{70}}{448} & 0 & \frac{11\sqrt{7}}{224} & 0 & -\frac{\sqrt{14}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{210}}{1344} & 0 & \frac{\sqrt{105}}{672} & 0 & \frac{5\sqrt{42}}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{42}}{192} & 0 & \frac{\sqrt{105}}{672} & 0 & -\frac{13\sqrt{210}}{1344} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{14}}{64} & 0 & \frac{11\sqrt{7}}{224} & 0 & \frac{5\sqrt{70}}{448} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{210}}{64} & 0 & -\frac{\sqrt{21}}{32} & 0 & -\frac{23\sqrt{42}}{1344} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{7\sqrt{6}}{64} & 0 & \frac{7\sqrt{3}}{96} & 0 & \frac{\sqrt{30}}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x468}}$ $\mathbb{G}_{5,1}^{(1,0;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{192} & 0 & -\frac{5\sqrt{2}i}{64} & 0 & -\frac{3\sqrt{30}i}{64} & 0 & -\frac{5\sqrt{6}i}{192} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{30}i}{960} & 0 & \frac{13\sqrt{6}i}{192} & 0 & \frac{9\sqrt{10}i}{320} & 0 & -\frac{\sqrt{210}i}{192} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}i}{160} & 0 & \frac{11\sqrt{5}i}{160} & 0 & -\frac{\sqrt{3}i}{96} & 0 & \frac{9\sqrt{15}i}{160} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{15}i}{160} & 0 & \frac{\sqrt{3}i}{96} & 0 & -\frac{11\sqrt{5}i}{160} & 0 & -\frac{3\sqrt{105}i}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{192} & 0 & -\frac{9\sqrt{10}i}{320} & 0 & -\frac{13\sqrt{6}i}{192} & 0 & \frac{23\sqrt{30}i}{960} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{192} & 0 & \frac{3\sqrt{30}i}{64} & 0 & \frac{5\sqrt{2}i}{64} & 0 & -\frac{\sqrt{42}i}{192} \\
-\frac{\sqrt{42}i}{192} & 0 & -\frac{3\sqrt{105}i}{160} & 0 & -\frac{\sqrt{210}i}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{23\sqrt{30}i}{960} & 0 & \frac{9\sqrt{15}i}{160} & 0 & -\frac{5\sqrt{6}i}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{2}i}{64} & 0 & -\frac{11\sqrt{5}i}{160} & 0 & \frac{9\sqrt{10}i}{320} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{6}i}{192} & 0 & -\frac{\sqrt{3}i}{96} & 0 & -\frac{3\sqrt{30}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{30}i}{64} & 0 & \frac{\sqrt{3}i}{96} & 0 & \frac{13\sqrt{6}i}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{9\sqrt{10}i}{320} & 0 & \frac{11\sqrt{5}i}{160} & 0 & -\frac{5\sqrt{2}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{6}i}{192} & 0 & -\frac{9\sqrt{15}i}{160} & 0 & -\frac{23\sqrt{30}i}{960} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{210}i}{192} & 0 & \frac{3\sqrt{105}i}{160} & 0 & \frac{\sqrt{42}i}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x469}}$ $\mathbb{G}_{5,2}^{(1,0;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{192} & 0 & \frac{5\sqrt{2}}{64} & 0 & -\frac{3\sqrt{30}}{64} & 0 & \frac{5\sqrt{6}}{192} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{30}}{960} & 0 & -\frac{13\sqrt{6}}{192} & 0 & \frac{9\sqrt{10}}{320} & 0 & \frac{\sqrt{210}}{192} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{160} & 0 & \frac{11\sqrt{5}}{160} & 0 & \frac{\sqrt{3}}{96} & 0 & \frac{9\sqrt{15}}{160} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{15}}{160} & 0 & \frac{\sqrt{3}}{96} & 0 & \frac{11\sqrt{5}}{160} & 0 & -\frac{3\sqrt{105}}{160} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{192} & 0 & \frac{9\sqrt{10}}{320} & 0 & -\frac{13\sqrt{6}}{192} & 0 & -\frac{23\sqrt{30}}{960} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{192} & 0 & -\frac{3\sqrt{30}}{64} & 0 & \frac{5\sqrt{2}}{64} & 0 & \frac{\sqrt{42}}{192} \\
\frac{\sqrt{42}}{192} & 0 & -\frac{3\sqrt{105}}{160} & 0 & \frac{\sqrt{210}}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{23\sqrt{30}}{960} & 0 & \frac{9\sqrt{15}}{160} & 0 & \frac{5\sqrt{6}}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{2}}{64} & 0 & \frac{11\sqrt{5}}{160} & 0 & \frac{9\sqrt{10}}{320} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{6}}{192} & 0 & \frac{\sqrt{3}}{96} & 0 & -\frac{3\sqrt{30}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{30}}{64} & 0 & \frac{\sqrt{3}}{96} & 0 & -\frac{13\sqrt{6}}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{9\sqrt{10}}{320} & 0 & \frac{11\sqrt{5}}{160} & 0 & \frac{5\sqrt{2}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{6}}{192} & 0 & \frac{9\sqrt{15}}{160} & 0 & -\frac{23\sqrt{30}}{960} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{210}}{192} & 0 & -\frac{3\sqrt{105}}{160} & 0 & \frac{\sqrt{42}}{192} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x470}}$ $\mathbb{G}_{5,1}^{(1,0;a)}(E_g,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{96} & 0 & -\frac{5\sqrt{6}i}{96} & 0 & \frac{\sqrt{10}i}{32} & 0 & \frac{5\sqrt{2}i}{32} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{10}i}{480} & 0 & \frac{13\sqrt{2}i}{96} & 0 & -\frac{\sqrt{30}i}{160} & 0 & \frac{\sqrt{70}i}{32} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{80} & 0 & \frac{11\sqrt{15}i}{240} & 0 & -\frac{i}{48} & 0 & -\frac{3\sqrt{5}i}{80} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{80} & 0 & \frac{i}{48} & 0 & -\frac{11\sqrt{15}i}{240} & 0 & \frac{\sqrt{35}i}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{32} & 0 & \frac{\sqrt{30}i}{160} & 0 & -\frac{13\sqrt{2}i}{96} & 0 & \frac{23\sqrt{10}i}{480} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{32} & 0 & -\frac{\sqrt{10}i}{32} & 0 & \frac{5\sqrt{6}i}{96} & 0 & -\frac{\sqrt{14}i}{96} \\
-\frac{\sqrt{14}i}{96} & 0 & \frac{\sqrt{35}i}{80} & 0 & \frac{\sqrt{70}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{23\sqrt{10}i}{480} & 0 & -\frac{3\sqrt{5}i}{80} & 0 & \frac{5\sqrt{2}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{6}i}{96} & 0 & -\frac{11\sqrt{15}i}{240} & 0 & -\frac{\sqrt{30}i}{160} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{2}i}{96} & 0 & -\frac{i}{48} & 0 & \frac{\sqrt{10}i}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{10}i}{32} & 0 & \frac{i}{48} & 0 & \frac{13\sqrt{2}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{30}i}{160} & 0 & \frac{11\sqrt{15}i}{240} & 0 & -\frac{5\sqrt{6}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{2}i}{32} & 0 & \frac{3\sqrt{5}i}{80} & 0 & -\frac{23\sqrt{10}i}{480} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{70}i}{32} & 0 & -\frac{\sqrt{35}i}{80} & 0 & \frac{\sqrt{14}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x471}}$ $\mathbb{G}_{5,2}^{(1,0;a)}(E_g,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{96} & 0 & \frac{5\sqrt{6}}{96} & 0 & \frac{\sqrt{10}}{32} & 0 & -\frac{5\sqrt{2}}{32} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{10}}{480} & 0 & -\frac{13\sqrt{2}}{96} & 0 & -\frac{\sqrt{30}}{160} & 0 & -\frac{\sqrt{70}}{32} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{80} & 0 & \frac{11\sqrt{15}}{240} & 0 & \frac{1}{48} & 0 & -\frac{3\sqrt{5}}{80} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{80} & 0 & \frac{1}{48} & 0 & \frac{11\sqrt{15}}{240} & 0 & \frac{\sqrt{35}}{80} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{32} & 0 & -\frac{\sqrt{30}}{160} & 0 & -\frac{13\sqrt{2}}{96} & 0 & -\frac{23\sqrt{10}}{480} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{32} & 0 & \frac{\sqrt{10}}{32} & 0 & \frac{5\sqrt{6}}{96} & 0 & \frac{\sqrt{14}}{96} \\
\frac{\sqrt{14}}{96} & 0 & \frac{\sqrt{35}}{80} & 0 & -\frac{\sqrt{70}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{23\sqrt{10}}{480} & 0 & -\frac{3\sqrt{5}}{80} & 0 & -\frac{5\sqrt{2}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{6}}{96} & 0 & \frac{11\sqrt{15}}{240} & 0 & -\frac{\sqrt{30}}{160} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{2}}{96} & 0 & \frac{1}{48} & 0 & \frac{\sqrt{10}}{32} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{10}}{32} & 0 & \frac{1}{48} & 0 & -\frac{13\sqrt{2}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{30}}{160} & 0 & \frac{11\sqrt{15}}{240} & 0 & \frac{5\sqrt{6}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{2}}{32} & 0 & -\frac{3\sqrt{5}}{80} & 0 & -\frac{23\sqrt{10}}{480} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{70}}{32} & 0 & \frac{\sqrt{35}}{80} & 0 & \frac{\sqrt{14}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x472}}$ $\mathbb{M}_{5}^{(a)}(A_{1g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x473}}$ $\mathbb{M}_1^{(a)}(A_{2g}) =$

$$\begin{bmatrix}
\frac{5\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{196} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}}{196} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}}{196} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{28}
\end{bmatrix}$$

$\boxed{\text{x474}}$ $\mathbb{M}_3^{(a)}(A_{2g}) =$

$$\begin{bmatrix}
-\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6}
\end{bmatrix}$$

$\boxed{\text{x475}}$ $\mathbb{M}_5^{(a)}(A_{2g},1) =$

$$\begin{bmatrix}
\frac{\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{42}}{588} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{42}}{588} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84}
\end{bmatrix}$$

$\boxed{\text{x476}}$ $\mathbb{M}_5^{(a)}(A_{2g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x477}}$ $\mathbb{M}_3^{(a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 \\
-\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 \\
0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 \\
0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} \\
0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{i}{14} & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 \\
-\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 \\
0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{3i}{14} & 0 \\
0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} \\
0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x478}}$ $\mathbb{M}_5^{(a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 \\
-\frac{i}{14} & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 \\
0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 \\
0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} \\
0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{42} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 \\
-\frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 \\
0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & \frac{5\sqrt{3}i}{42} & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 \\
0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} \\
0 & 0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x479}}$ $\mathbb{M}_3^{(a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 \\
-\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 \\
0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 \\
0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & \frac{\sqrt{42}}{42} \\
0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 \\
0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 \\
-\frac{\sqrt{6}}{21} & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 \\
0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & \frac{3}{14} & 0 \\
0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & \frac{\sqrt{105}}{42} \\
0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x480}}$ $\mathbb{M}_5^{(a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 \\
\frac{1}{14} & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & \frac{2\sqrt{6}}{21} & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 \\
0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & \frac{2\sqrt{6}}{21} & 0 \\
0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} \\
0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 \\
\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 \\
0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 \\
0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} \\
0 & 0 & 0 & \frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x481}}$ $\mathbb{M}_{1,1}^{(a)}(E_g) = \begin{bmatrix}
0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & \frac{\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{70}}{49} & 0 & \frac{4\sqrt{7}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{196} & 0 & \frac{\sqrt{21}}{196} & 0 & 0 & 0 & 0 \\
0 & \frac{4\sqrt{7}}{49} & 0 & \frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{196} & 0 & \frac{\sqrt{42}}{196} & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{14}}{49} & 0 & \frac{4\sqrt{7}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{196} & 0 & \frac{\sqrt{70}}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{4\sqrt{7}}{49} & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{196} & 0 & \frac{\sqrt{105}}{196} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{196} & 0 & \frac{\sqrt{3}}{28} \\
-\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & \frac{3\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{7}}{196} & 0 & -\frac{\sqrt{70}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{98} & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{21}}{196} & 0 & -\frac{\sqrt{42}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & \frac{3\sqrt{14}}{49} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}}{196} & 0 & -\frac{\sqrt{21}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{49} & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{70}}{196} & 0 & -\frac{\sqrt{7}}{196} & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & \frac{3\sqrt{42}}{98} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{42}}{98} & 0 & \frac{3\sqrt{2}}{28} \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0
\end{bmatrix}$

$\boxed{\text{x482}}$ $\mathbb{M}_{1,2}^{(a)}(E_g) = \begin{bmatrix}
0 & \frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & \frac{\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{70}i}{49} & 0 & \frac{4\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{196} & 0 & \frac{\sqrt{21}i}{196} & 0 & 0 & 0 & 0 \\
0 & -\frac{4\sqrt{7}i}{49} & 0 & \frac{3\sqrt{14}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{196} & 0 & \frac{\sqrt{42}i}{196} & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{14}i}{49} & 0 & \frac{4\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{196} & 0 & \frac{\sqrt{70}i}{196} & 0 & 0 \\
0 & 0 & 0 & -\frac{4\sqrt{7}i}{49} & 0 & \frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{196} & 0 & \frac{\sqrt{105}i}{196} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{196} & 0 & \frac{\sqrt{3}i}{28} \\
-\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}i}{196} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & \frac{3\sqrt{42}i}{98} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{7}i}{196} & 0 & -\frac{\sqrt{70}i}{196} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{98} & 0 & \frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{21}i}{196} & 0 & -\frac{\sqrt{42}i}{196} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & \frac{3\sqrt{14}i}{49} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}i}{196} & 0 & -\frac{\sqrt{21}i}{196} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{49} & 0 & \frac{3\sqrt{210}i}{196} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}i}{196} & 0 & -\frac{\sqrt{7}i}{196} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & \frac{3\sqrt{42}i}{98} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{42}i}{98} & 0 & \frac{3\sqrt{2}i}{28} \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0
\end{bmatrix}$

$\boxed{\text{x483}}$ $\mathbb{M}_{3,1}^{(a)}(E_g,1) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{15}}{28} & 0 & -\frac{5\sqrt{30}}{168} & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{6}}{28} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 \\
\frac{\sqrt{15}}{28} & 0 & -\frac{\sqrt{6}}{56} & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{2}}{28} & 0 & -\frac{\sqrt{30}}{56} & 0 & 0 \\
0 & -\frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{5\sqrt{30}}{168} & \frac{\sqrt{35}}{56} & 0 & \frac{\sqrt{15}}{56} & 0 & -\frac{1}{56} & 0 & -\frac{3\sqrt{5}}{56} & 0 \\
-\frac{5\sqrt{30}}{168} & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & \frac{1}{56} & 0 & -\frac{\sqrt{15}}{56} & 0 & -\frac{\sqrt{35}}{56} \\
0 & -\frac{5\sqrt{3}}{42} & 0 & -\frac{\sqrt{6}}{56} & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & \frac{\sqrt{30}}{56} & 0 & -\frac{\sqrt{2}}{28} & 0 & -\frac{\sqrt{10}}{56} & 0 \\
0 & 0 & -\frac{5\sqrt{30}}{168} & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{14}}{56} \\
-\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{35}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}}{56} & 0 & \frac{3\sqrt{5}}{56} & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & \frac{1}{28} & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 \\
\frac{\sqrt{6}}{28} & 0 & \frac{\sqrt{15}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & 0 & \frac{1}{28} & 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 \\
0 & \frac{\sqrt{2}}{28} & 0 & \frac{1}{56} & 0 & \frac{\sqrt{10}}{56} & -\frac{\sqrt{105}}{84} & 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{\sqrt{15}}{21} & 0 \\
-\frac{\sqrt{10}}{56} & 0 & -\frac{1}{56} & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & -\frac{\sqrt{3}}{14} & 0 & -\frac{\sqrt{5}}{28} & 0 & -\frac{\sqrt{105}}{84} \\
0 & -\frac{\sqrt{30}}{56} & 0 & -\frac{\sqrt{15}}{56} & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{1}{28} & 0 \\
0 & 0 & -\frac{3\sqrt{5}}{56} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & \frac{1}{28} & 0 & \frac{\sqrt{21}}{28} \\
0 & 0 & 0 & -\frac{\sqrt{35}}{56} & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{21}}{28} & 0
\end{bmatrix}$$

$\boxed{\text{x484}}$ $\mathbb{M}_{3,2}^{(a)}(E_g,1) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{15}i}{28} & 0 & \frac{5\sqrt{30}i}{168} & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & \frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 \\
-\frac{\sqrt{15}i}{28} & 0 & -\frac{\sqrt{6}i}{56} & 0 & \frac{5\sqrt{3}i}{42} & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{2}i}{28} & 0 & \frac{\sqrt{30}i}{56} & 0 & 0 \\
0 & \frac{\sqrt{6}i}{56} & 0 & -\frac{\sqrt{3}i}{14} & 0 & \frac{5\sqrt{30}i}{168} & \frac{\sqrt{35}i}{56} & 0 & -\frac{\sqrt{15}i}{56} & 0 & -\frac{i}{56} & 0 & \frac{3\sqrt{5}i}{56} & 0 \\
-\frac{5\sqrt{30}i}{168} & 0 & \frac{\sqrt{3}i}{14} & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & \frac{3\sqrt{5}i}{56} & 0 & -\frac{i}{56} & 0 & -\frac{\sqrt{15}i}{56} & 0 & \frac{\sqrt{35}i}{56} \\
0 & -\frac{5\sqrt{3}i}{42} & 0 & \frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & \frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{2}i}{28} & 0 & -\frac{\sqrt{10}i}{56} & 0 \\
0 & 0 & -\frac{5\sqrt{30}i}{168} & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{14}i}{56} \\
-\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{35}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{3\sqrt{5}i}{56} & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & \frac{i}{28} & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 \\
-\frac{\sqrt{6}i}{28} & 0 & \frac{\sqrt{15}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 & 0 & -\frac{i}{28} & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{5\sqrt{3}i}{42} & 0 & 0 \\
0 & -\frac{\sqrt{2}i}{28} & 0 & \frac{i}{56} & 0 & -\frac{\sqrt{10}i}{56} & -\frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{5}i}{28} & 0 & -\frac{\sqrt{3}i}{14} & 0 & \frac{\sqrt{15}i}{21} & 0 \\
-\frac{\sqrt{10}i}{56} & 0 & \frac{i}{56} & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & \frac{\sqrt{3}i}{14} & 0 & -\frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{105}i}{84} \\
0 & -\frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{15}i}{56} & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & -\frac{5\sqrt{3}i}{42} & 0 & \frac{\sqrt{5}i}{28} & 0 & \frac{i}{28} & 0 \\
0 & 0 & -\frac{3\sqrt{5}i}{56} & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & -\frac{i}{28} & 0 & \frac{\sqrt{21}i}{28} \\
0 & 0 & 0 & -\frac{\sqrt{35}i}{56} & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{\sqrt{21}i}{28} & 0
\end{bmatrix}$$

$\boxed{\text{x485}}$ $\mathbb{M}_{3,1}^{(a)}(E_g,2) =$

$$\begin{bmatrix}
0 & \frac{5}{28} & 0 & \frac{5\sqrt{2}}{56} & 0 & 0 & -\frac{\sqrt{210}}{168} & 0 & \frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 \\
\frac{5}{28} & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & \frac{5\sqrt{6}}{168} & 0 & \frac{\sqrt{30}}{84} & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 \\
0 & -\frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{5\sqrt{2}}{56} & -\frac{\sqrt{21}}{56} & 0 & \frac{5}{56} & 0 & -\frac{\sqrt{15}}{168} & 0 & \frac{3\sqrt{3}}{56} & 0 \\
\frac{5\sqrt{2}}{56} & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & -\frac{3\sqrt{3}}{56} & 0 & \frac{\sqrt{15}}{168} & 0 & -\frac{5}{56} & 0 & \frac{\sqrt{21}}{56} \\
0 & \frac{\sqrt{5}}{14} & 0 & -\frac{\sqrt{10}}{56} & 0 & \frac{5}{28} & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & -\frac{\sqrt{30}}{84} & 0 & -\frac{5\sqrt{6}}{168} & 0 \\
0 & 0 & \frac{5\sqrt{2}}{56} & 0 & \frac{5}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{210}}{168} \\
-\frac{\sqrt{210}}{168} & 0 & -\frac{\sqrt{21}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{6}}{168} & 0 & -\frac{3\sqrt{3}}{56} & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & \frac{\sqrt{15}}{84} & 0 & \frac{1}{7} & 0 & 0 & 0 \\
\frac{\sqrt{10}}{28} & 0 & \frac{5}{56} & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & -\frac{5\sqrt{3}}{84} & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\
0 & \frac{\sqrt{30}}{84} & 0 & \frac{\sqrt{15}}{168} & 0 & -\frac{\sqrt{6}}{56} & \frac{\sqrt{7}}{28} & 0 & -\frac{5\sqrt{3}}{84} & 0 & -\frac{\sqrt{5}}{14} & 0 & \frac{1}{7} & 0 \\
\frac{\sqrt{6}}{56} & 0 & -\frac{\sqrt{15}}{168} & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & \frac{1}{7} & 0 & -\frac{\sqrt{5}}{14} & 0 & -\frac{5\sqrt{3}}{84} & 0 & \frac{\sqrt{7}}{28} \\
0 & \frac{3\sqrt{2}}{56} & 0 & -\frac{5}{56} & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & -\frac{5\sqrt{3}}{84} & 0 & \frac{\sqrt{15}}{84} & 0 \\
0 & 0 & \frac{3\sqrt{3}}{56} & 0 & -\frac{5\sqrt{6}}{168} & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & \frac{\sqrt{15}}{84} & 0 & \frac{\sqrt{35}}{28} \\
0 & 0 & 0 & \frac{\sqrt{21}}{56} & 0 & \frac{\sqrt{210}}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & \frac{\sqrt{35}}{28} & 0
\end{bmatrix}$$

$\boxed{\text{x486}}$ $\mathbb{M}_{3,2}^{(a)}(E_g,2) =$

$$\begin{bmatrix}
0 & \frac{5i}{28} & 0 & -\frac{5\sqrt{2}i}{56} & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{10}i}{28} & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 \\
-\frac{5i}{28} & 0 & -\frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & -\frac{5\sqrt{6}i}{168} & 0 & \frac{\sqrt{30}i}{84} & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 \\
0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{5\sqrt{2}i}{56} & -\frac{\sqrt{21}i}{56} & 0 & -\frac{5i}{56} & 0 & -\frac{\sqrt{15}i}{168} & 0 & -\frac{3\sqrt{3}i}{56} & 0 \\
\frac{5\sqrt{2}i}{56} & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & -\frac{3\sqrt{3}i}{56} & 0 & -\frac{\sqrt{15}i}{168} & 0 & -\frac{5i}{56} & 0 & -\frac{\sqrt{21}i}{56} \\
0 & \frac{\sqrt{5}i}{14} & 0 & \frac{\sqrt{10}i}{56} & 0 & \frac{5i}{28} & 0 & 0 & -\frac{3\sqrt{2}i}{56} & 0 & \frac{\sqrt{30}i}{84} & 0 & -\frac{5\sqrt{6}i}{168} & 0 \\
0 & 0 & \frac{5\sqrt{2}i}{56} & 0 & -\frac{5i}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{10}i}{28} & 0 & \frac{\sqrt{210}i}{168} \\
-\frac{\sqrt{210}i}{168} & 0 & \frac{\sqrt{21}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{6}i}{168} & 0 & \frac{3\sqrt{3}i}{56} & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & \frac{\sqrt{15}i}{84} & 0 & -\frac{i}{7} & 0 & 0 & 0 \\
-\frac{\sqrt{10}i}{28} & 0 & \frac{5i}{56} & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & -\frac{\sqrt{15}i}{84} & 0 & -\frac{5\sqrt{3}i}{84} & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 \\
0 & -\frac{\sqrt{30}i}{84} & 0 & \frac{\sqrt{15}i}{168} & 0 & \frac{\sqrt{6}i}{56} & \frac{\sqrt{7}i}{28} & 0 & \frac{5\sqrt{3}i}{84} & 0 & -\frac{\sqrt{5}i}{14} & 0 & -\frac{i}{7} & 0 \\
\frac{\sqrt{6}i}{56} & 0 & \frac{\sqrt{15}i}{168} & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & \frac{i}{7} & 0 & \frac{\sqrt{5}i}{14} & 0 & -\frac{5\sqrt{3}i}{84} & 0 & -\frac{\sqrt{7}i}{28} \\
0 & \frac{3\sqrt{2}i}{56} & 0 & \frac{5i}{56} & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & \frac{5\sqrt{3}i}{84} & 0 & \frac{\sqrt{15}i}{84} & 0 \\
0 & 0 & \frac{3\sqrt{3}i}{56} & 0 & \frac{5\sqrt{6}i}{168} & 0 & 0 & 0 & 0 & \frac{i}{7} & 0 & -\frac{\sqrt{15}i}{84} & 0 & \frac{\sqrt{35}i}{28} \\
0 & 0 & 0 & \frac{\sqrt{21}i}{56} & 0 & -\frac{\sqrt{210}i}{168} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & -\frac{\sqrt{35}i}{28} & 0
\end{bmatrix}$$

$\boxed{\text{x487}}$ $\mathbb{M}_{5,1}^{(a)}(E_g,1) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{210}}{784} & 0 & -\frac{\sqrt{105}}{168} & 0 & \frac{3\sqrt{42}}{112} & -\frac{5}{224} & 0 & \frac{25\sqrt{21}}{1568} & 0 & -\frac{5\sqrt{35}}{224} & 0 & \frac{15\sqrt{7}}{224} & 0 \\
\frac{\sqrt{210}}{784} & 0 & -\frac{5\sqrt{21}}{392} & 0 & \frac{5\sqrt{42}}{336} & 0 & 0 & \frac{23\sqrt{35}}{1568} & 0 & -\frac{65\sqrt{7}}{1568} & 0 & \frac{\sqrt{105}}{224} & 0 & \frac{3\sqrt{5}}{32} \\
0 & -\frac{5\sqrt{21}}{392} & 0 & \frac{5\sqrt{42}}{392} & 0 & -\frac{\sqrt{105}}{168} & \frac{\sqrt{10}}{32} & 0 & -\frac{11\sqrt{210}}{1568} & 0 & \frac{5\sqrt{14}}{1568} & 0 & \frac{3\sqrt{70}}{224} & 0 \\
-\frac{\sqrt{105}}{168} & 0 & \frac{5\sqrt{42}}{392} & 0 & -\frac{5\sqrt{21}}{392} & 0 & 0 & -\frac{3\sqrt{70}}{224} & 0 & -\frac{5\sqrt{14}}{1568} & 0 & \frac{11\sqrt{210}}{1568} & 0 & -\frac{\sqrt{10}}{32} \\
0 & \frac{5\sqrt{42}}{336} & 0 & -\frac{5\sqrt{21}}{392} & 0 & \frac{\sqrt{210}}{784} & -\frac{3\sqrt{5}}{32} & 0 & -\frac{\sqrt{105}}{224} & 0 & \frac{65\sqrt{7}}{1568} & 0 & -\frac{23\sqrt{35}}{1568} & 0 \\
\frac{3\sqrt{42}}{112} & 0 & -\frac{\sqrt{105}}{168} & 0 & \frac{\sqrt{210}}{784} & 0 & 0 & -\frac{15\sqrt{7}}{224} & 0 & \frac{5\sqrt{35}}{224} & 0 & -\frac{25\sqrt{21}}{1568} & 0 & \frac{5}{224} \\
-\frac{5}{224} & 0 & \frac{\sqrt{10}}{32} & 0 & -\frac{3\sqrt{5}}{32} & 0 & 0 & \frac{5\sqrt{6}}{224} & 0 & -\frac{\sqrt{30}}{48} & 0 & \frac{3\sqrt{2}}{32} & 0 & 0 \\
0 & \frac{23\sqrt{35}}{1568} & 0 & -\frac{3\sqrt{70}}{224} & 0 & -\frac{15\sqrt{7}}{224} & \frac{5\sqrt{6}}{224} & 0 & -\frac{5\sqrt{14}}{196} & 0 & \frac{\sqrt{210}}{672} & 0 & \frac{3\sqrt{42}}{112} & 0 \\
\frac{25\sqrt{21}}{1568} & 0 & -\frac{11\sqrt{210}}{1568} & 0 & -\frac{\sqrt{105}}{224} & 0 & 0 & -\frac{5\sqrt{14}}{196} & 0 & \frac{\sqrt{70}}{1568} & 0 & \frac{5\sqrt{42}}{336} & 0 & \frac{3\sqrt{2}}{32} \\
0 & -\frac{65\sqrt{7}}{1568} & 0 & -\frac{5\sqrt{14}}{1568} & 0 & \frac{5\sqrt{35}}{224} & -\frac{\sqrt{30}}{48} & 0 & \frac{\sqrt{70}}{1568} & 0 & \frac{5\sqrt{42}}{392} & 0 & \frac{\sqrt{210}}{672} & 0 \\
-\frac{5\sqrt{35}}{224} & 0 & \frac{5\sqrt{14}}{1568} & 0 & \frac{65\sqrt{7}}{1568} & 0 & 0 & \frac{\sqrt{210}}{672} & 0 & \frac{5\sqrt{42}}{392} & 0 & \frac{\sqrt{70}}{1568} & 0 & -\frac{\sqrt{30}}{48} \\
0 & \frac{\sqrt{105}}{224} & 0 & \frac{11\sqrt{210}}{1568} & 0 & -\frac{25\sqrt{21}}{1568} & \frac{3\sqrt{2}}{32} & 0 & \frac{5\sqrt{42}}{336} & 0 & \frac{\sqrt{70}}{1568} & 0 & -\frac{5\sqrt{14}}{196} & 0 \\
\frac{15\sqrt{7}}{224} & 0 & \frac{3\sqrt{70}}{224} & 0 & -\frac{23\sqrt{35}}{1568} & 0 & 0 & \frac{3\sqrt{42}}{112} & 0 & \frac{\sqrt{210}}{672} & 0 & -\frac{5\sqrt{14}}{196} & 0 & \frac{5\sqrt{6}}{224} \\
0 & \frac{3\sqrt{5}}{32} & 0 & -\frac{\sqrt{10}}{32} & 0 & \frac{5}{224} & 0 & 0 & \frac{3\sqrt{2}}{32} & 0 & -\frac{\sqrt{30}}{48} & 0 & \frac{5\sqrt{6}}{224} & 0
\end{bmatrix}$$

$\boxed{\text{x488}}$ $\mathbb{M}_{5,2}^{(a)}(E_g,1) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{210}i}{784} & 0 & \frac{\sqrt{105}i}{168} & 0 & \frac{3\sqrt{42}i}{112} & \frac{5i}{224} & 0 & \frac{25\sqrt{21}i}{1568} & 0 & \frac{5\sqrt{35}i}{224} & 0 & \frac{15\sqrt{7}i}{224} & 0 \\
-\frac{\sqrt{210}i}{784} & 0 & -\frac{5\sqrt{21}i}{392} & 0 & -\frac{5\sqrt{42}i}{336} & 0 & 0 & -\frac{23\sqrt{35}i}{1568} & 0 & -\frac{65\sqrt{7}i}{1568} & 0 & -\frac{\sqrt{105}i}{224} & 0 & \frac{3\sqrt{5}i}{32} \\
0 & \frac{5\sqrt{21}i}{392} & 0 & \frac{5\sqrt{42}i}{392} & 0 & \frac{\sqrt{105}i}{168} & \frac{\sqrt{10}i}{32} & 0 & \frac{11\sqrt{210}i}{1568} & 0 & \frac{5\sqrt{14}i}{1568} & 0 & -\frac{3\sqrt{70}i}{224} & 0 \\
-\frac{\sqrt{105}i}{168} & 0 & -\frac{5\sqrt{42}i}{392} & 0 & -\frac{5\sqrt{21}i}{392} & 0 & 0 & -\frac{3\sqrt{70}i}{224} & 0 & \frac{5\sqrt{14}i}{1568} & 0 & \frac{11\sqrt{210}i}{1568} & 0 & \frac{\sqrt{10}i}{32} \\
0 & \frac{5\sqrt{42}i}{336} & 0 & \frac{5\sqrt{21}i}{392} & 0 & \frac{\sqrt{210}i}{784} & \frac{3\sqrt{5}i}{32} & 0 & -\frac{\sqrt{105}i}{224} & 0 & -\frac{65\sqrt{7}i}{1568} & 0 & -\frac{23\sqrt{35}i}{1568} & 0 \\
-\frac{3\sqrt{42}i}{112} & 0 & -\frac{\sqrt{105}i}{168} & 0 & -\frac{\sqrt{210}i}{784} & 0 & 0 & \frac{15\sqrt{7}i}{224} & 0 & \frac{5\sqrt{35}i}{224} & 0 & \frac{25\sqrt{21}i}{1568} & 0 & \frac{5i}{224} \\
-\frac{5i}{224} & 0 & -\frac{\sqrt{10}i}{32} & 0 & -\frac{3\sqrt{5}i}{32} & 0 & 0 & \frac{5\sqrt{6}i}{224} & 0 & \frac{\sqrt{30}i}{48} & 0 & \frac{3\sqrt{2}i}{32} & 0 & 0 \\
0 & \frac{23\sqrt{35}i}{1568} & 0 & \frac{3\sqrt{70}i}{224} & 0 & -\frac{15\sqrt{7}i}{224} & -\frac{5\sqrt{6}i}{224} & 0 & -\frac{5\sqrt{14}i}{196} & 0 & -\frac{\sqrt{210}i}{672} & 0 & \frac{3\sqrt{42}i}{112} & 0 \\
-\frac{25\sqrt{21}i}{1568} & 0 & -\frac{11\sqrt{210}i}{1568} & 0 & \frac{\sqrt{105}i}{224} & 0 & 0 & \frac{5\sqrt{14}i}{196} & 0 & \frac{\sqrt{70}i}{1568} & 0 & -\frac{5\sqrt{42}i}{336} & 0 & \frac{3\sqrt{2}i}{32} \\
0 & \frac{65\sqrt{7}i}{1568} & 0 & -\frac{5\sqrt{14}i}{1568} & 0 & -\frac{5\sqrt{35}i}{224} & -\frac{\sqrt{30}i}{48} & 0 & -\frac{\sqrt{70}i}{1568} & 0 & \frac{5\sqrt{42}i}{392} & 0 & -\frac{\sqrt{210}i}{672} & 0 \\
-\frac{5\sqrt{35}i}{224} & 0 & -\frac{5\sqrt{14}i}{1568} & 0 & \frac{65\sqrt{7}i}{1568} & 0 & 0 & \frac{\sqrt{210}i}{672} & 0 & -\frac{5\sqrt{42}i}{392} & 0 & \frac{\sqrt{70}i}{1568} & 0 & \frac{\sqrt{30}i}{48} \\
0 & \frac{\sqrt{105}i}{224} & 0 & -\frac{11\sqrt{210}i}{1568} & 0 & -\frac{25\sqrt{21}i}{1568} & -\frac{3\sqrt{2}i}{32} & 0 & \frac{5\sqrt{42}i}{336} & 0 & -\frac{\sqrt{70}i}{1568} & 0 & -\frac{5\sqrt{14}i}{196} & 0 \\
-\frac{15\sqrt{7}i}{224} & 0 & \frac{3\sqrt{70}i}{224} & 0 & \frac{23\sqrt{35}i}{1568} & 0 & 0 & -\frac{3\sqrt{42}i}{112} & 0 & \frac{\sqrt{210}i}{672} & 0 & \frac{5\sqrt{14}i}{196} & 0 & \frac{5\sqrt{6}i}{224} \\
0 & -\frac{3\sqrt{5}i}{32} & 0 & -\frac{\sqrt{10}i}{32} & 0 & -\frac{5i}{224} & 0 & 0 & -\frac{3\sqrt{2}i}{32} & 0 & -\frac{\sqrt{30}i}{48} & 0 & -\frac{5\sqrt{6}i}{224} & 0
\end{bmatrix}$$

$\boxed{\text{x489}}$ $\mathbb{M}_{5,1}^{(a)}(E_g,2) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{6}}{112} & 0 & \frac{3\sqrt{3}}{56} & 0 & \frac{\sqrt{30}}{112} & -\frac{\sqrt{35}}{224} & 0 & \frac{5\sqrt{15}}{224} & 0 & \frac{45}{224} & 0 & \frac{5\sqrt{5}}{224} & 0 \\
\frac{\sqrt{6}}{112} & 0 & -\frac{\sqrt{15}}{56} & 0 & -\frac{3\sqrt{30}}{112} & 0 & 0 & \frac{23}{224} & 0 & -\frac{13\sqrt{5}}{224} & 0 & -\frac{9\sqrt{3}}{224} & 0 & \frac{5\sqrt{7}}{224} \\
0 & -\frac{\sqrt{15}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & \frac{3\sqrt{3}}{56} & -\frac{9\sqrt{14}}{224} & 0 & -\frac{11\sqrt{6}}{224} & 0 & \frac{\sqrt{10}}{224} & 0 & -\frac{27\sqrt{2}}{224} & 0 \\
\frac{3\sqrt{3}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & -\frac{\sqrt{15}}{56} & 0 & 0 & \frac{27\sqrt{2}}{224} & 0 & -\frac{\sqrt{10}}{224} & 0 & \frac{11\sqrt{6}}{224} & 0 & \frac{9\sqrt{14}}{224} \\
0 & -\frac{3\sqrt{30}}{112} & 0 & -\frac{\sqrt{15}}{56} & 0 & \frac{\sqrt{6}}{112} & -\frac{5\sqrt{7}}{224} & 0 & \frac{9\sqrt{3}}{224} & 0 & \frac{13\sqrt{5}}{224} & 0 & -\frac{23}{224} & 0 \\
\frac{\sqrt{30}}{112} & 0 & \frac{3\sqrt{3}}{56} & 0 & \frac{\sqrt{6}}{112} & 0 & 0 & -\frac{5\sqrt{5}}{224} & 0 & -\frac{45}{224} & 0 & -\frac{5\sqrt{15}}{224} & 0 & \frac{\sqrt{35}}{224} \\
-\frac{\sqrt{35}}{224} & 0 & -\frac{9\sqrt{14}}{224} & 0 & -\frac{5\sqrt{7}}{224} & 0 & 0 & \frac{\sqrt{210}}{224} & 0 & \frac{3\sqrt{42}}{112} & 0 & \frac{\sqrt{70}}{224} & 0 & 0 \\
0 & \frac{23}{224} & 0 & \frac{27\sqrt{2}}{224} & 0 & -\frac{5\sqrt{5}}{224} & \frac{\sqrt{210}}{224} & 0 & -\frac{\sqrt{10}}{28} & 0 & -\frac{3\sqrt{6}}{224} & 0 & \frac{\sqrt{30}}{112} & 0 \\
\frac{5\sqrt{15}}{224} & 0 & -\frac{11\sqrt{6}}{224} & 0 & \frac{9\sqrt{3}}{224} & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{2}}{224} & 0 & -\frac{3\sqrt{30}}{112} & 0 & \frac{\sqrt{70}}{224} \\
0 & -\frac{13\sqrt{5}}{224} & 0 & -\frac{\sqrt{10}}{224} & 0 & -\frac{45}{224} & \frac{3\sqrt{42}}{112} & 0 & \frac{\sqrt{2}}{224} & 0 & \frac{\sqrt{30}}{56} & 0 & -\frac{3\sqrt{6}}{224} & 0 \\
\frac{45}{224} & 0 & \frac{\sqrt{10}}{224} & 0 & \frac{13\sqrt{5}}{224} & 0 & 0 & -\frac{3\sqrt{6}}{224} & 0 & \frac{\sqrt{30}}{56} & 0 & \frac{\sqrt{2}}{224} & 0 & \frac{3\sqrt{42}}{112} \\
0 & -\frac{9\sqrt{3}}{224} & 0 & \frac{11\sqrt{6}}{224} & 0 & -\frac{5\sqrt{15}}{224} & \frac{\sqrt{70}}{224} & 0 & -\frac{3\sqrt{30}}{112} & 0 & \frac{\sqrt{2}}{224} & 0 & -\frac{\sqrt{10}}{28} & 0 \\
\frac{5\sqrt{5}}{224} & 0 & -\frac{27\sqrt{2}}{224} & 0 & -\frac{23}{224} & 0 & 0 & \frac{\sqrt{30}}{112} & 0 & -\frac{3\sqrt{6}}{224} & 0 & -\frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{210}}{224} \\
0 & \frac{5\sqrt{7}}{224} & 0 & \frac{9\sqrt{14}}{224} & 0 & \frac{\sqrt{35}}{224} & 0 & 0 & \frac{\sqrt{70}}{224} & 0 & \frac{3\sqrt{42}}{112} & 0 & \frac{\sqrt{210}}{224} & 0
\end{bmatrix}$$

$\boxed{\text{x490}}$ $\mathbb{M}_{5,2}^{(a)}(E_g,2) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{6}i}{112} & 0 & -\frac{3\sqrt{3}i}{56} & 0 & \frac{\sqrt{30}i}{112} & \frac{\sqrt{35}i}{224} & 0 & \frac{5\sqrt{15}i}{224} & 0 & -\frac{45i}{224} & 0 & \frac{5\sqrt{5}i}{224} & 0 \\
-\frac{\sqrt{6}i}{112} & 0 & -\frac{\sqrt{15}i}{56} & 0 & \frac{3\sqrt{30}i}{112} & 0 & 0 & -\frac{23i}{224} & 0 & -\frac{13\sqrt{5}i}{224} & 0 & \frac{9\sqrt{3}i}{224} & 0 & \frac{5\sqrt{7}i}{224} \\
0 & \frac{\sqrt{15}i}{56} & 0 & \frac{\sqrt{30}i}{56} & 0 & -\frac{3\sqrt{3}i}{56} & -\frac{9\sqrt{14}i}{224} & 0 & \frac{11\sqrt{6}i}{224} & 0 & \frac{\sqrt{10}i}{224} & 0 & \frac{27\sqrt{2}i}{224} & 0 \\
\frac{3\sqrt{3}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 & \frac{27\sqrt{2}i}{224} & 0 & \frac{\sqrt{10}i}{224} & 0 & \frac{11\sqrt{6}i}{224} & 0 & -\frac{9\sqrt{14}i}{224} \\
0 & -\frac{3\sqrt{30}i}{112} & 0 & \frac{\sqrt{15}i}{56} & 0 & \frac{\sqrt{6}i}{112} & \frac{5\sqrt{7}i}{224} & 0 & \frac{9\sqrt{3}i}{224} & 0 & -\frac{13\sqrt{5}i}{224} & 0 & -\frac{23i}{224} & 0 \\
-\frac{\sqrt{30}i}{112} & 0 & \frac{3\sqrt{3}i}{56} & 0 & -\frac{\sqrt{6}i}{112} & 0 & 0 & \frac{5\sqrt{5}i}{224} & 0 & -\frac{45i}{224} & 0 & \frac{5\sqrt{15}i}{224} & 0 & \frac{\sqrt{35}i}{224} \\
-\frac{\sqrt{35}i}{224} & 0 & \frac{9\sqrt{14}i}{224} & 0 & -\frac{5\sqrt{7}i}{224} & 0 & 0 & \frac{\sqrt{210}i}{224} & 0 & -\frac{3\sqrt{42}i}{112} & 0 & \frac{\sqrt{70}i}{224} & 0 & 0 \\
0 & \frac{23i}{224} & 0 & -\frac{27\sqrt{2}i}{224} & 0 & -\frac{5\sqrt{5}i}{224} & -\frac{\sqrt{210}i}{224} & 0 & -\frac{\sqrt{10}i}{28} & 0 & \frac{3\sqrt{6}i}{224} & 0 & \frac{\sqrt{30}i}{112} & 0 \\
-\frac{5\sqrt{15}i}{224} & 0 & -\frac{11\sqrt{6}i}{224} & 0 & -\frac{9\sqrt{3}i}{224} & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & \frac{\sqrt{2}i}{224} & 0 & \frac{3\sqrt{30}i}{112} & 0 & \frac{\sqrt{70}i}{224} \\
0 & \frac{13\sqrt{5}i}{224} & 0 & -\frac{\sqrt{10}i}{224} & 0 & \frac{45i}{224} & \frac{3\sqrt{42}i}{112} & 0 & -\frac{\sqrt{2}i}{224} & 0 & \frac{\sqrt{30}i}{56} & 0 & \frac{3\sqrt{6}i}{224} & 0 \\
\frac{45i}{224} & 0 & -\frac{\sqrt{10}i}{224} & 0 & \frac{13\sqrt{5}i}{224} & 0 & 0 & -\frac{3\sqrt{6}i}{224} & 0 & -\frac{\sqrt{30}i}{56} & 0 & \frac{\sqrt{2}i}{224} & 0 & -\frac{3\sqrt{42}i}{112} \\
0 & -\frac{9\sqrt{3}i}{224} & 0 & -\frac{11\sqrt{6}i}{224} & 0 & -\frac{5\sqrt{15}i}{224} & -\frac{\sqrt{70}i}{224} & 0 & -\frac{3\sqrt{30}i}{112} & 0 & -\frac{\sqrt{2}i}{224} & 0 & -\frac{\sqrt{10}i}{28} & 0 \\
-\frac{5\sqrt{5}i}{224} & 0 & -\frac{27\sqrt{2}i}{224} & 0 & \frac{23i}{224} & 0 & 0 & -\frac{\sqrt{30}i}{112} & 0 & -\frac{3\sqrt{6}i}{224} & 0 & \frac{\sqrt{10}i}{28} & 0 & \frac{\sqrt{210}i}{224} \\
0 & -\frac{5\sqrt{7}i}{224} & 0 & \frac{9\sqrt{14}i}{224} & 0 & -\frac{\sqrt{35}i}{224} & 0 & 0 & -\frac{\sqrt{70}i}{224} & 0 & \frac{3\sqrt{42}i}{112} & 0 & -\frac{\sqrt{210}i}{224} & 0
\end{bmatrix}$$

$\boxed{\text{x491}}$ $\mathbb{M}_{5,1}^{(a)}(E_g,3) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{2}}{56} & 0 & -\frac{1}{28} & 0 & -\frac{3\sqrt{10}}{56} & -\frac{\sqrt{105}}{336} & 0 & \frac{5\sqrt{5}}{112} & 0 & -\frac{5\sqrt{3}}{112} & 0 & -\frac{5\sqrt{15}}{112} & 0 \\
\frac{\sqrt{2}}{56} & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & \frac{23\sqrt{3}}{336} & 0 & -\frac{13\sqrt{15}}{336} & 0 & \frac{3}{112} & 0 & -\frac{5\sqrt{21}}{112} \\
0 & -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{10}}{28} & 0 & -\frac{1}{28} & \frac{\sqrt{42}}{112} & 0 & -\frac{11\sqrt{2}}{112} & 0 & \frac{\sqrt{30}}{336} & 0 & \frac{3\sqrt{6}}{112} & 0 \\
-\frac{1}{28} & 0 & \frac{\sqrt{10}}{28} & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & -\frac{3\sqrt{6}}{112} & 0 & -\frac{\sqrt{30}}{336} & 0 & \frac{11\sqrt{2}}{112} & 0 & -\frac{\sqrt{42}}{112} \\
0 & \frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{5}}{28} & 0 & \frac{\sqrt{2}}{56} & \frac{5\sqrt{21}}{112} & 0 & -\frac{3}{112} & 0 & \frac{13\sqrt{15}}{336} & 0 & -\frac{23\sqrt{3}}{336} & 0 \\
-\frac{3\sqrt{10}}{56} & 0 & -\frac{1}{28} & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & \frac{5\sqrt{15}}{112} & 0 & \frac{5\sqrt{3}}{112} & 0 & -\frac{5\sqrt{5}}{112} & 0 & \frac{\sqrt{105}}{336} \\
-\frac{\sqrt{105}}{336} & 0 & \frac{\sqrt{42}}{112} & 0 & \frac{5\sqrt{21}}{112} & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{14}}{56} & 0 & -\frac{\sqrt{210}}{112} & 0 & 0 \\
0 & \frac{23\sqrt{3}}{336} & 0 & -\frac{3\sqrt{6}}{112} & 0 & \frac{5\sqrt{15}}{112} & \frac{\sqrt{70}}{112} & 0 & -\frac{\sqrt{30}}{42} & 0 & \frac{\sqrt{2}}{112} & 0 & -\frac{3\sqrt{10}}{56} & 0 \\
\frac{5\sqrt{5}}{112} & 0 & -\frac{11\sqrt{2}}{112} & 0 & -\frac{3}{112} & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & \frac{\sqrt{6}}{336} & 0 & \frac{\sqrt{10}}{56} & 0 & -\frac{\sqrt{210}}{112} \\
0 & -\frac{13\sqrt{15}}{336} & 0 & -\frac{\sqrt{30}}{336} & 0 & \frac{5\sqrt{3}}{112} & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{6}}{336} & 0 & \frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{2}}{112} & 0 \\
-\frac{5\sqrt{3}}{112} & 0 & \frac{\sqrt{30}}{336} & 0 & \frac{13\sqrt{15}}{336} & 0 & 0 & \frac{\sqrt{2}}{112} & 0 & \frac{\sqrt{10}}{28} & 0 & \frac{\sqrt{6}}{336} & 0 & -\frac{\sqrt{14}}{56} \\
0 & \frac{3}{112} & 0 & \frac{11\sqrt{2}}{112} & 0 & -\frac{5\sqrt{5}}{112} & -\frac{\sqrt{210}}{112} & 0 & \frac{\sqrt{10}}{56} & 0 & \frac{\sqrt{6}}{336} & 0 & -\frac{\sqrt{30}}{42} & 0 \\
-\frac{5\sqrt{15}}{112} & 0 & \frac{3\sqrt{6}}{112} & 0 & -\frac{23\sqrt{3}}{336} & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & \frac{\sqrt{2}}{112} & 0 & -\frac{\sqrt{30}}{42} & 0 & \frac{\sqrt{70}}{112} \\
0 & -\frac{5\sqrt{21}}{112} & 0 & -\frac{\sqrt{42}}{112} & 0 & \frac{\sqrt{105}}{336} & 0 & 0 & -\frac{\sqrt{210}}{112} & 0 & -\frac{\sqrt{14}}{56} & 0 & \frac{\sqrt{70}}{112} & 0
\end{bmatrix}$$

$\boxed{\text{x492}}$ $\mathbb{M}_{5,2}^{(a)}(E_g,3) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{2}i}{56} & 0 & \frac{i}{28} & 0 & -\frac{3\sqrt{10}i}{56} & \frac{\sqrt{105}i}{336} & 0 & \frac{5\sqrt{5}i}{112} & 0 & \frac{5\sqrt{3}i}{112} & 0 & -\frac{5\sqrt{15}i}{112} & 0 \\
-\frac{\sqrt{2}i}{56} & 0 & -\frac{\sqrt{5}i}{28} & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & -\frac{23\sqrt{3}i}{336} & 0 & -\frac{13\sqrt{15}i}{336} & 0 & -\frac{3i}{112} & 0 & -\frac{5\sqrt{21}i}{112} \\
0 & \frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{10}i}{28} & 0 & \frac{i}{28} & \frac{\sqrt{42}i}{112} & 0 & \frac{11\sqrt{2}i}{112} & 0 & \frac{\sqrt{30}i}{336} & 0 & -\frac{3\sqrt{6}i}{112} & 0 \\
-\frac{i}{28} & 0 & -\frac{\sqrt{10}i}{28} & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & -\frac{3\sqrt{6}i}{112} & 0 & \frac{\sqrt{30}i}{336} & 0 & \frac{11\sqrt{2}i}{112} & 0 & \frac{\sqrt{42}i}{112} \\
0 & \frac{\sqrt{10}i}{56} & 0 & \frac{\sqrt{5}i}{28} & 0 & \frac{\sqrt{2}i}{56} & -\frac{5\sqrt{21}i}{112} & 0 & -\frac{3i}{112} & 0 & -\frac{13\sqrt{15}i}{336} & 0 & -\frac{23\sqrt{3}i}{336} & 0 \\
\frac{3\sqrt{10}i}{56} & 0 & -\frac{i}{28} & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & -\frac{5\sqrt{15}i}{112} & 0 & \frac{5\sqrt{3}i}{112} & 0 & \frac{5\sqrt{5}i}{112} & 0 & \frac{\sqrt{105}i}{336} \\
-\frac{\sqrt{105}i}{336} & 0 & -\frac{\sqrt{42}i}{112} & 0 & \frac{5\sqrt{21}i}{112} & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & \frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{210}i}{112} & 0 & 0 \\
0 & \frac{23\sqrt{3}i}{336} & 0 & \frac{3\sqrt{6}i}{112} & 0 & \frac{5\sqrt{15}i}{112} & -\frac{\sqrt{70}i}{112} & 0 & -\frac{\sqrt{30}i}{42} & 0 & -\frac{\sqrt{2}i}{112} & 0 & -\frac{3\sqrt{10}i}{56} & 0 \\
-\frac{5\sqrt{5}i}{112} & 0 & -\frac{11\sqrt{2}i}{112} & 0 & \frac{3i}{112} & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & \frac{\sqrt{6}i}{336} & 0 & -\frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{210}i}{112} \\
0 & \frac{13\sqrt{15}i}{336} & 0 & -\frac{\sqrt{30}i}{336} & 0 & -\frac{5\sqrt{3}i}{112} & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{6}i}{336} & 0 & \frac{\sqrt{10}i}{28} & 0 & -\frac{\sqrt{2}i}{112} & 0 \\
-\frac{5\sqrt{3}i}{112} & 0 & -\frac{\sqrt{30}i}{336} & 0 & \frac{13\sqrt{15}i}{336} & 0 & 0 & \frac{\sqrt{2}i}{112} & 0 & -\frac{\sqrt{10}i}{28} & 0 & \frac{\sqrt{6}i}{336} & 0 & \frac{\sqrt{14}i}{56} \\
0 & \frac{3i}{112} & 0 & -\frac{11\sqrt{2}i}{112} & 0 & -\frac{5\sqrt{5}i}{112} & \frac{\sqrt{210}i}{112} & 0 & \frac{\sqrt{10}i}{56} & 0 & -\frac{\sqrt{6}i}{336} & 0 & -\frac{\sqrt{30}i}{42} & 0 \\
\frac{5\sqrt{15}i}{112} & 0 & \frac{3\sqrt{6}i}{112} & 0 & \frac{23\sqrt{3}i}{336} & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & \frac{\sqrt{2}i}{112} & 0 & \frac{\sqrt{30}i}{42} & 0 & \frac{\sqrt{70}i}{112} \\
0 & \frac{5\sqrt{21}i}{112} & 0 & -\frac{\sqrt{42}i}{112} & 0 & -\frac{\sqrt{105}i}{336} & 0 & 0 & \frac{\sqrt{210}i}{112} & 0 & -\frac{\sqrt{14}i}{56} & 0 & -\frac{\sqrt{70}i}{112} & 0
\end{bmatrix}$$

$\boxed{\text{x493}}$ $\mathbb{M}_5^{(1,-1;a)}(A_{1g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{77} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{385} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}i}{385} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{1155}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{1155}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{154} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{154} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{154} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{1155}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}i}{154} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x494}}$ $\mathbb{M}_7^{(1,-1;a)}(A_{1g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{130}i}{52} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{546}i}{52} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{546}i}{52} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{130}i}{52} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{130}i}{52} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{546}i}{52} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{546}i}{52} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{130}i}{52} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x495}}$ $\mathbb{M}_1^{(1,-1;a)}(A_{2g}) = \begin{bmatrix}
-\frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}}{49} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14}
\end{bmatrix}$

$\boxed{\text{x496}}$ $\mathbb{M}_3^{(1,-1;a)}(A_{2g}) = \begin{bmatrix}
\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{70}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{70}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{147} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28}
\end{bmatrix}$

$\boxed{\text{x497}}$ $\mathbb{M}_5^{(1,-1;a)}(A_{2g},1) =$

$$\begin{bmatrix}
-\frac{\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{385}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{385}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{5\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{539} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{385}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{9\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{385}}{1078} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{385}}{1078} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{385}}{1078} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{385}}{1078} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{385}}{1078} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{154}
\end{bmatrix}$$

$\boxed{\text{x498}}$ $\mathbb{M}_5^{(1,-1;a)}(A_{2g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}}{77} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{385} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{1155}}{385} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{1155}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{1155}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{154} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}}{154} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{1155}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}}{154} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x499}}$ $\mathbb{M}_7^{(1,-1;a)}(A_{2g},1) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{1716} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{858}}{1716} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{858}}{572} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{35\sqrt{858}}{1716} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{35\sqrt{858}}{1716} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{858}}{572} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{858}}{1716} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{1716}
\end{bmatrix}$

$\boxed{\text{x500}}$ $\mathbb{M}_7^{(1,-1;a)}(A_{2g},2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{130}}{52} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{546}}{52} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{546}}{52} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{130}}{52} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{130}}{52} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{546}}{52} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{546}}{52} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{130}}{52} & 0 & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x501}}$ $\mathbb{M}_3^{(1,-1;a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{21}i}{98} & 0 & 0 & -\frac{2\sqrt{5}i}{21} & 0 & 0 & 0 & -\frac{10\sqrt{7}i}{147} & 0 & 0 & 0 \\
\frac{\sqrt{105}i}{98} & 0 & 0 & 0 & \frac{\sqrt{21}i}{98} & 0 & 0 & -\frac{\sqrt{70}i}{147} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 \\
0 & \frac{\sqrt{21}i}{98} & 0 & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 & \frac{\sqrt{210}i}{147} & 0 & 0 & 0 & \frac{\sqrt{70}i}{147} & 0 \\
0 & 0 & -\frac{\sqrt{21}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{7}i}{147} & 0 & 0 & 0 & \frac{2\sqrt{5}i}{21} \\
0 & 0 & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}i}{147} & 0 & 0 & 0 \\
0 & \frac{2\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{70}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 \\
\frac{4\sqrt{35}i}{147} & 0 & 0 & 0 & -\frac{10\sqrt{7}i}{147} & 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 \\
0 & \frac{10\sqrt{7}i}{147} & 0 & 0 & 0 & -\frac{4\sqrt{35}i}{147} & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{196} & 0 \\
0 & 0 & \frac{\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} \\
0 & 0 & 0 & -\frac{\sqrt{70}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}i}{196} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x502}}$ $\mathbb{M}_5^{(1,-1;a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{330}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{66}i}{924} & 0 & 0 & -\frac{\sqrt{770}i}{385} & 0 & 0 & 0 & \frac{2\sqrt{22}i}{77} & 0 & 0 & 0 \\
\frac{\sqrt{330}i}{924} & 0 & 0 & 0 & -\frac{5\sqrt{66}i}{924} & 0 & 0 & \frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & \frac{2\sqrt{165}i}{385} & 0 & 0 \\
0 & -\frac{5\sqrt{66}i}{924} & 0 & 0 & 0 & \frac{\sqrt{330}i}{924} & 0 & 0 & -\frac{2\sqrt{165}i}{385} & 0 & 0 & 0 & -\frac{8\sqrt{55}i}{385} & 0 \\
0 & 0 & \frac{5\sqrt{66}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{22}i}{77} & 0 & 0 & 0 & \frac{\sqrt{770}i}{385} \\
0 & 0 & 0 & -\frac{\sqrt{330}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{77} & 0 & 0 & 0 \\
0 & \frac{\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{165}i}{385} & 0 & 0 & -\frac{5\sqrt{77}i}{154} & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{77} & 0 & 0 & 0 \\
\frac{\sqrt{110}i}{77} & 0 & 0 & 0 & \frac{2\sqrt{22}i}{77} & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 & 0 & 0 & \frac{2\sqrt{55}i}{77} & 0 & 0 \\
0 & -\frac{2\sqrt{22}i}{77} & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & 0 & 0 & \frac{2\sqrt{55}i}{77} & 0 & 0 & 0 & \frac{3\sqrt{165}i}{154} & 0 \\
0 & 0 & -\frac{2\sqrt{165}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{77} & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{154} \\
0 & 0 & 0 & \frac{8\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{154} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x503}}$ $\mathbb{M}_7^{(1,-1;a)}(B_{1g},1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{88} & 0 & 0 & 0 & \frac{\sqrt{66}i}{24} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & -\frac{\sqrt{66}i}{24} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{88} & 0 & 0 & 0 & \frac{\sqrt{770}i}{88} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{88} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{88} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{24} & 0 & 0 & 0 & \frac{\sqrt{22}i}{88} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x504}}$ $\mathbb{M}_7^{(1,-1;a)}(B_{1g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{26}i}{104} & 0 & 0 & 0 & -\frac{\sqrt{78}i}{24} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2730}i}{312} & 0 & 0 & 0 & \frac{\sqrt{78}i}{24} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{26}i}{104} & 0 & 0 & 0 & \frac{\sqrt{910}i}{104} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2730}i}{312} & 0 & 0 & 0 & -\frac{\sqrt{910}i}{104} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{910}i}{104} & 0 & 0 & 0 & \frac{\sqrt{2730}i}{312} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{910}i}{104} & 0 & 0 & 0 & -\frac{\sqrt{26}i}{104} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{78}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{2730}i}{312} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{78}i}{24} & 0 & 0 & 0 & \frac{\sqrt{26}i}{104} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x505}}$ $\mathbb{M}_3^{(1,-1;a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & -\frac{2\sqrt{5}}{21} & 0 & 0 & 0 & \frac{10\sqrt{7}}{147} & 0 & 0 & 0 \\
\frac{\sqrt{105}}{98} & 0 & 0 & 0 & -\frac{\sqrt{21}}{98} & 0 & 0 & -\frac{\sqrt{70}}{147} & 0 & 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 \\
0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 & -\frac{\sqrt{70}}{147} & 0 \\
0 & 0 & -\frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{7}}{147} & 0 & 0 & 0 & -\frac{2\sqrt{5}}{21} \\
0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{70}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 \\
\frac{4\sqrt{35}}{147} & 0 & 0 & 0 & \frac{10\sqrt{7}}{147} & 0 & 0 & -\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 \\
0 & \frac{10\sqrt{7}}{147} & 0 & 0 & 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 \\
0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} \\
0 & 0 & 0 & -\frac{\sqrt{70}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x506}}$ $\mathbb{M}_5^{(1,-1;a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & -\frac{\sqrt{330}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{77} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{66}}{924} & 0 & 0 & \frac{\sqrt{770}}{385} & 0 & 0 & 0 & \frac{2\sqrt{22}}{77} & 0 & 0 & 0 \\
-\frac{\sqrt{330}}{924} & 0 & 0 & 0 & -\frac{5\sqrt{66}}{924} & 0 & 0 & -\frac{8\sqrt{55}}{385} & 0 & 0 & 0 & \frac{2\sqrt{165}}{385} & 0 & 0 \\
0 & \frac{5\sqrt{66}}{924} & 0 & 0 & 0 & \frac{\sqrt{330}}{924} & 0 & 0 & \frac{2\sqrt{165}}{385} & 0 & 0 & 0 & -\frac{8\sqrt{55}}{385} & 0 \\
0 & 0 & -\frac{5\sqrt{66}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{22}}{77} & 0 & 0 & 0 & \frac{\sqrt{770}}{385} \\
0 & 0 & 0 & \frac{\sqrt{330}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{77} & 0 & 0 & 0 \\
0 & \frac{\sqrt{770}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{8\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{165}}{385} & 0 & 0 & \frac{5\sqrt{77}}{154} & 0 & 0 & 0 & -\frac{2\sqrt{55}}{77} & 0 & 0 & 0 \\
-\frac{\sqrt{110}}{77} & 0 & 0 & 0 & \frac{2\sqrt{22}}{77} & 0 & 0 & -\frac{3\sqrt{165}}{154} & 0 & 0 & 0 & \frac{2\sqrt{55}}{77} & 0 & 0 \\
0 & \frac{2\sqrt{22}}{77} & 0 & 0 & 0 & -\frac{\sqrt{110}}{77} & 0 & 0 & -\frac{2\sqrt{55}}{77} & 0 & 0 & 0 & \frac{3\sqrt{165}}{154} & 0 \\
0 & 0 & \frac{2\sqrt{165}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{77} & 0 & 0 & 0 & -\frac{5\sqrt{77}}{154} \\
0 & 0 & 0 & -\frac{8\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}}{154} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{770}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{154} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x507}}$ $\mathbb{M}_7^{(1,-1;a)}(B_{2g},1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x508}}$ $\mathbb{M}_7^{(1,-1;a)}(B_{2g},2) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{429}}{286} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5005}}{286} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{429}}{286} & 0 & 0 & 0 & -\frac{\sqrt{15015}}{286} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5005}}{286} & 0 & 0 & 0 & \frac{\sqrt{15015}}{286} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15015}}{286} & 0 & 0 & 0 & -\frac{\sqrt{5005}}{286} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15015}}{286} & 0 & 0 & 0 & \frac{\sqrt{429}}{286} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}}{286} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{429}}{286} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x509}}$ $\mathbb{M}_{1,1}^{(1,-1;a)}(E_g) = \begin{bmatrix}
0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & -\frac{\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{70}}{98} & 0 & -\frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & -\frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 \\
0 & -\frac{2\sqrt{7}}{49} & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & -\frac{\sqrt{42}}{49} & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & -\frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{7}}{49} & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{49} & 0 & -\frac{\sqrt{105}}{49} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{49} & 0 & -\frac{\sqrt{3}}{7} \\
\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{7}}{49} & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & \frac{\sqrt{210}}{98} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{21}}{49} & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{98} & 0 & \frac{2\sqrt{14}}{49} & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{42}}{49} & 0 & \frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{49} & 0 & \frac{\sqrt{210}}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & \frac{\sqrt{7}}{49} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{98} & 0 & \frac{\sqrt{42}}{49} & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & \frac{\sqrt{2}}{14} \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0
\end{bmatrix}$

$\boxed{\text{x510}}$ $\mathbb{M}_{1,2}^{(1,-1;a)}(E_g) = \begin{bmatrix}
0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & -\frac{\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{70}i}{98} & 0 & -\frac{2\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & -\frac{\sqrt{21}i}{49} & 0 & 0 & 0 & 0 \\
0 & \frac{2\sqrt{7}i}{49} & 0 & -\frac{3\sqrt{14}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{49} & 0 & -\frac{\sqrt{42}i}{49} & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{14}i}{98} & 0 & -\frac{2\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{49} & 0 & -\frac{\sqrt{70}i}{49} & 0 & 0 \\
0 & 0 & 0 & \frac{2\sqrt{7}i}{49} & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{49} & 0 & -\frac{\sqrt{105}i}{49} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{49} & 0 & -\frac{\sqrt{3}i}{7} \\
\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & \frac{\sqrt{42}i}{49} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{7}i}{49} & 0 & \frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{49} & 0 & \frac{\sqrt{210}i}{98} & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{21}i}{49} & 0 & \frac{\sqrt{42}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{98} & 0 & \frac{2\sqrt{14}i}{49} & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}i}{49} & 0 & \frac{\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}i}{49} & 0 & \frac{\sqrt{210}i}{98} & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{70}i}{49} & 0 & \frac{\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{98} & 0 & \frac{\sqrt{42}i}{49} & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{49} & 0 & \frac{\sqrt{2}i}{14} \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0
\end{bmatrix}$

$\boxed{\text{x511}}$ $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & -\frac{3\sqrt{14}}{196} & 0 & \frac{5\sqrt{7}}{196} & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & -\frac{\sqrt{35}}{49} & 0 & \frac{5\sqrt{21}}{294} & 0 & 0 & 0 \\
-\frac{3\sqrt{14}}{196} & 0 & \frac{3\sqrt{35}}{980} & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & -\frac{5\sqrt{21}}{294} & 0 & -\frac{\sqrt{105}}{147} & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 \\
0 & \frac{3\sqrt{35}}{980} & 0 & \frac{3\sqrt{70}}{490} & 0 & \frac{5\sqrt{7}}{196} & -\frac{5\sqrt{6}}{84} & 0 & -\frac{5\sqrt{14}}{196} & 0 & \frac{\sqrt{210}}{588} & 0 & \frac{5\sqrt{42}}{196} & 0 \\
\frac{5\sqrt{7}}{196} & 0 & \frac{3\sqrt{70}}{490} & 0 & \frac{3\sqrt{35}}{980} & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & -\frac{\sqrt{210}}{588} & 0 & \frac{5\sqrt{14}}{196} & 0 & \frac{5\sqrt{6}}{84} \\
0 & \frac{\sqrt{70}}{98} & 0 & \frac{3\sqrt{35}}{980} & 0 & -\frac{3\sqrt{14}}{196} & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & \frac{\sqrt{105}}{147} & 0 & \frac{5\sqrt{21}}{294} & 0 \\
0 & 0 & \frac{5\sqrt{7}}{196} & 0 & -\frac{3\sqrt{14}}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{294} & 0 & \frac{\sqrt{35}}{49} & 0 & -\frac{\sqrt{15}}{42} \\
\frac{\sqrt{15}}{42} & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & -\frac{5\sqrt{2}}{56} & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{21}}{294} & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & \frac{\sqrt{210}}{392} & 0 & -\frac{5\sqrt{14}}{98} & 0 & 0 & 0 \\
-\frac{\sqrt{35}}{49} & 0 & -\frac{5\sqrt{14}}{196} & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 & \frac{\sqrt{210}}{392} & 0 & -\frac{5\sqrt{42}}{392} & 0 & -\frac{5\sqrt{70}}{196} & 0 & 0 \\
0 & -\frac{\sqrt{105}}{147} & 0 & -\frac{\sqrt{210}}{588} & 0 & -\frac{5\sqrt{21}}{294} & -\frac{5\sqrt{2}}{56} & 0 & -\frac{5\sqrt{42}}{392} & 0 & -\frac{3\sqrt{70}}{196} & 0 & -\frac{5\sqrt{14}}{98} & 0 \\
\frac{5\sqrt{21}}{294} & 0 & \frac{\sqrt{210}}{588} & 0 & \frac{\sqrt{105}}{147} & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 & -\frac{3\sqrt{70}}{196} & 0 & -\frac{5\sqrt{42}}{392} & 0 & -\frac{5\sqrt{2}}{56} \\
0 & \frac{5\sqrt{7}}{98} & 0 & \frac{5\sqrt{14}}{196} & 0 & \frac{\sqrt{35}}{49} & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 & -\frac{5\sqrt{42}}{392} & 0 & \frac{\sqrt{210}}{392} & 0 \\
0 & 0 & \frac{5\sqrt{42}}{196} & 0 & \frac{5\sqrt{21}}{294} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 & \frac{\sqrt{210}}{392} & 0 & \frac{3\sqrt{10}}{56} \\
0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{56} & 0 & \frac{3\sqrt{10}}{56} & 0
\end{bmatrix}$$

$\boxed{\text{x512}}$ $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & -\frac{3\sqrt{14}i}{196} & 0 & -\frac{5\sqrt{7}i}{196} & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & -\frac{\sqrt{35}i}{49} & 0 & -\frac{5\sqrt{21}i}{294} & 0 & 0 & 0 \\
\frac{3\sqrt{14}i}{196} & 0 & \frac{3\sqrt{35}i}{980} & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & \frac{5\sqrt{21}i}{294} & 0 & -\frac{\sqrt{105}i}{147} & 0 & -\frac{5\sqrt{7}i}{98} & 0 & 0 \\
0 & -\frac{3\sqrt{35}i}{980} & 0 & \frac{3\sqrt{70}i}{490} & 0 & -\frac{5\sqrt{7}i}{196} & -\frac{5\sqrt{6}i}{84} & 0 & \frac{5\sqrt{14}i}{196} & 0 & \frac{\sqrt{210}i}{588} & 0 & -\frac{5\sqrt{42}i}{196} & 0 \\
\frac{5\sqrt{7}i}{196} & 0 & -\frac{3\sqrt{70}i}{490} & 0 & \frac{3\sqrt{35}i}{980} & 0 & 0 & -\frac{5\sqrt{42}i}{196} & 0 & \frac{\sqrt{210}i}{588} & 0 & \frac{5\sqrt{14}i}{196} & 0 & -\frac{5\sqrt{6}i}{84} \\
0 & \frac{\sqrt{70}i}{98} & 0 & -\frac{3\sqrt{35}i}{980} & 0 & -\frac{3\sqrt{14}i}{196} & 0 & 0 & -\frac{5\sqrt{7}i}{98} & 0 & -\frac{\sqrt{105}i}{147} & 0 & \frac{5\sqrt{21}i}{294} & 0 \\
0 & 0 & \frac{5\sqrt{7}i}{196} & 0 & \frac{3\sqrt{14}i}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{294} & 0 & -\frac{\sqrt{35}i}{49} & 0 & -\frac{\sqrt{15}i}{42} \\
\frac{\sqrt{15}i}{42} & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & \frac{5\sqrt{2}i}{56} & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{21}i}{294} & 0 & \frac{5\sqrt{42}i}{196} & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & \frac{\sqrt{210}i}{392} & 0 & \frac{5\sqrt{14}i}{98} & 0 & 0 & 0 \\
\frac{\sqrt{35}i}{49} & 0 & -\frac{5\sqrt{14}i}{196} & 0 & \frac{5\sqrt{7}i}{98} & 0 & 0 & -\frac{\sqrt{210}i}{392} & 0 & -\frac{5\sqrt{42}i}{392} & 0 & \frac{5\sqrt{70}i}{196} & 0 & 0 \\
0 & \frac{\sqrt{105}i}{147} & 0 & -\frac{\sqrt{210}i}{588} & 0 & \frac{5\sqrt{21}i}{294} & -\frac{5\sqrt{2}i}{56} & 0 & \frac{5\sqrt{42}i}{392} & 0 & -\frac{3\sqrt{70}i}{196} & 0 & \frac{5\sqrt{14}i}{98} & 0 \\
\frac{5\sqrt{21}i}{294} & 0 & -\frac{\sqrt{210}i}{588} & 0 & \frac{\sqrt{105}i}{147} & 0 & 0 & -\frac{5\sqrt{14}i}{98} & 0 & \frac{3\sqrt{70}i}{196} & 0 & -\frac{5\sqrt{42}i}{392} & 0 & \frac{5\sqrt{2}i}{56} \\
0 & \frac{5\sqrt{7}i}{98} & 0 & -\frac{5\sqrt{14}i}{196} & 0 & \frac{\sqrt{35}i}{49} & 0 & 0 & -\frac{5\sqrt{70}i}{196} & 0 & \frac{5\sqrt{42}i}{392} & 0 & \frac{\sqrt{210}i}{392} & 0 \\
0 & 0 & \frac{5\sqrt{42}i}{196} & 0 & -\frac{5\sqrt{21}i}{294} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{98} & 0 & -\frac{\sqrt{210}i}{392} & 0 & \frac{3\sqrt{10}i}{56} \\
0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{56} & 0 & -\frac{3\sqrt{10}i}{56} & 0
\end{bmatrix}$$

$\boxed{\text{x513}}$ $\mathbb{M}_{3,1}^{(1,-1;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{210}}{196} & 0 & -\frac{\sqrt{105}}{196} & 0 & 0 & \frac{5}{42} & 0 & -\frac{5\sqrt{21}}{147} & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 \\
-\frac{\sqrt{210}}{196} & 0 & \frac{\sqrt{21}}{196} & 0 & -\frac{\sqrt{42}}{98} & 0 & 0 & -\frac{5\sqrt{35}}{294} & 0 & -\frac{5\sqrt{7}}{147} & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 \\
0 & \frac{\sqrt{21}}{196} & 0 & \frac{\sqrt{42}}{98} & 0 & -\frac{\sqrt{105}}{196} & \frac{\sqrt{10}}{28} & 0 & -\frac{5\sqrt{210}}{588} & 0 & \frac{5\sqrt{14}}{588} & 0 & -\frac{3\sqrt{70}}{196} & 0 \\
-\frac{\sqrt{105}}{196} & 0 & \frac{\sqrt{42}}{98} & 0 & \frac{\sqrt{21}}{196} & 0 & 0 & \frac{3\sqrt{70}}{196} & 0 & -\frac{5\sqrt{14}}{588} & 0 & \frac{5\sqrt{210}}{588} & 0 & -\frac{\sqrt{10}}{28} \\
0 & -\frac{\sqrt{42}}{98} & 0 & \frac{\sqrt{21}}{196} & 0 & -\frac{\sqrt{210}}{196} & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & \frac{5\sqrt{7}}{147} & 0 & \frac{5\sqrt{35}}{294} & 0 \\
0 & 0 & -\frac{\sqrt{105}}{196} & 0 & -\frac{\sqrt{210}}{196} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & \frac{5\sqrt{21}}{147} & 0 & -\frac{5}{42} \\
\frac{5}{42} & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{56} & 0 & \frac{\sqrt{30}}{56} & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{35}}{294} & 0 & \frac{3\sqrt{70}}{196} & 0 & 0 & \frac{5\sqrt{6}}{56} & 0 & \frac{5\sqrt{14}}{392} & 0 & \frac{\sqrt{210}}{98} & 0 & 0 & 0 \\
-\frac{5\sqrt{21}}{147} & 0 & -\frac{5\sqrt{210}}{588} & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & \frac{5\sqrt{14}}{392} & 0 & -\frac{5\sqrt{70}}{392} & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 \\
0 & -\frac{5\sqrt{7}}{147} & 0 & -\frac{5\sqrt{14}}{588} & 0 & \frac{\sqrt{35}}{98} & \frac{\sqrt{30}}{56} & 0 & -\frac{5\sqrt{70}}{392} & 0 & -\frac{5\sqrt{42}}{196} & 0 & \frac{\sqrt{210}}{98} & 0 \\
-\frac{\sqrt{35}}{98} & 0 & \frac{5\sqrt{14}}{588} & 0 & \frac{5\sqrt{7}}{147} & 0 & 0 & \frac{\sqrt{210}}{98} & 0 & -\frac{5\sqrt{42}}{196} & 0 & -\frac{5\sqrt{70}}{392} & 0 & \frac{\sqrt{30}}{56} \\
0 & -\frac{\sqrt{105}}{98} & 0 & \frac{5\sqrt{210}}{588} & 0 & \frac{5\sqrt{21}}{147} & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & -\frac{5\sqrt{70}}{392} & 0 & \frac{5\sqrt{14}}{392} & 0 \\
0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & \frac{5\sqrt{35}}{294} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{98} & 0 & \frac{5\sqrt{14}}{392} & 0 & \frac{5\sqrt{6}}{56} \\
0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{56} & 0 & \frac{5\sqrt{6}}{56} & 0
\end{bmatrix}$$

$\boxed{\text{x514}}$ $\mathbb{M}_{3,2}^{(1,-1;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{210}i}{196} & 0 & \frac{\sqrt{105}i}{196} & 0 & 0 & -\frac{5i}{42} & 0 & -\frac{5\sqrt{21}i}{147} & 0 & \frac{\sqrt{35}i}{98} & 0 & 0 & 0 \\
\frac{\sqrt{210}i}{196} & 0 & \frac{\sqrt{21}i}{196} & 0 & \frac{\sqrt{42}i}{98} & 0 & 0 & \frac{5\sqrt{35}i}{294} & 0 & -\frac{5\sqrt{7}i}{147} & 0 & \frac{\sqrt{105}i}{98} & 0 & 0 \\
0 & -\frac{\sqrt{21}i}{196} & 0 & \frac{\sqrt{42}i}{98} & 0 & \frac{\sqrt{105}i}{196} & \frac{\sqrt{10}i}{28} & 0 & \frac{5\sqrt{210}i}{588} & 0 & \frac{5\sqrt{14}i}{588} & 0 & \frac{3\sqrt{70}i}{196} & 0 \\
-\frac{\sqrt{105}i}{196} & 0 & -\frac{\sqrt{42}i}{98} & 0 & \frac{\sqrt{21}i}{196} & 0 & 0 & \frac{3\sqrt{70}i}{196} & 0 & \frac{5\sqrt{14}i}{588} & 0 & \frac{5\sqrt{210}i}{588} & 0 & \frac{\sqrt{10}i}{28} \\
0 & -\frac{\sqrt{42}i}{98} & 0 & -\frac{\sqrt{21}i}{196} & 0 & -\frac{\sqrt{210}i}{196} & 0 & 0 & \frac{\sqrt{105}i}{98} & 0 & -\frac{5\sqrt{7}i}{147} & 0 & \frac{5\sqrt{35}i}{294} & 0 \\
0 & 0 & -\frac{\sqrt{105}i}{196} & 0 & \frac{\sqrt{210}i}{196} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{98} & 0 & -\frac{5\sqrt{21}i}{147} & 0 & -\frac{5i}{42} \\
\frac{5i}{42} & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{56} & 0 & -\frac{\sqrt{30}i}{56} & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{35}i}{294} & 0 & -\frac{3\sqrt{70}i}{196} & 0 & 0 & -\frac{5\sqrt{6}i}{56} & 0 & \frac{5\sqrt{14}i}{392} & 0 & -\frac{\sqrt{210}i}{98} & 0 & 0 & 0 \\
\frac{5\sqrt{21}i}{147} & 0 & -\frac{5\sqrt{210}i}{588} & 0 & -\frac{\sqrt{105}i}{98} & 0 & 0 & -\frac{5\sqrt{14}i}{392} & 0 & -\frac{5\sqrt{70}i}{392} & 0 & -\frac{5\sqrt{42}i}{196} & 0 & 0 \\
0 & \frac{5\sqrt{7}i}{147} & 0 & -\frac{5\sqrt{14}i}{588} & 0 & -\frac{\sqrt{35}i}{98} & \frac{\sqrt{30}i}{56} & 0 & \frac{5\sqrt{70}i}{392} & 0 & -\frac{5\sqrt{42}i}{196} & 0 & -\frac{\sqrt{210}i}{98} & 0 \\
-\frac{\sqrt{35}i}{98} & 0 & -\frac{5\sqrt{14}i}{588} & 0 & \frac{5\sqrt{7}i}{147} & 0 & 0 & \frac{\sqrt{210}i}{98} & 0 & \frac{5\sqrt{42}i}{196} & 0 & -\frac{5\sqrt{70}i}{392} & 0 & -\frac{\sqrt{30}i}{56} \\
0 & -\frac{\sqrt{105}i}{98} & 0 & -\frac{5\sqrt{210}i}{588} & 0 & \frac{5\sqrt{21}i}{147} & 0 & 0 & \frac{5\sqrt{42}i}{196} & 0 & \frac{5\sqrt{70}i}{392} & 0 & \frac{5\sqrt{14}i}{392} & 0 \\
0 & 0 & -\frac{3\sqrt{70}i}{196} & 0 & -\frac{5\sqrt{35}i}{294} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{98} & 0 & -\frac{5\sqrt{14}i}{392} & 0 & \frac{5\sqrt{6}i}{56} \\
0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & \frac{5i}{42} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{56} & 0 & -\frac{5\sqrt{6}i}{56} & 0
\end{bmatrix}$$

$\boxed{\text{x515}}$ $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & -\frac{5\sqrt{77}}{8624} & 0 & \frac{5\sqrt{154}}{3696} & 0 & -\frac{3\sqrt{385}}{1232} & \frac{\sqrt{330}}{1232} & 0 & -\frac{15\sqrt{770}}{8624} & 0 & \frac{5\sqrt{462}}{1232} & 0 & -\frac{3\sqrt{2310}}{1232} & 0 \\
-\frac{5\sqrt{77}}{8624} & 0 & \frac{5\sqrt{770}}{8624} & 0 & -\frac{5\sqrt{385}}{3696} & 0 & 0 & -\frac{23\sqrt{462}}{8624} & 0 & \frac{13\sqrt{2310}}{8624} & 0 & -\frac{3\sqrt{154}}{1232} & 0 & -\frac{3\sqrt{66}}{176} \\
0 & \frac{5\sqrt{770}}{8624} & 0 & -\frac{5\sqrt{385}}{4312} & 0 & \frac{5\sqrt{154}}{3696} & -\frac{\sqrt{33}}{88} & 0 & \frac{3\sqrt{77}}{392} & 0 & -\frac{\sqrt{1155}}{4312} & 0 & -\frac{3\sqrt{231}}{616} & 0 \\
\frac{5\sqrt{154}}{3696} & 0 & -\frac{5\sqrt{385}}{4312} & 0 & \frac{5\sqrt{770}}{8624} & 0 & 0 & \frac{3\sqrt{231}}{616} & 0 & \frac{\sqrt{1155}}{4312} & 0 & -\frac{3\sqrt{77}}{392} & 0 & \frac{\sqrt{33}}{88} \\
0 & -\frac{5\sqrt{385}}{3696} & 0 & \frac{5\sqrt{770}}{8624} & 0 & -\frac{5\sqrt{77}}{8624} & \frac{3\sqrt{66}}{176} & 0 & \frac{3\sqrt{154}}{1232} & 0 & -\frac{13\sqrt{2310}}{8624} & 0 & \frac{23\sqrt{462}}{8624} & 0 \\
-\frac{3\sqrt{385}}{1232} & 0 & \frac{5\sqrt{154}}{3696} & 0 & -\frac{5\sqrt{77}}{8624} & 0 & 0 & \frac{3\sqrt{2310}}{1232} & 0 & -\frac{5\sqrt{462}}{1232} & 0 & \frac{15\sqrt{770}}{8624} & 0 & -\frac{\sqrt{330}}{1232} \\
\frac{\sqrt{330}}{1232} & 0 & -\frac{\sqrt{33}}{88} & 0 & \frac{3\sqrt{66}}{176} & 0 & 0 & \frac{15\sqrt{55}}{1232} & 0 & -\frac{5\sqrt{11}}{88} & 0 & \frac{3\sqrt{165}}{176} & 0 & 0 \\
0 & -\frac{23\sqrt{462}}{8624} & 0 & \frac{3\sqrt{231}}{616} & 0 & \frac{3\sqrt{2310}}{1232} & \frac{15\sqrt{55}}{1232} & 0 & -\frac{5\sqrt{1155}}{1078} & 0 & \frac{5\sqrt{77}}{1232} & 0 & \frac{9\sqrt{385}}{616} & 0 \\
-\frac{15\sqrt{770}}{8624} & 0 & \frac{3\sqrt{77}}{392} & 0 & \frac{3\sqrt{154}}{1232} & 0 & 0 & -\frac{5\sqrt{1155}}{1078} & 0 & \frac{5\sqrt{231}}{8624} & 0 & \frac{5\sqrt{385}}{616} & 0 & \frac{3\sqrt{165}}{176} \\
0 & \frac{13\sqrt{2310}}{8624} & 0 & \frac{\sqrt{1155}}{4312} & 0 & -\frac{5\sqrt{462}}{1232} & -\frac{5\sqrt{11}}{88} & 0 & \frac{5\sqrt{231}}{8624} & 0 & \frac{15\sqrt{385}}{2156} & 0 & \frac{5\sqrt{77}}{1232} & 0 \\
\frac{5\sqrt{462}}{1232} & 0 & -\frac{\sqrt{1155}}{4312} & 0 & -\frac{13\sqrt{2310}}{8624} & 0 & 0 & \frac{5\sqrt{77}}{1232} & 0 & \frac{15\sqrt{385}}{2156} & 0 & \frac{5\sqrt{231}}{8624} & 0 & -\frac{5\sqrt{11}}{88} \\
0 & -\frac{3\sqrt{154}}{1232} & 0 & -\frac{3\sqrt{77}}{392} & 0 & \frac{15\sqrt{770}}{8624} & \frac{3\sqrt{165}}{176} & 0 & \frac{5\sqrt{385}}{616} & 0 & \frac{5\sqrt{231}}{8624} & 0 & -\frac{5\sqrt{1155}}{1078} & 0 \\
-\frac{3\sqrt{2310}}{1232} & 0 & -\frac{3\sqrt{231}}{616} & 0 & \frac{23\sqrt{462}}{8624} & 0 & 0 & \frac{9\sqrt{385}}{616} & 0 & \frac{5\sqrt{77}}{1232} & 0 & -\frac{5\sqrt{1155}}{1078} & 0 & \frac{15\sqrt{55}}{1232} \\
0 & -\frac{3\sqrt{66}}{176} & 0 & \frac{\sqrt{33}}{88} & 0 & -\frac{\sqrt{330}}{1232} & 0 & 0 & \frac{3\sqrt{165}}{176} & 0 & -\frac{5\sqrt{11}}{88} & 0 & \frac{15\sqrt{55}}{1232} & 0
\end{bmatrix}$$

$\boxed{\text{x516}}$ $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,1)$

$$= \begin{bmatrix}
0 & -\frac{5\sqrt{77}i}{8624} & 0 & -\frac{5\sqrt{154}i}{3696} & 0 & -\frac{3\sqrt{385}i}{1232} & -\frac{\sqrt{330}i}{1232} & 0 & -\frac{15\sqrt{770}i}{8624} & 0 & -\frac{5\sqrt{462}i}{1232} & 0 & -\frac{3\sqrt{2310}i}{1232} & 0 \\
\frac{5\sqrt{77}i}{8624} & 0 & \frac{5\sqrt{770}i}{8624} & 0 & \frac{5\sqrt{385}i}{3696} & 0 & 0 & \frac{23\sqrt{462}i}{8624} & 0 & \frac{13\sqrt{2310}i}{8624} & 0 & \frac{3\sqrt{154}i}{1232} & 0 & -\frac{3\sqrt{66}i}{176} \\
0 & -\frac{5\sqrt{770}i}{8624} & 0 & -\frac{5\sqrt{385}i}{4312} & 0 & -\frac{5\sqrt{154}i}{3696} & -\frac{\sqrt{33}i}{88} & 0 & -\frac{3\sqrt{77}i}{392} & 0 & -\frac{\sqrt{1155}i}{4312} & 0 & \frac{3\sqrt{231}i}{616} & 0 \\
\frac{5\sqrt{154}i}{3696} & 0 & \frac{5\sqrt{385}i}{4312} & 0 & \frac{5\sqrt{770}i}{8624} & 0 & 0 & \frac{3\sqrt{231}i}{616} & 0 & -\frac{\sqrt{1155}i}{4312} & 0 & -\frac{3\sqrt{77}i}{392} & 0 & -\frac{\sqrt{33}i}{88} \\
0 & -\frac{5\sqrt{385}i}{3696} & 0 & -\frac{5\sqrt{770}i}{8624} & 0 & -\frac{5\sqrt{77}i}{8624} & -\frac{3\sqrt{66}i}{176} & 0 & \frac{3\sqrt{154}i}{1232} & 0 & \frac{13\sqrt{2310}i}{8624} & 0 & \frac{23\sqrt{462}i}{8624} & 0 \\
\frac{3\sqrt{385}i}{1232} & 0 & \frac{5\sqrt{154}i}{3696} & 0 & \frac{5\sqrt{77}i}{8624} & 0 & 0 & -\frac{3\sqrt{2310}i}{1232} & 0 & -\frac{5\sqrt{462}i}{1232} & 0 & -\frac{15\sqrt{770}i}{8624} & 0 & -\frac{\sqrt{330}i}{1232} \\
\frac{\sqrt{330}i}{1232} & 0 & \frac{\sqrt{33}i}{88} & 0 & \frac{3\sqrt{66}i}{176} & 0 & 0 & \frac{15\sqrt{55}i}{1232} & 0 & \frac{5\sqrt{11}i}{88} & 0 & \frac{3\sqrt{165}i}{176} & 0 & 0 \\
0 & -\frac{23\sqrt{462}i}{8624} & 0 & -\frac{3\sqrt{231}i}{616} & 0 & \frac{3\sqrt{2310}i}{1232} & -\frac{15\sqrt{55}i}{1232} & 0 & -\frac{5\sqrt{1155}i}{1078} & 0 & -\frac{5\sqrt{77}i}{1232} & 0 & \frac{9\sqrt{385}i}{616} & 0 \\
\frac{15\sqrt{770}i}{8624} & 0 & \frac{3\sqrt{77}i}{392} & 0 & -\frac{3\sqrt{154}i}{1232} & 0 & 0 & \frac{5\sqrt{1155}i}{1078} & 0 & \frac{5\sqrt{231}i}{8624} & 0 & -\frac{5\sqrt{385}i}{616} & 0 & \frac{3\sqrt{165}i}{176} \\
0 & -\frac{13\sqrt{2310}i}{8624} & 0 & \frac{\sqrt{1155}i}{4312} & 0 & \frac{5\sqrt{462}i}{1232} & -\frac{5\sqrt{11}i}{88} & 0 & -\frac{5\sqrt{231}i}{8624} & 0 & \frac{15\sqrt{385}i}{2156} & 0 & -\frac{5\sqrt{77}i}{1232} & 0 \\
\frac{5\sqrt{462}i}{1232} & 0 & \frac{\sqrt{1155}i}{4312} & 0 & -\frac{13\sqrt{2310}i}{8624} & 0 & 0 & \frac{5\sqrt{77}i}{1232} & 0 & -\frac{15\sqrt{385}i}{2156} & 0 & \frac{5\sqrt{231}i}{8624} & 0 & \frac{5\sqrt{11}i}{88} \\
0 & -\frac{3\sqrt{154}i}{1232} & 0 & \frac{3\sqrt{77}i}{392} & 0 & \frac{15\sqrt{770}i}{8624} & -\frac{3\sqrt{165}i}{176} & 0 & \frac{5\sqrt{385}i}{616} & 0 & -\frac{5\sqrt{231}i}{8624} & 0 & -\frac{5\sqrt{1155}i}{1078} & 0 \\
\frac{3\sqrt{2310}i}{1232} & 0 & -\frac{3\sqrt{231}i}{616} & 0 & -\frac{23\sqrt{462}i}{8624} & 0 & 0 & -\frac{9\sqrt{385}i}{616} & 0 & \frac{5\sqrt{77}i}{1232} & 0 & \frac{5\sqrt{1155}i}{1078} & 0 & \frac{15\sqrt{55}i}{1232} \\
0 & \frac{3\sqrt{66}i}{176} & 0 & \frac{\sqrt{33}i}{88} & 0 & \frac{\sqrt{330}i}{1232} & 0 & 0 & -\frac{3\sqrt{165}i}{176} & 0 & -\frac{5\sqrt{11}i}{88} & 0 & -\frac{15\sqrt{55}i}{1232} & 0
\end{bmatrix}$$

$\boxed{\text{x517}}$ $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,2) = \begin{bmatrix}
0 & -\frac{\sqrt{55}}{1232} & 0 & -\frac{3\sqrt{110}}{1232} & 0 & -\frac{5\sqrt{11}}{1232} & \frac{\sqrt{462}}{1232} & 0 & -\frac{15\sqrt{22}}{1232} & 0 & -\frac{9\sqrt{330}}{1232} & 0 & -\frac{5\sqrt{66}}{1232} & 0 \\
-\frac{\sqrt{55}}{1232} & 0 & \frac{5\sqrt{22}}{1232} & 0 & \frac{15\sqrt{11}}{1232} & 0 & 0 & -\frac{23\sqrt{330}}{6160} & 0 & \frac{13\sqrt{66}}{1232} & 0 & \frac{27\sqrt{110}}{6160} & 0 & -\frac{\sqrt{2310}}{1232} \\
0 & \frac{5\sqrt{22}}{1232} & 0 & -\frac{5\sqrt{11}}{616} & 0 & -\frac{3\sqrt{110}}{1232} & \frac{9\sqrt{1155}}{3080} & 0 & \frac{3\sqrt{55}}{280} & 0 & -\frac{\sqrt{33}}{616} & 0 & \frac{27\sqrt{165}}{3080} & 0 \\
-\frac{3\sqrt{110}}{1232} & 0 & -\frac{5\sqrt{11}}{616} & 0 & \frac{5\sqrt{22}}{1232} & 0 & 0 & -\frac{27\sqrt{165}}{3080} & 0 & \frac{\sqrt{33}}{616} & 0 & -\frac{3\sqrt{55}}{280} & 0 & -\frac{9\sqrt{1155}}{3080} \\
0 & \frac{15\sqrt{11}}{1232} & 0 & \frac{5\sqrt{22}}{1232} & 0 & -\frac{\sqrt{55}}{1232} & \frac{\sqrt{2310}}{1232} & 0 & -\frac{27\sqrt{110}}{6160} & 0 & -\frac{13\sqrt{66}}{1232} & 0 & \frac{23\sqrt{330}}{6160} & 0 \\
-\frac{5\sqrt{11}}{1232} & 0 & -\frac{3\sqrt{110}}{1232} & 0 & -\frac{\sqrt{55}}{1232} & 0 & 0 & \frac{5\sqrt{66}}{1232} & 0 & \frac{9\sqrt{330}}{1232} & 0 & \frac{15\sqrt{22}}{1232} & 0 & -\frac{\sqrt{462}}{1232} \\
\frac{\sqrt{462}}{1232} & 0 & \frac{9\sqrt{1155}}{3080} & 0 & \frac{\sqrt{2310}}{1232} & 0 & 0 & \frac{15\sqrt{77}}{1232} & 0 & \frac{9\sqrt{385}}{616} & 0 & \frac{5\sqrt{231}}{1232} & 0 & 0 \\
0 & -\frac{23\sqrt{330}}{6160} & 0 & -\frac{27\sqrt{165}}{3080} & 0 & \frac{5\sqrt{66}}{1232} & \frac{15\sqrt{77}}{1232} & 0 & -\frac{5\sqrt{33}}{154} & 0 & -\frac{9\sqrt{55}}{1232} & 0 & \frac{15\sqrt{11}}{616} & 0 \\
-\frac{15\sqrt{22}}{1232} & 0 & \frac{3\sqrt{55}}{280} & 0 & -\frac{27\sqrt{110}}{6160} & 0 & 0 & -\frac{5\sqrt{33}}{154} & 0 & \frac{\sqrt{165}}{1232} & 0 & -\frac{45\sqrt{11}}{616} & 0 & \frac{5\sqrt{231}}{1232} \\
0 & \frac{13\sqrt{66}}{1232} & 0 & \frac{\sqrt{33}}{616} & 0 & \frac{9\sqrt{330}}{1232} & \frac{9\sqrt{385}}{616} & 0 & \frac{\sqrt{165}}{1232} & 0 & \frac{15\sqrt{11}}{308} & 0 & -\frac{9\sqrt{55}}{1232} & 0 \\
-\frac{9\sqrt{330}}{1232} & 0 & -\frac{\sqrt{33}}{616} & 0 & -\frac{13\sqrt{66}}{1232} & 0 & 0 & -\frac{9\sqrt{55}}{1232} & 0 & \frac{15\sqrt{11}}{308} & 0 & \frac{\sqrt{165}}{1232} & 0 & \frac{9\sqrt{385}}{616} \\
0 & \frac{27\sqrt{110}}{6160} & 0 & -\frac{3\sqrt{55}}{280} & 0 & \frac{15\sqrt{22}}{1232} & \frac{5\sqrt{231}}{1232} & 0 & -\frac{45\sqrt{11}}{616} & 0 & \frac{\sqrt{165}}{1232} & 0 & -\frac{5\sqrt{33}}{154} & 0 \\
-\frac{5\sqrt{66}}{1232} & 0 & \frac{27\sqrt{165}}{3080} & 0 & \frac{23\sqrt{330}}{6160} & 0 & 0 & \frac{15\sqrt{11}}{616} & 0 & -\frac{9\sqrt{55}}{1232} & 0 & -\frac{5\sqrt{33}}{154} & 0 & \frac{15\sqrt{77}}{1232} \\
0 & -\frac{\sqrt{2310}}{1232} & 0 & -\frac{9\sqrt{1155}}{3080} & 0 & -\frac{\sqrt{462}}{1232} & 0 & 0 & \frac{5\sqrt{231}}{1232} & 0 & \frac{9\sqrt{385}}{616} & 0 & \frac{15\sqrt{77}}{1232} & 0
\end{bmatrix}$

$\boxed{\text{x518}}$ $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,2)$

$$= \begin{bmatrix}
0 & -\frac{\sqrt{55}i}{1232} & 0 & \frac{3\sqrt{110}i}{1232} & 0 & -\frac{5\sqrt{11}i}{1232} & -\frac{\sqrt{462}i}{1232} & 0 & -\frac{15\sqrt{22}i}{1232} & 0 & \frac{9\sqrt{330}i}{1232} & 0 & -\frac{5\sqrt{66}i}{1232} & 0 \\
\frac{\sqrt{55}i}{1232} & 0 & \frac{5\sqrt{22}i}{1232} & 0 & -\frac{15\sqrt{11}i}{1232} & 0 & 0 & \frac{23\sqrt{330}i}{6160} & 0 & \frac{13\sqrt{66}i}{1232} & 0 & -\frac{27\sqrt{110}i}{6160} & 0 & -\frac{\sqrt{2310}i}{1232} \\
0 & -\frac{5\sqrt{22}i}{1232} & 0 & -\frac{5\sqrt{11}i}{616} & 0 & \frac{3\sqrt{110}i}{1232} & \frac{9\sqrt{1155}i}{3080} & 0 & -\frac{3\sqrt{55}i}{280} & 0 & -\frac{\sqrt{33}i}{616} & 0 & -\frac{27\sqrt{165}i}{3080} & 0 \\
-\frac{3\sqrt{110}i}{1232} & 0 & \frac{5\sqrt{11}i}{616} & 0 & \frac{5\sqrt{22}i}{1232} & 0 & 0 & -\frac{27\sqrt{165}i}{3080} & 0 & -\frac{\sqrt{33}i}{616} & 0 & -\frac{3\sqrt{55}i}{280} & 0 & \frac{9\sqrt{1155}i}{3080} \\
0 & \frac{15\sqrt{11}i}{1232} & 0 & -\frac{5\sqrt{22}i}{1232} & 0 & -\frac{\sqrt{55}i}{1232} & -\frac{\sqrt{2310}i}{1232} & 0 & -\frac{27\sqrt{110}i}{6160} & 0 & \frac{13\sqrt{66}i}{1232} & 0 & \frac{23\sqrt{330}i}{6160} & 0 \\
\frac{5\sqrt{11}i}{1232} & 0 & -\frac{3\sqrt{110}i}{1232} & 0 & \frac{\sqrt{55}i}{1232} & 0 & 0 & -\frac{5\sqrt{66}i}{1232} & 0 & \frac{9\sqrt{330}i}{1232} & 0 & -\frac{15\sqrt{22}i}{1232} & 0 & -\frac{\sqrt{462}i}{1232} \\
\frac{\sqrt{462}i}{1232} & 0 & -\frac{9\sqrt{1155}i}{3080} & 0 & \frac{\sqrt{2310}i}{1232} & 0 & 0 & \frac{15\sqrt{77}i}{1232} & 0 & -\frac{9\sqrt{385}i}{616} & 0 & \frac{5\sqrt{231}i}{1232} & 0 & 0 \\
0 & -\frac{23\sqrt{330}i}{6160} & 0 & \frac{27\sqrt{165}i}{3080} & 0 & \frac{5\sqrt{66}i}{1232} & -\frac{15\sqrt{77}i}{1232} & 0 & -\frac{5\sqrt{33}i}{154} & 0 & \frac{9\sqrt{55}i}{1232} & 0 & \frac{15\sqrt{11}i}{616} & 0 \\
\frac{15\sqrt{22}i}{1232} & 0 & \frac{3\sqrt{55}i}{280} & 0 & \frac{27\sqrt{110}i}{6160} & 0 & 0 & \frac{5\sqrt{33}i}{154} & 0 & \frac{\sqrt{165}i}{1232} & 0 & \frac{45\sqrt{11}i}{616} & 0 & \frac{5\sqrt{231}i}{1232} \\
0 & -\frac{13\sqrt{66}i}{1232} & 0 & \frac{\sqrt{33}i}{616} & 0 & -\frac{9\sqrt{330}i}{1232} & \frac{9\sqrt{385}i}{616} & 0 & -\frac{\sqrt{165}i}{1232} & 0 & \frac{15\sqrt{11}i}{308} & 0 & \frac{9\sqrt{55}i}{1232} & 0 \\
-\frac{9\sqrt{330}i}{1232} & 0 & \frac{\sqrt{33}i}{616} & 0 & -\frac{13\sqrt{66}i}{1232} & 0 & 0 & -\frac{9\sqrt{55}i}{1232} & 0 & -\frac{15\sqrt{11}i}{308} & 0 & \frac{\sqrt{165}i}{1232} & 0 & -\frac{9\sqrt{385}i}{616} \\
0 & \frac{27\sqrt{110}i}{6160} & 0 & \frac{3\sqrt{55}i}{280} & 0 & \frac{15\sqrt{22}i}{1232} & -\frac{5\sqrt{231}i}{1232} & 0 & -\frac{45\sqrt{11}i}{616} & 0 & -\frac{\sqrt{165}i}{1232} & 0 & -\frac{5\sqrt{33}i}{154} & 0 \\
\frac{5\sqrt{66}i}{1232} & 0 & \frac{27\sqrt{165}i}{3080} & 0 & -\frac{23\sqrt{330}i}{6160} & 0 & 0 & -\frac{15\sqrt{11}i}{616} & 0 & -\frac{9\sqrt{55}i}{1232} & 0 & \frac{5\sqrt{33}i}{154} & 0 & \frac{15\sqrt{77}i}{1232} \\
0 & \frac{\sqrt{2310}i}{1232} & 0 & -\frac{9\sqrt{1155}i}{3080} & 0 & \frac{\sqrt{462}i}{1232} & 0 & 0 & -\frac{5\sqrt{231}i}{1232} & 0 & \frac{9\sqrt{385}i}{616} & 0 & -\frac{15\sqrt{77}i}{1232} & 0
\end{bmatrix}$$

$\boxed{\text{x519}}$ $\mathbb{M}_{5,1}^{(1,-1;a)}(E_g,3) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{165}}{1848} & 0 & \frac{\sqrt{330}}{1848} & 0 & \frac{5\sqrt{33}}{616} & \frac{\sqrt{154}}{616} & 0 & -\frac{5\sqrt{66}}{616} & 0 & \frac{3\sqrt{110}}{616} & 0 & \frac{15\sqrt{22}}{616} & 0 \\
-\frac{\sqrt{165}}{1848} & 0 & \frac{5\sqrt{66}}{1848} & 0 & -\frac{5\sqrt{33}}{1848} & 0 & 0 & -\frac{23\sqrt{110}}{3080} & 0 & \frac{13\sqrt{22}}{616} & 0 & -\frac{3\sqrt{330}}{3080} & 0 & \frac{3\sqrt{770}}{616} \\
0 & \frac{5\sqrt{66}}{1848} & 0 & -\frac{5\sqrt{33}}{924} & 0 & \frac{\sqrt{330}}{1848} & -\frac{3\sqrt{385}}{1540} & 0 & \frac{\sqrt{165}}{140} & 0 & -\frac{\sqrt{11}}{308} & 0 & -\frac{9\sqrt{55}}{1540} & 0 \\
\frac{\sqrt{330}}{1848} & 0 & -\frac{5\sqrt{33}}{924} & 0 & \frac{5\sqrt{66}}{1848} & 0 & 0 & \frac{9\sqrt{55}}{1540} & 0 & \frac{\sqrt{11}}{308} & 0 & -\frac{\sqrt{165}}{140} & 0 & \frac{3\sqrt{385}}{1540} \\
0 & -\frac{5\sqrt{33}}{1848} & 0 & \frac{5\sqrt{66}}{1848} & 0 & -\frac{\sqrt{165}}{1848} & -\frac{3\sqrt{770}}{616} & 0 & \frac{3\sqrt{330}}{3080} & 0 & -\frac{13\sqrt{22}}{616} & 0 & \frac{23\sqrt{110}}{3080} & 0 \\
\frac{5\sqrt{33}}{616} & 0 & \frac{\sqrt{330}}{1848} & 0 & -\frac{\sqrt{165}}{1848} & 0 & 0 & -\frac{15\sqrt{22}}{616} & 0 & -\frac{3\sqrt{110}}{616} & 0 & \frac{5\sqrt{66}}{616} & 0 & -\frac{\sqrt{154}}{616} \\
\frac{\sqrt{154}}{616} & 0 & -\frac{3\sqrt{385}}{1540} & 0 & -\frac{3\sqrt{770}}{616} & 0 & 0 & \frac{5\sqrt{231}}{616} & 0 & -\frac{\sqrt{1155}}{308} & 0 & -\frac{15\sqrt{77}}{616} & 0 & 0 \\
0 & -\frac{23\sqrt{110}}{3080} & 0 & \frac{9\sqrt{55}}{1540} & 0 & -\frac{15\sqrt{22}}{616} & \frac{5\sqrt{231}}{616} & 0 & -\frac{5\sqrt{11}}{77} & 0 & \frac{\sqrt{165}}{616} & 0 & -\frac{15\sqrt{33}}{308} & 0 \\
-\frac{5\sqrt{66}}{616} & 0 & \frac{\sqrt{165}}{140} & 0 & \frac{3\sqrt{330}}{3080} & 0 & 0 & -\frac{5\sqrt{11}}{77} & 0 & \frac{\sqrt{55}}{616} & 0 & \frac{5\sqrt{33}}{308} & 0 & -\frac{15\sqrt{77}}{616} \\
0 & \frac{13\sqrt{22}}{616} & 0 & \frac{\sqrt{11}}{308} & 0 & -\frac{3\sqrt{110}}{616} & -\frac{\sqrt{1155}}{308} & 0 & \frac{\sqrt{55}}{616} & 0 & \frac{5\sqrt{33}}{154} & 0 & \frac{\sqrt{165}}{616} & 0 \\
\frac{3\sqrt{110}}{616} & 0 & -\frac{\sqrt{11}}{308} & 0 & -\frac{13\sqrt{22}}{616} & 0 & 0 & \frac{\sqrt{165}}{616} & 0 & \frac{5\sqrt{33}}{154} & 0 & \frac{\sqrt{55}}{616} & 0 & -\frac{\sqrt{1155}}{308} \\
0 & -\frac{3\sqrt{330}}{3080} & 0 & -\frac{\sqrt{165}}{140} & 0 & \frac{5\sqrt{66}}{616} & -\frac{15\sqrt{77}}{616} & 0 & \frac{5\sqrt{33}}{308} & 0 & \frac{\sqrt{55}}{616} & 0 & -\frac{5\sqrt{11}}{77} & 0 \\
\frac{15\sqrt{22}}{616} & 0 & -\frac{9\sqrt{55}}{1540} & 0 & \frac{23\sqrt{110}}{3080} & 0 & 0 & -\frac{15\sqrt{33}}{308} & 0 & \frac{\sqrt{165}}{616} & 0 & -\frac{5\sqrt{11}}{77} & 0 & \frac{5\sqrt{231}}{616} \\
0 & \frac{3\sqrt{770}}{616} & 0 & \frac{3\sqrt{385}}{1540} & 0 & -\frac{\sqrt{154}}{616} & 0 & 0 & -\frac{15\sqrt{77}}{616} & 0 & -\frac{\sqrt{1155}}{308} & 0 & \frac{5\sqrt{231}}{616} & 0
\end{bmatrix}$$

$\boxed{\text{x520}}$ $\mathbb{M}_{5,2}^{(1,-1;a)}(E_g,3) =$

$$\begin{bmatrix}
0 & -\frac{\sqrt{165}i}{1848} & 0 & -\frac{\sqrt{330}i}{1848} & 0 & \frac{5\sqrt{33}i}{616} & -\frac{\sqrt{154}i}{616} & 0 & -\frac{5\sqrt{66}i}{616} & 0 & -\frac{3\sqrt{110}i}{616} & 0 & \frac{15\sqrt{22}i}{616} & 0 \\
\frac{\sqrt{165}i}{1848} & 0 & \frac{5\sqrt{66}i}{1848} & 0 & \frac{5\sqrt{33}i}{1848} & 0 & 0 & \frac{23\sqrt{110}i}{3080} & 0 & \frac{13\sqrt{22}i}{616} & 0 & \frac{3\sqrt{330}i}{3080} & 0 & \frac{3\sqrt{770}i}{616} \\
0 & -\frac{5\sqrt{66}i}{1848} & 0 & -\frac{5\sqrt{33}i}{924} & 0 & -\frac{\sqrt{330}i}{1848} & -\frac{3\sqrt{385}i}{1540} & 0 & -\frac{\sqrt{165}i}{140} & 0 & -\frac{\sqrt{11}i}{308} & 0 & \frac{9\sqrt{55}i}{1540} & 0 \\
\frac{\sqrt{330}i}{1848} & 0 & \frac{5\sqrt{33}i}{924} & 0 & \frac{5\sqrt{66}i}{1848} & 0 & 0 & \frac{9\sqrt{55}i}{1540} & 0 & -\frac{\sqrt{11}i}{308} & 0 & -\frac{\sqrt{165}i}{140} & 0 & -\frac{3\sqrt{385}i}{1540} \\
0 & -\frac{5\sqrt{33}i}{1848} & 0 & -\frac{5\sqrt{66}i}{1848} & 0 & -\frac{\sqrt{165}i}{1848} & \frac{3\sqrt{770}i}{616} & 0 & \frac{3\sqrt{330}i}{3080} & 0 & \frac{13\sqrt{22}i}{616} & 0 & \frac{23\sqrt{110}i}{3080} & 0 \\
-\frac{5\sqrt{33}i}{616} & 0 & \frac{\sqrt{330}i}{1848} & 0 & \frac{\sqrt{165}i}{1848} & 0 & 0 & \frac{15\sqrt{22}i}{616} & 0 & -\frac{3\sqrt{110}i}{616} & 0 & -\frac{5\sqrt{66}i}{616} & 0 & -\frac{\sqrt{154}i}{616} \\
\frac{\sqrt{154}i}{616} & 0 & \frac{3\sqrt{385}i}{1540} & 0 & -\frac{3\sqrt{770}i}{616} & 0 & 0 & \frac{5\sqrt{231}i}{616} & 0 & \frac{\sqrt{1155}i}{308} & 0 & -\frac{15\sqrt{77}i}{616} & 0 & 0 \\
0 & -\frac{23\sqrt{110}i}{3080} & 0 & -\frac{9\sqrt{55}i}{1540} & 0 & -\frac{15\sqrt{22}i}{616} & -\frac{5\sqrt{231}i}{616} & 0 & -\frac{5\sqrt{11}i}{77} & 0 & -\frac{\sqrt{165}i}{616} & 0 & -\frac{15\sqrt{33}i}{308} & 0 \\
\frac{5\sqrt{66}i}{616} & 0 & \frac{\sqrt{165}i}{140} & 0 & -\frac{3\sqrt{330}i}{3080} & 0 & 0 & \frac{5\sqrt{11}i}{77} & 0 & \frac{\sqrt{55}i}{616} & 0 & -\frac{5\sqrt{33}i}{308} & 0 & -\frac{15\sqrt{77}i}{616} \\
0 & -\frac{13\sqrt{22}i}{616} & 0 & \frac{\sqrt{11}i}{308} & 0 & \frac{3\sqrt{110}i}{616} & -\frac{\sqrt{1155}i}{308} & 0 & -\frac{\sqrt{55}i}{616} & 0 & \frac{5\sqrt{33}i}{154} & 0 & -\frac{\sqrt{165}i}{616} & 0 \\
\frac{3\sqrt{110}i}{616} & 0 & \frac{\sqrt{11}i}{308} & 0 & -\frac{13\sqrt{22}i}{616} & 0 & 0 & \frac{\sqrt{165}i}{616} & 0 & -\frac{5\sqrt{33}i}{154} & 0 & \frac{\sqrt{55}i}{616} & 0 & \frac{\sqrt{1155}i}{308} \\
0 & -\frac{3\sqrt{330}i}{3080} & 0 & \frac{\sqrt{165}i}{140} & 0 & \frac{5\sqrt{66}i}{616} & \frac{15\sqrt{77}i}{616} & 0 & \frac{5\sqrt{33}i}{308} & 0 & -\frac{\sqrt{55}i}{616} & 0 & -\frac{5\sqrt{11}i}{77} & 0 \\
-\frac{15\sqrt{22}i}{616} & 0 & -\frac{9\sqrt{55}i}{1540} & 0 & -\frac{23\sqrt{110}i}{3080} & 0 & 0 & \frac{15\sqrt{33}i}{308} & 0 & \frac{\sqrt{165}i}{616} & 0 & \frac{5\sqrt{11}i}{77} & 0 & \frac{5\sqrt{231}i}{616} \\
0 & -\frac{3\sqrt{770}i}{616} & 0 & \frac{3\sqrt{385}i}{1540} & 0 & \frac{\sqrt{154}i}{616} & 0 & 0 & \frac{15\sqrt{77}i}{616} & 0 & -\frac{\sqrt{1155}i}{308} & 0 & -\frac{5\sqrt{231}i}{616} & 0
\end{bmatrix}$$

$\boxed{\text{x521}}$ $\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6006}}{27456} & 0 & -\frac{3\sqrt{30030}}{9152} & 0 & \frac{3\sqrt{2002}}{832} & 0 & -\frac{\sqrt{858}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6006}}{27456} & 0 & -\frac{35\sqrt{286}}{9152} & 0 & \frac{21\sqrt{4290}}{9152} & 0 & -\frac{7\sqrt{858}}{832} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{35\sqrt{286}}{9152} & 0 & \frac{35\sqrt{1430}}{9152} & 0 & -\frac{63\sqrt{858}}{9152} & 0 & \frac{3\sqrt{2002}}{832} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30030}}{9152} & 0 & \frac{35\sqrt{1430}}{9152} & 0 & -\frac{175\sqrt{858}}{27456} & 0 & \frac{21\sqrt{4290}}{9152} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{21\sqrt{4290}}{9152} & 0 & -\frac{175\sqrt{858}}{27456} & 0 & \frac{35\sqrt{1430}}{9152} & 0 & -\frac{3\sqrt{30030}}{9152} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2002}}{832} & 0 & -\frac{63\sqrt{858}}{9152} & 0 & \frac{35\sqrt{1430}}{9152} & 0 & -\frac{35\sqrt{286}}{9152} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{858}}{832} & 0 & \frac{21\sqrt{4290}}{9152} & 0 & -\frac{35\sqrt{286}}{9152} & 0 & \frac{5\sqrt{6006}}{27456} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{64} & 0 & \frac{3\sqrt{2002}}{832} & 0 & -\frac{3\sqrt{30030}}{9152} & 0 & \frac{5\sqrt{6006}}{27456} & 0
\end{bmatrix}$$

$\boxed{\text{x522}}$ $\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6006}i}{27456} & 0 & \frac{3\sqrt{30030}i}{9152} & 0 & \frac{3\sqrt{2002}i}{832} & 0 & \frac{\sqrt{858}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6006}i}{27456} & 0 & -\frac{35\sqrt{286}i}{9152} & 0 & -\frac{21\sqrt{4290}i}{9152} & 0 & -\frac{7\sqrt{858}i}{832} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{35\sqrt{286}i}{9152} & 0 & \frac{35\sqrt{1430}i}{9152} & 0 & \frac{63\sqrt{858}i}{9152} & 0 & \frac{3\sqrt{2002}i}{832} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30030}i}{9152} & 0 & -\frac{35\sqrt{1430}i}{9152} & 0 & -\frac{175\sqrt{858}i}{27456} & 0 & -\frac{21\sqrt{4290}i}{9152} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{21\sqrt{4290}i}{9152} & 0 & \frac{175\sqrt{858}i}{27456} & 0 & \frac{35\sqrt{1430}i}{9152} & 0 & \frac{3\sqrt{30030}i}{9152} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2002}i}{832} & 0 & -\frac{63\sqrt{858}i}{9152} & 0 & -\frac{35\sqrt{1430}i}{9152} & 0 & -\frac{35\sqrt{286}i}{9152} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{858}i}{832} & 0 & \frac{21\sqrt{4290}i}{9152} & 0 & \frac{35\sqrt{286}i}{9152} & 0 & \frac{5\sqrt{6006}i}{27456} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}i}{64} & 0 & -\frac{3\sqrt{2002}i}{832} & 0 & -\frac{3\sqrt{30030}i}{9152} & 0 & -\frac{5\sqrt{6006}i}{27456} & 0
\end{bmatrix}$$

$\boxed{\text{x523}}$ $\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{26}}{832} & 0 & \frac{\sqrt{130}}{832} & 0 & -\frac{25\sqrt{78}}{832} & 0 & -\frac{\sqrt{182}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{26}}{832} & 0 & -\frac{3\sqrt{546}}{832} & 0 & -\frac{\sqrt{910}}{832} & 0 & \frac{25\sqrt{182}}{832} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{546}}{832} & 0 & \frac{3\sqrt{2730}}{832} & 0 & \frac{3\sqrt{182}}{832} & 0 & -\frac{25\sqrt{78}}{832} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{130}}{832} & 0 & \frac{3\sqrt{2730}}{832} & 0 & -\frac{15\sqrt{182}}{832} & 0 & -\frac{\sqrt{910}}{832} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{910}}{832} & 0 & -\frac{15\sqrt{182}}{832} & 0 & \frac{3\sqrt{2730}}{832} & 0 & \frac{\sqrt{130}}{832} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{25\sqrt{78}}{832} & 0 & \frac{3\sqrt{182}}{832} & 0 & \frac{3\sqrt{2730}}{832} & 0 & -\frac{3\sqrt{546}}{832} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{25\sqrt{182}}{832} & 0 & -\frac{\sqrt{910}}{832} & 0 & -\frac{3\sqrt{546}}{832} & 0 & \frac{3\sqrt{26}}{832} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{182}}{64} & 0 & -\frac{25\sqrt{78}}{832} & 0 & \frac{\sqrt{130}}{832} & 0 & \frac{3\sqrt{26}}{832} & 0
\end{bmatrix}$$

$\boxed{\text{x524}}$ $\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{26}i}{832} & 0 & -\frac{\sqrt{130}i}{832} & 0 & -\frac{25\sqrt{78}i}{832} & 0 & \frac{\sqrt{182}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{26}i}{832} & 0 & -\frac{3\sqrt{546}i}{832} & 0 & \frac{\sqrt{910}i}{832} & 0 & \frac{25\sqrt{182}i}{832} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{546}i}{832} & 0 & \frac{3\sqrt{2730}i}{832} & 0 & -\frac{3\sqrt{182}i}{832} & 0 & -\frac{25\sqrt{78}i}{832} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{130}i}{832} & 0 & -\frac{3\sqrt{2730}i}{832} & 0 & -\frac{15\sqrt{182}i}{832} & 0 & \frac{\sqrt{910}i}{832} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{910}i}{832} & 0 & \frac{15\sqrt{182}i}{832} & 0 & \frac{3\sqrt{2730}i}{832} & 0 & -\frac{\sqrt{130}i}{832} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{25\sqrt{78}i}{832} & 0 & \frac{3\sqrt{182}i}{832} & 0 & -\frac{3\sqrt{2730}i}{832} & 0 & -\frac{3\sqrt{546}i}{832} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{25\sqrt{182}i}{832} & 0 & -\frac{\sqrt{910}i}{832} & 0 & \frac{3\sqrt{546}i}{832} & 0 & \frac{3\sqrt{26}i}{832} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{182}i}{64} & 0 & \frac{25\sqrt{78}i}{832} & 0 & \frac{\sqrt{130}i}{832} & 0 & -\frac{3\sqrt{26}i}{832} & 0
\end{bmatrix}$$

$\boxed{\text{x525}}$ $\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,3) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{64} & 0 & \frac{3\sqrt{5}}{64} & 0 & \frac{5\sqrt{3}}{64} & 0 & \frac{\sqrt{7}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{64} & 0 & -\frac{\sqrt{21}}{64} & 0 & -\frac{3\sqrt{35}}{64} & 0 & -\frac{5\sqrt{7}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{64} & 0 & \frac{\sqrt{105}}{64} & 0 & \frac{9\sqrt{7}}{64} & 0 & \frac{5\sqrt{3}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{64} & 0 & \frac{\sqrt{105}}{64} & 0 & -\frac{5\sqrt{7}}{64} & 0 & -\frac{3\sqrt{35}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{64} & 0 & -\frac{5\sqrt{7}}{64} & 0 & \frac{\sqrt{105}}{64} & 0 & \frac{3\sqrt{5}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{64} & 0 & \frac{9\sqrt{7}}{64} & 0 & \frac{\sqrt{105}}{64} & 0 & -\frac{\sqrt{21}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{64} & 0 & -\frac{3\sqrt{35}}{64} & 0 & -\frac{\sqrt{21}}{64} & 0 & \frac{1}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{64} & 0 & \frac{5\sqrt{3}}{64} & 0 & \frac{3\sqrt{5}}{64} & 0 & \frac{1}{64} & 0
\end{bmatrix}$

$\boxed{\text{x526}}$ $\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,3) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{64} & 0 & -\frac{3\sqrt{5}i}{64} & 0 & \frac{5\sqrt{3}i}{64} & 0 & -\frac{\sqrt{7}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{64} & 0 & -\frac{\sqrt{21}i}{64} & 0 & \frac{3\sqrt{35}i}{64} & 0 & -\frac{5\sqrt{7}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{64} & 0 & \frac{\sqrt{105}i}{64} & 0 & -\frac{9\sqrt{7}i}{64} & 0 & \frac{5\sqrt{3}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{64} & 0 & -\frac{\sqrt{105}i}{64} & 0 & -\frac{5\sqrt{7}i}{64} & 0 & \frac{3\sqrt{35}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{64} & 0 & \frac{5\sqrt{7}i}{64} & 0 & \frac{\sqrt{105}i}{64} & 0 & -\frac{3\sqrt{5}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{64} & 0 & \frac{9\sqrt{7}i}{64} & 0 & -\frac{\sqrt{105}i}{64} & 0 & -\frac{\sqrt{21}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{64} & 0 & -\frac{3\sqrt{35}i}{64} & 0 & \frac{\sqrt{21}i}{64} & 0 & \frac{i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{64} & 0 & -\frac{5\sqrt{3}i}{64} & 0 & \frac{3\sqrt{5}i}{64} & 0 & -\frac{i}{64} & 0
\end{bmatrix}$

$\boxed{\text{x527}}$ $\mathbb{M}_{7,1}^{(1,-1;a)}(E_g,4) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{143}}{9152} & 0 & -\frac{19\sqrt{715}}{9152} & 0 & \frac{\sqrt{429}}{832} & 0 & \frac{\sqrt{1001}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{143}}{9152} & 0 & -\frac{15\sqrt{3003}}{9152} & 0 & \frac{19\sqrt{5005}}{9152} & 0 & -\frac{\sqrt{1001}}{832} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{3003}}{9152} & 0 & \frac{15\sqrt{15015}}{9152} & 0 & -\frac{57\sqrt{1001}}{9152} & 0 & \frac{\sqrt{429}}{832} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{19\sqrt{715}}{9152} & 0 & \frac{15\sqrt{15015}}{9152} & 0 & -\frac{75\sqrt{1001}}{9152} & 0 & \frac{19\sqrt{5005}}{9152} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{19\sqrt{5005}}{9152} & 0 & -\frac{75\sqrt{1001}}{9152} & 0 & \frac{15\sqrt{15015}}{9152} & 0 & -\frac{19\sqrt{715}}{9152} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{429}}{832} & 0 & -\frac{57\sqrt{1001}}{9152} & 0 & \frac{15\sqrt{15015}}{9152} & 0 & -\frac{15\sqrt{3003}}{9152} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1001}}{832} & 0 & \frac{19\sqrt{5005}}{9152} & 0 & -\frac{15\sqrt{3003}}{9152} & 0 & \frac{15\sqrt{143}}{9152} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1001}}{64} & 0 & \frac{\sqrt{429}}{832} & 0 & -\frac{19\sqrt{715}}{9152} & 0 & \frac{15\sqrt{143}}{9152} & 0
\end{bmatrix}$

$\boxed{\text{x528}}$ $\mathbb{M}_{7,2}^{(1,-1;a)}(E_g,4) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{143}i}{9152} & 0 & \frac{19\sqrt{715}i}{9152} & 0 & \frac{\sqrt{429}i}{832} & 0 & -\frac{\sqrt{1001}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{143}i}{9152} & 0 & -\frac{15\sqrt{3003}i}{9152} & 0 & -\frac{19\sqrt{5005}i}{9152} & 0 & -\frac{\sqrt{1001}i}{832} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{3003}i}{9152} & 0 & \frac{15\sqrt{15015}i}{9152} & 0 & \frac{57\sqrt{1001}i}{9152} & 0 & \frac{\sqrt{429}i}{832} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{19\sqrt{715}i}{9152} & 0 & -\frac{15\sqrt{15015}i}{9152} & 0 & -\frac{75\sqrt{1001}i}{9152} & 0 & -\frac{19\sqrt{5005}i}{9152} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{19\sqrt{5005}i}{9152} & 0 & \frac{75\sqrt{1001}i}{9152} & 0 & \frac{15\sqrt{15015}i}{9152} & 0 & \frac{19\sqrt{715}i}{9152} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{429}i}{832} & 0 & -\frac{57\sqrt{1001}i}{9152} & 0 & -\frac{15\sqrt{15015}i}{9152} & 0 & -\frac{15\sqrt{3003}i}{9152} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1001}i}{832} & 0 & \frac{19\sqrt{5005}i}{9152} & 0 & \frac{15\sqrt{3003}i}{9152} & 0 & \frac{15\sqrt{143}i}{9152} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1001}i}{64} & 0 & -\frac{\sqrt{429}i}{832} & 0 & -\frac{19\sqrt{715}i}{9152} & 0 & -\frac{15\sqrt{143}i}{9152} & 0
\end{bmatrix}$

$\boxed{\text{x529}}$ $\mathbb{M}_5^{(1,1;a)}(A_{1g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & -\frac{3\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{308} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}i}{924} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3003}i}{462} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3003}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{3003}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1001}i}{2002} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{858}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}i}{2002} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}i}{2002} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}i}{2002} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}i}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{1430}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}i}{2002} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{858}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}i}{2002} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3003}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1001}i}{2002} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x530}}$ $\mathbb{M}_1^{(1,1;a)}(A_{2g}) =$

$$\begin{bmatrix}
\frac{2\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{6\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{2\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{2\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{6\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{294} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{294} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{294} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{105}}{294} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42}
\end{bmatrix}$$

$\boxed{\text{x531}}$ $\mathbb{M}_3^{(1,1;a)}(A_{2g}) = \begin{bmatrix} -\frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{539} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{77} \end{bmatrix}$

$\boxed{\text{x532}}$ $\mathbb{M}_5^{(1,1;a)}(A_{2g},1)$

$$= \begin{bmatrix}
\frac{\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6006}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{10\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{10\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{5\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6006}}{6468} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1001}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{6006}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{1001}}{14014} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{1001}}{14014} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{1001}}{14014} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{1001}}{14014} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{1001}}{14014} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6006}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{1001}}{14014} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1001}}{2002}
\end{bmatrix}$$

$\boxed{\text{x533}}$ $\mathbb{M}_5^{(1,1;a)}(A_{2g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & \frac{3\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{308} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{924} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3003}}{462} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3003}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{3\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{308} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{3003}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}}{2002} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}}{2002} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}}{2002} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1001}}{2002} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1001}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{1430}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2145}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{858}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2145}}{2002} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{3003}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1001}}{2002} & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x534}}$ $\mathbb{M}_3^{(1,1;a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & \frac{5\sqrt{462}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{2310}i}{294} & 0 & 0 & -\frac{\sqrt{22}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{196} & 0 & 0 & 0 \\
-\frac{5\sqrt{462}i}{294} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{294} & 0 & 0 & -\frac{\sqrt{77}i}{196} & 0 & 0 & 0 & -\frac{\sqrt{231}i}{196} & 0 & 0 \\
0 & -\frac{\sqrt{2310}i}{294} & 0 & 0 & 0 & -\frac{5\sqrt{462}i}{294} & 0 & 0 & \frac{\sqrt{231}i}{196} & 0 & 0 & 0 & \frac{\sqrt{77}i}{196} & 0 \\
0 & 0 & \frac{\sqrt{2310}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{196} & 0 & 0 & 0 & \frac{\sqrt{22}i}{28} \\
0 & 0 & 0 & \frac{5\sqrt{462}i}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{98} & 0 & 0 & 0 \\
0 & \frac{\sqrt{22}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{77}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}i}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{231}i}{196} & 0 & 0 & \frac{\sqrt{55}i}{77} & 0 & 0 & 0 & -\frac{2\sqrt{77}i}{539} & 0 & 0 & 0 \\
\frac{\sqrt{154}i}{98} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{196} & 0 & 0 & \frac{3\sqrt{231}i}{539} & 0 & 0 & 0 & \frac{2\sqrt{77}i}{539} & 0 & 0 \\
0 & \frac{\sqrt{770}i}{196} & 0 & 0 & 0 & -\frac{\sqrt{154}i}{98} & 0 & 0 & \frac{2\sqrt{77}i}{539} & 0 & 0 & 0 & \frac{3\sqrt{231}i}{539} & 0 \\
0 & 0 & \frac{\sqrt{231}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{77}i}{539} & 0 & 0 & 0 & \frac{\sqrt{55}i}{77} \\
0 & 0 & 0 & -\frac{\sqrt{77}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}i}{539} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{77} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x535}}$ $\mathbb{M}_5^{(1,1;a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{858}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{286}i}{924} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{4290}i}{154} & 0 & 0 & -\frac{\sqrt{2002}i}{924} & 0 & 0 & 0 & \frac{\sqrt{1430}i}{462} & 0 & 0 & 0 \\
-\frac{\sqrt{858}i}{154} & 0 & 0 & 0 & \frac{\sqrt{4290}i}{154} & 0 & 0 & \frac{2\sqrt{143}i}{231} & 0 & 0 & 0 & \frac{\sqrt{429}i}{462} & 0 & 0 \\
0 & \frac{\sqrt{4290}i}{154} & 0 & 0 & 0 & -\frac{\sqrt{858}i}{154} & 0 & 0 & -\frac{\sqrt{429}i}{462} & 0 & 0 & 0 & -\frac{2\sqrt{143}i}{231} & 0 \\
0 & 0 & -\frac{\sqrt{4290}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{462} & 0 & 0 & 0 & \frac{\sqrt{2002}i}{924} \\
0 & 0 & 0 & \frac{\sqrt{858}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{286}i}{924} & 0 & 0 & 0 \\
0 & \frac{\sqrt{2002}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}i}{2002} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{143}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{429}i}{2002} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{429}i}{462} & 0 & 0 & \frac{\sqrt{5005}i}{2002} & 0 & 0 & 0 & \frac{2\sqrt{143}i}{1001} & 0 & 0 & 0 \\
\frac{5\sqrt{286}i}{924} & 0 & 0 & 0 & \frac{\sqrt{1430}i}{462} & 0 & 0 & -\frac{3\sqrt{429}i}{2002} & 0 & 0 & 0 & -\frac{2\sqrt{143}i}{1001} & 0 & 0 \\
0 & -\frac{\sqrt{1430}i}{462} & 0 & 0 & 0 & -\frac{5\sqrt{286}i}{924} & 0 & 0 & -\frac{2\sqrt{143}i}{1001} & 0 & 0 & 0 & -\frac{3\sqrt{429}i}{2002} & 0 \\
0 & 0 & -\frac{\sqrt{429}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{143}i}{1001} & 0 & 0 & 0 & \frac{\sqrt{5005}i}{2002} \\
0 & 0 & 0 & \frac{2\sqrt{143}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{429}i}{2002} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{2002}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}i}{2002} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x536}}$ $\mathbb{M}_3^{(1,1;a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & -\frac{5\sqrt{462}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{98} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2310}}{294} & 0 & 0 & -\frac{\sqrt{22}}{28} & 0 & 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 \\
-\frac{5\sqrt{462}}{294} & 0 & 0 & 0 & \frac{\sqrt{2310}}{294} & 0 & 0 & -\frac{\sqrt{77}}{196} & 0 & 0 & 0 & \frac{\sqrt{231}}{196} & 0 & 0 \\
0 & -\frac{\sqrt{2310}}{294} & 0 & 0 & 0 & \frac{5\sqrt{462}}{294} & 0 & 0 & \frac{\sqrt{231}}{196} & 0 & 0 & 0 & -\frac{\sqrt{77}}{196} & 0 \\
0 & 0 & \frac{\sqrt{2310}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 & -\frac{\sqrt{22}}{28} \\
0 & 0 & 0 & \frac{5\sqrt{462}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{98} & 0 & 0 & 0 \\
0 & -\frac{\sqrt{22}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{77} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{231}}{539} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{231}}{196} & 0 & 0 & \frac{\sqrt{55}}{77} & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 & 0 \\
\frac{\sqrt{154}}{98} & 0 & 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & \frac{3\sqrt{231}}{539} & 0 & 0 & 0 & -\frac{2\sqrt{77}}{539} & 0 & 0 \\
0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 & \frac{\sqrt{154}}{98} & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 & 0 & -\frac{3\sqrt{231}}{539} & 0 \\
0 & 0 & \frac{\sqrt{231}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{77}}{539} & 0 & 0 & 0 & -\frac{\sqrt{55}}{77} \\
0 & 0 & 0 & -\frac{\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{231}}{539} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{77} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x537}}$ $\mathbb{M}_5^{(1,1;a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{858}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{286}}{924} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{4290}}{154} & 0 & 0 & \frac{\sqrt{2002}}{924} & 0 & 0 & 0 & \frac{\sqrt{1430}}{462} & 0 & 0 & 0 \\
\frac{\sqrt{858}}{154} & 0 & 0 & 0 & \frac{\sqrt{4290}}{154} & 0 & 0 & -\frac{2\sqrt{143}}{231} & 0 & 0 & 0 & \frac{\sqrt{429}}{462} & 0 & 0 \\
0 & -\frac{\sqrt{4290}}{154} & 0 & 0 & 0 & -\frac{\sqrt{858}}{154} & 0 & 0 & \frac{\sqrt{429}}{462} & 0 & 0 & 0 & -\frac{2\sqrt{143}}{231} & 0 \\
0 & 0 & \frac{\sqrt{4290}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}}{462} & 0 & 0 & 0 & \frac{\sqrt{2002}}{924} \\
0 & 0 & 0 & -\frac{\sqrt{858}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{286}}{924} & 0 & 0 & 0 \\
0 & \frac{\sqrt{2002}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}}{2002} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{2\sqrt{143}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{429}}{2002} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{429}}{462} & 0 & 0 & -\frac{\sqrt{5005}}{2002} & 0 & 0 & 0 & \frac{2\sqrt{143}}{1001} & 0 & 0 & 0 \\
-\frac{5\sqrt{286}}{924} & 0 & 0 & 0 & \frac{\sqrt{1430}}{462} & 0 & 0 & \frac{3\sqrt{429}}{2002} & 0 & 0 & 0 & -\frac{2\sqrt{143}}{1001} & 0 & 0 \\
0 & \frac{\sqrt{1430}}{462} & 0 & 0 & 0 & -\frac{5\sqrt{286}}{924} & 0 & 0 & \frac{2\sqrt{143}}{1001} & 0 & 0 & 0 & -\frac{3\sqrt{429}}{2002} & 0 \\
0 & 0 & \frac{\sqrt{429}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{143}}{1001} & 0 & 0 & 0 & \frac{\sqrt{5005}}{2002} \\
0 & 0 & 0 & -\frac{2\sqrt{143}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{429}}{2002} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{2002}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5005}}{2002} & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x538}}$ $\mathbb{M}_{1,1}^{(1,1;a)}(E_g) =$

$$\begin{bmatrix}
0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & -\frac{\sqrt{210}}{392} & 0 & 0 & 0 & 0 & 0 \\
\frac{2\sqrt{21}}{49} & 0 & \frac{4\sqrt{210}}{245} & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{392} & 0 & -\frac{3\sqrt{70}}{392} & 0 & 0 & 0 & 0 \\
0 & \frac{4\sqrt{210}}{245} & 0 & \frac{6\sqrt{105}}{245} & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{196} & 0 & -\frac{3\sqrt{35}}{196} & 0 & 0 & 0 \\
0 & 0 & \frac{6\sqrt{105}}{245} & 0 & \frac{4\sqrt{210}}{245} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{196} & 0 & -\frac{5\sqrt{21}}{196} & 0 & 0 \\
0 & 0 & 0 & \frac{4\sqrt{210}}{245} & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{392} & 0 & -\frac{15\sqrt{14}}{392} & 0 \\
0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{392} & 0 & -\frac{3\sqrt{10}}{56} \\
\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{15\sqrt{14}}{392} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{210}}{392} & 0 & \frac{5\sqrt{21}}{196} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{70}}{392} & 0 & \frac{3\sqrt{35}}{196} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & -\frac{2\sqrt{105}}{147} & 0 & 0 & 0 \\
0 & 0 & -\frac{3\sqrt{35}}{196} & 0 & \frac{3\sqrt{70}}{392} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{147} & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 \\
0 & 0 & 0 & -\frac{5\sqrt{21}}{196} & 0 & \frac{\sqrt{210}}{392} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & -\frac{\sqrt{35}}{49} & 0 \\
0 & 0 & 0 & 0 & -\frac{15\sqrt{14}}{392} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & -\frac{\sqrt{15}}{42} \\
0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0
\end{bmatrix}$$

$\boxed{\text{x539}}$ $\mathbb{M}_{1,2}^{(1,1;a)}(E_g) =$

$$\begin{bmatrix}
0 & \frac{2\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{56} & 0 & -\frac{\sqrt{210}i}{392} & 0 & 0 & 0 & 0 & 0 \\
-\frac{2\sqrt{21}i}{49} & 0 & \frac{4\sqrt{210}i}{245} & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}i}{392} & 0 & -\frac{3\sqrt{70}i}{392} & 0 & 0 & 0 & 0 \\
0 & -\frac{4\sqrt{210}i}{245} & 0 & \frac{6\sqrt{105}i}{245} & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{196} & 0 & -\frac{3\sqrt{35}i}{196} & 0 & 0 & 0 \\
0 & 0 & -\frac{6\sqrt{105}i}{245} & 0 & \frac{4\sqrt{210}i}{245} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{196} & 0 & -\frac{5\sqrt{21}i}{196} & 0 & 0 \\
0 & 0 & 0 & -\frac{4\sqrt{210}i}{245} & 0 & \frac{2\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{392} & 0 & -\frac{15\sqrt{14}i}{392} & 0 \\
0 & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{392} & 0 & -\frac{3\sqrt{10}i}{56} \\
\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{15\sqrt{14}i}{392} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & -\frac{\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{210}i}{392} & 0 & \frac{5\sqrt{21}i}{196} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{49} & 0 & -\frac{5\sqrt{7}i}{98} & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{70}i}{392} & 0 & \frac{3\sqrt{35}i}{196} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{98} & 0 & -\frac{2\sqrt{105}i}{147} & 0 & 0 & 0 \\
0 & 0 & \frac{3\sqrt{35}i}{196} & 0 & \frac{3\sqrt{70}i}{392} & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}i}{147} & 0 & -\frac{5\sqrt{7}i}{98} & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{21}i}{196} & 0 & \frac{\sqrt{210}i}{392} & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{98} & 0 & -\frac{\sqrt{35}i}{49} & 0 \\
0 & 0 & 0 & 0 & \frac{15\sqrt{14}i}{392} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{49} & 0 & -\frac{\sqrt{15}i}{42} \\
0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0
\end{bmatrix}$$

$\boxed{\text{x540}}$ $\mathbb{M}_{3,1}^{(1,1;a)}(E_g, 1) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{385}}{98} & 0 & -\frac{5\sqrt{770}}{588} & 0 & 0 & \frac{\sqrt{66}}{112} & 0 & -\frac{3\sqrt{154}}{392} & 0 & \frac{\sqrt{2310}}{784} & 0 & 0 & 0 \\
\frac{\sqrt{385}}{98} & 0 & -\frac{\sqrt{154}}{196} & 0 & -\frac{5\sqrt{77}}{147} & 0 & 0 & -\frac{\sqrt{2310}}{784} & 0 & -\frac{\sqrt{462}}{392} & 0 & \frac{3\sqrt{770}}{784} & 0 & 0 \\
0 & -\frac{\sqrt{154}}{196} & 0 & -\frac{\sqrt{77}}{49} & 0 & -\frac{5\sqrt{770}}{588} & -\frac{\sqrt{165}}{112} & 0 & -\frac{3\sqrt{385}}{784} & 0 & \frac{\sqrt{231}}{784} & 0 & \frac{3\sqrt{1155}}{784} & 0 \\
-\frac{5\sqrt{770}}{588} & 0 & -\frac{\sqrt{77}}{49} & 0 & -\frac{\sqrt{154}}{196} & 0 & 0 & -\frac{3\sqrt{1155}}{784} & 0 & -\frac{\sqrt{231}}{784} & 0 & \frac{3\sqrt{385}}{784} & 0 & \frac{\sqrt{165}}{112} \\
0 & -\frac{5\sqrt{77}}{147} & 0 & -\frac{\sqrt{154}}{196} & 0 & \frac{\sqrt{385}}{98} & 0 & 0 & -\frac{3\sqrt{770}}{784} & 0 & \frac{\sqrt{462}}{392} & 0 & \frac{\sqrt{2310}}{784} & 0 \\
0 & 0 & -\frac{5\sqrt{770}}{588} & 0 & \frac{\sqrt{385}}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{784} & 0 & \frac{3\sqrt{154}}{392} & 0 & -\frac{\sqrt{66}}{112} \\
\frac{\sqrt{66}}{112} & 0 & -\frac{\sqrt{165}}{112} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}}{154} & 0 & \frac{\sqrt{55}}{154} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2310}}{784} & 0 & -\frac{3\sqrt{1155}}{784} & 0 & 0 & -\frac{3\sqrt{11}}{154} & 0 & -\frac{\sqrt{231}}{1078} & 0 & \frac{2\sqrt{385}}{539} & 0 & 0 & 0 \\
-\frac{3\sqrt{154}}{392} & 0 & -\frac{3\sqrt{385}}{784} & 0 & -\frac{3\sqrt{770}}{784} & 0 & 0 & -\frac{\sqrt{231}}{1078} & 0 & \frac{\sqrt{1155}}{1078} & 0 & \frac{5\sqrt{77}}{539} & 0 & 0 \\
0 & -\frac{\sqrt{462}}{392} & 0 & -\frac{\sqrt{231}}{784} & 0 & -\frac{\sqrt{2310}}{784} & \frac{\sqrt{55}}{154} & 0 & \frac{\sqrt{1155}}{1078} & 0 & \frac{3\sqrt{77}}{539} & 0 & \frac{2\sqrt{385}}{539} & 0 \\
\frac{\sqrt{2310}}{784} & 0 & \frac{\sqrt{231}}{784} & 0 & \frac{\sqrt{462}}{392} & 0 & 0 & \frac{2\sqrt{385}}{539} & 0 & \frac{3\sqrt{77}}{539} & 0 & \frac{\sqrt{1155}}{1078} & 0 & \frac{\sqrt{55}}{154} \\
0 & \frac{3\sqrt{770}}{784} & 0 & \frac{3\sqrt{385}}{784} & 0 & \frac{3\sqrt{154}}{392} & 0 & 0 & \frac{5\sqrt{77}}{539} & 0 & \frac{\sqrt{1155}}{1078} & 0 & -\frac{\sqrt{231}}{1078} & 0 \\
0 & 0 & \frac{3\sqrt{1155}}{784} & 0 & \frac{\sqrt{2310}}{784} & 0 & 0 & 0 & 0 & \frac{2\sqrt{385}}{539} & 0 & -\frac{\sqrt{231}}{1078} & 0 & -\frac{3\sqrt{11}}{154} \\
0 & 0 & 0 & \frac{\sqrt{165}}{112} & 0 & -\frac{\sqrt{66}}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{154} & 0 & -\frac{3\sqrt{11}}{154} & 0
\end{bmatrix}$$

$\boxed{\text{x541}}$ $\mathbb{M}_{3,2}^{(1,1;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{385}i}{98} & 0 & \frac{5\sqrt{770}i}{588} & 0 & 0 & -\frac{\sqrt{66}i}{112} & 0 & -\frac{3\sqrt{154}i}{392} & 0 & -\frac{\sqrt{2310}i}{784} & 0 & 0 & 0 \\
-\frac{\sqrt{385}i}{98} & 0 & -\frac{\sqrt{154}i}{196} & 0 & \frac{5\sqrt{77}i}{147} & 0 & 0 & \frac{\sqrt{2310}i}{784} & 0 & -\frac{\sqrt{462}i}{392} & 0 & -\frac{3\sqrt{770}i}{784} & 0 & 0 \\
0 & \frac{\sqrt{154}i}{196} & 0 & -\frac{\sqrt{77}i}{49} & 0 & \frac{5\sqrt{770}i}{588} & -\frac{\sqrt{165}i}{112} & 0 & \frac{3\sqrt{385}i}{784} & 0 & \frac{\sqrt{231}i}{784} & 0 & -\frac{3\sqrt{1155}i}{784} & 0 \\
-\frac{5\sqrt{770}i}{588} & 0 & \frac{\sqrt{77}i}{49} & 0 & -\frac{\sqrt{154}i}{196} & 0 & 0 & -\frac{3\sqrt{1155}i}{784} & 0 & \frac{\sqrt{231}i}{784} & 0 & \frac{3\sqrt{385}i}{784} & 0 & -\frac{\sqrt{165}i}{112} \\
0 & -\frac{5\sqrt{77}i}{147} & 0 & \frac{\sqrt{154}i}{196} & 0 & \frac{\sqrt{385}i}{98} & 0 & 0 & -\frac{3\sqrt{770}i}{784} & 0 & -\frac{\sqrt{462}i}{392} & 0 & \frac{\sqrt{2310}i}{784} & 0 \\
0 & 0 & -\frac{5\sqrt{770}i}{588} & 0 & -\frac{\sqrt{385}i}{98} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{784} & 0 & -\frac{3\sqrt{154}i}{392} & 0 & -\frac{\sqrt{66}i}{112} \\
\frac{\sqrt{66}i}{112} & 0 & \frac{\sqrt{165}i}{112} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{11}i}{154} & 0 & -\frac{\sqrt{55}i}{154} & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2310}i}{784} & 0 & \frac{3\sqrt{1155}i}{784} & 0 & 0 & \frac{3\sqrt{11}i}{154} & 0 & -\frac{\sqrt{231}i}{1078} & 0 & -\frac{2\sqrt{385}i}{539} & 0 & 0 & 0 \\
\frac{3\sqrt{154}i}{392} & 0 & -\frac{3\sqrt{385}i}{784} & 0 & \frac{3\sqrt{770}i}{784} & 0 & 0 & \frac{\sqrt{231}i}{1078} & 0 & \frac{\sqrt{1155}i}{1078} & 0 & -\frac{5\sqrt{77}i}{539} & 0 & 0 \\
0 & \frac{\sqrt{462}i}{392} & 0 & -\frac{\sqrt{231}i}{784} & 0 & \frac{\sqrt{2310}i}{784} & \frac{\sqrt{55}i}{154} & 0 & -\frac{\sqrt{1155}i}{1078} & 0 & \frac{3\sqrt{77}i}{539} & 0 & -\frac{2\sqrt{385}i}{539} & 0 \\
\frac{\sqrt{2310}i}{784} & 0 & -\frac{\sqrt{231}i}{784} & 0 & \frac{\sqrt{462}i}{392} & 0 & 0 & \frac{2\sqrt{385}i}{539} & 0 & -\frac{3\sqrt{77}i}{539} & 0 & \frac{\sqrt{1155}i}{1078} & 0 & -\frac{\sqrt{55}i}{154} \\
0 & \frac{3\sqrt{770}i}{784} & 0 & -\frac{3\sqrt{385}i}{784} & 0 & \frac{3\sqrt{154}i}{392} & 0 & 0 & \frac{5\sqrt{77}i}{539} & 0 & -\frac{\sqrt{1155}i}{1078} & 0 & -\frac{\sqrt{231}i}{1078} & 0 \\
0 & 0 & \frac{3\sqrt{1155}i}{784} & 0 & -\frac{\sqrt{2310}i}{784} & 0 & 0 & 0 & 0 & \frac{2\sqrt{385}i}{539} & 0 & \frac{\sqrt{231}i}{1078} & 0 & -\frac{3\sqrt{11}i}{154} \\
0 & 0 & 0 & \frac{\sqrt{165}i}{112} & 0 & \frac{\sqrt{66}i}{112} & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{154} & 0 & \frac{3\sqrt{11}i}{154} & 0
\end{bmatrix}$$

$\boxed{\text{x542}}$ $\mathbb{M}_{3,1}^{(1,1;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & \frac{5\sqrt{231}}{294} & 0 & \frac{5\sqrt{462}}{588} & 0 & 0 & \frac{\sqrt{110}}{112} & 0 & -\frac{\sqrt{2310}}{392} & 0 & -\frac{3\sqrt{154}}{784} & 0 & 0 & 0 \\
\frac{5\sqrt{231}}{294} & 0 & -\frac{\sqrt{2310}}{588} & 0 & \frac{\sqrt{1155}}{147} & 0 & 0 & -\frac{5\sqrt{154}}{784} & 0 & -\frac{\sqrt{770}}{392} & 0 & -\frac{3\sqrt{462}}{784} & 0 & 0 \\
0 & -\frac{\sqrt{2310}}{588} & 0 & -\frac{\sqrt{1155}}{147} & 0 & \frac{5\sqrt{462}}{588} & \frac{3\sqrt{11}}{112} & 0 & -\frac{5\sqrt{231}}{784} & 0 & \frac{\sqrt{385}}{784} & 0 & -\frac{9\sqrt{77}}{784} & 0 \\
\frac{5\sqrt{462}}{588} & 0 & -\frac{\sqrt{1155}}{147} & 0 & -\frac{\sqrt{2310}}{588} & 0 & 0 & \frac{9\sqrt{77}}{784} & 0 & -\frac{\sqrt{385}}{784} & 0 & \frac{5\sqrt{231}}{784} & 0 & -\frac{3\sqrt{11}}{112} \\
0 & \frac{\sqrt{1155}}{147} & 0 & -\frac{\sqrt{2310}}{588} & 0 & \frac{5\sqrt{231}}{294} & 0 & 0 & \frac{3\sqrt{462}}{784} & 0 & \frac{\sqrt{770}}{392} & 0 & \frac{5\sqrt{154}}{784} & 0 \\
0 & 0 & \frac{5\sqrt{462}}{588} & 0 & \frac{5\sqrt{231}}{294} & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{784} & 0 & \frac{\sqrt{2310}}{392} & 0 & -\frac{\sqrt{110}}{112} \\
\frac{\sqrt{110}}{112} & 0 & \frac{3\sqrt{11}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{154} & 0 & -\frac{\sqrt{33}}{154} & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{154}}{784} & 0 & \frac{9\sqrt{77}}{784} & 0 & 0 & -\frac{\sqrt{165}}{154} & 0 & -\frac{\sqrt{385}}{1078} & 0 & -\frac{2\sqrt{231}}{539} & 0 & 0 & 0 \\
-\frac{\sqrt{2310}}{392} & 0 & -\frac{5\sqrt{231}}{784} & 0 & \frac{3\sqrt{462}}{784} & 0 & 0 & -\frac{\sqrt{385}}{1078} & 0 & \frac{5\sqrt{77}}{1078} & 0 & -\frac{\sqrt{1155}}{539} & 0 & 0 \\
0 & -\frac{\sqrt{770}}{392} & 0 & -\frac{\sqrt{385}}{784} & 0 & \frac{3\sqrt{154}}{784} & -\frac{\sqrt{33}}{154} & 0 & \frac{5\sqrt{77}}{1078} & 0 & \frac{\sqrt{1155}}{539} & 0 & -\frac{2\sqrt{231}}{539} & 0 \\
-\frac{3\sqrt{154}}{784} & 0 & \frac{\sqrt{385}}{784} & 0 & \frac{\sqrt{770}}{392} & 0 & 0 & -\frac{2\sqrt{231}}{539} & 0 & \frac{\sqrt{1155}}{539} & 0 & \frac{5\sqrt{77}}{1078} & 0 & -\frac{\sqrt{33}}{154} \\
0 & -\frac{3\sqrt{462}}{784} & 0 & \frac{5\sqrt{231}}{784} & 0 & \frac{\sqrt{2310}}{392} & 0 & 0 & -\frac{\sqrt{1155}}{539} & 0 & \frac{5\sqrt{77}}{1078} & 0 & -\frac{\sqrt{385}}{1078} & 0 \\
0 & 0 & -\frac{9\sqrt{77}}{784} & 0 & \frac{5\sqrt{154}}{784} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{231}}{539} & 0 & -\frac{\sqrt{385}}{1078} & 0 & -\frac{\sqrt{165}}{154} \\
0 & 0 & 0 & -\frac{3\sqrt{11}}{112} & 0 & -\frac{\sqrt{110}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{154} & 0 & -\frac{\sqrt{165}}{154} & 0
\end{bmatrix}$$

$\boxed{\text{x543}}$ $\mathbb{M}_{3,2}^{(1,1;a)}(E_g,2) = \begin{bmatrix}
0 & \frac{5\sqrt{231}i}{294} & 0 & -\frac{5\sqrt{462}i}{588} & 0 & 0 & -\frac{\sqrt{110}i}{112} & 0 & -\frac{\sqrt{2310}i}{392} & 0 & \frac{3\sqrt{154}i}{784} & 0 & 0 & 0 \\
-\frac{5\sqrt{231}i}{294} & 0 & -\frac{\sqrt{2310}i}{588} & 0 & -\frac{\sqrt{1155}i}{147} & 0 & 0 & \frac{5\sqrt{154}i}{784} & 0 & -\frac{\sqrt{770}i}{392} & 0 & \frac{3\sqrt{462}i}{784} & 0 & 0 \\
0 & \frac{\sqrt{2310}i}{588} & 0 & -\frac{\sqrt{1155}i}{147} & 0 & -\frac{5\sqrt{462}i}{588} & \frac{3\sqrt{11}i}{112} & 0 & \frac{5\sqrt{231}i}{784} & 0 & \frac{\sqrt{385}i}{784} & 0 & \frac{9\sqrt{77}i}{784} & 0 \\
\frac{5\sqrt{462}i}{588} & 0 & \frac{\sqrt{1155}i}{147} & 0 & -\frac{\sqrt{2310}i}{588} & 0 & 0 & \frac{9\sqrt{77}i}{784} & 0 & \frac{\sqrt{385}i}{784} & 0 & \frac{5\sqrt{231}i}{784} & 0 & \frac{3\sqrt{11}i}{112} \\
0 & \frac{\sqrt{1155}i}{147} & 0 & \frac{\sqrt{2310}i}{588} & 0 & \frac{5\sqrt{231}i}{294} & 0 & 0 & \frac{3\sqrt{462}i}{784} & 0 & -\frac{\sqrt{770}i}{392} & 0 & \frac{5\sqrt{154}i}{784} & 0 \\
0 & 0 & \frac{5\sqrt{462}i}{588} & 0 & -\frac{5\sqrt{231}i}{294} & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}i}{784} & 0 & -\frac{\sqrt{2310}i}{392} & 0 & -\frac{\sqrt{110}i}{112} \\
\frac{\sqrt{110}i}{112} & 0 & -\frac{3\sqrt{11}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{154} & 0 & \frac{\sqrt{33}i}{154} & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{154}i}{784} & 0 & -\frac{9\sqrt{77}i}{784} & 0 & 0 & \frac{\sqrt{165}i}{154} & 0 & -\frac{\sqrt{385}i}{1078} & 0 & \frac{2\sqrt{231}i}{539} & 0 & 0 & 0 \\
\frac{\sqrt{2310}i}{392} & 0 & -\frac{5\sqrt{231}i}{784} & 0 & -\frac{3\sqrt{462}i}{784} & 0 & 0 & \frac{\sqrt{385}i}{1078} & 0 & \frac{5\sqrt{77}i}{1078} & 0 & \frac{\sqrt{1155}i}{539} & 0 & 0 \\
0 & \frac{\sqrt{770}i}{392} & 0 & -\frac{\sqrt{385}i}{784} & 0 & -\frac{3\sqrt{154}i}{784} & -\frac{\sqrt{33}i}{154} & 0 & -\frac{5\sqrt{77}i}{1078} & 0 & \frac{\sqrt{1155}i}{539} & 0 & \frac{2\sqrt{231}i}{539} & 0 \\
-\frac{3\sqrt{154}i}{784} & 0 & -\frac{\sqrt{385}i}{784} & 0 & \frac{\sqrt{770}i}{392} & 0 & 0 & -\frac{2\sqrt{231}i}{539} & 0 & -\frac{\sqrt{1155}i}{539} & 0 & \frac{5\sqrt{77}i}{1078} & 0 & \frac{\sqrt{33}i}{154} \\
0 & -\frac{3\sqrt{462}i}{784} & 0 & -\frac{5\sqrt{231}i}{784} & 0 & \frac{\sqrt{2310}i}{392} & 0 & 0 & -\frac{\sqrt{1155}i}{539} & 0 & -\frac{5\sqrt{77}i}{1078} & 0 & -\frac{\sqrt{385}i}{1078} & 0 \\
0 & 0 & -\frac{9\sqrt{77}i}{784} & 0 & -\frac{5\sqrt{154}i}{784} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{231}i}{539} & 0 & \frac{\sqrt{385}i}{1078} & 0 & -\frac{\sqrt{165}i}{154} \\
0 & 0 & 0 & -\frac{3\sqrt{11}i}{112} & 0 & \frac{\sqrt{110}i}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{154} & 0 & \frac{\sqrt{165}i}{154} & 0
\end{bmatrix}$

$\boxed{\text{x544}}$ $\mathbb{M}_{5,1}^{(1,1;a)}(E_g,1)$

$$= \begin{bmatrix}
0 & \frac{3\sqrt{5005}}{4312} & 0 & -\frac{\sqrt{10010}}{616} & 0 & \frac{9\sqrt{1001}}{616} & \frac{5\sqrt{858}}{14784} & 0 & -\frac{25\sqrt{2002}}{34496} & 0 & \frac{5\sqrt{30030}}{14784} & 0 & -\frac{5\sqrt{6006}}{4928} & 0 \\
\frac{3\sqrt{5005}}{4312} & 0 & -\frac{15\sqrt{2002}}{4312} & 0 & \frac{5\sqrt{1001}}{616} & 0 & 0 & -\frac{23\sqrt{30030}}{103488} & 0 & \frac{65\sqrt{6006}}{103488} & 0 & -\frac{\sqrt{10010}}{4928} & 0 & -\frac{\sqrt{4290}}{704} \\
0 & -\frac{15\sqrt{2002}}{4312} & 0 & \frac{15\sqrt{1001}}{2156} & 0 & -\frac{\sqrt{10010}}{616} & -\frac{\sqrt{2145}}{1056} & 0 & \frac{\sqrt{5005}}{1568} & 0 & -\frac{5\sqrt{3003}}{51744} & 0 & -\frac{\sqrt{15015}}{2464} & 0 \\
-\frac{\sqrt{10010}}{616} & 0 & \frac{15\sqrt{1001}}{2156} & 0 & -\frac{15\sqrt{2002}}{4312} & 0 & 0 & \frac{\sqrt{15015}}{2464} & 0 & \frac{5\sqrt{3003}}{51744} & 0 & -\frac{\sqrt{5005}}{1568} & 0 & \frac{\sqrt{2145}}{1056} \\
0 & \frac{5\sqrt{1001}}{616} & 0 & -\frac{15\sqrt{2002}}{4312} & 0 & \frac{3\sqrt{5005}}{4312} & \frac{\sqrt{4290}}{704} & 0 & \frac{\sqrt{10010}}{4928} & 0 & -\frac{65\sqrt{6006}}{103488} & 0 & \frac{23\sqrt{30030}}{103488} & 0 \\
\frac{9\sqrt{1001}}{616} & 0 & -\frac{\sqrt{10010}}{616} & 0 & \frac{3\sqrt{5005}}{4312} & 0 & 0 & \frac{5\sqrt{6006}}{4928} & 0 & -\frac{5\sqrt{30030}}{14784} & 0 & \frac{25\sqrt{2002}}{34496} & 0 & -\frac{5\sqrt{858}}{14784} \\
\frac{5\sqrt{858}}{14784} & 0 & -\frac{\sqrt{2145}}{1056} & 0 & \frac{\sqrt{4290}}{704} & 0 & 0 & -\frac{15\sqrt{143}}{16016} & 0 & \frac{\sqrt{715}}{1144} & 0 & -\frac{3\sqrt{429}}{2288} & 0 & 0 \\
0 & -\frac{23\sqrt{30030}}{103488} & 0 & \frac{\sqrt{15015}}{2464} & 0 & \frac{5\sqrt{6006}}{4928} & -\frac{15\sqrt{143}}{16016} & 0 & \frac{5\sqrt{3003}}{14014} & 0 & -\frac{\sqrt{5005}}{16016} & 0 & -\frac{9\sqrt{1001}}{8008} & 0 \\
-\frac{25\sqrt{2002}}{34496} & 0 & \frac{\sqrt{5005}}{1568} & 0 & \frac{\sqrt{10010}}{4928} & 0 & 0 & \frac{5\sqrt{3003}}{14014} & 0 & -\frac{\sqrt{15015}}{112112} & 0 & -\frac{5\sqrt{1001}}{8008} & 0 & -\frac{3\sqrt{429}}{2288} \\
0 & \frac{65\sqrt{6006}}{103488} & 0 & \frac{5\sqrt{3003}}{51744} & 0 & -\frac{5\sqrt{30030}}{14784} & \frac{\sqrt{715}}{1144} & 0 & -\frac{\sqrt{15015}}{112112} & 0 & -\frac{15\sqrt{1001}}{28028} & 0 & -\frac{\sqrt{5005}}{16016} & 0 \\
\frac{5\sqrt{30030}}{14784} & 0 & -\frac{5\sqrt{3003}}{51744} & 0 & -\frac{65\sqrt{6006}}{103488} & 0 & 0 & -\frac{\sqrt{5005}}{16016} & 0 & -\frac{15\sqrt{1001}}{28028} & 0 & -\frac{\sqrt{15015}}{112112} & 0 & \frac{\sqrt{715}}{1144} \\
0 & -\frac{\sqrt{10010}}{4928} & 0 & -\frac{\sqrt{5005}}{1568} & 0 & \frac{25\sqrt{2002}}{34496} & -\frac{3\sqrt{429}}{2288} & 0 & -\frac{5\sqrt{1001}}{8008} & 0 & -\frac{\sqrt{15015}}{112112} & 0 & \frac{5\sqrt{3003}}{14014} & 0 \\
-\frac{5\sqrt{6006}}{4928} & 0 & -\frac{\sqrt{15015}}{2464} & 0 & \frac{23\sqrt{30030}}{103488} & 0 & 0 & -\frac{9\sqrt{1001}}{8008} & 0 & -\frac{\sqrt{5005}}{16016} & 0 & \frac{5\sqrt{3003}}{14014} & 0 & -\frac{15\sqrt{143}}{16016} \\
0 & -\frac{\sqrt{4290}}{704} & 0 & \frac{\sqrt{2145}}{1056} & 0 & -\frac{5\sqrt{858}}{14784} & 0 & 0 & -\frac{3\sqrt{429}}{2288} & 0 & \frac{\sqrt{715}}{1144} & 0 & -\frac{15\sqrt{143}}{16016} & 0
\end{bmatrix}$$

$\boxed{\text{x545}}$ $\mathbb{M}_{5,2}^{(1,1;a)}(E_g,1)$

$$= \begin{bmatrix}
0 & \frac{3\sqrt{5005}i}{4312} & 0 & \frac{\sqrt{10010}i}{616} & 0 & \frac{9\sqrt{1001}i}{616} & -\frac{5\sqrt{858}i}{14784} & 0 & -\frac{25\sqrt{2002}i}{34496} & 0 & -\frac{5\sqrt{30030}i}{14784} & 0 & -\frac{5\sqrt{6006}i}{4928} & 0 \\
-\frac{3\sqrt{5005}i}{4312} & 0 & -\frac{15\sqrt{2002}i}{4312} & 0 & -\frac{5\sqrt{1001}i}{616} & 0 & 0 & \frac{23\sqrt{30030}i}{103488} & 0 & \frac{65\sqrt{6006}i}{103488} & 0 & \frac{\sqrt{10010}i}{4928} & 0 & -\frac{\sqrt{4290}i}{704} \\
0 & \frac{15\sqrt{2002}i}{4312} & 0 & \frac{15\sqrt{1001}i}{2156} & 0 & \frac{\sqrt{10010}i}{616} & -\frac{\sqrt{2145}i}{1056} & 0 & -\frac{\sqrt{5005}i}{1568} & 0 & -\frac{5\sqrt{3003}i}{51744} & 0 & \frac{\sqrt{15015}i}{2464} & 0 \\
-\frac{\sqrt{10010}i}{616} & 0 & -\frac{15\sqrt{1001}i}{2156} & 0 & -\frac{15\sqrt{2002}i}{4312} & 0 & 0 & \frac{\sqrt{15015}i}{2464} & 0 & -\frac{5\sqrt{3003}i}{51744} & 0 & -\frac{\sqrt{5005}i}{1568} & 0 & -\frac{\sqrt{2145}i}{1056} \\
0 & \frac{5\sqrt{1001}i}{616} & 0 & \frac{15\sqrt{2002}i}{4312} & 0 & \frac{3\sqrt{5005}i}{4312} & -\frac{\sqrt{4290}i}{704} & 0 & \frac{\sqrt{10010}i}{4928} & 0 & \frac{65\sqrt{6006}i}{103488} & 0 & \frac{23\sqrt{30030}i}{103488} & 0 \\
-\frac{9\sqrt{1001}i}{616} & 0 & -\frac{\sqrt{10010}i}{616} & 0 & -\frac{3\sqrt{5005}i}{4312} & 0 & 0 & -\frac{5\sqrt{6006}i}{4928} & 0 & -\frac{5\sqrt{30030}i}{14784} & 0 & -\frac{25\sqrt{2002}i}{34496} & 0 & -\frac{5\sqrt{858}i}{14784} \\
\frac{5\sqrt{858}i}{14784} & 0 & \frac{\sqrt{2145}i}{1056} & 0 & \frac{\sqrt{4290}i}{704} & 0 & 0 & -\frac{15\sqrt{143}i}{16016} & 0 & -\frac{\sqrt{715}i}{1144} & 0 & -\frac{3\sqrt{429}i}{2288} & 0 & 0 \\
0 & -\frac{23\sqrt{30030}i}{103488} & 0 & -\frac{\sqrt{15015}i}{2464} & 0 & \frac{5\sqrt{6006}i}{4928} & \frac{15\sqrt{143}i}{16016} & 0 & \frac{5\sqrt{3003}i}{14014} & 0 & \frac{\sqrt{5005}i}{16016} & 0 & -\frac{9\sqrt{1001}i}{8008} & 0 \\
\frac{25\sqrt{2002}i}{34496} & 0 & \frac{\sqrt{5005}i}{1568} & 0 & -\frac{\sqrt{10010}i}{4928} & 0 & 0 & -\frac{5\sqrt{3003}i}{14014} & 0 & -\frac{\sqrt{15015}i}{112112} & 0 & \frac{5\sqrt{1001}i}{8008} & 0 & -\frac{3\sqrt{429}i}{2288} \\
0 & -\frac{65\sqrt{6006}i}{103488} & 0 & \frac{5\sqrt{3003}i}{51744} & 0 & \frac{5\sqrt{30030}i}{14784} & \frac{\sqrt{715}i}{1144} & 0 & \frac{\sqrt{15015}i}{112112} & 0 & -\frac{15\sqrt{1001}i}{28028} & 0 & \frac{\sqrt{5005}i}{16016} & 0 \\
\frac{5\sqrt{30030}i}{14784} & 0 & \frac{5\sqrt{3003}i}{51744} & 0 & -\frac{65\sqrt{6006}i}{103488} & 0 & 0 & -\frac{\sqrt{5005}i}{16016} & 0 & \frac{15\sqrt{1001}i}{28028} & 0 & -\frac{\sqrt{15015}i}{112112} & 0 & -\frac{\sqrt{715}i}{1144} \\
0 & -\frac{\sqrt{10010}i}{4928} & 0 & \frac{\sqrt{5005}i}{1568} & 0 & \frac{25\sqrt{2002}i}{34496} & \frac{3\sqrt{429}i}{2288} & 0 & -\frac{5\sqrt{1001}i}{8008} & 0 & \frac{\sqrt{15015}i}{112112} & 0 & \frac{5\sqrt{3003}i}{14014} & 0 \\
\frac{5\sqrt{6006}i}{4928} & 0 & -\frac{\sqrt{15015}i}{2464} & 0 & -\frac{23\sqrt{30030}i}{103488} & 0 & 0 & \frac{9\sqrt{1001}i}{8008} & 0 & -\frac{\sqrt{5005}i}{16016} & 0 & -\frac{5\sqrt{3003}i}{14014} & 0 & -\frac{15\sqrt{143}i}{16016} \\
0 & \frac{\sqrt{4290}i}{704} & 0 & \frac{\sqrt{2145}i}{1056} & 0 & \frac{5\sqrt{858}i}{14784} & 0 & 0 & \frac{3\sqrt{429}i}{2288} & 0 & \frac{\sqrt{715}i}{1144} & 0 & \frac{15\sqrt{143}i}{16016} & 0
\end{bmatrix}$$

$\boxed{\text{x546}}$ $\mathbb{M}_{5,1}^{(1,1;a)}(E_g,2)$

$$
= \begin{bmatrix}
0 & \frac{3\sqrt{143}}{616} & 0 & \frac{9\sqrt{286}}{616} & 0 & \frac{3\sqrt{715}}{616} & \frac{\sqrt{30030}}{14784} & 0 & -\frac{5\sqrt{1430}}{4928} & 0 & -\frac{15\sqrt{858}}{4928} & 0 & -\frac{5\sqrt{4290}}{14784} & 0 \\
\frac{3\sqrt{143}}{616} & 0 & -\frac{3\sqrt{1430}}{616} & 0 & -\frac{9\sqrt{715}}{616} & 0 & 0 & -\frac{23\sqrt{858}}{14784} & 0 & \frac{13\sqrt{4290}}{14784} & 0 & \frac{9\sqrt{286}}{4928} & 0 & -\frac{5\sqrt{6006}}{14784} \\
0 & -\frac{3\sqrt{1430}}{616} & 0 & \frac{3\sqrt{715}}{308} & 0 & \frac{9\sqrt{286}}{616} & \frac{3\sqrt{3003}}{2464} & 0 & \frac{\sqrt{143}}{224} & 0 & -\frac{\sqrt{2145}}{7392} & 0 & \frac{9\sqrt{429}}{2464} & 0 \\
\frac{9\sqrt{286}}{616} & 0 & \frac{3\sqrt{715}}{308} & 0 & -\frac{3\sqrt{1430}}{616} & 0 & 0 & -\frac{9\sqrt{429}}{2464} & 0 & \frac{\sqrt{2145}}{7392} & 0 & -\frac{\sqrt{143}}{224} & 0 & -\frac{3\sqrt{3003}}{2464} \\
0 & -\frac{9\sqrt{715}}{616} & 0 & -\frac{3\sqrt{1430}}{616} & 0 & \frac{3\sqrt{143}}{616} & \frac{5\sqrt{6006}}{14784} & 0 & -\frac{9\sqrt{286}}{4928} & 0 & -\frac{13\sqrt{4290}}{14784} & 0 & \frac{23\sqrt{858}}{14784} & 0 \\
\frac{3\sqrt{715}}{616} & 0 & \frac{9\sqrt{286}}{616} & 0 & \frac{3\sqrt{143}}{616} & 0 & 0 & \frac{5\sqrt{4290}}{14784} & 0 & \frac{15\sqrt{858}}{4928} & 0 & \frac{5\sqrt{1430}}{4928} & 0 & -\frac{\sqrt{30030}}{14784} \\
\frac{\sqrt{30030}}{14784} & 0 & \frac{3\sqrt{3003}}{2464} & 0 & \frac{5\sqrt{6006}}{14784} & 0 & 0 & -\frac{3\sqrt{5005}}{16016} & 0 & -\frac{9\sqrt{1001}}{8008} & 0 & -\frac{\sqrt{15015}}{16016} & 0 & 0 \\
0 & -\frac{23\sqrt{858}}{14784} & 0 & -\frac{9\sqrt{429}}{2464} & 0 & \frac{5\sqrt{4290}}{14784} & -\frac{3\sqrt{5005}}{16016} & 0 & \frac{\sqrt{2145}}{2002} & 0 & \frac{9\sqrt{143}}{16016} & 0 & -\frac{3\sqrt{715}}{8008} & 0 \\
-\frac{5\sqrt{1430}}{4928} & 0 & \frac{\sqrt{143}}{224} & 0 & -\frac{9\sqrt{286}}{4928} & 0 & 0 & \frac{\sqrt{2145}}{2002} & 0 & -\frac{\sqrt{429}}{16016} & 0 & \frac{9\sqrt{715}}{8008} & 0 & -\frac{\sqrt{15015}}{16016} \\
0 & \frac{13\sqrt{4290}}{14784} & 0 & \frac{\sqrt{2145}}{7392} & 0 & \frac{15\sqrt{858}}{4928} & -\frac{9\sqrt{1001}}{8008} & 0 & -\frac{\sqrt{429}}{16016} & 0 & -\frac{3\sqrt{715}}{4004} & 0 & \frac{9\sqrt{143}}{16016} & 0 \\
-\frac{15\sqrt{858}}{4928} & 0 & -\frac{\sqrt{2145}}{7392} & 0 & -\frac{13\sqrt{4290}}{14784} & 0 & 0 & \frac{9\sqrt{143}}{16016} & 0 & -\frac{3\sqrt{715}}{4004} & 0 & -\frac{\sqrt{429}}{16016} & 0 & -\frac{9\sqrt{1001}}{8008} \\
0 & \frac{9\sqrt{286}}{4928} & 0 & -\frac{\sqrt{143}}{224} & 0 & \frac{5\sqrt{1430}}{4928} & -\frac{\sqrt{15015}}{16016} & 0 & \frac{9\sqrt{715}}{8008} & 0 & -\frac{\sqrt{429}}{16016} & 0 & \frac{\sqrt{2145}}{2002} & 0 \\
-\frac{5\sqrt{4290}}{14784} & 0 & \frac{9\sqrt{429}}{2464} & 0 & \frac{23\sqrt{858}}{14784} & 0 & 0 & -\frac{3\sqrt{715}}{8008} & 0 & \frac{9\sqrt{143}}{16016} & 0 & \frac{\sqrt{2145}}{2002} & 0 & -\frac{3\sqrt{5005}}{16016} \\
0 & -\frac{5\sqrt{6006}}{14784} & 0 & -\frac{3\sqrt{3003}}{2464} & 0 & -\frac{\sqrt{30030}}{14784} & 0 & 0 & -\frac{\sqrt{15015}}{16016} & 0 & -\frac{9\sqrt{1001}}{8008} & 0 & -\frac{3\sqrt{5005}}{16016} & 0
\end{bmatrix}
$$

$\boxed{\text{x547}}$ $\mathbb{M}_{5,2}^{(1,1;a)}(E_g,2)$

$$= \begin{bmatrix}
0 & \frac{3\sqrt{143}i}{616} & 0 & -\frac{9\sqrt{286}i}{616} & 0 & \frac{3\sqrt{715}i}{616} & -\frac{\sqrt{30030}i}{14784} & 0 & -\frac{5\sqrt{1430}i}{4928} & 0 & \frac{15\sqrt{858}i}{4928} & 0 & -\frac{5\sqrt{4290}i}{14784} & 0 \\
-\frac{3\sqrt{143}i}{616} & 0 & -\frac{3\sqrt{1430}i}{616} & 0 & \frac{9\sqrt{715}i}{616} & 0 & 0 & \frac{23\sqrt{858}i}{14784} & 0 & \frac{13\sqrt{4290}i}{14784} & 0 & -\frac{9\sqrt{286}i}{4928} & 0 & -\frac{5\sqrt{6006}i}{14784} \\
0 & \frac{3\sqrt{1430}i}{616} & 0 & \frac{3\sqrt{715}i}{308} & 0 & -\frac{9\sqrt{286}i}{616} & \frac{3\sqrt{3003}i}{2464} & 0 & -\frac{\sqrt{143}i}{224} & 0 & -\frac{\sqrt{2145}i}{7392} & 0 & -\frac{9\sqrt{429}i}{2464} & 0 \\
\frac{9\sqrt{286}i}{616} & 0 & -\frac{3\sqrt{715}i}{308} & 0 & -\frac{3\sqrt{1430}i}{616} & 0 & 0 & -\frac{9\sqrt{429}i}{2464} & 0 & -\frac{\sqrt{2145}i}{7392} & 0 & -\frac{\sqrt{143}i}{224} & 0 & \frac{3\sqrt{3003}i}{2464} \\
0 & -\frac{9\sqrt{715}i}{616} & 0 & \frac{3\sqrt{1430}i}{616} & 0 & \frac{3\sqrt{143}i}{616} & -\frac{5\sqrt{6006}i}{14784} & 0 & -\frac{9\sqrt{286}i}{4928} & 0 & \frac{13\sqrt{4290}i}{14784} & 0 & \frac{23\sqrt{858}i}{14784} & 0 \\
-\frac{3\sqrt{715}i}{616} & 0 & \frac{9\sqrt{286}i}{616} & 0 & -\frac{3\sqrt{143}i}{616} & 0 & 0 & -\frac{5\sqrt{4290}i}{14784} & 0 & \frac{15\sqrt{858}i}{4928} & 0 & -\frac{5\sqrt{1430}i}{4928} & 0 & -\frac{\sqrt{30030}i}{14784} \\
\frac{\sqrt{30030}i}{14784} & 0 & -\frac{3\sqrt{3003}i}{2464} & 0 & \frac{5\sqrt{6006}i}{14784} & 0 & 0 & -\frac{3\sqrt{5005}i}{16016} & 0 & \frac{9\sqrt{1001}i}{8008} & 0 & -\frac{\sqrt{15015}i}{16016} & 0 & 0 \\
0 & -\frac{23\sqrt{858}i}{14784} & 0 & \frac{9\sqrt{429}i}{2464} & 0 & \frac{5\sqrt{4290}i}{14784} & \frac{3\sqrt{5005}i}{16016} & 0 & \frac{\sqrt{2145}i}{2002} & 0 & -\frac{9\sqrt{143}i}{16016} & 0 & -\frac{3\sqrt{715}i}{8008} & 0 \\
\frac{5\sqrt{1430}i}{4928} & 0 & \frac{\sqrt{143}i}{224} & 0 & \frac{9\sqrt{286}i}{4928} & 0 & 0 & -\frac{\sqrt{2145}i}{2002} & 0 & -\frac{\sqrt{429}i}{16016} & 0 & -\frac{9\sqrt{715}i}{8008} & 0 & -\frac{\sqrt{15015}i}{16016} \\
0 & -\frac{13\sqrt{4290}i}{14784} & 0 & \frac{\sqrt{2145}i}{7392} & 0 & -\frac{15\sqrt{858}i}{4928} & -\frac{9\sqrt{1001}i}{8008} & 0 & \frac{\sqrt{429}i}{16016} & 0 & -\frac{3\sqrt{715}i}{4004} & 0 & -\frac{9\sqrt{143}i}{16016} & 0 \\
-\frac{15\sqrt{858}i}{4928} & 0 & \frac{\sqrt{2145}i}{7392} & 0 & -\frac{13\sqrt{4290}i}{14784} & 0 & 0 & \frac{9\sqrt{143}i}{16016} & 0 & \frac{3\sqrt{715}i}{4004} & 0 & -\frac{\sqrt{429}i}{16016} & 0 & \frac{9\sqrt{1001}i}{8008} \\
0 & \frac{9\sqrt{286}i}{4928} & 0 & \frac{\sqrt{143}i}{224} & 0 & \frac{5\sqrt{1430}i}{4928} & \frac{\sqrt{15015}i}{16016} & 0 & \frac{9\sqrt{715}i}{8008} & 0 & \frac{\sqrt{429}i}{16016} & 0 & \frac{\sqrt{2145}i}{2002} & 0 \\
\frac{5\sqrt{4290}i}{14784} & 0 & \frac{9\sqrt{429}i}{2464} & 0 & -\frac{23\sqrt{858}i}{14784} & 0 & 0 & \frac{3\sqrt{715}i}{8008} & 0 & \frac{9\sqrt{143}i}{16016} & 0 & -\frac{\sqrt{2145}i}{2002} & 0 & -\frac{3\sqrt{5005}i}{16016} \\
0 & \frac{5\sqrt{6006}i}{14784} & 0 & -\frac{3\sqrt{3003}i}{2464} & 0 & \frac{\sqrt{30030}i}{14784} & 0 & 0 & \frac{\sqrt{15015}i}{16016} & 0 & -\frac{9\sqrt{1001}i}{8008} & 0 & \frac{3\sqrt{5005}i}{16016} & 0
\end{bmatrix}$$

$\boxed{\text{x548}}$ $\mathbb{M}_{5,1}^{(1,1;a)}(E_g,3) =$

$$\begin{bmatrix}
0 & \frac{\sqrt{429}}{308} & 0 & -\frac{\sqrt{858}}{308} & 0 & -\frac{3\sqrt{2145}}{308} & \frac{\sqrt{10010}}{7392} & 0 & -\frac{5\sqrt{4290}}{7392} & 0 & \frac{5\sqrt{286}}{2464} & 0 & \frac{5\sqrt{1430}}{2464} & 0 \\
\frac{\sqrt{429}}{308} & 0 & -\frac{\sqrt{4290}}{308} & 0 & \frac{\sqrt{2145}}{308} & 0 & 0 & -\frac{23\sqrt{286}}{7392} & 0 & \frac{13\sqrt{1430}}{7392} & 0 & -\frac{\sqrt{858}}{2464} & 0 & \frac{5\sqrt{2002}}{2464} \\
0 & -\frac{\sqrt{4290}}{308} & 0 & \frac{\sqrt{2145}}{154} & 0 & -\frac{\sqrt{858}}{308} & -\frac{\sqrt{1001}}{1232} & 0 & \frac{\sqrt{429}}{336} & 0 & -\frac{\sqrt{715}}{3696} & 0 & -\frac{3\sqrt{143}}{1232} & 0 \\
-\frac{\sqrt{858}}{308} & 0 & \frac{\sqrt{2145}}{154} & 0 & -\frac{\sqrt{4290}}{308} & 0 & 0 & \frac{3\sqrt{143}}{1232} & 0 & \frac{\sqrt{715}}{3696} & 0 & -\frac{\sqrt{429}}{336} & 0 & \frac{\sqrt{1001}}{1232} \\
0 & \frac{\sqrt{2145}}{308} & 0 & -\frac{\sqrt{4290}}{308} & 0 & \frac{\sqrt{429}}{308} & -\frac{5\sqrt{2002}}{2464} & 0 & \frac{\sqrt{858}}{2464} & 0 & -\frac{13\sqrt{1430}}{7392} & 0 & \frac{23\sqrt{286}}{7392} & 0 \\
-\frac{3\sqrt{2145}}{308} & 0 & -\frac{\sqrt{858}}{308} & 0 & \frac{\sqrt{429}}{308} & 0 & 0 & -\frac{5\sqrt{1430}}{2464} & 0 & -\frac{5\sqrt{286}}{2464} & 0 & \frac{5\sqrt{4290}}{7392} & 0 & -\frac{\sqrt{10010}}{7392} \\
\frac{\sqrt{10010}}{7392} & 0 & -\frac{\sqrt{1001}}{1232} & 0 & -\frac{5\sqrt{2002}}{2464} & 0 & 0 & -\frac{\sqrt{15015}}{8008} & 0 & \frac{\sqrt{3003}}{4004} & 0 & \frac{3\sqrt{5005}}{8008} & 0 & 0 \\
0 & -\frac{23\sqrt{286}}{7392} & 0 & \frac{3\sqrt{143}}{1232} & 0 & -\frac{5\sqrt{1430}}{2464} & -\frac{\sqrt{15015}}{8008} & 0 & \frac{\sqrt{715}}{1001} & 0 & -\frac{\sqrt{429}}{8008} & 0 & \frac{3\sqrt{2145}}{4004} & 0 \\
-\frac{5\sqrt{4290}}{7392} & 0 & \frac{\sqrt{429}}{336} & 0 & \frac{\sqrt{858}}{2464} & 0 & 0 & \frac{\sqrt{715}}{1001} & 0 & -\frac{\sqrt{143}}{8008} & 0 & -\frac{\sqrt{2145}}{4004} & 0 & \frac{3\sqrt{5005}}{8008} \\
0 & \frac{13\sqrt{1430}}{7392} & 0 & \frac{\sqrt{715}}{3696} & 0 & -\frac{5\sqrt{286}}{2464} & \frac{\sqrt{3003}}{4004} & 0 & -\frac{\sqrt{143}}{8008} & 0 & -\frac{\sqrt{2145}}{2002} & 0 & -\frac{\sqrt{429}}{8008} & 0 \\
\frac{5\sqrt{286}}{2464} & 0 & -\frac{\sqrt{715}}{3696} & 0 & -\frac{13\sqrt{1430}}{7392} & 0 & 0 & -\frac{\sqrt{429}}{8008} & 0 & -\frac{\sqrt{2145}}{2002} & 0 & -\frac{\sqrt{143}}{8008} & 0 & \frac{\sqrt{3003}}{4004} \\
0 & -\frac{\sqrt{858}}{2464} & 0 & -\frac{\sqrt{429}}{336} & 0 & \frac{5\sqrt{4290}}{7392} & \frac{3\sqrt{5005}}{8008} & 0 & -\frac{\sqrt{2145}}{4004} & 0 & -\frac{\sqrt{143}}{8008} & 0 & \frac{\sqrt{715}}{1001} & 0 \\
\frac{5\sqrt{1430}}{2464} & 0 & -\frac{3\sqrt{143}}{1232} & 0 & \frac{23\sqrt{286}}{7392} & 0 & 0 & \frac{3\sqrt{2145}}{4004} & 0 & -\frac{\sqrt{429}}{8008} & 0 & \frac{\sqrt{715}}{1001} & 0 & -\frac{\sqrt{15015}}{8008} \\
0 & \frac{5\sqrt{2002}}{2464} & 0 & \frac{\sqrt{1001}}{1232} & 0 & -\frac{\sqrt{10010}}{7392} & 0 & 0 & \frac{3\sqrt{5005}}{8008} & 0 & \frac{\sqrt{3003}}{4004} & 0 & -\frac{\sqrt{15015}}{8008} & 0
\end{bmatrix}$$

$\boxed{\text{x549}}$ $\mathbb{M}_{5,2}^{(1,1;a)}(E_g,3)$

$$= \begin{bmatrix}
0 & \frac{\sqrt{429}i}{308} & 0 & \frac{\sqrt{858}i}{308} & 0 & -\frac{3\sqrt{2145}i}{308} & -\frac{\sqrt{10010}i}{7392} & 0 & -\frac{5\sqrt{4290}i}{7392} & 0 & -\frac{5\sqrt{286}i}{2464} & 0 & \frac{5\sqrt{1430}i}{2464} & 0 \\
-\frac{\sqrt{429}i}{308} & 0 & -\frac{\sqrt{4290}i}{308} & 0 & -\frac{\sqrt{2145}i}{308} & 0 & 0 & \frac{23\sqrt{286}i}{7392} & 0 & \frac{13\sqrt{1430}i}{7392} & 0 & \frac{\sqrt{858}i}{2464} & 0 & \frac{5\sqrt{2002}i}{2464} \\
0 & \frac{\sqrt{4290}i}{308} & 0 & \frac{\sqrt{2145}i}{154} & 0 & \frac{\sqrt{858}i}{308} & -\frac{\sqrt{1001}i}{1232} & 0 & -\frac{\sqrt{429}i}{336} & 0 & -\frac{\sqrt{715}i}{3696} & 0 & \frac{3\sqrt{143}i}{1232} & 0 \\
-\frac{\sqrt{858}i}{308} & 0 & -\frac{\sqrt{2145}i}{154} & 0 & -\frac{\sqrt{4290}i}{308} & 0 & 0 & \frac{3\sqrt{143}i}{1232} & 0 & -\frac{\sqrt{715}i}{3696} & 0 & -\frac{\sqrt{429}i}{336} & 0 & -\frac{\sqrt{1001}i}{1232} \\
0 & \frac{\sqrt{2145}i}{308} & 0 & \frac{\sqrt{4290}i}{308} & 0 & \frac{\sqrt{429}i}{308} & \frac{5\sqrt{2002}i}{2464} & 0 & \frac{\sqrt{858}i}{2464} & 0 & \frac{13\sqrt{1430}i}{7392} & 0 & \frac{23\sqrt{286}i}{7392} & 0 \\
\frac{3\sqrt{2145}i}{308} & 0 & -\frac{\sqrt{858}i}{308} & 0 & -\frac{\sqrt{429}i}{308} & 0 & 0 & \frac{5\sqrt{1430}i}{2464} & 0 & -\frac{5\sqrt{286}i}{2464} & 0 & -\frac{5\sqrt{4290}i}{7392} & 0 & -\frac{\sqrt{10010}i}{7392} \\
\frac{\sqrt{10010}i}{7392} & 0 & \frac{\sqrt{1001}i}{1232} & 0 & -\frac{5\sqrt{2002}i}{2464} & 0 & 0 & -\frac{\sqrt{15015}i}{8008} & 0 & -\frac{\sqrt{3003}i}{4004} & 0 & \frac{3\sqrt{5005}i}{8008} & 0 & 0 \\
0 & -\frac{23\sqrt{286}i}{7392} & 0 & -\frac{3\sqrt{143}i}{1232} & 0 & -\frac{5\sqrt{1430}i}{2464} & \frac{\sqrt{15015}i}{8008} & 0 & \frac{\sqrt{715}i}{1001} & 0 & \frac{\sqrt{429}i}{8008} & 0 & \frac{3\sqrt{2145}i}{4004} & 0 \\
\frac{5\sqrt{4290}i}{7392} & 0 & \frac{\sqrt{429}i}{336} & 0 & -\frac{\sqrt{858}i}{2464} & 0 & 0 & -\frac{\sqrt{715}i}{1001} & 0 & -\frac{\sqrt{143}i}{8008} & 0 & \frac{\sqrt{2145}i}{4004} & 0 & \frac{3\sqrt{5005}i}{8008} \\
0 & -\frac{13\sqrt{1430}i}{7392} & 0 & \frac{\sqrt{715}i}{3696} & 0 & \frac{5\sqrt{286}i}{2464} & \frac{\sqrt{3003}i}{4004} & 0 & \frac{\sqrt{143}i}{8008} & 0 & -\frac{\sqrt{2145}i}{2002} & 0 & \frac{\sqrt{429}i}{8008} & 0 \\
\frac{5\sqrt{286}i}{2464} & 0 & \frac{\sqrt{715}i}{3696} & 0 & -\frac{13\sqrt{1430}i}{7392} & 0 & 0 & -\frac{\sqrt{429}i}{8008} & 0 & \frac{\sqrt{2145}i}{2002} & 0 & -\frac{\sqrt{143}i}{8008} & 0 & -\frac{\sqrt{3003}i}{4004} \\
0 & -\frac{\sqrt{858}i}{2464} & 0 & \frac{\sqrt{429}i}{336} & 0 & \frac{5\sqrt{4290}i}{7392} & -\frac{3\sqrt{5005}i}{8008} & 0 & -\frac{\sqrt{2145}i}{4004} & 0 & \frac{\sqrt{143}i}{8008} & 0 & \frac{\sqrt{715}i}{1001} & 0 \\
-\frac{5\sqrt{1430}i}{2464} & 0 & -\frac{3\sqrt{143}i}{1232} & 0 & -\frac{23\sqrt{286}i}{7392} & 0 & 0 & -\frac{3\sqrt{2145}i}{4004} & 0 & -\frac{\sqrt{429}i}{8008} & 0 & -\frac{\sqrt{715}i}{1001} & 0 & -\frac{\sqrt{15015}i}{8008} \\
0 & -\frac{5\sqrt{2002}i}{2464} & 0 & \frac{\sqrt{1001}i}{1232} & 0 & \frac{\sqrt{10010}i}{7392} & 0 & 0 & -\frac{3\sqrt{5005}i}{8008} & 0 & \frac{\sqrt{3003}i}{4004} & 0 & \frac{\sqrt{15015}i}{8008} & 0
\end{bmatrix}$$

$\boxed{\text{x550}}$ $\mathbb{T}_2^{(1,0;a)}(A_{1g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{84} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x551}}$ $\mathbb{T}_4^{(1,0;a)}(A_{1g}, 1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{44} & 0 & 0 & 0 & -\frac{\sqrt{330}i}{132} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{33} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{22} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{44} & 0 & 0 & 0 & -\frac{\sqrt{77}i}{44} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{44} & 0 & 0 & 0 & -\frac{\sqrt{55}i}{44} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{22} & 0 & 0 & 0 & -\frac{\sqrt{66}i}{33} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{132} & 0 & 0 & 0 & \frac{\sqrt{110}i}{44} & 0 \\
0 & 0 & 0 & -\frac{\sqrt{77}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{110}i}{44} & 0 & 0 & 0 & -\frac{\sqrt{22}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{66}i}{33} & 0 & 0 & 0 & -\frac{\sqrt{330}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{55}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{55}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{330}i}{132} & 0 & 0 & 0 & \frac{\sqrt{66}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{22}i}{22} & 0 & 0 & 0 & -\frac{\sqrt{110}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{77}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x552}}$ $\mathbb{T}_4^{(1,0;a)}(A_{1g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}i}{308} & 0 & 0 & 0 & \frac{\sqrt{462}i}{132} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{231} & 0 & 0 & 0 & \frac{\sqrt{770}i}{110} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{308} & 0 & 0 & 0 & \frac{7\sqrt{55}i}{220} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{55}i}{220} & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{308} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{110} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{231} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{132} & 0 & 0 & 0 & \frac{5\sqrt{154}i}{308} & 0 \\
0 & 0 & 0 & \frac{7\sqrt{55}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{154}i}{308} & 0 & 0 & 0 & \frac{\sqrt{770}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2310}i}{231} & 0 & 0 & 0 & \frac{\sqrt{462}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{77}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{462}i}{132} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{770}i}{110} & 0 & 0 & 0 & -\frac{5\sqrt{154}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{7\sqrt{55}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x553}}$ $\mathbb{T}_6^{(1,0;a)}(A_{1g},1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{264} & 0 & 0 & 0 & -\frac{7\sqrt{11}i}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{88} & 0 & 0 & 0 & \frac{7\sqrt{165}i}{264} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}i}{264} & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{66}i}{264} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{165}i}{264} & 0 & 0 & 0 & \frac{\sqrt{55}i}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{11}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{33}i}{264} & 0 \\
0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{33}i}{264} & 0 & 0 & 0 & \frac{7\sqrt{165}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{55}i}{88} & 0 & 0 & 0 & -\frac{7\sqrt{11}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{66}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{7\sqrt{11}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{55}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{7\sqrt{165}i}{264} & 0 & 0 & 0 & \frac{\sqrt{33}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x554}}$ $\mathbb{T}_6^{(1,0;a)}(A_{1g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{264} & 0 & 0 & 0 & \frac{\sqrt{77}i}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{264} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}i}{264} & 0 & 0 & 0 & \frac{\sqrt{330}i}{264} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{462}i}{264} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{264} & 0 & 0 & 0 & \frac{\sqrt{385}i}{88} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{231}i}{264} & 0 \\
0 & 0 & 0 & \frac{\sqrt{330}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{231}i}{264} & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{385}i}{88} & 0 & 0 & 0 & \frac{\sqrt{77}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{5\sqrt{462}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{5\sqrt{462}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{77}i}{88} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{1155}i}{264} & 0 & 0 & 0 & \frac{\sqrt{231}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{330}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x555}}$ $\mathbb{T}_4^{(1,0;a)}(A_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{55} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{110} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{55} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{1155}}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{55} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{330}}{55} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{1155}}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x556}}$ $\mathbb{T}_6^{(1,0;a)}(A_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{132} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2310}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x557}}$ $\mathbb{T}_2^{(1,0;a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{21} & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & \frac{\sqrt{35}i}{21} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{35}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{70}i}{84} & 0 & 0 & 0 & \frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & \frac{\sqrt{70}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{35}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x558}}$ $\mathbb{T}_4^{(1,0;a)}(B_{1g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{330}i}{220} & 0 & 0 & 0 & \frac{\sqrt{462}i}{308} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{1155}i}{1540} & 0 & 0 & 0 & -\frac{17\sqrt{385}i}{1540} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{385}i}{1540} & 0 & 0 & 0 & -\frac{9\sqrt{1155}i}{1540} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{308} & 0 & 0 & 0 & \frac{3\sqrt{330}i}{220} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{154} & 0 & 0 & 0 \\
0 & \frac{3\sqrt{330}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{9\sqrt{1155}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{17\sqrt{385}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & \frac{\sqrt{462}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{462}i}{308} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{17\sqrt{385}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{1155}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{330}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x559}}$ $\mathbb{T}_6^{(1,0;a)}(B_{1g}, 1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{24} & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{24} & 0 & 0 & 0 & \frac{\sqrt{35}i}{24} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 & \frac{\sqrt{14}i}{24} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{24} & 0 & 0 & 0 & -\frac{i}{24} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & \frac{\sqrt{7}i}{24} & 0 & 0 & 0 \\
0 & -\frac{i}{24} & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{14}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{7}i}{24} & 0 & 0 & 0 & \frac{\sqrt{35}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{14}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & \frac{i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x560}}$ $\mathbb{T}_6^{(1,0;a)}(B_{1g},2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{264} & 0 & 0 & 0 & \frac{\sqrt{11}i}{8} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{264} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{264} & 0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & -\frac{\sqrt{770}i}{264} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{264} & 0 & 0 & 0 & \frac{\sqrt{55}i}{264} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{385}i}{264} & 0 & 0 & 0 \\
0 & \frac{\sqrt{55}i}{264} & 0 & 0 & 0 & \frac{\sqrt{11}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{770}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{385}i}{264} & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{77}i}{264} & 0 & 0 & 0 & \frac{\sqrt{385}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{2310}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{770}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{11}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{55}i}{264} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x561}}$ $\mathbb{T}_2^{(1,0;a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{21} & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & \frac{\sqrt{35}}{21} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 & \frac{\sqrt{10}}{12} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 \\
0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{35}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{70}}{84} & 0 & 0 & 0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{14}}{21} & 0 & 0 & 0 & \frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{35}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x562}}$ $\mathbb{T}_4^{(1,0;a)}(B_{2g}) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{154} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{330}}{220} & 0 & 0 & 0 & -\frac{\sqrt{462}}{308} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{1155}}{1540} & 0 & 0 & 0 & \frac{17\sqrt{385}}{1540} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{385}}{1540} & 0 & 0 & 0 & \frac{9\sqrt{1155}}{1540} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{308} & 0 & 0 & 0 & -\frac{3\sqrt{330}}{220} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{154} & 0 & 0 & 0 \\
0 & -\frac{3\sqrt{330}}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{9\sqrt{1155}}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{17\sqrt{385}}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2310}}{154} & 0 & 0 & 0 & -\frac{\sqrt{462}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{462}}{308} & 0 & 0 & 0 & -\frac{\sqrt{2310}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{17\sqrt{385}}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{9\sqrt{1155}}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{3\sqrt{330}}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x563}}$ $\mathbb{T}_6^{(1,0;a)}(B_{2g}, 1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x564}}$ $\mathbb{T}_6^{(1,0;a)}(B_{2g},2) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{66} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{66} & 0 & 0 & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{66} & 0 & 0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 & 0 & -\frac{\sqrt{154}}{66} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 & \frac{\sqrt{11}}{66} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{66} & 0 & 0 & 0 \\
0 & \frac{\sqrt{11}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{154}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{77}}{66} & 0 & 0 & 0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{385}}{66} & 0 & 0 & 0 & \frac{\sqrt{77}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{462}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{154}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{11}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x565}}$ $\mathbb{T}_{2,1}^{(1,0;a)}(E_g) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{24} & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{168} & 0 & \frac{11\sqrt{14}}{168} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & \frac{\sqrt{7}}{12} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{12} & 0 & \frac{\sqrt{105}}{84} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{14}}{168} & 0 & -\frac{\sqrt{70}}{168} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{168} & 0 & -\frac{5\sqrt{2}}{24} \\
\frac{5\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{70}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{42}}{168} & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{11\sqrt{14}}{168} & 0 & -\frac{\sqrt{7}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{7}}{12} & 0 & -\frac{11\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$

$\boxed{\text{x566}}$ $\mathbb{T}_{2,2}^{(1,0;a)}(E_g) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{24} & 0 & \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{168} & 0 & \frac{11\sqrt{14}i}{168} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & \frac{\sqrt{7}i}{12} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{12} & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{14}i}{168} & 0 & -\frac{\sqrt{70}i}{168} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & 0 & -\frac{5\sqrt{2}i}{24} \\
\frac{5\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{70}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{42}i}{168} & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{11\sqrt{14}i}{168} & 0 & -\frac{\sqrt{7}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{7}i}{12} & 0 & -\frac{11\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x567}}$ $\mathbb{T}_{4,1}^{(1,0;a)}(E_g, 1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{176} & 0 & \frac{5\sqrt{22}}{88} & 0 & \frac{\sqrt{330}}{176} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{330}}{880} & 0 & -\frac{\sqrt{66}}{88} & 0 & \frac{7\sqrt{110}}{880} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1155}}{880} & 0 & \frac{\sqrt{55}}{880} & 0 & -\frac{7\sqrt{33}}{176} & 0 & -\frac{\sqrt{165}}{880} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{880} & 0 & \frac{7\sqrt{33}}{176} & 0 & -\frac{\sqrt{55}}{880} & 0 & -\frac{3\sqrt{1155}}{880} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{110}}{880} & 0 & \frac{\sqrt{66}}{88} & 0 & \frac{13\sqrt{330}}{880} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{176} & 0 & -\frac{5\sqrt{22}}{88} & 0 & -\frac{\sqrt{462}}{176} \\
\frac{\sqrt{462}}{176} & 0 & \frac{3\sqrt{1155}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{330}}{880} & 0 & \frac{\sqrt{165}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{22}}{88} & 0 & \frac{\sqrt{55}}{880} & 0 & -\frac{7\sqrt{110}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{66}}{88} & 0 & \frac{7\sqrt{33}}{176} & 0 & -\frac{\sqrt{330}}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{330}}{176} & 0 & -\frac{7\sqrt{33}}{176} & 0 & \frac{\sqrt{66}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{7\sqrt{110}}{880} & 0 & -\frac{\sqrt{55}}{880} & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{165}}{880} & 0 & \frac{13\sqrt{330}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{1155}}{880} & 0 & -\frac{\sqrt{462}}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x568}}$ $\mathbb{T}_{4,2}^{(1,0;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{176} & 0 & \frac{5\sqrt{22}i}{88} & 0 & -\frac{\sqrt{330}i}{176} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{330}i}{880} & 0 & -\frac{\sqrt{66}i}{88} & 0 & -\frac{7\sqrt{110}i}{880} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1155}i}{880} & 0 & -\frac{\sqrt{55}i}{880} & 0 & -\frac{7\sqrt{33}i}{176} & 0 & \frac{\sqrt{165}i}{880} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{880} & 0 & -\frac{7\sqrt{33}i}{176} & 0 & -\frac{\sqrt{55}i}{880} & 0 & \frac{3\sqrt{1155}i}{880} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{110}i}{880} & 0 & -\frac{\sqrt{66}i}{88} & 0 & \frac{13\sqrt{330}i}{880} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{176} & 0 & \frac{5\sqrt{22}i}{88} & 0 & -\frac{\sqrt{462}i}{176} \\
\frac{\sqrt{462}i}{176} & 0 & -\frac{3\sqrt{1155}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{330}i}{880} & 0 & -\frac{\sqrt{165}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{22}i}{88} & 0 & \frac{\sqrt{55}i}{880} & 0 & \frac{7\sqrt{110}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{66}i}{88} & 0 & \frac{7\sqrt{33}i}{176} & 0 & \frac{\sqrt{330}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{330}i}{176} & 0 & \frac{7\sqrt{33}i}{176} & 0 & \frac{\sqrt{66}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{7\sqrt{110}i}{880} & 0 & \frac{\sqrt{55}i}{880} & 0 & -\frac{5\sqrt{22}i}{88} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\frac{\sqrt{165}i}{880} & 0 & -\frac{13\sqrt{330}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{3\sqrt{1155}i}{880} & 0 & \frac{\sqrt{462}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x569}}$ $\mathbb{T}_{4,1}^{(1,0;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{176} & 0 & \frac{5\sqrt{154}}{616} & 0 & -\frac{\sqrt{2310}}{176} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2310}}{6160} & 0 & -\frac{\sqrt{462}}{616} & 0 & -\frac{7\sqrt{770}}{880} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{21\sqrt{165}}{880} & 0 & \frac{\sqrt{385}}{6160} & 0 & -\frac{\sqrt{231}}{176} & 0 & \frac{\sqrt{1155}}{880} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{880} & 0 & \frac{\sqrt{231}}{176} & 0 & -\frac{\sqrt{385}}{6160} & 0 & \frac{21\sqrt{165}}{880} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{770}}{880} & 0 & \frac{\sqrt{462}}{616} & 0 & \frac{13\sqrt{2310}}{6160} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{176} & 0 & -\frac{5\sqrt{154}}{616} & 0 & -\frac{\sqrt{66}}{176} \\
\frac{\sqrt{66}}{176} & 0 & -\frac{21\sqrt{165}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{2310}}{6160} & 0 & -\frac{\sqrt{1155}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{5\sqrt{154}}{616} & 0 & \frac{\sqrt{385}}{6160} & 0 & \frac{7\sqrt{770}}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{462}}{616} & 0 & \frac{\sqrt{231}}{176} & 0 & \frac{\sqrt{2310}}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2310}}{176} & 0 & -\frac{\sqrt{231}}{176} & 0 & \frac{\sqrt{462}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{7\sqrt{770}}{880} & 0 & -\frac{\sqrt{385}}{6160} & 0 & -\frac{5\sqrt{154}}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{1155}}{880} & 0 & \frac{13\sqrt{2310}}{6160} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{21\sqrt{165}}{880} & 0 & -\frac{\sqrt{66}}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x570}}$ $\mathbb{T}_{4,2}^{(1,0;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{176} & 0 & \frac{5\sqrt{154}i}{616} & 0 & \frac{\sqrt{2310}i}{176} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{2310}i}{6160} & 0 & -\frac{\sqrt{462}i}{616} & 0 & \frac{7\sqrt{770}i}{880} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{21\sqrt{165}i}{880} & 0 & -\frac{\sqrt{385}i}{6160} & 0 & -\frac{\sqrt{231}i}{176} & 0 & -\frac{\sqrt{1155}i}{880} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{880} & 0 & -\frac{\sqrt{231}i}{176} & 0 & -\frac{\sqrt{385}i}{6160} & 0 & -\frac{21\sqrt{165}i}{880} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{770}i}{880} & 0 & -\frac{\sqrt{462}i}{616} & 0 & \frac{13\sqrt{2310}i}{6160} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{176} & 0 & \frac{5\sqrt{154}i}{616} & 0 & -\frac{\sqrt{66}i}{176} \\
\frac{\sqrt{66}i}{176} & 0 & \frac{21\sqrt{165}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{13\sqrt{2310}i}{6160} & 0 & \frac{\sqrt{1155}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{5\sqrt{154}i}{616} & 0 & \frac{\sqrt{385}i}{6160} & 0 & -\frac{7\sqrt{770}i}{880} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{462}i}{616} & 0 & \frac{\sqrt{231}i}{176} & 0 & -\frac{\sqrt{2310}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2310}i}{176} & 0 & \frac{\sqrt{231}i}{176} & 0 & \frac{\sqrt{462}i}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{7\sqrt{770}i}{880} & 0 & \frac{\sqrt{385}i}{6160} & 0 & -\frac{5\sqrt{154}i}{616} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{\sqrt{1155}i}{880} & 0 & -\frac{13\sqrt{2310}i}{6160} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{21\sqrt{165}i}{880} & 0 & \frac{\sqrt{66}i}{176} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x571}}$ $\mathbb{T}_{6,1}^{(1,0;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{1056} & 0 & \frac{\sqrt{154}}{352} & 0 & -\frac{\sqrt{2310}}{352} & 0 & -\frac{\sqrt{462}}{96} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1056} & 0 & -\frac{5\sqrt{462}}{1056} & 0 & \frac{3\sqrt{770}}{352} & 0 & \frac{\sqrt{330}}{96} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{176} & 0 & \frac{\sqrt{385}}{176} & 0 & \frac{5\sqrt{231}}{528} & 0 & -\frac{\sqrt{1155}}{176} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{176} & 0 & -\frac{5\sqrt{231}}{528} & 0 & -\frac{\sqrt{385}}{176} & 0 & \frac{\sqrt{165}}{176} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{96} & 0 & -\frac{3\sqrt{770}}{352} & 0 & \frac{5\sqrt{462}}{1056} & 0 & \frac{\sqrt{2310}}{1056} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{96} & 0 & \frac{\sqrt{2310}}{352} & 0 & -\frac{\sqrt{154}}{352} & 0 & -\frac{\sqrt{66}}{1056} \\
\frac{\sqrt{66}}{1056} & 0 & -\frac{\sqrt{165}}{176} & 0 & -\frac{\sqrt{330}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2310}}{1056} & 0 & \frac{\sqrt{1155}}{176} & 0 & \frac{\sqrt{462}}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{154}}{352} & 0 & \frac{\sqrt{385}}{176} & 0 & -\frac{3\sqrt{770}}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{462}}{1056} & 0 & -\frac{5\sqrt{231}}{528} & 0 & \frac{\sqrt{2310}}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2310}}{352} & 0 & \frac{5\sqrt{231}}{528} & 0 & \frac{5\sqrt{462}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{770}}{352} & 0 & -\frac{\sqrt{385}}{176} & 0 & -\frac{\sqrt{154}}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{462}}{96} & 0 & -\frac{\sqrt{1155}}{176} & 0 & \frac{\sqrt{2310}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{330}}{96} & 0 & \frac{\sqrt{165}}{176} & 0 & -\frac{\sqrt{66}}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x572}}$ $\mathbb{T}_{6,2}^{(1,0;a)}(E_g,1) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{1056} & 0 & \frac{\sqrt{154}i}{352} & 0 & \frac{\sqrt{2310}i}{352} & 0 & -\frac{\sqrt{462}i}{96} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1056} & 0 & -\frac{5\sqrt{462}i}{1056} & 0 & -\frac{3\sqrt{770}i}{352} & 0 & \frac{\sqrt{330}i}{96} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{176} & 0 & -\frac{\sqrt{385}i}{176} & 0 & \frac{5\sqrt{231}i}{528} & 0 & \frac{\sqrt{1155}i}{176} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{176} & 0 & \frac{5\sqrt{231}i}{528} & 0 & -\frac{\sqrt{385}i}{176} & 0 & -\frac{\sqrt{165}i}{176} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{96} & 0 & -\frac{3\sqrt{770}i}{352} & 0 & -\frac{5\sqrt{462}i}{1056} & 0 & \frac{\sqrt{2310}i}{1056} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{96} & 0 & \frac{\sqrt{2310}i}{352} & 0 & \frac{\sqrt{154}i}{352} & 0 & -\frac{\sqrt{66}i}{1056} \\
\frac{\sqrt{66}i}{1056} & 0 & \frac{\sqrt{165}i}{176} & 0 & -\frac{\sqrt{330}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{2310}i}{1056} & 0 & -\frac{\sqrt{1155}i}{176} & 0 & \frac{\sqrt{462}i}{96} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{154}i}{352} & 0 & \frac{\sqrt{385}i}{176} & 0 & \frac{3\sqrt{770}i}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{462}i}{1056} & 0 & -\frac{5\sqrt{231}i}{528} & 0 & -\frac{\sqrt{2310}i}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{2310}i}{352} & 0 & -\frac{5\sqrt{231}i}{528} & 0 & \frac{5\sqrt{462}i}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{3\sqrt{770}i}{352} & 0 & \frac{\sqrt{385}i}{176} & 0 & -\frac{\sqrt{154}i}{352} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{462}i}{96} & 0 & -\frac{\sqrt{1155}i}{176} & 0 & -\frac{\sqrt{2310}i}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{330}i}{96} & 0 & \frac{\sqrt{165}i}{176} & 0 & \frac{\sqrt{66}i}{1056} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x573}}$ $\mathbb{T}_{6,1}^{(1,0;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{64} & 0 & \frac{\sqrt{21}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 & \frac{\sqrt{7}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{64} & 0 & -\frac{5\sqrt{7}}{64} & 0 & -\frac{\sqrt{105}}{64} & 0 & -\frac{\sqrt{5}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{64} & 0 & \frac{\sqrt{210}}{64} & 0 & \frac{5\sqrt{14}}{64} & 0 & \frac{\sqrt{70}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{64} & 0 & -\frac{5\sqrt{14}}{64} & 0 & -\frac{\sqrt{210}}{64} & 0 & -\frac{\sqrt{10}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{64} & 0 & \frac{\sqrt{105}}{64} & 0 & \frac{5\sqrt{7}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{64} & 0 & -\frac{\sqrt{35}}{64} & 0 & -\frac{\sqrt{21}}{64} & 0 & -\frac{1}{64} \\
\frac{1}{64} & 0 & \frac{\sqrt{10}}{64} & 0 & \frac{\sqrt{5}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{35}}{64} & 0 & -\frac{\sqrt{70}}{64} & 0 & -\frac{\sqrt{7}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{21}}{64} & 0 & \frac{\sqrt{210}}{64} & 0 & \frac{\sqrt{105}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{7}}{64} & 0 & -\frac{5\sqrt{14}}{64} & 0 & -\frac{\sqrt{35}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{35}}{64} & 0 & \frac{5\sqrt{14}}{64} & 0 & \frac{5\sqrt{7}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}}{64} & 0 & -\frac{\sqrt{210}}{64} & 0 & -\frac{\sqrt{21}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{7}}{64} & 0 & \frac{\sqrt{70}}{64} & 0 & \frac{\sqrt{35}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{5}}{64} & 0 & -\frac{\sqrt{10}}{64} & 0 & -\frac{1}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x574}}$ $\mathbb{T}_{6,2}^{(1,0;a)}(E_g,2) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{64} & 0 & \frac{\sqrt{21}i}{64} & 0 & -\frac{\sqrt{35}i}{64} & 0 & \frac{\sqrt{7}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{64} & 0 & -\frac{5\sqrt{7}i}{64} & 0 & \frac{\sqrt{105}i}{64} & 0 & -\frac{\sqrt{5}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{64} & 0 & -\frac{\sqrt{210}i}{64} & 0 & \frac{5\sqrt{14}i}{64} & 0 & -\frac{\sqrt{70}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{64} & 0 & \frac{5\sqrt{14}i}{64} & 0 & -\frac{\sqrt{210}i}{64} & 0 & \frac{\sqrt{10}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{64} & 0 & \frac{\sqrt{105}i}{64} & 0 & -\frac{5\sqrt{7}i}{64} & 0 & \frac{\sqrt{35}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{64} & 0 & -\frac{\sqrt{35}i}{64} & 0 & \frac{\sqrt{21}i}{64} & 0 & -\frac{i}{64} \\
\frac{i}{64} & 0 & -\frac{\sqrt{10}i}{64} & 0 & \frac{\sqrt{5}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{35}i}{64} & 0 & \frac{\sqrt{70}i}{64} & 0 & -\frac{\sqrt{7}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{21}i}{64} & 0 & \frac{\sqrt{210}i}{64} & 0 & -\frac{\sqrt{105}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{7}i}{64} & 0 & -\frac{5\sqrt{14}i}{64} & 0 & \frac{\sqrt{35}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{35}i}{64} & 0 & -\frac{5\sqrt{14}i}{64} & 0 & \frac{5\sqrt{7}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{105}i}{64} & 0 & \frac{\sqrt{210}i}{64} & 0 & -\frac{\sqrt{21}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{7}i}{64} & 0 & \frac{\sqrt{70}i}{64} & 0 & -\frac{\sqrt{35}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{\sqrt{5}i}{64} & 0 & -\frac{\sqrt{10}i}{64} & 0 & \frac{i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x575}}$ $\mathbb{T}_{6,1}^{(1,0;a)}(E_g,3) =$

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{2112} & 0 & \frac{\sqrt{1155}}{2112} & 0 & -\frac{9\sqrt{77}}{704} & 0 & \frac{\sqrt{385}}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{2112} & 0 & -\frac{5\sqrt{385}}{2112} & 0 & \frac{9\sqrt{231}}{704} & 0 & -\frac{5\sqrt{11}}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{22}}{704} & 0 & \frac{5\sqrt{462}}{2112} & 0 & \frac{5\sqrt{770}}{2112} & 0 & -\frac{9\sqrt{154}}{704} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{704} & 0 & -\frac{5\sqrt{770}}{2112} & 0 & -\frac{5\sqrt{462}}{2112} & 0 & \frac{9\sqrt{22}}{704} \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{64} & 0 & -\frac{9\sqrt{231}}{704} & 0 & \frac{5\sqrt{385}}{2112} & 0 & \frac{5\sqrt{77}}{2112} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{64} & 0 & \frac{9\sqrt{77}}{704} & 0 & -\frac{\sqrt{1155}}{2112} & 0 & -\frac{\sqrt{55}}{2112} \\
\frac{\sqrt{55}}{2112} & 0 & -\frac{9\sqrt{22}}{704} & 0 & \frac{5\sqrt{11}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{77}}{2112} & 0 & \frac{9\sqrt{154}}{704} & 0 & -\frac{\sqrt{385}}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{1155}}{2112} & 0 & \frac{5\sqrt{462}}{2112} & 0 & -\frac{9\sqrt{231}}{704} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{385}}{2112} & 0 & -\frac{5\sqrt{770}}{2112} & 0 & \frac{9\sqrt{77}}{704} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{9\sqrt{77}}{704} & 0 & \frac{5\sqrt{770}}{2112} & 0 & \frac{5\sqrt{385}}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{9\sqrt{231}}{704} & 0 & -\frac{5\sqrt{462}}{2112} & 0 & -\frac{\sqrt{1155}}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{\sqrt{385}}{64} & 0 & -\frac{9\sqrt{154}}{704} & 0 & \frac{5\sqrt{77}}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{11}}{64} & 0 & \frac{9\sqrt{22}}{704} & 0 & -\frac{\sqrt{55}}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$\boxed{\text{x576}}$ $\mathbb{T}_{6,2}^{(1,0;a)}(E_g,3) = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{2112} & 0 & \frac{\sqrt{1155}i}{2112} & 0 & \frac{9\sqrt{77}i}{704} & 0 & \frac{\sqrt{385}i}{64} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{2112} & 0 & -\frac{5\sqrt{385}i}{2112} & 0 & -\frac{9\sqrt{231}i}{704} & 0 & -\frac{5\sqrt{11}i}{64} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{22}i}{704} & 0 & -\frac{5\sqrt{462}i}{2112} & 0 & \frac{5\sqrt{770}i}{2112} & 0 & \frac{9\sqrt{154}i}{704} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}i}{704} & 0 & \frac{5\sqrt{770}i}{2112} & 0 & -\frac{5\sqrt{462}i}{2112} & 0 & -\frac{9\sqrt{22}i}{704} \\
0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{11}i}{64} & 0 & -\frac{9\sqrt{231}i}{704} & 0 & -\frac{5\sqrt{385}i}{2112} & 0 & \frac{5\sqrt{77}i}{2112} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{64} & 0 & \frac{9\sqrt{77}i}{704} & 0 & \frac{\sqrt{1155}i}{2112} & 0 & -\frac{\sqrt{55}i}{2112} \\
\frac{\sqrt{55}i}{2112} & 0 & \frac{9\sqrt{22}i}{704} & 0 & \frac{5\sqrt{11}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{5\sqrt{77}i}{2112} & 0 & -\frac{9\sqrt{154}i}{704} & 0 & -\frac{\sqrt{385}i}{64} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{1155}i}{2112} & 0 & \frac{5\sqrt{462}i}{2112} & 0 & \frac{9\sqrt{231}i}{704} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{385}i}{2112} & 0 & -\frac{5\sqrt{770}i}{2112} & 0 & -\frac{9\sqrt{77}i}{704} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{9\sqrt{77}i}{704} & 0 & -\frac{5\sqrt{770}i}{2112} & 0 & \frac{5\sqrt{385}i}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{9\sqrt{231}i}{704} & 0 & \frac{5\sqrt{462}i}{2112} & 0 & -\frac{\sqrt{1155}i}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{\sqrt{385}i}{64} & 0 & -\frac{9\sqrt{154}i}{704} & 0 & -\frac{5\sqrt{77}i}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{5\sqrt{11}i}{64} & 0 & \frac{9\sqrt{22}i}{704} & 0 & \frac{\sqrt{55}i}{2112} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$

Cluster SAMB

- Site cluster

** Wyckoff: `2a`

$\boxed{\text{y1}}$ $\mathbb{Q}_0^{(s)}(A_{1g}) = [1]$

- Bond cluster

** Wyckoff: `4b@4c`

$\boxed{\text{y2}}$ $\mathbb{Q}_0^{(s)}(A_{1g}) = \left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$

$\boxed{\text{y3}}$ $\mathbb{Q}_2^{(s)}(B_{1g}) = \left[\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right]$

$\boxed{\text{y4}}$ $\mathbb{T}_{1,1}^{(s)}(E_u) = [0, -i]$

$\boxed{\text{y5}}$ $\mathbb{T}^{(s)}_{1,2}(E_u) = [i,\, 0]$

Site and Bond

Table 4: Orbital of each site

#	site	orbital
1	U	$\lvert\frac{3}{2},\frac{3}{2};d\rangle$, $\lvert\frac{3}{2},\frac{1}{2};d\rangle$, $\lvert\frac{3}{2},-\frac{1}{2};d\rangle$, $\lvert\frac{3}{2},-\frac{3}{2};d\rangle$, $\lvert\frac{5}{2},\frac{5}{2};d\rangle$, $\lvert\frac{5}{2},\frac{3}{2};d\rangle$, $\lvert\frac{5}{2},\frac{1}{2};d\rangle$, $\lvert\frac{5}{2},-\frac{1}{2};d\rangle$, $\lvert\frac{5}{2},-\frac{3}{2};d\rangle$, $\lvert\frac{5}{2},-\frac{5}{2};d\rangle$, $\lvert\frac{5}{2},\frac{5}{2};f\rangle$, $\lvert\frac{5}{2},\frac{3}{2};f\rangle$, $\lvert\frac{5}{2},\frac{1}{2};f\rangle$, $\lvert\frac{5}{2},-\frac{1}{2};f\rangle$, $\lvert\frac{5}{2},-\frac{3}{2};f\rangle$, $\lvert\frac{5}{2},-\frac{5}{2};f\rangle$ $\lvert\frac{7}{2},\frac{7}{2};f\rangle$, $\lvert\frac{7}{2},\frac{5}{2};f\rangle$, $\lvert\frac{7}{2},\frac{3}{2};f\rangle$, $\lvert\frac{7}{2},\frac{1}{2};f\rangle$, $\lvert\frac{7}{2},-\frac{1}{2};f\rangle$, $\lvert\frac{7}{2},-\frac{3}{2};f\rangle$, $\lvert\frac{7}{2},-\frac{5}{2};f\rangle$, $\lvert\frac{7}{2},-\frac{7}{2};f\rangle$

Table 5: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	U	U	[1]	[d,f]	[d,f]

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 6: 'U' (#1) site cluster (2a), 4/mmm

SL	position ($\boldsymbol{s}$)	mapping
1	[0.00000, 0.00000, 0.00000]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 7: 1-th 'U'-'U' [1] (#1) bond cluster (4b@4c), ND, $|\boldsymbol{v}|$= 4.126 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.00000, 1.00000, 0.00000]	[0.00000, 0.50000, 0.00000]	[1,-2,5,-6,-9,10,-13,14]	(1,1)	(1,1)	[-1,0,-1]
2	[-1.00000, 0.00000, 0.00000]	[0.50000, 0.00000, 0.00000]	[3,-4,-7,8,-11,12,15,-16]	(1,1)	(1,1)	[0,1,1]